



syngo MR B19

Operator Manual – *syngo* MR

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The CE mark applies exclusively to medical equipment and products that are released under the relevant EU guidelines mentioned above.

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Information on this Operator Manual

To operate the MR system accurately and safely, the operating personnel have to have the necessary expertise as well as knowledge of the entire operator manual. It has to be read carefully prior to starting up the MR system.

Layout of Operator Manual

To improve readability, your complete operator manual has been broken down into several individual operator manuals with thematically distinct content:

- ☐ Hardware components (system, coils, etc.)
- ☐ Software (measurement, evaluation, etc.)

Information for the operator of the MR system is another part of the complete operator manual.

The volume of smaller operator manuals that you receive depends on your system configuration, and may vary.



All parts of the complete operator manual may contain safety information requiring strict adherence.



The operator manual for hardware and software are intended for authorized users. They assume a basic knowledge of working with PCs and software.

This manual includes descriptions covering standard as well as optional software. Consult the respective Siemens sales organization to determine the type of software available for your system. The description of an option is not legally binding for providing this option.

The graphics, figures, and medical images used in this operator manual are examples only. The actual display and design of these may be slightly different on your system.

For the sake of linguistic simplicity, male and female patients are referred to as "the patient".

The term "Siemens Service" includes Service personnel authorized by Siemens.

Configuration

This manual consists of multiple parts (Part A, Part B, etc.). A comprehensive table of contents can be found at the beginning of each part.



Please observe the content of the safety chapter during your daily activities.

→ [Chapter A.1](#), *Safety Instructions*

The other parts contain descriptive and instructive chapters. The descriptive chapters explain the design, application, and function of the individual MR system components. The instructive chapters explain the operation of the individual components.

In addition, the software includes an Online Help derived from the Operator Manual.

Operator manuals syngo MR and syngo

The **Operator Manual - syngo MR** describes only MR-specific applications. General *syngo* applications, such as e.g. **Viewing** or **Filming**, also available for MR systems, are described in the **syngo Operator Manual**.

Refer to:

→ [Chapter A.3](#), *Differences between syngo and syngo MR*

Important icons

To improve readability, certain special content is identified accordingly. The following presents the icons used and their meaning:

- ✧ Requests for action
- ☐ List items
- Additional information for the user



Notes for optimal use of the MR system.



WARNING/CAUTION

Safety information



Indications for use

Your system is indicated for use as a magnetic resonance diagnostic device (MRDD) that produces transverse, sagittal, coronal and oblique cross sectional images, spectroscopic images and/or spectra, and that displays the internal structure and/or function of the head, body, or extremities.

Other physical parameters derived from the images and/or spectra may also be produced. Depending on the region of interest, contrast agents may be used. These images and/or spectra and the physical parameters derived from the images and/or spectra when interpreted by a trained physician yield information that may assist in diagnosis.

Your system may also be used for imaging during interventional procedures when performed with MR compatible devices such as in-room display and MR-safe biopsy needles.

Federal law restricts this device to sale, distribution and use by or on the order of a physician.

Authorized operating personnel

The system must be operated according to the intended use and only by qualified persons with the necessary knowledge in accord with country-specific regulations, e.g., physicians, trained radiological technicians or technologists, subsequent to the necessary user training.

This user training has to include basics in MR technology as well as safe handling of MR systems. The user has to be familiar with potential hazard and safety guidelines the same way he is familiar with emergency and rescue scenarios. In addition, the user has to have read and understood the contents of the operator manual.

Open source software included

The software delivery volume includes a separate Open Source DVD. This DVD contains the Open Source software packages as well as the associated licensing information installed in this product. For additional information, please read the following files on DVD:

- ❑ release-notes.pdf
- ❑ release-notes-linux.pdf

Data recovery in case of database problems

If database problems occur (e.g., a measurement is terminated or post-processing failed) when reconstructing an application, the images of this application are buffered as standardized DICOM files.

When starting the MR system again, you'll receive the following message:

"Not all images of the last job could be stored. The images have been buffered and are available after you start the system again."

✧ Click **OK** to close the message.

After a new start, the images are appended to the correct study in the database. Use the **Patient Browser** to select the images of the respective application again.



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In this chapter you will find all the general safety instructions for the software manuals **syngo MR**, **Argus**, **Vessel View**, **Neuro 3D** and **Spectroscopy**. Further safety instructions will be found in the descriptions of the individual applications.



CAUTION

Missing or incorrect display of small anatomical structures as a result of motion correction!

Incorrect diagnosis and omitted treatment due to false negative diagnosis.

Incorrect diagnosis and unnecessary biopsies due to false positive diagnosis.

- ✧ For diagnosis, always use the original image in addition to the corrected image.



WARNING

Unapproved software!

System error

- ✧ Use only software authorized by Siemens.
 - ✧ Never copy or install unauthorized software onto the MR system.
-



CAUTION

Inaccurate display of temperature changes when using temperature sensitive sequences!

Incorrect diagnosis, wrong therapy planning

- ✧ Consider the measurement inaccuracy of this method (also refer to relevant literature).
 - ✧ Ensure that the patient does not move during the acquisition of images (also assess anatomical images), and check for sufficiently high SNR.
 - ✧ Consider carefully the temporal and spatial resolution when planning the therapy.
 - ✧ Perform preliminary examinations for analyzing the tissue (e.g. in case of fat myoma), as temperature sensitive sequences use fat suppression by default.
-

CHAPTER **A.2**

Coordinate System

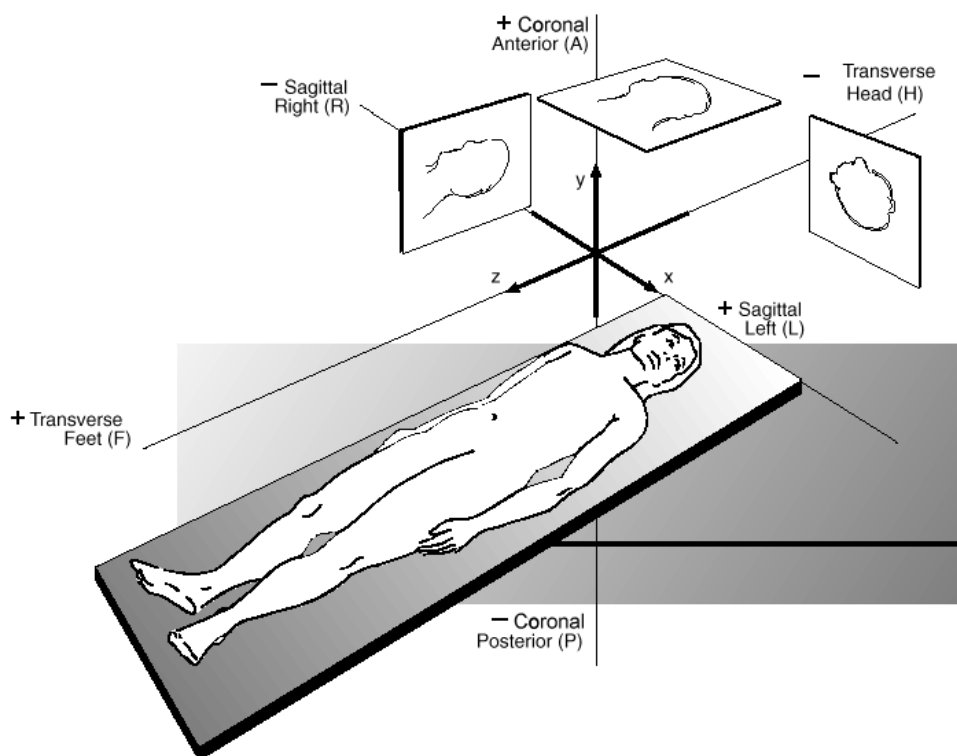
The software **syngo MR** and its predecessor software **Numaris** (including **Numaris 3.5**) use different patient coordinate systems.

Please make a note of this, especially when you diagnose **Numaris** acquired MR images under **syngo MR**.

If you have any questions about converting the different coordinate systems, please contact your applications specialist.

Coordinate system for Numaris 3 and Numaris 3.5 (+FAL)

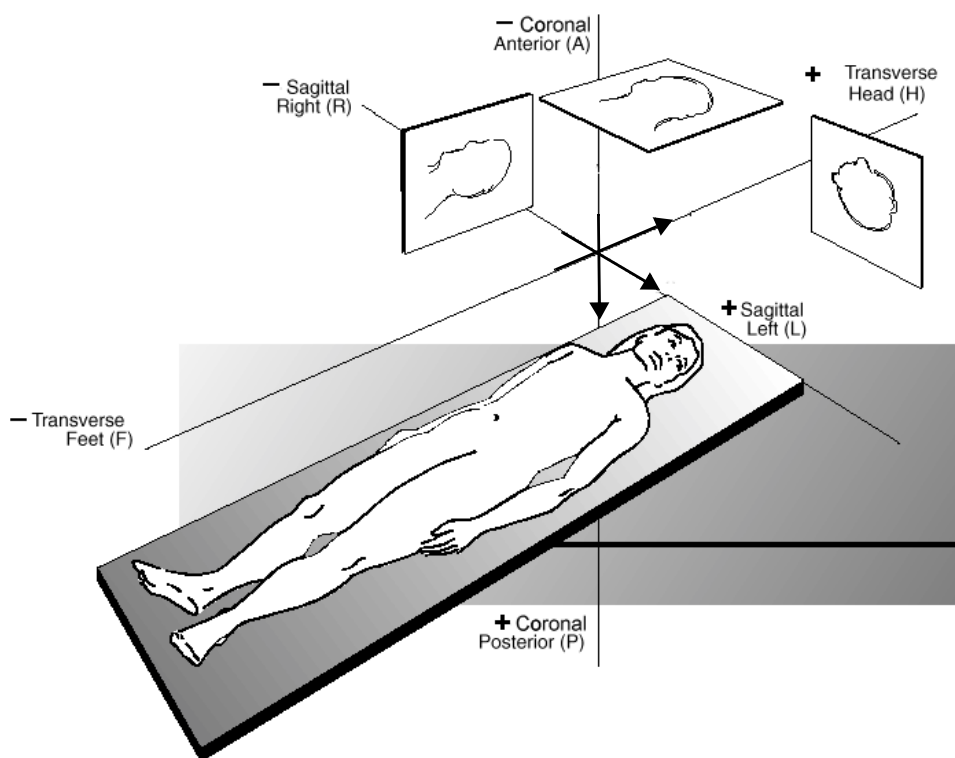
Slice positions as seen from the magnet isocenter into the direction of Feet, Anterior and Left are positive.



MR images based on this patient-oriented coordinate system are only converted when exporting them into a DICOM format.

Coordinate system for syngo MR (DICOM standard, +LPH)

Slice positions as seen from the magnet isocenter in the direction of **L**eft, **P**osterior and **H**ead are positive.





The following overview shows how the direction of the slice position changes as it moves from minus to plus.

Direction of slice position	
Numaris 3 and Numaris 3.5	+FAL
syngo MR	+LPH

If the slice position ascends, the slices move toward the positive coordinates:

Numaris 3 and Numaris 3.5

- ❑ The transverse slice moves in the (F) direction of the feet.
- ❑ The coronal slice moves in the anterior (A) direction.
- ❑ The sagittal slice moves in the left (L) direction.

syngo MR

- ❑ The transverse slice moves in the (H) direction of the head.
- ❑ The coronal slice moves in the posterior (P) direction.
- ❑ The sagittal slice moves in the left (L) direction.

! In **syngo MR**, position information in the PCS is shown in the form of abbreviations L=Left, R=Right, A=Anterior, P=Posterior, H=Head, F=Feet. They replace the algebraic signs "+" or "-" used in previous **Numaris** versions.



Whole Body Patient Coordinate System

Beginning with software version **syngo MR B13**, all measurements provide their table position (TP) and slice position (SP) in the *Whole Body Patient Coordinate System*.

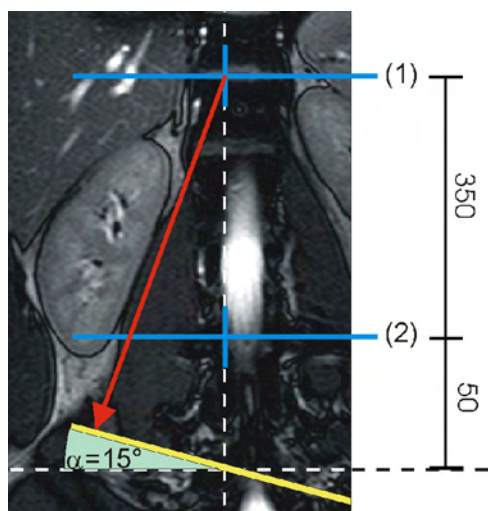
Origin

The origin of the *Whole Body Patient Coordinate System* is located at the magnet isocenter during the first measurement of a new series block (the first table position). In practical application, this is usually the location that was marked with the laser light localizer prior to starting a new study.

Slice position (SP)

Beginning with software version **syngo MR B13**, the slice position (SP) for all measurements is shown relative to the origin of the *Whole Body Patient Coordinate System* of the current series block.

In the following figure you see the computation method for determining a slice position in the Whole Body Patient Coordinate System. In this example, the slice is tilted by 15° and offset by 50 mm from the magnet isocenter.



- (1) Starting table position (TP=0). Origin of the *Whole Body Patient Coordinate System*
- (2) Current table position (TP=F350)

The table positions are computed from the vertical vector from the origin of the *Whole Body Patient Coordinate System* to the measurement slice:

$$(350+50) \cdot \cos 15 = 386$$

TP: F350

SP: F386

Differences between *syngo* and *syngo MR*

The ***syngo MR*** software on your system differs in some aspects from the basic *syngo* version as described in the *syngo* Operator Manual. This chapter provides information regarding these differences.

All applications or topics are presented that show deviations from the original *syngo* version. The following differences are described:

- ❑ *syngo* functions that are not available on your system with ***syngo MR***
- ❑ ***syngo MR*** functions that operate differently than the original *syngo* version
- ❑ ***syngo MR*** functions named different than in the original *syngo* version
- ❑ Functions that are not mentioned in the Operator Manual of the original *syngo* version

Common functions

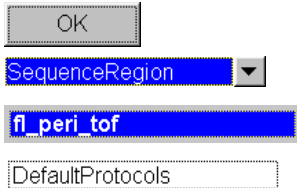
Using the keyboard

Setting the keyboard focus

You can control and operate the input and operating elements of the user interface with the keyboard. In this case, you set a focus on the required operating element, which then reacts to subsequent input from the keyboard.

Within an object (e.g., a window, card, or special area), move the keyboard focus as follows:

- ✧ To move the keyboard focus to the next or previous element, press the **Tab** key or the **Shift + Tab** keys.
- ✧ Move the keyboard focus back and forth until you reach the element you want to operate.



Elements with the keyboard focus are highlighted.

- ☐ Screen keys are highlighted by a dashed frame.
- ☐ List entries and entries in the program control change to a blue background and are highlighted by dashed frames.
- ☐ Entries on the program card are highlighted by a dashed frame.

Keyboard focus in the image area



The keyboard focus may be set only on the entire image area, that is not in individual image segments. The image area is then indicated by a thin white frame. Within the image area you will see the input focus (dashed blue frame) indicating the image segment your entries will effect.

Scrolling within card stacks

If the keyboard focus is on a card stack, you can move individual cards to the foreground via keyboard commands.

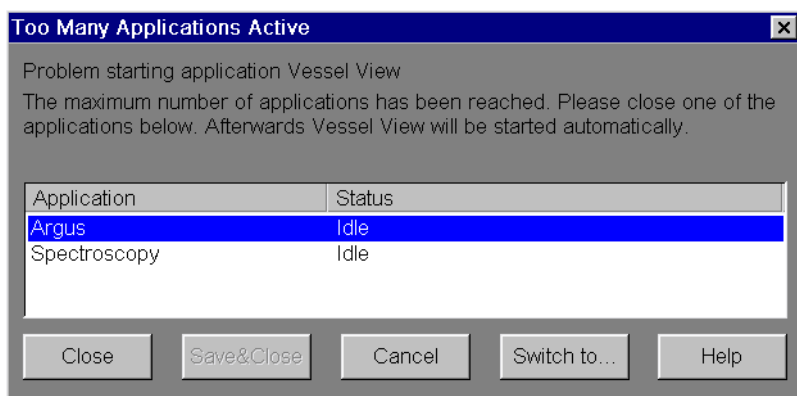
- ✧ Press the **Ctrl** + **←** or **Ctrl** + **→** keys to move the keyboard focus to the left or right.

If the card in foreground is subdivided into cards, e.g., such as the **System** parameter card, you can also move these cards to the foreground with the keyboard.

- ✧ Press the keys **Shift** + **Ctrl** + **←** or the **Shift** + **Ctrl** + **→** keys to move the keyboard focus to the left or right.

Too many active applications

You may only open six applications simultaneously on your MR system. If this number is exceeded, the **Too Many Applications Active** dialog window is displayed.



- ✧ To start the new application, close an application in the list that you are no longer using.

Configuring image text

The following text supplements the description of the **Image Text Editor** in the syngo Operator Manual. You access the **image text editor** via the **NUMARIS/4 – Configuration Panel** dialog window which you select with **Options > Configuration....**

Highlighting image text

You can highlight selected image text. This can be helpful if text would otherwise appear too small in certain layouts (20 and 35 layout).

- ✧ Click the required image text in the image text overview.

The selected text is highlighted in gray and the **B** screen button is now active.

- ✧ Click **B**.

The marked text is displayed in bold.

- ⇒ Image texts can be highlighted only if the **Customized Text** option is selected. You can also cancel text highlighting in the same manner.
- ⇒ Depending on your system configuration, some texts may not be able to be displayed.

B

-
- ! If the orientation marks are deselected, the patient name is not displayed.
-

Configuring the MR system

As authorized user or system administrator, you set **syngo** MR to the configuration of your system. For example, this includes specifying the system type, the optional system components, and the ambient conditions.

- ✧ Select **Options > Customer Service > Local Service**.
- ✧ Enter your password, and press the **OK** button in the **Authentication** dialog window.

Multi-Monitor operation

Multi-Monitor operation is not available on your MR system.

Patient registration



CAUTION

Incorrect input of patient-related data!

Wrong identification and marking

- ✧ Verify that the patient name has been entered correctly.
 - ✧ Prior to the MR examination, correct the data for patient orientation, especially if the patient is to be repositioned during the examination.
-

Emergency registration

Emergency registrations are not intended for MR systems. The corresponding functions are not available.

Missing or contradictory patient data

The following information supplements the description of patient registration in the *syngo* Operator Manual.

Missing weight or age

If the weight or age of the patient was not entered and you click **Exam**, a dialog window appears that points out the missing input.

- The **Exam** task card can be opened only when you entered the weight *and* age of the patient.

Contradictory entries

If contradictory data was provided in several studies, e.g., with respect to patient positioning, a dialog window appears after you click **Exam**. You will be informed about the contradictory information.

- The **Exam** task card can be opened only after all entries for the studies are consistent.

Registering the patient to continue the examination

During registration, you can link a new examination with an already existing study (e.g. from the previous day).

- ✧ Open the **Patient Browser**.
- ✧ Select the study of the patient in the local database.
- ✧ Open **Patient Registration**.

Most data are already set by linking the study with the previous day.

- ✧ Enter missing examination data.
- ✧ Apply data for the examination.



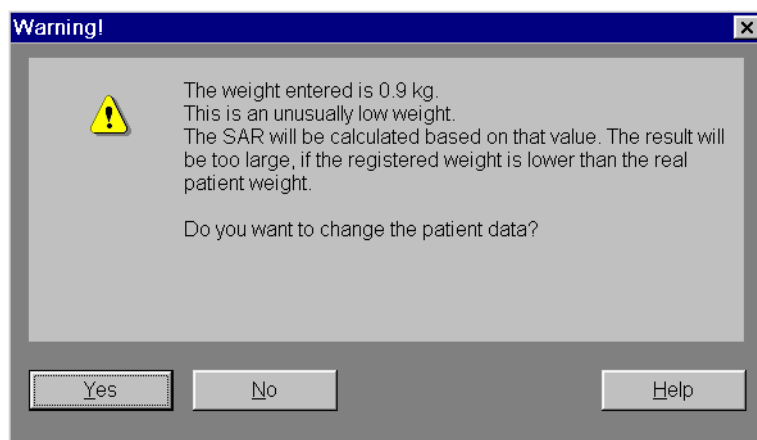
The system does not create a new study. The images measured are added to the study selected during registration.

Checking the weight during registration

Since SAR determination depends on the weight of the patient, special emphasis is put on the correct acquisition of patient data. The following information supplements the description of patient registration in the *syngo* Operator Manual.

-
- ! Register the patient as accurately as possible, e.g., date of birth/age, sex and especially the weight. Depending on the examination, large deviations may lead to a greatly overestimated SAR - which may, e.g., lead to slice reductions- or an underestimated SAR.
-

During registration, the system checks whether the relationship between age and weight is realistic. A warning window is displayed if there are discrepancies.



Yes

No

✧ If the entries have to be corrected, click **Yes**.

You return to the **Patient Registration** dialog window.

✧ To register and examine the patient with these settings, click **No**.

Patient position MR system

The systems use the typical patient positioning as described in the *syngo* Operator Manual.

Orientation

- ☐ **Head first**

Patient lying with his head pointing toward the examination unit.

- ☐ **Feet first**

Patient lying with his feet pointing toward the magnet bore.

Position

- ☐ **Left lateral position**

The patient is lying on his left side.

- ☐ **Right lateral position**

The patient is lying on his right side.

- ☐ **Supine position**

The patient is lying on his back.

- ☐ **Prone position**

The patient is lying on his stomach.

Patient Browser

Splitting mosaic images

Some post-processing functions, e.g. **3D**, cannot use mosaic images. In special cases it may be necessary to split them into single images again.

Splitting a mosaic image

- ✧ In the **Patient Browser** select *one* mosaic image.
- ✧ Select **Applications > Mosaic > Split to one Series**.

The images of the selected mosaic image are saved in a new series under the same study.

Splitting several mosaic images

- ✧ Select several mosaic images or all mosaic images of a series in the **Patient Browser**.
- ✧ Select **Applications > Mosaic > Split to one Series**.

The images of *all* select mosaic images are saved in *one* new series.

Or

- ✧ Select **Applications > Mosaic > Split to one Series per Mosaic**.

A new series is created with the split images for each mosaic image selected.

Canceling splitting

-
- ! Splitting images may take some time. The process runs in background and is completed even if the current user logs off or another user logs on.
-

✧ Select **Applications > Mosaic > Abort Splitting process....**

-
- ! Canceling a splitting operation in progress after a user change is only possible if the new user has the necessary access rights.
-

syngo MR Workplace

The following important information on working with a *syngo* MR Workplace supplements the description of the Patient Browser in the *syngo* Operator Manual.

-
- ! If you select the Patient Browser at a *syngo* MR Workplace, the Browser is not updated automatically.

✧ In this case, select **View > Refresh**.



Transferring images to the 3D task card

If you want to apply the images selected in the Patient Browser to the **3D** task card, please consider the following regarding **syngo MR**:

The corresponding menu items are available under **Applications > 3D** and *not* under **Patient**.

Procedure tracking with MPPS

On MR systems, the **Modality Performed Procedure Step** dialog window, which you can select in the **Patient Browser** for procedure tracking, does not contain a subtask card with information regarding the **radiation dose**.

Media Browser

The **Media Browser** is not supported by your MR System.

Pixel Lens in Viewing

Basics

In an MR system, **Viewing** includes enhanced **Pixel Lens** functions.

The **Pixel Lens** displays the average value of gray scales of a small 5×5 pixel area. In *syngo* MR the patient coordinates are displayed together with this value.

The displayed coordinates refer to the **Whole Body patient coordinate system**. The zero point of the coordinate system is at the table position located in the magnet isocenter during the first measurement of a series block (in practice, it is the position marked by the light localizer prior to the beginning of a new MR examination).

→ [Chapter A.2, Coordinate System](#)

Prerequisites for coordinate display

The patient coordinates are displayed only if the reference image meets the following conditions:

- ☐ The image is a Siemens MR image
- ☐ The image is not a mosaic image
- ☐ The image has a defined DICOM image plane

Restrictions

- ❑ The images used *have to* belong to the same series block.
→ [Page C.1–6, Series block](#)
- ❑ The result images may contain distortion artifacts as a result of the non-linearity of the gradient field inside the magnet. Image distortion artifacts in the visible image are *not* corrected by the coordinates displayed.
→ [Chapter E.2, Correcting Distortion Artifacts](#)
- ❑ The patient coordinate is displayed only if the inaccuracy of the specified image coordinates lies within a 7 mm tolerance margin.
- ❑ For Composing images, only the transverse part of the **Pixel Lens** coordinates is shown in the image.
- ❑ The user is responsible for using the correct (mutually compatible) images for comparing the **Pixel Lens** coordinates.



WARNING

Coordinate data when using the pixel lens function!

Incorrect diagnosis

- ✧ Use the pixel lens function only in the center of MR images.
 - ✧ Do not use the pixel lens function to compare coordinates between images with distortion correction and those without.
 - ✧ Reference the coordinates of the pixel lens function only to the table position in the lower right in the image.
-

! Do not copy Pixel Lens markers from image to image!

If a **Pixel Lens** marker is copied, it does not appear at the identical patient coordinate in the images, but rather at the same distance to the upper left corner of the images!

! Do NOT use the Pixel Lens when multiple images are selected!

If several images are selected, the **Pixel Lens** marker does not appear at the identical patient coordinate in the images, but rather at the same distance to the upper left corner of the images!

Using the Pixel Lens

The **Pixel Lens** enables you to use gray scale and coordinate values as follows:

- ☐ Displaying the values of the Pixel Lens at the mouse pointer position
- ☐ Inserting the Pixel Lens marker at a fixed position
- ☐ Selecting, moving, or deleting the Pixel Lens marker
- ☐ Saving the Pixel Lens marker with the reference image

Activating / deactivating the Pixel Lens



When the **Pixel Lens** is activated, the values of the current mouse pointer position are displayed.

✧ Select **Extras > Pixel Lens**.

- Or -

Click this button of the **Extras** subtask card.

The cursor changes shape. The grayscale value is shown in the first line. The second line shows patient coordinates.

✧ Deactivate the **Pixel Lens** in the same manner as you activated it.

Inserting a marker at a fixed position

You can also enter the pixel values in an image.

- ✧ Activate the **Pixel Lens** function.
- ✧ Click the pixel of interest with the left mouse button.

The pixel is marked with a small circle. The grayscale value is shown in the first line. The second line shows patient coordinates.



The circular marker of the **Pixel Lens** does not match the area measured to determine the grayscale value. It has a constant size and is independent of the zoom factor of the image.

The gray scale value is always determined by a 5 x 5 pixel area.

Selecting and moving the marker

You can move a fixed **Pixel Lens** marking that has been inserted.

- ✧ Deactivate the **Pixel Lens** function.
→ [Page A.3–19, Activating / deactivating the Pixel Lens](#)

- ✧ Move the mouse pointer to the marking.

The arrow cross is displayed in enlarged format.

- ✧ Press and hold the mouse button and move the marking with the mouse to the required position.

The grayscale value and patient coordinates are updated.

Application examples:

- ❑ **How to proceed with the application examples:**

The procedure is identical for the three examples presented:

- ✧ Ensure that only one image is selected.

! Do NOT use the Pixel Lens when multiple images are selected!

If several images are selected, the Pixel Lens marker does not appear at the identical patient coordinate in the images, but rather at the same distance to the upper left corner of the images!

-
- ✧ With the **Pixel Lens** activated, mark the required vertebra in the first image via mouse click.

! Do not copy Pixel Lens markers from image to image!

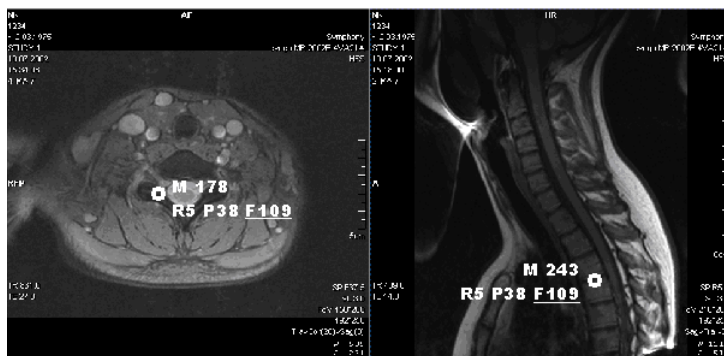
If a Pixel Lens marker is copied, it does not appear at the identical patient coordinate in the images, but rather at the same distance to the upper left corner of the images!

-
- ✧ Move the marker in the second image until the patient coordinates are the same as shown in the first image.

-
- ! Please note the inaccuracies that may be displayed when using this application.
-

❑ Example 1

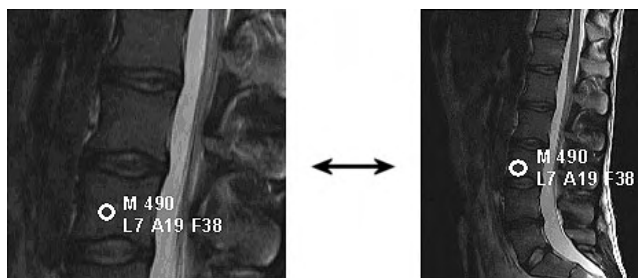
You are searching for the patient coordinates from the transverse image to the left in the second, sagittal image to the right.



□ Example 2

You compare two sagittal images that belong to the same series block but have a different FOV.

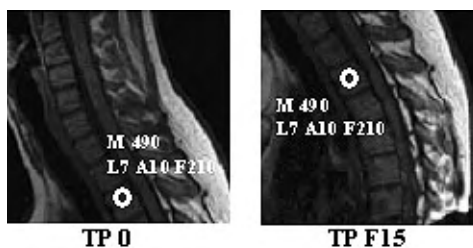
The first image shows only parts of the lumbar vertebrae, the second image shows the entire length of the lumbar vertebrae.



□ Example 3

You compare two sagittal images of the same series block, but at different table positions.

The first image shows the upper section, the second image shows the lower section of the cervical spine.



Deleting a marker



You delete a marking when it is no longer required.

✧ Click the marker to select it.

The marker changes shape.

✧ Press **Del**.

The marker of the **Pixel Lens** is deleted.

Saving the marker with the image

When images are saved, the **Pixel Lens** markers in the images are saved as superimposed graphics.

→ **Viewer/Store images** in the *syngo* Operator Manual

The same applies to exporting and filming images.

Viewing

2D Evaluation

Adhere to the following information regarding 2D evaluation of MR images:

-
- ! For optimum results, use quantitative analysis on distortion-corrected images only.

This is especially important when evaluating along the image margin or when using a large FOV.

-
- ! Note the following special feature for explicitly selected images of a different matrix:

If you draw a graphic in an image, this graphic is not displayed exactly at the same position or at the same size in the other images.

Windowing with the keyboard

The following information supplements the description of windowing with the keyboard in the *syngo* Operator Manual:



If a dialog window is open and active in **Viewing**, the keys of the symbol keypad may not respond.

✧ In this case, click the image area to deactivate the dialog window or just close it.

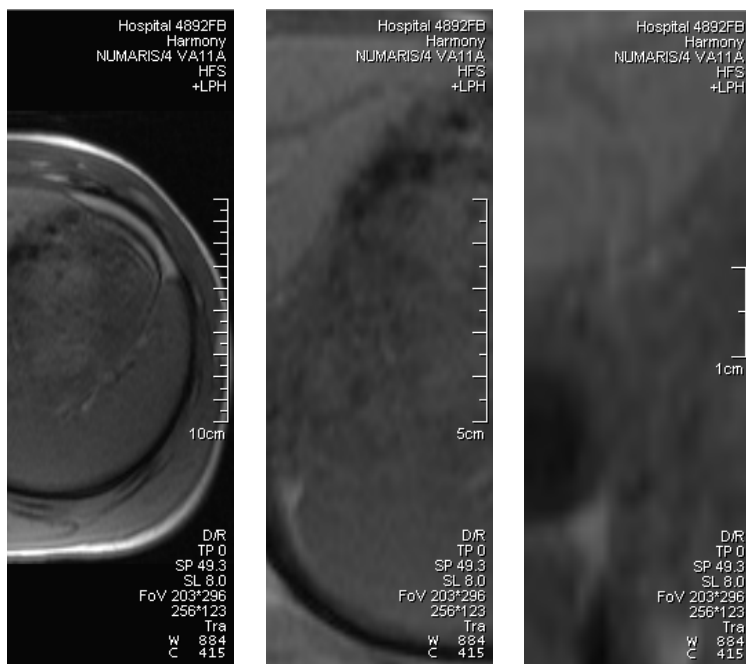
Displaying images with LUTs

The use of look-up tables (LUT) for the display of images is not supported by your MR system.

Scale in MR images

The scale is displayed at the right border of the segment and clarifies the proportions in the image.

The scale adapts to the current proportions in the image.



- ☞ The scale bar is not displayed if it would otherwise hide the image text.

Saving and exchanging data

MOD storage medium

Your MR system supports neither reading from nor writing to MOD data storage media.

A **MOD Reading station** was specially developed for reading archived data from MOD data media.

If you have any questions, please contact your Siemens sales organization.

Storage media CD and DVD

Your MR system does not provide **archiving functions**.



The term **archiving** used in the syngo Operator Manual should be interpreted as **Save** for MR applications.

CDs and DVD-Rs can be kept only for a limited time, and are therefore only suited for limited data storage. For this reason, you can only use CDs and DVDs as storage medium with **syngo MR**, but not for archiving.

Displaying images with a rectangular matrix

The description of displaying images with a rectangular matrix in the *syngo* Operator Manual *does not* apply to ***syngo* MR**. The following section replaces that part of the *syngo* Operator Manual.

In some circumstances, images imported from an external storage medium or hard drive may be displayed with a reduced pixel matrix.

-
- ! When transferring edited, non-quadratic matrices/Viewer segments, diagnostically relevant image areas can be lost.

To prevent this, the original image should be included when edited images are exported or sent to another workstation.

We strongly recommend to include original images for the final diagnosis. It should not be made based solely on edited or modified images.

Behavior and possible occurrence

With the exception of the **3D** task card, this behavior occurs in all task cards when exporting or sending modified images.

It is caused by a mismatch between the size of the image matrix and the copy segment.

Example: A rectangular image matrix should be displayed in a square image segment or vice versa.

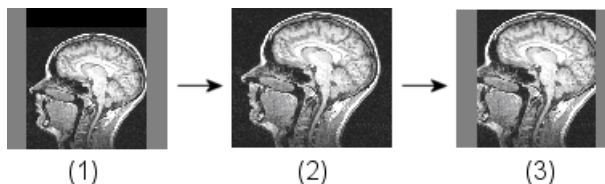
If an image has been zoomed, reduced, panned or rotated in a task card, the modified image is displayed correctly, i.e., it fills the entire image segment.

After being saved (using the **Save As** function), the visible pixel matrix is reduced when this image is exported or sent, and then displayed again. At the edges, the image information of the newly displayed image is replaced by black borders.

If the image contains an ROI that is partially covered by black borders, the statistical information refers to the complete ROI and not just to the area that remains visible.

Example 1

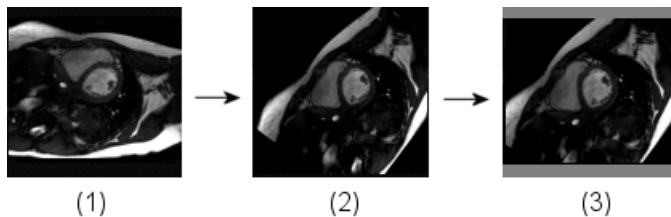
Zooming and panning rectangular images:



- (1) Original image, Workplace 1
- (2) Zoomed and panned image, Workplace 1
- (3) Exported / sent image, Workplace 2

Example 2

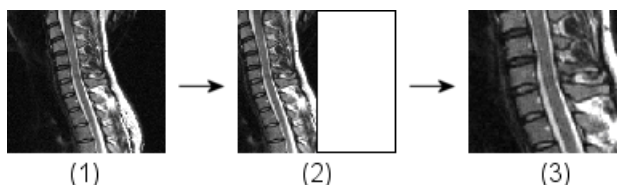
Rotating rectangular images:



- (1) Original image, Workplace 1
- (2) Rotated image, Workplace 1
- (3) Exported / sent image, Workplace 2

Example 3

Display of a square matrix in a rectangular Viewer segment:



- (1) Original image, square matrix
- (2) Image in a rectangular Viewer segment
- (3) Image on Workplace 2

Running multiple movies

If you are working in study compare mode, you can use Auto Movie to simultaneously run all loaded studies and series as a movie. Note the following features with **syngo MR**:

The menu items for the corresponding functions are available under **Scroll > Sorting** and *not* under **Scroll > Link**.

Filming

Automatic filming

Automatic filming during the examination is not supported by your MR system.

Displaying the reference image

The following section replaces the section of the same name in the *syngo* Operator Manual:

You can display a small reference image (Mother) corresponding to the image (Child) in the film sheet segment. This simplifies the orientation and overview of the measured region.

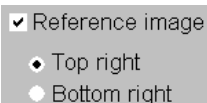
You set the display of the reference image in the **Film Properties** dialog window, or when configuring a film sheet.

✧ Select **Filming > Properties....**

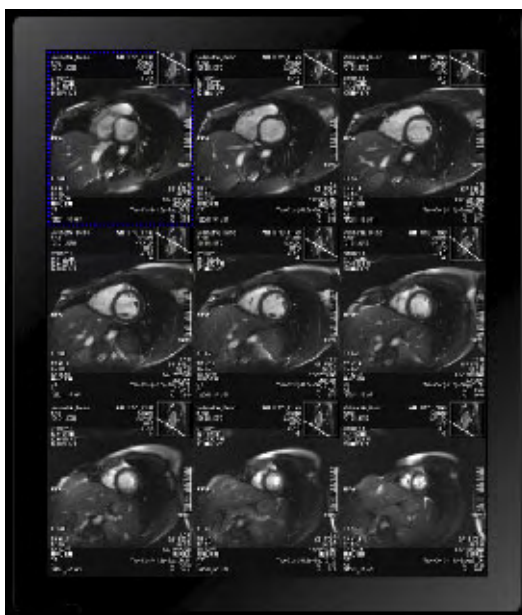
The **Film Properties** dialog window opens.

✧ Activate the **Reference image** checkbox.

✧ Select an option for the position (**Top right** or **Bottom right**) of the reference image.



The system searches the database for a reference image used for slice positioning of the images to be filmed, and displays it in all segments of the current film sheet. The image text is moved to the left.



Displaying cut lines

To enable the position of the image (Child) to be displayed as a cut line in the reference image (Mother), the Mother and Child have to meet the same prerequisites as reference images.

→ [Page C.4–4, Prerequisites](#)

Similar rules apply for the display of cut lines as they do for graphical slice positioning:

- ☐ A perpendicular slice is represented by a solid line.
- ☐ An oblique slice is represented by a dashed line.
- ☐ No cut line is displayed if the angle between the Mother and Child is less than 30°.

Loading another image as the reference image

Instead of the displayed reference image, you can use any other image of the same examination as the mother as long as it was measured at the same table position.

- ✧ Activate the **Reference Image** option and select the position of the reference image.
- ✧ Open the **Patient Browser**.
- ✧ Drag a suitable image to the film sheet with the mouse.
- ✧ If additional images from the series to which the child belongs are loaded on the film sheet, the new reference image is also displayed there automatically.

Subsequently windowing the reference image

The small reference image may be displayed on the film sheet with unsuitable windowing values.

You cannot window the small reference image directly on the film sheet, but rather in **Viewing** only:

- ✧ Load the reference image into the **Viewing** task card.
- ✧ Window the reference image and close the patient.

The new windowing values are permanently applied to the reference image.

- ✧ Drag the modified reference image from the **Patient Browser** to the film sheet with the mouse.

Reorganizing film sheets

The following section supplements the description on film sheet reorganization in the *syngo* Operator Manual.

You can change the order of images of a series on the **Filming** task card.

- ✧ Place the input focus on an image from this series that you want to select.
- ✧ Select **Edit > Select Series**.
 - Or -
 - Select **Select Series** in the context menu.

- ✧ Delete the selected series from the film job.
 - **Filming**/*Scrolling through film and selecting images* in the syngo Operator Manual
- ✧ Change the image sort in the **Patient Browser**.
 - **Patient Browser**/*Sort Data* in the syngo Operator Manual
- ✧ Load the series again into the **Filming** task card.
 - **Filming**/*Copying images to filming* in the syngo Operator Manual

The images are displayed in the new sort order.

3D evaluation

Preferred orientation

Menu item **Orientation Control** to activate automatic default alignment is available under **Orientation** but *not* under **Options**.

Clip planes

Clip planes on the **3D** task card are not supported by your MR system.

For series containing a large number of images, it is often useful to display the position of images in an overview. The cut lines are calculated for acquired images with the original reference images or other suitable images.

The following two variants are available for position display:

☐ Offline position display

→ [Page A.4–5](#)

☐ Inline position display

→ [Page A.4–15](#)

You may display single images, complete series, or several series during offline in the position display. You may configure the display and label cut lines yourself.



For computing position display, you need to have full access rights to the data to be evaluated.

Detailed information regarding all security settings in *syngo* are included in the *syngo* Security Package brochure.

Basics

The position display shows the cut lines of acquired images in other images belonging to a new series.

Definition of terms

Term	Description
Data image	Position line of an image will be displayed
Data series	Series containing the data image
Reference image	Image on which the position line of a data image will be displayed
Reference series	Series containing the reference image

Prerequisites

For calculating the position lines, the images have to meet the following prerequisites:

General remarks

- ☐ The images are **MR images** only.
- ☐ The images are not Secondary Capture images (e.g., Alpha images).
- ☐ The images have an image plane.
- ☐ The images are not mosaic images.
- ☐ The images are part of the current examination of the current patient.

Images from software versions prior to syngo MR B15:

- ☐ The images must have the same frame of reference.
→ [Page C.1–7, Frame of Reference \(FoR\)](#)
- ☐ Reference images may not be distortion corrected.
→ [Page E.2–7, Resetting 2D distortion correction](#)

Images from software versions beginning with syngo MR B15:

- ❑ The images must have the same frame of reference. Beginning with **syngo MR B15**, the type of distortion correction is part of the FoR.
→ [Page C.1–7, Frame of Reference \(FoR\)](#)
- ❑ The images are part of the same series block.
- ❑ The images are of type *ND*, *DIS2D* or *DIS3D*.
- ❑ For reference images of type *ND*, all images were measured at the same table position. These conditions do not apply to reference images of type *DIS2D* and *DIS3D*.
- ❑ Type *DIS2D** images cannot be used as reference images for position display.
→ [Page P.2–17, List of image types](#)

Offline position display

You create the offline position display on the **Viewing** task card by defining the data and reference images and subsequently starting with the calculation.

Defining data and reference images

You've loaded the images or series for position display into the **Viewing** task card.

The sequence used for selecting images explicitly or implicitly in the next step automatically determines the images to be used as data images or reference images.

→ [Page A.4–13](#), *Calculation examples, Position Display*

Selected images are part of series S1

Example 1: You have selected images from a S1 series.

The system identifies the reference series R1 used for slice positioning of S1 and uses R1 as a reference series for position display.



The calculations for reference series R1, R2, etc. have to be completed prior to calculating the slice position for series S1.

- If the S1 series was planned based on images from different reference R1 and R2 series, the sum $R1 + R2$ will be used as the reference series in position display.
- If no reference series exists for S1, the images of S1 will be mapped onto themselves, if it is possible to calculate cut lines.

Selection includes several series (S1, S2, and S3)

Example 2: You have selected series S1, S2, and S3.

All images from the series selected *last* (S3) and/or all images from the series that contains the last image selected, will be used as reference images.

The images of series S1 and S2 will be used as data images.

Starting calculation

Single images or all images of one or more series may be displayed in the position display.

Displaying entire series



- ✧ Select **Evaluation > Position Display > Series**.

- Or -

Click the **Patient** subtask card using this button.

The cut lines with the reference images are calculated and displayed for all images of the data series.

Displaying individual images



- ✧ Select **Evaluation > Position Display > Selection**.

- Or -

Click the **Patient** subtask card using this button.

The cut lines are calculated and displayed with the reference images only for selected images of the data series.

- ⇒ A message will notify you if you did not select suitable images and it is not possible to calculate cut lines.

The calculation runs in background. The status bar shows messages during the course of the measurement. During calculation, you can continue working in the **Viewing** task card.

The new Position Display series

Calculating the position display generates a new series. It includes reference images for which at least one cut line can be calculated and displayed.

The image at the top of the stack in the new series is the image with the greatest number of perpendicular (or nearly perpendicular) cuts.

All images of the new series are fitted to the segment size, even if they had been zoomed or panned prior to calculating the position display.

You may edit images in the position display series.

→ **Basics**/Zooming and panning images in the syngo Operator Manual

Name of the new series

The name of the new series consists of **PosDisp:**, the series number set in square brackets, plus the name of the first data series.

Example: **PosDisp: [2] t1_se_sag**

Number of the new series

The number of the new series is the highest series number assigned thus far +1.

If a series number has not yet been assigned, e.g., after system installation, the **PosDisp** series will be given the number 5000.

Labeling and display of cut lines

The slice positions of the data images are drawn in the **PosDisp** images and labeled as cut lines.

The length of the cut lines does not always represent the spatial extent. It may be cut off at the edges of the images to make space for image text and cut line labels. Cut lines may also be truncated for legends.

Labeling rules

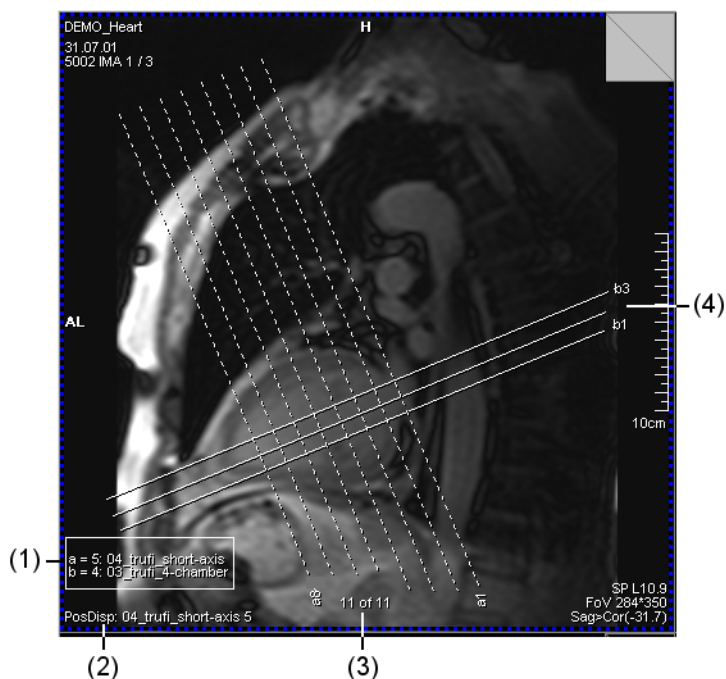
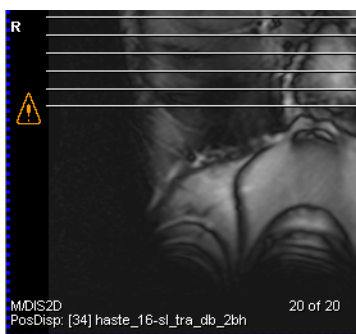


Image legend

- (1) Allocation of letters to series (offline only)
- (2) Name of the new position display series
- (3) Line 1: Number of images displayed in the reference image and number of the total images in (a) data series
Line 2: Image commentary on the reference image
- (4) Image numbers



For strongly distorted images (Type *DIS2D**) a triangular warning symbol appears in the left segment margin.

Due to possible distortion artifacts of *DIS2D** images, geometric allocation of cut lines may be inaccurate.

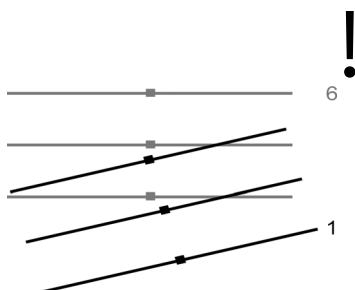
The cut lines are labeled with the number of the image, if the **PosDisp** series shows images of *one* series only.

→ [Page C.6–36, Numbering reconstructed images](#)

If the **PosDisp** series shows images of *several* series, the cut lines are labeled with a lower-case letter to identify the series, plus the number of the image. If multiple cut lines are covered, the lowest image number is displayed followed by three periods.

The comment line of the **PosDisp images** (3) shows how many data images are displayed as cut lines in the **PosDisp** image.

If you have entered comments in the reference image, these comments are shown in the second comment line of the **PosDisp** image.



If you have configured the Position Display so that not all cut lines are labeled, the allocation of overlapping cut lines to cut-line numbers that are no longer visible may be incorrect.

In cases of this kind, select the labeling option **All** to be on the safe side.

→ [Page A.4–18](#)

Line display

Cut line	Display
Perpendicular cut	Solid
Oblique cuts (angle $\geq 30^\circ$)	Dashed
Oblique cuts (angle $< 30^\circ$)	None
General cut lines	Always with a shaded line

Displaying in the segment

The **PosDisp** images are optimally adapted to the available segment: For every image, an attempt is made to display the relevant part where the cut line is shown fully. In the case of reduced display (e.g., for data series with combined images), the zoom factor may not fall below 60%.

Calculation examples, Position Display

Example 1 (Series)

All 15 images of a transverse series are to be displayed on the original reference images. The original reference series consists of a transverse, a coronal, and a sagittal image.

- ✧ Load the transverse series to the **Viewing** task card.
- ✧ Place the input focus on an image from this series.
- ✧ Select **Evaluation > Position Display > Series**.
 - Or -
 - Click the **Patient** subtask card using this button.



The position display is calculated. The new **PosDisp** series contains the sagittal and coronal reference images, each with 15 cut lines.



Example 2 (selection)



A coronal and a transverse series are loaded into the **Viewing** task card. The cut lines of selected coronal images are to be displayed in the transverse images.

- ✧ Explicitly select the coronal images of interest.
- ✧ Select an image from the transverse series.
- ✧ Select **Evaluation > Position Display > Selection**.
 - Or -
 - Click this button on the **Patient** subtask card.

The position display is calculated. The new **PosDisp** series contains all images of the transverse series for which cut lines of the selected coronal images exist.

Example 3 (several series)



A coronal, transverse as well as sagittal series are loaded into the **Viewing** task card. The cut lines of all transverse and sagittal images are to be displayed in the coronal images.

- ✧ Explicitly select a transverse and a sagittal image.
- ✧ Select an image from the coronal series.
- ✧ Select **Evaluation > Position Display > Series**.
 - Or -
 - Click this button on the **Patient** subtask card.

The position display is calculated. The new **PosDisp** series contains all images of the coronal series including the cut lines of the transverse and sagittal images.

Inline position display

You create an Inline position display in the **Exam** task card by activating the protocol properties of the respective sequence prior to the measurement.

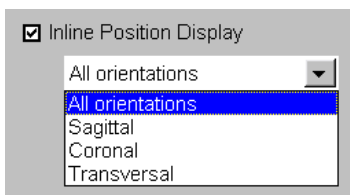
In this case, a position display is created parallel to the data series in the *Inline* measurement. Different from the offline position display, only *one* data series per overview series is generated.

Activating Inline position display

- ✧ Select a protocol in the **Exam Explorer**.
 - Or -
 - Select the pending protocol from the job list.
- ✧ Select the **Protocol properties** dialog window with the **Auto Load** subtask card.
 - [Page C.8–1](#), *Editing the properties for individual protocols*
- ✧ Activate the **Inline Position Display** checkbox.
- ✧ For protocols that do not support Inline position display, the **Inline Position Display** checkbox is hidden.
- ✧ From the corresponding selection list, select the reference images and their orientation that you would like to use for position display.



☒ Inline Position Display



The following options are available for selection:

☐ **All orientations**

Position display considers all available reference images, independent of their orientation.

☐ **Sagittal**

Position display only considers reference images of sagittal orientation

☐ **Coronal**

Position display only considers reference images of coronal orientation.

☐ **Transverse**

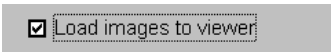
Position display only considers reference images of transverse orientation.

⇒ If a reference series does not contain images of the selected orientation, an Inline position display is generated on the basis of all reference images available.

✧ Confirm your settings with **OK**.

When measuring the edited protocol, a position display is created only parallel to the data series. It is also appended directly to the data series in the **Exam** task card.

⇒ Reference images without visible cut lines are not shown in the Inline position display.



You are also able to automatically load the Inline position display into **Viewing**.

- ✧ Go to the **Auto load** subtask card and activate the **Protocol properties** dialog window and then the **Load images to viewer** checkbox.

Configuring the position display

You can configure the number of slices to be drawn into the reference images, the location of cut line labels, as well as how many cut lines should be numbered.

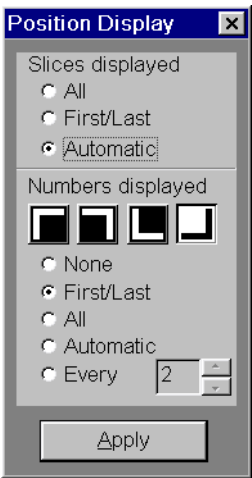
Selecting position display

To change the default setting for position display without a **PosDisp** series loaded in the **Viewing** task card, open the **Position Display** dialog window via the menu.

- ✧ Select **Evaluation > Position Display > Settings**.
- Or -
- ✧ Place the input focus on the **PosDisp** series.
- ✧ Click one of these buttons in the **Patient** subtask card.



The **Position Display** dialog window opens.



Configuring the display of cut lines

You can choose the number of cut lines to be displayed.

Option	Description
All	All cut lines are displayed.
First/Last	Only the first and last cut lines of a parallel line group are displayed.
Automatic	Cut lines that are too close to one another are automatically hidden. Overlapping cut line labels are avoided.

✧ Select an option under **Slices**.



The system automatically switches from **First/last** to **Automatic** as soon as the first and last cut line of a group *cannot* be calculated.

This does not affect any other line groups.

Determining the position of the cut line annotation



You may set a preferred position for the cut line annotations in the **PosDisp** image.

✧ Click the appropriate button to place the annotations on the left, on the right, below, or above.

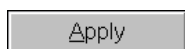
Determining the display of the cut line annotation

If you are editing a **PosDisp** series containing several slices, it is not advisable to label every cut line. The following display options are possible:

Option	Result
None	No cut lines are labeled.
First/Last	Only the first and last cut lines from each parallel and equidistant line group are labeled.
All	All cut lines are labeled.
Automatic	All cut lines are labeled as long as they do not overlap. Individual annotations are hidden if they do overlap.
Every	Only every nth cut line is labeled. Sorting is based on image number and starts at 1.

- ✧ Select the desired option.
- ✧ If you have selected the **Every** option, enter an increment in the adjacent field.

Applying the settings



How to save configuration changes:

- ✧ Click the **Apply** button in the **Position Display** dialog window.

The dialog window will stay open. The images selected in the **Viewer** segment are calculated anew with the configuration settings.

- The saved settings also apply after the system is restarted.



When saving the configuration settings while loading a large number of images into **Viewing** at the same time, the loading of images is given priority. For this reason, the new series calculated with the current configuration settings cannot be stored. The following message appears.

Complete the configuration setting only after the loading process running simultaneously in **Viewing** has ended.

Configuring automatic data transfer

All newly reconstructed images saved to the database can be sent automatically to specific network nodes. You can configure your MR system accordingly.

→ **Basics**/Sending data to network nodes in the syngo Operator Manual

The configuration established applies to all images of your MR system.

Do not transfer PosDisp images automatically

You can exclude **PosDisp** images from automatic data transfer.

- ✧ Select **Evaluation > Position Display > Disable Auto Transfer**.
- ☞ The setting is saved and remains even if you restart the MR system.

Introduction

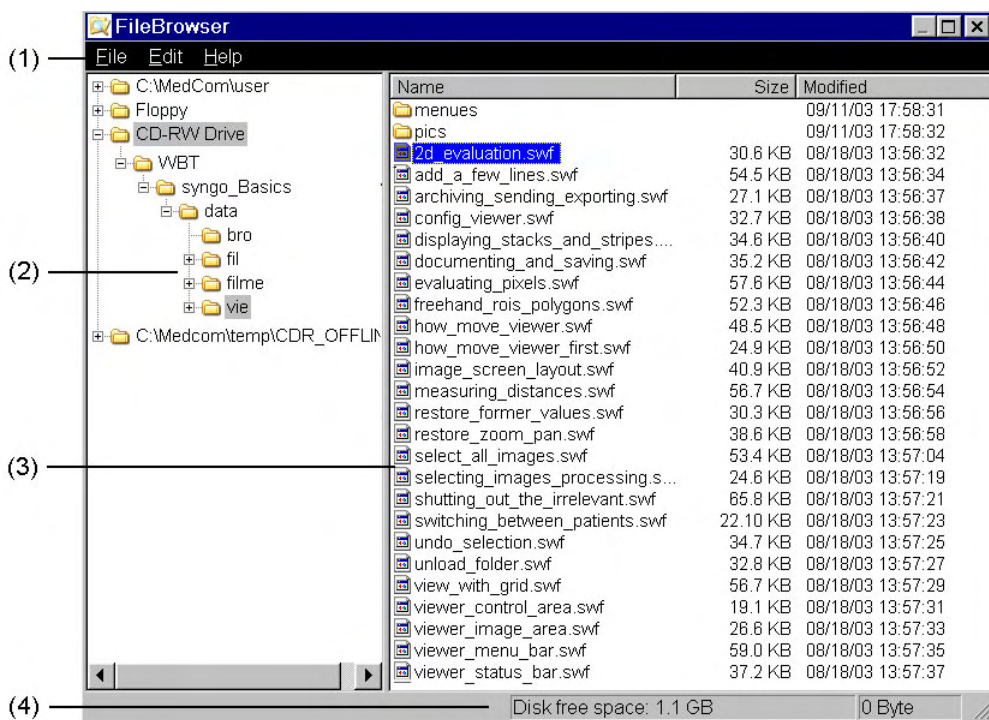
The **File Browser** provides you with the opportunity to access and manage a clearly defined portion of the NUMARIS/4 file system.

The **File Browser** enables:

- ❑ Copying or moving files (DICOM image data), i.e., to the CD folder for export
- ❑ Access to generated reports and AVI files
- ❑ Access to files downloaded via NUMARIS *life*
- If Security (HIPAA) is activated, you need export authorization for the **File Browser**. Detailed information regarding all security settings in *syngo* is included in the *syngo* Security Package brochure.

Selecting the File Browser ✧ Select **Options > File Browser** from the main menu.

The File Browser opens.



- (1) Menu bar
- (2) View of the directory
- (3) Directory view with subdirectories and files
- (4) Status bar with display of free disk space

✧ You cannot open more than one **File Browser** at a time.

Closing the File Browser ✧ Select **File > Close** in the **File Browser** menu.

Functions of the File Browser

Opening files and folders

✧ Double-click the desired file or folder.

Or

✧ Select a file or folder.

✧ Select **File > Open**.

Exporting files and folders

You are able to export files and folders to the following destinations:

- ☐ Floppy disk (A:\)
- ☐ Memory stick (depending on the system setting)
- ☐ CD burn folder
- ☐ Additional released folders (depending on the system setting)

✧ Select a file or folder.

✧ Select the export destination under **File > Send to**.

The file or folder is copied to the export target.

Managing files and folders

The **File Browser** provides the standard functions of the usual file explorer.

Function	Comment
Cut	You move an object (file or folder) via cut and paste.
Copy	You copy an object (file or folder) via copy and paste.
Paste	Inserts a previously cut or copied object (file or folder) into the current directory.
Rename	Displays the name of the file or folder to be processed in an input field.
New folder	Creates a new folder and displays an input field where you can enter a new name.
Delete	Deletes the selected file or folder once you confirm the safety query.

- ✧ Select files or folders.
- ✧ Select the desired function from the **File** or **Edit** menu.
 - Or -
 - Select the desired function in the context menu of the **File Browser**.
- ✧ When creating or renaming a file or folder, enter a new name.
- ✧ Preinstalled folders and files cannot be deleted, moved, or renamed.

The *syngo* **Control-i** license activates a software interface at the MR console to which a compatible third-party medical device may connect. Following authentication and user confirmation, such third-party device can then guide the MR workflow and synchronize it with its own workflow.

For instance, a therapeutic third-party device that uses MR for imaging during the treatment procedure may exchange information with the MR system in order to synchronize patient registration, protocol management and measurements.



If a remote connection is established, this icon is displayed in the status bar.



In contrast, the red slash indicates that no connection is available.

Additionally, the following applies:

- ☐ The characteristics of the MR system are not changed by *syngo* **Control-i**.
- ☐ To allow a third-party device to connect, a device's request must be authenticated and confirmed by the user.
- ☐ At any one time, connection to one third-party device is possible.
- ☐ The connection may be severed by the user from the MR console at any time.
- ☐ The connection is constantly monitored. If it fails, the user is notified and the MR console maintains normal functionality (e.g. with real-time DICOM image transfer discontinued).
- ☐ Measurement stop functionality is available at the MR host at all times, as well as from the third-party device.

- ❑ Safety relevant messages are passed to the third-party device.
- ❑ All pop-ups of the MR system are maintained on the MR console, irrespective of whether a third-party device is connected. If a third-party device is connected, the user is notified via the third-party's user interface that a pop-up needs to be observed on the MR console.
- ❑ All images are maintained on the MR console, irrespective of whether a third-party device is connected.
- ❑ If a third-party device is connected, copies of the MR images may be sent to it in real time in DICOM format.
- ❑ The third-party device and its combined use with the MR system must have all required approvals stipulated by local regulatory provisions in order to be used for diagnostic or therapeutic procedures on humans.
- Further details will be provided by the third-party device manufacturer.

Managing and Adjusting the System

B.1 *System Manager*

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B.2 *Adjusting the System*

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System Manager

Your MR system comprises the following components:

- ❑ Magnetic resonance system (MR Scanner)
- ❑ Image reconstruction system for image reconstruction
- ❑ *syngo* Acquisition Workplace for operating **syngo MR**
- ❑ Optional: the *syngo* MR Workplace

The **syngo MR System Manager** is a module that allows you to monitor, as well as start and stop the individual components of the measurement system.

-
- ! Siemens Service is able to restart the image reconstruction system and MR Scanner via remote control.
-

Starting the system

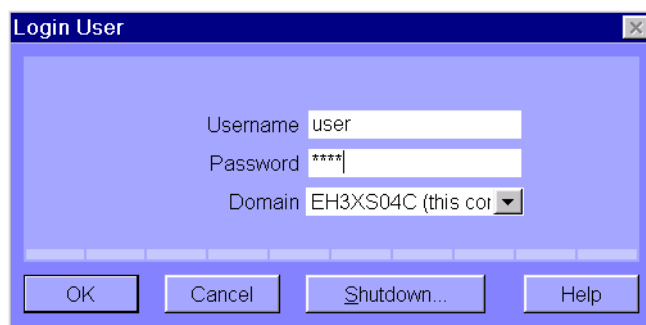
The MR system can be switched on via the alarm box. No preliminary examination steps may be performed at the MR system during switch on.

- ✧ Turn the key switch at the alarm box to the right.
- ✧ Press the **SYSTEM ON** button.

The **System On** LED lights up. The MR system is switched on.

Without user administration The computer system boots automatically. The software starts at the *syngo* Acquisition Workplace.

With user administration If you have installed the *User administration (HIPAA)* option, the **Login User** dialog window opens.



- ✧ Enter your user name and password.
- ✧ You switch on the *syngo* MR Workplace separately from the *syngo* Acquisition Workplace.

Controlling the MR system

You control and operate your MR system from the *syngo* Acquisition Workplace. The program **syngo MR** available for this purpose runs under Windows XP®. The **syngo MR** program comprises individual applications, for example **Examination** or **BOLD**.

You control the MR system via the **System Manager** dialog window and **End Session**.

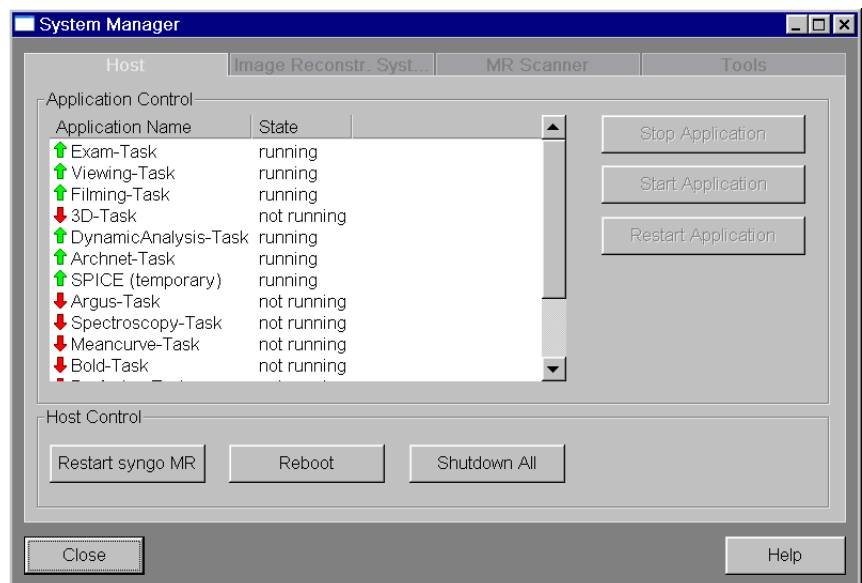
Selecting control dialogs

Selecting the System Manager dialog window

- ✧ Select **System > Control**.

The **System Manager** dialog window opens.

- ✧ Open the **Host** subtask card.



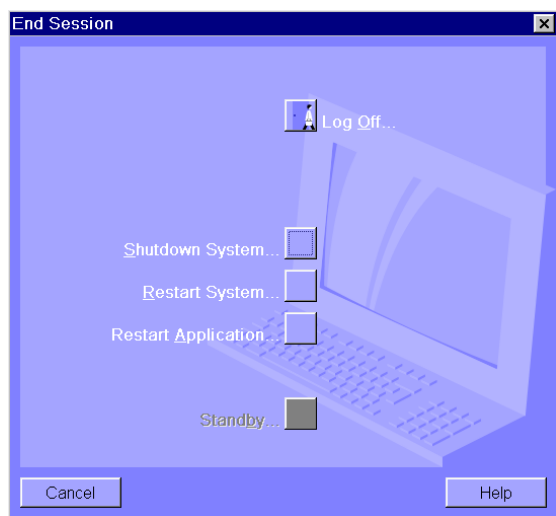
Selecting the End session dialog window

✧ Select **System > End Session**.

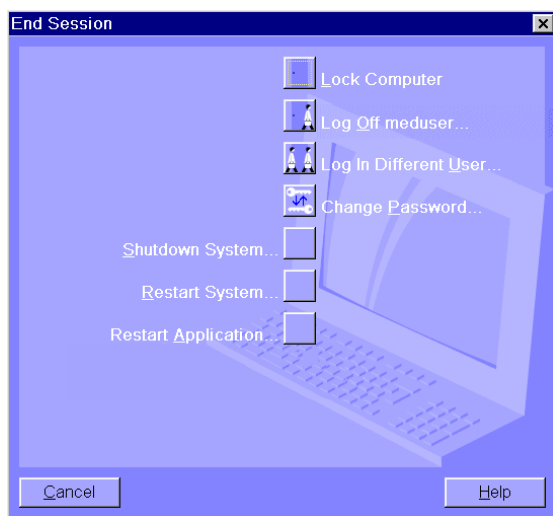
The **End Session** dialog window opens.

When user administration is active, additional functions are available.

→ [Page B.1–9, User administration \(additional functions\)](#)



Dialog window without user administration

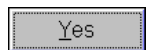
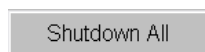


Dialog window with user administration (HIPAA)

Switching off/starting the system again

Depending on the circumstances, you can switch off the entire MR system or start the control processor or **syngo MR** again.

Shutting down the entire system



At the end of the workday, you switch off the entire system, that is, the MR scanner and the image reconstruction system at the *syngo* Acquisition Workplace.

✧ Click this key in the **End Session** dialog window.

Or

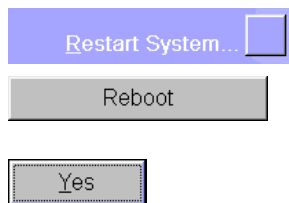
✧ Click **Shutdown All** in the **Host** subtask card of the **System Manager**.

✧ To shut off the system, confirm the safety query with **Yes**.

➤ Once you have closed down all programs and the operating system, you have to switch off the components of the measurement system via the alarm box. This is the only way to ensure that the **Image reconstruction system** will boot when you start the system.

Refer to the System Manual.

Starting the control processor again



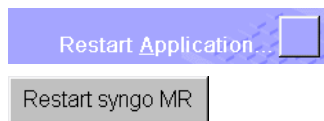
Due to an error in the operating system or in **syngo MR** you may have to shut down the *syngo* Acquisition Workplace and restart it. The other components of the system such as the MR Scanner are not affected.

- ✧ Click this key in the **End Session** dialog window.
 - Or -Click **Reboot** in the **Host** subtask card of the **System Manager**.

- ✧ To shut off the *syngo* Acquisition Workplace, confirm the safety query with **Yes**.

The **syngo MR** program and the operating system are shut down and then started again.

Rebooting *syngo MR*



When **syngo MR** does not react, you can start it separately.

- ✧ Click this key in the **End Session** dialog window.
 - Or -Click **Restart syngo MR** in the **Host** subtask card of the **System Manager**.

The program **syngo MR** shuts down and restarts. The operating system is *not* shut down.



Controlling applications

If one of the applications no longer reacts, you may close the applications separately without shutting down **syngo MR** or the operating system.

The applications that run under **syngo MR** on your *syngo* Acquisition Workplace are listed in the **Application Name** list on the **Host** subtask card in the **System Manager**.

The status of each application is also displayed:

Icon	Status	Comment
	Running	The application is running
	Not running	The application is not running
	Waiting	The application status has not been received shortly after starting the System Manager
	Not responding	It is not possible to communicate with the software components
	Unknown name	Certain components of the application are not available
	Stop in progress	The application is being shut down
	Startup in progress	The application is booting

Stopping an application

A rectangular button with a light gray background and a thin blue border. The text "Stop Application" is centered in a dark gray font.

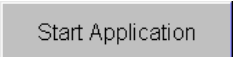
✧ Select the application from the list.

✧ Click **Stop Application**.

Initially, the application shows the status **Stop in progress**. Then it changes to **Not running**.

A message window keeps you informed about the course and progress made by the action.

Starting an application

A rectangular button with a light gray background and a thin blue border. The text "Start Application" is centered in a dark gray font.

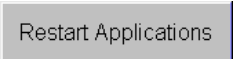
You may explicitly start applications that you have stopped or that do not start automatically.

✧ Select the application from the list.

✧ Click **Start Application**.

✧ Initially, the application shows the status **Startup in progress**. Then it changes to **Running**.

Restart application

A rectangular button with a light gray background and a thin blue border. The text "Restart Applications" is centered in a dark gray font.

You can shut down and restart the applications, e.g., if they are no longer responding.

✧ Select the application from the list.

✧ Click **Restart Applications**.

User administration (additional functions)

The **End Session** dialog window contains a number of additional functions when *User Administration (HIPAA)* is active.

✧ Select **System > End Session**.

The **End Session** dialog window opens.

➤ Refer to the *syngo* Security Package brochure for detailed information regarding *User administration (HIPAA)*.

Locking the computer



If you have to leave the MR system for a short time, you can protect the data on the computer from unauthorized access without having to log off.

✧ Click this button.

A registration window appears that covers the program surface.

Logging off

When you want to end your work at the MR system, please log off.

! When a user logs off, unsaved data will be lost irretrievably.

Always check for unsaved data and save any data you want to keep before logging off.



✧ Click this button.

Opened patient data are unloaded. The log-off window is displayed in full screen size.

Another user can log onto the system.

Logging on a new user



A new user can log onto the system even if the current user has not logged off.

- ✧ Click this button.

The registration window is displayed. It hides the user interface of the program.

- ✧ Enter the new user name.



If you have the same user rights as the previous user, you can continue working with the data loaded. If you have fewer rights, the data used by the previous user will be unloaded. Unsaved data will be lost.

If a measurement or image reconstruction of the previous user is still running in background, the current user cannot start until the previous user's measurement or image reconstruction has finished.

Changing your password



- ✧ Click this button.

A dialog window opens.

- ✧ Enter the old password in the first field for authentication.
- ✧ Enter the new password in the next field.
- ✧ Enter the new password in the third field for verification.
- ☞ Remember, passwords are case sensitive!

Monitoring the image reconstruction system

The raw data supplied by the MR system are reconstructed with the **Image Reconstruction System**. As with the control processor, you can also shut down and restart this processor.

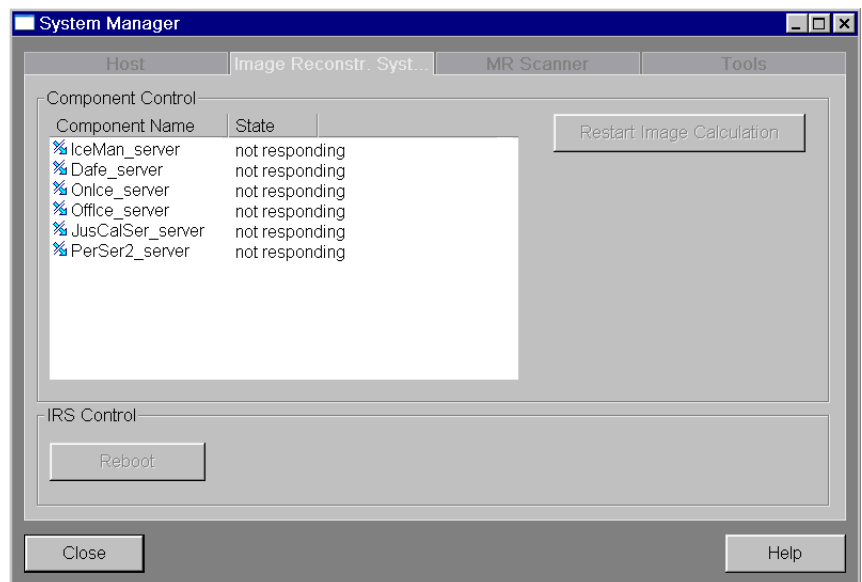
- ✧ The Image Reconstruction System can be put into the standby status together with the MR Scanner.

→ [Page B.1–16](#), *Switching the power supply to standby*

- ✧ Select **System > Control**.

The **System Manager** dialog window opens.

- ✧ Open the **Image Reconstr. Syst.** subtask card.





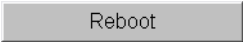
Status of the image reconstruction system

The individual status of the components of the image reconstruction system is shown in the **Image Reconstruction System** subtask card of the **System Manager**:

Icon	Definition
	Component is ready
	Component is not ready
	Component is operational to a limited extent
	Component status has not been reported.

Rebooting the image reconstruction system

In the event of errors, you can shut down and reboot the **Image Reconstruction System**.



✧ Click **Reboot**.



Restarting software components

The **Component Control** area of the **Image Reconstruction System** subtask card shows the status of the software components.

In case a software error occurred during image reconstruction, you can restart the software component responsible for this task.



✧ Click **Restart Image Calculation**.

All software components are shut down and then restarted.

The system is ready for acquisitions again once the components of the image processor have been restarted.

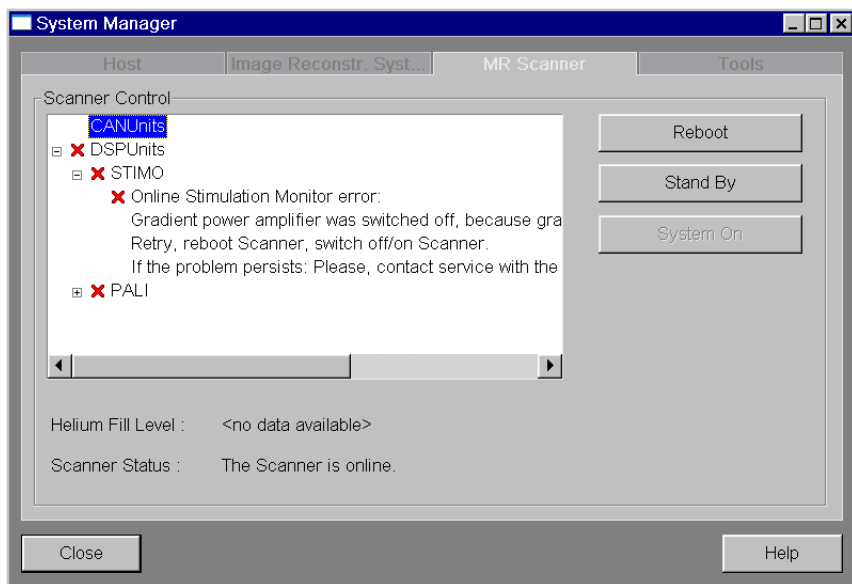
Controlling the MR system

The computer system of the MR Scanner has an operating system as well. You can display the current status of individual components of the MR Scanner and switch the power supply on or off.

✧ Select **System > Control**.

The **System Manager** dialog window opens.

✧ Open the **MR Scanner** subtask card.



Operating modes

The **Component Name** list displays the status of individual measurement system components hierarchically. The main components include CANUnits, PALIs, and e.g., the Stimulation Monitor.



Icon	Definition
	The unit is ready.
	The unit is not ready for operation.
	The unit is operational within limits.
	Component status has not been reported.

⇒ The list is updated automatically when a connection to the MR Scanner exists (MR Scanner is online).

Helium fill level

Helium Fill Level : ... %

The helium fill level is shown in percent below the list on the **MR Scanner** subtask card.

Overall status of the MR Scanner

The overall status of the **MR Scanner** is displayed below the helium fill level:

	The MR Scanner is online.
	The MR Scanner is offline or is booting.


Reboot

Switching the power supply to standby

Stand By

Rebooting the operating system

If an error occurs in the operating system of the MR Scanner during operation, you can eliminate the error by shutting down and rebooting the operating system.

✧ Click **Reboot**.

Switching the power supply to standby

If you do not intend to perform measurements for a longer period of time at the *syngo* Acquisition Workplace or only want to edit or evaluate images, you can save energy by switching off the voltage supply to the MR Scanner and the Image Reconstruction System.



As soon as you are in the standby mode, you are no longer able to perform measurements.

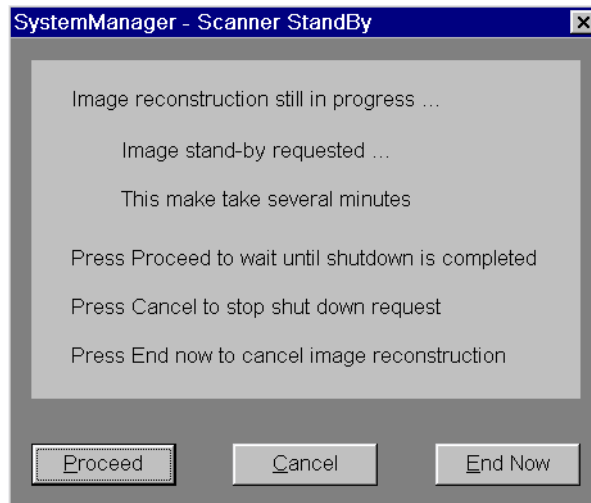
✧ Click **Stand By**.

- Or -

Switch the MR scanner and image reconstruction system to standby via the Alarm Box.

If the Image Reconstruction System no longer contains jobs in the job list, it is immediately switched to standby followed by the MR Scanner.

If the Image Reconstruction System still contains jobs in the job list, the **Scanner StandBy** dialog window appears:



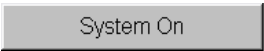


Select the required approach:

Option	Consequences
Proceed	The Image Reconstruction System is shut down only after reconstructing current jobs. This may take several minutes. There is no associated loss in data. The dialog window is closed and cannot be selected again.
Cancel	Image Reconstruction System shut-down is terminated.
End now	Image reconstruction in progress is ended immediately. Data may be lost in this case.

✦ Click the key for the desired option.

Switching the power supply on again



When you end the standby mode and continue with acquisitions, please switch on the power supply.

- ✦ Click **System On**.
- ☞ You can also use the alarm box to switch the power supply back on for the **MR Scanner** (*refer to the System Manual*).

Central status display in the status line

A measurement job consists of a measurement sequence and an image reconstruction or image post-processing job. The measurement job is either in the **Acquisition Mode** or **Background Mode**.

The measurement job status is clearly displayed via icons on the system status bar.

Statuses in the Acquisition Mode

As long as the sequence has not completed its job, the measurement remains in **Acquisition Mode**. Only one measurement may be in Acquisition Mode at any time. Possible statuses:



There is at least one system error. Measurements are not possible until the error is corrected.



The previous measurement has at least one acquisition error. The error does not affect subsequent measurements.

Action: A new measurement may be started. If a new measurement is started, all errors in the previous measurement are reset. The pictogram for "no errors" is displayed, refer to the next line.



There are no errors.

Statuses in Background Mode

As soon as the measurement job is completed and the images from image reconstruction or post-processing have not been saved to the database, the measurement is in **Background Mode**. Multiple measurements may be in background mode. Possible statuses:



No errors have been reported.



Errors were reported that occurred sometime between image reconstruction or post-processing and the images being saved to the database.

Example:



Explanation:

- ☐ The current measurement is error-free.
- ☐ In one of the previous jobs that has not yet been saved, an error occurred after reconstruction.
- Double-clicking the icon in the status bar opens a dialog window with information.

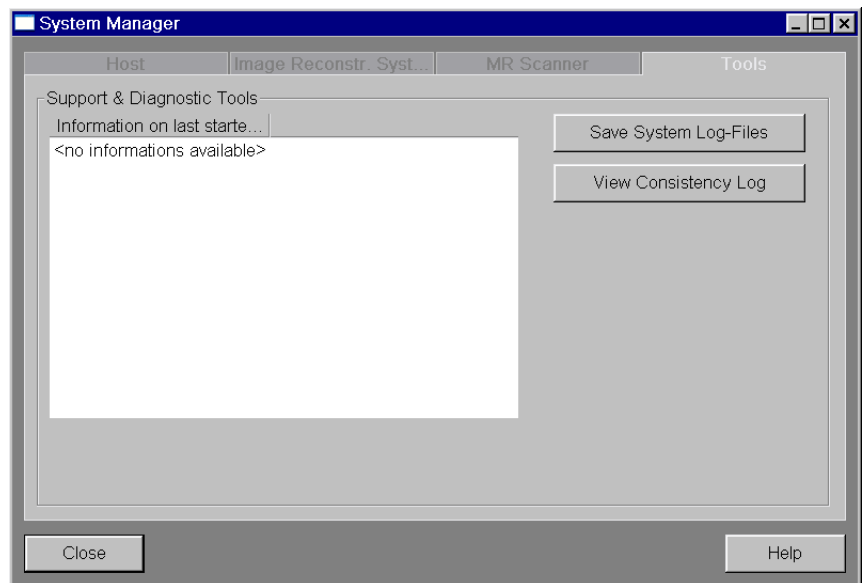
Tools

The **Tools** subtask card is used to acquire diagnostic information or to check important system files.

- ✧ Select **System > Control**.

The **System Manager** dialog window opens.

- ✧ Open the **Tools** subtask card.



Gathering diagnostic information

In case of errors, it is useful to create an MrLog data set. This data set collects information regarding the status and history of the MR system at the time of error. The information is needed for error analysis. These include:

- ☐ Information with respect to the databases
 - ☐ System information
 - ☐ Log files
- ⇒ The diagnostic information contained in it can be used to help locate the error.

Generating the MrLog data set

A rectangular button with a light gray background and a thin black border. The text "Save System Log-Files" is centered in a dark gray font.

- ✧ Click **Save System Log-Files**.

- Or -

- ✧ Select **Programs > MrSaveLog** via the start menu.

The MR system is now collecting data for the new MrLog data set. The programs in progress are shown in the **MrSaveLog** DOS window.

After a short time, the **Save Log MRLog...** dialog window is opened as well.

The MrLog files generated are stored in the directory C:\Med-Com\MriDiagnostic. The MrLog files of a data set have a common name master:

MrLog_Computername_Date_Time_Datatype.zip
(or *.txt)

Acquiring own error information

You can acquire additional information about the error via the **Save Log MrLog...** dialog window.

Save Log MrLog_YKTJ002109_20071116_121816

SYSTEM

Save Log ID: MrLog_YKTJ002109_20071116_121816

Serial Number: 10009

Hospital Name: YKTJ002109 /2D96048783636D/FEA31B/

Product Name: Verio

SW Labels: COEM VE10B, syngo VE31G, isvnao Vx69A

USER

Reported by: Petro Miller

Time: 11/16/2007, 10:58 AM

Summary: Viewing task card is without reaction

Details: Scenario:
1. Transfer of the patient data to syngo MM Workplace
2. Scan
3. Mean Curve task card was started

Workaround:

OK

Additionally, you can add the following information in the **User** area:

- ☐ Name of the person registering the error
 - ☐ Time of error
 - ☐ Brief description of the error
 - ☐ Detailed description of the error
 - ☐ Description of the workaround
- ✧ Enter the respective information into the fields.

✧ Click **OK**.

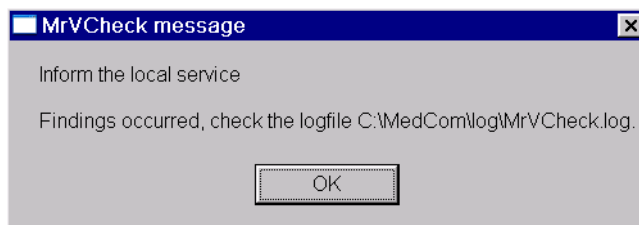
Your additional information is added to the current MrLog data set and stored. The dialog window closes.

Displaying changed files

The most important system files are checked for changes every time the system boots. The result is saved in a consistency log file.

Message

If changes are noticed, the following message appears:



✧ Click **OK** to close the message.

Viewing the consistency log file



- ✧ Click **View Consistency Log** in the **Tools** subtask card.

The contents of the log file are shown in the **Support & Diagnosis Tools** window.

- ✧ Please do not install additional software on the system. It may replace existing system files. This may also adversely affect the system's ability to acquire data.

Displaying system information

The **syngo MR** program provides you with information regarding your measurement system.

Selecting the dialog

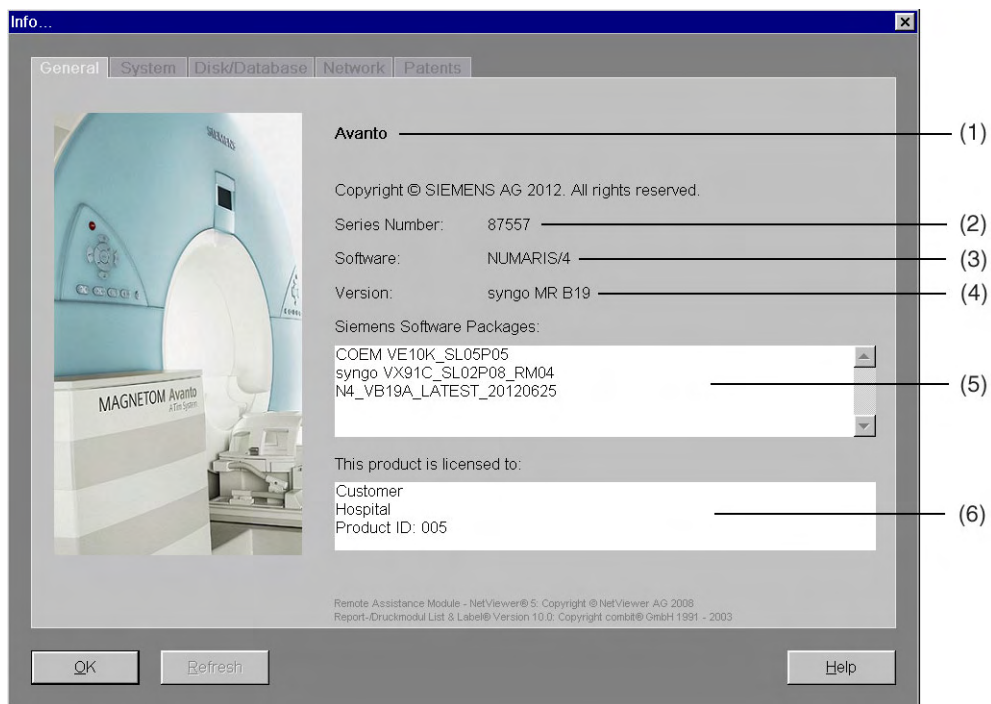
- ✧ Select **Help > Info ...**

The **Info ...** dialog window opens.

General system information

General system information is required, e.g., when you contact Siemens Service regarding problems.

- ✧ Click the **General** subtask card.



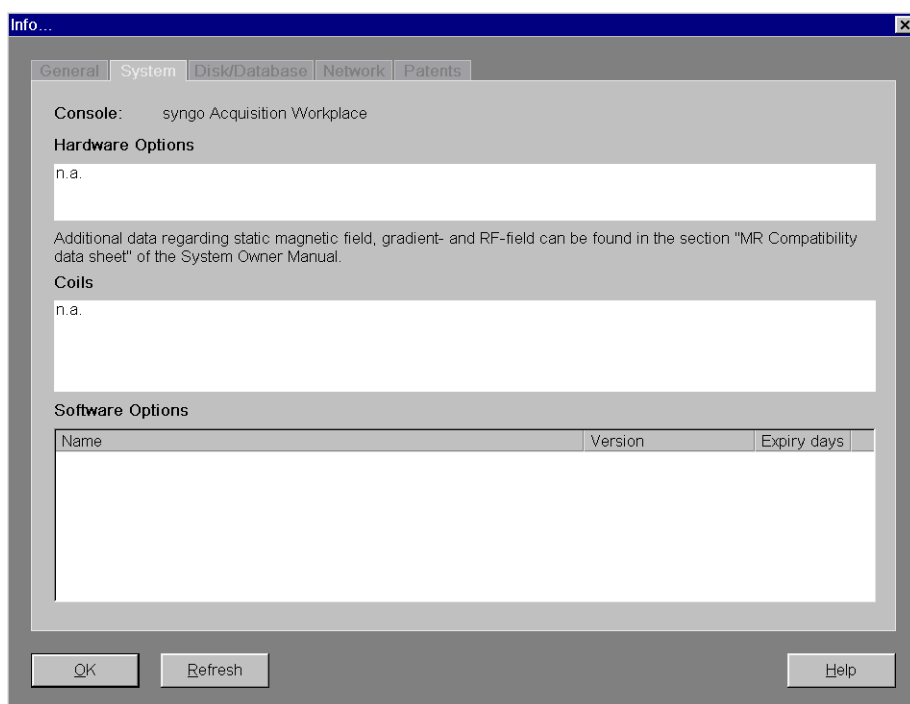
Example: MAGNETOM Avanto

- (1) Product name
- (2) Series Number
- (3) Software used
- (4) Software version
- (5) Software name
- (6) Information about the licensee

Information about the system

The **System** subtask card includes the hardware and software options.

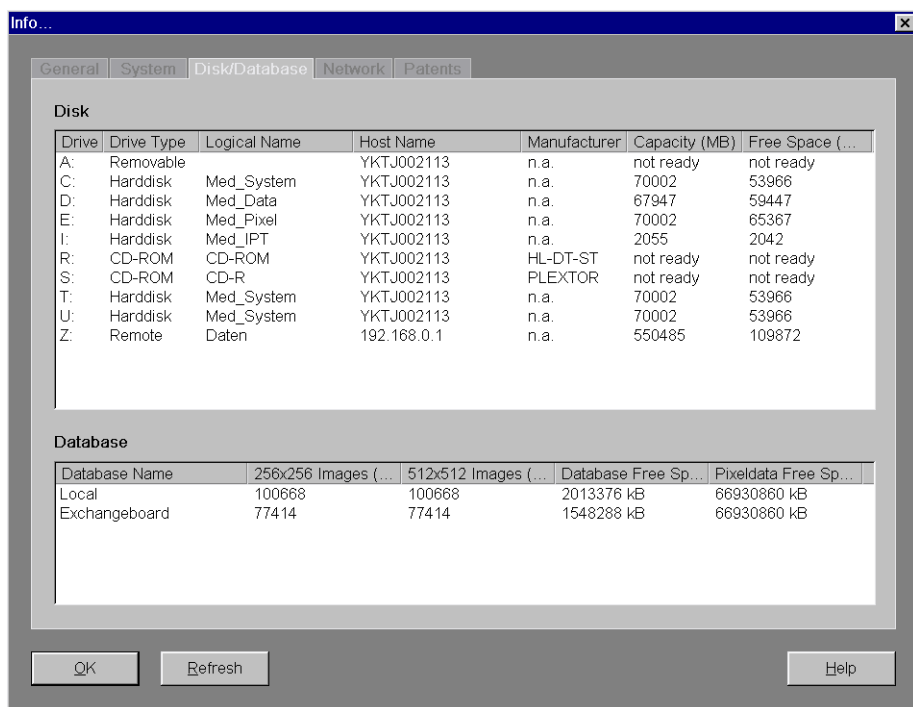
✧ Open the **System** subtask card.



Drive and database information

Information regarding locally available drives and databases are included on the **Disk/Database** subtask card.

✧ Open the **Disk/Database** subtask card.



Drive

The **Drive** list displays the following information for all physical drives (e.g., CD-ROM) and all logical drives (e.g., network drives):

- ☐ Drive letter
- ☐ Type, e.g., CD-R
- ☐ Logical name
- ☐ Name of the processor associated with the drive
- ☐ Manufacturer
- ☐ Capacity in MB
- ☐ Free storage space available, in MB

Database

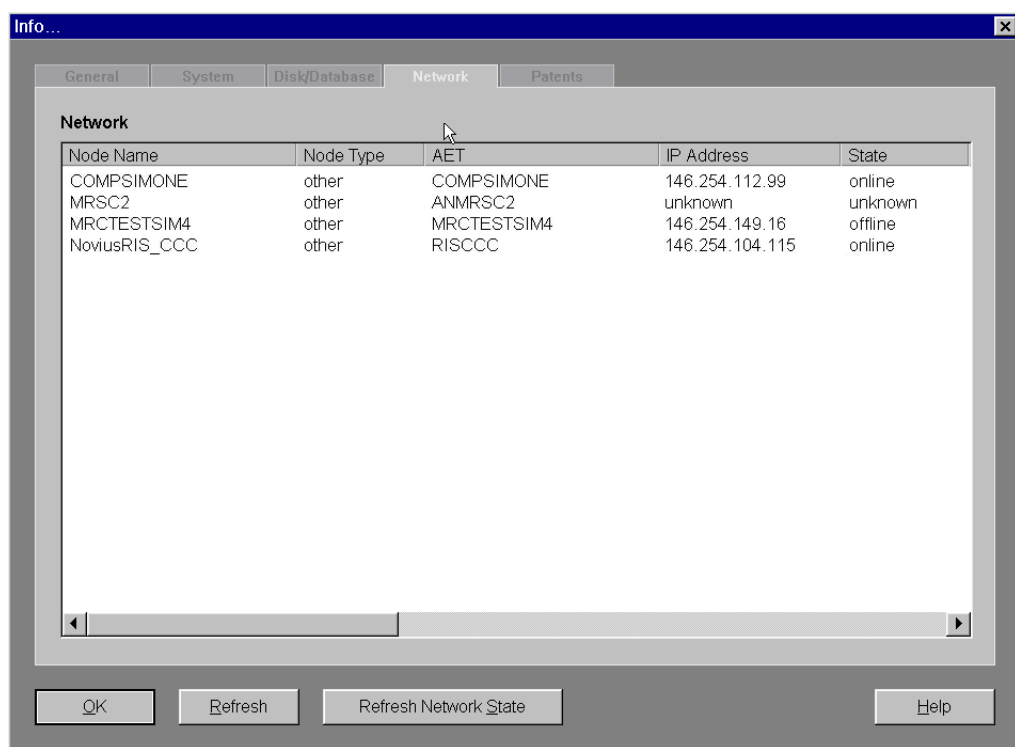
The **Database** list contains all local databases that contain header as well as pixel data. The following information is displayed:

- ☐ Name of the database
- ☐ Number of images in 256×256 matrix size that can still be stored
- ☐ Number of images in 512×512 matrix size that can still be stored
- ☐ Free space in the database partition for header data (in kB)
- ☐ Free space in the pixel data partition (in kB)

Network information

If your *syngo* Acquisition Workplace or *syngo* MR Workplace is connected to a network, you may view network-specific information on the **Network** subtask card.

✧ Open the **Network** subtask card.



Updating the network status

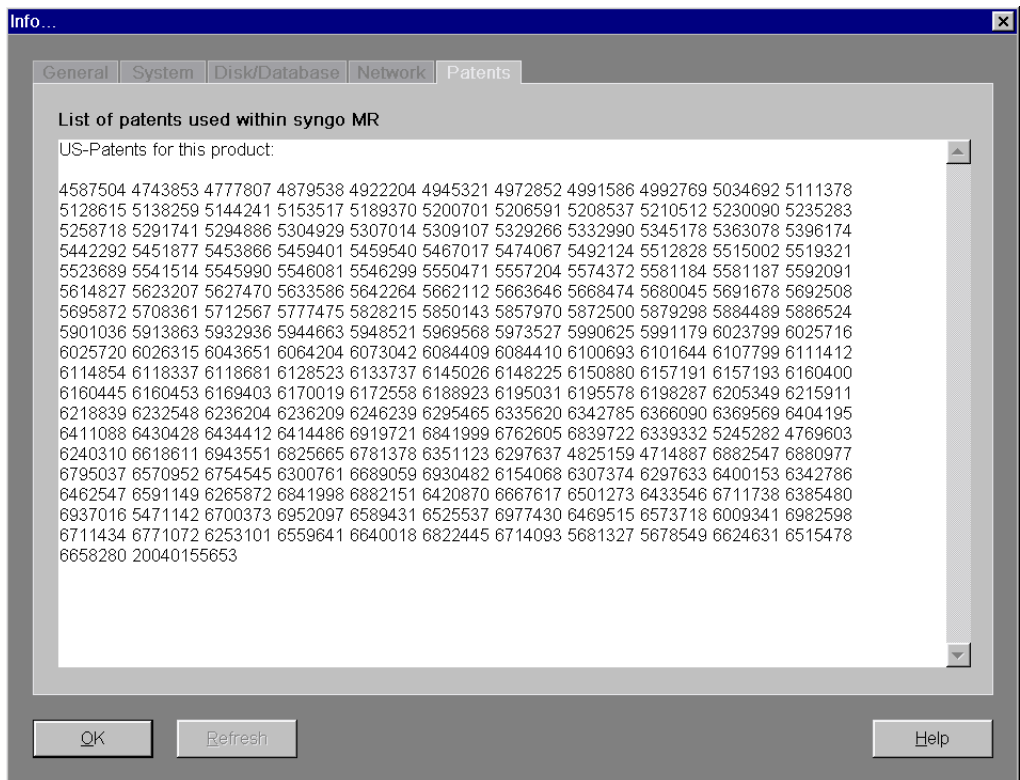


✧ Click the **Refresh Network State** button.

Displaying patents

The **Patents** subtask card lists all patents used by **syngo MR**.

✧ Open the **Patents** subtask card.



Updating the display

Information on the **System**, **Drives/Databases** and **Network** subtask cards can be updated.

A rectangular button with a light gray background and a thin blue border. The word "Refresh" is written in a dark gray, sans-serif font.

✧ Click **Refresh**.

⇒ The softkey is not active on the other subtask cards.

Automatic adjustment

To consistently obtain high-quality images, it is important to adjust the MR system prior to each protocol. For this purpose, individual settings are available for automatic adjustments prior to each protocol. During measurement with a changed table position, adjustment values of similar table positions may be used by determining adjustment tolerances in place of new adjustment values.

An adjustment volume is defined for each protocol. Most adjustments are referenced to this volume which you can change as required.

Manual adjustment

For special measurement procedures, as for example, **Spectroscopy**, it is at times required to manually fine-tune automatic adjustments of the MR system prior to the measurement. Through manual shimming, experienced users can compensate for magnetic field inhomogeneities.

Inline adjustment

Inline adjustment is performed automatically before each image acquisition. The user has no influence on this adjustment.



Configuring automatic adjustment

In most cases, it is not necessary to change the settings during routine operation. You can check these parameters, however, you should not modify them under normal circumstances.

Opening a protocol

- ✧ Select the requested protocol in the program control area.
→ [Page C.6–25, Opening a protocol](#)

The parameters of the protocol are shown.

You may view and edit the automatic adjustment parameters in the **System** parameter card.

Setting the adjustment parameter card

✧ Click the **System – Adjustments** parameter card.

(1) Shim mode Standard

(2) Adj. water suppr. ☒

(3) Adjust with body coil ☐

(4) Confirm freq. adjustment ☐

(5) Assume Silicone ☐

(6) Tx Ref [Nucleus] Ref.
Ref. amplitude 1H 111.000

Reset

Adjustment Tolerance Auto (None)

Position Isocenter

Orientation Transversal

Rotation 0.00 deg

R >> L 300 mm

A >> P 300 mm

F >> H 3008 mm

Reset

Program Routine Contrast Resolution Geometry System Physio Sequence

Parameter card for stationary measurements

Confirm freq. adjustment ☐

Assume Silicone ☐

(9) Perform CTM adjustments ☒

Tx Ref [Nucleus] Ref.
? Ref. amplitude 1H 0.000

Reset

Position Isocenter

Orientation Transversal

Rotation 0 deg

R >> L 350 mm

A >> P 263 mm

F >> H 350 mm

Reset

Program Routine Contrast Resolution Geometry System Sequence

Parameter card for TimCT overview measurements

- 
- (1) Select shim mode
 - (2) Perform water suppression adjustment
 - (3) Perform adjustment with the body coil
 - (4) Confirm adjustment of the resonance frequency
 - (5) Adjustment in silicone mode
 - (6) Change the reference amplitude
 - (7) Define the adjustment tolerance
 - (8) Geometry of the adjustment volume
 - (9) Additional adjustment after TimCT overview measurements

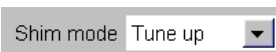
! All system adjustment settings made on the **System** parameter card *only* apply to the protocol currently open.

Set the shim mode

For the 3D-shim, several modes are offered in the protocol. The range of options depends on the measurement sequence.

- Each protocol is preset to the shim mode required for optimum image quality.

Tune-up	Adjustment measurements and evaluations are not performed. The system uses the values set during system tune-up by Siemens Service. This setting is sufficient for standard imaging protocols without special requirements.
Standard	Adjustment measurements and evaluation are performed in the standard mode. This mode is suitable for imaging protocols with special requirements, for example, for fat saturation or EPI.
Advanced	Adjustment measurements and evaluation are performed in the advanced mode. This mode is used mainly for spectroscopy protocols. The advanced mode is both time-consuming and only necessary for measurements that place the greatest demand on the homogeneity of the magnetic field.
Cardiac	Adjustment measurements and evaluation are performed in the cardiac mode. This mode is suitable for imaging protocols with special requirements due to the beating heart.
Foot/Ankle	Adjustment measurement and evaluation are optimized for Foot/Ankle exams.
Brain	Adjustment measurement and evaluation are optimized for brain spectroscopy.
Breast	Adjustment measurement and evaluation are optimized for breast spectroscopy.
Prostate	Adjustment measurement and evaluation are optimized for prostate spectroscopy.



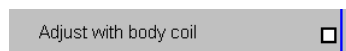
- ✧ Select a shim mode from the selection list.

It is also possible to use the manual interactive shimming method, independent of this protocol setting.

→ [Page B.2–49, Interactive Shim](#)

Performing adjustments with the body coil

Usually, all adjustments are made by using the coil elements established in the respective protocol. In addition, it is possible to receive with the body coil only. This means that new adjustments are omitted when changing the selection of coil elements.



- ✧ Activate the **Adjust with body coil** checkbox.

✧ For measurements with the transmit coils, it is not possible to perform adjustments using the body coil.

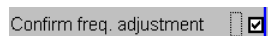
Performing water suppression adjustment



- ✧ Activate the **Adj. water suppr.** checkbox.

For spectroscopy measurements, special RF pulse trains are used that suppress the water signal.

Confirming or changing the resonance frequency



- ✧ Activate the **Confirm freq. adjustment** checkbox.

→ [Page B.2–64, Changing the resonance frequency after inline adjustment](#)

Adjustment in silicone mode

Assume Silicone ☐

The **Assume Silicone** option optimizes measurements of patients having silicone implants.

- ✦ Activate the **Assume Silicone** checkbox.

Adjustment to dominant fat

Assume Dominant Fat ☐

The **Assume Dominant Fat** option optimizes measurements of patients having dominant fat tissue.

- ✦ Activate the **Assume Dominant Fat** checkbox.

Change the reference amplitude

Tx Ref [Nucleus]	Ref.
Ref. amplitude [1H]	200.000

The **TxRef/Ref** display field shows a list of reference amplitudes for the selected primary or secondary nucleus.

- ✦ Enter new values for the reference amplitude.
All values changed manually are marked by an exclamation mark ("!").

You can undo your manual settings at any time.

Reset

- ✦ Click **Reset** on the left side of the parameter card.
The automatically computed values are shown again.

Define the adjustment tolerance

If you would like to use the adjustment results for similar table positions, you can set them as default values via the **Adjustment Tolerance** input field:

Option	Description
None	Adjustment results are only used again at the identical table position.
Maximum	Adjustment results are used as well on similar table positions.
Auto (None)	The adjustment tolerance is determined automatically by the system. "None" is selected.
Auto (Maximum)	The adjustment tolerance is determined automatically by the system. "Maximum" is selected.

Adjustment Tolerance

Auto (None) ▾

✦ Select the desired tolerance option.



Adapting the adjustment volume

A localized volume, the adjustment volume, is defined for each protocol for tuning the transmitter and receiver. The adjustment volume defines the space in the magnet for which the parameters will be optimized. It is determined automatically but may also be adjusted manually.

Imaging protocols

The default setting of the adjustment volume depends on the slices and slabs. In most cases, an orthogonal cuboid completely encloses the slices or slabs to be measured.

Spectroscopy protocols

For spectroscopy protocols, the default setting corresponds to the adjustment volume of the VOI.



Protocols that do not require shim adjustment measurements (e.g., **Tune-Up** mode) or water suppression adjustments use a fixed adjustment volume in the form of a large central cuboid. This saves time since new adjustment procedures are not necessary after slice positioning. It is of course possible to manually manipulate the adjustment volume and then perform the adjustment as such.

Editing the adjustment volume graphically



During graphic slice positioning, you may display the adjustment volume as an additional graphic object in the reference images.

✧ Click this button on the **Position Toolbar**.

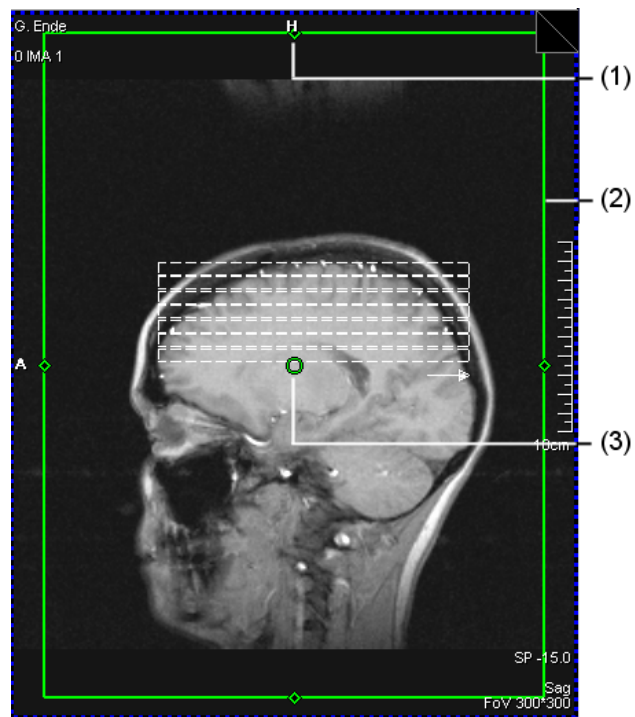
The adjustment volume is shown as a green rectangle in each reference image and may now be edited graphically.



Selecting an adjustment volume

✧ Click the contour line of the adjustment volume.

The contour line is highlighted in color. The center and the handles for changing the size are shown.



- (1) Handle
- (2) Contour line
- (3) Center

Moving the adjustment volume



- ✧ Click the center of the adjustment volume and drag it with the mouse into the desired location.
- Or -
- Click the contour line or one of the handles of the adjustment volume and drag it into the desired position while keeping the **Shift** key and the mouse pressed.

Enlarging/reducing the adjustment volume



- ✧ Click the required handle and drag it toward the outside to enlarge the adjustment volume. To reduce it, just drag the handle toward the inside.

As long as you keep the mouse button pressed, the width is shown in mm right next to the mouse pointer.

The size of the adjustment volume changes symmetrically. The center remains unchanged.

- ✧ You are able to change the size asymmetrically as well (into one direction only) by keeping the **ALT** key pressed as well. This means that the center is shifting as well.

Rotating the adjustment volume



The volume element can be tilted about an axis that runs vertically to the reference image and through the center.

- ✧ Click the contour line and drag the mouse into the required direction.

Tilting the adjustment volume

The volume element can be tilted about an axis that runs parallel to the reference image and through the center point.

- ✧ Click the contour line, keep the **ALT** key pressed and drag the mouse into the required direction.

Freely turning or tilting the adjustment volume

This is how you turn or tilt the adjustment volume about its center.

- ✧ Click the contour line, keep the **Ctrl** key pressed and drag the mouse into the required direction(s).

Numerically setting the adjustment volume

On the **System Adjustment** parameter card, you can numerically set the adjustment volume via the **Orientation**, **Position** and **Rotation** input fields.



- ✧ Select the desired orientation in the **Orientation** selection list.

- ✧ You are also able to enter additional orientations via the **Orientation** dialog window. You open the window via this button.

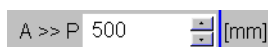


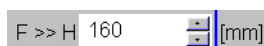
The **Position** field displays the current position of the slice or slab center.

- ✧ Select the desired position from the **Position** selection list.

- ✧ You are also able to enter additional positions via the **Position** dialog window. You open the window via this button.


 A horizontal input field with the label "Rotation" on the left and the value "90" in the center. There are small up and down arrow buttons on the right side of the field.


 A horizontal input field with the label "A >> P" on the left and the value "500" in the center. To the right of the value is a unit indicator "[mm]". There are small up and down arrow buttons on the right side of the field.


 A horizontal input field with the label "F >> H" on the left and the value "160" in the center. To the right of the value is a unit indicator "[mm]". There are small up and down arrow buttons on the right side of the field.


 A horizontal input field with the label "R >> L" on the left and the value "500" in the center. To the right of the value is a unit indicator "[mm]". There are small up and down arrow buttons on the right side of the field.

- ✧ Enter the angle for rotating the adjustment volume in the **Rotation** input field.
- ✧ Enter the edge length of the adjustment volume in the **A >> P** (anterior/posterior) direction.
- ✧ Enter the edge length of the adjustment volume in the **F >> H** (feet/head) direction.
- ✧ Finally, enter a translation of the edge length of the adjustment volume in the **R >> L** (right/left) direction.

Applying the settings

- ✧ Click **ENTER** to accept the settings.
 - Or -
 - Click in the image area.
- The graphic display is updated.

Restoring the default settings


 A rectangular button with the text "Reset" in the center.

This is how you enable automatic calculations of the adjustment volume and discard the manual settings:

- ✧ Click **Reset** on the right side of the parameter card.
- The automatically computed values are shown again and the exclamation marks disappear.



TimCT measurement

For each TimCT measurement, adjustment data have to be available for the table move covered during the measurement. If the adjustment data are not at all or only partially available, the system performs an automatic adjustment immediately before the measurement.

Additional adjustment after the TimCT localizer

Perform CTM adjustments ☒

This is how you select an additional adjustment after the TimCT localizer:

✧ Activate the **Perform CTM adjustments** checkbox.

After the localizer measurement, adjustment is performed across the entire table range of the localizer.

✧ For subsequent measurements *within* this area, no additional adjustments are performed.



Do not use the **Table Positioning** buttons during the TimCT measurement.

Displaying adjustment parameters in the protocol

The **System Transmitter/Receiver** parameter card shows the results of the last automatic and successful adjustment.

✦ Click the **System Transmitter/Receiver** parameter card.

The screenshot displays the 'System Transmitter/Receiver' parameter card within a software interface. The card is divided into two main sections: 'Transmitter' and 'Receiver'. The 'Transmitter' section includes fields for 'Frequency 1H' (63.800000 MHz), 'Ref. amplitude 1H' (111.000 V), and 'Correction factor' (1). Below these is a table for 'Puls' and 'Amplitude' with one row showing 'SRFExcit 1H' at 82.810. The 'Receiver' section includes a 'Gain' dropdown menu set to 'Low' and a table for 'Coil elements' and 'FFT scale factor' with one row showing 'BC' at 0.420. Both sections have a 'Reset' button at the bottom. The interface also features a top navigation bar with tabs for 'Coils', 'Miscellaneous', 'Adjustments', and 'Transmitter/Receiver', and a bottom navigation bar with tabs for 'Program', 'Routine', 'Contrast', 'Resolution', 'Geometry', 'System', 'Physio', and 'Sequence'.

Transmitter		Receiver																					
(1) Frequency 1H	63.800000 MHz	I Gain	Low																				
(2) Ref. amplitude 1H	111.000 V																						
(3) Correction factor	1																						
<table border="1"> <thead> <tr> <th>Puls</th> <th>Amplitude</th> </tr> </thead> <tbody> <tr> <td>SRFExcit 1H</td> <td>82.810</td> </tr> <tr> <td></td> <td></td> </tr> <tr> <td></td> <td></td> </tr> <tr> <td></td> <td></td> </tr> </tbody> </table>		Puls	Amplitude	SRFExcit 1H	82.810							<table border="1"> <thead> <tr> <th>Coil elements</th> <th>FFT scale factor</th> </tr> </thead> <tbody> <tr> <td>BC</td> <td>0.420</td> </tr> <tr> <td></td> <td></td> </tr> <tr> <td></td> <td></td> </tr> <tr> <td></td> <td></td> </tr> </tbody> </table>		Coil elements	FFT scale factor	BC	0.420						
Puls	Amplitude																						
SRFExcit 1H	82.810																						
Coil elements	FFT scale factor																						
BC	0.420																						
(4)		(6)																					

- (1) System frequency
- (2) Reference amplitude
- (3) Correction factor for water suppression
- (4) Amplitude of the RF pulses used in the protocol
- (5) Receiver gain
- (6) FFT-scaling factors for individual images of each coil element



The left side of the parameter card displays the values of the last successful transmitter tuning.

- ⇒ All pulse amplitudes are automatically calculated based on the reference amplitude and, if necessary, the correction factor.

The right side of the parameter card shows the values for the receiver gain automatically calculated by the system.

- ⇒ You are able to overwrite the automatically computed settings. All values changed manually are marked by an exclamation mark ("!").

Restoring the default settings



- ✧ Click **Reset** on the **System Transmitter/Receiver** parameter card.

The automatically computed values are shown again and the exclamation marks disappear.

Automatic sequence of adjustment steps

To avoid performing a complete adjustment each time, the system uses the adjustment results of a previously measured protocol. Usually, the results can be applied only if the following parameters of the two protocols remain unchanged:

- ☐ Table position
- ☐ Coil elements used
- ☐ Shim mode (Standard/Advanced/Tune-up)
- ☐ Adjustment volume

Dependencies between adjustments

When adjusting the system, the sequence of adjustment operations is important since some adjustments may invalidate the results of previously performed adjustments.

When you establish adjustment tolerances, you avoid additional adjustment measurements if you have previous adjustment results for similar table positions.

The adjustment tolerance depends on the adjustment type, on the MR system and/or individual sequences. However, it is automatically limited by both patient safety and the image quality desired.

Validity of the adjustment parameters

In the following cases, adjustment parameters for a successful adjustment cannot be used for additional protocols:

- ☐ A new patient is registered
- ☐ the parameters are rendered invalid by hand
→ [Page B.2–63, Rejecting results](#)
- ☐ The coils on the patient table are changed
- ☐ the patient table is moved into the Home position and the **Patient is unchanged in his scan position** checkbox is deactivated
→ [Page C.10–6, Messages when changing patients](#)
- ☐ The patient table is lowered

The dependency on coil elements can be avoided by selecting **Adjust with body coil**. Dependency on the adjustment volume can be avoided by using the shim mode **Tune-Up** without water suppression.

→ [Page B.2–3, Setting the adjustment parameter card](#)

Manual adjustment

Manual adjustments are intended primarily for test purposes, for error analysis, or for special applications such as **Spectroscopy**.

- ✧ Manual adjustment of the system should only be performed by experienced users and should not be performed during routine operation.

The following adjustment options are available (depending on the system):

- ☐ Optimizing the resonance frequency
- ☐ Optimizing the multi-nuclear resonance frequency
- ☐ Tuning the transmitter
- ☐ Tuning the multi-nuclear resonance frequency
- ☐ Performing 3D shim and interactive shim
- ☐ Optimizing water suppression

Opening a protocol

- ✧ Select the requested protocol in the program control.
The parameters of the protocol are shown.

Open the adjustment dialog

- ✧ Select **Options > Adjustments**.
The **Manual Adjustments** dialog window is opened.

Optimizing the resonance frequency

In this manner, you determine the exact proton resonance frequency. For this purpose, you enter various parameters and accept the frequency calculated for the MR system following adjustment.

- ✧ Open the **Frequency** parameter card in the **Manual Adjustments** dialog window.

Manual adjustments

No.	Freq. [MHz]	Ampl. [V]	Pos.	Diff. [Hz]	C. freq. [MHz]	A.
1	63.681700	400.0	265	88	63.681788	N
2	63.681788	400.0	256	0	63.681788	Y

(1) —

(2) — Real 303

(3) — Imag 187

(4) — Ampl 132665 5000Hz

(5) — Phase 180

(6) — Apply

(7) — Frequency (temp) [MHz]: 63.681788

(8) — Frequency (syst) [MHz]: 63.681788

(9) — Amplitude [V]: 400.0

(10) — Sequence & Resolution: STEAM 10 Hz

(11) — Receiver gain: Low High

(12) — Ch. Coil Con. 1 HE2 3

Go Abort Reset

Frequency Transmitter 3D Shim Interactive shim Water suppr. Show

Adjustment converged Close Help

- 
- (1) Numeric results
 - (2) Real part of the measurement signal in the time domain
 - (3) Imaginary part of the measurement signal in the time domain
 - (4) Amplitude in the frequency domain
 - (5) Phase in the frequency domain
 - (6) Temporary transmit frequency
 - (7) System transmitter frequency
 - (8) Transmit amplitude
 - (9) Sequence and resolution
 - (10) Receiver gain
 - (11) Coil measurement signal for graphic displays
 - (12) Range display

Entering the frequency

A different transmit frequency may be used for each iteration.

Frequency (temp) [MHz]: 63.623318 

- ✧ Enter the new transmit frequency in the **Frequency (temp)** input field.
- ⇒ The temporary frequency you entered is used only for the adjustment measurement. The frequency currently used is displayed in the field underneath for comparison.

- ⇒ By graphically displaying the frequency domain, you can determine the frequency graphically and enter it into the input field.

→ [Page B.2–26, Graphic frequency determination](#)

- ⇒ After successful adjustment, the temporary frequency is automatically accepted as the system frequency. However, you can also accept it manually without adjustment measurements.

→ [Page B.2–27, Entering the frequency](#)

Entering the amplitude

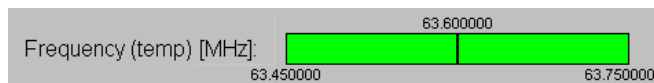


Amplitude [V]: 450.0

- ✧ Enter the transmit amplitude for the adjustment to be performed in the **Amplitude [V]** input field.

Range display

Using the range display, you can check whether the entered frequency and/or amplitude is/are in the acceptable range. If not, you have to correct the value.



Selecting sequence and resolution

If the frequency lies outside the narrowband standard range (no resonance line in the spectrum), it is recommended to search a wide frequency band first to limit the frequency. To subsequently adjust the transmit frequency, you will have to work with a narrow frequency band.

The following options are available:

- ☐ Broadband not volume-selective (FID 400 Hz)
- ☐ Narrowband not volume-selective (FID 10 Hz)
- ☐ Narrowband volume-selective (STEAM 10 Hz)
- ✧ Following each non-volume-selective adjustment measurement, it is necessary to perform at least one volume-selective adjustment.

Sequence & Resolution : STEAM 10 Hz ▼

- ✧ Select a sequence and resolution from the list.

Selecting the amplification (gain)

Low

High

You set the receiver gain based on the intensity of the measured signal.

- ✧ Click **Low** if the receive signal is strong.

Or

- ✧ Click **High** if the receive signal is weak.

- ✧ If necessary, the system will adjust this setting automatically.



Starting the measurement

Go

✦ Click **Go**.

Numeric results

The results are listed below the running numbers. The following parameters are displayed:

Parameters	Comment
No.	Running number of the adjustment measurement
Freq. [MHz]	Transmit frequency in MHz
Ampl. [V]	Transmit amplitude of the adjustment measurement
Pos.	Position of the highest amplitude peak in the frequency domain
Diff. [Hz]	Difference between the transmit frequency and the calculated (better) frequency
C.Freq. [MHz]	Calculated (better) frequency (Calculated Frequency)
A.	Transmit frequency successfully adjusted (Yes/No)

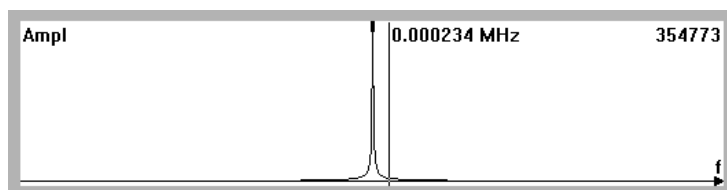
Graphic display of the results

The results are displayed graphically in both the time and the frequency domain. The maximum values of the raw data set are displayed numerically in the graphics.

If available, the resonance peak is highlighted.

Graphic frequency determination

A vertical marker is shown when you position the mouse pointer inside the graphic display of the frequency domain. It indicates the delta frequency of the calculated resonance frequency. If no peak was detected during the adjustment, the absolute frequency is indicated instead.



- ⇒ The position of the resonance frequency is indicated by a small line at the top of the graphic.

To select a frequency graphically, proceed as follows:

- ✧ Use the mouse to drag the vertical marker to the frequency desired.

The frequency is entered in the **Frequency (temp)** field.

Inverting the frequency axis and zooming the graphic

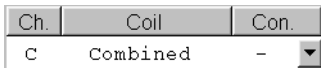


A context menu is available inside the graphic display of the frequency domain.

You can invert the display of all frequency axes (from left to right) or zoom into the detected peak. If no peak is available, the program zooms into the center of the graphic.

- When you select a new patient, the default display of the frequency axes is restored. The zoom applied is reset when you perform a new measurement.

Selecting the receive channel



The receive channel is used to select the signals of the local coils.

- ✧ Select the receive channel whose signal you want to have displayed.
- The **Combined** entry is used to select the combined signal of all coils listed.

Entering the frequency

The frequency is automatically transmitted to the MR scanner after successful adjustment ("Y" in the numeric input field). This may require several iterations.

This is how you manually correct the system frequency (without successful adjustment):

- ✧ Enter the new transmit frequency in the **Frequency (temp)** input field.
 - Or -
 - Graphically determine the frequency.
→ [Page B.2–26](#)

A rectangular button with a thin black border and the word "Apply" centered in a sans-serif font.

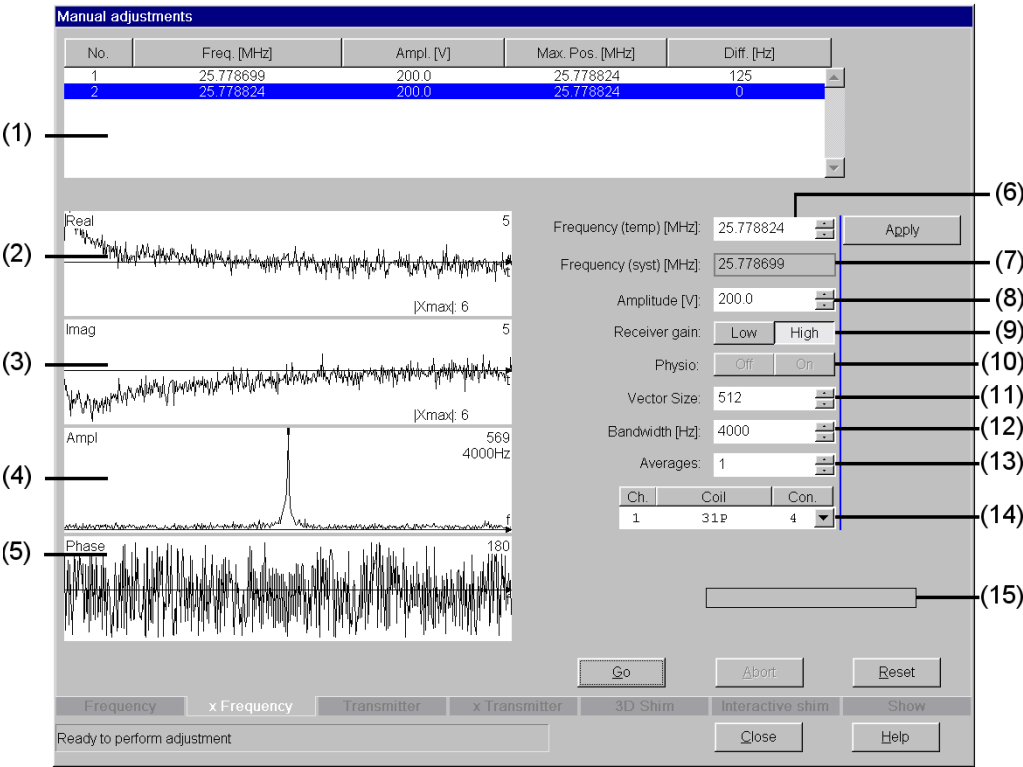
- ✧ Click **Apply**.

The amplitude is applied as the system reference amplitude.

Optimizing the multinuclear frequency

With multinuclear frequency optimization, you can calculate the exact resonance frequency for the nucleus determined by the protocol. For this purpose, you enter various parameters and accept the multinuclear frequency calculated for the MR system following adjustment.

- ✧ Open the **X-Frequency** parameter card in the **Manual adjustments** dialog window.



- (1) Numeric results
- (2) Real part of the measurement signal in the time domain
- (3) Imaginary part of the measurement signal in the time domain
- (4) Amplitude in the frequency domain
- (5) Phase in the frequency domain
- (6) Temporary transmit frequency
- (7) System transmitter frequency
- (8) Transmit amplitude
- (9) Receive gain
- (10) Physiological parameters
- (11) Number of measurement points
- (12) Bandwidth for measurement
- (13) Number of averages
- (14) Coil measurement signal for graphic displays
- (15) Range display

Entering the multinuclear frequency

A different transmit frequency may be used for each iteration.

Frequency (temp) [MHz]: 25.745506

- ✧ Enter the new transmit frequency in the **Frequency (temp)** input field.
- Or -
- Graphically determine the frequency.
- [Page B.2–26](#)

- ✧ The temporary frequency you entered is used only for the adjustment measurement. The frequency currently used is displayed in the field underneath for comparison.
- ✧ After successful adjustment, the temporary frequency is automatically accepted as the system frequency. However, you can also accept it manually without adjustment measurements.
→ [Page B.2–34, Entering the frequency](#)

Entering the amplitude

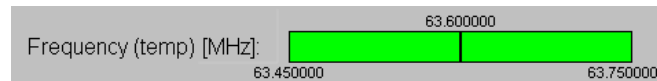


Amplitude [V]: 100.0

- ✧ Enter the transmit amplitude for the multinuclear adjustment to be performed in the **Amplitude** input field.

Range display

Using the range display, you can check whether the entered frequency and/or amplitude is/are in the acceptable range. If not, you have to correct the value.



Selecting the amplification (gain)



Low



High

You set the receiver gain based on the intensity of the measured signal.

- ✧ Click **Low** if the receive signal is strong.

Or

- ✧ Click **High** if the receive signal is weak.

- ✧ If necessary, the system will adjust this setting automatically.

Accepting physiological parameters



For the adjustment measurement, you may apply physiological measurement parameters from the relevant parameter card of the protocol.

✧ Click the **On** button under **Physio**.

✧ When the **Off** button is activated, the physiological measurement parameters are not accepted.

Entering the number of measurement points



You determine the number of measurement points to be used during a measurement.

✧ Enter the number of measurement points in the **Vector Size** field.

Entering the measurement bandwidth



The **Bandwidth** parameter determines the bandwidth of the frequency range to be measured. The transmit frequency set is at the center of this range.

✧ Enter the measurement bandwidth.

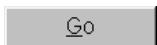
Entering the number of averages



The individual measurements are averaged to determine the resonance frequency. You can keep the noise portion low in this manner.

✧ Enter the number of averages.

Starting the measurement



✧ Click **Go**.

Numeric results

The results are listed below the running numbers. The following parameters are displayed:

Parameters	Comment
No.	Running number of the adjustment measurement
Freq. [MHz]	Transmit frequency in MHz
Ampl. [V]	Transmit amplitude of the adjustment measurement
Max.pos. [MHz]	Position of the highest amplitude peak in the frequency domain (in MHz)
Diff. [Hz]	Difference between the transmit frequency and the calculated (better) frequency

Graphic display of the results

The results are displayed graphically in both the time and the frequency domain. The maximum values of the raw data set are displayed numerically in the graphics.

If available, the resonance peak is highlighted.

Graphic frequency determination

The graphical display of the frequency domain lets you graphically select frequencies by using a vertical marker.
→ [Page B.2–26](#), *Graphic frequency determination*

Inverting the frequency axis and zooming the graphic

You can invert the display of all frequency axes (from left to right) or zoom into the detected peak.

→ [Page B.2–27](#), *Inverting the frequency axis and zooming the graphic*

Selecting the receive channel

Ch.	Coil	Con.
C	Combined	-

The receive channel is used to select the signals of the local coils.

- ✧ Select the receive channel whose signal you want to have displayed.
- ✧ The **Combined** entry is used to select the combined signal of all coils listed.

Correcting the frequency when adjustment was not successful

If the system did not locate the correct frequency, you are able to manually correct the frequency.

- ✧ Enter the new transmit frequency in the **Frequency (temp)** input field.
- Or -
- Graphically determine the frequency.
- [Page B.2–26](#)

Entering the frequency



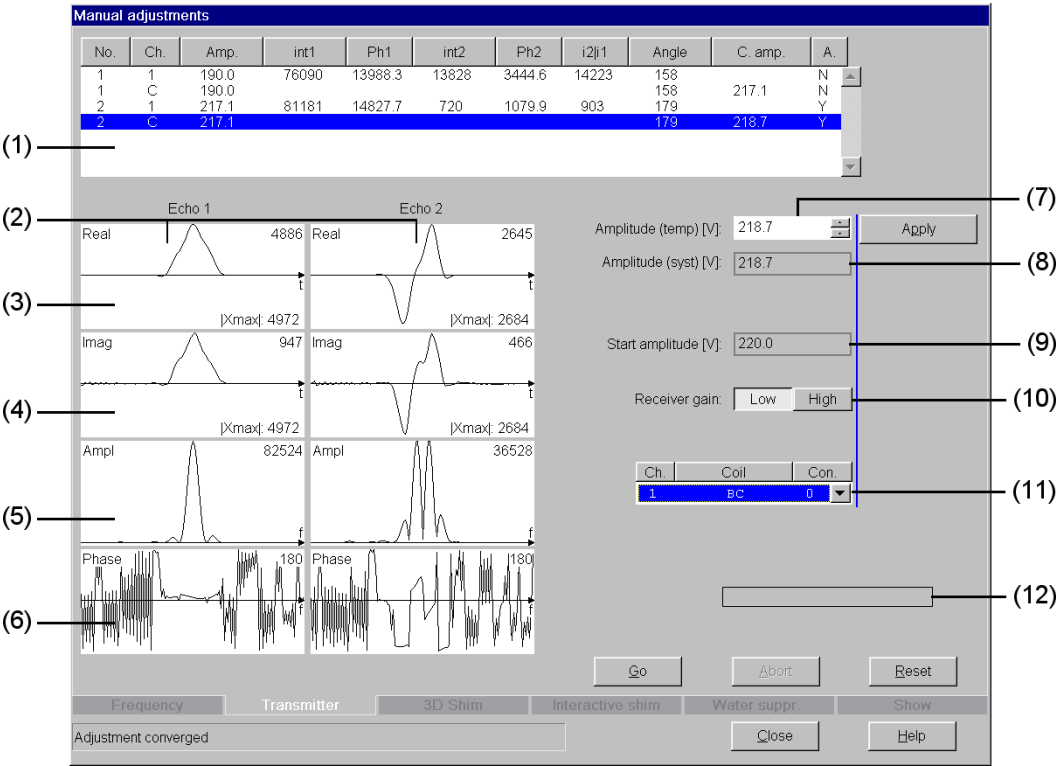
- ✧ To accept the frequency as the system frequency, click **Apply**.



Transmitter adjustment

Transmitter adjustment is used for calibration of the transmitter. It does not refer to the adjustment volume, but rather to a range at the isocenter. The reference amplitude is continuously varied until the spins are rotated 180° (flip angle) by the RF pulse. Two echoes are generated, one normal echo and one stimulated echo. The magnitude and the phase relation of both echoes is directly related to the flip angle.

- ✧ Open the **Transmitter** parameter card in the **Manual Adjustment** dialog window.



- 
- (1) Numeric results
 - (2) Normal echo (Echo 1) and stimulated echo (Echo 2)
 - (3) Real part of the measurement signal in the time domain
 - (4) Imaginary part of the measurement signal in the time domain
 - (5) Amplitude of the measurement signal in the frequency domain
 - (6) Phase of the measurement signal in the frequency domain
 - (7) Temporary reference amplitude
 - (8) System reference amplitude
 - (9) Start amplitude
 - (10) Receiver gain
 - (11) Coil measurement signal for graphic displays
 - (12) Range display

Entering the amplitude

Determine a temporary reference amplitude for the adjustment measurement.



Amplitude (temp) [V]: 370.0

- ✧ Enter a value for the reference amplitude into the **Amplitude (temp)** input field.

Using the range display, you can check whether the amplitude is in the acceptable range. If not, you have to correct the value.



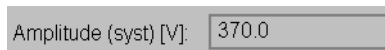
- ✧ The amplitude currently used is displayed for comparison in the field underneath the system reference amplitude for measurements.

After successful adjustment, the temporary reference amplitude is automatically accepted as the system reference amplitude. However, you can also accept it manually without adjustment measurements.

→ [Page B.2–39, Applying the amplitude](#)

Start amplitude

The **Start amplitude** field shows the amplitude value (specific parameter of the transmitter coil) used for the first iteration.



Amplitude (syst) [V]: 370.0

Selecting the amplification (gain)

Select the receiver gain according to the intensity of your signal.
→ [Page B.2–24, Selecting the amplification \(gain\)](#)

Starting the measurement

A rectangular button with a thin border and the word "Go" in a sans-serif font.

✧ Click **Go**.

Numeric results

The results are listed below the running number of the iteration.
The following parameters are displayed:

Parameters	Comment
No.	Running number of the adjustment measurement
Ch.	Receive channel used to determine results
Amp.	Transmit amplitude of the adjustment measurement
int1	Integral of the first echo
Ph1	Phase of the first echo
int2	Integral of the second echo
Ph2	Phase of the second echo
i2 / i1	Projection of the second echo on the first echo
Angle	Measured flip angle
C. Amp.	Calculated (better) amplitude
A.	Adjustment successful (Yes/No)

Graphic display of the results

The following results are displayed graphically:

- ☐ Real and imaginary part of the echo in the time domain
- ☐ Amplitude and phase of the respective echo in the frequency domain

The graphics show the maximum value of each raw data set.

Selecting the receive channel

Ch.	Coil	Con.
1	Body Coil	- 

The receive channel is used to select the signals of the local coils.

- ✧ Select the receive channel whose signal you want to have displayed.

Applying the amplitude



The frequency is automatically transmitted to the MR scanner after successful adjustment ("Y" in the numeric input field).

This is how you manually transfer a temporary reference amplitude without successful adjustment into the MR system:

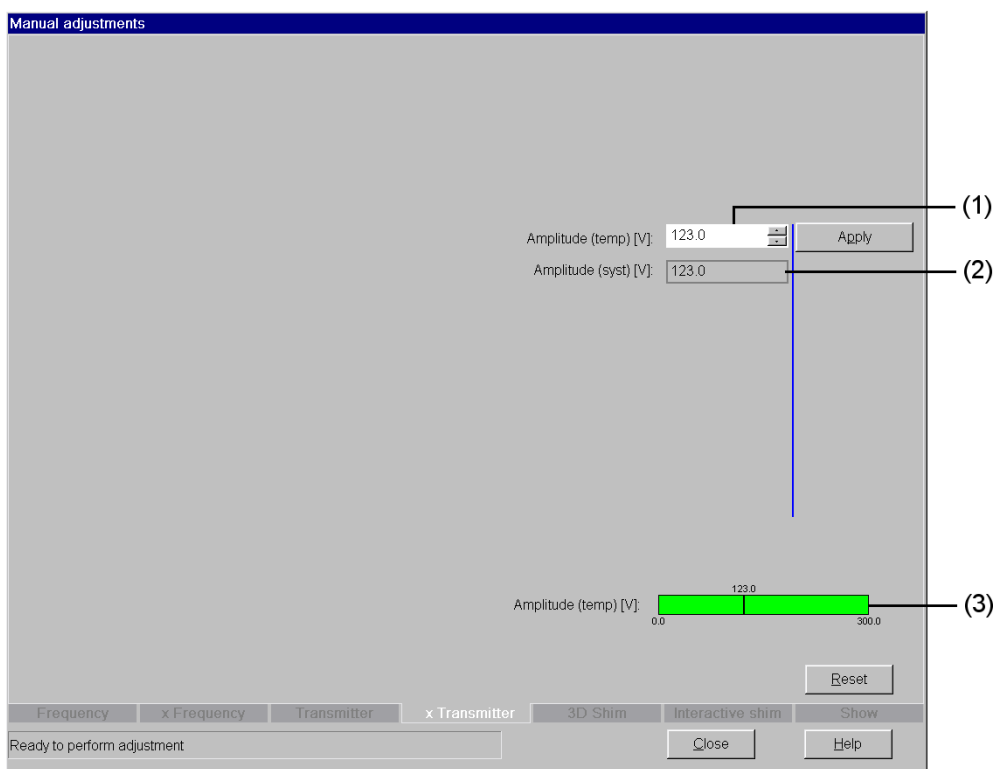
- ✧ Click **Apply**.

The amplitude is applied as the system reference amplitude.

Multinuclear transmitter adjustment

Since the signal strength is minimal, transmit adjustment measurements are not possible during multinuclear measurements. However, you can define a reference amplitude.

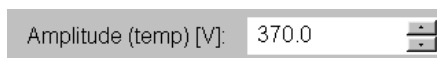
- ✧ Open the **X-Transmitter** parameter card in the **Manual adjustments** dialog window.



- (1) Temporary reference amplitude
- (2) System reference amplitude
- (3) Range display

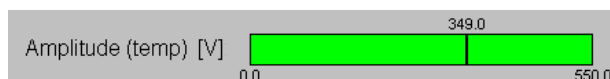
Entering the amplitude

You can enter a temporary reference amplitude for the type of nucleus used.

A screenshot of a software interface showing a text input field labeled "Amplitude (temp) [V]". The field contains the numerical value "370.0". To the right of the text field is a small icon consisting of two horizontal bars with arrows pointing outwards, likely representing a range or scale.

- ✧ Enter a value in the **Amplitude (temp)** input field.

Using the range display, you can check whether the amplitude is in the acceptable range. If not, you have to correct the value.



- ✧ The temporary reference amplitude used is displayed for comparison in the field underneath the currently used system reference amplitude for measurements.

Applying the amplitude

You can accept the temporary reference amplitude as the system reference amplitude.

A screenshot of a button labeled "Apply". The button is rectangular with a light gray background and a thin black border.

- ✧ Click **Apply**.

3D Shim

The 3D shim allows you to correct inhomogeneities of the magnetic field.

Performing adjustments

First enter the shim parameters and check the selected shim mode. The subsequent adjustment scan to shim the magnetic field requires the following steps:

- ❑ As a first step, a field map (three-dimensional description of the magnetic field) is created by acquiring a 3D data set (**Measure** button).
- ❑ Subsequently, the shim parameters are computed (**Calculate** button).
- After changing the shim setting, you have to readjust the frequency. If necessary, the system will perform this adjustment automatically.
- ✧ Open the **3D Shim** parameter card in the **Manual adjustments** dialog window.

The screenshot shows the 'Manual adjustments' window. At the top is a table of shim parameters. Below it is a large black image area for field maps. To the right of the image area are controls for amplitude, coil, temperature, mode, and receiver gain. At the bottom right are input fields for shim parameters (A00, A11, B11, A10, A20, A21, B21, A22, B22) and buttons for 'Calculate', 'Load Tune Up', 'Load System', and 'Apply'. At the bottom are tabs for 'Frequency', 'Transmitter', '3D Shim', 'Interactive shim', 'Water suppr.', and 'Show', along with a status bar and 'Close'/'Help' buttons.

No.	A00 1 [Hz]	A11 X [μT/m]	B11 Y [μT/m]	A10 Z [μT/m]	A20 Z2 [μT/m]	A21 ZX [μT/m]	B21 ZY [μT/m]	A22 X2-Y2 [μT/m]	B22 XY [μT/m]
1	63681788	3.04	4.26	23.77	-138.78	-11.01	-37.02	0.10	-12.58
2	63681788	3.08	4.20	23.71	-136.15	-10.94	-37.25	-1.70	-13.49

Callouts in the image:

- (1) Points to the table of shim parameters.
- (2) Points to the image area for displaying the field map.
- (3) Points to the slider for windowing the images.
- (4) Points to the switch-over between magnitude and phase display.
- (5) Points to the transmit amplitude control.
- (6) Points to the coil measurement signal for graphic displays.
- (7) Points to the name of the field map (temporary and system).
- (8) Points to the shim mode (standard and advanced).
- (9) Points to the receive gain control.
- (10) Points to the temporary shim parameter set.
- (11) Points to the system shim parameter set.

- (1) Numeric results
- (2) Image area for displaying the field map
- (3) Slider for windowing the images
- (4) Switch-over between magnitude and phase display
- (5) Transmit amplitude
- (6) Coil measurement signal for graphic displays
- (7) Name of the field map (temporary and system)
- (8) Shim mode (standard and advanced)
- (9) Receive gain
- (10) Temporary shim parameter set
- (11) System shim parameter set

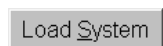
Entering the amplitude



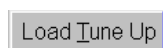
Amplitude [V]: 450.0

- ✧ Enter the transmit amplitude for the adjustment to be performed in the **Amplitude** input field.

Loading shim parameters



Load System



Load Tune Up

The nine shim parameters are part of a parameter set that describes field correction. You can load the shim parameters used by the system or those from the most recent tune-up:

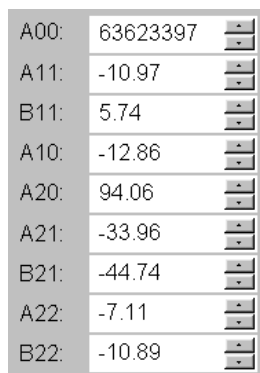
- ✧ Click **Load System**.

Or

- ✧ Click **Load Tune Up**.

The data loaded are transferred to the fields of the temporary shim parameter set.

Entering the shim parameters



A00:	63623397
A11:	-10.97
B11:	5.74
A10:	-12.86
A20:	94.06
A21:	-33.96
B21:	-44.74
A22:	-7.11
B22:	-10.89

You can also manually enter the shim parameters.

- ✧ Enter the transmit frequency for the measurement into input field **A00**.
- ✧ You can enter the components of the linear gradients in the **A11**, **B11**, and **A10** input fields.
- ✧ Depending on the configuration of your system, you can enter the second order shimming currents in input fields **A20**, **A21**, **B21**, **A22** and **B22**.

These shim parameters are used for field map acquisition.

Select shim mode

The shim mode used to perform the adjustment is defined during protocol generation.

→ [Page B.2–5](#), *Set the shim mode*

The activated button shows the current mode.



✧ To change the mode, click the button for the mode you would like to select next.

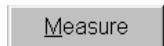
Selecting the amplification (gain)

Select the receiver gain according to the intensity of your signal.

→ [Page B.2–24](#), *Selecting the amplification (gain)*

Starting the measurement

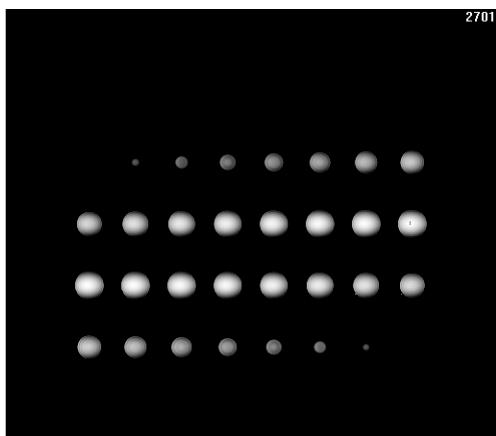
The measurement takes approximately 20 seconds in the standard mode. In the expanded mode, it will take approximately 40 seconds.



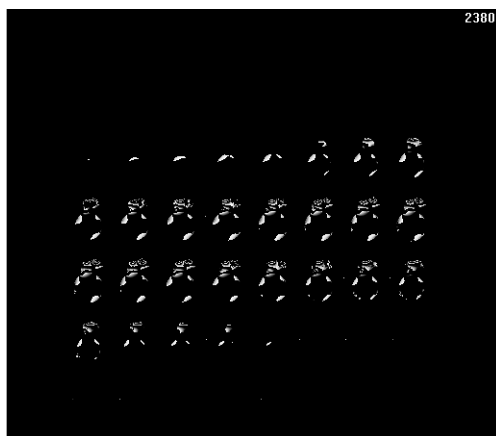
✧ Click **Measure**.

Displaying the field map

The field map is scaled accordingly and shown in the image area of the dialog window. A maximum of 64 images with a maximum resolution of 48×64 pixels are displayed.



Field map in magnitude display



Field map in phase display

The tomographic images are displayed row by row.

Selecting image type

Magnitude

✧ Click **Magnitude** to display magnitude images.

Or

Phase

✧ Click **Phase** to display phase images.

Windowing images

To set the window values for the image, use the two sliders at the bottom to the right in the image area.

- ✧ Move the vertical slider to adjust the contrast.
- ✧ Move the horizontal slider to adjust the width.

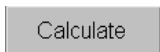
Selecting the receive channel

Ch.	Coil	Con.
C	Combined	-

The receive channel is used to select the signals of the local coils.

- ✧ Select the receive channel whose signal you want to have displayed.
- ✧ The **Combined** entry is used to select the combined signal of all coils listed.

Calculating the shim parameters



After the measurement, you can calculate the new shim parameters.

- ✧ Click **Calculate**.

The parameters are recalculated. They are used as the basis for the next field map measurement. They are displayed in the list of numeric results.

Numeric results

The results are listed below the running number of the iteration. The following parameters are displayed:

Parameters	Comment
No.	Running number of the adjustment calculation
A00	Transmitter frequency
A11, B11, A10	Linear gradients
A20, A21, B21, A22, B22	Second order shim currents



The three-dimensional field map measurement always shows the entire homogeneity volume of the magnet independent of the adjustment volume. The adjustment volume is taken into consideration and the field homogeneity is optimized while calculating the shim parameters. For this reason, changing the adjustment volume also results in different shim parameter settings for the same field map.

Applying the field map



Transfer the current temporary field map valid for adjustment to the MR system:

- ✦ Click **Apply** in the upper part of the **3D-Shim** parameter card.

Applying the shim parameters



Accept the temporarily calculated parameters for the MR system:

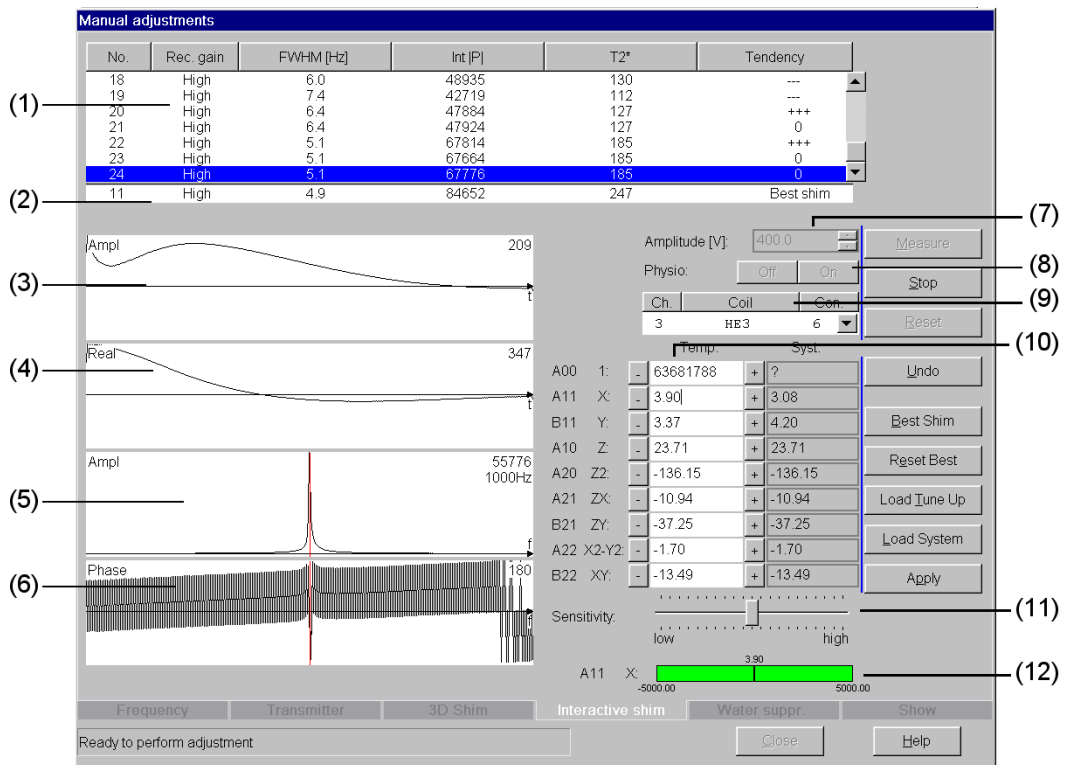
- ✧ Click **Apply** in the lower part of the **3D-Shim** parameter card.
Both parameters and field map are applied.
- ✧ When the adjustment was unsuccessful, you can change the system parameters by accepting manually entered parameters for the MR system.

Interactive Shim

During interactive shim, inhomogeneities of the magnetic field are compensated for as well. You can enter the shim parameters and monitor the measurement of the MR signal in real time.

If you enter new parameters, the changes are reflected in the results of the following MR signal measurement. You may continue shimming the magnetic field until it meets your requirements.

- ✧ After interactive shimming, you have to readjust the frequency. If necessary, the system will perform this adjustment automatically.
- ✧ Open the **Interactive Shim** parameter card in the **Manual adjustments** dialog window.
- ✧ Each time you open the **Manual Adjustment** dialog window, the tune-up parameters are loaded as temporary and thus as currently valid shim parameters.
→ [Page B.2–51, Loading shim parameters](#)



- (1) Numeric results
- (2) Best shim results
- (3) Amplitude of the measurement signal in the time domain
- (4) Real part of the measurement signal in the time domain
- (5) Amplitude of the measurement signal in the frequency domain
- (6) Phase of the measurement signal in the frequency domain
- (7) Transmit amplitude
- (8) Physiological parameters
- (9) Coil measurement signal for graphic displays
- (10) Temporary and system shim parameters
- (11) Sensitivity to shim parameter changes
- (12) Range display

Accepting physiological parameters



For the adjustment measurement, you may apply physiological measurement parameters from the relevant parameter card of the protocol.

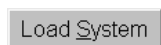
- ✧ Click the **On** button under **Physio**.
- ✧ When the **Off** button is activated, the physiological measurement parameters are not accepted.

Entering the amplitude



- ✧ Enter the transmit amplitude for performing the adjustment into the **Amplitude** input field.

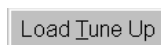
Loading shim parameters



The nine shim parameters are part of a parameter set that describes field correction. You can load the shim parameters used by the system or those from the most recent tune-up:

- ✧ Click **Load System**.

Or



- ✧ Click **Load Tune Up**.

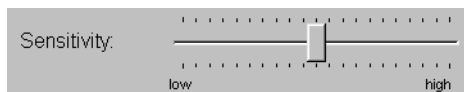
The data loaded are transferred to the fields of the temporary shim parameter set.

Entering the shim parameters

A00	1:	-	63623378	+
A11	X:	-	-10.97	+
B11	Y:	-	5.74	+
A10	Z:	-	-12.86	+
A20	Z2:	-	94.06	+
A21	ZX:	-	-33.96	+
B21	ZY:	-	-44.74	+
A22	X2-Y2:	-	-7.11	+
B22	XY:	-	-10.89	+

- ✧ Enter the transmit frequency for the measurement into input field **A00**.
- ✧ You can enter the components of the linear gradients in the **A11**, **B11**, and **A10** input fields.
- ✧ Depending on the configuration of your system, you can enter gradient parts of the second order in input fields **A20**, **A21**, **B21**, **A22** and **B22**.

As an alternative for numerical input, you can decrease or increase the shim parameters step-by-step with the **minus** or **plus** keys. Use the **Sensitivity** slider to set the increments.



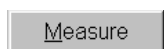
- ✧ Set the slider to the position desired between **high** and **low**.
- ✧ Adjust the setting during the course of the shim process as required (rather high in the beginning, and rather low toward the end).

Range display

Using the range display, you can check whether the transmit amplitude and/or the shim parameters fall within the acceptable range. If not, you have to correct the value.



Starting the measurement



✧ Click **Measure**.

The measurement is performed continuously using the current parameters (endless loop measurement).

Displaying results

The results for each measurement are displayed numerically and graphically.

After the measurement, the following parameters are displayed numerically:

Parameters	Comment
No.	Running number of the adjustment measurement
Rec. gain	Gain setting used
FWHM (Hz)	Width of the pulse received (frequency domain) at half the maximum amplitude (Full Width Half Maximum) During spectroscopy measurements, please note that the line width will decrease considerably in the post-processed real part spectrum.
Int P 	Integral of the amplitude in the time domain
T2*	Transverse relaxation time (approximation)
Tendency	indicates whether the parameter changes result in shim improvement (+++) or degradation (---) or whether the status remains the same (o)

After the measurement, the following results are displayed graphically:

- ☐ Amplitude of the measurement signal in the time domain
- ☐ Real part of the measurement signal in the time domain
- ☐ Amplitude of the measurement signal in the frequency domain
- ☐ Phase of the measurement signal in the frequency domain

Selecting the receive channel

Ch.	Coil	Con.
C	Combined	-

The receive channel is used to select the signals of the local coils.

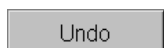
- ✧ Select the receive channel whose signal you want to have displayed.
- ✧ The **Combined** entry is used to select the combined signal of all coils listed. In this case, only the magnitude amplitudes are displayed graphically.

Change parameters

Once the measurement results are displayed, you can make changes while the measurement is in progress. The **Tendency** column provides valuable information.

- ✧ Change the parameters so that full width at half maximum (**FWHM**) decreases and **T2*** increases.
- ✧ In the numeric display, the line with the highest **T2*** value is marked **Best Shim**.

Undoing changes



To undo your changes in the shim channel last used:

- ✧ Click **Undo**.

Restoring the default settings



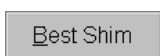
You can undo your changes and set the parameters that appear when you open the dialog window.

- ✧ Click **Reset**.
- ✧ System parameters that were changed during the adjustment cannot be restored to their initial settings before the adjustment.

Loading the best shim

The best shim parameters determined with the current measurement can be read in as temporary shim parameters and used for the next adjustment measurement.

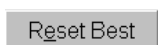
This is especially useful if you have moved away from the most optimal values while varying parameters. The most optimal shim parameters are determined based on the maximum transverse relaxation time (**T2***).



✧ Click **Best Shim**.

Deleting the best shim

You may delete the best shim parameters obtained in the current measurement, making them invalid.



✧ Click **Reset Best**.

Applying parameters

To use the temporary shim parameters for examinations, transfer them to the MR system.



✧ Click **Apply**.

Stopping the measurement



✧ To end the measurement, click **Stop**.

Manual water suppression

There are special RF pulses that are used for water suppression during measurements. With this adjustment, you determine a correction factor for the transmitter amplitude of these RF pulses. Within the adjustment volume, the flip angles are set for optimal suppression of the water signal.

- ✧ Open the **Water suppr.** parameter card in the **Manual adjustments** dialog window.

The screenshot shows the 'Manual adjustments' dialog window. It contains a table at the top, several plots on the left, and adjustment controls on the right. Numbered callouts (1) through (12) point to specific elements:

- (1) Points to the table header.
- (2) Points to the 'Ampl' plot.
- (3) Points to the 'Phase' plot.
- (4) Points to the 'Ampl' plot showing a peak.
- (5) Points to the 'Phase' plot.
- (6) Points to the 'CorrFac - Delta' button.
- (7) Points to the 'Apply' button.
- (8) Points to the 'Corr. factor (sys):' input field.
- (9) Points to the 'Delta transmitter:' input field.
- (10) Points to the 'Receiver gain:' buttons (Low/High).
- (11) Points to the 'Ch' dropdown menu.
- (12) Points to the 'Go' button.

No.	CorrFac	Signal (CorrFac-Delta)	Signal (CorrFac)	Signal (CorrFac+Delta)	New CorrFac	A.
1	1.0000	117.591469	22.169137	49.619967	1.0277	Y

Ampl 23.95
Phase 167
Ampl 3002
Phase 177

Corr. factor (temp): 1.0277
Corr. factor (sys): 1.0277
Delta transmitter: 0.1000
Receiver gain: Low High
Ch: 2 Coil: HE4 Con: 3

Go Abort Reset

Frequency Transmitter 3D Shim Interactive shim Water suppr. Show

Adjustment converged Close Help

- (1) Numeric results
- (2) Amplitude of the measurement signal in the time domain
- (3) Phase of the measurement signal in the time domain
- (4) Amplitude of the measurement signal in the frequency domain
- (5) Phase of the measurement signal in the frequency domain
- (6) Selecting a signal
- (7) Temporary correction factor
- (8) System correction factor
- (9) Increment for transmit amplitude
- (10) Receiver gain
- (11) Coil measurement signal for graphic displays
- (12) Range display

Entering the correction factor

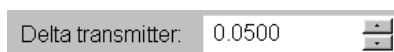


- ✧ Enter the correction factor desired in the **Corr. factor (temp)** input field.

For comparison, the correction factor currently used by the system is displayed below.

Enter the increment for the transmit amplitude

To determine the correction factor, enter an increment so that multiple measurements can be performed with different correction factors. The correction factors used for the measurements differ by one increment exactly.

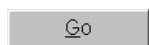
A screenshot of a software interface showing a label "Delta transmitter:" followed by a text input field containing the value "0.0500". To the right of the input field is a small vertical slider control.

- ✧ Enter an increment into the **Delta transmitter** input field.

Selecting the amplification (gain)

Select the receiver gain according to the intensity of your signal.
→ [Page B.2–24, Selecting the amplification \(gain\)](#)

Starting the measurement

A screenshot of a button labeled "Go" with a right-pointing arrow.

- ✧ Click **Go**.
The measurement completes in a few seconds.
- ✧ Repeat the measurements using different parameters until the system has optimized the correction factor.
- ⇒ Successful adjustments are identified by a **Y** in the numerical results.

Numeric results

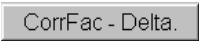
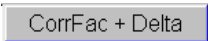
The results are listed below the running number of the iteration.
The following parameters are displayed:

Parameters	Comment
No.	Running number of the adjustment measurement
Corr. factor	Correction factor of the adjustment measurements
Signal (corr. factor - delta)	Signal strength for measurements with correction factor minus increment
Signal (corr. factor)	Signal strength for measurements with correction factor
Signal (corr. factor + delta)	Signal strength for measurements with correction factor plus increment
New corr. factor	Calculated (better) correction factor
A.	Adjustment successful (Yes/No)

Graphic display of the results

Amplitude and phase of the respective measurement signal are graphically displayed in the time and frequency domains.

Use the following keys to select among three different signals:

Button	Comment
	Displays the signal of the measurement <i>Correction factor minus increment</i> .
	Displays the signal of the measurement <i>Correction factor</i> .
	Displays the signal of the measurement <i>Correction factor plus increment</i> .

Selecting the receive channel

The receive channel is used to select the signals of the local coils.

Ch.	Coil	Con.
C	Combined	- 

- ✧ Select the receive channel whose signal you want to have displayed.
- ✧ The **Combined** entry is used to select the combined signal of all coils listed. In this case, only the magnitude amplitudes are displayed graphically.

Accepting the correction factor



The frequency is automatically transmitted to the MR scanner after successful adjustment ("Y" in the numeric input field).

This is how you manually (without successful adjustment) transfer a correction factor within the permissible range into the MR system:

- ✧ Click **Apply**.

The correction factor is transferred as the system reference amplitude.

Displaying adjustment results

You can display an overview of all adjustment results.

- ✧ Open the **Show** parameter card in the **Manual adjustments** dialog window.

Manual adjustments

Adjustment Information

System values

Adjust Frequency		adjusted
system frequency	63.681785 MHz	

Adjust Transmitter		adjusted
reference amplitude	215.3 V	

Adjust Shim		adjusted
gradient offsets	2.85 $\mu\text{T/m}$ = 35.95 dac (x / A11) 4.27 $\mu\text{T/m}$ = 53.85 dac (y / B11) 23.50 $\mu\text{T/m}$ = 260.11 dac (z / A10)	
shim currents	-125.89 $\mu\text{T/m}^2$ = -864.65 mA (A20) -10.26 $\mu\text{T/m}^2$ = -160.24 mA (A21) -38.12 $\mu\text{T/m}^2$ = -595.65 mA (B21) 1.32 $\mu\text{T/m}^2$ = 22.33 mA (A22) -11.70 $\mu\text{T/m}^2$ = -197.62 mA (B22)	

Latest values

Adjust Frequency		
system frequency	63.681785 MHz	

Adjust Transmitter		
reference amplitude	215.3 V	

Adjust Shim		
gradient offsets	2.85 $\mu\text{T/m}$ = 35.95 dac (x / A11) 4.27 $\mu\text{T/m}$ = 53.85 dac (y / B11) 23.50 $\mu\text{T/m}$ = 260.11 dac (z / A10)	
shim currents	-125.89 $\mu\text{T/m}^2$ = -864.65 mA (A20) -10.26 $\mu\text{T/m}^2$ = -160.24 mA (A21) -38.12 $\mu\text{T/m}^2$ = -595.65 mA (B21) 1.32 $\mu\text{T/m}^2$ = 22.33 mA (A22) -11.70 $\mu\text{T/m}^2$ = -197.62 mA (B22)	

Information Depth:

- ☐ Customer
- ☐ Service
- ☐ Developer

Adj. Vector:

- ☒ System
- ☒ Latest

Adj. Type:

- Frequency
- Transmitter
- Dico
- Shimming
- Coil Position

Invalidate All

Adjust All

Abort

Update Information

Frequency Transmitter 3D Shim Interactive shim Show

Close Help

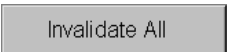
- ✧ This list is primarily intended for diagnostic purposes and accessible by Siemens Service.

Updating the display

Update Information

- ✧ To update results immediately, click **Update Information**.

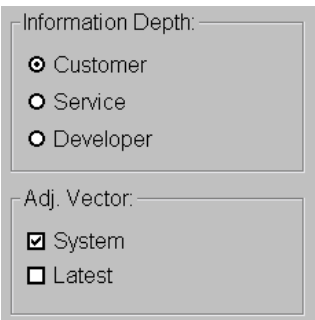
Rejecting results



How to reject all adjustments already made:

- ✦ Click **Invalidate All**.

Determining the display of parameters

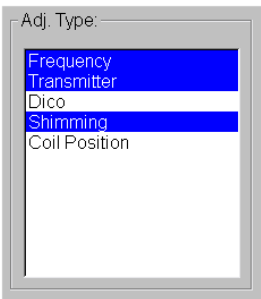


In the **Customer** mode, information relevant to the customer is displayed.

Go to the **Adj. Vector** group and select the parameter sets to be displayed:

Abbreviations	Comment
System	Adjustment parameters applicable to the system
Latest	Parameters used during the last measurement

- ✦ Activate the respective parameter sets.



Go to the list **Adj. Type** and select the types of adjustments to be displayed.

- ✦ Click the entry to either select or deselect a particular adjustment type.

Restoring the default settings

Aside from the **3D Shim** parameter card, you are able to reset the manual adjustments of the parameters after you completed the adjustment. The parameters displayed when the dialog window was opened are re-loaded.

- ✧ It is not possible to reset system parameters that were changed in the course of the adjustment.
- ✧ To reset parameters, click **Reset** in the respective parameter card.

A rectangular button with a light gray background and a thin black border. The word "Reset" is centered in a black, sans-serif font.

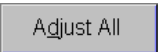
Starting protocol adjustments manually

All adjustments planned in the selected protocol may be started manually without starting the following protocol.

- ✧ Open the protocol.
- ✧ Open the **Show** parameter card in the **Manual adjustments** dialog window.
- ✧ Click **Adjust All**.

The adjustments are performed.

- ✧ If an error occurs during the measurement, the adjustment will be aborted and an error message is displayed.

A rectangular button with a light gray background and a thin black border. The text "Adjust All" is centered in a black, sans-serif font.

Changing the resonance frequency after inline adjustment

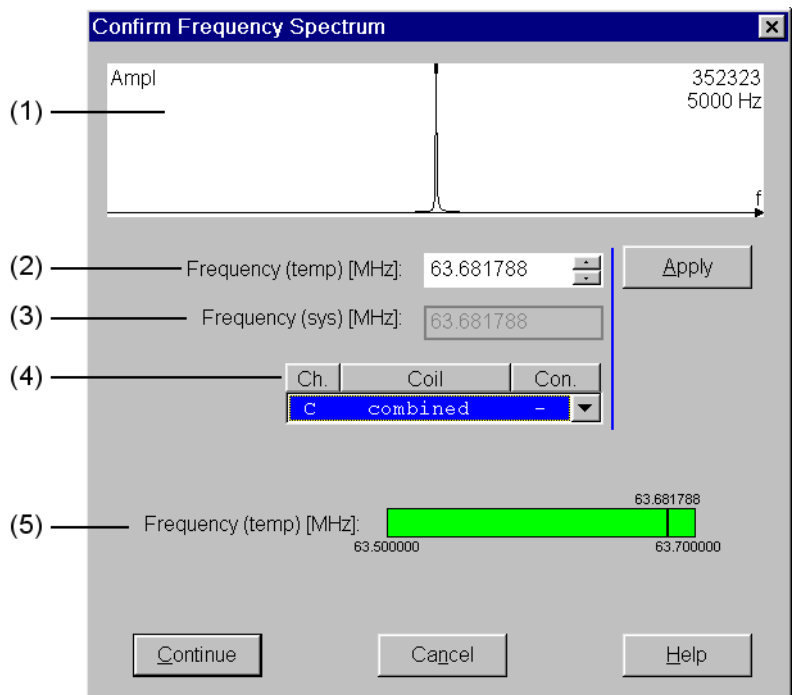
An inline adjustment is performed automatically prior to each measurement. You can pause the system to confirm or change the resonance frequency calculated by the adjustment.

Confirm freq. adjustment ☒

- ✧ Activate the **Confirm freq. adjustment** checkbox on the **System Adjustment** parameter card.

→ [Page B.2–3, Setting the adjustment parameter card](#)

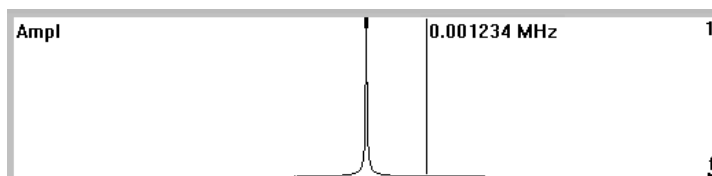
After each Inline adjustment, the **Confirm Frequency Spectrum** dialog window is shown.



- (1) Amplitude for the adjustment measurement
- (2) Temporary frequency
- (3) System frequency
- (4) Receive channel
- (5) Range display

Graphic display of the adjustment scan

A graphic is shown together with the applicable frequency spectrum.



Using a vertical marker, you can graphically select the frequency in this display.

→ [Page B.2–26](#), *Graphic frequency determination*

If no adjustment graphics are available, you have to generate the data for graphic display via a manual frequency adjustment.

→ [Page B.2–21](#)

Entering the transmit frequency

Frequency (temp) [MHz]: 63.623318

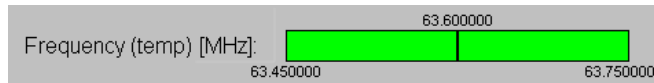
✧ Enter the new transmit frequency in the **Frequency (temp)** field.

- Or -

Determine the transmit frequency graphically.

→ [Page B.2–26](#)

Using the range display, you can check whether the frequency is in the acceptable range. If not, you have to correct the value.



- ✧ For comparison, the system frequency used for the examination is shown in the field below the temporary frequency.

Selecting the receive channel

Ch.	Coil	Con.
C	Combined	-

The receive channel is used to select the signals of the local coils.

- ✧ Select the receive channel whose signal you want to have displayed.
- ✧ The **Combined** entry is used to select the combined signal of all coils listed.

Starting the measurement

Continue

After you have corrected the resonance frequency you are ready to start the measurement:

- ✧ Click **Continue**.



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Introduction to the MR Examination

After you register a patient, you'll make the necessary preparations in the examination room.

Preparations in the examination room

Position the patient in the orientation entered during patient registration. Observe the information regarding patient safety provided in the *System Manual*. Position the coil(s) and connect the coil cables to the connectors on the patient table. Subsequently identify the center of the examination region with the light localizer and move the patient into the magnet isocenter. *refer to the System Manual*

Working at the syngo Acquisition Workplace

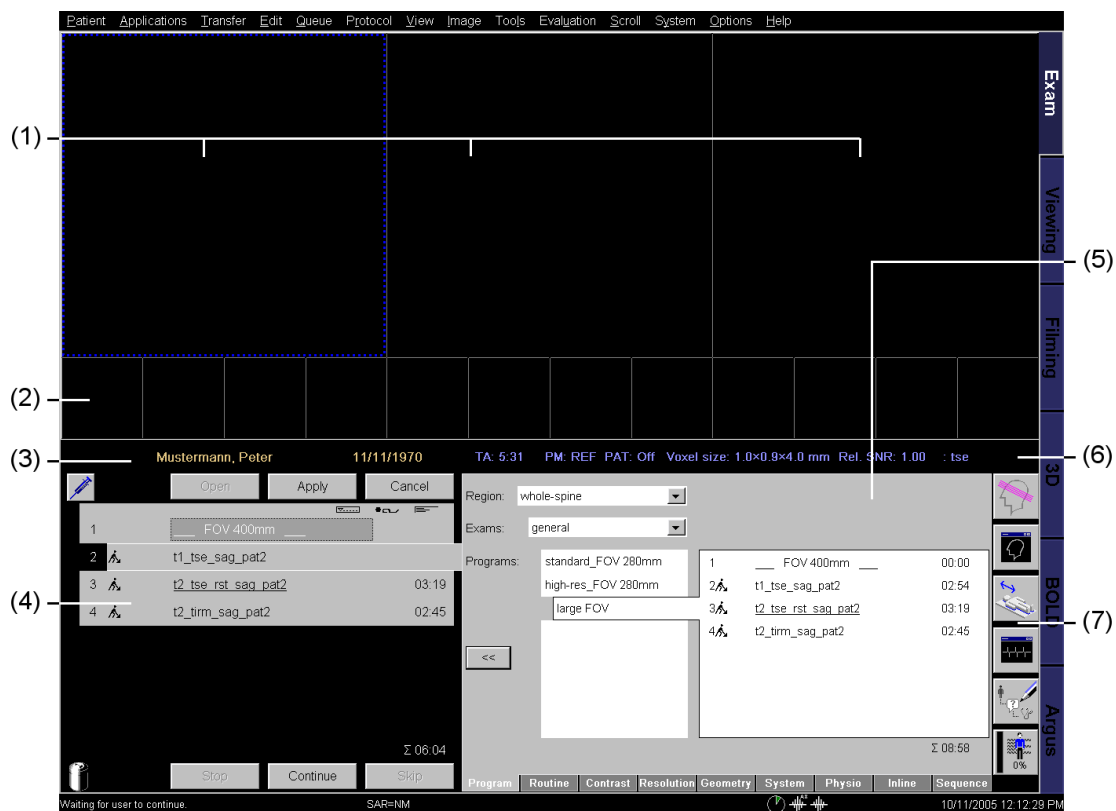
Then, you will start the actual measurement at the *syngo* Acquisition Workplace in the control room. For this purpose, use the **Exam** task card. This card enables you to select, control, and monitor examination procedures.

You are able to observe the patient through the window in the control room during the examination. Using the intercom system you are able to hear the patients or give them instructions.

The next chapters walk you through an MR examination. You will find detailed descriptions on how to perform special measurements, organize and manage measurement programs, and monitor thresholds relevant to the health of the patient.

The following pages provide an overview of the **Exam** task card, the measurement procedure, as well as important terms.

Layout of the Exam task card



- (1) Reference image segments
- (2) Stamp segments (Maestro Layout only)
- (3) Patient data
- (4) Program control
- (5) Parameter cards
- (6) Protocol info line
- (7) Toolbar



Reference image segments	The reference images used to position graphic measurement objects (slices, slabs) are displayed.
Stamp segments	Previously measured images are loaded into the stamp segments of the Maestro layout for further use. → Page C.2–2, Maestro layout
Patient data	Above the program control, you'll find information concerning the current patient.
Program control	After loading, the program control contains a list of the protocols to be measured (wait list). It also includes buttons for controlling the program sequence.
Parameter cards	The Program parameter card is used to select a measurement program or protocol. You may use the other cards to adapt the parameters for the protocol currently open.
Protocol info line	Information regarding the open protocol is located in the protocol info line. → Page C.7–2, Protocol info line
Toolbar	The buttons on the tool bar are used to open dialog windows for editing and controlling the measurement.

Important terms

The following terms are of essential importance.

Sequence

A **Sequence** is the chronological order of RF pulses and gradient pulses for excitation of the volume to be measured, for signal generation, and for position encoding.

Your MR system is equipped with a large number of sequences that allow you to perform different MR imaging techniques.

Measurement protocol

A sequence that is characterized by a full set of optimized parameters, is known as the **Measurement protocol (Protocol)**.

Measurement pause

Measurement pauses (Pauses) are planned interruptions in an examination procedure, e.g., for contrast agent administration.

Program steps

Measurement protocols and measurement pauses are known collectively as **Program steps**.



Measurement program

A **measurement program (Program)** contains a list of optimized program steps suitable for clarifying diagnostic problems in a specific region of the body.

For greater clarity, the numerous measurement programs are assigned to specific examinations and regions.

Examination

An **examination** contains one or several measurement programs for clarifying the medical question involved.

Region

A **Region** describes the hierarchical level in the Exam Explorer. It contains all MR examinations for an anatomical region of the patient.

In the Exam Explorer you can identify a region defined with a DICOM standardized examination region (DICOM Body Part Examined)

Examination region

A **Body Part Examined** (DICOM Body Part Examined) is used as an anatomical identification for patient images that have been measured. This information is not visible in the image text but can be used, for example, in combined examinations to filter data and for billing purposes.

Series block

A series block groups together all series measured during the examination of a patient, and whose coordinate system refers to the same zero point. The zero point is defined by the initial table position of the first measurement of a series block.

The series of a series block meet the following minimum prerequisites:

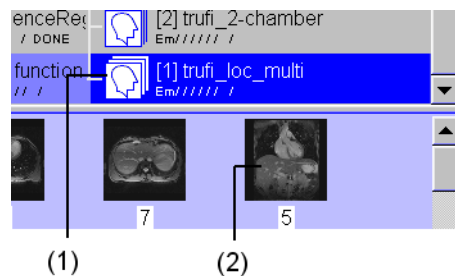
- ☐ Same patient
- ☐ Same date of examination
- ☐ Same whole body patient coordinate system
→ [Page A.2–5, Whole Body Patient Coordinate System](#)
- ☐ Same position

A series block ends when you:

- ☐ Close the current patient
- ☐ Reregister a new patient and/or the current patient (since you have changed the patient's position, for example)
- ☐ Lower the patient table
- ☐ Set a new mark with the laser light localizer

Series and images

A series combines all images acquired while measuring the same protocol.



Series (1) and images (2) in the patient list

Frame of Reference (FoR)

The FoR defines a common area of validity for various images in a series block.

The following images have the same FoR:

- ☐ Non distortion-corrected images (image type ND) of a single table position have the same FoR.
- ☐ All distortion-corrected images (image type DIS2D) of a series block have the same FoR.

The following images have different FoR's:

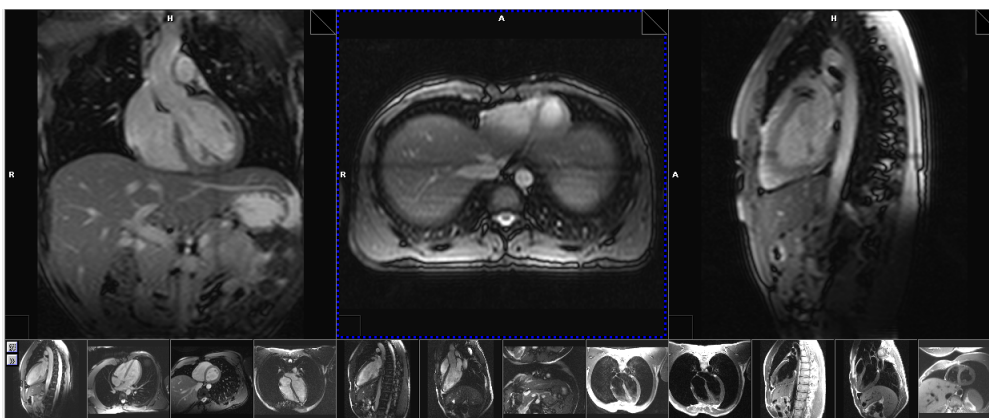
- ☐ Non distortion-corrected images (image type ND) from different table positions.
- ☐ Distortion-corrected images with strong distortions (image type DIS2D*) have a different FoR.

Possibilities for images with the same FoR:

- ☐ They can be combined on the **3D** task card.
- ☐ They can generally be post-processed together.
- ☐ They are post-processing compatible in DICOM.

Reference image

The images in the reference image segments (large segments in the **Exam** task card) are reference images. You can use the reference images to plan the slices for the actual protocol in graphical format.



- ⇒ Images in stamp segments of the Maestro layout are not reference images, instead they represent additional series.

→ [Page C.2–2, Maestro layout](#)



Isocenter of the system

Magnetic field lines run parallel to the isocenter. Distortions can occur outside this area.

Isocenter

The isocenter identifies the center of the magnet core. Prior to the measurement, the patient table is moved so that the planned protocol isocenter and the magnet isocenter are identical. The planned isocenter of a protocol corresponds to the protocol table position.

Isocentered protocols calculate their table position such that the center of all slices of a protocol are at the magnet isocenter. As a result, all slices of this type of protocol are as close to the isocenter as possible.



Procedure for routine scanning

Selecting and transferring a program	You select a measurement program on the Program parameter card and apply it in the program control. → Page C.3–1
Acquiring reference images	The measurement of reference images is the first program step, and usually starts automatically. The measured reference images are displayed in the image area. → Page C.4–1
Preparing a protocol	Subsequently, you prepare the next protocol in the program control. Slices, slabs and regional presaturation as well as navigators can be positioned graphically. Use the parameter cards to check or change the measurement parameters of the protocol. → Page C.7–1
Running a protocol	Start the protocol. You prepare the next protocol during the measurement. → Page C.6–1
Saving and transferring images	You are able to save the images from the Examination task card as well as export them and transfer them to other task cards. → Page C.4–1



You are able to interrupt a protocol in progress, repeat measurements, change the sequence of the program steps, and insert pauses (e.g., for contrast agent administration).

To control the images measured you can use the **Inline Display** or movie display. You can also use the small or large segments.

For physiologically triggered measurements, the **Physiological Display** dialog window is available.

During the measurement, the system automatically monitors the SAR and dB/dt load on the patient (RF exposure and time-varying magnetic fields).

Use the **Exam Explorer** to organize examinations, measurement programs, and protocols.



Maestro Layout and Movie Display (Option)

The Maestro layout of the **Exam** task card provides stamp segments in addition to the two or three large GSP image segments. The stamp segments help you to orient yourself during examinations and manage the acquired series.

Stamp segments *do not* allow graphic slice positioning. While you can window images in stamp segments, you cannot zoom or pan them.

You can load the series from the stamp segments into the large image segments and view them in movie display. This provides you with a live impression of the morphology and physiology of the region to be examined, enabling you to obtain the first diagnostic results early in the examination.

There are numerous ways of configuring and controlling movie display.

Maestro layout

This is how you work with stamp segments:

- ☐ You load the series either automatically or manually into the stamp segments.
- ☐ For slice positioning you drag the suitable series from the stamp segments to the large image segments via the mouse.
- ☐ To repeat a measurement, drag the measured series from the stamp segments into the queue.
- ☐ To apply a series for post-processing to another task card, drag the series from the stamp segments to the tab of the desired application.
- ☐ You store series from the stamp segments together to another new series or append them to existing series.
- ☐ You transfer series from the stamp segments for filming.
- ☐ You are exporting series from stamp segments.

Selecting the maestro layout and loading series

- ✧ Select **View > Maestro UI**.

The **Exam** task card is displayed in the Maestro layout.

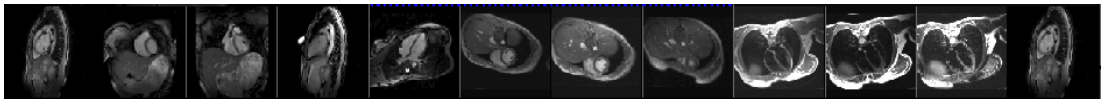
Manually loading series

- ✧ Select the series in the program control or in the **Patient Browser**.
- ✧ Use the mouse to drag the selection to the stamp segments.

Automatically loading series

Use the protocol properties to ensure that the images of the protocol are loaded automatically into the stamp segments immediately after reconstruction.

→ [Page C.8–8, Automatically storing and loading](#)



Each stamp segment represents a series.

You can change the layout of the series in the stamp segments as desired.

- ✧ If you leave the mouse pointer at an occupied stamp segment, a brief message appears informing you about the series loaded in it.

Transferring series from stamp segments into image segments

You can load series from stamp segments into the large image segments to position slices or start movie display.

- ✧ Mark the large image segment where you want to load the series.
- ✧ Double-click the required stamp segment.
 - Or -
 - Drag the series from the stamp segment to the large segment using the mouse.

The series from the stamp segment is loaded into the selected GSP image segment.

Repeating the measurement of a series

If required, you can repeat a measurement.

- ✧ Drag the series from the stamp segment to the queue using the mouse.
 - Or -
 - Select **Append To Queue** in the context menu.

A copy of the protocol is generated including the current slice position and converted into an executable program step (Phoenix function).

Saving, filming, exporting, and transferring series for post-processing

You can save the series directly from the stamp segments to a new series, append them to an existing series, film them, export them or load them into post-processing task cards.

You proceed the same way as you would for images or series of large image segments.

→ [Page C.4–31](#), *Saving, exporting, and transferring images*

Navigating in stamp segments

You may load any number of series into the stamp segments. A maximum of 12 series are displayed.

If you loaded more than 12 series, small arrow keys are displayed in the far left segment.



Navigating from series to series

- ✧ To set the input focus on a stamp segment, click the desired segment.
- Or -
- Use the cursor keys to shift the input focus to the required position.

Shifting all series by one segment



You are able to shift all series by one position to the left or right:

- ✧ Click the respective arrow key in the outer left segment.

Shifting series by 12 segments

You can shift all series in blocks of 12 segments. This hides all series previously visible.

- ✧ Press the **Image up** or **Image down** key.

Jumping to the first/last series

- ✧ Use the **Home** or **End** key to branch to the first or last series.

Scrolling from image to image in the stamp segment



Identical to a large image segment, you can scroll from image to image.

- ✧ Press the appropriate key on the symbol keypad.
- ✧ When you keep the key pressed, the images are shown as in a movie.

Changing the layout of series

You can remove series from the stamp segments and repeat the measurement. You load the new series either automatically or manually back into the stamp segments.

Removing series

The **Patient Browser** is not affected by removing series via stamp segments.

- ✧ Select the requested stamp segments.
 - **Viewing/Changing the image display** in the *syngo* Operator Manual
- ✧ When you select stamp segments, the entire series is selected automatically.



- ✧ To delete series and keep the empty segments, select **Editing > Clear Document(s)**.

Or

- ✧ To delete series without empty segments remaining, select **Remove Segment** in the context menu.

Inserting an empty stamp segment

You are inserting an empty stamp segment to correctly position a series during reload with the mouse.

- ✧ Select the segment in front of which you want to insert an empty segment.
- ✧ Select **Insert Segment** from the context menu.
- ✧ Repeat inserting in case you need several empty segments.

Reloading series with the mouse

You may reload series from the program control, the **Patient Browser**, or the large image segments.

- ✧ Select the series to be loaded.
- ✧ Use the mouse to drag the selection to an empty stamp segments.



If the destination segment is not empty, the old series will be replaced by the first series after a new selection.

Movie display

During a measurement or measurement pause, you can start the movie display and check the quality of the images being acquired, and gather preliminary diagnostic information.

A large image segment is used for the movie display. The images of a series are always shown in the form of a movie.

You can optimize the movie display according to your own requirements by:

- ☐ Setting the speed
- ☐ Selecting the **LOOP** or **Yo-Yo** mode
- ☐ Changing the order of images
- ☐ Skipping to the next group or series during movie display
- ☐ Setting the real-time mode
- ☐ Changing image display during a movie pause

⇒ You are able to perform additional measurements or repeat them while the movie is running.

→ [Page C.2–4](#), *Repeating the measurement of a series*



The movie quality may be affected by processes running in the background.

Groups within a series

Groups are formed for the movie display to provide a better overview within series. Movies display the images of one group. However, it is possible to jump from group to group.

One series – one group

A series is treated as *one* group if all images in the series differ according to the same sorting criterion.

Example: Single-slice multi-phase series (cardiac imaging)

Sorting SP >> TT >> TA. All images of the initial series have the same slice position. However, they have different trigger times.

→ [Page C.4–14](#), *Sort criteria*

One series – multiple groups

A series is divided into *several* groups, if the current display mode contains images in consecutive order that can be differentiated only when using the secondary search criteria.

Example: Multi-slice multi-phase series (cardiac imaging)

Sorting SP >> TT >> TA. This series is divided into n groups, because it contains images of n different slice positions and trigger times.

The first group contains all images with slice position 1 and different trigger times. The second group contains all images with slice position 2 and different trigger times.

→ [Page C.4–14](#), *Sort criteria*

Starting the movie display

Start the movie display either manually or automatically.

Manual start

- ✧ Place the input focus on the segment you want to use for movie display.
- ✧ Select **Scroll > Movie > Movie ON**.
 - Or -
 - Ensure that the input focus is on the image area and press the space bar.
 - [Page A.3–3](#), *Keyboard focus in the image area*

Auto start

Using the properties of a protocol enables you to automatically start the movie display of a series immediately after reconstruction. The property is already set by default for some of the protocols.

→ [Page C.8–8](#), *Automatically storing and loading*

If the protocol generates more than one series or multiple groups within the series, only the first series or group will be shown in movie display. All other series or groups can be displayed with **Scroll > Series Next**.

Controlling movie display (basic functions)

You start, interrupt and end the movie display using the keyboard, mouse or menu commands.

⇒ For extended control, use the **Movie Control** dialog window.
→ [Page C.2–13](#), *Controlling movie display via the dialog window*



Movie display has three statuses:

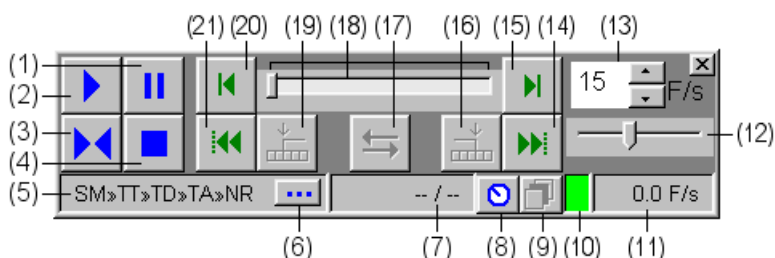
- ☐ **running**
Movie display is active and running. You can advance to the next group or series.
- ☐ **paused**
Movie display is active, but paused. You can scroll from image to image.
- ☐ **stopped**
Movie display is not active. Series divided into groups are ungrouped.

Overview of basic functions:

Function	Menu	Keyboard	Mouse
Starting the movie display	Scroll > Movie > Movie ON In the context menu: Movie ON	Space bar	–
Beginning or ending the pause	Scroll > Movie > Movie Paused	Space bar	Mouse click in the movie segment
Stop movie display	Scroll > Movie > Movie ON In the context menu: Movie ON	ESC	–

Controlling movie display via the dialog window

✧ To open the **Movie Control** dialog window, select **Scroll > Movie > Movie Control**.



- | | |
|---|--|
| (1) Pausing movie display | (15) Scroll forward by one image, status automatically set to paused |
| (2) LOOP mode | (16) Set end marker to limit range |
| (3) Yo-Yo mode | (17) Reverse display order (in paused status only) |
| (4) Stop movie display | (18) Position of the current image in the series |
| (5) Current sort order | (19) Set "beginning" marker to limit range (in paused status only) |
| (6) Superposing the Movie Properties dialog window | (20) Scroll back by one image, status automatically set to paused |
| (7) Index of the current image/total number of images | (21) Running/paused status: Scrolling backward by one group. Stopped status: Scroll back by one series |
| (8) Real-time mode | |
| (9) Skip images (real-time mode) | |
| (10) LED display | |
| (11) Current speed in images per second | |
| (12) Speed regulator | |
| (13) Input field for speed | |
| (14) Running/paused status: Scrolling forward by one group. Stopped status: Scrolling forward by one series | |

- ⇨ It is also possible to automatically set the dialog window each time movie display is started.

→ [Page C.2–20](#), *Configuring the movie display*

LOOP and Yo-Yo mode

In **LOOP** mode, the images of a series are displayed in a loop. After the last image of the series, the first image is displayed again, etc.

In the **Yo-Yo** mode, the direction of movie display is changed after the last image. The second to last image is displayed, etc.



- ⇨ To start movie display in the **LOOP** mode, click this button.

Or



- ⇨ Click this button to start movie display in the **Yo-Yo** mode.

Order of the images

The image sequence within a series is determined by the sort set by you in the **Display sequence** dialog window.

→ [Page C.4–13](#), *Defining the scroll sequence*

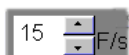
SM»TT»TD»TA»NR

This is how you reverse the image sequence in movie display:



- ⇨ To reverse the order of images in movie display, click this button.

Speed of movie display



You can set the number of images displayed per second in movie display either numerically or with a slider.

✧ Enter a whole number between 1 and 99 and press the **Enter** key.

- Or -

Drag the slider to the desired new position.

The far left position of the slider corresponds to a speed of 1 image per second. The far right position corresponds to the maximum speed obtainable.

✧ If you enter a higher speed in the input field, the value range of the slider is adjusted accordingly.

The position of the slider does not change in movie display, that is, the slider always shows the speed set by the user.

The actual speed is shown in the bottom right dialog window.

15.0 F/s

Setting the real-time mode



Series with images that differ in acquisition time can be displayed in real-time mode. In this mode, images are shown according to the time interval they were measured.

✧ To start the real-time mode, click this button.



If the time interval between images is very short, you may find that the real-time speed is greater than the maximum speed technically possible. The display is then delayed and no longer corresponds to real-time.

When you skip images in movie display, the display remains in real-time even at very high speeds.

- ✦ To skip images in the real-time mode, click this button.
- ➡ If you manually change the speed while real-time mode is active, real-time mode will automatically stop.

LED display

The color of the LED lets you differentiate between the different statuses of the movie display.

Color	Status
	Stopped or paused
	Running, all images are displayed
	Running, the Skip images option is enabled and at least two images are skipped
	Running, the Skip images option is disabled, the current speed is lower than the currently set one

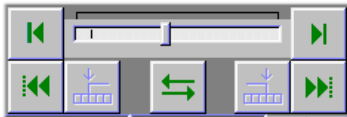
Editing images during movie display

Images in movie display can be windowed, zoomed, reduced and panned.

→ [Page C.4–20](#), *Zooming and panning reference images*

Movie display is paused during image editing while pressing and holding the left mouse button. Movie display resumes when the mouse button is released.

Navigating in movie display



The area display in the center of the **Movie Control** dialog window shows the area of the current group. The black bracket above the slider symbolizes the beginning and end of the group displayed.

In the **paused** status, the position of the slider indicates the position of the current image within the group. The status line also displays the position of the current image.

✧ To stop movie display and show the first image of the group, pull the slider all the way to the left.

Or

✧ To stop movie display and show the last image of the group, pull the slider all the way to the right.

✧ You can drag the slider beyond the boundary of the group. If you subsequently change the **running** status, movie display is set to the beginning and end of the group belonging to the current image index.

Scrolling from image to image

In the movie segment, you can scroll all images in a series across group boundaries. To do this, you work with the button in the **Movie Control** dialog window, with menu commands, with the keys on the symbol key block or yo-yo with dog ears.

→ [Page C.4–18](#), *Scrolling through images or series*

✧ In the **stopped** status, the buttons in the **Movie Control** dialog window are not available for scrolling.

✧ To scroll forward by one image in the **running** or **paused** status, click this button.

Or

✧ To scroll backward by one image in the **running** or **paused** status, click this button.

Movie display is stopped for this purpose.



Scrolling from group to group

In the **running** or **paused** status, you can scroll from group to group with the buttons, menu commands or keys on the symbol keyboard. You may also move from the last group of series A to the first group of series B.

✧ If the series in the movie segment contain only one group, you will scroll from series to series.

✧ To scroll forward by one group, select **Scrolling > Series Next**.

- Or -

Click this button.





- ✧ To scroll back by one group, select **Scrolling > Series Previous**.

- Or -

Click this button.

As compared to normal series scrolling, the proxy image of the next group is not shown, but rather the image corresponding to the index appears.

→ [Page C.4–12](#), *Representative image*

- ✧ In the **stopped** status you always scroll from series to series since groups are cancelled.

Changing the movie display range



After you started movie display, the beginning marker is located at the first image of the group to be displayed and the end marker is at the last image of the group to be displayed.

If a series contains one group only, the beginning marker is at the far left and the end marker at the far right of the range displayed.

Proceed as follows to establish the beginning or end markers in the **running** or **paused** status:

- ✧ Move the slider so that the first image of the new movie sequence is displayed in the movie segment.
- ✧ Click this button.



The beginning marker is assigned to this position.



✧ Move the slider so that the last image of the new movie sequence is displayed in the movie segment.

✧ Click this button.

The end marker will be assigned to this position.

If you adjust the marker in **running** status, movie display will immediately start within the new range.

✧ When you set a marker for the second time to the same position, the beginning marker is automatically set to the beginning and the end marker to the end of the group.

Configuring the movie display

You are provided with four default parameter sets for movie display. You can modify these parameter sets or define your own parameter settings.

Select the Movie properties dialog window

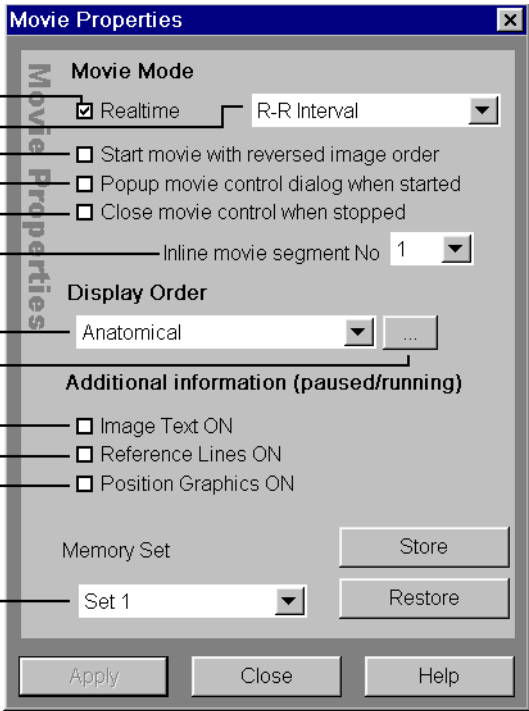


✧ Select **Scroll > Movie > Properties....**

- Or -

Click this button in the **Movie Control** dialog window.

The **Movie Properties** dialog window is superimposed. The current movie display settings and the parameter set last used are displayed.



(1) Setting the real-time mode
→ [Page C.2–15](#)

(2) Select a time frame for real-time mode

(3) Reverse display order

(4) Open the **Movie Control** dialog window automatically at the start

(5) Close the **Movie Control** dialog window automatically on stopping

(6) Select the segment for movie display for automatic movie start

(7) Select sort scheme

(8) Select the **Display Order**
→ [Page C.4–13](#)

(9) Switching image text on/off
→ [Page C.2–23](#)

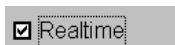
(10) Switching image text on/off
→ [Page C.2–23](#)

(11) Switching image text on/off
→ [Page C.2–23](#)

(12) Selecting default parameter sets
→ [Page C.2–23](#)

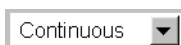
☞ To display brief information for a dialog element, leave the mouse pointer on the dialog element.

Setting the real-time mode



- ✧ Activate the **Realtime** checkbox.

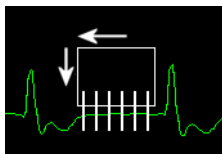
If the current group is sorted according to trigger time (TT) and you want movie display to run in **Loop** mode, you can also select a time frame for the real-time display.



- ✧ In the list next to **Realtime**, select **Continuous**.

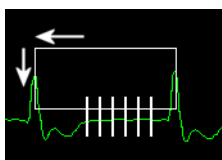
The image sequence will be repeated without delay.

Or



- ✧ In the list next to **Realtime**, select **R-R Interval**.

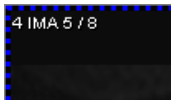
A delay will be added between image sequences so that the duration of an image sequence corresponds to the mean R-R interval.



Displaying additional information

In movie display, you can show or hide image text, reference lines, and GSP objects.

- ✧ For optimal movie quality, we recommend hiding this additional information.

☒ Image Text ON


When the image text is switched on, the following information is displayed in the movie segment in the **running** or **paused** status:

☐ Image number, top left

☐ Trigger time TT, bottom right

☐ Slice position SP, bottom right

☒ Reference Lines ON

When the reference lines are switched on, the current image in the movie segment will be displayed through a reference line in the other image segment and continuously updated in the **running** or **paused** status.

☒ Position Graphics ON

When the positioning graphics are activated, all GSP objects as well as all orientation aids selected via the menu are shown in the movie segment in the **running** or **paused** status.

→ [Page C.5–28, Display of orientation aids](#)

✧ You can perform graphic slice positioning in the movie display. The movie display is paused while you do so.

Displaying the existing parameter set



The existing parameter sets are included in the **Memory Set** list. These parameter sets can be modified, but not renamed.

✧ To load the setting of the parameter set into the **Movie Properties** dialog window, select the parameter set in the list.

Applying the parameter set



When the parameter setting is suitable for the current movie display:

- ✧ Click **Apply**.

The movie display is now performed with the applied settings.

Changing a predefined parameter set

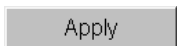


You can change parameter sets and permanently save them so that they are available when you start the system again.

- ✧ Select a parameter set from the list.
- ✧ Change the settings in the **Movie Properties** dialog window.
- ✧ Click **Store**.

The current properties in the **Movie Properties** dialog window are stored to the selected parameter set. The current movie display is not updated.

Applying temporary settings



You can change the parameters and apply them for the current movie display without storing them.

- ✧ Make the required settings in the **Movie Properties** dialog window.
- ✧ Click **Apply**.

The current movie display is updated. The parameter set displayed in the **Memory Set** list is *not* changed.

Loading the Measurement Program

Automatic loading (Chorus)

In the HIS/RIS system, examinations for a patient can be planned in detail.

During preregistration, patient data and detailed examination data are stored in the scheduler database.

After registering the patient via the scheduler, the planned program steps are transferred to the wait queue.

➤ Additional information is included in the **syngo Chorus MR** online reference.

Prerequisites in the HIS/RIS system

- ☐ The patient was registered in the HIS/RIS system.
- ☐ The patient was allocated detailed examination data including program steps.

Prerequisites at the syngo Acquisition Workplace

- ❑ A HIS/RIS query was performed in the scheduler.
- ❑ The patient was registered via the scheduler.
- ⇒ Detailed information on the topic of *patient registration* is included *in the syngo Operator Manual*.
→ **Patient registration/***Registering a known patient*

Follow-up examinations

For follow-up examinations, program steps for the patient do not have to be created again in the HIS/RIS system. In the **FollowUp** card, all program steps of the previously performed examination can be applied.

Study Split

Multiple examination requirements planned in the HIS/RIS system (e.g., head and cervical spine) can be performed in one session. For each Requested Procedure, a study is created.

Advantages of several studies:

- ❑ direct allocation of the individual studies in the documentation and invoicing
- ❑ easier distribution of the studies to the reporting radiologist (reporting stations)

Prerequisites in the HIS/RIS system

- ❑ The patient was registered in the HIS/RIS system.
- ❑ The patient has more than one examination request, each with individual accession numbers.

Study Split at the Acquisition Workplace

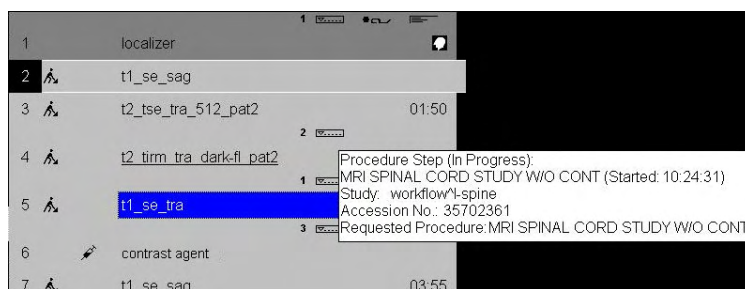
- ✧ A HIS/RIS query is performed in the scheduler.
- ✧ Select the patient.
 - Or -
 - Select all applicable requirements.
- ✧ Register the patient.
- ⇒ The same patient positioning has to be entered for all studies.
- ⇒ Detailed information on the topic of *patient registration* is included in the syngo Operator Manual.
 - **Patient registration/***Registering a known patient*

If you have a license for *syngo* Chorus, the program steps and markers for procedure steps are automatically applied to the wait queue.

Otherwise, you can transfer the program steps and markers for procedure steps manually into the wait queue.

→ [Page C.3–16](#), *Applying the measurement program to the program control*

→ [Page C.15–5](#), *Editing the instruction range Procedure step*



The individual studies are identified through work steps in the queue.

✧ The procedure steps and individual studies can be differentiated using the tool tip for the procedure step icon.

Changing the assignment to studies

Prior to the measurement, one can change the allocation of program steps to the studies:

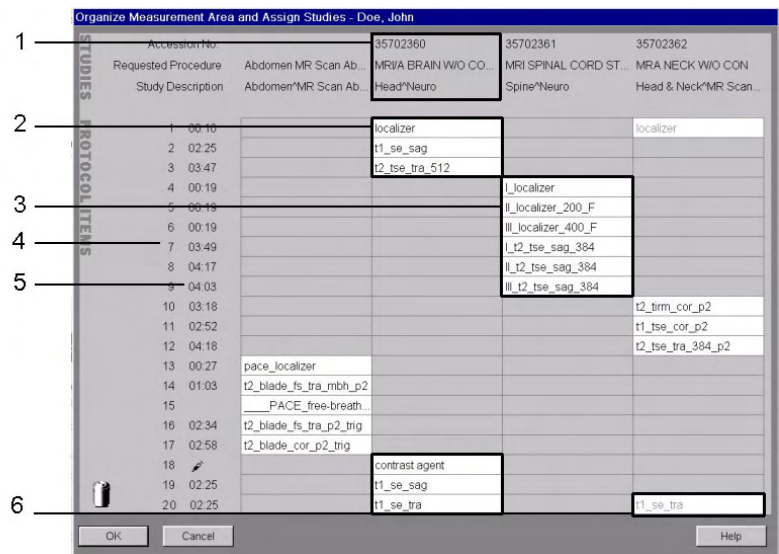
✧ Right-click the black area below the wait queue and select **Assign Studies** in the context menu.

Moving scope markers



Scope markers can be moved by drag & drop or copied by **Ctrl** drag & drop.

The **Organize Measurement Area and Assign Studies** dialog window is displayed:



- (1) Study with Accession No., Requested Procedure and Study Description
- (2) Program steps assigned to this study (1)
- (3) Sequence number of the program step
- (4) Number of program step(s)
- (5) Acquisition time of a program step during measurement
- (6) Program step assigned to several studies



In this dialog window you have several possibilities to change the assignment of program steps within the study or between different studies:

- ☐ Assign a program step to a study.
- ☐ Move a program step within a study to change the measuring sequence.

Moving a program step

You can move program steps to any position by using the mouse. When dragging protocols, a line indicating the insertion position is displayed. Program steps can be moved within the study or to another study. The selection of multiple program steps is possible by using the **Shift** or **Ctrl** key.

Assigning program steps to several studies

If you need a program step several times in different studies, you can assign it accordingly. It is not possible to insert duplicates in the original study.

- ✧ Keep the **Ctrl** key pressed and move the program step with the mouse.



If you drag the program step horizontally, it is inserted at the same position in the target study. If you drag it diagonally, the program step is not only duplicated, but the step order is rearranged as well.

- ⇒ For duplicated program steps, the instructions of the first study (left column) are relevant. The duplicates in the other columns are grayed and are not measured twice. Duplicates can be deleted.
- ⇒ Multiple assignments are not visible in the **Program Control** dialog window, but only in the **Organize Measurement Area and Assign Studies** dialog window.

Deleting duplicated program steps

You can delete duplicated program steps (displayed in gray) from the respective study.

- ✧ Move the program step(s) to the recycle bin in the dialog window or use the **Delete** key.

Terminating an acquisition

- ⇒ Terminated program steps cannot be changed any more. However, after the examination you can make corrections for the assignment of program steps in the **Patient Browser**.
→ [Page C.3–8, Study Split in the Patient Browser](#)
- ✧ Click **OK** to close the dialog window.

The **Organize Measurement Area and Assign Studies** dialog window is closed. The assignments are transferred to the program control.

Study Split in the Patient Browser

Alternatively, the assignment of series can also be modified via the **Patient Browser**. This enables you, for instance, to assign the series to separate (different) studies even after completing the examination.

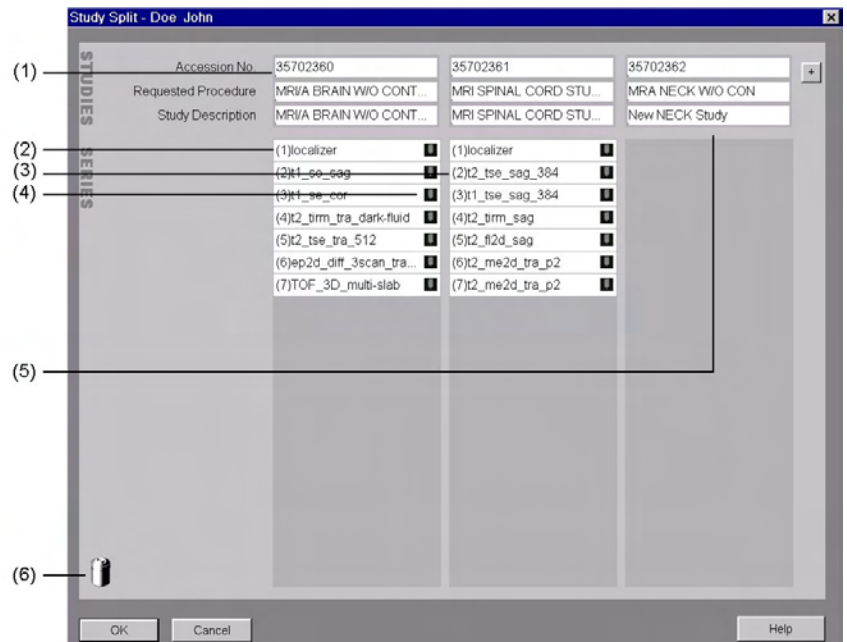
- ✧ You can only assign series via the **Patient Browser** if the respective examination of the patient is terminated and the patient data is not loaded in any application card.
- ✧ The series of sent, archived, exported, and printed image data can only be copied to a target study. Sent, archived, exported, and printed series are grayed and cannot be modified any more.

Opening the Study Split dialog window



- ✧ Call up **Applications > Study Split** via the **Patient Browser**.
- Or
- ✧ Click the **Study Split** icon.

The **Study Split** dialog window is displayed.



- (1) Study with Accession No., Requested Procedure, and Study Description
- (2) Series stored in this study (1)
- (3) Series stored in another study
- (4) Image stamps of a program step's image series
- (5) Entry fields for Accession No., Requested Procedure, and Study Description of a newly generated study
- (6) Recycle bin



Moving series

Series can be moved via the mouse to another study. Multiple series selection is possible.

- ✧ Move the series to the target study.

Copying a series

If you need a series in different studies, you can copy it.

- ✧ Keep the **Ctrl** key pressed and move the series with the mouse.

A copy of the series is inserted in the target study.

Displaying the image stamps of a series

Image stamps for the individual series are provided in the dialog window. You can enlarge (zoom) an image stamp by moving the mouse pointer over it.

As soon as you move the mouse pointer to another position, the image is minimized again.

Generating a new study



You can generate new studies on the right side of the **Study Split** dialog window.

✧ Click this icon on the top right side.

Several input fields are displayed. There are no mandatory fields, but it is advisable to assign at least a Study Description. Accession Numbers and Requested Procedures are offered from the Scheduler, when you click the respective field.

✧ Select the requested accession number from the list or enter a study number manually.

✧ Select a procedure description from the list or enter a procedure description manually.

You can assign a series at any time.

Closing the dialog window

✧ Click **OK** to close the dialog window.

The **Study Split** dialog window is closed. The assignments are stored in the **Patient Browser** and in the database. If you have configured an autosend-rule they are transferred to the PACS.

Checking information according to the region and examination

After you start the **Exam** task card, the **Program** parameter card opens. The region and examination selected during patient registration are entered in this parameter card.

(1) Region: head

(2) Exams: general

(3) Programs: standard_1

(4) standard_2

(5)

1	localizer	00:09
2	t1_se_sag	03:16
3	pd+t2_tse_tra	05:03
4	contrast_agent	
5	t1_se_sag	2 03:16

(6) Σ 11:44

Program Routine Contrast Resolution Geometry System Physio Sequence

- (1) Region
- (2) Available examinations for the selected region
- (3) Selected measurement program
- (4) Available measurement programs for the selected examination
- (5) Program steps for the selected measurement program
- (6) Sum of the measurement times for the individual protocols of the measurement program selected

Changing the region

A screenshot of a software interface showing a dropdown menu labeled 'Region:'. The menu is open, and the word 'head' is visible in the selection box. A small downward arrow is visible on the right side of the box.

- ✧ Select a different examination region in the **Region** selection list.

Changing the examination

As soon as you select an examination region, the **Exams** selection list contains the examinations available for this region.

A screenshot of a software interface showing a dropdown menu labeled 'Exams:'. The menu is open, and the word 'sella' is visible in the selection box. A small downward arrow is visible on the right side of the box.

- ✧ Select another examination.

Displaying properties

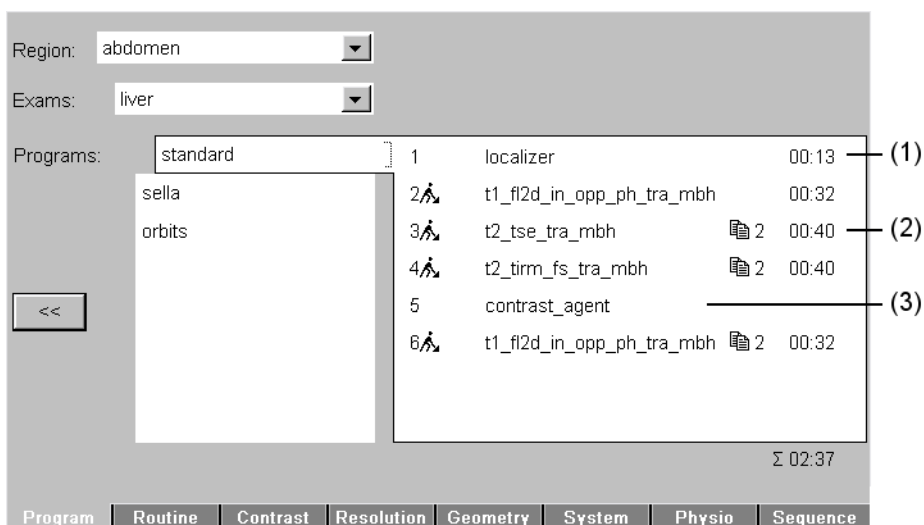
The properties of the examination region or examination selected can be displayed in the **Region properties** or in the **Exam properties** dialog window.

- ✧ Select **Edit > Properties**.
- Or -
- Go to the context menu of the selected region or examination and select **Properties**.

Selecting a measurement program

After you have selected an examination, **Program** will display the measurement programs that are available for this examination.

The contents of the first measurement program, that is, the chronological sequence of the individual program steps, is shown in the right window of the parameter card.



- (1) Overview measurement **localizer**
- (2) Measurement protocol
- (3) Measurement pause (in this case, for contrast agent administration)

Selecting a measurement program on the program parameter card

Select another one, if the preselected measurement program is not suitable for the examination.

- ✧ Click the program in the list of available measurement programs you want to use for the examination.

The program steps of the newly selected measurement program are shown in the right corner of the parameter card.

- ✧ To see other information in the **Exam properties** dialog window (brief description, time of last change, etc.), select **Properties** in the context window.

Selecting a measurement program in the Exam Explorer

If you do not find a suitable measurement program on the **Program** parameter card, change to the **Exam Explorer**. There all examinations and measurement programs are listed in a comprehensive manner. When needed, you can also access the Siemens tree.

→ [Page C.13–1](#), *Managing Measurement Programs*



✧ Select **View > Exam Explorer**.

- Or -

Click this button.

The **Exam Explorer** dialog window is displayed.

Look for the requested measurement program in the Exam Explorer and apply it to the program control.

→ [Page C.13–10](#), *Searching for entries*

→ [Page C.13–54](#), *Applying the measurement program to the program control*

Applying the measurement program to the program control

You select a measurement program to be measured from the **Program** parameter card (or from the **Exam Explorer**) and apply it to the program control.

Transferring a routine measurement program

✧ Select the measurement program.

→ [Page C.3–14](#), *Selecting a measurement program*

✧ Select **Queue > Append**.

- Or -

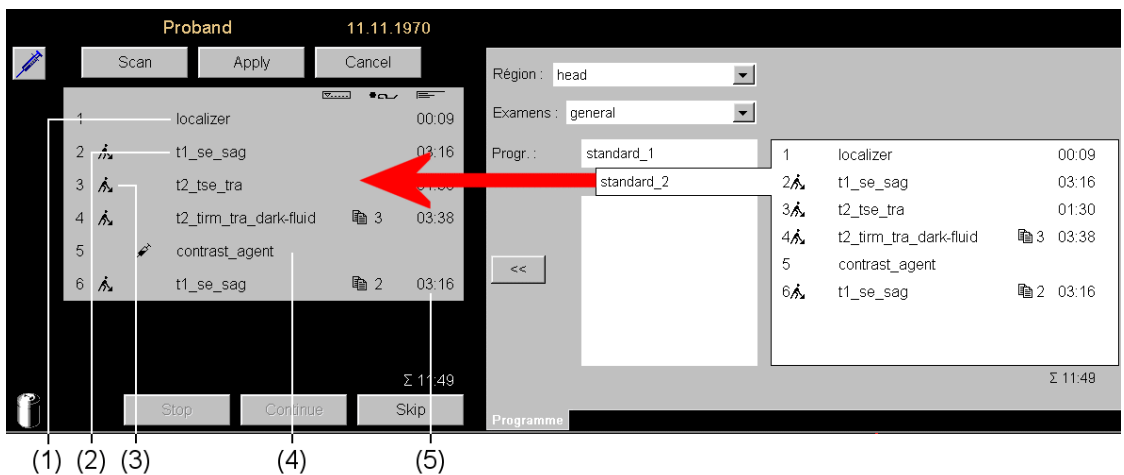
Click this button << on the **Program** parameter card.



The program steps of the selected measurement program are applied to the wait queue of the program control.

- The first program step is started automatically if it does not have a construction worker or flagman icon.
- If no reference image has been loaded, protocols with the **ISO** and **REF** setting in the **Positioning mode** parameter are measured at the current table position.
→ [Page P.1–99](#), *Parameter Positioning mode*

You can append additional program steps to the end of the queue at any time.



- (1) Localizer
- (2) Measurement protocol
- (3) Start option of protocols (protocol properties)
- (4) Measurement pause
- (5) Measurement time of protocol

Transferring individual program steps

You may also transfer individual protocols from one or more measurement programs.

- ✧ Select the program step(s) you would like to transfer.
→ **Basics/Selecting Images** in the *syngo* Operator Manual

- ✧ Select **Queue > Append**.

- Or -

Click this button on the **Program** parameter card.

The selected program steps will be transferred to the program control. You can append additional program steps to the end of the queue at any time.



Viewing protocol parameters

- ✧ Select the protocol from the **Program** parameter card.

- ✧ Select **Edit > View Protocol**.

- Or -

Double-click the protocol entry.

The **View Protocol** dialog window opens.

- ✧ In the Protocol view, you may view (but not change) the measurement parameters of the protocol on the parameter cards.

- 
- Protocol or pause properties** This is how you display information about a protocol (e.g., start options) or a measurement pause:
- ✧ Select the protocol or measurement pause from the **Program** parameter card.
 - ✧ Select **Edit > Properties**.
 - Or -
Select **Properties...** from the context menu.
- Either the **Protocol properties** or the **Pause properties** dialog window will be displayed.



Measuring, Editing, and Managing Reference Images

Most measurement programs will prompt you to perform a localizer measurement first. These localizer measurements provide you with reference images that let you position slices or determine the reconstruction section (Field of View, FoV) for the following measurements.

→ [Page C.1–10](#), *Procedure for routine scanning*

Instead of localizer images, you may also use other patient images as reference images acquired during the exam of the current patient.

You are viewing the reference images in the reference image segments, improve image display and determine the most optimal images for slice positioning.

You can use the reference image segments of the **Exam** task card for more than graphic slice positioning. For example, they allow you to view, evaluate, and edit the acquired images.

You can load images directly from the **Exam** task card into other task cards for post-processing, image exporting or saving them in the **Patient Browser**.



To perform the examination, you need to have full access rights.

Detailed information regarding all security settings in *syngo* are included in the *syngo* Security Package brochure.

Measuring reference images

Reference images from localizer

The localizer starts automatically as the first program step after you load a measurement program into the program control.

While the measurement is in progress, the localizer protocol has a white background in the program control.



The status bar successively displays the processes in progress:

Adjustment (coil)...	Adjusting the system to the parameters set and the coils of the program instruction
Preparing	
Measuring...	The measurement is in progress, the remaining measurement time of the protocol is shown



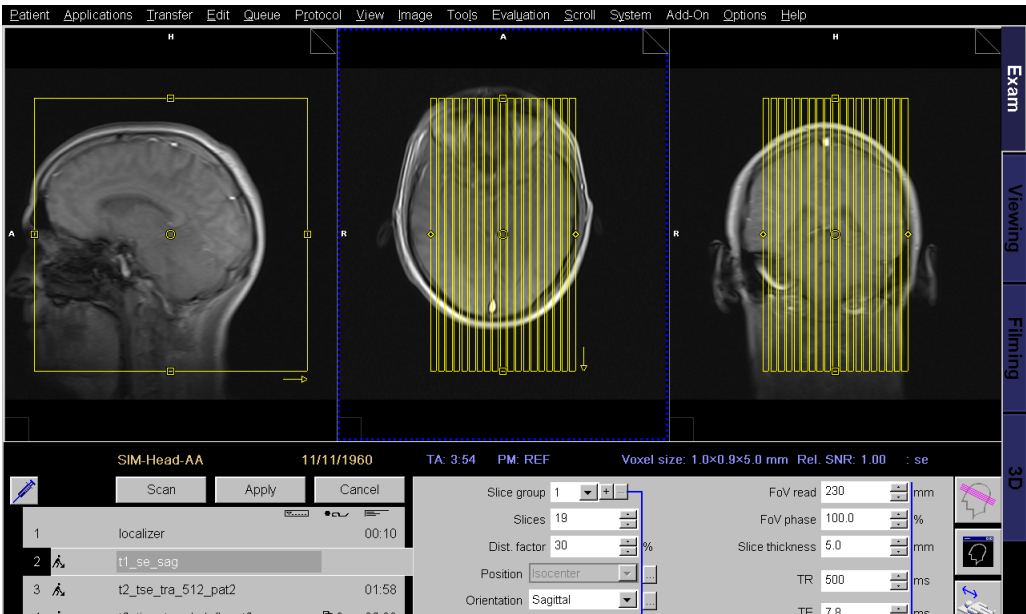
The series symbol is displayed as soon as the measurement and image calculation have been completed.

1	localizer	
2	t1_se_sag	



➤ This series icon indicates that the images of the series or several series are not fully available.

These reference images are displayed in the image area of the **Exam** task card once the localizer measurement and image calculation have been completed.



The reference images show the organ or part of the body to be examined, usually in the three main orientations (sagittal, transverse, and coronal).

The next program step (e.g., **t1_se_sag**) is normally incomplete and is opened for graphic slice positioning.



Reference images from other protocols

If you already measured protocols in the course of your examination, you can use images from these protocols for graphic slice positioning.

- ✎ Try to avoid loading large data series (e.g., from epi sequences) if they are not needed for additional graphic positioning in the exam card. Otherwise, the system may slow down.

Prerequisites

The images have to meet the following prerequisites:

- ☐ The images are part of the current examination of the current patient.
- ☐ All images must be part of the same series block.
→ [Page C.1–6, Series block](#)

Depending on how the table position of the protocol is determined, the prerequisites to be met by the reference images differ.

→ [Page P.1–99, Parameter Positioning mode](#)

Positioning mode of protocol	Reference images
Fixed table position Positioning mode = FIX	Non distortion-corrected images (image type ND) with identical table position Distortion-corrected images (image type DIS2D/ DIS3D) with random table position
Table position of reference image Positioning mode = REF	Non distortion-corrected images (image type ND) with identical table position Distortion-corrected images (image type DIS2D/ DIS3D) with identical table position
Isocentric protocols Positioning mode = ISO	All distortion-corrected images (image type DIS2D or DIS3D) → Page P.1–52 , Parameter Card <i>Resolution - Filter</i>

The following images cannot be used as reference images:

- ☐ Mosaic images
- ☐ Secondary capture images (e.g., alpha images of the BOLD evaluation)
- ☐ Curved cuts of the 3D evaluation
- ☐ Distortion-corrected images with too much distortion (image type DIS2D* or DIS3D*)

Changing the distortion correction of reference images

You can subsequently change the distortion correction of reference images. This changes the distortion correction of the entire series.

- ✧ Please note that requirements with respect to distortion correction of the reference images depend on the type of protocol loaded.

→ [Page C.4–4, Prerequisites](#)

Generating/changing distortion correction

- ✧ Mark the reference image segment or stamp segment for the series where you want to change distortion correction.
- ✧ Select **Evaluation > 2D distortion correction**.
 - Or -
 - Select **Evaluation > 3D distortion correction**.

A new series with distortion-corrected images is generated (image type DIS2D or DIS3D). If the image type is suitable for the current protocol, the images of this series are loaded automatically as reference images.

Resetting 2D distortion correction

✧ Mark the reference image segment or stamp segment for the series where you want to reset the distortion correction.

✧ Select **Evaluation > Inverse Distortion 2D Correction**.

A new series with non distortion-corrected images is generated (image type ND). If the image type is suitable for the current protocol, the images of this series are loaded automatically as reference images.

✧ 3D distortion correction cannot be reset.

Loading reference images into the image area

You are able to load images and series from the program control or **Patient Browser** as reference images into the image area of the **Exam** task card.

Please note the dependencies among the positioning mode of the protocol, the image type and the table positions of the reference images.

→ [Page C.4–26](#), *Dependencies between the table position, reference image, and protocol*

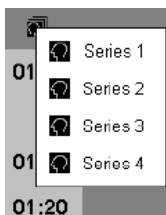
Loading a series from the program control

You can also load images or series from the program control, even though the underlying protocol has not been fully processed at this point in time.

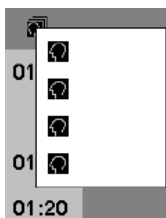
- ✧ Select a protocol in the program control that has been processed either in part or in full.
- ✧ Select **Queue > Load Series**.
- Or -
- Double-click the series icon.
- [Page C.6–8, Series icons in the program step](#)

The series is loaded into the image area of the **Exam** task card.

If the protocol generates several result series, a dialog window allows you to select the series to be transferred.



Dialog window with serial number:
Series reconstruction has been completed.



Dialog window without serial number:
Series are not yet fully reconstructed.

- ✧ Click the series so that you can load its images.

Loading the series from the Patient Browser

You can also load images for graphical slice positioning via the **Patient Browser**.

- ✧ Open the **Patient Browser**.
- ✧ Select the images or series in the **Patient Browser**.
→ **Patient Browser**/Searching for and selecting patient data in the *syngo* Operator Manual
- ✧ Use the mouse and drag the images or series into one of the reference image segments.
- Or -
To load *an* image or *a* series, double-click the selected image or series.

Displaying loaded reference images

You can choose between different layouts to display the loaded images.

Three-segment layout

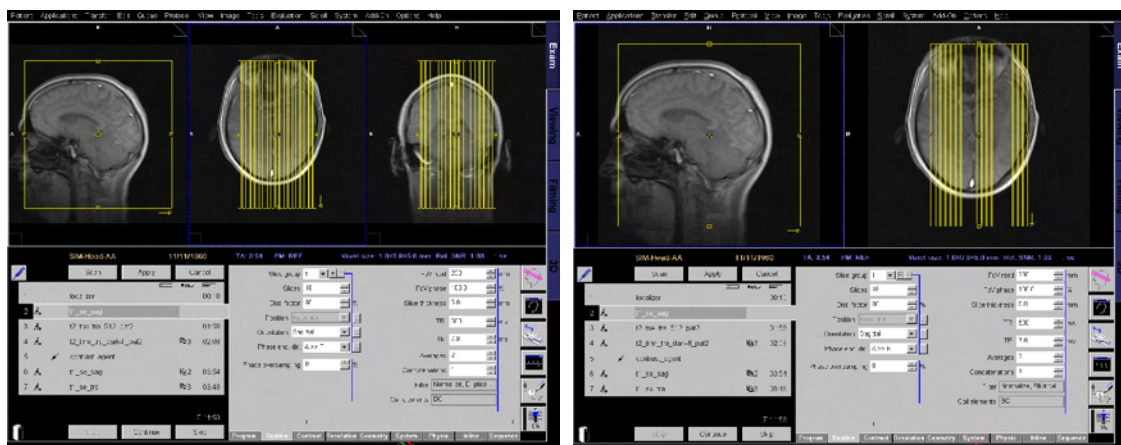
Select a three-segment layout if you want to display three reference images side by side.

- ✧ Select **View > 3 Segments**.

Two-segment layout

Select a two-segment layout if you want to display two reference images side by side. The reference images are displayed larger than in the three-segment layout.

✧ Select **View > 2 Segments**.

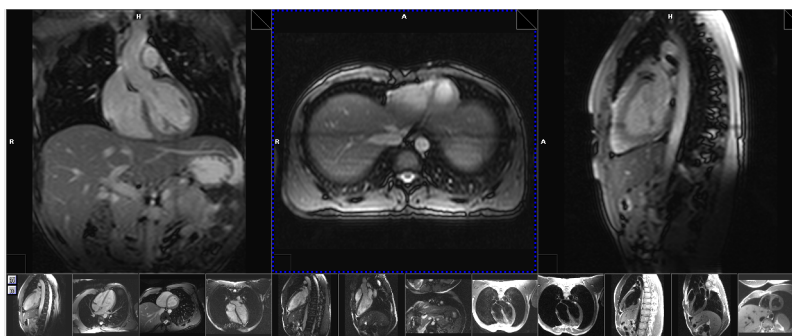


Maestro layout (optional)

Select the Maestro layout to gain a better overview of the series generated in the study and to facilitate selection of the images generated as reference images.

The series measured are shown in stamp segments below the three reference image segments.

→ [Page C.2–2, Maestro layout](#)

**Distributing images across segments**

The following rules apply:

- ☐ If the image area is empty, the loaded images will be distributed across the image area so that every segment displays at least one image.
- ☐ If only *one* series was loaded, the images of this series are automatically distributed across the (2 or 3) segments available.
- ☐ If *several* series were loaded, one series each per segment will be represented by its representative image.
- ☐ If all reference image segments are occupied, the series to be loaded will be loaded in the background. You can move these series into the foreground by scrolling with **Series-/Series+**.
→ [Page C.4–17, Scrolling in the image segments](#)

Representative image

The anatomical center image is used as the representative image for a series with different slice positions.

The first image from the current display series is the representative image for a series where the images *do not* differ in slice position.

→ [Page C.4–13](#), *Defining the scroll sequence*

Example 1

All three reference image segments are empty. A localizer with one sagittal, two coronal, and three transverse overview images has just been completed.

The images are distributed as follows:



The series is duplicated twice and displayed in all three segments. The left segment shows the sagittal image, the center segment shows the first coronal image, and the right segment shows the first transverse image.

Example 2

All three reference image segments are empty. Two series from the current study are dragged from the **Patient Browser** to the center segment by using the mouse.

The images are distributed as follows:



The first series is loaded into the target segment (center segment), the second series is loaded into the right segment. In addition, the second series is duplicated and displayed in the left segment. The system automatically searches for another suitable image in the second series to be used as the representative image for the first segment.

Defining the scroll sequence

You have to differentiate between scrolling *within* a series and scrolling *between* series.

If multiple series are loaded into an image segment, you can scroll between series. The order of the series is defined by the series number. This number depends on the date and time of reconstruction. The order of the series *cannot* be modified.

However, you can define the order of the images within a series. You can select from various default sort options in the menu. You can also define your own sort schemes for the order of images.

The sort used also determines the order of images for movie display.

→ [Page C.2–8, Movie display](#)

Sort criteria

The following sort criteria are available:

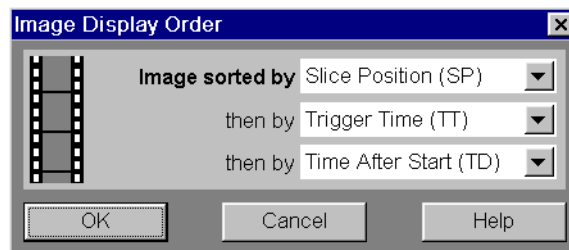
Abbreviation	Example
MO	Sorting according to main orientation sag – cor – tra
SP	Sorting according to a) main orientation sag – cor – tra, b) allocation to a 3D group, c) depending on the main orientation, the portion of the center vector of the group in the sequence tra – cor – sag, d) the portion of the center vector of the slice corresponding to the main orientation (within the 3D group) in the sequence tra – cor – sag
TT	Sorting according to trigger time
TA	Sorting according to acquisition time
TE	Sorting according to echo time
TD	Sorting according to time after delay (perfusion or diffusion time)
BV	Sorting according to b-value
NR	Sorting according to image numbers
--	Sorting according to load order. This criterion does not allow any sub-criteria.

Defining your own sort scheme

You can establish up to 3 sort criteria for images in this series.

- ✧ Select **Scroll > Display Order > User Defined**.

The **Image Display Order** dialog window opens.



- ✧ Select the first sort criterion in the upper selection list.
- ✧ Select the second sort criterion in the center selection list.
- ✧ Select the third sort criterion in the lower selection list.

The images are sorted according to the first sort criterion. Images with an identical value in the first sort criterion are sorted according to the second sort criterion. Images with an identical value in the second sort criterion are sorted according to the third sort criterion.

- ✧ Use three (different) sort criteria. When you selected the first sort criterion **Load Order**, other sort criteria are not necessary. The selection lists **then by** are now inactive.
- ✧ Click **OK**.

Using a Siemens sort scheme

You may select from five different Siemens sort schemes for the order of images within a series.

Name	Sorting scheme	Application range
Anatomical	SP»TT»TA»NR	Standard sort, localizer, anatomical examinations, multi-slice/multi-phase measurements (grouped according to phases), multi-slice real-time measurements
Triggered	MO»TT»TD»SP	Multi-slice/single phase measurements or multi-slice/multi-phase measurements (when a slice package is anatomically viewed at a defined trigger time). Perfusion and diffusion multi-slice measurements
Chronological	MO»TA»SP»NR	Sorting by time of acquisition (TA), images with the same TA are sorted by SP, images with the same TA and SP are sorted by NR.
As numbered	NR»SP	Sorting according to image number (NR). Images with the same NR are sorted according to SP.
As loaded		Sorted by load order

✧ Select a sort scheme under **Scroll > Display Order**.

The sort scheme selected is applied until you select another scheme.

✧ The **Anatomical** sort scheme is selected by default. This is the optimal scheme for most applications.

Scrolling in the image segments

As soon as you have loaded images into the image area, you can scroll through these images in a reference image segment. This helps you select the images most suitable for slice positioning.

Please note the dependencies among the positioning mode of the protocol, the image type and the table positions of the reference images.

→ [Page C.4–26](#), *Dependencies between the table position, reference image, and protocol*

Setting the input focus

The **Input Focus** indicates the active segment of the image area. It is highlighted by a dashed blue border. All processing steps which you start via the keyboard (e.g., scrolling with **Image+**/**Image-** keys) are always performed in this segment.

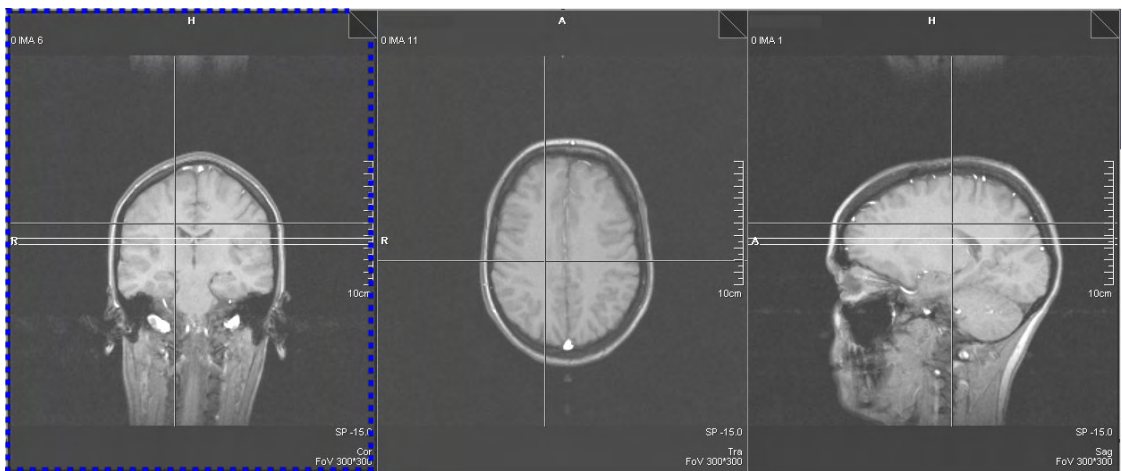


Image segment with input focus



You can move the input focus into another image segment by pressing the **Left** or **Right** cursor key or clicking the other segment.

Scrolling through images or series

Sorting via the menu using dog ears is described in detail in the *syngo* Operator Manual.

→ **Viewing/Changing image settings**

Scrolling in parallel

To compare images of two or three series, scroll in parallel through images in these series.

✧ Explicitly select all images in the reference image segments where you want to scroll in parallel.

→ **Basics/Selecting images** in the *syngo* Operator Manual

✧ Scroll from image to image using menu commands, the dog-ears, or symbol keys.

If the series do not contain the same number of images, parallel-scrolling stops at the end of the shorter series. Subsequently, the program scrolls through the longer series first before it begins again with the first image of the shorter series.

✧ Please note that the response time during parallel scrolling depends on the number of images and processes running in the background.

Editing reference images

You can edit reference images as follows:

- ☐ Set contrast and brightness (windowing)
 - ☐ Zoom and reduce images and pan image content
 - ☐ Fit to segment size
 - ☐ Flip and rotate
 - ☐ Display pixel coordinates, measure distances and angles
 - ☐ Show/hide image text
- ⇒ Prior to applying these functions, you have to select the images to be edited. For details, please refer to the *syngo* Operator Manual.
→ **Basics/Selecting images**

Changing window values

Windowing changes the grayscale values, the brightness, and the contrast of the reference images.

You are provided with the following possibilities for changing window values:

- ☐ Windowing using the mouse
- ☐ Windowing with the keyboard
- ☐ Automatic windowing
- ☐ Windowing the complete series

For additional information, refer to the *syngo* Operator Manual.
→ **Basics/Windowing images**

The following operating elements for windowing are located on the **Exam** task card:

Menu item	Function
Image > Auto Windowing	Automatic windowing
Image > Windowing On Series	Single image windowing/series windowing

Zooming and panning reference images

For additional information, refer to the syngo Operator Manual.
→ **Basics/Zooming and panning images**

In addition, you can change the image size or image position on the **Exam** task card:

- ☐ Zooming/panning the entire series
- ☐ Zooming to segment size
- ☐ Zooming to segment height

Zooming/panning the entire series

You are able to determine that changes in one image are automatically transferred to all images of the respective series.

✧ Select **Image > Zoom/Pan On Series**.

When you deselected the option, you only change the current image.

➤ The setting is retained after you start the system again.

Zooming to segment size

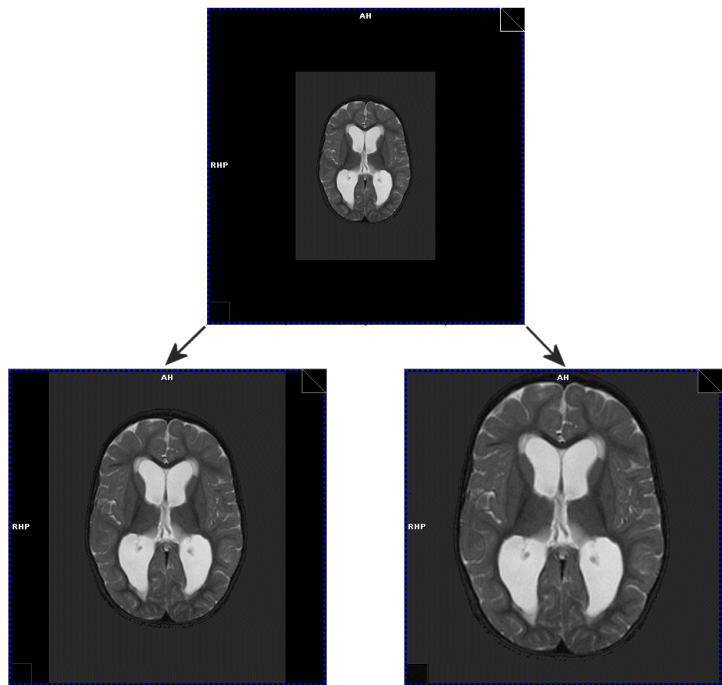
You can zoom the image automatically for optimally utilizing the image content of the segment.

- ✧ Select the image.
- ✧ Select **Image > Fit to segment**.
 - Or -
 - Double-click the image with the *left* mouse key.

Zooming to segment height

- ✧ Select the image.
- ✧ Select **Image > Fit to Segment Height**.

The image height is adjusted to the segment height.



Zoomed to segment size

Zoomed to segment height

- ⇒ When you use the two-segment layout, alignment to segment size and segment height lead to the same result.

Flipping and rotating images

You are not able to mirror images horizontally or vertically or rotate them clockwise by 90°.

→ **Viewing/Rotating and flipping** in the *syngo* Operator Manual

Showing/hiding image text

- ✧ Select **View > Image Text on/off** in the main menu.
 - Or -
 - Click this button on the **Position Toolbar**.



Pixel coordinates, distances, and angles

You can display the position of the mouse pointer as coordinates in the *whole body patient coordinate system*.

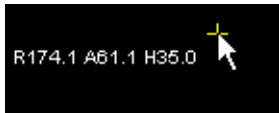
- ✧ Select **Image > Modify Graphics**.

The Zooming/Panning mode is switched off and the process Modify Graphics is switched on.

- ✧ Click the area of interest in the image and keep the mouse button pressed.

The coordinates in the *whole body patient coordinate system* are displayed without an arithmetic sign.

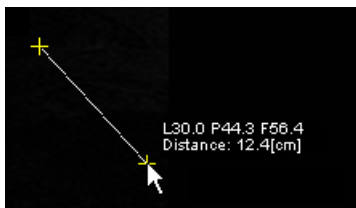
- ✧ The coordinate display continually updates if you move the mouse with the mouse button pressed.



Measuring the distance

This is how you measure the distance between two image positions:

- ✧ Select **Image > Modify Graphics**.
- ✧ Click the image and hold the mouse button down.
- ✧ Move the mouse pointer to the start of the measurement.



- ✧ To start the measurement, press **Shift**.

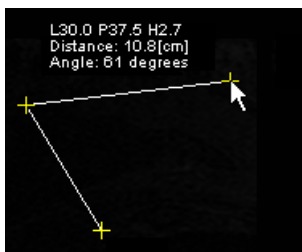
Release the mouse button to display the distance line. You can now release the **Shift** key.

- ✧ Move the mouse pointer to the end point of the measurement.

The distance to the first point is displayed below the coordinates.

The coordinates and the distance disappear when you release the mouse button.

Measuring angles



- ✧ Select **Image > Modify Graphics**.

- ✧ Press and hold the mouse button and measure the distance from the apex of the angle and any point along the first leg. Continue pressing the mouse button.

- ✧ Press **Ctrl** and move the mouse to a point at the second leg.

You can release the **Ctrl** key as soon as the second side is visible.

The angle is displayed below the coordinates and the distance.

The coordinates, distance, and angle disappear when you release the mouse button.



Removing images from the image area

Removing images from all segments

This is how you remove images from all segments:

- ✧ Select **Edit > Clear All Segments**.

Removing images from reference image segments

This is how you remove images from the segments:

- ✧ Select **Edit > Clear Graphic Segments**.

Repeating scans (Phoenix)

You can use the Phoenix function to repeat the measurement. You generate a copy of the protocols measured from the images and append it at the end of the queue.

→ [Page C.13–42](#), *Generating protocols from a series or an image (Phoenix)*

Dependencies between the table position, reference image, and protocol

Beginning with software version **syngo MR B15V**, you can plan your measurements on non-distortion-corrected (image type ND) and distortion-corrected reference images (image type DIS2D or DIS3D).

- ❑ When you are using non-distortion-corrected reference images (image type ND), the table position of the protocol has to match that of the table position of the reference images. This enables you to use a FIX protocol with an identical table position or a REF protocol with an automatic propagation of the table position.
- ❑ For distortion-corrected reference images (image type DIS2D or DIS3D) you can plan protocols with freely-selectable positioning modes (REF, FIX or ISO). The table position of the protocol does not have to agree with the reference images. Isocenter measurements can be planned for distortion-corrected reference images only (protocol with "ISO" positioning mode). In these protocols, the table position is automatically determined by the system so that the common center of all positioned slices is located as completely as possible in the magnet isocenter during a later measurement.



To this end the relationship between the table positions of the reference images and the protocol parameters of the table position has to be known.
→ [Page P.1–98](#), Parameter Card System - Miscellaneous

Reference images while opening a protocol

Depending on the **Positioning mode** of the protocol to be opened, the effects with respect to the reference images displayed and the table position of the protocol are different.

	FIX Protocol	REF Protocol	ISO Protocol
Open protocol with loaded reference images	The table position of the protocol remains unchanged. Suitable reference images are sought.	The protocol takes the table position of the reference image selected. Reference images remain unchanged.	<i>DIS images are loaded:</i> Reference images remain unchanged. <i>ND images are loaded:</i> Reference images are unloaded. The table position of the protocol remains unchanged.
Open protocol without loaded reference images	The table position of the protocol remains unchanged. Suitable reference images are sought.	The current hardware table position is propagated to the protocol. Suitable reference images are sought.	The table position of the protocol remains unchanged. Suitable reference images are sought.

Loading reference images while the protocol is open

The image type of the reference images to be loaded has varying effects on the table position of the respective protocol.

	FIX Protocol	REF Protocol	ISO Protocol
DIS images with other table positions are loaded.	DIS images are loaded. The table position of the protocol remains unchanged.	DIS images are loaded. The protocol takes the table position of the reference image.	DIS images are loaded. The table position of the protocol remains unchanged.
ND images with other table positions are loaded.	ND images are loaded. The protocol takes the table position of the reference image.	ND images are loaded. The protocol takes the table position of the reference image.	ND images are cancelled. ISO protocols cannot be planned with ND images.
Composed images are loaded	Composed images are loaded. The table position of the protocol remains unchanged.	Composed images are loaded. The table position of the protocol remains unchanged.	Composed images are loaded. The table position of the protocol remains unchanged.

⇒ Composed images can only be loaded simultaneously with DIS images as the reference image.

Navigating in reference images

Depending on the **Positioning mode** of the protocol, navigation with **Series+/-** differs due to the reference images.

	FIX Protocol	REF Protocol	ISO Protocol
DIS Reference Image;	Freely-selectable navigation between all DIS series at all table positions. The table position of the protocol remains unchanged.	Freely-selectable navigation between all DIS series at all table positions. The protocol takes the respective table position of the reference image.	Freely-selectable navigation between all DIS series at all table positions. The table position of the protocol remains unchanged.
ND reference images of the same table position	Freely-selectable navigation between all ND series with identical table positions. The table position of the protocol remains unchanged.	Freely-selectable navigation between all ND series with identical table positions. The table position of the protocol remains unchanged.	not applicable

Manually changing the table position of the protocol

Depending on the **Positioning Mode** of the protocol, manual changes in table position may differ.

	FIX Protocol	REF Protocol	ISO Protocol
Change table position while DIS images are loaded:	Reference images remain unchanged. Protocol shifts to the newly entered table position.	Table position cannot be edited.	Reference images remain unchanged. Protocol shifts to the newly entered table position.
Change table position while ND images are loaded:	The system looks for suitable reference images for the new table position. Protocol shifts to the newly entered table position.	Table position cannot be edited.	not applicable
Change table position when no images are loaded	The system looks for suitable reference images for the new table position. Protocol shifts to the newly entered table position.	Table position cannot be edited.	Protocol shifts to the newly entered table position.

- When you change a table position via **Queue > Copy Parameter...** in a REF protocol or via a copy reference, the REF protocol is set to **FIX**.

Changing the mode of the protocol

Depending on the **Positioning mode** of the protocol, changes in the **Positioning mode** occur.

	FIX Protocol	REF Protocol	ISO Protocol
Changing the Positioning mode	Previously ISO or REF: Switch to FIX without affecting the protocol table position or the reference images.	Previously ISO or FIX : The protocol takes the table position of the reference image.	Previously FIX or REF: The isocentric table position is calculated. The slice positions remain constant.

Saving, exporting, and transferring images

You may save, export or transfer images to other task cards from the **Exam** task card.

The images in the reference image segments of the **Exam** task card are reference images (may include graphic objects).

The images in the stamp segments of the Maestro layout are not reference images, instead they represent additional series.

!

When a user logs off, unsaved data will be lost irretrievably.

Always check for unsaved data and save any data you want to keep before logging off.

Rules for saving, exporting, and filming

The following rules apply for reference images:

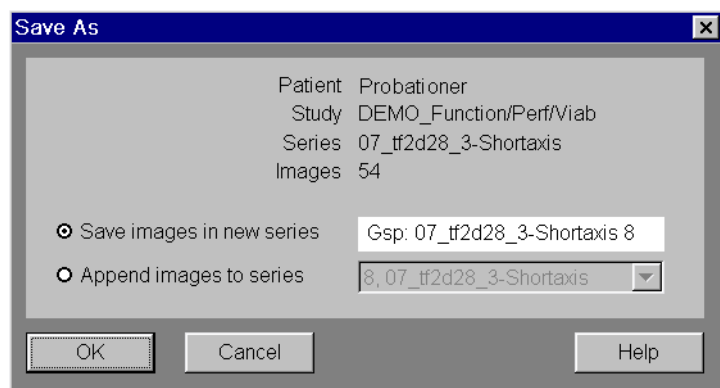
- ☐ Graphic slice positioning objects are saved with reference images. They are displayed in the non-selected version (without handles) with shadow lines and monochromatic.
- ☐ Distance lines, angles, etc. are *not* saved.
- ✧ Objects of graphical slice positioning in reference images cannot be edited in the **Viewing** task card.

Storing images and reference images

The **Exam** task card allows you to group a number of series and save them in a new series. These series have to belong to the same series block and have the same frame of reference.

- ✧ Select the series or images explicitly.
- ✧ Select **Patient > Save As**.

The **Save As** dialog window opens.



- ✧ Select the **Save images in new series** option.
- ✧ Change the suggested series name as needed.
- ✧ Click **OK**.

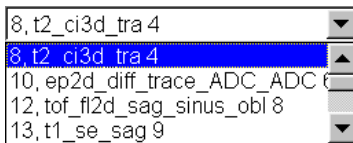
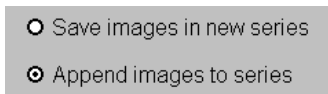
Appending images to an existing series

You can append individual GSP images to an existing series.

- ✧ Select the GSP images explicitly.
- ✧ Select **Patient > Save As**.

The **Save As** dialog window opens.

- ✧ Select the **Append images to series** option.



- ✧ Go to the list and select the series where you would like to append the GSP images and click **OK**.
- ✧ The new images of the series have image numbers 5000 and up. This enables you to distinguish between old and new images within the series.

Appending an image to a reference series

This is how you append a GSP image directly to the existing reference series.

- ✧ Select the segment with the desired image.
- ✧ Select **Patient > Copy Selection**.

Transferring images to other task cards

You can directly transfer images and series from the **Exam** task card to other open task cards for post-processing.

- ✧ Explicitly select the images or series on the **Exam** task card.
- ✧ Drag your selection via *drag & drop* onto the tab of the other task card.

The selected images will be transferred to the task card. The **Exam** task card will remain in foreground.

Filming, sending, and exporting images

For additional information, refer to the syngo Operator Manual.

→ **Filming**

For additional information, refer to the syngo Operator Manual.

→ **Patient Browser**/Storing and sending data

Copying images to the film sheet



You can copy images directly from the **Exam** task card to the virtual film sheet when editing or after having completed an evaluation.

✧ Select the images or series explicitly.

✧ Select **Patient > Copy to Film Sheet**.

- Or -

Press this key on the symbol keypad.

☞ If you have not explicitly selected an image, the image from the segment with the input focus is copied to the film sheet.

Sending images



If your system is connected to a hospital computer network (HIS/RIS), you can send images to other network addresses directly from the **Exam** task card.

✧ Select the images or series explicitly.

✧ Select **Transfer > Export to....**

- Or -

Press this key on the symbol keypad.

Exporting images

You can save images on the hard disk as single files.

- ✧ Select the images or series explicitly.
- ✧ Select **Transfer > Export to Off-line....**

The **Export to Off-line** dialog window opens.

- ✧ As destination enter **C:\TEMP**.
- ✧ Click **OK**.

The images are exported to the target directory.

Checking the data transfer

You can check and control processing of the export and network jobs:

- ✧ Select **Transfer > Local Job Status** or **Transfer > Network Job Status**.

- Or -

Click the corresponding icon on the status bar.



A dialog window for checking and controlling the jobs is displayed.

→ **Patient Browser** | *Storing and sending data* in the syngo Operator Manual

Based on the reference images measured, you determine the position, extent and orientation of the region under examination. As soon as you open a protocol, these settings are shown as graphic objects in the reference images.

- ☐ Slices or slabs to be measured from which images will be reconstructed at a later date,
- ☐ Saturation regions for avoiding motion artifacts and
- ☐ Navigator objects when the protocol is based on a navigator sequence.

You adapt the graphic objects of the protocol quickly and comfortably to the anatomical facts of the patient using the tools for graphic slice positioning.

You are able to numerically enter the parameters of the graphic objects into the **Geometry** parameter card.

→ [Page C.7–9](#), *Adjusting the orientation and position of graphic objects*

Prepare for positioning

After completing the localizer, the next protocol automatically opens, allowing you to position the slices to be measured.

Opening a protocol

You can edit every subsequent protocol in the queue.

- ✧ Select the requested protocol in the program control.
→ [Page C.6–25, Opening a protocol](#)

The protocol opens and its parameters are displayed on the parameter cards.

The graphic objects of the protocol (slices, slabs, navigators, and saturation regions) are displayed in the reference images in the image area.

The information line above the parameter cards displays a wealth of information regarding the open protocol.

→ [Page C.7–2, Protocol info line](#)

- ✧ Please note the dependencies among the positioning mode of the protocol, the image type, and the table positions of the reference images.
→ [Page C.4–26, Dependencies between the table position, reference image, and protocol](#)

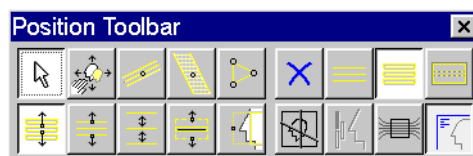
Size of the measurement region

All slices to be examined with a protocol should be as close to the center of the system as possible, because the magnetic field lines run in parallel, ensuring optimum image quality. Distortions can occur outside the isocenter.

You can divide protocols with large examination regions into several protocols that fit fully in the isocenter. ISO protocols automatically calculate the optimum table position and automatically move the table to the isocenter during the measurement.

→ [Page C.9–1](#), *Measuring Large Examination Regions*

Position tool bar



For positioning the objects, superpose the **Position Toolbar**.

- ✧ Select **View > Position Toolbar**.
- Or -
- Click this button.



Graphic objects

Depending on the default sequence technique set in the protocol, various graphics are shown in the reference images.

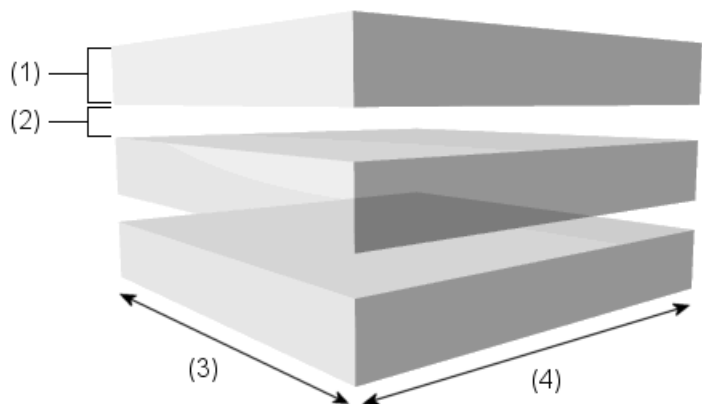
- ❑ Slice groups (for 2D measurements)
→ [Page C.5–5](#)
- ❑ 3D/Slabs (for 3D measurements)
→ [Page C.5–7](#)
- ❑ Saturation regions (possible for both 2D and 3D measurements)
→ [Page C.5–9](#)
- ❑ Navigator objects (for Navigator sequences)
→ [Page C.5–14](#)
- ❑ Spectroscopy volume elements
(e.g., for SVS protocols)
- ❑ Spectroscopy CSI slice
(e.g., for CSI protocols)
- The protocols and objects of spectroscopy are explained in a separate manual
→ *Refer to the MR Spectroscopy Operator Manual*

Slices and slice groups

In 2D measurement, you use slice groups to define the extent, position, and orientation of the examination area to be measured. Slice groups are composed of individual slices used for subsequent reconstruction of one or more tomographic images.

Slices always belong to a slice group. During graphical slice positioning you have to select an entire slice group. You are not able to select individual slices. The only exception is a slice group that consists of one slice only.

All slices in a group are located in parallel to one another, have the same slice thickness, the same slice distance, the same field of view (FoV) and the same phase oversampling factor (for reducing artifacts).



- (1) Slice thickness
- (2) Slice distance
- (3) Phase FoV
- (4) Read FoV

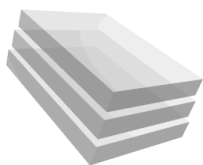
Field of view (FoV)

The field of view of a slice group is defined by its length and width. The length and width correspond to the read-out and phase-encoding directions in the magnetic field.

→ [Page C.5–71](#), *Changing the field of view*

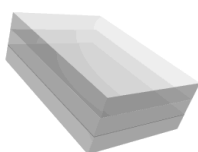
Slice thickness and slice distance

The distance between the slices of a slice group is a percentage of the slice thickness. As soon as you change the slice thickness of a group, the absolute slice distance changes as well.



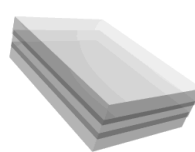
Distance factor > 0%

There is a distance between the slices of a group.



Distance factor = 0%

The slices of the group are contiguous.



Distance factor < 0%

The slices of the group overlap.

→ [Page P.1–3](#), Parameter *Dist. factor*

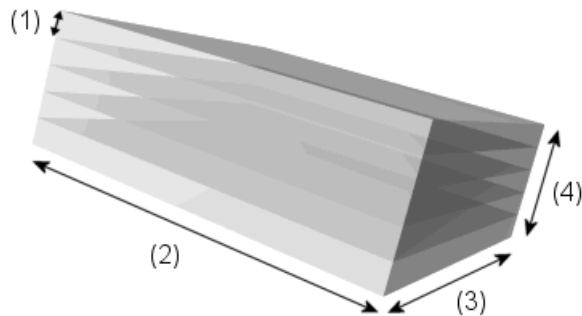
→ [Page C.5–70](#), *Changing the distance between slices/slabs*

→ [Page C.5–69](#), *Changing the thickness of slices/slabs*

Slabs and slab groups

For 3D measurements, phase encoding is applied to the spins in the slice selection direction. This allows you to measure entire volume ranges. In graphic slice positioning, these volume ranges are defined as slabs.

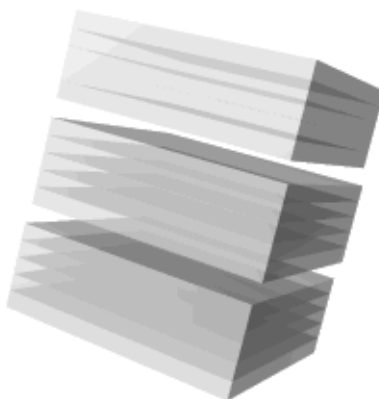
Slabs are made up of contiguous slices. The thickness of a slab is the product of the effective slice thickness and the number of slices.



- (1) Effective slice thickness
- (2) Read FoV
- (3) Phase FoV
- (4) Slab thickness

Slab groups

A slab group consists of several parallel slabs that have the same thickness and distance between them.



The slab distance is a percentage of the slab thickness.

Distance	Comment
> 0%	There is a distance between the slabs.
= 0%	The slabs are contiguous.
< 0%	The slabs overlap.

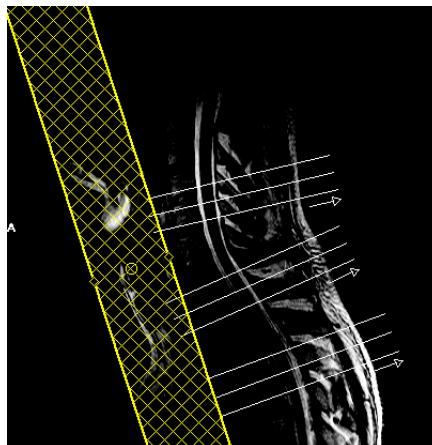
→ [Page C.5–70](#), *Changing the distance between slices/slabs*

Saturation regions

A saturation region (sat for short) is a volume where the MR signal is suppressed by an RF saturation pulse.

Regular saturation regions

You may freely define the position, orientation, and thickness of regular saturation regions. Several saturation regions can be planned in a single protocol with freely-selectable geometry. Standard saturation regions are not associated with a slice or slab group.

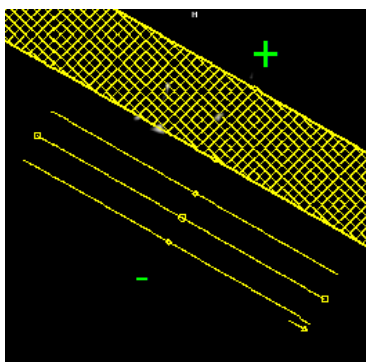


Parallel saturation regions

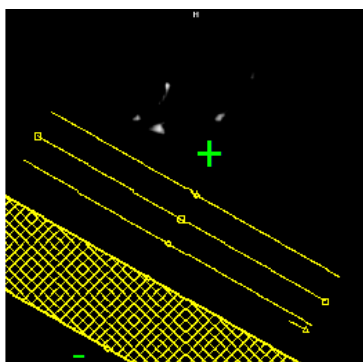
Parallel saturation regions may be positioned in pairs or individually (positive or negative sat). They are associated with and parallel to a slice or slab group. The distance from the saturation region to the far-end slices is set via a parameter in the protocol.

→ [Page P.1–71](#), Parameter Card *Geometry - Saturation*

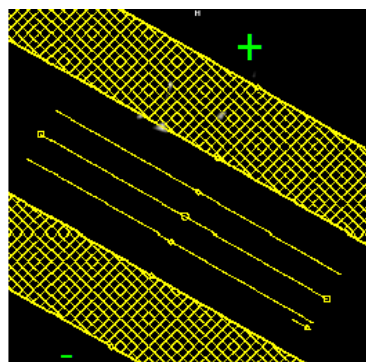
If you move, rotate, or change the particular slice or slab group, the two saturation regions are adjusted automatically.



Positive parallel sat



Negative parallel sat



Paired parallel saturation regions

The designations for parallel sat regions used in the **Geometry – Saturation** parameter card depend on the preferred direction (orientation) of the slice/slab orientations.

→ [Page P.1–71](#), Parameter Card *Geometry - Saturation*



Orientation of the slice group	Negative parallel sat	Positive parallel sat
Sagittal preferred	Parallel R (right)	Parallel L (left)
Coronal preferred	Parallel A (anterior)	Parallel P (posterior)
Transverse preferred	Parallel F (feet)	Parallel H (head)

Tracking saturation regions

You are able to plan tracking sat regions only for certain protocols. These protocols use sequences for sequential multi-slice measurements and measure one slice group or slab group only.

A tracking sat region is associated with and parallel to the slices of a slice or slab group. They can be on either side of the group.

The tracking sat region passes through the slice/slab group in either the ascending or descending direction. The name of the saturation regions depends on the preferred direction of the slice/slab group. Suitable saturation regions may be selected from the list on the **Geometry – Saturation** parameter card.

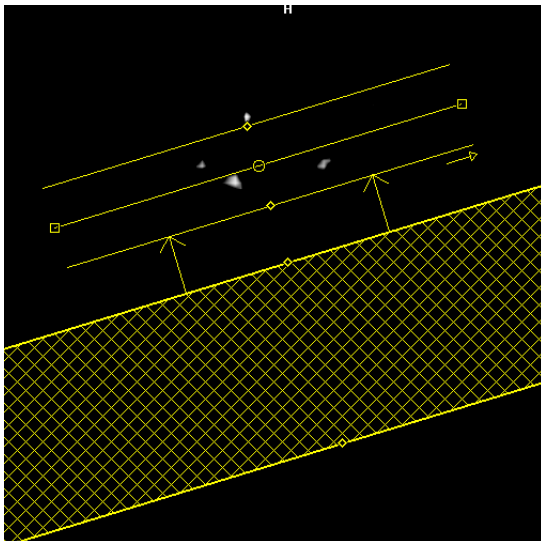
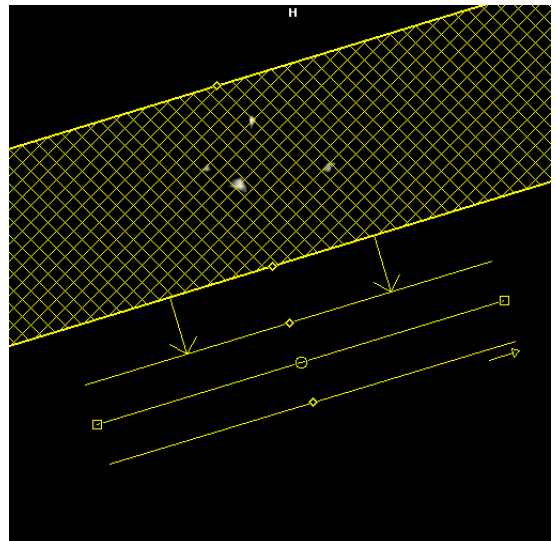
→ [Page P.1–71](#), Parameter Card *Geometry - Saturation*



Orientation of the slice group	Order of RF excitation	Sat name
Sagittal preferred	Ascending	Tracking R -> L
Sagittal preferred	Descending	Tracking L -> R
Coronal preferred	Ascending	Tracking A -> P
Coronal preferred	Descending	Tracking P -> A
Transverse preferred	Ascending	Tracking F -> H
Transverse preferred	Descending	Tracking H -> F

During acquisition, the saturation region tracks the most recently measured slice. The distance between the saturation region and this particular slice is the same as at the beginning of the measurement (distance to the first slice of the slice or slab group).

With transverse images, an ascending saturation region is located below and a descending saturation region is located above the slice group.

Tracking F $\rightarrow$ HTracking H $\rightarrow$ F

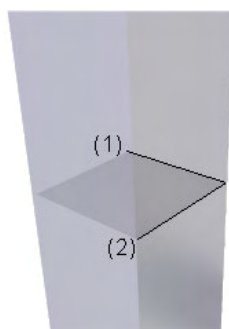
The arrows between the slice groups and sat regions show the standard vector.

- ⇒ The thickness and distance of a tracking sat region can be set only on the **Geometry – Saturation** parameter card.
→ [Page P.1–71](#), Parameter Card *Geometry - Saturation*

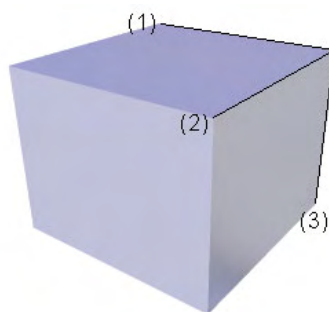
Navigator objects

Navigator objects are used in special navigator sequences, e.g., to record the movement of the diaphragm and uptake of contrast media, and to trigger the signal that starts the measurement.

There are two types of navigator objects:



Navigator pen: infinitely long "pen"



Navigator cuboid: finite cuboid volume

- (1) Base vector 1 (normalized direction)
- (2) Base vector 2 (phase encoding direction)
- (3) Extent in the readout direction (navigator direction)



Parameters of navigator objects

Base vector 1 and base vector 2 define the basic area of the navigator objects.

As compared to the cuboids, the pens are of infinite extent in the navigator direction.

A two-dimensional excitation profile may be defined within the base area. This excitation profile is defined by the sequence. Depending on the application, it may have a rectangular, trapezoidal, or elliptical contour.

⇒ New navigator objects can be added only on the **Geometry – Navigator** parameter card.

→ [Page P.1–79](#), Parameter Card *Geometry - Navigator*

Display of graphic objects

As soon as you have measured or loaded reference images, the graphic display shows the intersection or projection of your planned slices or slabs together with the reference images.

The display used for these objects shows you their position and orientation with respect to the reference images.

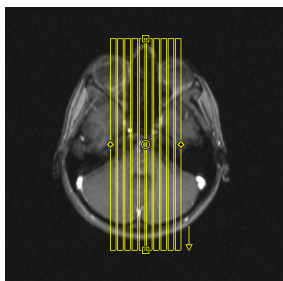
Intersecting or projection areas

If a graphic object intersects the reference image at an angle $\geq 30^\circ$ (limit angle), the lines or areas of this intersection are displayed.

Example: Sagittal slices in a transverse image

✧ For CSI slices, the limit angle is 45° . You can have the limit angles changed by Siemens Service.

When a graphic object penetrates the reference image at a shallow angle ($\leq 30^\circ$), it will be projected on the reference image. Instead of intersecting lines, a projection surface is shown.



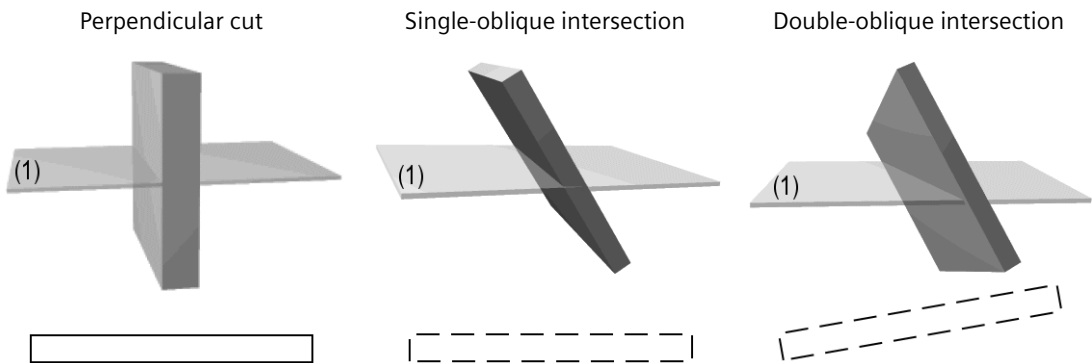
In a graphic object located parallel to the reference image, the projection area corresponds exactly to the dimensions of the field of view (FoV).

Example: Sagittal slices in a sagittal image.

If several graphic objects intersect the image at a shallow angle, only the projection area of the selected object is displayed.

Angle of intersection

The display of the intersecting or projection lines in the reference images shows you whether a graphic object is located vertically, obliquely or double-obliquely **to the reference image**.

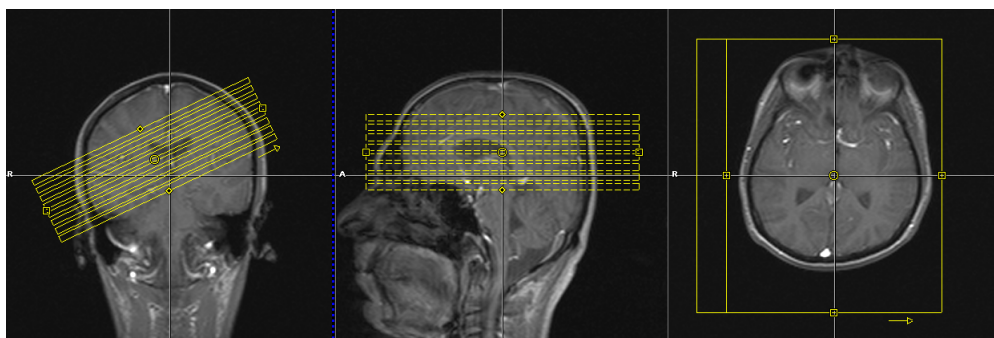


(1) Reference image

- ⇒ The line type provides no information regarding the position of the object with respect to standard anatomical views because reference images can be freely oriented.

Single-oblique intersection

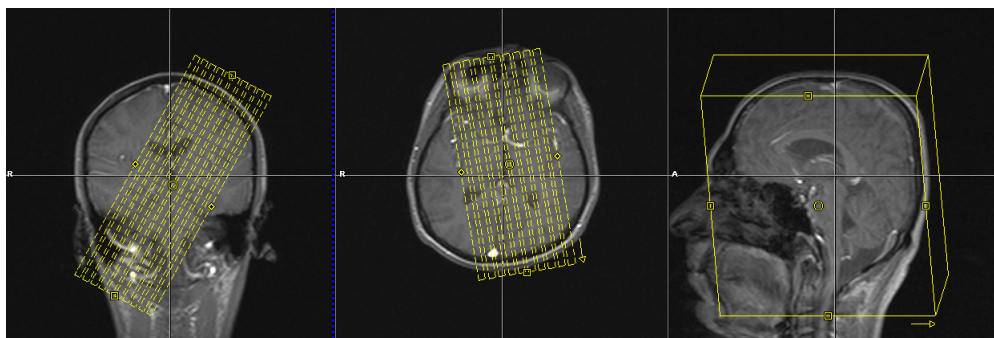
With single oblique intersections, one side of the field of view (FoV) is located in parallel with the intersecting line. The projection also maps a side area of the slice group.



➤ The arrows indicate the phase-encoding direction.

Double-oblique intersection

With a double-oblique intersection, neither of the two sides of the field of view (FoV) runs parallel to the common intersecting line with the reference image. Accordingly, two side areas are displayed on the projection area, providing a spatial impression.

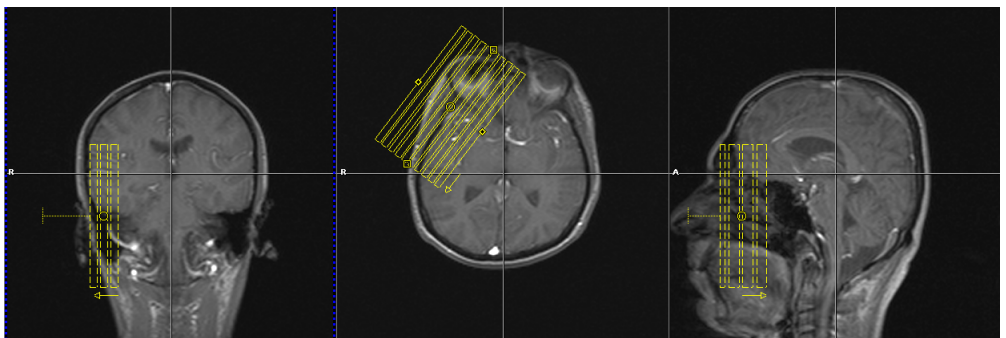


Slices without intersecting the reference image

With oblique intersections, not all intersections of a slice or slab group will be intersecting the reference image. Slices located completely in front or after the reference image are not drawn in.

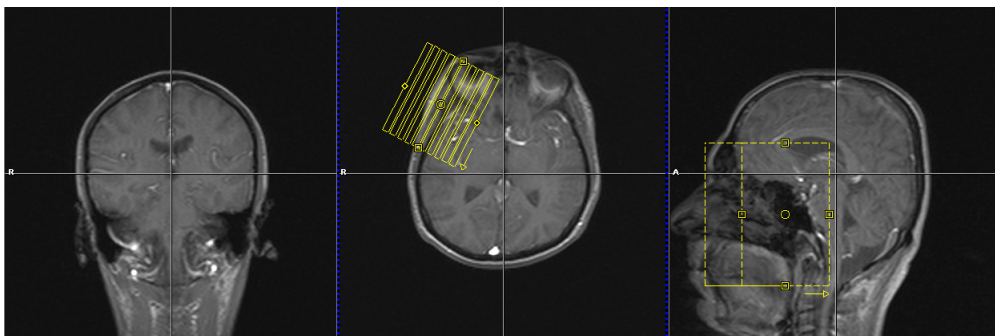


For this type of partial views, a bar indicates the location of the hidden slices.



Partial view of a slice group in the coronal reference image

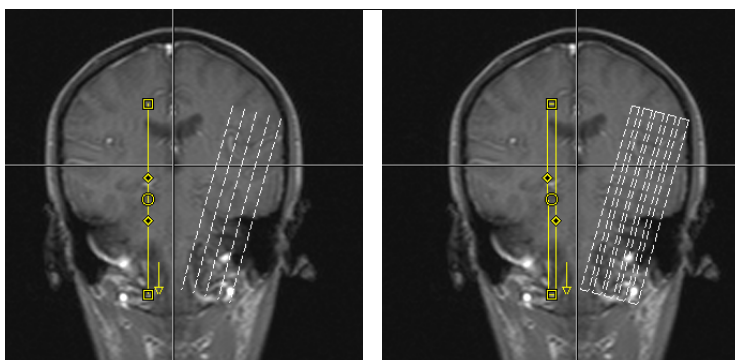
However, the projection surface of a slice group is still not visible, if none of the slices intersects the reference image. However, the edge is then shown as a dotted line.



Slice group without intersecting the coronal or sagittal reference image

Display in the line or box mode

A slice intersecting through a reference image can be displayed either as a line (line mode) or as a rectangle (box mode).



Line Mode

Box Mode

The box mode shows the actual intersection surface and therefore the anatomical region covered by the slice. In the line mode you see the center line of the intersection area.

This is how you change the display mode:

✧ Select/deselect **View > Graphics > Line Mode On**.

- Or -

Go to the **Position Toolbar** and select the button of the desired mode.



Displaying in the package mode

When your slice group contains a considerable amount of individual slices, you can hide the individual intersections to provide for a better overview. For that purpose, switch to the so-called package mode.

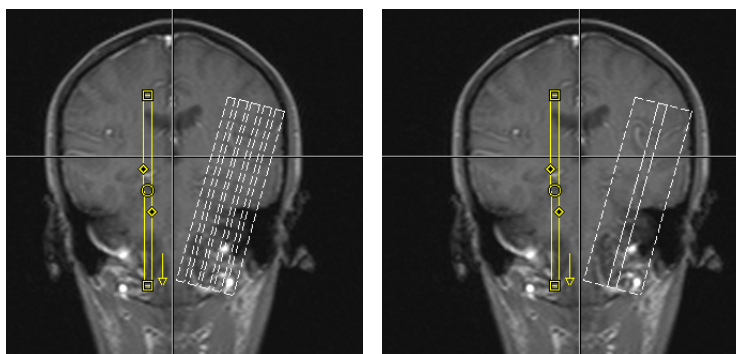
✧ Go to **View > Graphics** and select the **Package mode On/Off** option.

- Or -

Go to the **Position Toolbar** and select the **Package mode** button.



You only see the outline of the intersection for the entire group. The inner lines show the thickness of the individual slices and/or individual slabs.



Box Mode

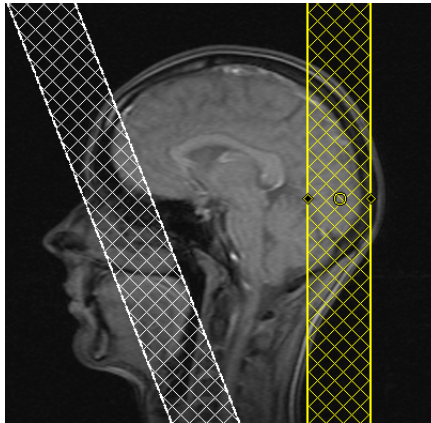
Package Mode

- ⇒ Slice or slab groups are automatically displayed in the package mode when the elements of the group overlap and are closely located to one another (depending on the magnification set for the reference image).

Display of saturation regions

Saturation regions that intersect the reference image are displayed as cross-hatched bars.

Saturation regions located vertically to the reference image show a solid outline, oblique ones show a dotted outline.

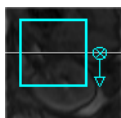


As you select a saturation region, the bar is highlighted. Handles are displayed.

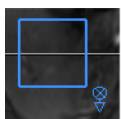
A saturation region running parallel to the reference image completely covers the reference image with its cross-hatched shading if the region includes the image plane of the reference image.

➤ It is not possible to project saturation regions onto reference images.

Displaying navigator objects



Non-selected navigator cuboids are displayed in turquoise.

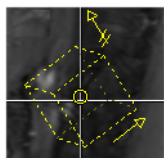


Non-selected navigator pens are displayed in blue.

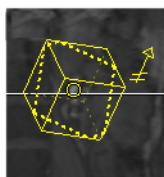


Selected navigator objects are shown in yellow and include handles.

The navigator objects are always shown with projection lines. This gives you a spatial impression of the position and extent of the navigator object.



- ☐ If the navigator does *not* have a common intersection with the reference image, the projection lines will be shown as dashed lines.



- ☐ If the navigator *does* have a common intersection with the reference image, the projection lines will be shown as continuous lines. The intersection area, in this case, is shown as dashed lines.



Next to the navigator object, arrows show the direction of basic vector 2 (phase-encoding direction) and the navigation direction.

The arrow without lines shows the direction of base vector 2. The arrow with the dotted lines points into the direction of the navigator.

This display lets you recognize special situations.



☐ The arrow for navigator direction points away from the viewer. The navigator direction continues *into* the image plane.



☐ The arrow for navigator direction points toward the viewer. The navigator direction runs *out of* the image plane.



☐ The arrow for the direction of base vector 2 is not shown, if the angle between the reference image and base vector 2 lies between 45° and 90°.

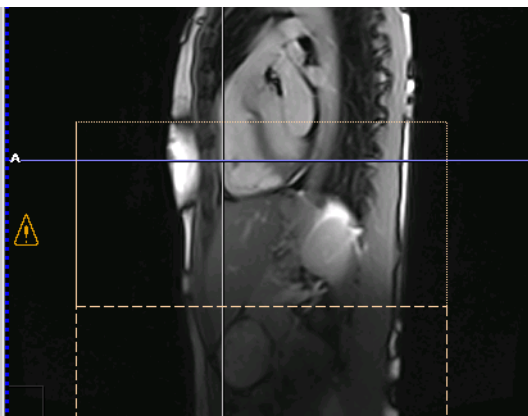
Display of objects outside the isocenter

If you plan the slice, slab or FoV too far outside the isocenter (i.e., strong distortions can be expected), the corresponding object is highlighted in color:

- ☐ Selected: orange
- ☐ Not selected: pink

In addition, an orange warning symbol is displayed in the reference image segments.

- ⇒ The results of such measurements cannot be fully corrected using 2D or 3D distortion correction (image type DIS2D*).

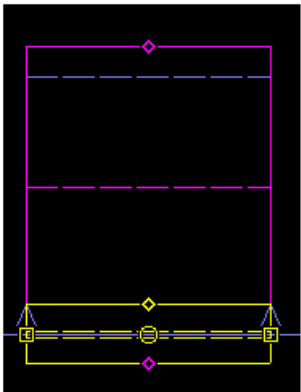


- ⇒ DIS2D* images may show strong image distortion. As a result, they may lead to incorrect results during post-processing. The results of this type of post-processing should not be used for diagnosis or if then only after checking them carefully.



Displaying TimCT slices

If you plan slices that will be measured with continuous table move (TimCT), the measurement region is displayed as follows:



Area		Color	Line type
FoV read		yellow	solid
Measuring range	general	magenta	solid
Measuring range	when selecting a variable resolution, e.g., area of an increased resolution → Page P.1–90, Variable resolution	magenta	dashed
Table position	at the start of the measurement.	blue	solid
Table position	at the end of the measurement	blue	dashed

- The table positions are in the isocenter of the magnet. The table position prior to the measurement can be displayed or hidden.
→ [Page C.5–32](#)

Display of orientation aids

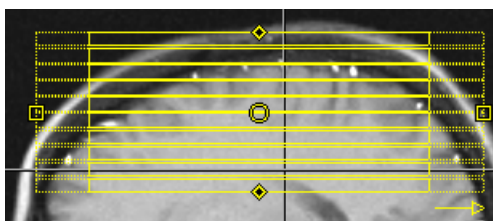
The following orientation aids facilitate the processing and positioning of graphic objects.

- ☐ Display of phase oversampling
- ☐ Display of slice oversampling
- ☐ Display of coil elements
- ☐ Display of the adjustment volume
- ☐ Display of table position
- ☐ Projection display
- ☐ Shadow lines
- ☐ Reference lines

Display of phase oversampling

Phase oversampling is used to prevent aliasing artifacts. In this case, the measured area along the phase-encoding direction is expanded symmetrically on both sides of the field of view (FoV).

When you set phase oversampling, the additional area to be measured is displayed as dashed lines in the reference images.

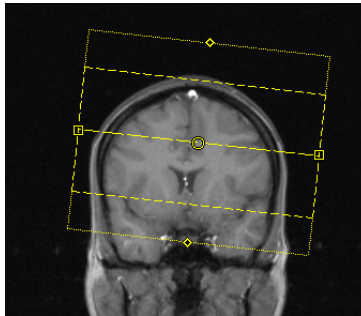


Display of 30% phase oversampling (2x15%) in the box mode

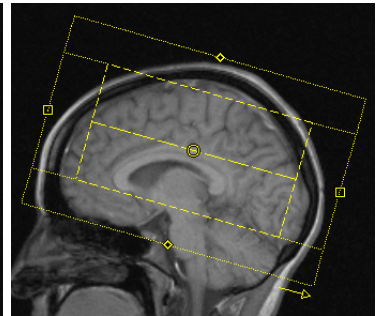
→ [Page C.5–72, Changing phase oversampling](#)

Displaying slice oversampling

For 3D measurements, you can set slice oversampling to prevent aliasing artifacts. In this case, the measured area along the slice selection direction is expanded symmetrically on both sides of the FoV.



Display of 50% slice oversampling
(2x25%) in the box mode



Displaying a combination of slice
and phase oversampling in the
box mode

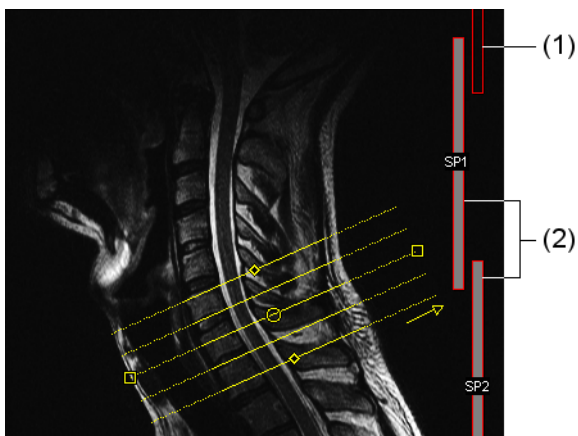
Showing/hiding slice oversampling

✧ Select **View > Graphics > Hide Slice Oversampling**.

The option is activated or deactivated and the view is changed accordingly.

Displaying coil elements

You can display the coils and their coil elements that have been connected for the examination. This facilitates your selecting the correct coil elements for the region of the body to be examined.



(1) coil is not selected

(2) coil is selected

- Only coils with a known table position may be displayed. These are either coils with fixed table positions or coils whose position was determined dynamically via application adjustments.

Activating/deactivating the coil elements



✧ Select **View > Coils on/off**.

- Or -

Click this button on the **Position Toolbar**.

The option is activated or deactivated and the view is changed accordingly.

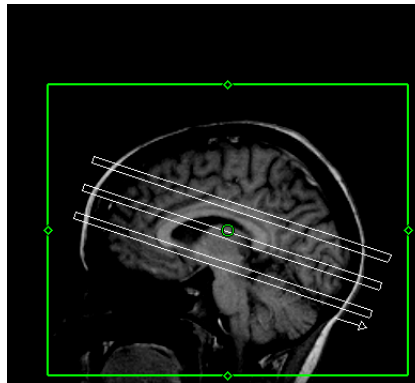
✧ Coil elements are *only* displayed on sagittal or coronal reference images.

You are able to select or deselect coil elements directly in the reference image or on the **System – Coils** parameter card.

→ [Page P.1–91](#), Parameter Card Geometry - Inversion

Showing adjustment volume

In some cases it may be useful to display the adjustment volume in the reference images.



This is how you show or hide the adjustment volume:

✧ Select **View > Adjustment Volume on/off**.

- Or -

Click this button on the **Position Toolbar**.



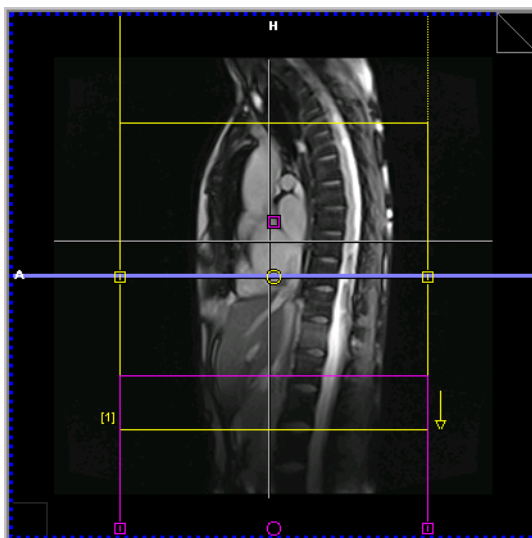
The option is activated or deactivated and the view is changed accordingly.

→ [Page B.2–10](#), *Editing the adjustment volume graphically*

Displaying the table position of the protocol

When working with isocenter protocols, it may be helpful to show the table position of the protocol as a blue line:

✧ Select **View > Protocol Table Position On**.



Activating the projection display

Normally, the projection display of slices/slabs is active only when the angle between the graphic object and the reference image is less than 30°.

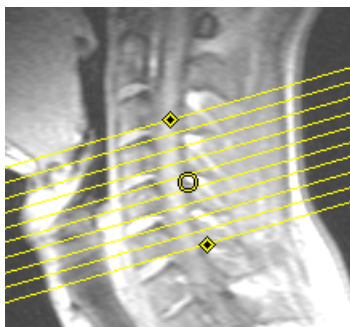
→ [Page C.5–16](#), *Intersecting or projection areas*

However, you can generally switch on projection display.

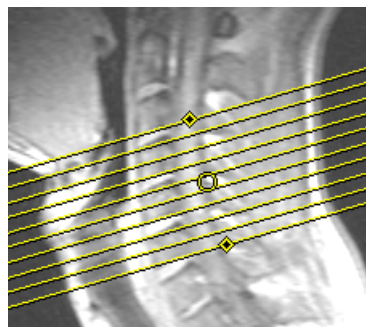
✧ Select **View > Graphics > Projection On/Off**.

Displaying shadow lines

Shadows improve the visibility of graphic objects in light images.



without shadow lines



with shadow lines

This is how you switch on/off shadow lines:

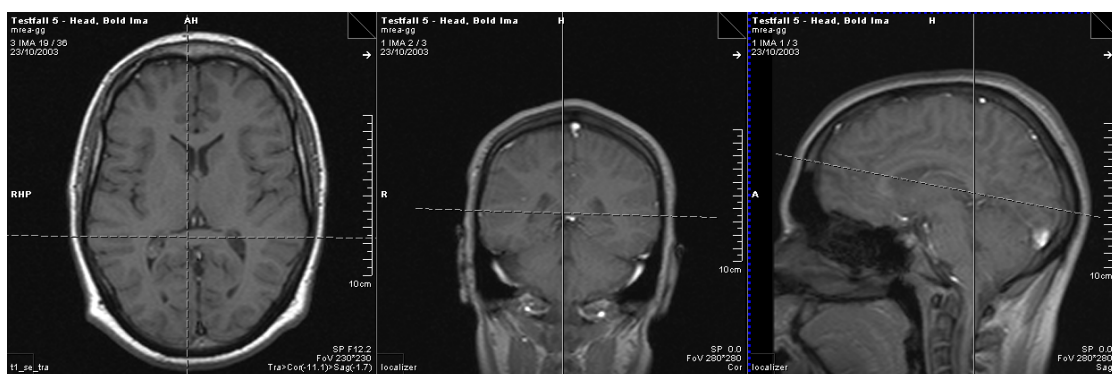
✧ Select **View > Graphics > Shadow On/Off**.

The option is activated or deactivated and the view is changed accordingly.

- ✧ When saving or filming, shadow lines are always applied regardless of the current setting.

Displaying reference lines

The reference lines show the positions of other reference images and facilitate spatial orientation.



This is how you show or hide reference lines:

- ✧ Select **View > Reference Lines On**.

- Or -

Click this button on the **Position Toolbar**.



The option is activated or deactivated and the view is changed accordingly.

Introduction to geometry parameters

This section describes important geometry parameters and their interconnections. The information provided will help you to efficiently plan measurements with the help of graphical slice positioning.

Basic resolution

Using basic resolution you determine the spatial resolution (number of pixels) in the readout direction.

The basic resolution is the reference value for showing resolution as a percentage in the phase-encoding direction.

⇒ The basic resolution also determines the image matrix size which can be doubled in size by selecting interpolation.

With 256 readout steps and a phase resolution of 100% as well as a square FoV (FoV Phase = 100%) you obtain a matrix size of 256×256 .

The size of the pixels depends on the size of the field of view (FoV) you have selected in the readout direction.

For an FoV of 256 mm in the readout direction, the pixels are 1×1 mm in size, and for an FoV of 396 mm in the readout direction, the pixels are 1.5×1.5 mm.

Phase resolution

Phase resolution is the resolution of slices in the phase-encoding direction expressed as a percentage value of the basic resolution.

If phase resolution is 100%, the resolution in the readout and phase-encoding directions will have the same value and the pixels are square. At 75% the pixels are rectangular. Resolution is decreasing.

If you reduce the FoV in the phase-encoding direction, the number of pixels has to be reduced as well to keep the resolution ratio constant. As a result, the number of phase-encoding steps is reduced.

FoV readout (mm)	FoV Phase (%)	Phase resolution (%)	Matrix size	Pixels (mm)
256	100	100	256 × 256	1 × 1
256	75	100	192 × 256	1 × 1
256	100	75	192 × 256	1.33 × 1
256	75	75	144 × 256	1.33 × 1

The pixel size is determined by both the basic resolution and the phase resolution, as well as by the FoV in the readout direction. Basic resolution determines the number of pixels in the readout direction, and phase resolution determines the aspect ratio of the pixels (square or rectangular).

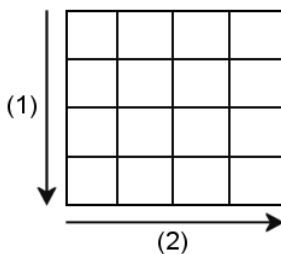
Changing the FoV in the readout direction changes the pixel size, but not the aspect ratio. For very large pixels, resolution of the reconstructed images decreases because the signal is averaged across a larger area.

FoV readout and FoV phase

The field of view (FoV) defines the anatomical region to be displayed (extent of measurement in the readout and phase-encoding direction) and its resolution (pixel size).

The FoV in the phase-encoding direction (**FoV Phase**) is defined as a percentage ratio to the FoV in the readout direction (**FoV read**).

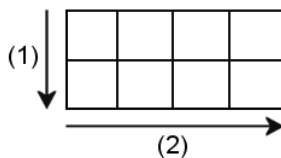
Square FoV



For a square FoV, the **FoV Phase** is set to 100 %. The same number of pixels is measured in the readout direction as it is in the phase-encoding direction.

- (1) Phase-encoding direction 4 steps
- (2) Readout direction 4 steps

Rectangular FoV



When examining elongated parts of the body, it may be helpful to work with a rectangular field of view to reduce the measurement time.

For a rectangular FoV, the **FoV Phase** parameter is smaller than 100%.

Example:

- (1) Phase-encoding direction 2 steps
- (2) Readout direction 4 steps

Changing the FoV in the readout direction

Changing the FoV in the readout direction also changes the FoV in the phase-encoding direction. This changes the pixel size.

Changing the FoV in the phase-encoding direction

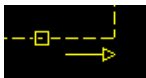
The gradients for encoding the phases are switched in the phase-encoding direction (**FoV phase**).

You can only set the extent in the phase-encoding direction to be less than or equal to the extent in the readout direction. If you change the FoV in the phase-encoding direction, the number of phase-encoding steps is always adapted to keep the resolution ratio constant.

⇒ If you increase the number of phase-encoding steps, the measurement time will be adjusted accordingly.

⇒ The phase-encoding direction is identified by an arrow in the image.

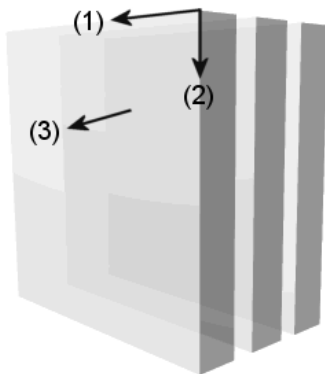
→ [Page C.5–39](#), *Phase-encoding direction*



Phase-encoding direction

The current phase-encoding direction (direction of the phase-encoding gradient) is identified by an arrow in the image. It is shown on the parameter card by using the main orientations of the *Whole Body Patient Coordination system*, e.g., **A >> P**.

You can change the phase-encoding direction. This will swap the phase-encoding and readout direction. Using this method allows you to prevent aliasing artifacts in the phase-encoding direction or change the direction of flow and motion artifacts.



- (1) Phase-encoding direction
- (2) Readout direction
- (3) Transverse slice orientation

Phase Enc. Dir. R >> L

Or

- (1) Readout direction
- (2) Phase-encoding direction
- (3) Transverse slice orientation

Phase Enc. Dir. A >> P

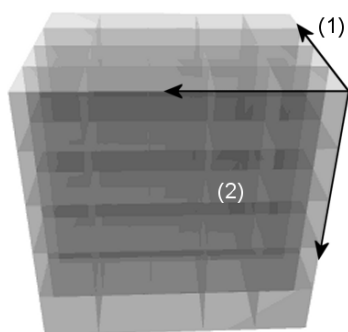
Slice resolution

In 3D measurements, phase-encoding is performed in addition to the slice selection and phase-encoding directions.

The **Slice resolution** parameter defines the resolution in the slice-selection direction. The resolution ratio is expressed as a percentage of the readout steps.

(1) Slice-selection direction

(2) field of view



With slice resolution at 100%, the phase-encoding table in the slice-selection direction has as many steps as the number of slices (partitions) to be reconstructed.

With slice resolution < 100%, fewer phase-encoded steps are measured in the slice-selection direction (resulting in a reduced measurement time). Additional slices are subsequently calculated using interpolation. The slice thickness displayed in the image text does not change. An "i" is simply appended to its numeric value to identify it (e.g., **SL 2.0i**).

Phase oversampling

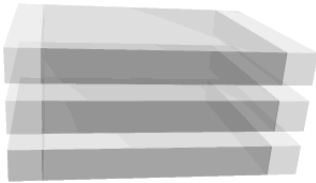
Phase oversampling increases the phase-encoded area symmetrically on both sides of the field of view (FoV). This extended FoV area is not displayed in the reconstructed image.

Phase oversampling is used to prevent overfolding artifacts. Aliasing artifacts are present when the excited body regions in the phase-encoding direction are larger than or extend outside the selected FoV. These regions are visibly "folded" on the opposite side of the image.

Phase oversampling increases the measurement time. The signal-to-noise ratio is improved.

Phase oversampling is always stated as a percentage of the FoV in the phase-encoding direction. Example: At 30% phase oversampling, the measurement area on both sides of the phase-encoding direction increases by 15% each.

⇒ Oversampling is automatically applied in the readout direction because it does not increase the measurement time.



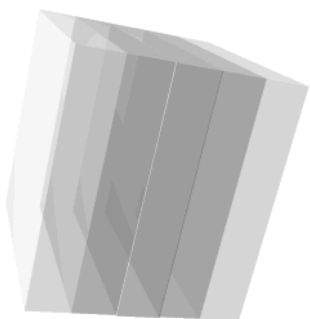
Slice oversampling

Slice oversampling increases the phase-encoded area symmetrically on both sides of the slab in the slice selection direction.

This creates slices that are reconstructed but not displayed in the image. Using this method prevents aliasing artifacts in the slice-selection direction for 3D measurements.

Slice oversampling is indicated as a percentage of the slab thickness.

- ⇒ Similar to phase oversampling, slice-oversampling increases the measurement time while improving the signal-to-noise ratio.



Graphic Slice Positioning (GSP)

Go to the reference images and use the **Position Toolbar** and the mouse to adjust the graphic objects (slices, slabs, saturation regions and navigators) to the anatomy of the patient. You may:

- ☐ Move
- ☐ Rotate
- ☐ Add
- ☐ Change parameters
- ☐ Delete

You may also use the parameter cards to manipulate graphic objects as described. In this case, you adjust the parameters numerically and work mainly in the **Geometry** parameter card.

- [Page P.1–60](#), Parameter Card *Geometry - Common*
- [Page P.1–71](#), Parameter Card *Geometry - Saturation*
- [Page P.1–79](#), Parameter Card *Geometry - Navigator*



If you plan the slices, slabs or FoV too far outside the isocenter (i.e., strong distortions can be expected), the corresponding object is highlighted in color and a warning icon is shown:

- [Page C.5–26](#), *Display of objects outside the isocenter*

Please avoid these types of measurements whenever possible.

- [Page C.9–1](#), *Measuring Large Examination Regions*
-

Selecting graphic objects

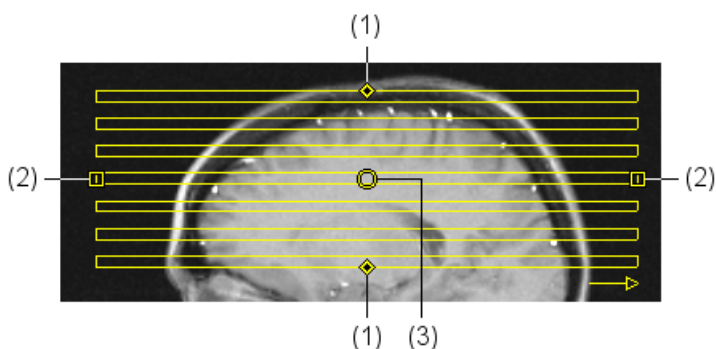
Selecting in the reference image

This is how you select a graphic object in the reference image:

- ✧ Move the mouse cursor across the object.
- ✧ Click the object at a location where the mouse pointer changes its appearance.

The object is highlighted and the handles are shown.

- ✧ Features in Set-n-Go protocols: When you select the object of a suitable step, it becomes the active step. You can change between the steps with Ctrl+T.



- (1) Extent handles
- (2) FoV handles and/or phase oversampling handles
- (3) Pivot handle

Selecting from the parameter card

You are also able to select objects via the parameter card:

- ✧ Go to the **Geometry** or **Routine** parameter card and select the object in the **Slice Group**, **Slab group** or in the **Navigator**.

Deselecting

This is how you deselect a previously selected object:

- ✧ Select **Edit > Deselect All**.
 - Or -
 - Select another graphic object.

Panning objects

- ✧ You can group graphic objects in order to rotate or move them together.
 - [Page C.5–65](#), *Grouping graphic objects*

You can pan objects in different ways:

- ☐ With the mouse
- ☐ Via menu commands
- ☐ By changing position values (numerically)
 - [Page C.7–9](#), *Adjusting the orientation and position of graphic objects*

Moving via the mouse



- ✧ Select the graphic object.

The pivot handle for panning is shown in the center of the image.

- ✧ Click and hold the pivot handle with the mouse and drag the object to the desired position.

The object position is updated on all reference images.

When the pivot handle of an object is not visible in the reference image, proceed as follows:

- ✧ Press **Shift** and keep the key pressed.
- ✧ Click and hold the intersecting line with the mouse and drag the object to the desired position.

- ✧ Parallel and tracking saturation regions can be moved together only with slices or slabs that they have been assigned to.

Panning by half the slice distance

You can move slice groups by half the distance between slices in the positive or negative slice-selection direction. When using manual breathhold measurements, you can pan the slice group in several steps so that part of the slices are congruent before and after panning.

✧ Select **Protocol > Gap filling +** or **Protocol > Gap filling –**

- Or -

Press the short cut **Ctrl + 6** (Gap filling +) or **Ctrl + 5** (Gap filling –).

Panning by group thickness

You are able to pan the slice group in one step by the thickness of the entire group. This function is especially suitable for multiple breathhold measurements.

✧ Select **Protocol > Stack +** or **Protocol > Stack –**.

- Or -

Press the short cut **Ctrl + 4** (Stack +) or **Ctrl + 3** (Stack –).

Moving centers in the reference plane

You are able to move the centers of slice/slab groups and navigator objects accurately into the reference plane (image plane of the reference image).

✧ Select the reference image to be used as the reference plane.

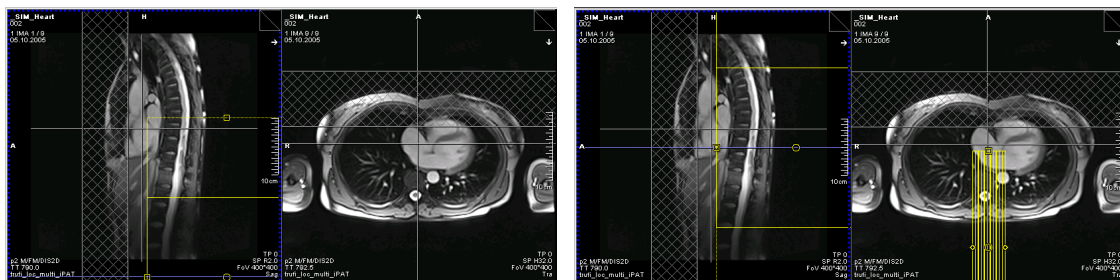
✧ Select **Protocol > Shift to Image Plane**.

- Or -

Select **Shift to Image Plane** in the context menu of the reference image.

The centers of the graphic objects (except sat. regions) are moved into the reference plane and, if necessary, into the visible area of the image.

✧ You are *not* able to use this function for slice groups and slab groups located in parallel to the reference image.



Reference images before and after selecting **Shift to Image Plane** with respect to the sagittal reference plane

Shifting the image in the Z direction

In distortion-corrected reference images (image type DIS2D or DIS 3D), you can move all graphic objects of the open protocol in the Z direction of the patient coordinate system (patient table movement direction). The position and orientation of the objects to one another are retained. The protocol takes the table position of the reference image.

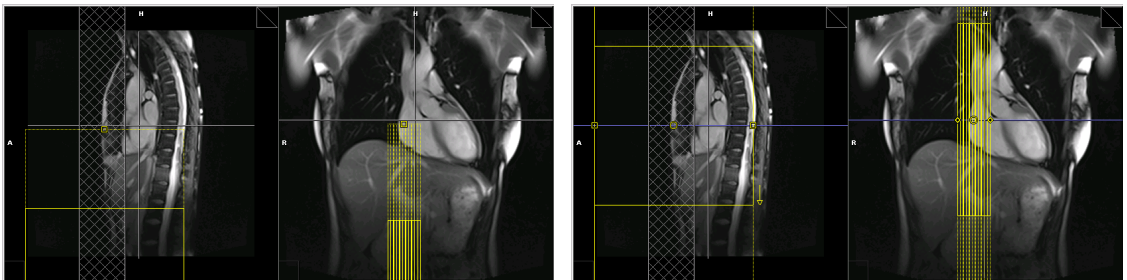
✧ Select the reference image to be used as the reference.

✧ Select **Protocol > Shift to Segment Center**.

- Or -

Go to the context menu and select **Shift to Segment Center** for the reference image.

The centers of the graphic objects of the protocol are moved in the reference image in the Z direction. The protocol now has the table position of the reference image.



Reference images before and after selecting the **Shift to Segment Center** function

Moving the FoV with overlapping

You can move the FoV overlapping the entire length of the patient coordinate system in the Z direction (patient table movement direction).

- ☐ Sagittal and coronal slices are moved in the readout or phase-encoding direction.
- ☐ Transverse slices are moved perpendicular to the slice plane.

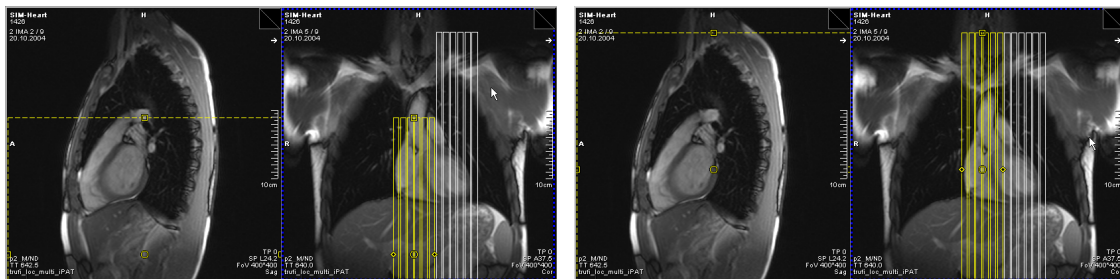
In this manner, you can plan multiple protocols whose FoVs overlap so that the resulting images can be reconstructed into a complete image.

✧ If necessary, use the **Protocol > Overlap...** to set the value for the overlapping required.

✧ Mark the FoV you want to move.

✧ Select **Protocol > FoV +** or **FoV -**.
- Or -

Press the short cut **Ctrl + 8** (FoV +) or **Ctrl + 7** (FoV -).



FoV before and after being moved with the **FoV+** function.

Rotating objects

You can rotate objects in different ways:

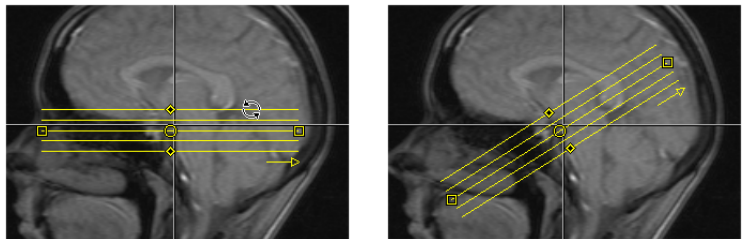
- ☐ With the mouse
- ☐ Via menu commands
- ☐ By changing the orientation values (numerically)
 - [Page C.7–10](#), *Entering the slice orientation*

Rotate using the mouse

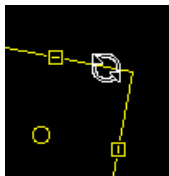


- ✧ Move the mouse pointer to a location of the object where it takes this particular shape.
- ✧ Click and hold the left mouse button at this position and rotate the object in the desired direction.

The object is rotated about its center. The rotational axis is now perpendicular to the reference image.



Slice group before and after rotation



Rotating sat regions and the navigator pens

When slices or slab groups are displayed as projection area (.....) rotate the object with the **Ctrl** key.

- ✧ Keep the **Ctrl** key pressed.
- ✧ Position the mouse pointer on the contour line of the object.
The mouse pointer changes shape for rotation.
- ✧ Click and hold the left mouse button at this position and rotate the object in the desired direction.

You are able to determine the center of rotation for regular sat regions and navigator pens:

- ✧ Click a position in the object to serve as the center of rotation.
The center (pivot) is displayed at the closest possible position to where you clicked.
- ✧ Click the object with the mouse button away from the center and rotate the object in the desired direction.
The object rotates about its center
- ✧ Parallel and tracking saturation regions may be rotated *only* with the slices or slabs assigned to them.

Combining rotate and shift functions



Using the **Shift** key you can freely combine rotation and panning without putting the object down.

- ✧ Grab the object with the mouse at a place for rotation.
- ✧ Keep the **Shift** key pressed and move the mouse to shift the object.
- ✧ Release the **Shift** key and move the mouse to rotate the object.
- ✧ Press the **Shift** key again and move the mouse if you want to shift the object again.

Rotation of oblique intersections

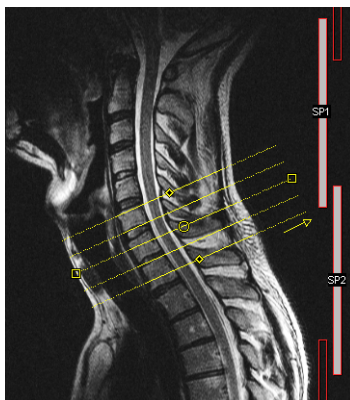
When you rotate slices the center of which is not in the plane of rotation (oblique and double-oblique cuts), the center of the slice moves as you rotate the slice.

Swapping the phase direction

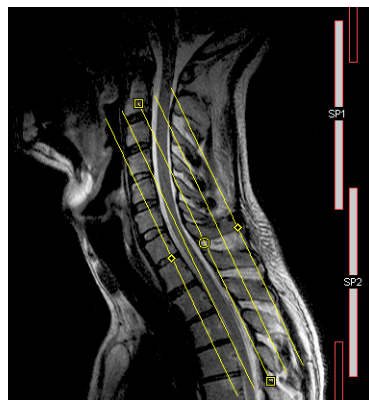
The main orientation of the object may change as it is rotated.

If the object has a rectangular FoV, you may find that the intersection lines in the resulting display are noticeably longer or shorter than the original, because of the realigned phase-encoding direction of the FoV.

Example:



Slice group prior to rotation:
transverse preferred



Slice group after rotation:
coronal preferred

Check the phase-encoding direction after this kind of rotation.

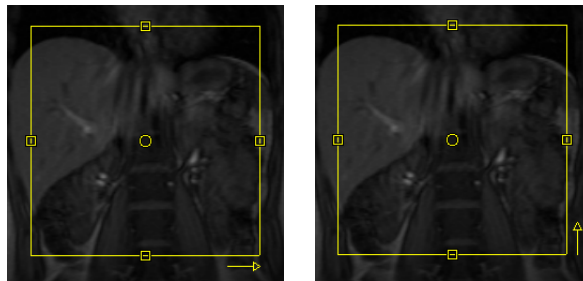
This is how you swap the phase-encoding direction of a slice group, a slab group or a Navigator object:

- ✧ Select the object.
- ✧ Select **Protocol > Swap Phase**.

- Or -

Go to the context menu and select **Swap Phase**.

The field of view of the selected object is rotated 90°. The phase-encoding and readout directions are swapped.



Example for swapping the phase direction (before/after)

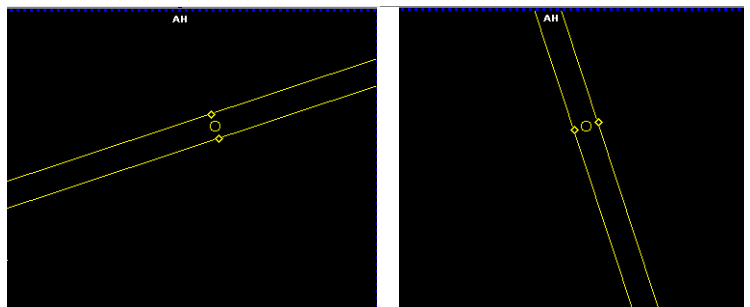
❖ You are *not* able to process saturation regions in this way.

Rotating by 90°

You can rotate slice and slab groups easily by exactly 90°:

- ❖ Select the group.
- ❖ Select **Protocol > Turn Group**.

The intersections are rotated by 90°. The center of the group remains unchanged.



Example for rotating by 90° (before/after)

Aligning objects

Use the menu to align objects perpendicular to the reference plane without rotating them manually.

You are also able to align slices or slab groups perpendicular to one another.

Aligning perpendicular to the reference plane

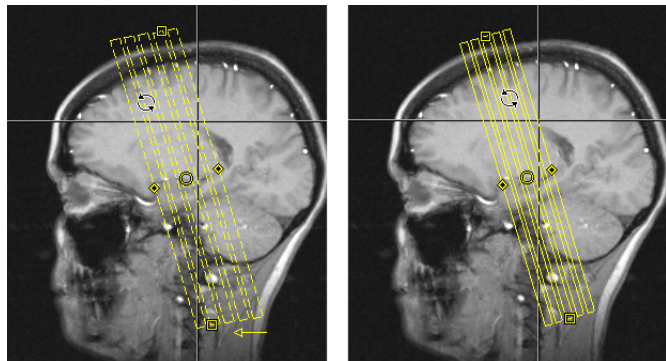
This is how you align objects at a right angle to the reference plane:

- ✧ Select the object to be aligned.
- ✧ Select **Protocol > Ortho Sync**.
 - Or -
 - Press **Ctrl + 1**.

The object is aligned. In addition, slices or slab groups are shifted with their center into the reference plane.

When you align a navigator at an angle of $\leq 45^\circ$, it is aligned in parallel instead of perpendicular to the reference plane. The navigator orientation is then in the image plane.

- ✧ If no object is selected, **Ortho Sync** affects all oblique slices and slab groups. Oblique saturation regions or navigators are *not* aligned in this case.



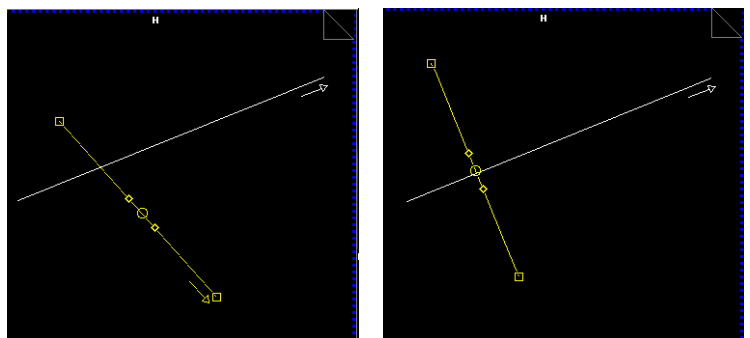
Example for aligning the reference plane (before/after)

Alignment perpendicular to the reference group

This is how you align a slice or slab group perpendicular to the other:

- ✧ Select the group to you want to align.
- ✧ Select **Protocol > Orthogonal**.
Or
Press **Ctrl + 2**.

The group is aligned and its center is moved to the reference group.



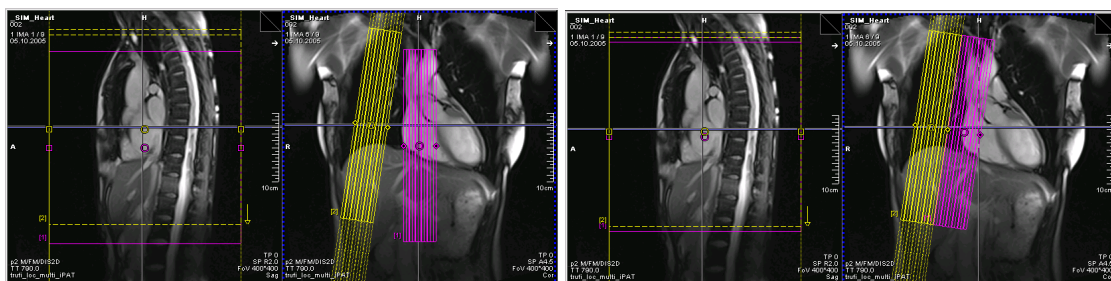
Example for alignment along a reference group (before/after)

Parallel alignment

You are able to align all slice groups of a protocol or all steps of a Set-n-Go protocol in parallel to one another in the slice-selection direction.

This function is suitable for aligning slice groups from several steps using one command.

- ✦ This function is available only, if you have a *Tim Planning Suite*.
- ✦ Mark the slice group used to align all other slice groups in parallel to it.
- ✦ Select the **Protocol > Align Parallel**.



Example for parallel alignment (before/after)

Align the FoV

You can align the FoV of all steps of a Set-n-Go protocol in the Z direction of the patient coordinate system (patient table movement direction) and have them overlap while positioned next to each other. The slices are moved in the readout or phase-encoding direction.



In this manner, you can plan the steps of Set-n-Go protocols whose FoV overlap so that the resulting images can be reconstructed into an overall image.

- This function is available only, if you have a *Tim Planning Suite*.
- ✧ If necessary, use the **Protocol > Overlap...** to set the value for the overlapping required.
- ✧ Mark the FoV to which the others should be aligned and positioned.
- ✧ Select the **Protocol > Align FoV**.

Adding slice and slab groups

You can add new slices and/or slab groups in different ways:

- ☐ Freely positionable with the mouse
- ☐ Positionable in the center of the reference image
- ☐ Adding in the 3 point mode

It depends on the protocol selected how many slices or slab groups you want to add to those already present.

→ [Page C.7–4](#), *Soft limits and extended limits*

The order used for drawing or processing new slices or slabs does not affect the sequence of the measurement and the numbering of the reconstructed image.

→ [Page C.6–36](#), *Numbering reconstructed images*

Adding using the reference groups

If you selected another group prior to adding, all characteristics of this reference group are propagated to the new group save for this position (number of slices and/or slabs, distance factor, swap, phase oversampling, slice-oversampling and orientation).

When it comes to the position, the new group is offset by one group thickness in the slice selection direction from the reference group.

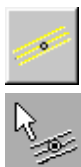
Adding without reference group

If *no* other group was selected prior to add, the new group has *one* slice and/or *one* slab and the center of the reference plane.

Positioning freely

You can add new slice or slab groups anywhere in the reference image via a mouse click or by using the orientation line.

- ✧ Click this button in the **Position Toolbar**.



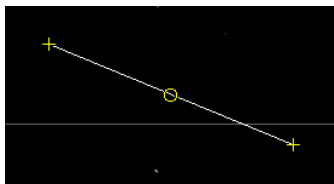
The slice or slab group drawing mode is activated. The cursor changes shape.

- ✧ Click the position in the reference image where the center of the new group should be located.

Or

- ✧ Use the mouse to draw a line where the new group should be positioned (orientation line).

The new slice/slab group is drawn in. Its center is in the image plane of the reference image.



- ⇒ If you have drawn an orientation line, the center of the new group is located at the center of the line.

You are able to move new groups, rotate them, or add new groups.

This is how you switch off the mode for drawing slice or slab groups:



- ✧ Click this button in the **Position Toolbar**.
 - Or -
 - Select another processing mode via the menu or in the **Position Toolbar**.
- ⇒ After you have drawn in the maximum possible number of slices and/or slabs, the mode is automatically switched off.

Positioning in the center of the segment

This is how you enter a new group in the center of the selected reference image segment:

- ✧ Select **Processing > Deselect All** to ensure that no group is selected.
- ✧ Select **Protocol > Add Slice (Slab) Group**.
 - The center of the new slice and/or slab group is located exactly in the image plane of the reference image.
- ⇒ If the center of the reference image segment already holds a slice group with the same alignment, the new slice group is shifted in parallel so that it joins the existing slice group equidistantly.

Adding in the 3 point mode

In the 3-point mode, you set 3 points in freely-selectable reference images and define a plane in this manner. The center slice or slab of this new group is in this plane.

- ✧ Select **Processing > Deselect All** to ensure that no group is selected.
- ✧ Click this button in the **Position Toolbar**.



The 3 point mode is activated. The cursor changes shape.

- ✧ Click three positions in the reference images.

You can scroll through the reference images or load new reference images while you are setting the three points.

If you have not set the third point, you can move the first and second points with the mouse or delete them by clicking again.

Once you set the 3rd point, the new group is displayed in the reference images.

- ✧ If a group is selected, a new group is not added when you set the third point; instead, the selected group is aligned with the three points.

How to deactivate the 3-point mode:

- ✧ Click this button in the **Position Toolbar**.
 - Or -
 - Press **Esc**.



- (1) First point
- (2) Second point
- (3) Third point

Creating radial slices

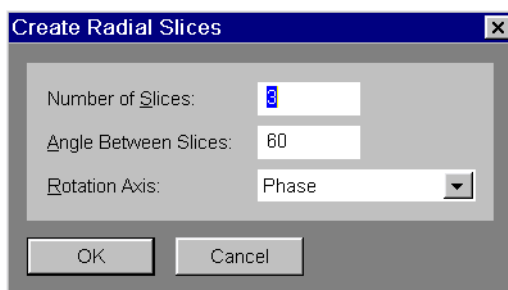


You are generating a random number of radial slices from a previously drawn slice group. The parameter settings of the slice group is transferred.

You are able to individually change the parameters of the radial slices.

- ✧ Radial slices replace all slices, slice groups, slabs and slab groups of the current protocol.
- ✧ Select a slice group, from which the radial slices are to be created.
- ✧ Select **Add-On > Create Radial Slices...**

The **Create Radial Slices** dialog window is shown.



- ✧ Specify the number or the angle of the slices to one another.
- ✧ The corresponding value is calculated automatically.



- ✧ Select the rotational axis.

- ✧ Click **OK**.

The radial slices are generated on the basis of the selected slice group.

- ✧ To move the radial slices as a group, you must activate **Tools > Coupled Graphics On**.

Grouping graphic objects

You can group graphic objects in order to rotate or move them together.

The following objects can be included in groups:

- ☐ Slice groups
- ☐ Slab groups
- ☐ Saturation

- ✧ To group the objects, select **Tools > Coupled Graphics On**.

- ✧ To eliminate a group, deselect **Tools > Coupled Graphics On**.

Adding saturation regions

You are able to add regular saturation regions using graphic slice positioning. Parallel and tracking saturation regions can be set only in the **Geometry – Saturation** parameter card.

→ [Page P.1–71](#), Parameter Card *Geometry - Saturation*

Positioning freely

You are able to freely position a regular sat region in the reference image using the mouse or an orientation line.

✧ Select **Tools > Create Sat.**

- Or -

Click this button in the **Position Toolbar**.



The mode for drawing regular saturation regions is activated. The cursor changes shape.

✧ Click the position in the reference image where you want to insert the new saturation region.

Or

✧ Use the mouse and draw a line in the reference image where the new sat region is to be located (orientation line).

The saturation region is located perpendicularly to the reference image and runs in the vertical direction. The thickness of the region is calculated using standard protocol parameters.



How to switch off the mode for drawing regular saturation regions:

✧ Click this button in the **Position Toolbar**.

- Or -

Press **Esc**.

✧ If you have drawn the maximum number of saturation regions and select a different editing mode, the mode switches off automatically.

Positioning in the center of the segment

✧ Select a reference image.

✧ Select **Protocol > Add Sat**.

The saturation region is located perpendicularly to the reference image and runs in the vertical direction. The thickness of the region is calculated using standard protocol parameters.

Changing parameters graphically

You can change the following object parameters graphically using the mouse:

- ☐ Thickness of slices/slabs and/or sat region
- ☐ Number of slices/slabs
- ☐ Distance of slices/slabs and/or sat region
- ☐ Size of slice oversampling for slab groups
- ☐ Size of field of view
- ☐ Size of phase oversampling
- ⇒ You can change this parameter numerically on the **Geometry** card.
 - [Page P.1–60](#), Parameter Card *Geometry - Common*
 - [Page P.1–71](#), Parameter Card *Geometry - Saturation*
 - [Page P.1–79](#), Parameter Card *Geometry - Navigator*
- ⇒ While you are changing a parameter, you can use the **ALT** key to quickly switch back and forth between an alternative mode.
 - [Page C.5–73](#), *Changes in mode during editing*

Changing the thickness of slices/slabs



- ✧ Select **Tools > Extent Mode > Change slice (slab) thickness**.
 - Or -
 - Click this button in the **Position Toolbar**.



- ✧ To increase the thickness, use the mouse to drag the **Extent** handle away from the group.

Or

- ✧ To decrease the thickness, use the mouse to drag the **Extent** handle to the center of the group.

As long as you keep the mouse button pressed, the thickness is shown in mm right next to the mouse pointer.

- ✧ Changing the slice thickness also changes the distance between slices since the distance factor remains constant.

Changing the thickness of slices/slabs



- ✧ Select **Tools > Extent Mode > Extent Slices (Slabs)**.

- Or -

Click this button in the **Position Toolbar**.



- ✧ To increase the number of the slices/slabs, use the mouse to drag the **Extent** handle away from the group.

Or

- ✧ To reduce the number of the slices/slabs, use the mouse to drag the **Extent** handle into the center of the group.

As long as you keep the mouse button pressed, the number is shown in mm right next to the mouse pointer.

- ✧ The maximum possible number of slices is limited by other protocol parameters (for example, by TR). Once you reach this limit when dragging the extent handle, the group is displayed with bold lines.

Changing the distance between slices/slabs



- ✧ Select **Tools > Extent Mode > Extent Distance**.

- Or -

Click this button in the **Position Toolbar**.

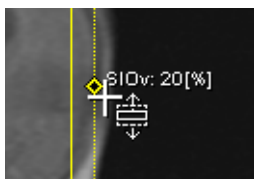
- ✧ To increase the distance of the slices/slabs, use the mouse to drag the **Extent** handle away from the group.

Or

- ✧ To reduce the distance of the slices/slabs, use the mouse to drag the **Extent** handle into the center of the group.

As long as you keep the mouse button pressed, the distance is displayed in percent right next to the mouse pointer.

Change slice oversampling



With 3D measurements, you can change the size of slice oversampling.

- ✧ Select **Tools > Extent Mode > Extent slice oversampling**.

- Or -

Click this button in the **Position Toolbar**.

- ✧ To increase slice oversampling, use the mouse to drag the **Extent** handle away from the group.

Or

- ✧ To reduce slice oversampling, use the mouse to drag the **Extent** handle into the center of the group.

As long as you keep the mouse button pressed, the size of slice oversampling is shown in percent right next to the mouse pointer.

Changing the field of view

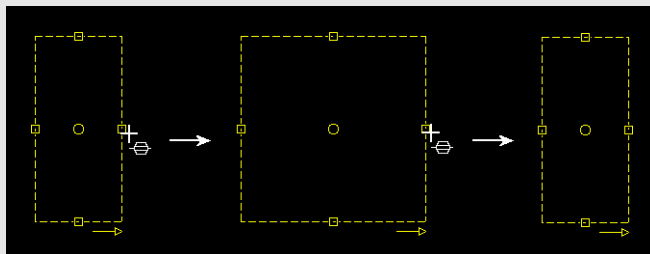
- ✧ Select the slice or slab group whose field of view (FoV) you want to change.
- ✧ Use the mouse and drag the FoV handle into the required direction.



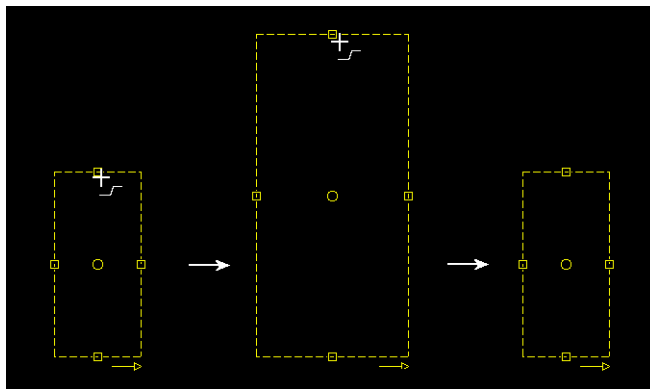
When you drag the FoV handle into the readout direction, the mouse pointer changes into this shape.

When you drag the FoV handle into the phase-encoding direction, the mouse pointer changes into this shape.

The dimensions of the current field of view (FoV) are displayed at the mouse pointer as long as you keep the mouse button pressed.

Notes on increasing/ reducing the field of view

You can increase the field of view to a square in the phase-encoding direction. You can reduce the FoV in the phase-encoding direction independent of the readout direction.



In the readout direction, you can increase or decrease the field of view only as a function of the FoV aspect ratio. The ratio between the length in the phase-encoding and readout direction remains constant. When you increase or decrease the field of view in the readout direction, the field of view grows or shrinks in the phase-encoding direction as well.

Changing phase oversampling



- ✧ Select **Tools > Extent Mode > Extent Phase-Oversampling On**.

- Or -

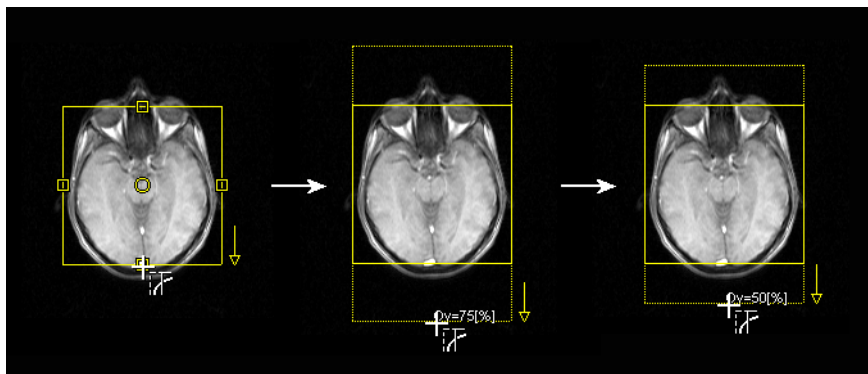
Click this button in the **Position Toolbar**.

- ✧ Select the slice or slab group whose phase oversampling factor you want to change.
- ✧ Drag the FoV handle with the mouse in the phase-encoding direction to increase phase oversampling.

Or

- ✧ Drag the FoV handle with the mouse opposite the phase-encoding direction to decrease phase oversampling.

The current value of the phase oversampling factor will be displayed at the mouse pointer as long as you keep the mouse button pressed.



- ✧ Changes in phase oversampling have a symmetric effect on the slice or slab group. Size and center of the field of view remain constant.

Changes in mode during editing

Use the **ALT** key to switch back and forth to another mode for editing. The following table shows you which alternative mode is available in what initial mode.

Initial mode	Alternative mode
Changing the slice (slab) thickness	Changing the number of slices (slabs)
Changing the number of slices (slabs)	Changing the slice (slab) thickness
Changing the slice (slab) distance	Changing the number of slices (slabs)
Changing slice oversampling	Changing the slice (slab) thickness
Changing the field of view	Changing phase oversampling
Changing phase oversampling	Changing the field of view

- ✧ To switch into the alternative mode, keep the **ALT** key pressed.
- ✧ Use the mouse to change the parameter in the alternative mode.
- ✧ To return to the initial mode, release the **ALT** key.

Changing regular saturation regions

You are able to move saturation regions graphically, or change their orientation or thickness with the mouse. The procedure is the same as for positioning slices and slab groups.

Other saturation region parameters may only be edited numerically on the **Geometry – Saturation** parameter card.

→ [Page P.1–71](#), Parameter Card *Geometry - Saturation*

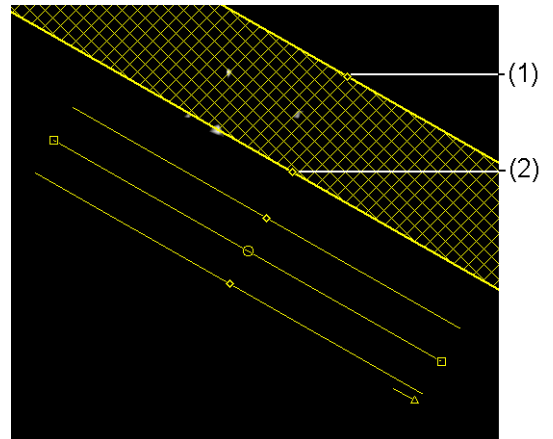
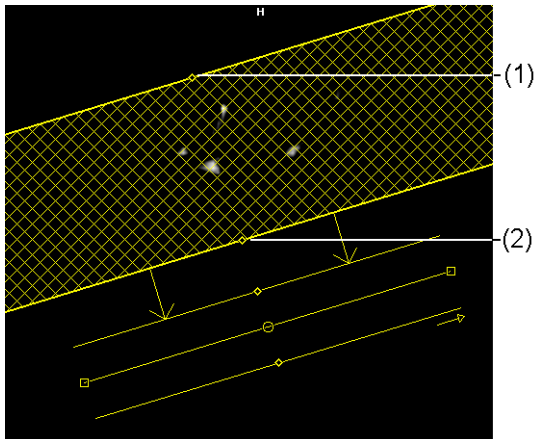
Changing parallel and tracking saturation regions

You can change the position and orientation of a parallel/tracking saturation region via the slice/slab group associated with this slice group. The distance between slice/slab groups remains constant.

→ [Page C.5–45](#), *Panning objects*

Depending on the extent handle clicked, you are able to change the thickness of the sat region or the distance to the slice/slab group.

⇒ With parallel saturation regions, modifying the slice thickness and the distance to the slice/slab group always affects both saturation regions symmetrically.



- (1) Handle for changing the sat thickness
(2) Handle for changing the distance from the slice/slab group

- ✧ To change the thickness of a parallel sat region, use the mouse to drag the **Extent** handle *not facing* the slice and/or slab group in the desired direction.
- ✧ To change the distance of a parallel sat region, use the mouse to drag the **Extent** handle *facing* the slice and/or slab group in the desired direction.
- ✧ The complete sat region is shifted when you press the **Shift** key as well while changing the slice thickness of the sat region. In this way, you can quickly toggle back and forth between changing the thickness or the position.

Deleting graphic objects

You can delete or replace graphic objects that you no longer need for measurements.

Deleting a selected object



✧ Select the object you would like to delete.

✧ Select **Edit > Delete**.

- Or -

Click this button on the **Position Toolbar**.

The selected object is deleted and another object is selected.

☞ If deletion causes the slices or slabs to fall below the minimum number required by the protocol, the object is not deleted, but the number of slices or slabs is reduced to the minimum.

☞ If you delete the last slice group of a step, the entire step is deleted. When a Set-n-Go protocol contains only one step with only one slice group, you cannot delete this slice group.

Deleting all objects

✧ Select **Protocol > Delete All Graphics**.

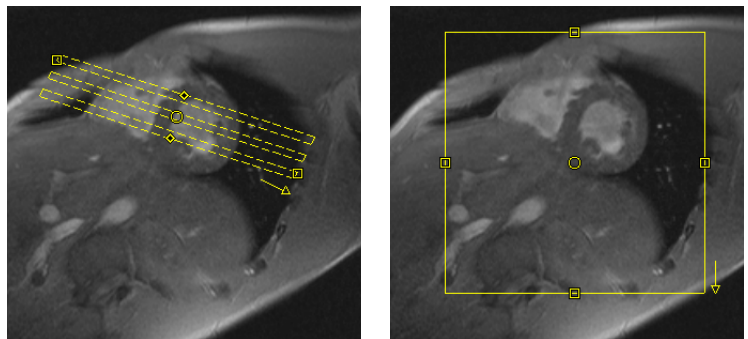
All objects of your protocol will be deleted except for the minimum number of slice/slab groups defined in the protocol. Only parallel saturation regions are retained.

Transferring the position of reference images

You can copy the position and orientation of a reference image to a selected slice or slab group.

- ✧ Select the required reference image in one of the reference image segments.
- ✧ Select the slice or slab group whose position you want to change.
- ✧ Select **Tools > Copy Image Position**.
 - Or -
 - Select **Copy Image Position** from the context menu.

The center of the group is moved into the plane of the selected reference image. The shift is perpendicular to the plane of the reference image. The orientation of the group is adjusted to that of the reference image.



Example for transferring the image position (before/after)

- ✧ If no group is selected, the position of the reference image is transferred to the first group of the protocol.



This is how you transfer the copied image position to additional slice and/or slab groups:

- ✧ Select the requested slice or slab group.
- ✧ Select **Tools > Paste Image Position**.

Nearest

Use the **Nearest** function to find the reference image best suited to the area of interest. This function is particularly useful in spectroscopy.

- ✧ Select the required reference image in the loaded reference image series.
- ✧ Select the graphic object.
- ✧ If the graphic object is to have the same orientation as the reference image, select **Tools > Copy Image Position**.
→ [Page C.5–77](#), *Transferring the position of reference images*
- ✧ Select **Scroll > Nearest**.

The reference images with planes nearest to the position of the selected graphic object are automatically displayed in the other two segments.



Apply slice settings

After you have correctly positioned the graphic objects, you may release the protocol for measurement.



✧ Click this button in the program control.

All graphic objects in the reference images are now removed.
The reference images remain displayed.

When you have finished graphical slice positioning and completed the protocol, it is loaded into the measurement system and run in the queue accordingly.

→ [Page C.6–1](#), *Performing a Routine Examination*

The next incomplete protocol of the program control is opened automatically for processing. You can now position the slices for this protocol and check the other parameter settings in the parameter cards.

→ [Page C.7–1](#), *Adapting the Measurement Parameters*



After completing exam-related preparations, you acquire a localizer and graphically planned the slice or slab groups for the examination in the reference images. You have modified the measurement parameters, as required.

→ [Page C.1–10](#), *Procedure for routine scanning*

Routine course of measurement

As soon as you complete modifying the first protocol and close it again, the system automatically starts with the measurement of this protocol. Subsequently the next protocol is automatically measured, if it is suitable for this task.

Utilizing measurement times

During the measurement, you can edit the program steps in the program control that have not been measured so far. You can complete the parameters of pending protocols and plan the examination sequence.

→ [Page C.6–25](#), *Editing the queue during measurements in progress*

You can also use this time to view the images of a previously measured series, evaluate them, comment on them or film them.

Stopping an examination

If the measurement procedure is interrupted (e.g., by patient movement) or you determine that all necessary images have been measured, you can end the measurement at any time. This does not affect the remaining program sequence.



Repeating measurements	Protocols that have been interrupted can be repeated at any time. It is your decision to repeat measuring the last protocol or an entire section of the queue.
Manually starting measurements	Breathhold measurements to prevent motion artifacts are started manually. In this case, the protocol is loaded automatically into the MR system, however, it waits until you provide a manual start command. For kinematic examinations, you can run protocols any number of times. For this purpose, you start a protocol manually and repeat it until all required measurements have been performed.
Contrast agent examinations	If you want to run an examination with contrast agent, plan a number of pauses in the measurement program for contrast examination. → Page C.16–1, Performing Contrast Agent Measurements Alternatively, you can perform contrast medium examinations using multiple measurements. → Page C.6–17, Measuring a protocol multiple times
Measurements with patient instructions	Before and after measurements, you can automatically add patient instructions in different languages via protocols or manually play back voice-overs. → Chapter C.12, Patient Instructions

Automatic positioning suggestion

If you have a license for **AutoAlign Head** or **AutoAlign Spine**, the MR system performs automatic and reproducible slice positioning for head or spine examinations independent of the patient's position.

→ [Page C.19–1](#), *Automatic Positioning Suggestion*

Jobs

If you have the respective license and your MR system is connected to a HIS (hospital information system), you can get jobs for your MR system from the HIS via the network. When you register a patient, a reference to the job is created in the form of steps that appear as a range marker in the queue. All services rendered are automatically entered in the job. The information of the procedure step is now available for further processing in the hospital.

→ [Page C.3–2](#), *Study Split*

→ **MPPS** in the *syngo* Operator Manual

Conflict management

Before and after measurements, inconsistencies may occur between the measurement program and the actual settings at the MR system. The MR system detects this type of inconsistencies and supports their removal by providing you with messages.

Displaying the queue in the program control

The program control allows you to trace and modify the course of an examination including all measurements. Program steps are shown in color depending on their processing status.



- (1) Completed measurement
- (2) Completed Set-n-Go measurement
- (3) Measurement in progress
- (4) Open protocol
- (5) Element not yet executed in queue: Contrast agent pause
- (6) Element not yet executed in queue: Protocol
- (7) Series icon
- (8) Display of the measurement time of the protocol
- (9) Display of the remaining overall measurement time

Sequence of a Set-n-Go protocol

When you are measuring a Set-n-Go protocol, the completed steps are displayed separately in the queue.



Set-n-Go protocol during measurement in progress: Two steps are completed

Brief information

When you leave the mouse pointer on the name of a program step, the following information is displayed:

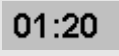
- ☐ The full name of the program step
- ☐ A brief comment from the properties of the program step

Icons to identify the protocols

Icons before or after the protocol names identify additional properties of the program steps.

Icon	Comment
 Construction worker	At this particular stage, the protocol cannot be started automatically (cannot be executed). The parameters (usually positioning of the slices to be measured) have to be completed → Page C.5–1, Positioning Slices
 Flagman	The protocol has to be started manually → Page C.6–15, Manually starting protocols
 Flagman (more than one flag)	The protocol can manually be started and measured multiple times in sequence. → Page C.6–17, Measuring a protocol multiple times
 Set-n-Go protocol	The protocol is a Set-n-Go protocol or a step within a Set-n-Go protocol.
 Set-n-Go protocol with flagman	The Set-n-Go protocol has to be started manually. Depending on the option selected, that is, Single measurement or Repeated measurement under the Wait for user to start protocol properties, the pause dialog appears prior to the first or all steps of a Set-n-Go protocol.
 Flash	The measurement was cancelled or skipped → Page C.6–17, Interrupting the measurement
 Image reconstruction system	Image reconstruction is started automatically after the protocol is measured.
 Syringe	Contrast agent pause → Page C.16–11, Inserting and planning contrast agent pauses
 Loudspeaker	Before or after measuring a protocol, a patient instruction is automatically inserted
 Voice output is missing	The protocol is linked to an automatic patient instruction which does not have a voice output. The symbol is shown following a corresponding system message → Page C.6–29, Patient instructions



Icon		Comment
	Copy reference	The protocol is the target of a copy reference. The number shows the source of the copy reference
	Defective copy reference	The protocol is the target of a copy reference which is ineffective due to an error
	Measurement time	Estimating the measurement time of the program step Protocols including the measurement time required are still waiting for editing and measurement. The measurement time displayed next to a multiple loading protocol indicates only the duration of a single repetition

Series icons in the program step

Once program steps are processed and the images have been calculated, the series icon is displayed to the right of the protocol name in the program control. The icon shows the status or event of the program step to be processed:

	Start of image reconstruction
	A number of images of the first series were reconstructed
	During the measurement, only one series was generated. All images of this series were fully reconstructed
	During the measurement, several series were generated. Only part of the images were reconstructed
	During the measurement, several series were generated. All series were completely reconstructed → Page C.4–8, Loading a series from the program control

Information in the status bar

The status bar shows messages during the course of the measurement.

- ☐ In the left part: Information regarding the current status of the program control
- ☐ In the center part: Results and error messages
- ☐ In the right part: Date and system time

Terminating a measurement due to system error

If an error occurs in the MR system during a measurement, the measurement is stopped automatically and identified as terminated.

	Terminating image reconstruction
	Terminating reconstruction via the MR scanner

Range markers

Within the queue, range markers provide information about different properties that are valid across multiple protocols.

	Image comment → Page C.15–10, Editing image comments
	Body region examined → Page C.15–2, Editing the instruction range Body Part Examined
	Work step → Page C.15–5, Editing the instruction range Procedure step
	Work step completed
	Work step canceled

Measurement process for routine examinations

For the most part, routine examinations run automatically.
→ [Page C.7–15, Applying measurement parameters](#)

Running complete protocols

The measurement of complete protocols is automatically started in the program control.

The images of the protocol are reconstructed in parallel to the measurement progress. After image reconstruction has been completed, the series icon is displayed. If the following protocol can be executed (construction worker icon is missing), it is subsequently started.

Completing and running a protocol with a construction worker icon

It is not possible to automatically start a protocol when the parameter settings are not complete or when the slice positions are not adjusted to the reference images.



This type of protocol is identified by a construction worker icon in the queue. As soon as an incomplete protocol takes its turn in the queue, it is opened automatically.

- ✧ Position the slices or slabs.
- ✧ Change the parameters as required on the parameter cards.
- ✧ Click **Apply**.

Apply

The protocol is closed and the measurement is started.

Manually set the protocol status to complete

Measurement protocols appearing with the construction worker icon can be considered operable (complete) without testing.

- ✧ Select the protocol from the queue.
- ✧ Select **Complete** from the context menu.

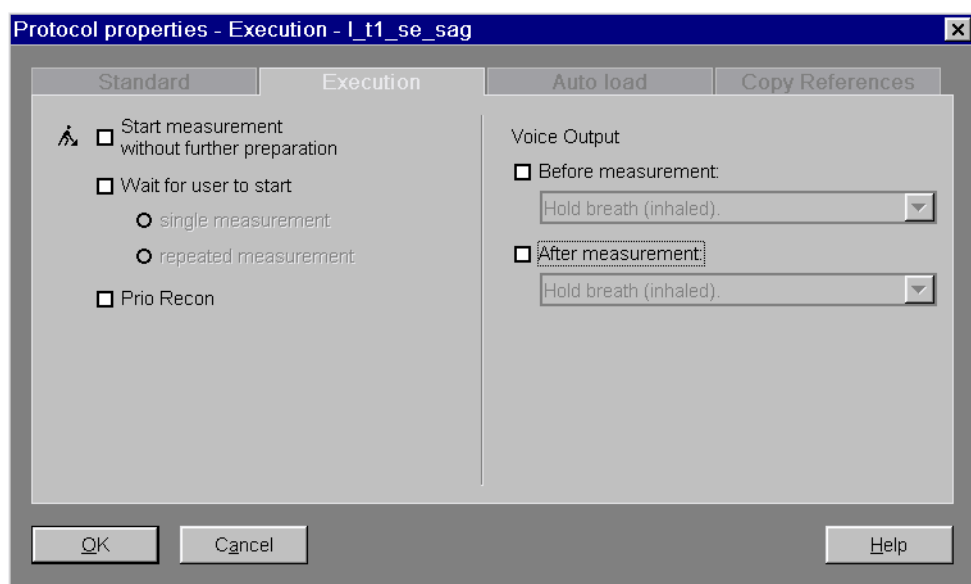
The protocol may be started immediately.

Or

- ✧ Select **Edit > Properties**.

The **Protocol properties** dialog window opens.

- ✧ Open the **Execution** subtask card.



☒ Start measurement without further preparation

✧ Select the **Start measurement without further preparation** checkbox.

✧ Click **OK**.

The protocol may be started immediately.

✧ If you do not want to automatically start an executable protocol, deselect the checkbox. The measurement process stops, when it is the protocol's turn in the measurement queue.



Set a protocol without checks to complete so it will start immediately. The table may move immediately to the table position entered in the protocol (if the table is not already there).

→ [Page C.6–42](#), *Inconsistencies resulting from changed table positions*

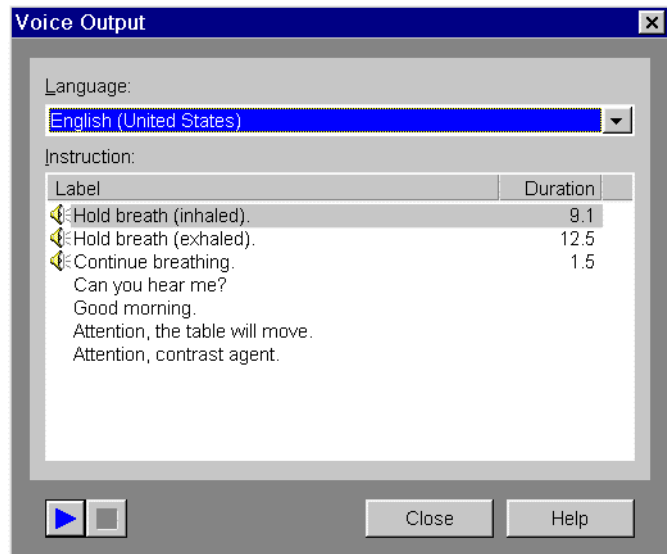
Manually playing back voice outputs

Via voice output, the patient can be provided with instructions in the examination room.

Playing back an existing voice output

✧ Select **Queue > Voice output > Play...**

The **Voice Output** dialog window opens.



✧ Select the language from the selection list.

✧ Select the required patient instruction.



✧ Click **Play**.

The selected voice output will be played back.

Recording patient instructions

If the required patient instruction is not on the list of voice outputs, you can record it yourself.

→ [Page C.12–9, Recording and editing voice outputs](#)

✧ Voice outputs can be recorded or edited only if no patient is currently registered. Close the registered patient one more time.

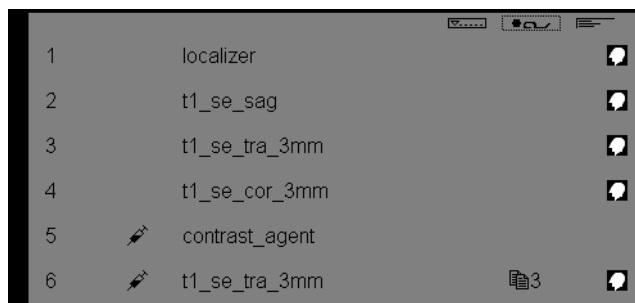
→ [Page C.6–15, Closing the patient](#)

Ending a routine examination

An examination is complete after all program steps in the program control have been processed and all images have been reconstructed.

All protocols completed

Now, all program steps show the series icon.



The examination is ended if you do not want to perform additional measurements or repeat protocols.

Closing the examination

As soon as the patient table is in the Home position (fully outside the magnet) and lowered, the current examination is completed.

→ [Chapter C.11](#), *Controlling the Patient Table and System Manual*

Closing the Work order

You can close the work order after all measurements of the examination have been completed, all planned post-processing has been made and the series have been copied to a film sheet and printed out, as required.

Closing the patient

✧ Select **Patient > Close Patient**.

- Or -

Select **Close Patient** from the patient context menu.

The queue is completely removed from the program control.

Manually starting protocols

Protocols that have to be started manually are identified in the queue by a flagman icon.

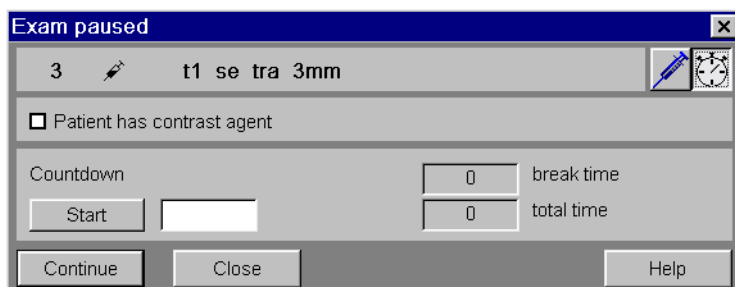
2		t1_se_sag	03:54
3		t1_se_tra_3mm	03:51

✧ You are able to determine the way a protocol is to be started in the **Execution** subtask card under the **Protocol properties** dialog window.

-
- ! Multiple breathhold measurements are always started from the **Inline Display**.
→ [Page C.18–16](#), *Starting multiple breathhold measurements*
-

Measuring protocols once

A protocol that you start manually and measure once, is automatically transferred to the MR system. After completion of adjustments, the program control waits for your start command. The **Exam paused** dialog window is displayed.



Starting the measurement



- ✧ To start the measurement immediately, close the **Exam paused** dialog window by clicking **Continue**.

Or



- ✧ Close the dialog window with **Close** and click **Continue** in the program control.
- ✧ As an alternative, you can also start the measurement with **F12**.

Interrupting the measurement



✧ If you do *not* want to measure the protocol loaded, click **Skip**.



The dialog window closes. The protocol is not measured. The next program step is loaded.

Measuring a protocol multiple times



Some protocols have to be started manually several times (e.g., a contrast agent examination of the liver using breathhold technique). A series of the region of interest is acquired at regular intervals to document the progress of contrast agent uptake. The protocol is repeated using a breathhold technique until all necessary data have been acquired.

As soon as the program control comes across this protocol while processing program steps, the **Exam paused** dialog window opens.

✧ Another application for multiple measurements includes kinematic examinations (e.g., when examining a joint).

Starting the protocol



✧ Click **Continue**.

- Or -

Press **F12**.

The **Exam paused** dialog window closes.

Starting the next measurement

Continue

The **Exam paused** dialog window opens again when the system is ready for the next measurement.

✧ The protocol parameters and the slice positioning cannot be changed.

✧ Start the protocol for the second time.

The same protocol is inserted for a second time in the wait queue and selected for measurement.

1		localizer	
2.1		t1_se_sag	
2.2		t1_se_sag	
2		t1_se_sag	
3		t2_tse_tra	02:54
4		contrast_agent	
5		t1_se_sag	03:16

The processed protocols are numbered and inserted in front of the multiple protocol.

Canceling the measurement

Skip



✧ If you do *not* want to measure the multiple protocol loaded, click **Skip**.

The multiple protocol is ended. The dialog window closes.



Performing measurements with countdown

Contrast examinations often require a precisely timed measurement sequence, especially following contrast agent administration.

Exam paused

5

contrast agent

☒ Patient has contrast agent

Volume

0.00

ml

Agent name

Catalog...

Total dose

ml

Active ingredient

Dilution

Concentration

mg / ml

Countdown

0

break time

Start

0

total time

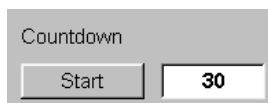
Continue

Close

Help

Three fields are available for planning the timing of the sequences for the following measurements:

Field name	Field type	Comment
Countdown	Input field	Length of pause in seconds, is counted down after start-up
Wait time	Display field	Shows the time beginning with the end of the previous measurement
Total time	Display field	Shows the time since the beginning of the first measurement of the current multiple measurement



✧ Enter the planned length of a pause in seconds in the **Count-down** field.

✧ Click **Start**.

Countdown starts and the **Start** button changes to **Stop**.

✧ If the planned pause is expired, countdown is continued in the minus area.

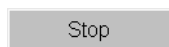
By clicking the **Stop** button, you can interrupt the time measurement in the **Countdown** field at any time and/or stop it after the end of the pause. This allows you to reenter a pause time.

Interrupting the measurement

✧ Select **Queue > Stop**.

- Or -

Click **Stop** in the program control.



The measurement will be stopped immediately. The protocol is identified as terminated.

✧ You can also interrupt a measurement in the examination room by using the STOP button on the control unit next to the magnet bore.

refer to the System Manual

Terminating of a measurement due to system error



If an error occurs in the MR system during a measurement, the measurement is stopped automatically and is also identified as terminated.

These icons appear in the status bar.
→ [Page C.6–9](#), *Terminating a measurement due to system error*

Terminating a measurement and starting the next measurement



Under special circumstances, a measurement can be terminated so the next measurement can be started immediately. (e.g., contrast agent inflow followed by starting a high-resolution protocol).

- ✧ To end the current measurement, click **Stop**.
- ✧ To start the next measurement, click **Continue**.
- ✧ You can also continue the measurement by using the START button at the magnet bore in the examination room.
refer to the System Manual

Skipping a program step



You are able to skip a program step opened for processing in the queue.

- ✧ Click **Skip**.
- Or -
- Go to the context menu and select **Skip Next Step** in the left margin of the program control.



The skipped program step is identified as terminated and highlighted in dark grey.

1		localizer	
2		t1_se_sag	
3		t1_se_tra_3mm	03:51

1		localizer	
2		t1_se_sag	
3		t1_se_tra_3mm	03:51

Program step 2 before and after being skipped

Repeating measurements

You can repeat already processed or terminated program steps individually or as entire sections. In individual cases it may be necessary to measure the entire queue again.

Repeating terminated protocols

- ✧ Select the canceled protocol from the queue.
 - ✧ Select **Queue > Rerun**.
 - Or -
 - Select **Rerun** from the context menu.
- The protocol is started automatically and processed.

Repeating processed protocols

- ✧ Select the protocol you would like to repeat.
 - ✧ Select **Queue > Append**.
 - Or -
 - Select **Append** from the context menu.
- The program step is copied and appended to the end of the queue as a protocol requiring processing.

Repeating multiple program steps

- ✧ Select the program step after which all protocols are to be repeated.
 - ✧ Select **Queue > Rerun**.
 - Or -
 - Select **Rerun** from the context menu.
- The selected protocol is automatically started and all subsequent, previously processed program steps are repeated.

- 
- ⇒ The selected protocol that was skipped the first time, may still be in the "incomplete" status. This protocol is automatically opened for completion.

Repeating the entire queue

To measure all previously completed protocols again, repeat the queue beginning with the first program step.

Defining a new center position

Resetting the table position to zero allows you to start a new series block. The current table position is defined as the center position.

Setting the table position to zero

- ✧ Select **Patient > Reset Table Position**.

- Or -

Select **Reset Table Position** from the context menu of the patient name.

All images planned in the graphic segments and all series icons in the program control are removed from protocols that have already been measured.

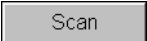
All protocols ready for measurement are reset to incomplete if indicated as such in the property dialog window. They have to be planned again.

Subsequently the queue is ready again.

Measuring an open protocol multiple times

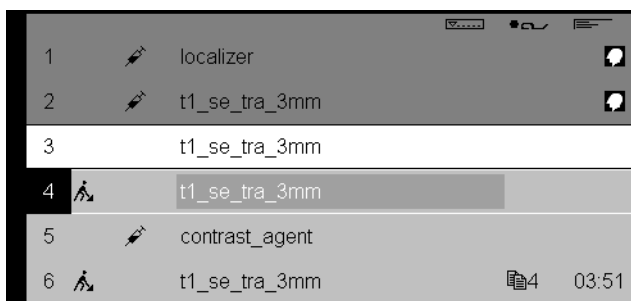
For different examinations where measurements are repeated by changing the protocol parameters slightly, e.g., cardio and angio applications, measurements can be performed with the protocol open. This saves you from closing the protocols and opening them again after each change in parameters.

- ✧ Change your parameters as required.
- ✧ Select **Queue > Scan opened protocol**.
- Or -
Click **Scan** in the program control.



A copy of the open protocol is generated and inserted in the queue below the open protocol.

The protocol is complete and ready to run immediately. The protocol will be performed as soon as it reaches its turn in the queue.



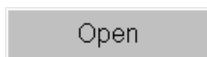
Editing the queue during measurements in progress

You can reduce the duration of the overall examination by effectively using the measurement time for post-processing and preparing the measurements.

Completing protocols

While the first protocols are being measured, you can check and complete the parameters for the subsequent protocols.

Opening a protocol



✧ Double-click the required protocol in the queue.

- Or -

Select the required protocol in the queue and click **Open**.

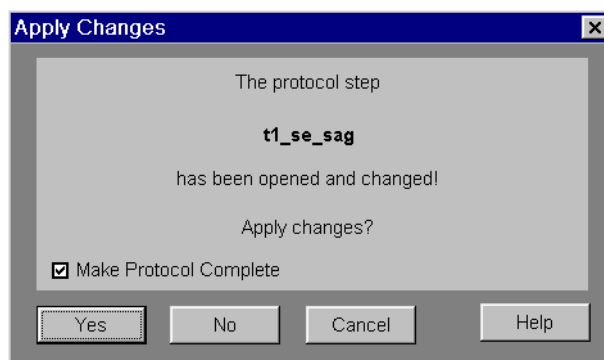
The measurement parameters of the protocol are shown in the parameter card stack.

⇒ The **Positioning mode** of the protocol to be opened has different effects on the reference images displayed and the table position of the protocol.

→ [Page C.4–26](#), *Dependencies between the table position, reference image, and protocol*

Opening the next protocol

When you open the next program step without having closed the previously opened one, the **Apply Changes** dialog window appears.



Processing of previously opened protocols is ended by using the **Apply Changes** dialog window. You are now able to edit the new protocol that opened.

- ✧ To assign the "complete" status, activate the **Make Protocol Complete** checkbox.
- ✧ If the checkbox is not activated, you have to open the protocol again prior to the measurement and switch it to executable.
- ✧ To save changes, click **Yes**.

Or

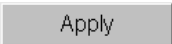
- ✧ To discard changes, click **No**.

The protocol is closed. The new protocol is opened.

Edit the protocol

- ✧ Add to and/or change the parameters of the open protocol according to requirements.

Saving the protocol



✧ To accept changes and close the protocol, click **Apply**.

It is now complete and executable.

Viewing parameters of previously measured protocols

Parameters of previously measured protocols can be viewed for the purpose of comparing them with pending program steps.

✧ Double-click the previously measured protocol.

- Or -

Select **Edit > View Protocol**.

The protocol parameters are displayed in a dialog window. They cannot be changed in the dialog window.

Applying parameters

There are two possibilities for accepting parameters of one protocol by another:

- (1) One-time transfer via copying
→ [Page C.7–15](#), *Copying measurement parameters*
- (2) Permanent transfer via copy reference
→ [Page C.8–12](#), *Setting a copy reference*

Copying parameters from an image in the GSP segment

You are able to select the copying of parameter settings in the GSP segment as well when it contains an image for the protocol which is still waiting in the queue.

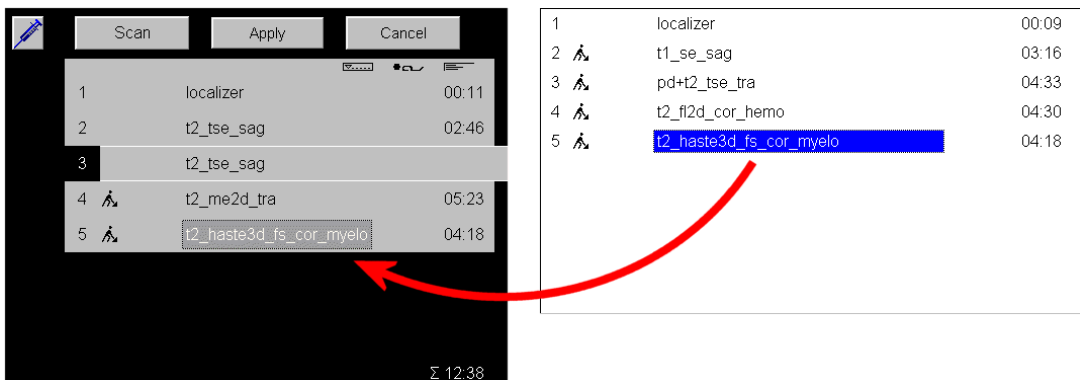
In this case open the dialog window to select the parameter group via **Copy Parameter** in the context menu of the respective GSP segment.

Inserting new program steps

You may add further program instructions to the queue in addition to the selected scan program.

Inserting a new protocol

- ✧ Select the required protocol from the **Program** parameter card or in the **Exam Explorer**.
→ [Page C.13–3, Selecting the Exam Explorer](#)
- ✧ Drag the protocol with the mouse to the required location in the section of the queue that still needs processing.
- ✧ When you drag the protocol into the empty part of the program control, the program step is appended at the end of the queue.





Alternatively, you can append the selected protocol to the end of the queue via the control elements listed in the following table. Drag the protocol with the mouse to the required location in the section of the queue that still needs processing.
→ [Page C.6–32, Changing the sequence of program steps](#)

In the Exam Explorer	From the Program parameter card
Main menu item Tools > Append to Queue	Main menu item Queue > Append
Context menu item Append to Queue	Context menu item Add
	Button <<

Patient instructions

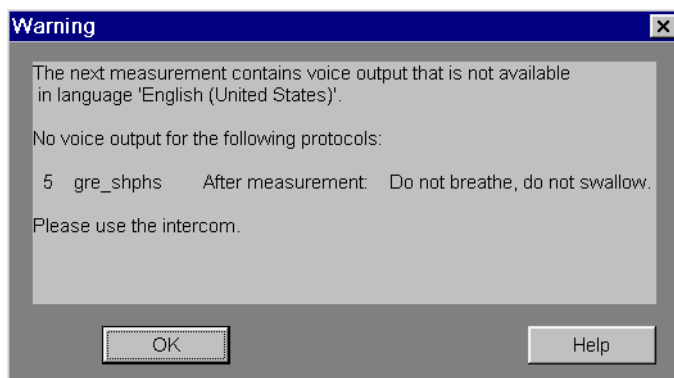
When adding protocols with voice output, the system checks whether the voice outputs are present or available in the language set.
→ [Page C.8–7, Playing back voice outputs](#)



If voice outputs are available, the protocol is marked in the queue with the loudspeaker icon.



If voice outputs are missing, the protocol is identified by using a crossed out loudspeaker and a warning.



- ✧ Click **OK** and announce your instructions over the intercom system.

Alternatively, you can set another language or record the voice output in the language required while keeping the patient file closed.

- [Page C.12–9, Recording and editing voice outputs](#)
- [Page C.12–6, Changing the properties of voice outputs](#)

- ☞ You will be informed about missing voice outputs when you start the protocol.

Inserting a new pause

- ✧ Select **Queue > New Pause...**
 - Or -
 - Select **New Pause...** in the context menu along the left margin of the program control.
- The **Pause Properties** dialog window is displayed.

The screenshot shows a dialog box titled "Pause properties - Standard - Pause". It has two tabs: "Standard" and "Contrast agent". The "Standard" tab is selected. Inside the dialog, there are four labeled fields: "Number" with the value "25", "Name" with the value "Pause", "Description" which is empty, and "Location" with the value "Queue". Below the "Description" field is a horizontal scrollbar. At the bottom of the dialog are three buttons: "OK", "Cancel", and "Help".

✧ Enter a **Name** in the respective field and, if needed, a **Description**.

✧ Click **OK**.

The new pause is appended at the end of the queue.

You can now shift the pause to the desired location.

→ [Page C.6–32, Changing the sequence of program steps](#)

✧ When entering a contrast agent pause, go to the **Contrast agent** subtask card and define the contrast agent administration.

→ [Page C.16–11, Inserting and planning contrast agent pauses](#)

Generating a protocol from the images acquired (Phoenix)

You can reconstruct a protocol from previously measured images or series and insert it into the queue.

→ [Page C.13–42](#), *Generating protocols from a series or an image (Phoenix)*

Changing the sequence of program steps

If you do not agree with the sequence order for a routine measurement program for the current examination, you are able to change it.

⇒ You can only modify the order in the part of the queue that has not been measured so far.

Moving a program step

✧ Drag the program step into the desired position using the mouse.

Or

✧ Select the program step that you want to move.

✧ Select **Queue > Move up** or **Move down**.

- Or -

Select **Move up** or **Move down** in the context menu.

Cut and insert program step

- ✧ Select the program step.
- ✧ Select **Edit > Cut**.
 - Or -
 - Select **Cut** from the context menu.

The selected input is cut and transferred to intermediate storage.
- ✧ Select the program step *in front of* which you want to insert the cut program step.
- ✧ Select **Edit > Paste** from the main menu.
 - Or -
 - Press **Ctrl + V**.

If you did not select a program step, the program step will be appended at the end of the queue.

Copying and inserting the program step

- ✧ Drag the program step into the desired position using the mouse and keeping the **Ctrl** key pressed.
- Or
- ✧ Select the program step.
- ✧ Select **Edit > Copy**.
 - Or -
 - Select **Copy** from the context menu.

The selected input is copied and transferred to intermediate storage.
- ✧ Select the program step *in front of* which you want to insert the copied program step.
- ✧ Select **Edit > Paste** from the main menu.
 - Or -
 - Press **Ctrl + V**.

Deleting program steps

You can delete individual steps or all program steps in the queue.

Deleting individual program steps



- ✧ Highlight the program step and select **Edit > Delete**.
 - Or -
 - Use the mouse to drag the program step to the wastepaper basket.

Deleting all program steps

- ✧ Select **Queue > Clear all**.
 - Or -
 - Select **Clear all** from the context menu.
- ➡ You can only delete the entire queue if no measurement is running.

Changing protocol and pause properties

During the examination, you are able to adjust the properties of protocols and pauses not yet processed in the queue.

You are still able to display or change the following characteristics although the protocol is already open:

Standard protocol properties

- ☐ **Number** (for info only)
- ☐ **Name**
- ☐ **Description**
- ☐ **Path** (for info only)
- ☐ **Protocol coding** (for info only)

Execution protocol properties

- ☐ **Start measurement without further preparation**
- ☐ **Wait for user to start**
- ☐ **Prio Recon**
- ☐ **Voice output: Before measurement**
- ☐ **Voice output: After measurement**

Auto load protocol properties

- ☐ **Auto store images** (can be modified depending on protocol)
- ☐ **Load images to viewer**
- ☐ **Load images to stamp segments**
- ☐ **Load images to graphic segments**
- ☐ **Inline movie**
- ☐ **Auto open Inline Display**
- ☐ **Auto close Inline Display**
- ☐ **Disable Auto Transfer**
- ☐ **Inline Position Display**
- ☐ **AutoAlign Spine**

✧ Select the desired program step from the queue.

✧ Select **Edit > Properties**.

- Or -

Select **Properties...** from the context menu.

For a detailed description, please refer to

→ [Chapter C.8](#), *Protocol and Pause Properties*

Storing the queue as a new program

If you want to save a modified measurement program for subsequent examinations, you can save it as a new program.

✧ Select **Queue > Save as program**.

- Or -

Select **Save As Program...** from the context menu.

The **Exam Explorer** is opened and displays the **Save Program As [Customer Tree]** dialog window. Perform the required settings.

→ [Page C.13–50](#), *Saving the measurement program under a new name*

Numbering reconstructed images

Numbering is determined by the orientation and position of the images and performed according to slice groups. It is *independent* of the slice positioning, excitation sequence and number of concatenations in multi-slice measurements.

Numbering rules

- (1) **Primary order**
are divided into sagittal, coronal, and transverse groups according to their main orientation.
- (2) **Secondary order**
the slices are numbered according to their position in ascending or descending order.

- ⇒ You can determine the primary and secondary order. In their basic setting, the images are numbered according to their primary sequence **S - C - T** and their secondary sequence **R >> L, A >> P, F >> H**.

As an alternative, sagittal images can be sorted with the options **Med >> Lat** or **Lat >> Med** according to their distance from or to the isocenter. Again numbering is performed according to slice groups.



When you activated the **Inherit Image numbering** setting of the **Queue** menu, the settings for image numbering via protocols are inherited as soon as they are opened for editing.

However, this applies only to consecutive protocols with similar table positions.

Example

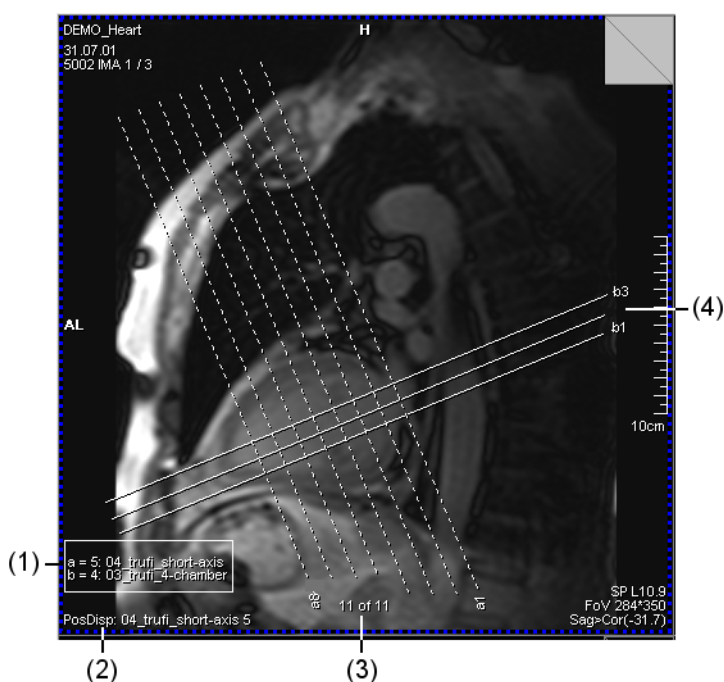


You are planning a measurement with 3 slice groups. You can move or rotate the slice groups without changing the numbers of the slice groups.

- ⇒ The number of the selected slice group is shown on the **Geometry** parameter card in the **Slice group** selection list. The number of the slice group does not affect the numbering of reconstructed images.

After the measurement, you may view the numbering of the reconstructed images in the **Position Display** of the **Viewing** task card, in the image area of the **Viewing** task card or in the **Patient Browser**.

The **Position Display** shows you the position and numbers of the reconstructed images.



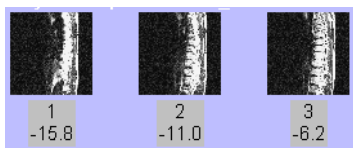
- (1) Allocation of letters to series
- (2) Name of the new position display series
- (3) Line 1: Number of actually visible slices in the current image section and number of total images in the data series
Line 2: Image commentary on the reference image
- (4) Image numbers



Configure the position overview so that all cut lines are labeled. In case slice cut lines are overlapping, they may be incorrectly allocated and their image numbers are not visible. Select a setting for cut line annotation so that *all* cut lines and image numbers are displayed.

→ [Page A.4–18](#), *Configuring the display of cut lines*

Images are displayed according to sort criteria in the **Patient Browser**. If you select the criterion **Image number** on the series level, the images will be displayed in the order of their slice positions.



By using the displayed image numbers, you are able to clearly allocate the images in the **Patient Browser** to the images in the **Position Display**.

Propagating numbering settings

You can propagate the image numbering set for consecutive protocols with similar table positions.

✧ For that purpose, activate the **Queue > Inherit Image Numbering**.

✧ Open a protocol with a similar table position.

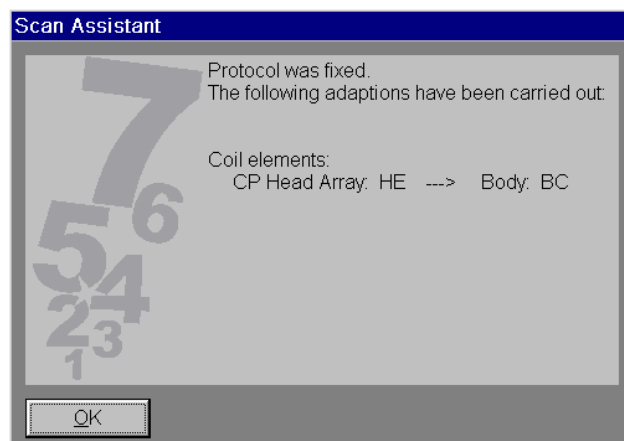
While you open it, the protocol automatically accepts the settings for image numbering of the previously processed and completed predecessor protocols.

Resolving conflicts and inconsistencies

Before and after measurements, inconsistencies may occur between the measurement program and the actual settings at the MR system.

Inconsistent coil configuration

The protocol requires coils different from the ones plugged into the patient table. A system message is displayed.

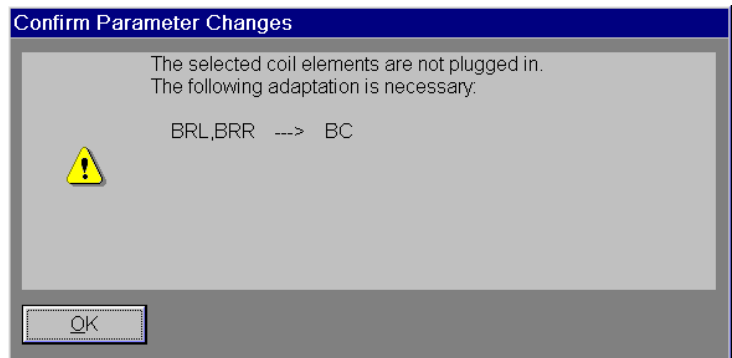


✧ Click **OK** to close the dialog window.

- ✧ Open the protocol.

Since the coils required for this protocol are not plugged in, the system automatically adjusts the coil configuration.

The message window **Confirm Parameter Changes** appears.



- ✧ Click **OK** to confirm the change.
- ✧ Connect the missing coils to the patient table.
- ✧ Open the **System - Coil** parameter card.
 - [Page P.1–91](#), Parameter Card *Geometry - Inversion*
- ✧ Select the coils you would like to use for measurement.
- ✧ To apply the settings, click **Apply**.



The protocol is closed and the measurement is started.

Inconsistencies resulting from changed table positions

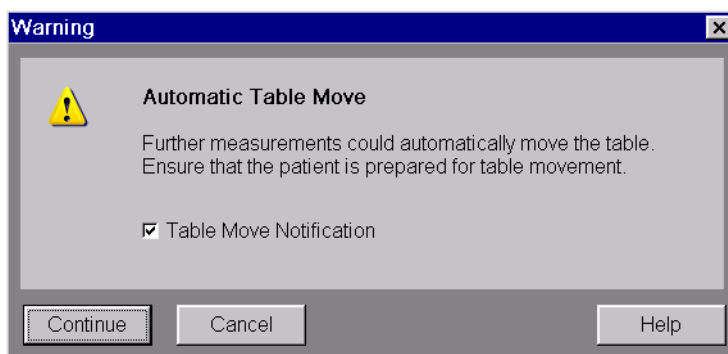
Prior to certain measurements, it may be necessary to move the patient table. Use the **Table Positioning** dialog window or the control unit at the magnet bore.

→ [Chapter C.11](#), *Controlling the Patient Table and System Manual*

When starting a measurement, the MR system automatically checks to see that the specifications for the table position in the following protocol match the current table position.

If this is not the case, a warning appears that the table may shift:

- ☐ Prior to the first measurement after patient registration.
- ☐ The current table position was set manually.
- ☐ The checkbox in the warning window is activated.



-
- ! Do not lower the table. Otherwise, the examination will be considered completed. You will not be able to continue with the measurement. Instead, you will have to start the entire examination again from the beginning.
-



Adapting the Measurement Parameters

Siemens Service installed routine measurement programs on your MR system. They have been configured so that only a few parameters have to be entered or changed prior to starting the measurement.

→ [Page C.1–10](#), *Procedure for routine scanning*

The parameter cards of the **Exam** task card contain all measurement parameters of a protocol sorted by main topics.

When working in the routine measurement mode, you usually need to check not more than the parameters of the **Routine** card for graphic slice positioning. With this card you can change the field of view (FoV) and other frequently adjusted settings.

In case of special questions and special anatomical conditions, you can adjust the measurement parameters of the **Contrast**, **Resolution** and **Geometry** cards.

⇒ The reference section contains detailed descriptions of the individual measurement parameters and information for settings.

→ [Page P.1–1](#), *Parameters of the Parameter Cards*

⇒ The parameter cards for spectroscopy measurements are described in the *Spectroscopy Operator Manual*.

Working in the parameter card stack

The input and orientation tools on and above the parameter cards help you to quickly and easily adapt a measurement protocol.

Protocol info line

As soon as you open a measurement protocol, you will see the protocol info line located directly above the parameter card stack. It includes the following information.

TA: 5:31	PM: REF	PAT: Off	Voxel size: 1.0×0.9×4.0 mm	Rel. SNR: 1.00	: tse
(1)	(2)	(3)	(4)	(5)	(6)

No.	Information	Comment
(1)	Measurement time	Total measurement time of the protocol with current settings
(2)	Type of protocol	Abbreviation for the positioning mode of the protocol → Page P.1–99 , <i>Parameter Positioning mode</i>
(3)	PAT acceleration factor	When PAT is switched on, the product from both acceleration factors PE and 3D is shown. If one or both factors exceed the max. recommended acceleration as provided by the system, the PAT acceleration factor is shown in the protocol info line as a button with red labeling. When you click this button, the Tim Assistant dialog window appears and resets the acceleration factors PE and 3D to the max. recommended acceleration. When PAT is switched off, OFF is shown.
(4)	Voxel size	Voxel size for current parameter settings. The voxel size is shown with one decimal point after the dot. When you leave the mouse pointer on the display, the exact voxel size is shown briefly.
(5)	Relative signal-to-noise ratio (SNR)	SNR for current parameter settings referenced to the relative SNR when opening the protocol. In this case you will see how much your parameter changes affected the signal-to-noise ratio either positively or negatively.
(6)	Sequence type	Abbreviation for the sequence type of the protocol opened

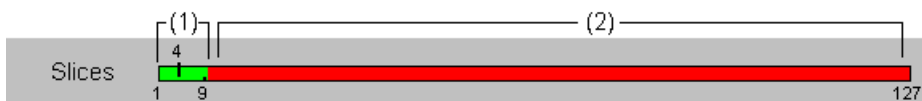
Changing parameter values and observing value ranges

Measurement settings can be changed in the parameter card stack by clicking options, selecting options from a list or changing values by using spin boxes.

However, certain values and settings cannot be freely defined or combined. The value ranges of a few parameters are limited by the technical possibilities of your MR system. Certain parameters depend on other measurement parameters. In this case, a change in one parameter value will require changes in other measurement parameters as well. Your MR system will point out the limits to you.

Soft limits and extended limits

As soon as you select or change a parameter via the input field, the range display including the acceptable limits is shown in the lower half of the parameter card.

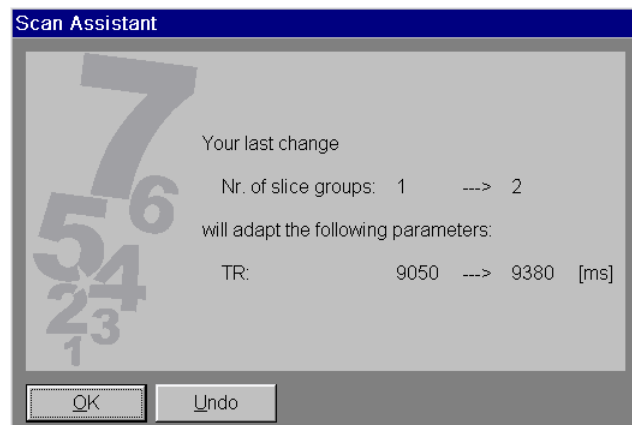


- (1) Soft limits
- (2) Extended limits

Within a narrower range, the so-called **Soft Limits** (green bar), you can set your parameter value as required without affecting other parameters.

Within an extended range, the so-called **Extended Limits** (red bar), your MR system will still allow parameter changes; however, these changes will require adjustments of other measurement parameters.

When you exceed the **Soft Limits**, the **Scan Assistant** dialog window appears.

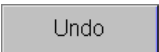


It informs you about the necessary adjustments to other measurement parameters and allows you to decide how to proceed:

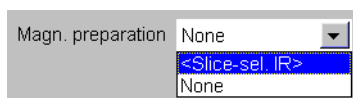
- ✧ Click **OK** to confirm your parameter change and to agree to the adjustment of other measurement parameters.

Or

- ✧ Click **Undo** to recall your parameter change.

A rectangular button with the text "Undo" in a standard font.

Restrictions when selecting options



Subsequent changes may be necessary also when checking or selecting options from lists.

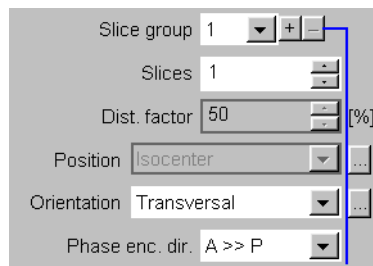
In selection lists, these entries can be identified by pointed brackets.

Once activated, checkboxes that involve additional parameter changes are shown in red.

The symbols on the +/- buttons are shown in red.

Displaying and editing parameter groups

Associated parameters are graphically combined into a group by using blue brackets.



A screenshot of a software interface showing parameter settings for a slice group. The settings are as follows:

Parameter	Value	Unit/Note
Slice group	1	
Slices	1	
Dist. factor	50	[%]
Position	Isocenter	
Orientation	Transversal	
Phase enc. dir.	A >> P	

Blue brackets on the right side of the interface group the 'Slices', 'Dist. factor', 'Position', 'Orientation', and 'Phase enc. dir.' parameters together.

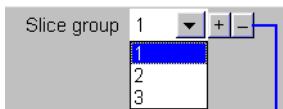
This example shows the parameter settings for a slice group.

✧ Open up the **Slice Group** selection list.

The list shows you how many slice groups your measurement protocol contains.

✧ To be able to edit the parameters of another slice group, select the number for that group.

The parameters of the group selected are shown on the parameter card.





Other input or output fields allow you to access parameter sets. In this example, the open measurement protocol has several echo times.

- ✧ Click the arrow buttons in the lower left corner to scroll the echo times for the protocol.

The echo time currently displayed is indicated by square brackets.

- ✧ Click the arrow keys on the spin box to change the value of the currently shown echo times.
- ✧ To apply the changes, press ENTER.
 - Or -
 - Change to another operating element.

Adding or deleting graphic objects

You generally use the graphic method to plan regions or volumes that will be stimulated when a protocol is run. You do this using the mouse and the graphic slice positioning tool bar. You are able to move the slice/slab groups or sat regions in the reference images and delete or add new ones.

→ [Page C.5–43, Graphic Slice Positioning \(GSP\)](#)

As an alternative, you can use the parameter cards to plan new objects or to delete objects no longer required.

New group or region



- ✧ Click this button right next to the selection list.

The new group or saturation region is inserted into the reference image selected. You can now adjust its properties (e.g., position or orientation) to the requirements at hand.

Deleting a group or region



- ✧ Select the slice/slab group or saturation region you want to delete from the selection lists.

- ✧ Click this button to remove the group from the measurement protocol.

- Or -

Select **Delete** in the context menu of graphic slice positioning.

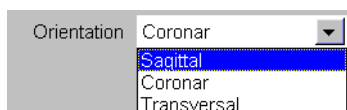
Adjusting the orientation and position of graphic objects

Identical to adding or deleting objects, you work with the mouse when positioning them in the reference images.

→ [Page C.5–43](#), *Graphic Slice Positioning (GSP)*

However, if this is not accurate enough for you, do not use the mouse, but rather enter the orientation and position numerically into the parameter cards.

Entering the slice orientation

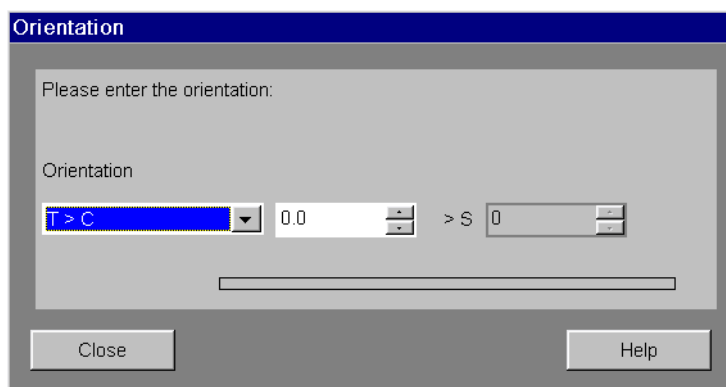


✧ Select the slice /slab group, or saturation region where you want to change the orientation.

✧ Go to the **Orientation** selection list and choose one of the three main orientations.

Or

✧ To open the **Orientation** dialog window, click this button next to the selection list.



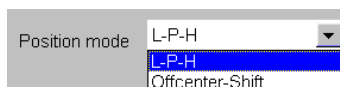
✧ Go to the selection list and choose the direction of tilt (single slice).

✧ Enter the requested tilt angle in the center field.

✧ For a double-oblique slice, enter the tilt angle for the third orientation in the right field.

→ [Page P.1–60](#), Parameter Card *Geometry - Common*

Entering the slice position

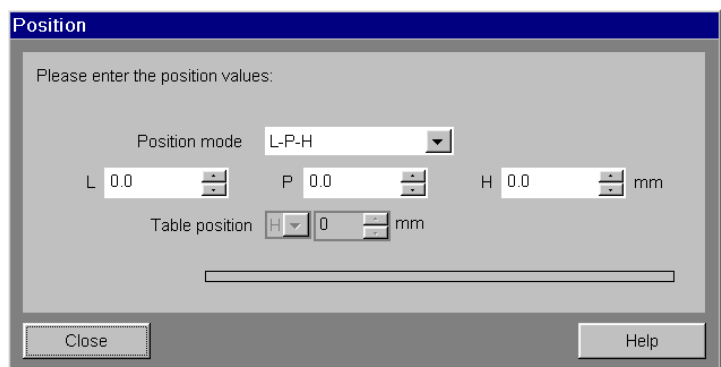


✧ Select the slice /slab group, or saturation region where you want to change the position.

✧ Next, select **Isocenter** from the **Position** list to position the slice or region in the magnet isocenter.

Or

✧ To open the **Position** dialog window, click this button next to the selection list.



✧ Under **Position mode** in the selection list, select the mode you want to use.

Depending on the position mode selected, the dialog window is displayed differently.

✧ During volume positioning (e.g., VOI, adjustment volume), you *always* work in **L-P-H** mode. The **Position mode** selection list is *not* available.

- ✧ Proceed by defining the offset of the center of the object from the magnet isocenter.
- ☐ In the **L-P-H** mode, enter the offset based on the *Whole Body Patient Coordinate System*:
 - L** – to the left
 - P** – to posterior
 - H** – towards the headEnter a negative value to move in the opposite direction (right, anterior, and feet).
- ☐ Enter the shift of the slice from the planned isocenter in the gradient direction in the **Offcenter-Shift** mode:
 - Phase** – in the phase-encoding direction
 - Read** – in the readout direction
 - Shift** – in slice-selection direction
- ✧ The selected position mode is maintained after the current positioning. When you open the **Position** dialog window the next time (even from another protocol), the position mode previously selected is still applied.
- ✧ If necessary, go to the **Table position** field and set the table position where the protocol will be measured.



Keyboard shortcuts

You can use both the mouse and the keyboard to select parameter cards from the stack or select and edit parameters.

The table below lists all actions in the parameter card stack that may be performed via the keyboard as well.

Action	Keyboard operation
Go to the next parameter card	Ctrl + right arrow key
Go to the previous parameter card	Ctrl + left arrow key
Go to the next parameter sub-card	Ctrl + Shift + right arrow key
Go to the previous sub-card	Ctrl + Shift + left arrow key
Go to the next parameter	Tab
Go to the previous parameter	Shift + Tab
Increase numerical value	Up arrow key
Decrease numerical value	Down arrow key
Accept numerical value	Enter or Tab (go to the next parameter)
Scroll to next parameter in the parameter set	Up arrow or right arrow key
Scroll to previous parameter in the parameter set	Down arrow key or left arrow key
Scroll to first parameter in the parameter set	Home
Scroll to last parameter in the parameter set	End
Activate/deactivate checkbox	Enter or space bar
Expand/collapse selection list	Enter
Select entry from selection list	Up/down/right/left arrow key or Home to first input or End to the last input



Action	Keyboard operation
Apply entry from selection list	Enter or Esc or Tab (go to the next parameter)
Select button (e.g., for coil selection)	Enter or space bar
Toggle checkbox or disable button selection	Press Esc key while Return or Space are pressed

Applying measurement parameters

After adapting the measurement parameters of the open protocol to your requirements, save the new settings and close the protocol.

- ✧ Select **Edit > Apply**.
- Or -
- Click **Apply**.



When the protocol reaches the top of the queue, it is loaded into the MR system. Depending on the start mode selected, the system starts the measurement automatically or waits for your start command.

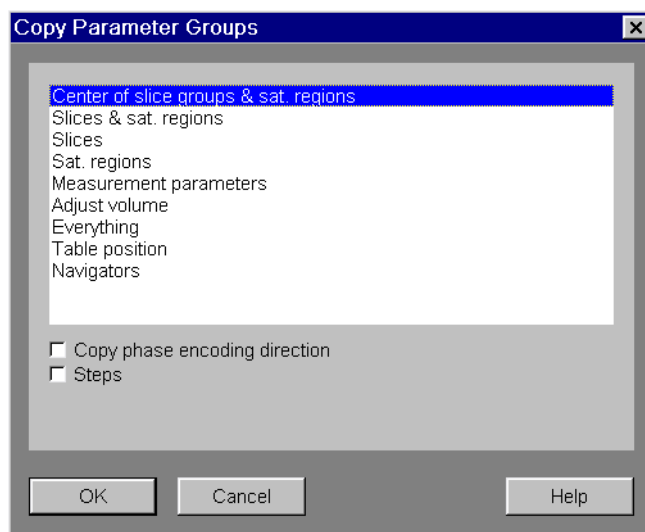
Copying measurement parameters

You can copy parameter groups from protocol to protocol. This saves time during parameter entry and completion of the protocols.

- ✧ Go to the part of the wait queue that has not been processed yet and open the protocol to which you want to copy the parameters (target protocol).

- ✧ Go to the wait queue and select the protocol from which you want to apply the parameters (source protocol).
- ✧ The source protocol can be behind the target protocol in the wait queue.
- ✧ Select **Queue > Copy parameter.**
 - Or -
 - Select **Copy Parameter** from the context menu.

The **Copy Parameter Groups** dialog window is displayed.





- ✧ Go to the list and select the parameter groups whose settings you would like to copy.

Depending on the parameter group selected, the following checkboxes are available:

☒ Copy phase encoding direction

- ✧ To copy the phase-encoding direction, activate the **Copy phase-encoding direction** checkbox.

- ✧ The phase-encoding direction of navigators does not respond, even if you activate the checkbox.

☒ Ignore measurements

- ✧ If you do not want to apply the number of measurements to the target protocol, activate the **Ignore measurements** checkbox.

☒ Ignore last step

- ✧ If you do not want to apply the setting of the parameter **Last Step** to the target protocol for Inline Composing, activate the **Ignore last step** checkbox.

☒ Steps

- ✧ If you want to copy all steps to the target protocol of a Set-n-Go protocol, activate the **Steps** checkbox.

- ✧ Click **OK**.

The parameters of the selected group are applied to the open protocol.

→ [Page C.8–12, Setting a copy reference](#)

Parameter groups

In the following table you'll find a list of parameters that can be copied via different parameter groups.

Parameter group	Parameters
Center of slice groups & Saturators	– Coil element selection (without matrix mode) – <i>Slice/slab groups</i> Position, Orientation and Rotation – <i>Sat. regions</i> Position, Orientation and Thickness – <i>Navigators</i> Position, Orientation, Rotation, Extension in phase-encoding and readout direction, Thickness.
Slices and sat. regions	– Coil element selection (without matrix mode) – Slice thickness – <i>Slice/slab groups</i> Number of slices/slab groups number of slices/slabs per group Position, orientation rotation, distance factor – <i>Sat. regions</i> Number, position, orientation and thickness of sat. regions, Number, thickness, and gap of parallel/tracking sat. regions sat. mode – <i>Navigators</i> Position, Orientation, Rotation, Extension in phase-encoding and readout direction, Thickness.

Parameter group	Parameters
Slices	– Coil element selection (without matrix mode) – Image numbering – Slice thickness – <i>Slice/slab groups</i> - Number of slices/slab groups - number of slices/slabs per group - Position, orientation rotation, distance factor – <i>Navigators</i> - Position, Orientation, Rotation, Extension in phase-encoding and readout direction, Thickness.
Saturation	– <i>Sat. regions</i> - Position, Orientation, Position and Thickness – <i>Parallel/tracking sat. regions</i> - Number, thickness and gap – Saturation mode

Parameter group	Parameters
Measurement parameters	– Coil element selection (without matrix mode) – Image numbering – <i>FoV / slice</i> FoV readout FoV phase basic resolution, phase resolution, slice resolution phase oversampling, slice oversampling slice thickness – Radial views, Radial interleaves, BLADE coverage – <i>Slice/slab groups</i> Number of slices/slab groups number of slices/slabs Position, orientation rotation, distance factor, phase-encoding direction – <i>Sat. regions</i> Number, position, orientation and thickness of sat. regions, Number, thickness, and gap of parallel/tracking sat. regions sat. mode – VOI for Spectroscopy – Interpolation resolution for CSI – <i>Navigators</i> Position, Orientation, Rotation, Extension in phase-encoding and readout direction, Thickness.



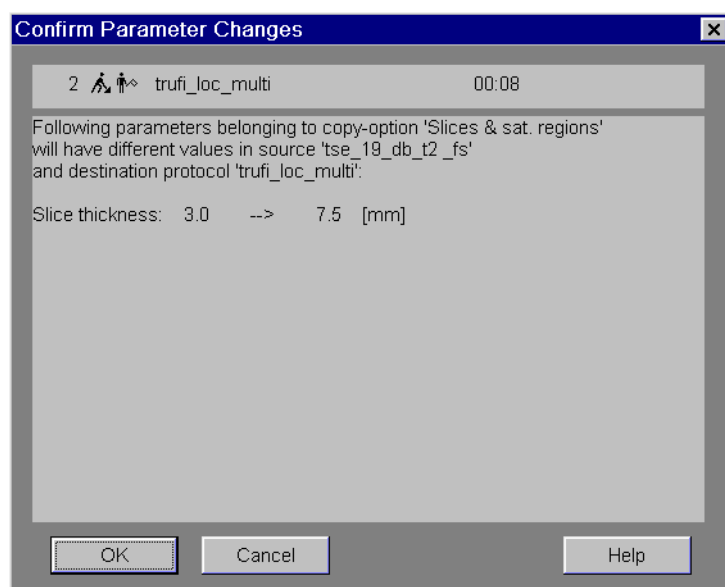
Parameter group	Parameters
Adjustment volume	– Shim mode – Position, orientation, rotation and extension of the adjustment volume
Everything	– All parameters including the sequence
Table position	– Table position of the protocol:
Navigators	– Position, orientation, rotation, extension in phase-encoding and readout directions, thickness

Resolving parameter conflicts

Due to the following, parameter conflicts may occur during copying:

- ☐ At least one of the parameters of the group to be copied was not or was only partially copied.
- ☐ At least one other parameter has to be adjusted to copy the parameters of the selected group (e.g., number of links).

The **Confirm Parameter Changes** dialog window is shown to resolve the parameter conflict.



- ✧ To confirm the required adjustments and resolve the conflict, click **OK**.

The new settings are applied.

Saving the queue

You can save all protocols in the job list as a new measurement program.

✧ Select **Queue > Save As Program**.

- Or -

Select **Save As Program** from the context menu of the program control.

The **Exam Explorer** is opened and displays the **Save Program As [Customer Tree]** dialog window. In this case, you assign your modified measurement program to an examination region as well as an examination and enter the new name.

→ [Page C.13–50](#), *Saving the measurement program under a new name*

Saving a single measurement protocol

Instead of the entire queue, you can also save individual protocols in the queue. This involves transferring the protocol to the Exam Explorer.

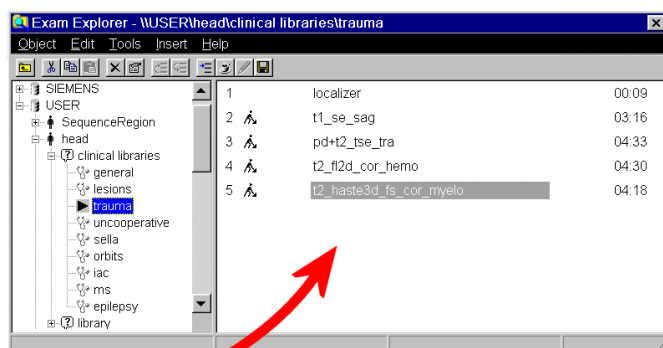
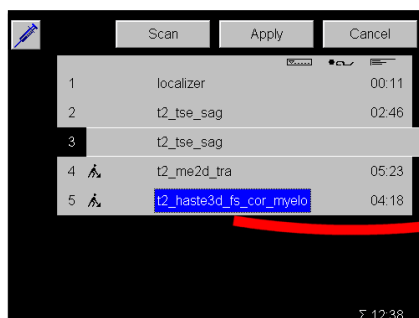
✧ Select the **Exam Explorer**.

→ [Page C.13–3](#), *Selecting the Exam Explorer*

- ✧ Open the data tree for the measurement program you want to save in the protocol in navigation area of the **Exam Explorer**.

→ [Page C.13–10, Searching for entries](#)

- ✧ You can only save the protocol in a measurement program of the customer tree (USER).
- ✧ Using the mouse, drag the protocol desired from the wait queue to the content area of the **Exam Explorer**.



- ✧ Select **Object > Save** in the **Exam Explorer**.

- Or -

Click this button on the tool bar.



The program is saved with the added protocol.

Protocol and Pause Properties

To optimize the course of examination, you change the properties of the protocols and pauses:

- ☐ Editing the properties for individual protocols
 - ☐ Simultaneously editing the properties for several protocols
 - ☐ Editing the properties for pauses
- ✧ The protocols and pauses that are still in the queue can be adjusted as late as during the examination.

Editing the properties for individual protocols

The properties of a protocol are changed in the **Protocol Properties** dialog window. You can select this dialog window from the **Exam Explorer** or from the queue.

Selecting via the Exam Explorer



- ✧ Select a protocol in the **Exam Explorer**.
- ✧ Select **Edit > Properties** in the **Exam Explorer**.
- Or -
- Click this button on the tool bar.

The **Protocol properties** dialog window opens.

Selecting via the queue

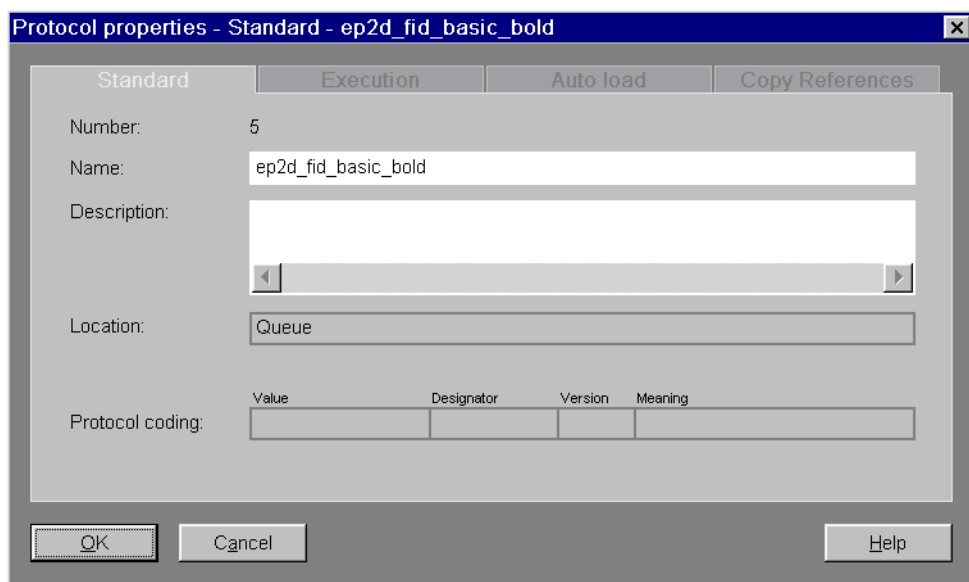
✧ Select the pending protocol from the queue.

✧ Select **Edit > Properties**.

- Or -

Select **Properties...** from the context menu.

The **Protocol properties** dialog window opens.



The image shows a screenshot of the 'Protocol properties' dialog window. The title bar reads 'Protocol properties - Standard - ep2d_fid_basic_bold'. The dialog has four tabs: 'Standard' (selected), 'Execution', 'Auto load', and 'Copy References'. The 'Standard' tab contains the following fields:

- Number:** 5
- Name:** ep2d_fid_basic_bold
- Description:** (empty text box with scrollbars)
- Location:** Queue
- Protocol coding:** A table with four columns: Value, Designator, Version, and Meaning. The table is currently empty.

At the bottom of the dialog are three buttons: 'OK', 'Cancel', and 'Help'.

Enter the protocol name and description

On the **Standard** subtask card of the **Protocol Properties** dialog window, enter the name and a description for the protocol.

- Give the protocol a unique name. The name of the protocol is used as a series description for the images acquired with this protocol and is shown, for example, in the **Patient Browser**.

Name:

- ✧ Provide the protocol with a meaningful name.

Description:

- ✧ If necessary, enter a description of the protocol.
- When you put the mouse pointer on a protocol entry in either the **Exam Explorer** or the program control, the protocol designation and description are shown temporarily.

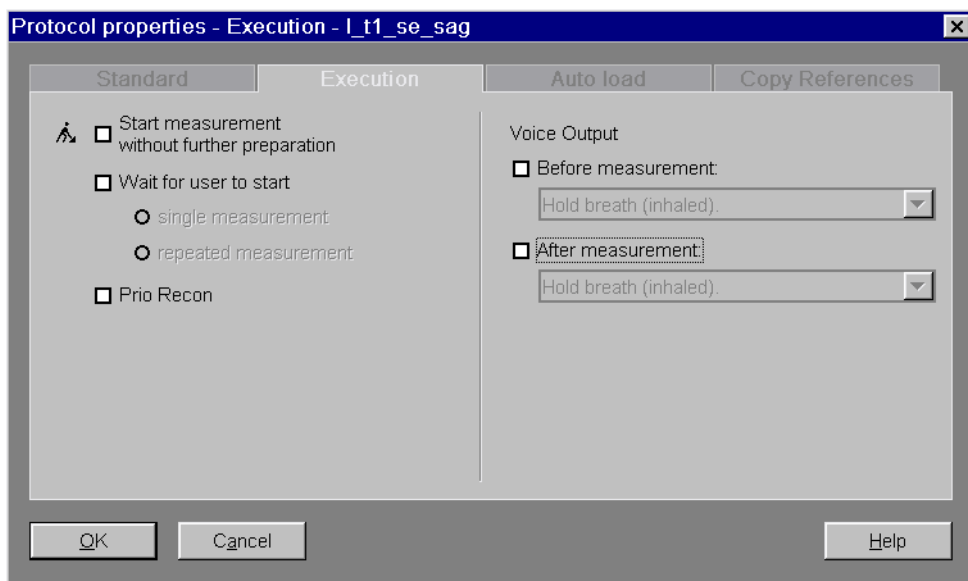
Defining start options

Localizer and protocols that do not require precise positioning can be started directly (automatically).

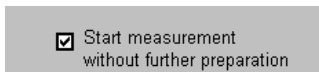
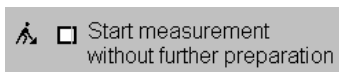
You can select the following start options on the **Execution** card:

- ☐ Starting a protocol without further preparation
- ☐ Starting a protocol manually, once or several times
- ☐ Performing image reconstruction with a higher priority
- ☐ Playing back voice output before/after the measurement

✧ Open the **Execution** subtask card.



Starting the measurement with/without preparation



- ✧ Deactivate the **Start measurement without further preparation** checkbox.

The protocol is provided with the construction worker icon.

Or

- ✧ Select the **Start measurement without further preparation** checkbox.

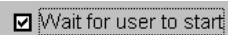
The construction worker icon is removed.

- ✧ Protocols that do not include the construction worker icon are started automatically as soon as it is their turn in the queue. Protocols including the construction worker symbol are "not executable", that is, they cannot be started automatically by the MR system.
- ✧ Localizers are usually started automatically without being checked.

Starting the measurement manually

For some examinations (e.g. protocols with breathhold commands), it is better to set manual start-up. In this case, the MR system stops after the adjustment. You can then give the breathhold command and start the measurement manually.

→ [Page C.6–15, Manually starting protocols](#)

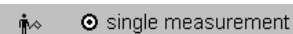
A checkbox with a checkmark inside, followed by the text "Wait for user to start".

✧ Select the **Wait for user to start** checkbox.

✧ If you have recorded an appropriate voice output on your system, you can perform this type of examination without manual start-up.

→ [Page C.8–7, Playing back voice outputs](#)

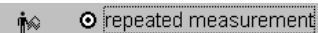
If you set manual start-up for the protocol, please decide whether the protocol should be measured once or several times.

An icon of a person with a single checkmark, followed by the text "single measurement".

✧ Select the **single measurement** option.

Protocols that allow for only one manual start-up are identified using this symbol in the measurement program.

Or

An icon of a person with a checkmark and a plus sign, followed by the text "repeated measurement".

✧ Select the **repeated measurement** option.

Protocols that allow for multiple loading and measurements are identified using this icon in the measurement program. Prior to each new measurement, a dialog window is displayed allowing you to start the protocol.

Prio Recon

☒ Prio Recon

- ✧ Activate the **Prio Recon** checkbox.

When the option is activated, the image is reconstructed at a higher priority, that is, reconstruction is performed parallel to the measurement and not appended to the back.

- ⇒ Since reconstruction of the images is performed parallel to the measurement, it is limited to a certain raw data volume. The next measurement is started only when all current raw data have been reconstructed.

Playing back voice outputs

Frequently occurring patient instructions are stored in the form of voice outputs.

You can use the **Execution** subtask card to ensure that a voice output is played prior to measuring the current protocol. Subsequently, the measurement is started automatically. After the end of the measurement, you can play a new voice output.
→ [Chapter C.12, Patient Instructions](#)

Voice Output
☒ Before measurement:
 Hold breath (inhaled).

- ✧ Activate the **Before measurement** checkbox.
- ✧ Select a suitable voice output from the selection list.

☒ After measurement:
 Continue breathing.

- ✧ Activate the **After measurement** checkbox.
- ✧ Select a suitable voice output from the selection list.

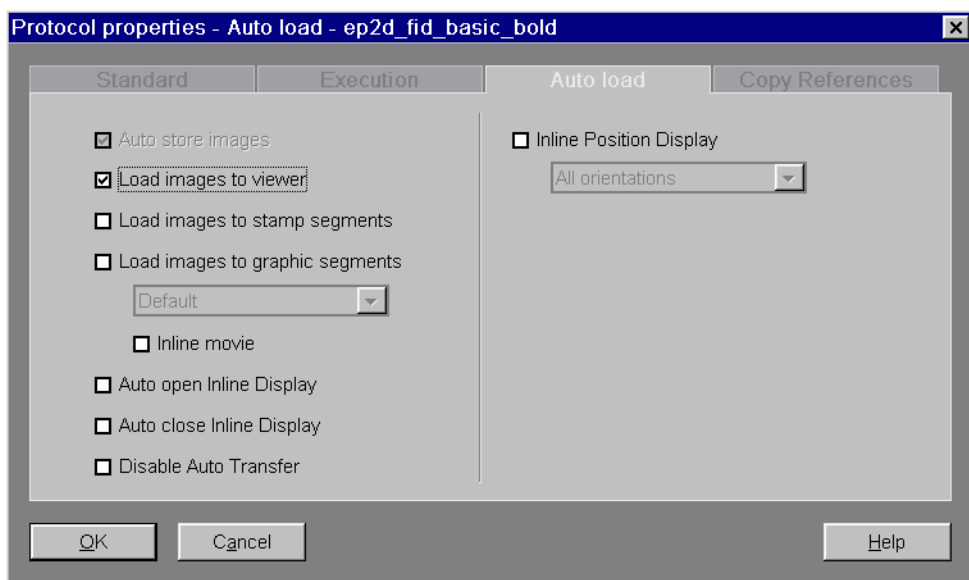


- ⇒ Protocols with voice outputs are marked with a loudspeaker icon in the measurement program.

Automatically storing and loading

Images and series generated during a measurement can be saved automatically and loaded into other applications.

✧ Open the **Auto load** subtask card.



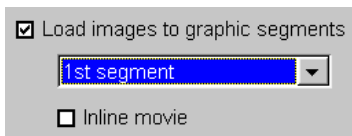
Checkbox/function	Comment
Auto store images	The images measured are stored automatically. This function cannot be deactivated for most protocols. In a few cases (e.g. for real-time protocols), you are able to switch off auto save and manually save individual images displayed in the Inline Display. → Chapter C.18, Inline Display
Load images to viewer	The series measured are automatically loaded into the Viewing task card. If you reconstruct more than 400 images for a protocol, automatic loading into Viewing is no longer possible. The images are loaded into the viewer according to the image numbers. You determine the image number in the protocol editor of the System - Miscellaneous parameter card. → Page P.1–98, System - Miscellaneous
Load images to stamp segments	The series measured are automatically loaded into the stamp segments of the Maestro layout of the Exam task card. → Page C.2–2, Maestro layout
Load images to graphic segments	The series measured are automatically loaded into a GSP segment of the Exam task card. Select the required GSP segment from the selection list. You can also activate or deactivate automatic start of movie display via the respective checkbox. → Page C.2–8, Movie display
Auto open Inline Display	The images can be viewed and checked immediately after reconstruction.
Auto close Inline Display	Automatically closes Inline Display at the end of the measurement.
Disable Auto Transfer	Prevents automatic DICOM transfer of images from the current protocol.



Checkbox/function	Comment
Inline Position Display	Creates a position display for every series measured. You can select the following orientations from the selection list: All orientations, Sagittal, Coronal or Transverse. If you activate the Load images to viewer checkbox, the position display of the series is automatically loaded into Viewing. → Page A.4–15, Activating Inline position display
AutoAlign Spine	Automatically activates the AutoAlign Spine algorithm for positioning the slices. → Chapter C.19, Automatic Positioning Suggestion



If you are performing a measurement and are simultaneously post-processing images, loading them automatically into **Viewing** may result in a mixed display of images from both processes. Please note that this affects the display only. The contents of the series are not mixed. To avoid this, deactivate the **Load images to Viewer** function.

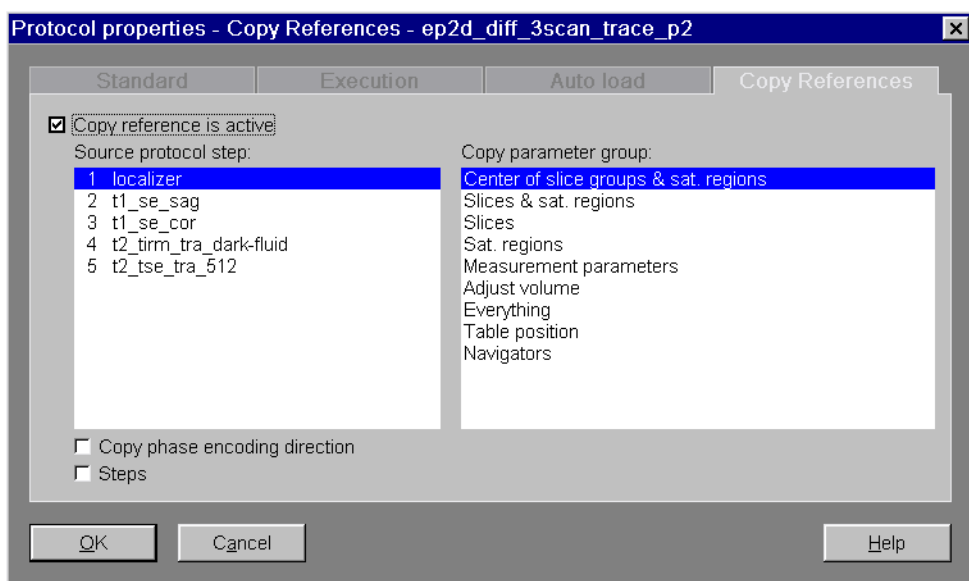


- ✧ To switch on (switch off) a function, activate (deactivate) the respective checkbox.
- ✧ When you activated the **Load images to graphic segment** checkbox, you also have to select the required setting in the segment selection list and the **Inline movie** checkbox.

Setting a copy reference

Using the copy reference, you are applying the permanent parameter settings from a source protocol to a target protocol.

- Information for applying the parameter settings once:
→ [Page C.7–15, Copying measurement parameters](#)
- Open the **Copy References** subtask card.



☒ Copy reference is active

- ✧ Activate the **Copy references is active** checkbox.

The list of source protocols provides you with all protocols located before the selected protocol in the measurement program and that are not already the target of the copy reference.

- ✧ Select the required protocol from the **Source protocol** list.
- ✧ Go to the **Parameter group** list and select the parameter settings you would like to apply from the source to the target protocol.
- ✧ You can only apply the settings of parameter groups, but not those of individual parameters.
→ [Page C.7–15, Copying measurement parameters](#)
- ✧ If you set the copy references from or to a Set-n-Go protocol, please observe the following information:
→ [Page C.9–18, Copying parameters](#)

Depending on the parameter group selected, the following checkboxes are available.

☒ Copy phase encoding direction

- ✧ To copy the phase-encoding direction, activate the **Copy phase encoding direction** checkbox.
- ✧ The phase encoding direction of navigators does not respond, even if you activate the checkbox.

☒ Ignore measurements

✧ If you do not want to apply the number of measurements to the target protocol, activate the **Ignore measurements** checkbox.

☒ Ignore last step

✧ If you do not want to apply the setting of the parameter **Last Step** to the target protocol for Inline Composing, activate the **Ignore last step** checkbox.

☒ Steps

✧ If you want to copy all steps to the target protocol of a Set-n-Go protocol, activate the **Steps** checkbox.



The target protocols of a copy reference are identified with the copy reference icon in the queue and the **Exam Explorer**.

The number after the icon informs you about the source protocol used for applying the parameter settings.

The designation of the applied parameter group is shown when you set the mouse pointer on the icon of the copy reference.

Updating a copy reference

To transfer defined parameters to a copy reference, update the copy reference in the queue.

✧ Select the target protocol with the reference icon.

✧ Select **Queue > Update Copy Reference**.

The copy reference will be updated and used in the measurement process.

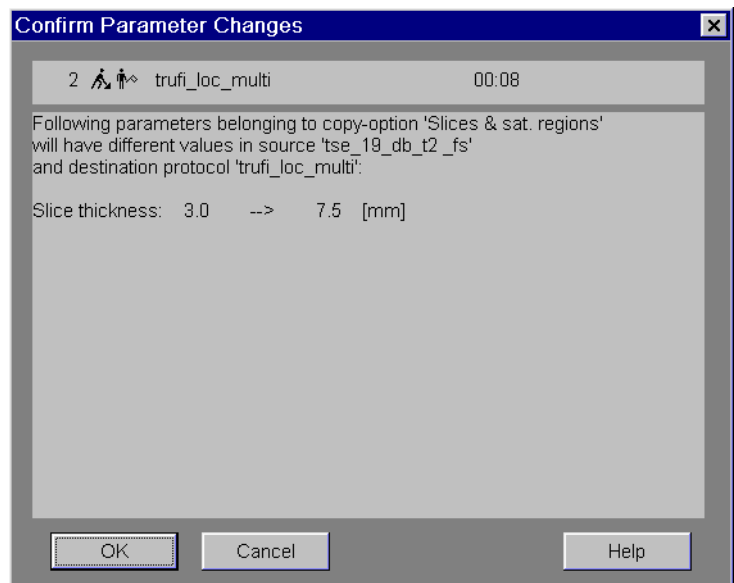
Conflicts may occur when updating parameters.

Resolving parameter conflicts

Due to the following, parameter conflicts may occur during copying:

- ☐ At least one of the parameters of the group to be copied was not or was only partially copied.
- ☐ At least one other parameter has to be adjusted to copy the parameters of the selected group (e.g. number of links).

The **Confirm Parameter Changes** dialog window is shown to resolve the parameter conflict.



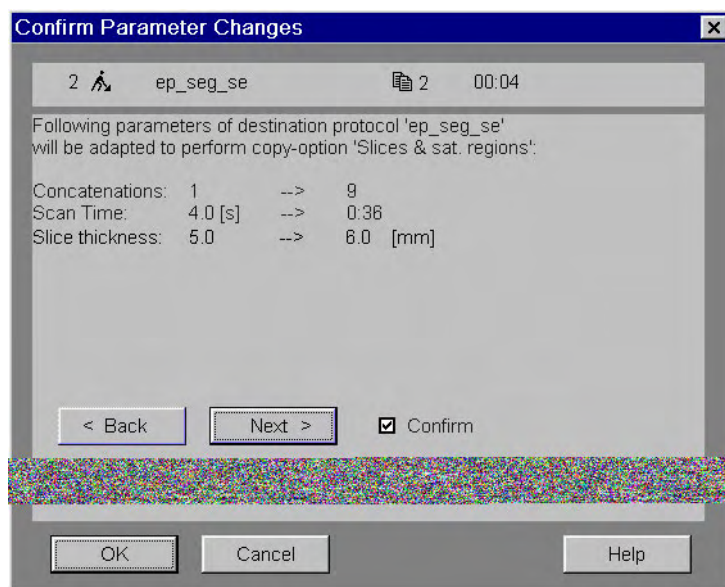
The window includes a proposal for resolving the conflict.

✧ To apply the proposed adjustments, click **OK**.

The new settings are applied and a valid copy reference is set.

Copy reference for multiple target protocols

Updating copy references may cause multiple conflicts when several target protocols are updated.



To apply changes to the target protocol displayed, activate the **Confirm** checkbox.

Or

✧ When you want to exclude this target protocol from the proposed parameter changes, deactivate the **Confirm** checkbox.

- ✧ Scroll with **Next >** and **< Back** to all protocols involved and apply the desired setting in the **Confirm** checkbox.
- ✧ To set the source protocol to complete, activate the **Make source protocol complete** checkbox.
- ✧ Click **OK**.

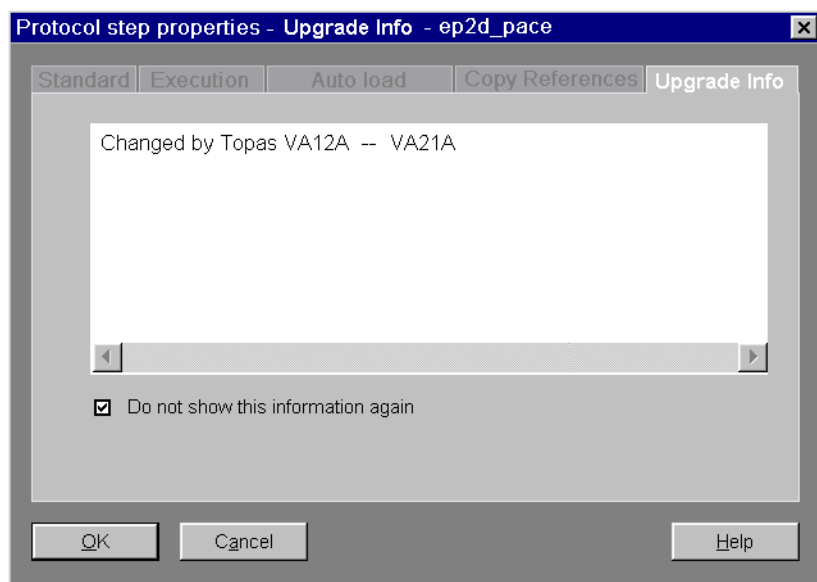
The target protocols intended for applying the parameters are updated and the dialog window is closed.

Displaying upgrade info

If a protocol has been modified by a software update, the **Protocol step properties** dialog window will include the **Upgrade Info** subtask card. It enables you to determine the day the changes were made as well as the changes.

Protocols including change information are underlined in the measurement program.

- ✧ To see the change information, open the **Upgrade Info** subtask card.



☒ Do not show this information again

- ✧ If you no longer need the change information, activate the **Do not show this information again** checkbox.

The underscoring of the name disappears. The **Upgrade Info** subtask card is no longer shown for this particular protocol.

Applying changed protocol properties

When you have completed editing the protocol properties for all subtask cards of the **Protocol properties** dialog window, apply the new settings.

✧ Click **OK**.

Your changes are applied and the dialog window closes.

Editing the properties for several protocols

If you want to set the properties of several protocols to a uniform setting, the Exam Explorer enables you to do this.

⇒ Before editing the protocol properties, you have to select the requested protocols with a search dialog.

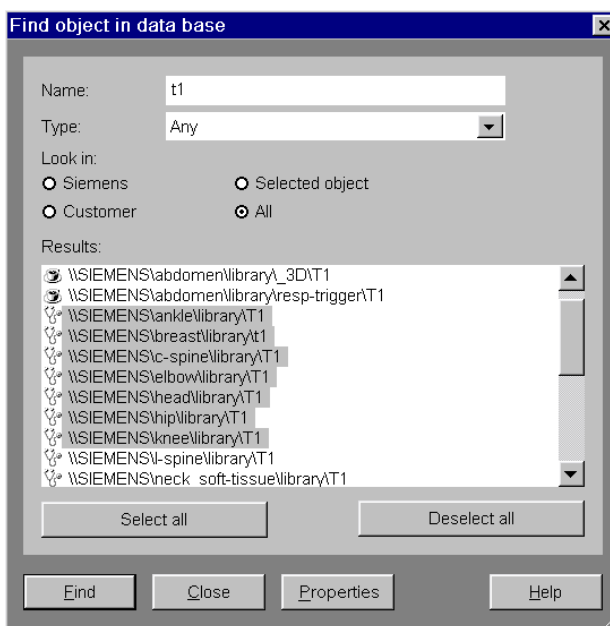
Selecting the search dialog

✧ Select **Tools > Search** in the **Exam Explorer**.
- Or -
Press key combination **Ctrl + F**.

The **Find object in database** dialog window appears.

- ✧ Perform the required settings for the search.
 - [Page C.13–10, Searching for entries.](#)

As a result you receive a list of the objects located including their path:



- ✧ To set common parameters in different protocols, it is helpful to limit the search result in the list to protocols. You can achieve this by selecting the object type **Protocol** for the search.

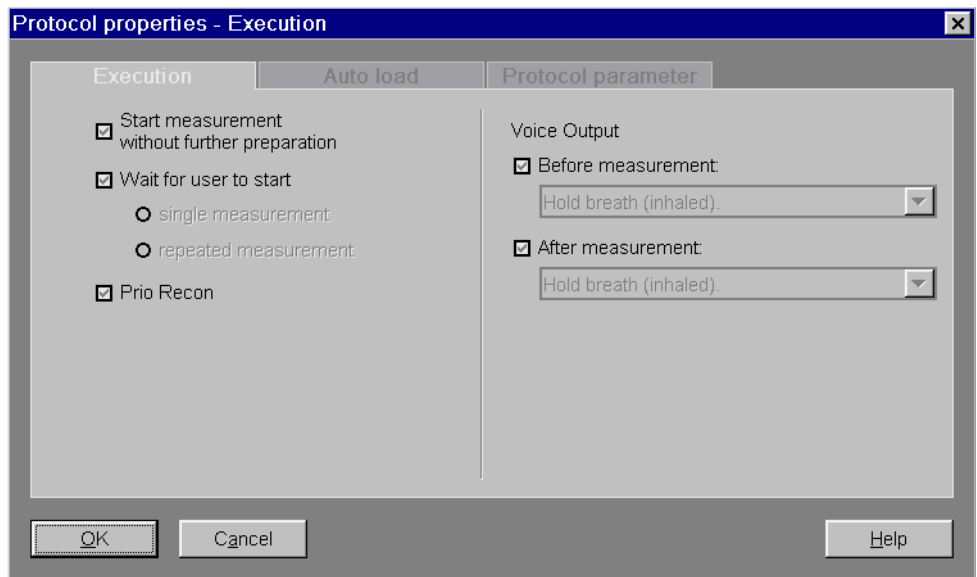
You can edit the properties together for all protocols marked in the list. For that purpose, you can mark or unmark single protocols.

✧ Keep the **Ctrl** key pressed and mark all requested protocols in the list.

Opening the properties dialog

✧ Click the **Properties** button in the **Find object in database** dialog window.

The **Protocol properties** dialog window opens.



You can set the same settings on the **Execute** and **Auto Load** subtask cards as you do for an individual protocol. However, the checkbox now shows three different statuses.

Checkbox**Description/status**
 Wait for user to start

A dimmed check mark indicates that this setting will not be changed for the selected protocols.

 Wait for user to start

When the checkbox is empty, the function is switched off for all selected protocols.

 Wait for user to start

When the checkbox is marked with a check mark, the function is switched on for all selected protocols.

Defining start options

You can set the following for all selected protocols on the **Execute** subtask card:

- ☐ Starting the measurement with/without preparation
- ☐ Starting scanning manually
- ☐ Setting the priority of image reconstruction (Prio Recon)
- ☐ Playing back voice outputs

✧ Activate or deactivate the requested functions.
→ [Page C.8–4, Defining start options](#)

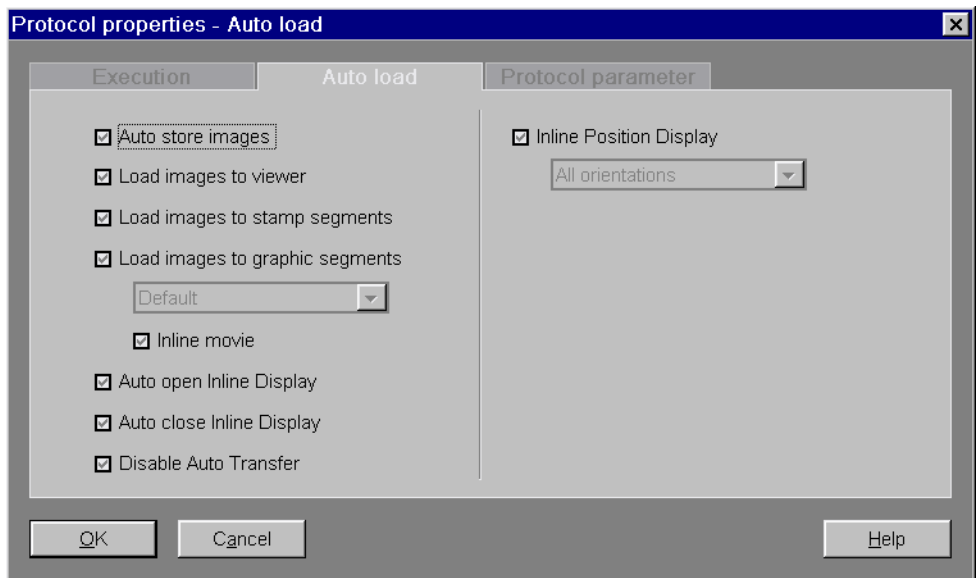
The setting chosen applies to all selected protocols.

Setting automatic saving/loading

✧ Open the **Auto load** subtask card.

You can set the following for all selected protocols on the **Auto Load** subtask card:

- ☐ Auto store images
- ☐ Load images to viewer
- ☐ Load images to stamp segments
- ☐ Load images to graphic segment
- ☐ Auto open Inline Display
- ☐ Auto close Inline Display
- ☐ Disable Auto Transfer
- ☐ Inline Position Display



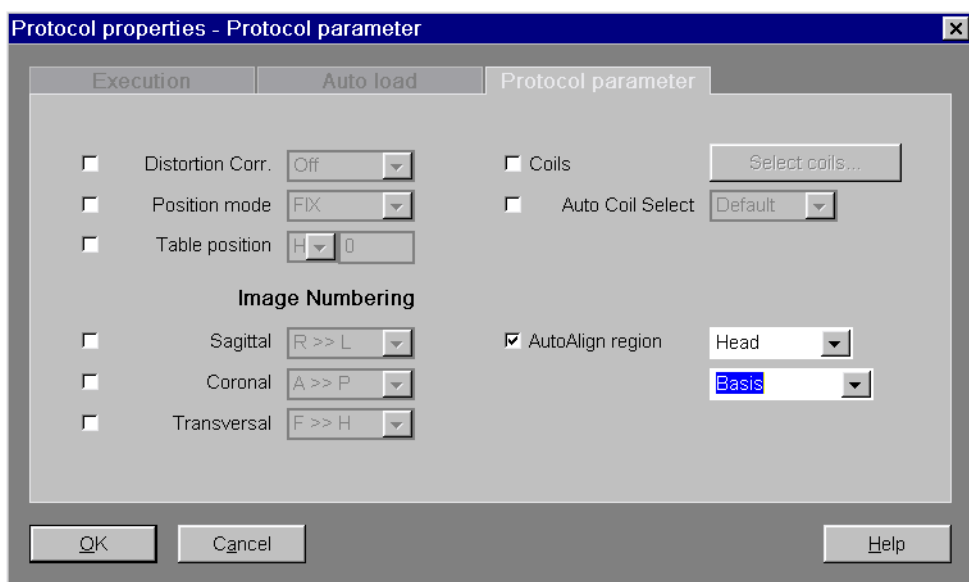
- ✧ Activate or deactivate the requested functions.
 - [Page C.8–8, Automatically storing and loading](#)

The setting chosen applies to all selected protocols.

Setting protocol parameters

Perform the following settings for the selected protocols on the **Protocol Parameter** subtask card.

- ✧ Open the **Protocol Parameter** subtask card.



The value to be set for the protocols selected is determined by activating the check mark or selecting a value from the selection list on this subtask card. The settings remain unchanged by deactivated the check mark.

Activating the setting

- ✧ Activate the checkbox prior to the requested function.
- ✧ Select the desired setting.
 - Or -
 - Select the requested virtual coils in the dialog following the **Coil** option.

Function	Fine adjustment, refer to
Distortion Corr.	→ Page P.1–57 , <i>Distortion Corr.</i>
Position mode	→ Page P.1–99 , <i>Positioning mode</i>
Table position	→ Page P.1–84 , <i>Table position</i>
Image numbering	→ Page P.1–100 , <i>Image numbering</i>
Sagittal	
Coronal	
Transversal	
Coils (virtual coils)	→ Page C.13–47 , <i>Selecting virtual coils</i>
AutoAlign	→ Page P.1–223 , <i>AutoAlign region</i>
Reference	→ Page P.1–223 , <i>AutoAlign reference</i>

For functions with activated checkboxes, the setting made is accepted by all selected protocols.

- ✧ Settings are not accepted when the checkbox is deactivated.

Editing pause properties

To change the properties of a pause, open the **Pause properties** dialog window via the **Exam Explorer** or the queue.

✧ You can only edit pause properties in the part of the queue that has not been measured so far.

Selecting via the Exam Explorer



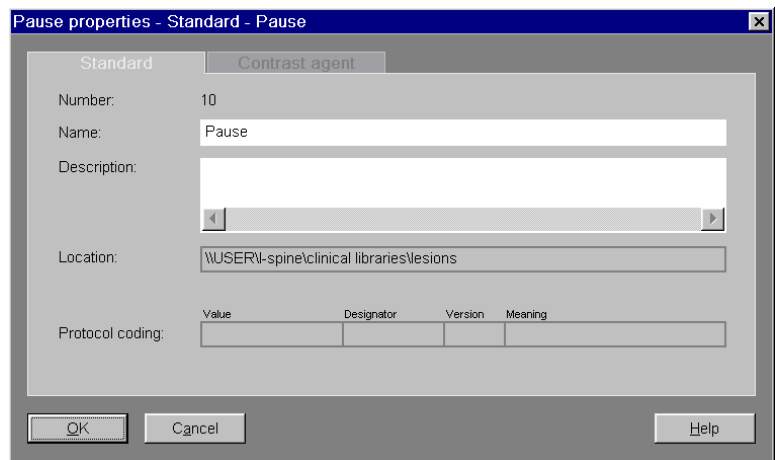
- ✧ Select a pause in the **Exam Explorer**.
 - ✧ Select **Edit > Properties** in the **Exam Explorer**.
 - Or -
 - Click this button on the tool bar.
- The **Pause properties** dialog window opens.

Selecting via the queue

- ✧ Select the pending pause from the queue.
 - ✧ Select **Edit > Properties**.
 - Or -
 - Select **Properties...** from the context menu.
- The **Pause properties** dialog window opens.

Entering the pause name and comment

After you opened the **Pause properties** dialog window, the **Standard** subtask card is in foreground.



Name:

✧ Enter a name for the pause that describes its purpose.

Description:

✧ Enter a comment, if required.

✧ If you set the mouse pointer to one of the pause entries in either the queue or the Exam Explorer, the pause name and comment are shown temporarily.

Planning contrast agent administration

If you planned a pause for administering contrast agent, establish the type and quantity of contrast agent on the **Contrast agent** subtask card.

→ [Page C.16–11](#), *Inserting and planning contrast agent pauses*

Applying changed pause properties

After you edited the pause properties, apply the new settings.

✧ Click **OK**.

Your changes are applied and the dialog window closes.

Measuring Large Examination Regions

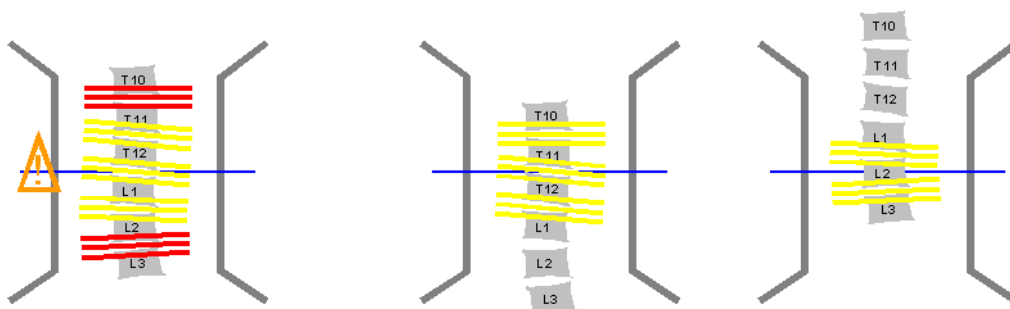
The idea behind scan@center and the Tim Planning Suite

For all measurements, the examination region should be as close to the center of the system as possible because in this region the magnetic field lines run in parallel, ensuring optimum image quality. Distortions can occur outside of the isocenter.

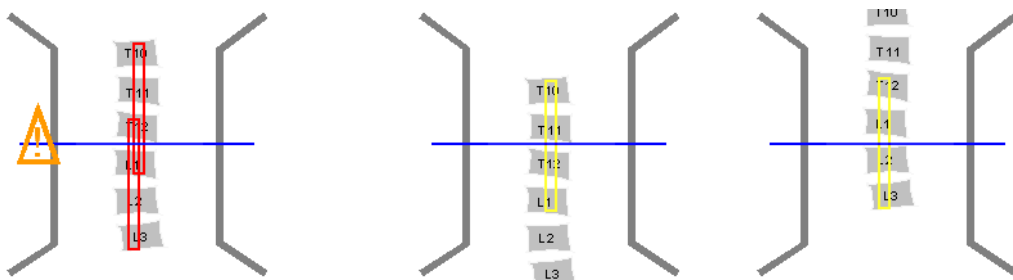
You can divide large examination regions into several sections that fit fully in the isocenter.

Measurement protocols can be configured so that the system automatically calculates the optimum table position (isocenter) and automatically moves the table to the isocenter during the measurement.

You can then automatically compose the images resulting from the individual protocols into an overall image.



This example shows a spine examination with slabs outside the isocenter. The examination is split into two isocentered measurements.



This example shows a spine examination with slices exceeding the isocenter. The examination is split into two isocentered measurements.

This procedure is supported by the *scan@center* concept and the *Tim Planning Suite*.

- ❑ *Set-n-Go protocols* support you in preparing the measurement and examining large regions of the body, since divided examination regions can be merged and graphically positioned at the same time.
- ❑ *Isocenter protocols* support this procedure through automatic determination of the optimum table position based on the location of the positioned slices.
- ❑ *Composing Groups* and the *Inline Composing* function support you in automatically composing images from divided examination regions.

Isocenter

The isocenter identifies the center of the magnet core. Prior to the measurement, the patient table is moved so that the planned protocol isocenter and the magnet isocenter are identical. The planned isocenter of a protocol corresponds to the protocol table position.

The table position is calculated based on a cuboid-shaped envelope across all slices of the protocol. Saturation regions, navigators, coil elements, adjustment volumes, and oversampling areas are ignored.

Automatic coil selection

Table movements also mean frequently-changing coil configurations. With the *Coil Memory* function and *Automatic coil selection*, you can have the coils selected automatically.

❑ Coil Memory

The coil selection from a previous protocol that was measured with the same or a similar table position is applied automatically.

❑ Automatic coil selection

All surrounding coils are selected automatically.

- ⇒ The maximum number of coils used is limited by the number of system channels.

For large measurement regions on systems with a low number of channels, it may happen that automatic coil selection does not supply the most optimal results.

- 
- ⇒ To ensure that automatic coil selection functions smoothly when using several Body Matrix Coils, the following has to be observed with respect to the numbers assigned to the coil sockets: Coils closer to the magnet have to be connected to a socket with a lower number.

Whole Body Patient Coordinate System

A uniform table position is entered into the Whole Body Patient Coordinate System for all images of a series block. The zero point of the coordinate system is set for a series block when the first measurement protocol starts.

The idea of TimCT

Applications such as angiography and metastasis screening require a field of view which is considerably larger than the isocenter. With the conventional *Set-n-Go* measurements, large examination regions are subdivided into several sub-regions that individually fit into the isocenter. This method requires several examination steps for positioning the patient table again, for planning additional measurement steps and for combining images.

The idea behind *TimCT* is to perform measurements of large examination regions in their entirety.

→ [Page C.9–27](#), *TimCT measurement*

TimCT provides this via the following special characteristics:

- ❑ Continuous table move during the measurement,
 - ❑ Localizer measurement across the entire region of examination,
 - ❑ Examinations that are both simple and intuitive,
 - ❑ Parameters with manageable settings.
- Since *TimCT* is able to measure large examination regions in their entirety via continuous table move, subsequent combination of the resulting images is no longer necessary.

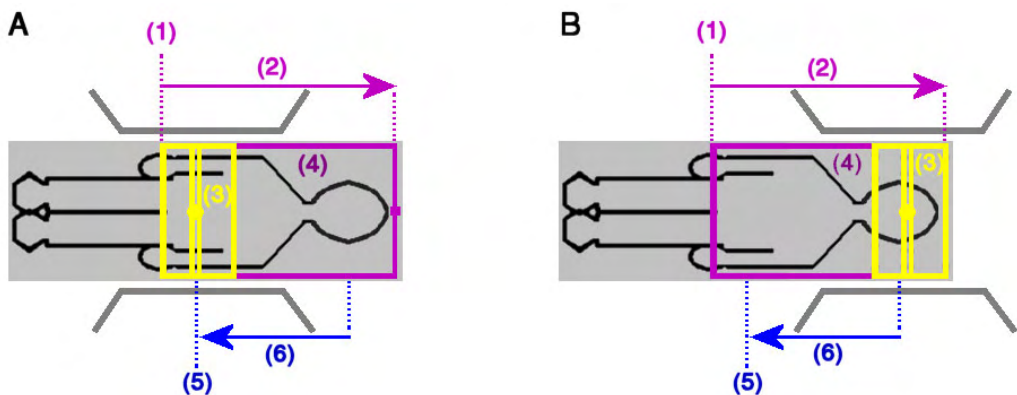
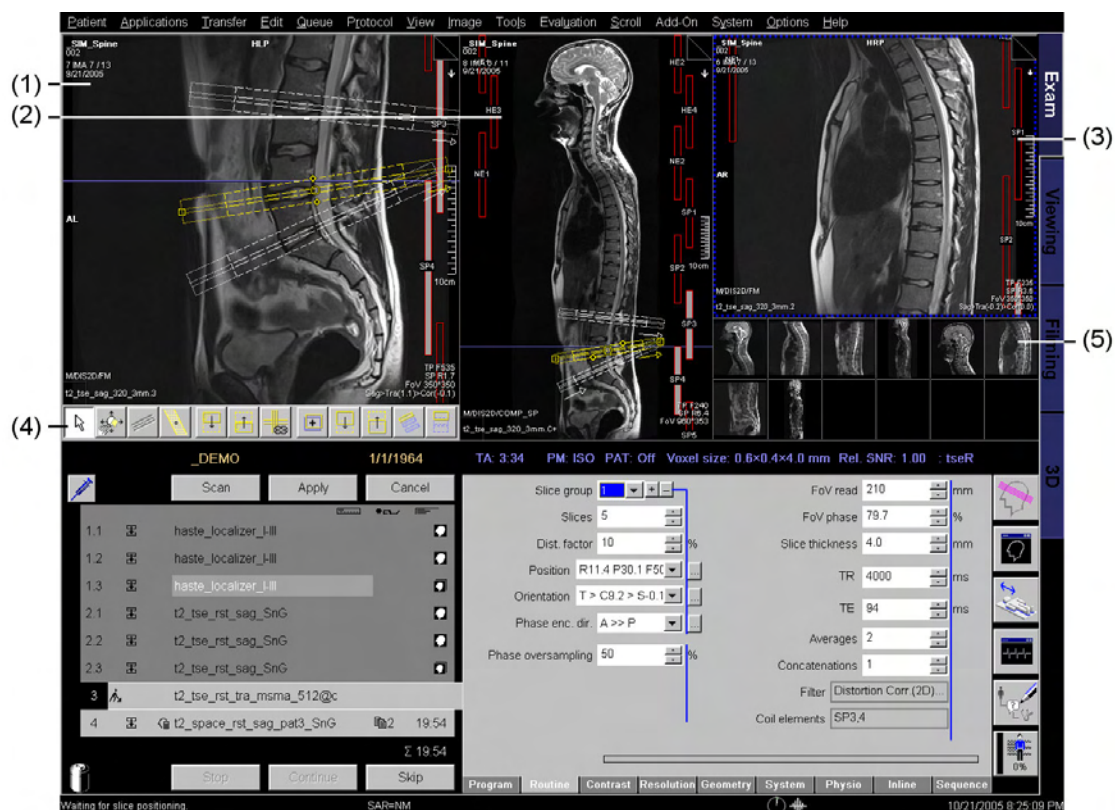


Diagram of a *TimCT* measurement

- (1) Range start
- (2) Entire examination region
- (3) Basic FoV
- (4) *TimCT* area
- (5) Start position of patient table
- (6) Table movement

The Tim Planning UI



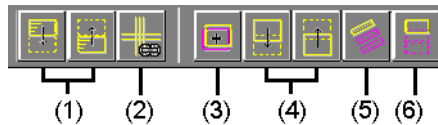
- (1) Segment of reference image
- (2) Segment of reference image
- (3) Segment of reference image
- (4) Tim Planning tool bar
- (5) Mini segments

Selecting the Tim Planning UI

✧ Select **View > Tim Planning UI**.

The **Exam** task card is displayed in the Tim Planning layout.

Tim Planning tool bar



- (1) Stack- / Stack+
→ [Page C.5–45](#), *Panning objects*
- (2) Coupled Graphics On/Off
→ [Page C.5–65](#), *Grouping graphic objects*
- (3) Adding a step
→ [Page C.9–14](#), *Adding a step*
- (4) FoV- / FoV+
→ [Page C.5–45](#), *Panning objects*
- (5) Parallel alignment
→ [Page C.5–56](#), *Aligning objects*
- (6) Aligning the FoV
→ [Page C.5–56](#), *Aligning objects*

Isocentric measurements

General

Isocenter protocols

Isocenter protocols (abbreviated: ISO protocols) calculate their table position such that the center of all slices of a protocol is in the magnet isocenter. As a result, all of the slices of this type of protocol are as close to the isocenter as possible.

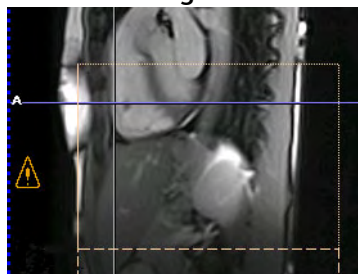
Distortion-corrected reference images

To plan isocenter protocols and plan Set-n-Go protocols, the reference images have to be distortion-corrected (image type DIS2D or DIS3D).

Table position line

During graphical slice positioning of an ISO protocol, a blue line marks the table position of the protocol (i.e., the planned isocenter). The blue line can be displayed or hidden using the **View > Protocol Table Position On** menu.

Off-center warning in reference images



If you plan the slice, slab or FoV too far outside the isocenter i.e., strong distortions can be expected. A warning icon is displayed along the left edge of the reference images. In addition, the corresponding objects are highlighted in red.

⇒ For some applications, it may be necessary to perform measurements outside the isocenter. For example, in the case of time-critical measurements, or examinations where the patient table may not be moved.

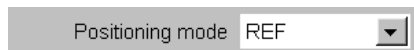
Loading the reference image for an ISO protocol

Isocentric measurements may only be planned using distortion-corrected images (image type DIS2D or DIS3D).

- ✧ Load a distortion-corrected reference image.
 - Or -
 - Start a localizer measurement that provides distortion-corrected images.
- ✧ On the **Resolution - Common** parameter card of the localizer, the **Distort. Corr.** filter has to be set to **2D** or **3D**.

Preparing the protocol and planning measurement slices

Preparing a protocol



- ✧ On the **System – Miscellaneous** parameter card, select the **ISO** mode in the **Positioning mode** parameter.
- ✧ In many cases it is recommended to activate distortion correction in the **Distortion Corr.** selection list on the **Geometry – Tim Planning Suite** parameter card or **Inline Composing** so that distortion-corrected images result.

Planning the measurement slices

- ✧ Plan the measurement slices.
 - [Page C.5–43](#), *Graphic Slice Positioning (GSP)*

Ensure that the area to be measured is not too large.

- ✧ If the measurement slices are outside the isocenter and strong distortions are expected, a warning icon is displayed on the left edge of the image and the slices are highlighted in red. Measurements of this kind result in strong image distortions, even after 2D or 3D distortion correction. They cannot be used as reference images for slice planning or position display, and are labeled with image type DIS2D* or DIS3D*.

Planning additional measurement slices

When you subdivide large examination regions into several sub-regions, you can plan additional measurement slices.

- ✧ Copy the measurement protocol.
- ✧ Pan the slices/FoV of the copied protocol and/or align them.
 - [Page C.9–15](#), *Moving/aligning slices/FoV*



Starting the isocentric measurement

Once you have set the measurement parameters, start the measurement.



- ✧ Click **Apply** in the program control.

The settings are confirmed. The protocol is now complete and starts automatically.

- ✧ Prior to moving the patient table to the isocenter for the first time, confirm in the dialog window that automatic table movements during the course of the examination are permitted.

As long as the **Table Move Notification** checkbox is activated, this dialog window appears prior to each table move. The pause can be used for patients at risk to announce every table movement individually via the intercom.

Set-n-Go protocols

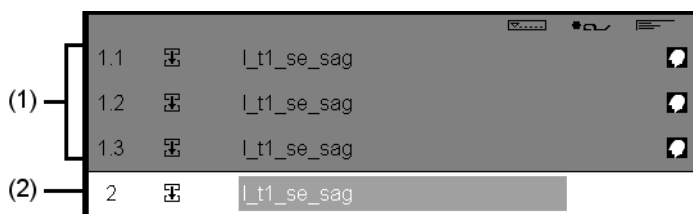
General

Set-n-Go protocols

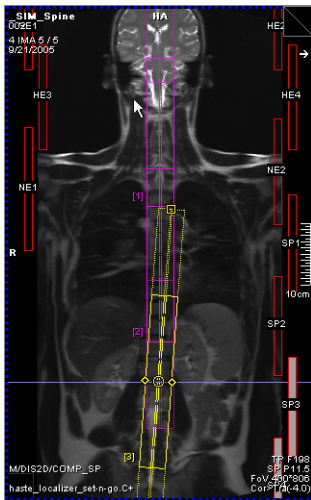
Set-n-Go protocols comprise multiple, coordinated, individual protocols (steps) and thereby enable easy measurement of large examination regions. Using the Set-n-Go protocol, you can open measurement steps together, then plan at different table positions, and edit them. From the images measured during the measurement steps, you are able to immediately compose images of the entire examination region (Inline Composing).

Measurement step

Set-n-Go protocols consist of at least two individual protocols (steps) that you can edit in parallel. Every measurement step contains a complete parameter data set that you can set individually.

Display in the program control

- (1) Previously measured Set-n-Go protocols:
The individual measurement steps are shown with an additional ordinal number.
- (2) Set-n-Go Protocols not yet measured:
When you leave the mouse pointer on the protocol, the program will show the number of measurement steps.



Display in graphic slice positioning

One measurement step is active, the others are passive. Objects of passive measurement steps are shown in magenta. Saturators and Navigators of passive measurement steps are not displayed.

Load reference image for the Set-n-Go protocol

Set-n-Go protocols can be planned only on composed images or on distortion-corrected reference images (Image type DIS2D or DIS3D).

- ✧ Load a composed image or a distortion-corrected reference image (Image type DIS2D or DIS3D).

Creating a Set-n-Go protocol

You create Set-n-Go protocols from the regular FIX or ISO protocols (**Positioning mode = FIX or ISO**).

Generating it

Set-n-Go Protocol



- ✧ Select the desired protocol in the program control.
- ✧ Activate the **Set-n-Go Protocol** checkbox in the **Geometry – Tim Planning Suite** parameter card.
 - Or -
 - Click this button in the Tim Planning UI.

The protocol is copied. The Set-n-Go protocol now consists of two steps, each containing the same sequence.
- ✧ Distortion correction is a prerequisite for Set-n-Go protocols. The **Distort. Corr.** parameter is automatically set to **2D**.

Adding a step

Step 2



- ✧ Select a step you would like to copy in the **Step** selection list.
 - Or -
 - Press **Ctrl+T** to select the desired step.
- ✧ Select **Protocol > Add Subprotocol**.
 - Or -
 - Click this button in the Tim Planning UI.

The selected measurement step is copied and appended at the end of the step list.
- ✧ The slices of the Step added are positioned according to the value in the **Overlap** dialog window.
 - [Page C.5–50, Moving the FoV with overlapping](#)

Moving/aligning slices/FoV

In the reference images, the slices and FoV of the newly created measurement step are exactly above those of the protocol from which they were derived.

Moving



- ✧ Select **Protocol > Stack +** or **Stack -**, protocol to shift the slices up or down.

- Or -

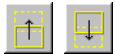
Click one of these buttons in the Tim Planning UI.

Or

- ✧ Select **Protocol > FoV+** or **FoV-** to move the FoV with overlapping either up or down.

- Or -

Click one of these buttons in the Tim Planning UI.



Alignment



- ✧ Select **Protocol > Align Parallel** to align the slice groups parallel to one another in the slice selection direction.

- Or -

Click this button in the Tim Planning UI.

Or

- ✧ Select **Protocol > Align FoV**, to align all FoVs of all steps in the Z-direction of the patient coordinate system (direction of movement for the patient table) and to position them adjacent so they overlap each other.

The slices are moved in the readout or phase-encoding direction.

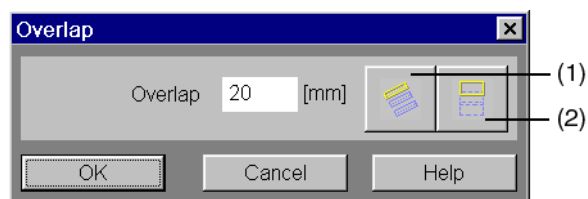
- Or -

Click this button in the Tim Planning UI.



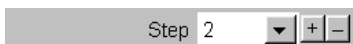
- ✧ Overlapping is necessary if the resulting images are to be composed into a complete image. The extent of overlapping can be set via the **Protocol > Overlap**.

The buttons have a slightly modified function in the **Overlap** dialog window:



The **Align parallel** (1) and **Align FoV** (2) functions are performed with the current overlap value entered.

Deleting Steps



✧ Select a step in the **Step** selection list.

✧ Click- next to the selection list

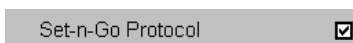
- Or -

Press the **Del** key.

The step is deleted.

⇒ If only one measurement step is left, the protocol automatically changes to a single step protocol.

Or

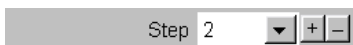


✧ Deactivate the **Set-n-Go Protocol** checkbox in the **Geometry – Tim Planning Suite** parameter card.

⇒ The current active step changes to a single step protocol. All additional steps are deleted.

Setting the step parameters

You change the parameters of a step independent of the parameters for the other steps.



✧ Select a step you would like to edit in the **Step** selection list.

- Or -

Press **Ctrl+T** to select the desired step.

✧ Adapt the step's measurement parameters to your requirements.

→ [Page C.7–1, Adapting the Measurement Parameters](#)

Setting the step protocol properties

The majority of the protocol properties for a step can be changed in the Set-n-Go protocol independent of the other measurement steps.

- ✧ Adapt the step's measurement parameters to your requirements.
→ [Page C.8–1](#), *Editing the properties for individual protocols*

Copying parameters

You can copy parameter groups from Set-n-Go Protocol to Set-n-Go Protocol. The parameter groups of all steps of the Set-n-Go protocol are copied.

→ [Page C.7–15](#), *Copying measurement parameters*

☐ Steps

When you deactivate the **Steps** checkbox and the number of steps between the source and the target protocol is not the same, the additional steps remain unchanged (in the target protocol) or are ignored (in the source protocol).

☒ Steps

- ✧ To match the number of steps between source and target protocol, activate the **Steps** checkbox.

Following copying, the target protocol has the same number of steps as the source protocol.



Starting the Set-n-Go protocol

Once you have set the measurement parameters, start the measurement.



- ✧ Click **Apply** in the program control.

The settings are confirmed. The protocol is now complete and starts automatically.

- ✧ Prior to moving the patient table to the isocenter for the first time, confirm in the dialog window that automatic table movements during the course of the examination are permitted.

As long as the **Table Move Notification** checkbox is activated, this dialog window appears prior to each table move. This pause can be set for at-risk patients to inform them of every table movement via the intercom.

Measure the Set-n-Go protocol multiple times

You can repeat a complete Set-n-Go protocol that has been processed in full or repeat individual steps only.

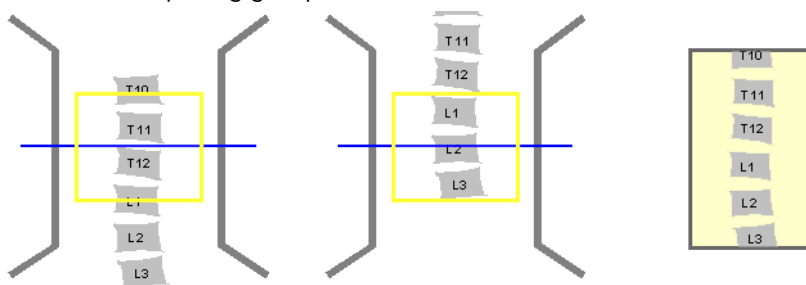
→ [Page C.6–22](#), *Repeating measurements*

Composed images from Set-n-Go protocols

General

Inline Composing

Through Inline Composing, the system reconstructs a composed image from the individual images of the protocol directly after the measurement of a Set-n-Go protocol or a composing group.



Composed images together with other distortion-corrected images (DIS2D / DIS3D) can be used for graphic slice positioning.



Limited use

-
- ! Images composed using Inline Composing may not be used for therapy planning (e.g., surgical procedures) or diagnosis (e.g., segmental instabilities).

Perform therapy planning and diagnosis on original images only.

-
- ! Composed images differ from original images with respect to image information, especially at the image transitions. Therefore, in terms of bones and vascular structures, composed images are only conditionally suitable for diagnosis.

If you send one or more composed images via network or forward them via a data medium, you have to inform the receiver of these limitations as well.

Additional Inline post-processing

You can combine Inline Composing with Inline post-processing functions, such as Subtraction, MIP and standard deviations.

Prerequisites for Inline Composing

Measured images have to meet various prerequisites for post-processing using Inline Composing:

Criterion	Description
Image orientation	All images have a primary orientation of sagittal or coronal (not transverse).
Overlapping	The images have to overlap.
Slice thickness	All images of a volume must have the same slice thickness.
Image type	All images have the same image type, e.g., MIP, SUB.
Distortion correction	All images have to be distortion-corrected (image type DIS2D or DIS3D).
Slice distance	All images of a volume must have the same slice distance.
Slice inclination (based on the first volume)	Only sagittal or coronal images can be used for Image Composing. The images can be tilted in axial direction, but they cannot be twisted between Sag >> Cor or Cor >> Sag (tolerance $\pm 1^\circ$). Graphic slice positioning may also be used if the image composed of individual planes shows no gaps at the joints and approximates a spatial plane.

Automatically composing images from Set-n-Go protocols

Distortion Corr. 2D

Inline Composing ☒

Composing Function Angio

- ✧ Open the Set-n-Go protocols.
- ✧ Select the **Tim Planning Suite** parameter card.
- ✧ Select the desired distortion correction in the **Distortion Corr** selection list (2D or 3D).
- ✧ Activate the **Inline Composing** checkbox.
- ✧ **Inline Composing** is only possible with distortion correction.
- ✧ Go to the **Composing Function** selection list and chose the method for composing images.

As soon as the Set-n-Go protocol is measured, a composed image is created from the images of the measurement step.

Composed images from several protocols

General

Composing Groups

You can compile multiple protocols into Composing groups to examine large regions. The images from protocols of a group automatically create composed images. This simplifies the planning of examinations requiring measurements at different table positions.

- ⇒ Please note the prerequisites that images have to meet in order to be processed using Inline Composing.
→ [Page C.9–22, Prerequisites for Inline Composing](#)
- ⇒ Please note the limitations on using composed images.
→ [Page C.9–21, Limited use](#)
- ⇒ Different from Set-n-Go protocols, you cannot open and plan Composing groups together.

Display in the program control



- (1) The protocols of a Composing group are identified via a post-processing icon.
- (2) If a Composing group has not been completed with the **Last Step** status, the post-processing icons for this group are crossed out.

Creating a composing group

Allocating the protocol to a group

Inline Composing ☒

- ✧ Select the desired protocol from the program control.
- ✧ Activate the **Inline Composing** checkbox on the **Geometry – Tim Planning Suite** parameter card.
 - Or -
 - Activate the **Inline Composing** checkbox in the **Inline – Composing** parameter card.
- ✧ If the **Inline Composing** checkbox is activated, the **Distortion Corr.** parameter is also activated
- ✧ When you are working on the **Geometry – Tim Planning Suite** parameter card, the **Set-n-Go Protocol** checkbox has to be deactivated.
- ✧ Go to the **Composing Function** selection list and choose the method for composing images.
- ✧ Select the composing group to which the protocol is to be assigned in the **Composing Group** selection list.

The protocol is now part of the selected composing group.

Composing Function Angio

Composing Group 1

Closing the group

The last protocol of a Composing Group has to be identified with the **Last Step** status. Only when this status has been set is the system able to successfully compose the images into an overall image.



As long as a Composing Group has not been completed with the **Last Step** status, the post-processing icons for this group are crossed out.

✧ Open the last protocol of the Composing group.

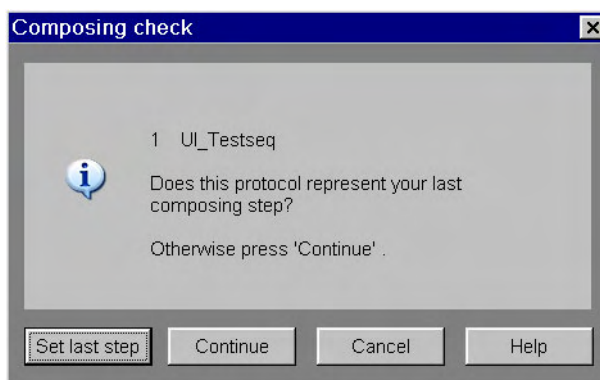
✧ Activate the **Last Step** checkbox.

Last Step



The post-processing icons of this group are now normal.

✧ When your composing group was not completed with the **Last Step** status, a dialog window appears at the beginning of the measurement where you can set the **Last Step** status for the current protocol or start the measurement anyhow with **Continue**.



TimCT measurement

General

TimCT

With *TimCT* (Continuous Table Move, powered by Tim) large examination regions can be measured with a single scan. Continuous table move during the measurement makes this possible.

→ [Page C.9–4](#), *The idea of TimCT*

TimCT-specific parameters

The *TimCT* functionality includes specific elements that are determined via the following global parameters in the protocol:

Parameters	Description
Tim CT mode	Activating/deactivating the TimCT functionality
Range start	Defines the start position of the measurement region (measured in the Whole Body Patient Coordinate System).
Total FoV	Defines the total length of the examination region.
Table Speed	Shows the computed speed of the table movement.
Recon. Mode	Defines whether the measurement results will be saved as an overall image and/or divided into several single images.
Variable resolution	Enables a different resolution to be set for parts of the examination region, e.g. a high resolution at the end of the examination region for contrast-enhanced angiography.

→ [Page P.1–86](#), *Geometry - Tim CT*

⇒ The settings of the TimCT-specific parameters are applied to all slice groups of the examination region.

Performing TimCT measurements

TimCT packages

In software version syngo MR **B17** TimCT is available in the following variants:

Package	Description
TimCT FastView	Fast scout
TimCT Angiography	Angiography examination with continuous table movement
TimCT Oncology	Oncology examination with continuous table movement

Additional adjustment

If the adjustment data for table move are only partially or not at all available, adjustment is performed automatically prior to the measurement.

To save time, you are able to determine the adjustment across the entire table range as part of the FastView localizer measurement with the **System - Adjustments** parameter card.

Perform CTM adjustments ☒

✧ Activate the **Perform CTM adjustments** checkbox.

After the FastView localizer measurement, adjustment is performed across the entire range of table movement.

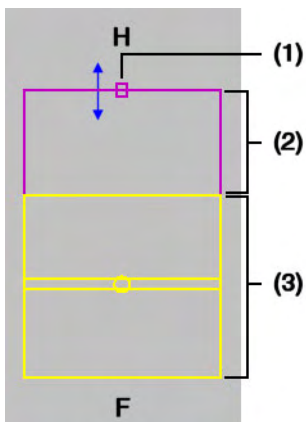
→ [Page B.2–15](#), *Additional adjustment after the TimCT localizer*

Localizer

The *TimCT-FastView* localizer enables you to perform a quick *TimCT* whole body measurement in advance.

You can then use the mouse and determine the *TimCT*-specific parameters in graphic slice positioning.

Changing the length of the *TimCT* area



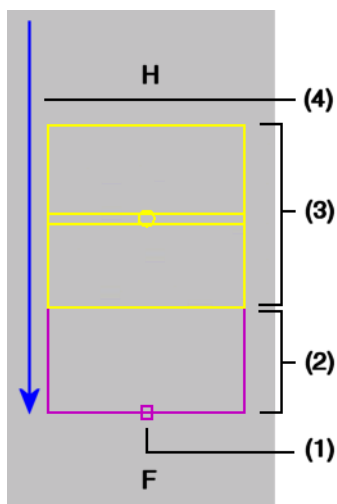
- (1) Handle to extend the *TimCT* area
- (2) *TimCT* area, orientation feet-head
- (3) Basic FoV

✧ Use the mouse and drag the handle away or to the basic FoV.

The length of the *TimCT* area increases or decreases. The orientation remains the same.

During the modification, the values for end position, distance traveled and length of area will be shown as image annotation.

Changing the orientation of the TimCT area



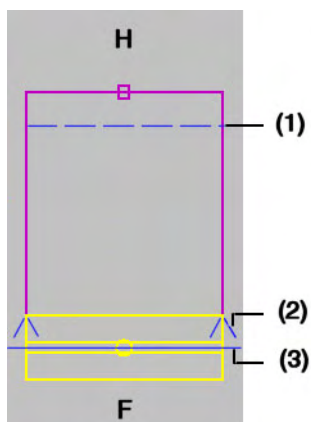
- (1) Handle to extend the *TimCT* area
- (2) *TimCT* area, orientation head-feet
- (3) Basic FoV
- (4) Movement across the basic FoV

✧ Use the mouse and drag the handle across the basic FoV.

The orientation of the *TimCT* area is reversed.

✧ The change in orientation is applied to all slice groups.

Displaying the table position



- (1) End position
After the measurement, this table position is in the isocenter of the magnet.
- (2) Direction of movement for the patient table
- (3) Start position
Prior to the measurement, this table position is in the isocenter of the magnet.

The display of the table position is used as an orientation aid.

✧ The start position can be shown or hidden.

→ [Page C-5,32](#)

Starting the TimCT measurement

- ✧ Prior to the measurement, adapt the protocol parameters to your requirements, as needed.
→ [Chapter C 7, Adapting the Measurement Parameters](#)
- ✧ Start the measurement.

Using breath-hold protocols



- ✧ Click this icon in the **Inline Display** to start a breath-hold measurement.



- ✧ Click this icon in the **Inline Display** to pause/resume a breath-hold measurement
→ [Page C.18–1, Inline Display](#)



Do not use the **Table Positioning** buttons during the *TimCT* measurement.



Physiological effects in the patient may occur during MR examinations. These effects are caused by radio-frequency fields or by magnetic fields varying over time.

- Detailed information regarding patient safety (especially regarding positioning) can be found in the "Safety" section of the system manual.

Temperature rise due to radio frequency

The patient is in a radio-frequency field during the examination. Part of this electromagnetic radiation is absorbed by the patient's body tissue and converted into heat.

The RF output absorbed per kg body weight is known as **Specific Absorption Rate (SAR)**. It is expressed in W/kg. If the SAR exceeds thermal regulation capacity of the patient, his/her body temperature will rise.

Stimulation due to varying magnetic fields

In addition to the RF field, electrical currents affect the patient during the examination. They are induced through temporal changes in the magnetic flux induced by the gradient coils and expressed mathematically as **dB/dt**.

Peripheral nerve stimulation may result if dB/dt (and therefore the strength of the electrical currents) rises above a certain characteristic value for each patient. Indications are prickling, tickling or slight muscle spasms in various body regions.

Avoiding physiological effects

Your MR system automatically monitors the SAR and the dB/dt load on the patient. Depending on the tolerance of the patient, different operating modes can be set to limit physiological effects.

Operating modes

The SAR and stimulation monitoring may be performed in two operating modes: The **Normal mode** and the controlled operating mode, the **First level** mode.

Normal mode

In the **Normal mode**, SAR and dB/dt are set to very low values. Warming of the patient is kept to a minimum and stimulation can be excluded.

Normal mode is the default setting at the beginning of an examination.

Controlled operating mode known as First Level

If the **First Level** operating mode is enabled, the patient may experience notable heat increases. If stimulation monitoring is performed in **First level** mode, increased dB/dt values may cause the patient to experience slight peripheral nerve stimulation.

- ⇒ The decision to switch to **First level** mode is within the discretion of the person assigned medical responsibility.
 - [Page C.10–11](#), *Switching the operating mode*
 - [Page C.10–27](#), *Exceeding the stimulation threshold*

SAR limits

The specific absorption rate is established and monitored for such aspects as **Whole-body**, **partial body** and **local** RF exposition. The limits applied are set in accordance with requirements defined in IEC 60601-2-33 and requirements from national regulations.

Different limit values are applied as a function of the type of RF exposition. A larger SAR may occur locally than measured across the entire body. For this reason, several limits are assigned to each operating mode.

The SAR limit values are also a function of the ambient temperature. At a room temperature of $> 25^{\circ}\text{C}$, the permitted SAR limits are decreased.

Limits according to stimulation models

Stimulation monitoring is based on stimulation models. These models are used to calculate the threshold values (limit of **Normal mode**) and stimulation limits (limit of **First level mode**) depending on the gradient coils.

❑ **SAFE model**

The threshold values and stimulation limits are determined by semi-empirical procedures that take the results of extensive clinical trials into consideration. Only very few patients will experience stimulation at the threshold. At the stimulation limit, the probability of stimulation is 50%.

❑ **dB/dt model**

The legally defined dB/dt restrictions are included in the calculation of the threshold values and stimulation limits.

Stimulation monitoring according to the SAFE model is performed on a regular basis. Depending on country-specific guidelines, the dB/dt model may be additionally applied.



The MR system software automatically monitors the SAR and dB/dt limits. Correct automatic monitoring presumes compliance with the prescribed sequence of operations.

Display of operating mode

The status bar shows you the operating mode used for the current measurement.

The abbreviation identifies the operating mode for the stimulation (*Stimu*) and SAR monitoring (*SAR*):

Stimu=NM SAR=NM

☐ **NM** (Normal mode)

The limit values of the operating mode **Normal mode** are maintained.

SAR=FL

☐ **FL** (First level mode)

The controlled operating mode **First level** is activated. The SAR values may exceed the limits for the normal mode.

Stimu= SP

☐ **SP** (Stimulation possible)

This mode is only possible with stimulation monitoring according to the SAFE model. Stimulations may occur during the measurement in progress.

⇒ Changing to the **First level** mode is permissible, e.g., within the framework of automatic SAR and/or stimulation monitoring.

→ [Page C.10–11, Switching the operating mode](#)

→ [Page C.10–27, Switching on the First Level operating mode](#)



Automatic SAR monitoring

The weight, age, sex, as well as positioning of the registered patient are used to calculate SAR monitoring. For this reason, it is important to correctly enter the patient data during patient registration.

The expected SAR values will be calculated automatically as soon as you load a new protocol. It is not possible to start the measurement unless all limit values are observed.

The MR system ensures that the limits are maintained during the measurement by continuously measuring the transmit power. If the permissible transmit power is exceeded, the SAR is switched off immediately. This eliminates the possibility of exceeding the limit values.

We recommend to set the temperature of the examination room to 21°C, since a higher ambient temperature may lead to SAR limits that are too low.

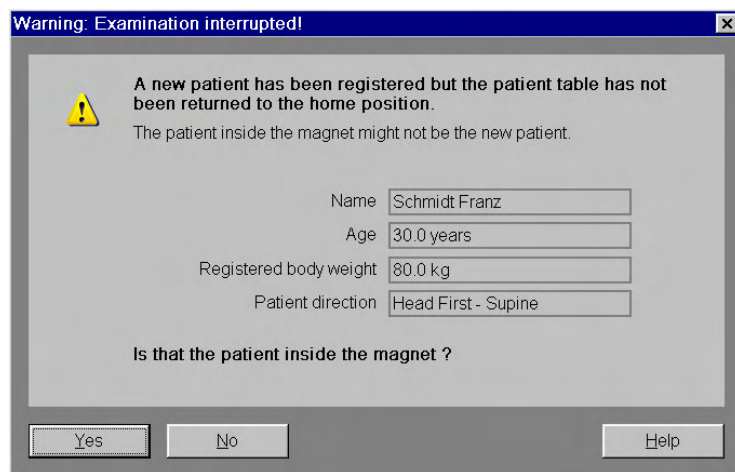
Messages when changing patients

Because correct patient registration is of utmost importance for SAR monitoring, the MR system monitors events that indicate a change in patients.

To ensure that the current examination data is still correct, the measurement is interrupted and a message window is displayed in the following cases:

- ☐ **1st Case:** A new patient is registered without moving the table to Home position.
- ☐ **2nd Case:** After moving to Home position, the measurement is continued without registering a new patient.

Message regarding 1st case



Yes

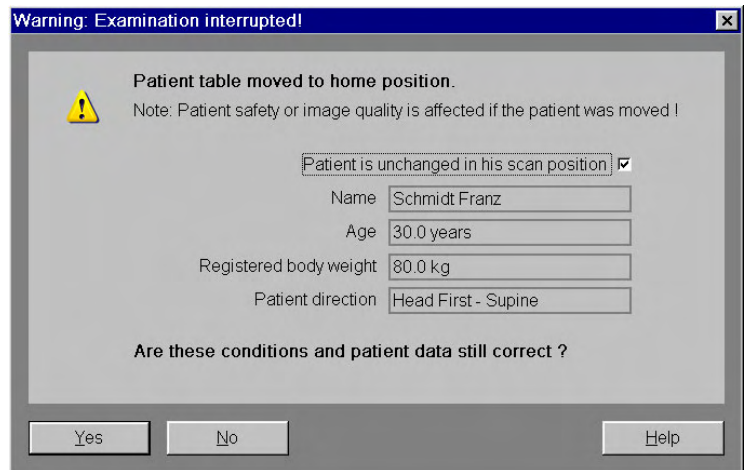
No

✧ To continue the measurement, click **Yes**.

Or

✧ To terminate the measurement, click **No**.

Message regarding 2nd case



Warning: Examination interrupted!

 **Patient table moved to home position.**
Note: Patient safety or image quality is affected if the patient was moved !

☒ Patient is unchanged in his scan position

Name

Age

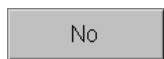
Registered body weight

Patient direction

Are these conditions and patient data still correct ?

If the patient data is not correct, register the current patient.
End the measurement for this purpose:

✧ Click **No**.



No

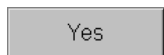
If the patient data is correct, continue the examination:

✧ In case the patient moved or was repositioned, deactivate this checkbox.



☒ Patient is unchanged in his scan position

✧ Click **Yes**.



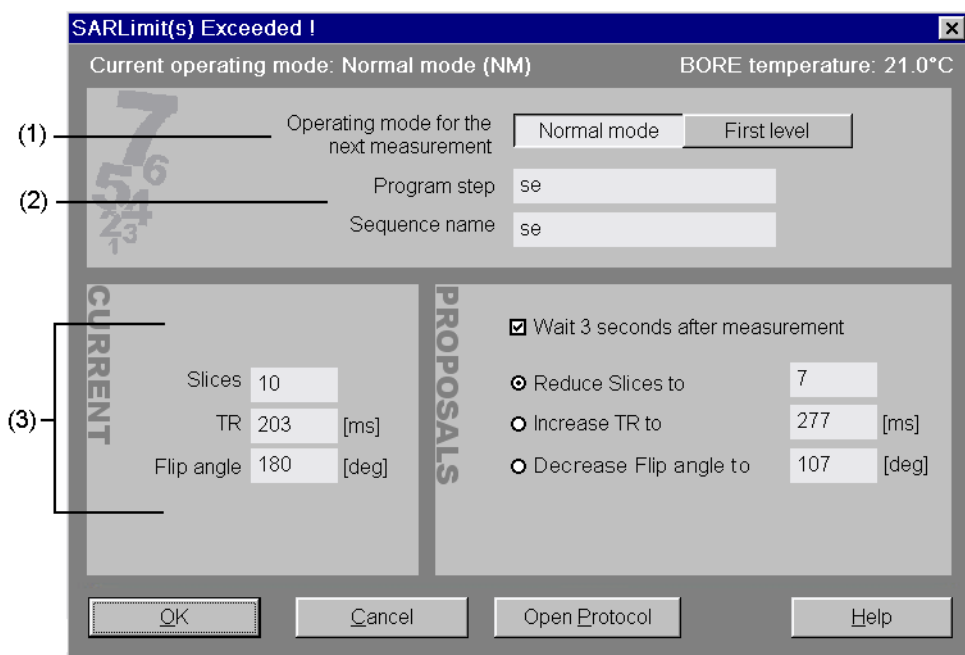
Yes

The message box closes.

✧ When the checkbox is deactivated, the reference images are unloaded. The measurement is not started automatically. Instead it has to be started with **Continue** in the program control. It is also recommended to reposition with the light localizer.

Computed SAR exceeds the limit

Protocols will not be loaded if your MR system calculated in advance that the measurement will exceed the SAR limit. The **SAR Limit(s) Exceeded!** dialog window appears on the **Exam** task card.



- (1) Buttons for selecting the SAR operating mode
- (2) Name of program step and sequence
- (3) Measurement parameters (actual to the left and recommended to the right)

Alternative measures

After successfully performing the following alternative measures, you may continue the examination as follows:

- ☐ You select an option for changing the measurement parameters in the dialog window so that the SAR remains below the limit.
- ☐ Switch to **First level** mode (only if appropriate for the physical condition of the patient).
→ [Page C–10,11](#)
- ☐ Open the protocol and change the measurement parameters.
→ [Page C–10,14](#)
- ☐ Skip the planned protocol and continue with the measurement of the subsequent protocol.
→ [Page C–10,15](#)

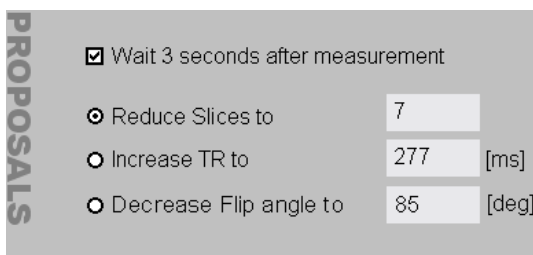
Changing the measurement parameters in the dialog window

The **SAR Limit(s) Exceeded!** window provides you with recommendations that describe how you can change the measurement parameters so as to avoid exceeding the SAR limits.

The recommendations are made for one parameter each (TR, slices or flip angle), assuming that the two other parameters are not to be changed.

- If an option is dimmed, no change proposal is available for that particular measurement parameter. When all options are dimmed, you have to select different measures.
→ [Page C.10–9, Alternative measures](#)

- ✧ Any changes you make to the parameters may increase the measurement time and affect image quality.



The image shows a dialog box titled "PROPOSALS" on the left side. It contains four options:

- ☒ Wait 3 seconds after measurement
- ☒ Reduce Slices to
- ☐ Increase TR to [ms]
- ☐ Decrease Flip angle to [deg]

If tolerated by the program, you are able to add a pause after the protocol has been measured.

- ✧ Select the **Wait... seconds after measurement** checkbox.
- ✧ Select a change proposal.
- ✧ Click **OK**.

The setting is applied. The measurement will start automatically.

Measurement is not possible

If it is not possible to perform a measurement with the parameters set, the **OK** button remains inactive and a message is shown in the lower part of the dialog window that will inform you how to proceed. You can change the parameters listed in the message via the protocol.

→ [Page C.10–14](#), *Changing measurement parameters in the protocol*

Switching the operating mode

In the **SAR Limit(s) Exceeded!** dialog window you can switch for subsequent measurement from **Normal mode** to **First level** mode. It may be possible to measure the protocol with the given parameters or the less parameter changes are required.



The increased RF load in **First level** mode may pose a hazard to the health of the patient.

- ☐ In the **First Level** operating mode, all patients have to be carefully monitored (e.g., squeeze ball, pulse sensor).
- ☐ Ensure that the temperature in the examination room is controlled accordingly.

You may not switch to **First level** mode in the following situations:

- ☐ Examining patients with impaired heat regulation or temperature.
 - ☐ Examining pregnant patients
 - ☐ Examining fetuses, testicles, and eyes with local transmitter coils. In addition, please note the special restrictions applying to these coils.
-



Normal mode is activated automatically as soon as you register a new patient.

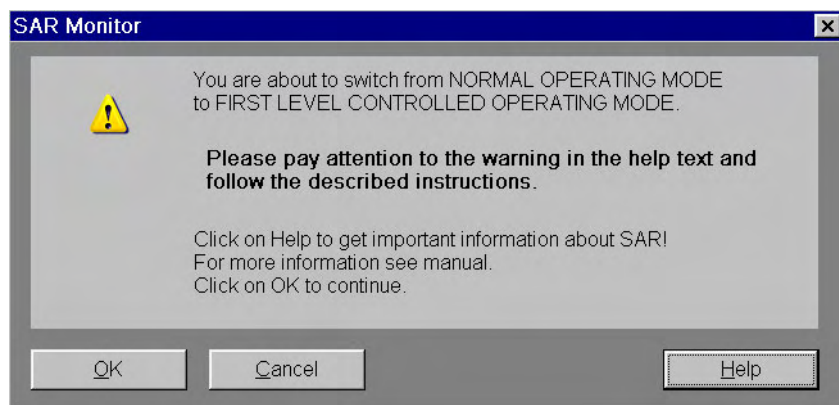
While examining a patient, **First level** mode remains active until you switch modes manually.

Activating the First Level operating mode

First level

- ✧ Click to **First level** in the **SAR Limit(s) Exceeded!** dialog window.

The **SAR Monitor** dialog window opens.



- ✧ Click **OK** in the **SAR Monitor** dialog window.
- ✧ If you want to work in the **Normal mode**, close the **SAR Monitor** dialog window with the **Cancel** button.



The **First level** button is switched on in the **SAR Limit(s) Exceeded!** dialog window. The parameter proposals are adjusted to this operating mode.

✧ Click **OK** to run the protocol in **First level** mode.

No proposals are active

In case change options are not active in the **First level** operating mode, the reason may be twofold:

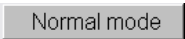
☐ No changes are required.

☐ The system cannot make recommendations.

Try to change parameters directly in the protocol, so that measurements are acceptable in the **First level** operating mode.

→ [Page C.7–1, Adapting the Measurement Parameters](#)

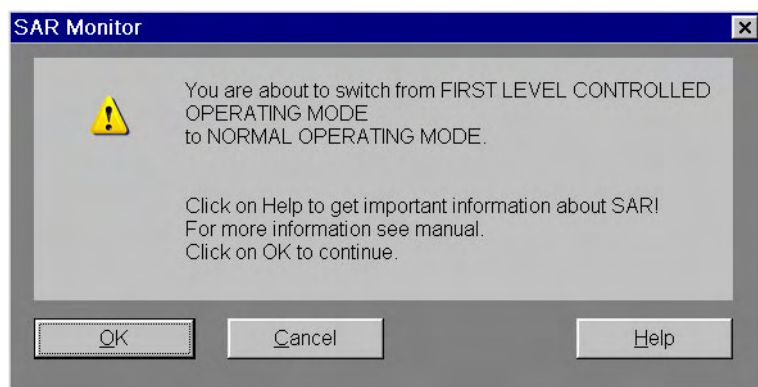
Switching back to Normal mode



Normal mode

✧ Click **Normal mode**.

The **SAR Monitor** dialog window opens.



- ✧ Click **OK**.

The operating mode is switched to **Normal mode**.

Changing measurement parameters in the protocol

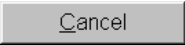
You can open the protocol and perform suitable changes on the parameter cards. This is especially useful when the **SAR Limit(s) Exceeded!** dialog window does not contain active change proposals.

For example, a reduction in numbers and thickness of saturation slices decreases the SAR. You can also set the RF pulse type **Low SAR** on the **Sequence - Part 2** parameter card.

- ✧ Click **Open Protocol**.
- ✧ Change the appropriate parameters on the parameter cards.
 - [Page C.7–1, Adapting the Measurement Parameters](#)

Open Protocol

Skipping the measurement

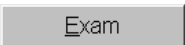
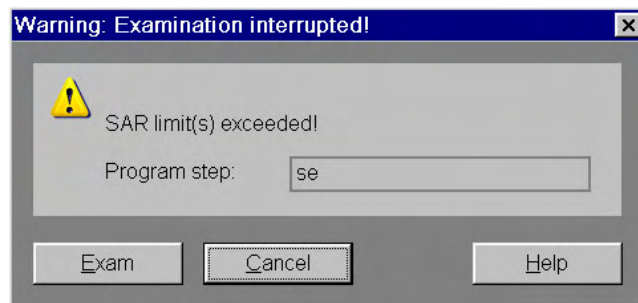
A rectangular button with a light gray background and a thin black border. The text "Cancel" is centered in a dark gray font.

- ✧ To skip the measurement, click **Cancel** in the **SAR Limit(s) Exceeded!** dialog window.

The protocol is marked as canceled in the queue. The next protocol is loaded.

SAR message on other task cards

If you opened a task card other than **Exam**, the measurement is cancelled if the SAR is too high or it is not even started. The following warning is shown:

A rectangular button with a light gray background and a thin black border. The text "Exam" is centered in a dark gray font.

- ✧ To change to the **Exam** task card, click **Exam**.

The **SAR Limit(s) Exceeded!** dialog window appears and you are able to adjust the parameters.

→ [Page C.10–8](#), *Computed SAR exceeds the limit*

Displaying SAR information

Your MR system enables you to view data relevant to SAR monitoring in the *current* protocol.

SAR display



The **SAR Information** button in the tool bar of the **Exam** task card includes the fill level and percentages. This indicator lets you determine to what degree the estimated highest and therefore limiting SAR value of the measurement approximates the critical SAR limit.

- ✧ The SAR display is not updated during the measurement; for this reason it does *not* show the current SAR load for the patient.

SAR Information dialog window



The **SAR Information** dialog window provides detailed information for the SAR monitoring parameters.

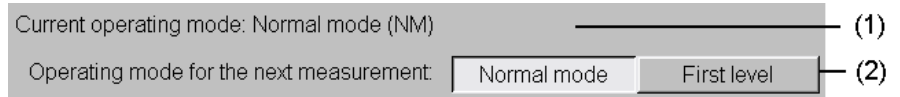
- ✧ Select **Options > SAR Information...**

- Or -

Click this button.

The **SAR Information** dialog window opens.

Independent of additionally open subtask cards, the **SAR Information** dialog window displays the operating mode currently set.



- (1) Current operating mode set
- (2) Buttons to switch over to the operating mode for the next measurement

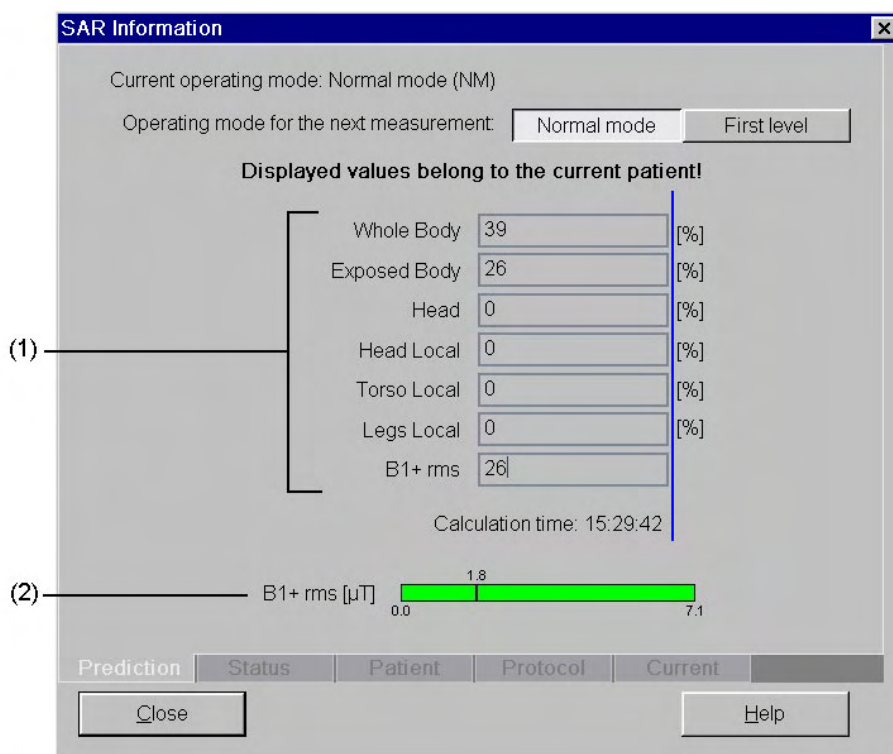
You can change the operating mode for the next measurement via the **Normal mode** or **First level** mode.

→ [Page C.10–11](#), *Switching the operating mode*

Displaying the SAR and B1+ rms values predicted

The predicted SAR and B1+ rms values are shown for the current measurement on the **Prediction** subtask card.

✧ Open the **Prediction** subtask card.



- (1) Relative SAR of the monitored SAR aspects and relative B1+ rms values
- (2) Displaying the range

Monitored SAR aspects

The display of monitored SAR aspects depends on the transmit coil used and the SAR guidelines applicable for your MR system.

Relative SAR values

The percentages of the relative SAR values are based on the applicable SAR limits.

If one of the values exceeds 100%, the respective SAR limit value is already exceeded. This may occur when you select a high-performance RF protocol in the **Normal mode**.

Whole Body	114	[%]
Exposed Body	42	[%]
Head	0	[%]

(1) (2) (3)

- (1) Region is without SAR load or is not monitored
- (2) Predicted SAR is below the limit value
- (3) SAR limit value has been exceeded

⇒ Within the framework of SAR monitoring, the different limit values for the individual SAR aspects have to be observed.

→ [Page C.10–3, SAR limits](#)

Relative B1+ rms values

The relative value is the relation of the actual value to the corresponding limit.

B1+ rms is the root mean square value of the MR effective component of the B1 field (B1+). B1+ hereby is the average value of the total axial, central slab of up to a few centimeters thickness.



Displaying the range

A bar diagram shows detailed information about the item selected in area (2). In this example, B1 +rms has been selected. The absolute value of the selected item is displayed above the green bar, and the corresponding limit is displayed below at the right end of the green bar.

If the SAR value is outside the green range, the permissible limit value has been exceeded for the operating mode currently selected.

Displaying the SAR status

✧ Open the **Status** subtask card.

The screenshot shows a window titled "SAR Information" with a close button (X) in the top right corner. The window contains the following text and controls:

- Current operating mode: Normal mode (NM)
- Operating mode for the next measurement:
- Displayed values belong to the current patient!
- (1) — Supervision mode
- (2) — Result of last SAR check
- (3) — Selected guideline

At the bottom of the window, there is a tabbed interface with the following tabs: **Prediction**, **Status** (selected), **Patient**, **Protocol**, **Current**, and an unlabeled tab. Below the tabs are two buttons: and .

- (1) Current mode (patient operation or service)
- (2) Results of the last SAR check
- (3) Legal guidelines for SAR monitoring

Querying SAR-relevant patient data

The patient data has to be correct for determining the SAR limits. If the data is incorrect, you have to register the patient again.

✧ Open the **Patient** subtask card.

The screenshot shows a window titled "SAR Information" with a close button (X) in the top right corner. Inside the window, it displays the current operating mode as "Normal mode (NM)" and allows selecting the next measurement mode between "Normal mode" and "First level". A section titled "Displayed values belong to the current patient!" contains seven data entry fields, each with a number in parentheses to its left: (1) Name (Mustermann^Erika), (2) Sex (Female), (3) Age (30.0 [years]), (4) Registered body weight (60.0 [kg]), (5) SAR body weight (60.0 [kg]), (6) Patient direction (Head First), and (7) Specific energy dose (813 [J/kg]). At the bottom, there are tabs for "Prediction", "Status", "Patient" (which is selected), "Protocol", and "Current". Below the tabs are "Close" and "Help" buttons.

Field Number	Field Name	Value	Unit
(1)	Name	Mustermann^Erika	
(2)	Sex	Female	
(3)	Age	30.0	[years]
(4)	Registered body weight	60.0	[kg]
(5)	SAR body weight	60.0	[kg]
(6)	Patient direction	Head First	
(7)	Specific energy dose	813	[J/kg]

- (1) Patient Name
- (2) Sex of patient
- (3) Age of patient
- (4) Body weight entered during registration
- (5) Body weight considered for SAR check
- (6) Patient orientation with respect to magnet
- (7) Accumulated specific RF dose

SAR weight

The **SAR body weight** may deviate from the Registered body weight.

The system compares the registered weight with the statistic growth data (percentiles) and corrects it in case of strong deviations. In case of a correction, the **SAR body weight** corresponds to the 5 percent or 95 percent percentile (depending on the direction of deviation) of the registered age. The **SAR body weight** is used for the SAR computation.

Showing data with respect to the protocol

✧ Open the **Protocol** subtask card.

The screenshot shows a window titled "SAR Information" with a close button (X) in the top right corner. Inside the window, the text "Current operating mode: Normal mode (NM)" is displayed. Below it, "Operating mode for the next measurement:" is followed by two buttons: "Normal mode" and "First level". A bold heading "Displayed values belong to the current patient!" is centered. Below this heading, four rows of data are shown, each with a label and a text box: "Program step" with "se_15b130", "Sequence" with "se_15b130", "Run time" with "80" and "[sec]" to its right, and "Time averaged RF power" with "10.6" and "[W]" to its right. A vertical blue line is positioned to the right of the text boxes. At the bottom of the window, there are two tabs: "Prediction" and "Status", and a "Patient" tab is selected. Below the tabs are two buttons: "Close" and "Help".

	Program step	Sequence	Run time	Time averaged RF power
(1)	se_15b130	se_15b130	80 [sec]	10.6 [W]
(2)				
(3)				
(4)				

- (1) Name of current program step
- (2) Sequence used
- (3) Measurement duration
- (4) Medium RF power

Displaying current SAR values

The **Current** subtask card contains the same SAR aspects as the **Prediction** subtask card. However, the relative SAR values shown are not the predicted but rather the current SAR load for the patient or parts of his body.

→ [Page C.10–19, Monitored SAR aspects](#)

→ [Page C.10–20, Displaying the range](#)

✧ Open the **Current** subtask card.

The screenshot shows a software window titled "SAR Information" with a close button (X) in the top right corner. The window displays the following information:

- Current operating mode: Normal mode (NM)
- Operating mode for the next measurement:
- Displayed values belong to the current patient!
- A table of SAR values for different body parts:

Body Part	Value	Unit
Whole Body	4	[%]
Exposed Body	2	[%]
Head	0	[%]
Head Local	0	[%]
Torso Local	0	[%]
Legs Local	0	[%]

Time of the last update: 16:32:35

Whole Body [W/lb] 0.0  0.9

At the bottom, there are tabs for "Prediction", "Status", "Patient", "Protocol", "Current", and a "Close" button. The "Current" tab is selected.

The values are updated whenever the **Current** subtask card is selected.

Automatic Stimulation Monitoring

Your MR system includes automatic stimulation monitoring. When you load a new protocol, the MR system uses the stimulation models to check whether stimulation may occur during the measurement.

When stimulation monitoring is active in the **Normal mode**, it is not possible to start a measurement that would exceed the stimulation threshold. Nerve stimulation is not expected in **Normal mode**.

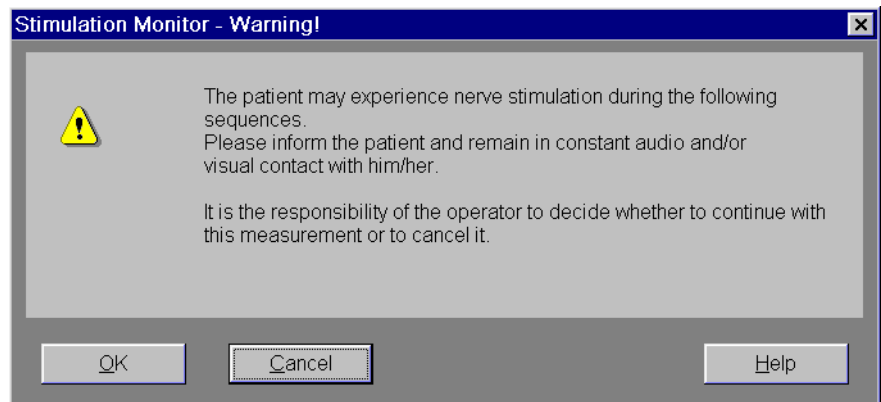
When the MR system switches into the **Stimulation possible (SP)** operating mode, the higher limits are taken into account. In this case, nerve stimulation may occur during the course of the measurement. However, it is not possible to start the measurement, if the MR system predicts that the stimulation limit will be exceeded. In this case, the dB/dt values have to be decreased by e.g., changing the measurement parameters accordingly.

Even during the measurement, stimulation monitoring is active and controls the actual values of the gradient signals. If the stimulation limits are exceeded, the active measurement is stopped immediately and the gradient currents are reset. This eliminates the possibility of exceeding the limit values.

- ✧ Unlike in SAR monitoring, patient data such as age, weight and size are not used for the stimulation models. With respect to stimulation monitoring, a protocol in the same slice orientation may either be run for all or none of the patients.

Exceeding the stimulation threshold

The following dialog window appears if the MR system determines that the stimulation threshold is exceeded for the pending measurement:



Switching on the First Level operating mode

You may start the measurement despite possible stimulation.

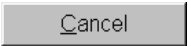
- ✧ Click **OK**.

Stimulation monitoring is now performed in the controlled **First level** operating mode.

The abbreviation **SP** appears in the status bar.

→ [Page C.10–4, Display of operating mode](#)

Skipping a measurement

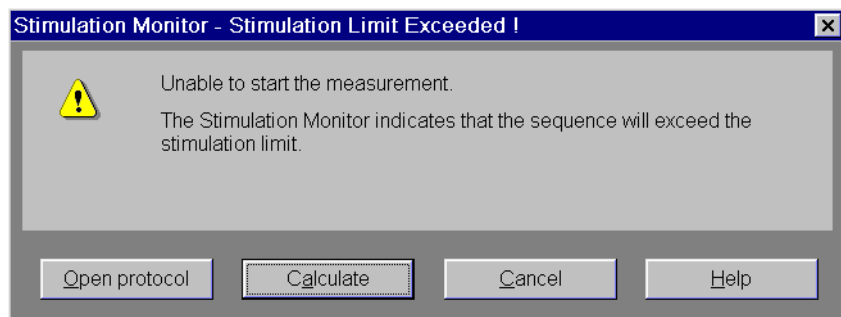
A rectangular button with a light gray background and a thin black border. The text "Cancel" is centered on the button.

✧ Click **Cancel**.

The protocol is marked as canceled in the queue. The next protocol is loaded.

Stimulation limit has been reached

A protocol is not loaded if the stimulation limit (**First level** mode limit) will be exceeded during the measurement. The **Stimulation Monitor - Stimulation Limit Exceeded!** dialog window appears.

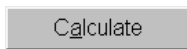




The protocol can be run if you:

- ☐ Have the MR system compute suitable parameter changes (display in the **Stimulation Monitor - Recommendations** dialog window) and apply these.
- ☐ Change the gradient mode of the protocol in the **Stimulation Monitor - Recommendations** dialog window.
- ☐ Open the protocol and manually adjust the measurement parameters in the parameter cards.

Calculating parameter changes



You let the MR system calculate parameter changes that lead to values below the stimulation limit:

- ✧ Click on **Calculate** in the **Stimulation Monitor - Stimulation Limit Exceeded!** dialog window.

The **Stimulation Monitor - Recommendations** dialog window is displayed.

Stimulation Monitor - Recommendations

Program step: ep2d_diff

Sequence name: ep2d_diff

Current parameters

FoV read: 334 mm

Slice thickness: 4 mm

Phase enc. dir.:

Recommended changes

☐ Increase FoV read to: ??? mm

☐ Increase Slice thickness to: mm

☐ Change Phase enc. dir.:

Change gradient mode

TE: 201 [ms] ☐ Adapt to: 205 [ms]

TR: 280 [ms] ☐ Change gradient mode: 290 [ms]

TA: 00:02 ☐ Change gradient mode: 00:02

OK Cancel Open protocol Help

- (1) Name of protocol and sequence
- (2) Current and recommended measurement parameters
- (3) Current parameters and those modified by changing the gradient mode

☒ Increase Read FoV to: 330 [mm]

☐ Increase Slice Thickness to: 4.0 [mm]

☐ Change Phase Direction: L → R

✧ Select one of the recommended changes.

Changing a gradient mode

A rectangular button with a circular icon containing a right-pointing arrow and the text "Adapt to".

You may also set a new, slower gradient mode. As a result, changes are made to TE, TR, and the measurement time.

✧ To change the gradient mode, select the **Adapt to** option.

The modified gradient mode is identified with an asterisk in the **Sequence – Part 2** parameter card.

→ [Page P.1–207](#), Parameter card *Sequence - Part 2*

✧ To apply the new settings, click **OK**.

The protocol starts automatically.

Editing the protocol

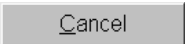
A rectangular button with the text "Open Protocol".

If no change proposals are made for the pending protocol, manually adjust the measurement parameters in a suitable way to the parameter cards.

→ [Page C.7–1](#), *Adapting the Measurement Parameters*

✧ Click **Open Protocol**.

Skipping a measurement

A rectangular button with the text "Cancel".

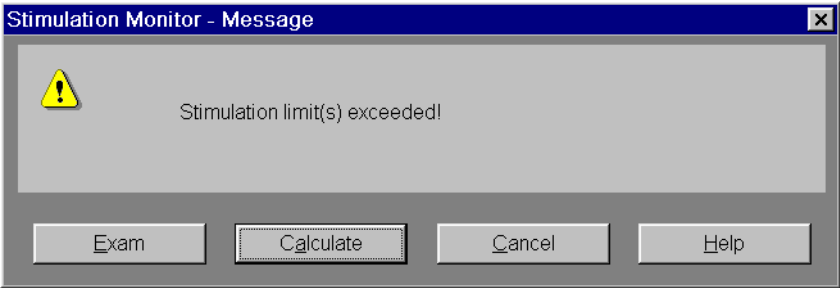
You can skip measurements for which exceeding stimulation limits have been determined.

✧ Click **Cancel**.

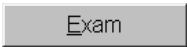
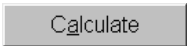
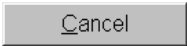
The protocol is marked as canceled in the queue. The next protocol is loaded.

Stimulation message on other task cards

If you are currently not working in the **Exam** task card while the MR system is precalculating that the stimulation limit will be exceeded, the measurement is not started and the **Stimulation Monitor - Message** is shown.



You are provided with the following possibilities for continuing:

Button	Function
	You immediately switch to the Exam task card. The Stimulation Monitor - Stimulation Limit Exceeded! dialog window is displayed. → Page C.10–28 , <i>Stimulation limit has been reached</i>
	The MR system calculates suitable parameter changes. The previous application remains open. The system informs you as soon as change proposals are ready. Should you then switch to the Exam task card, the Stimulation Monitor - Recommendations window opens including change proposals.
	The measurement is skipped and identified as cancelled in the queue. The next protocol is loaded.

✧ Click the button with the desired function.



Information about Stimulation Monitoring

You may query the threshold/limit values and other parameters important for automatic stimulation monitoring of the current protocol.

✧ Select **Options > Stimulation Info....**

The **Stimulation Monitor - Information** dialog window is shown together with the **SAFE** subtask card and (depending on country-specific approval) the **dB/dt** card.

The SAFE subtask card

The **SAFE** subtask card contains a list of the calculated and measured parameters that have to be factored into the SAFE model.

→ [Page C.10–3](#), *Limits according to stimulation models*

- ✧ Open the **SAFE** subtask card to select the stimulation parameters according to the SAFE model.

The left-hand side of the subtask card displays the precalculated parameter values of the stimulation monitoring. The measured data is shown to the right. The data refers to the measurement in progress or the measurement just completed.

Stimulation Monitor - Information

SAFE

Displayed values belong to the last patient!

Predicted values	Measured values
(1) Lookahead safety factor 20.00	
(2) Stimulation level 20.00 [%]	Stimulation level 20.87 [%]
(3) Stimulation threshold exceeded? Yes	Stimulation threshold exceeded? Yes
Stimulation limit exceeded? No	Stimulation limit exceeded? Yes

Close Help

- (1) Safety factors for the stimulation prediction.
- (2) Maximal approximation of the stimulation limit (in%)
- (3) Exceeding the stimulation threshold

The dB/dt subtask card

The country-specific regulations determine whether monitoring is performed by taking the statutory dB/dt limit values into account on your system. In this case, you may also use the **dB/dt** subtask card.



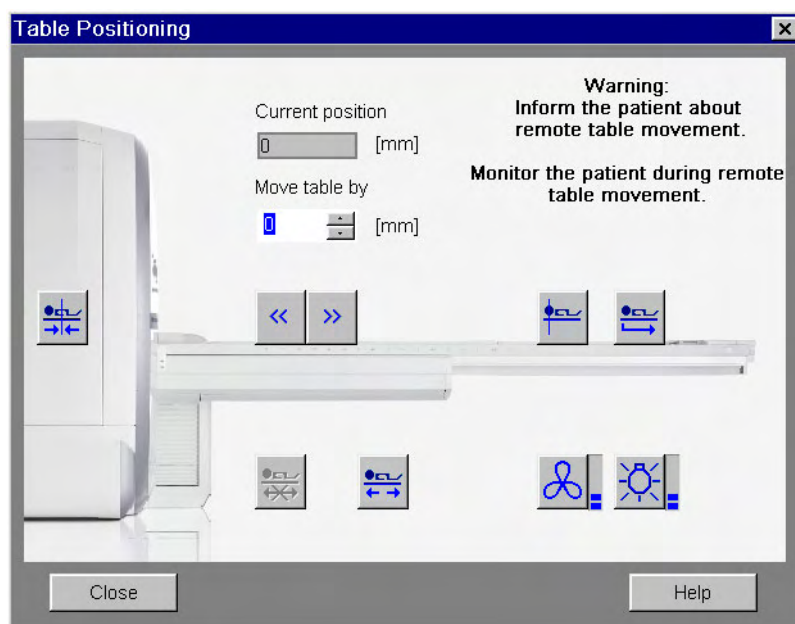
Controlling the Patient Table

You can control the patient table movement by using either the control panel next to the magnet bore or the software at the syngo Acquisition Workplace in the control room.

Selecting table positioning

✧ Select **View > Table Positioning....**

The **Table Positioning** dialog window opens.



The current table position is displayed. You are able to control various horizontal movements of the patient table.

-
- ! When performing table movements at the console, ensure that the table is *not* simultaneously moved by another person via the operating panel at the magnet bore.
-

Current table position

In this case, the distance between the magnet isocenter and the slice marked using the light localizer is displayed.

Moving the table by a certain distance

You may position the patient table with millimeter precision in the **Table Positioning** dialog window.

- ✧ Enter the distance you want to move the patient table in the **Move table by** field.
- ✧ Click this button to move the patient table into the magnet by the distance specified.
- ✧ Click this button to move the patient table out of the magnet by the distance specified.

Current position
1073 [mm]

Move table by
100 [mm]



Moving the table into the center of the measurement field



- ✧ Click this key to move the patient table precisely into the center of the measurement field.

The center of the measurement field is generally the position of the last measurement. If a measurement has not yet been performed, it is the position marked by the light localizer. If the marked position and the position of the last measurement are not identical, the table moves to the following position when you press the respective button:

- ☐ **Position of the last measurement**
if the table is at the position identified by the light barrier
- ☐ **Position marked by the light localizer**
if the table is at the position of the last measurement
- ☐ **Measurement plane of the head coil**
if the measurement or identification with the light localizer was not performed



- ✧ When you press this button during the examination workflow, the table is alternately moved between the last measurement position (first click) and the zero point of the examination (next click).

Moving the table almost completely out of the magnet



- ✧ Click this button to move the patient table almost completely out of the magnet.

Canceling a table movement



- ✧ To stop table movement, click this button.

The patient table will stop at the point it has reached.

Canceling table STOP

Before you can resume table positioning, you have to first cancel the table STOP.



- ✧ Click this button.

Setting the tunnel lighting and ventilation

This is how you regulate both lighting and ventilation in the magnet bore from the *syngo* Acquisition Workplace in the control room:



- ✧ Click the respective button to switch on the lighting or ventilation in the magnet bore.
- ✧ To increase/decrease the lighting or ventilation, click this button again until you have reached the appropriate level.

Sequence of levels:

Off > low > medium > high > medium > low > off ...



The current setting for tunnel ventilation is shown to the right of the button.



Closing the examination

- ✧ Click this button to move the patient table completely out of the magnet.



Do not lower the table when you want to continue the examination!

When the table is lowered, the examination is considered complete.

The system assumes that you now want to register and examine the next patient. All reference images are removed from the image area.

Two essential factors in obtaining useful MR images are patient communication and cooperation.

Patient information and instructions are given during an examination via manual or automatic voice outputs, e.g., breathing control instructions.

Demo files for voice output in two languages are delivered together with your MR system. You may record additional voice outputs in various languages. You can assign these voice outputs to a protocol as a property and automate patient communication. You are also able to play back these voice outputs manually without reference to a scan.

⇒ Use of voice output for patient instructions is not suitable during multiple measurements, e.g., **Abdomen**, multi-breath-hold.

Terms

Two special terms are frequently used in the following descriptions:

- ❑ **Voice output:** A single spoken patient instruction. Instructions are available in various languages.
- ❑ **Label:** Unambiguous identification of the individual language outputs. The label is displayed in the selection list and is used to identify a voice output.



Please do not confuse the automatic voice outputs with music played for the patient. The intercom includes a separate input for connecting a CD player, radio, etc.



Do not insert a commercial music CD into the CD drive of the *syngo* Acquisition Workplace. The copy protection on the CDs could cause the system to crash and damage the drive.

When using automatic voice outputs to give instructions to the patient, it is advisable not to play music at the same time.

Playing back an existing voice output

Existing voice outputs can be played manually or automatically. You are able to individually configure your voice outputs.

→ [Page C.12–7, Changing the volume](#)

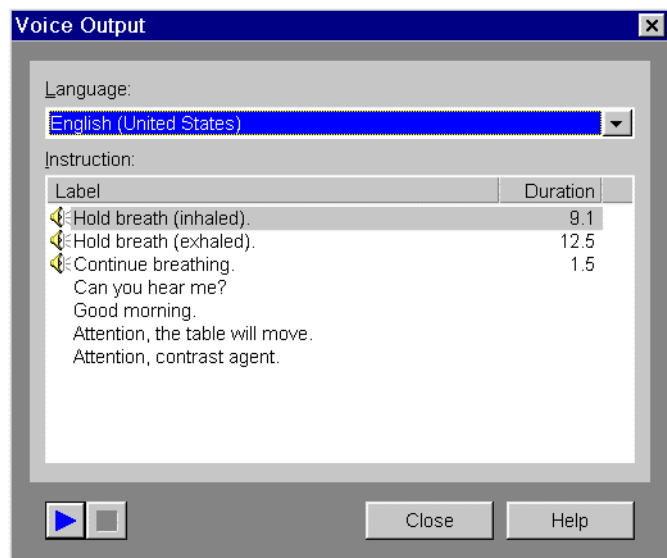
→ [Page C.12–8, CV/CBT switch](#)

✧ Select **Queue > Voice output > Play.**

- Or -

Select **Voice Output** from the context menu for the program control.

The **Voice Output** dialog window opens.



Existing voice outputs are identified with a loudspeaker symbol and the play-back time. The language of the label does not correspond to the language set.

- ✧ The dialog window cannot be opened as long as a measurement is in progress.

Manually playing back voice outputs

Voice outputs are available in several languages. They can be played back as often as necessary and at any time.

- ✧ Voice output can be played back before and after but not during a measurement.

Languages are set by default in the **Language** selection list. Whenever required, you can change the setting.



- ✧ Select the required language from the **Language** selection list.
- ✧ You can select a language only if at least one language output has been stored for it.

Instruction:

Label	Duration
🔊 Hold breath (exhaled).	2.5
🔊 Continue breathing.	3.2
🔊 Hold breath (inhaled).	
Can you hear me?	1.5
🔊 Do you understand	2.0
Attention, the table will move.	



- ✧ Select the required language from the **Instruction** selection list.
- ✧ Click this button to play the voice output.

The patient listens to the instructions via the headphone or loudspeaker in the examination room.

Stopping voice output



- ✧ Click this button.

Automatically playing back voice outputs

For automatically playing back voice outputs prior or after the measurement, go to the protocol and references for the required voice output.

- ☐ You either establish references when you install the program in the **Exam Explorer**.

Or

- ☐ You establish references during the examination when you already applied the program to the program control.
 - [Page C.8–7](#), *Playing back voice outputs*

If a protocol with a referenced language output is in the wait queue and selected while the language output is not available, a system message appears. Issue your commands via the intercom.

→ [Page C.6–29](#), *Patient instructions*

Changing the properties of voice outputs

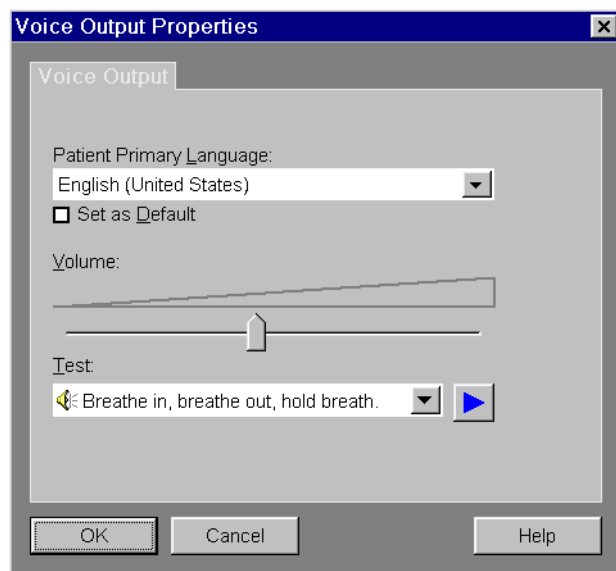
You can adjust the voice outputs to the requirements of the patient.

✧ Select **Patient > Voice Output Properties**.

- Or -

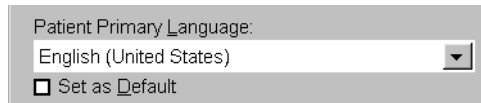
Select **Voice Output Properties** from the context menu of the patient name.

The **Voice Output Properties** dialog window opens.



✧ When the **Test** selection list and the **Play** button are inactive, your MR system is not equipped for language outputs.

Changing the language



- ✧ Select the required language from the **Patient Primary Language** selection list.
- ✧ You can select any language that is included on your MR system. It is not possible to add new languages.
- ✧ It is not possible to switch between languages during measurements.

Changing the default language

You set the default language (standard language) for all patients in the **Exam UI Configuration - voice output** dialog window.

→ [Page C.12–17, Configuring voice outputs](#)

As an alternative, the **Voice Output Properties** dialog window provides the possibility of using the language selected for the current patient as the default language:

☒ Set as Default

- ✧ Activate the **Set as Default** checkbox.
- ✧ The default language for patient instructions does not have to be the same as the language of the user interface.

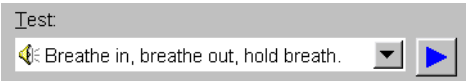
Changing the volume



- ✧ Slide the **Volume** regulator to the right (left) to increase (decrease) the volume or to the left to reduce the volume of the voice output.

Testing voice output

Play a voice output to check the settings.



- ✧ Select **Test** in the selection list and choose a voice output.
- ✧ To play the voice output, click this button.



CV/CBT switch

The **CV/CBT** switch at the intercom allows you to determine where the patient instructions will be heard:

Position	Comment
CV	Patient instructions are played in the control room and in the examination room. This setting has to be selected for the examination.
CBT	Patient instructions are played in the control room only. You select this setting when you want to record and test new voice outputs. This setting cannot be selected for examinations.

- ✧ Set the **CV/CBT** switch to the required position on the intercom.

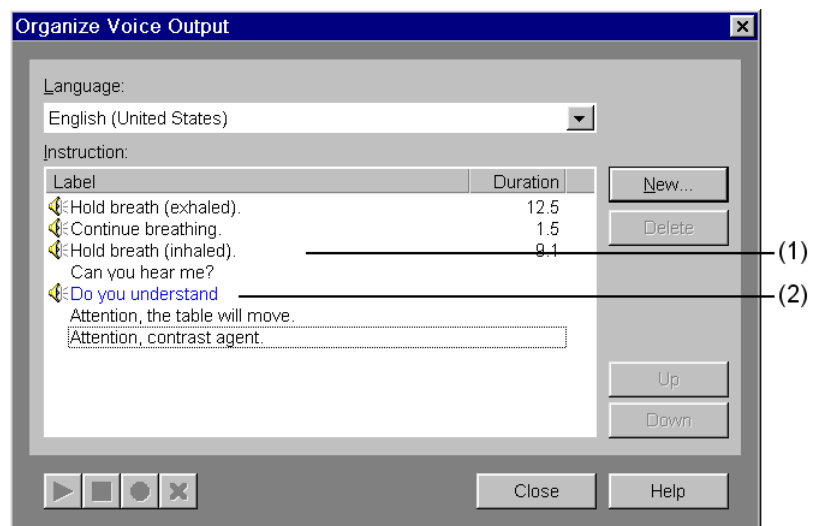
Recording and editing voice outputs

You may record or edit patient instructions only when you do not have a patient registered.

✧ If a patient is registered, close it.
→ [Page C.6–15](#), *Closing the patient*

✧ Select **Queue > Voice Output > Organize**.

The **Organize Voice Output** dialog window opens.



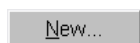
(1) Default voice output (black)

(2) User-defined voice output (blue)

Recording new voice outputs

The **Instructions** selection list under the **Organize Voice Outputs** dialog window uses a loudspeaker symbol to identify files that are available on your MR system. You can now record additional commands.

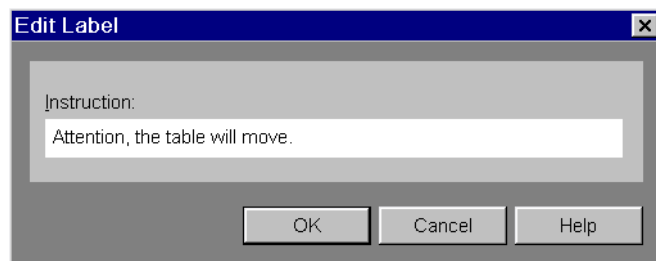
Establishing labels



- ✧ Select the language for the new instructions in the **Organize Voice Output** dialog window.

- ✧ Click **New...**

The **Edit Label** dialog window opens.



- ✧ Enter a label for the new voice output.
- ✧ The label cannot contain more than 31 characters and has to start with a letter. The following characters cannot be used: ?, ", *, |, :, \, /, >, <.
- ✧ Click **OK**.

The new label is appended at the end of the list. It is available in all languages.

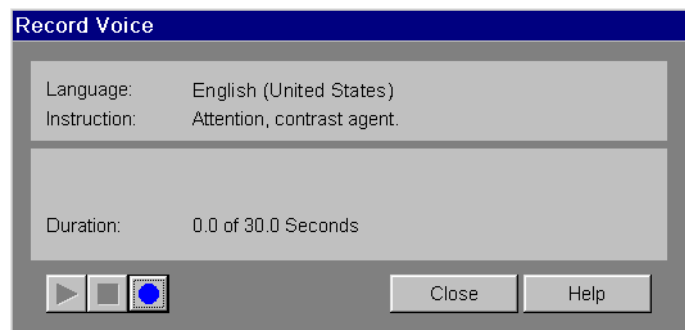
Recording new voice outputs



For this purpose, you need the intercom.

- ✧ Click this button in the **Organize Voice Output** dialog window.

The **Record Voice** dialog window opens.



The dialog window shows the language selected as well as the label for the voice output that is to be recorded.

- ✧ If your *syngo* Acquisition Workplace includes a volume display, the dialog window includes a bar that lets you monitor the volume.
- ✧ Depress and hold this button at the intercom.



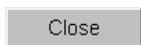
- ✧ To start recording, click this button in the **Record Voice** dialog window.
- ✧ Record your new instructions via the microphone of the intercom.

The duration of recording is displayed (max. 30 seconds).



To ensure the quality of your recording, we recommend that you begin recording in a normal voice at 30 cm from the intercom. The volume slider in the exam room has no effect on the recording volume.

→ [Page C.12–20, Setting the volume](#)



✧ To end recording, click this button in the **Record Voice** dialog window.

✧ Release this button at the intercom.

✧ Click **Close** to close the dialog window.

Testing voice output



✧ To check this voice output, click this button.

Identical to examinations, the voice outputs are played back via the loudspeaker or the headphones.

→ [Page C.12–8, CV/CBT switch](#)

Changing voice outputs

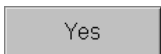
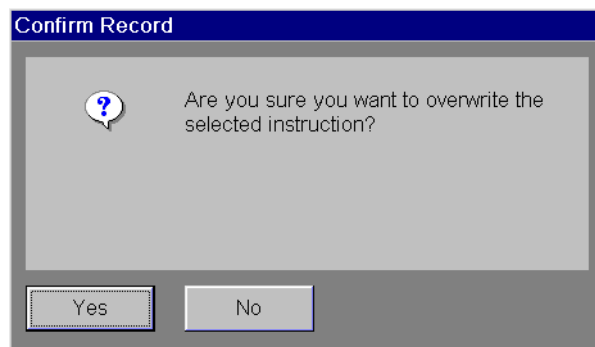
- ✧ Go to the **Instruction** selection list of the **Organize Voice Output** dialog window and select the instructions you want to change.

→ [Page C.12–9](#)



- ✧ Click this button.

The **Confirm Record** dialog window is displayed.



- ✧ Click **Yes**.

The **Record Voice** dialog window opens.

You can now record the patient instructions.

→ [Page C.12–11](#), *Recording new voice outputs*

Sorting voice outputs

- ✧ Select the required language from the **Instructions** selection list of the **Organize Voice Output** dialog window.
- ✧ To move the selected voice output toward the top in the selection list, click **Up**.

Or

- ✧ To move the selected voice output toward the bottom in the selection list, click **Down**.

Deleting user-defined voice outputs

Voice outputs that are no longer required can be deleted.

- ✧ If you delete a voice output needed to automatically play program steps, you have to speak the patient instructions via the intercom while processing the respective program steps.
- ✧ Default voice output may not be deleted.
- ✧ The changes you make in the **Organize Voice Output** dialog window are implemented immediately. It is not possible to undo them.

Deleting a voice output from the current language

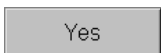
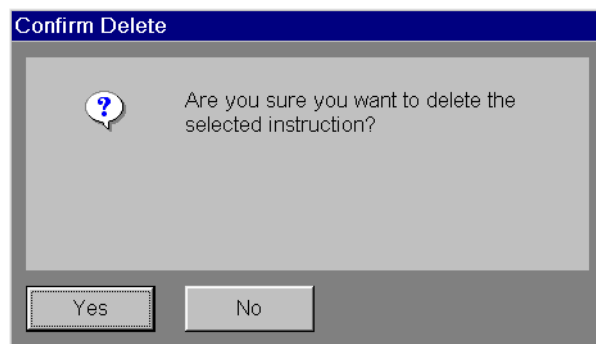


- ✧ Go to the **Instruction** selection list of the **Organize Voice Output** dialog window and select the instructions you want to delete.
→ [Page C.12–9](#)

- ✧ Click this button.

- ✧ The button is active only when you locate a voice output for the respective instruction.

The **Confirm Delete** message window opens.



- ✧ Click **Yes**.

The loudspeaker icon in front of the label disappears and the voice output is no longer available in the selected language. Deleting voice outputs in the current language does not automatically delete them in other languages.

Deleting a voice output from all languages

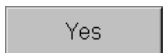
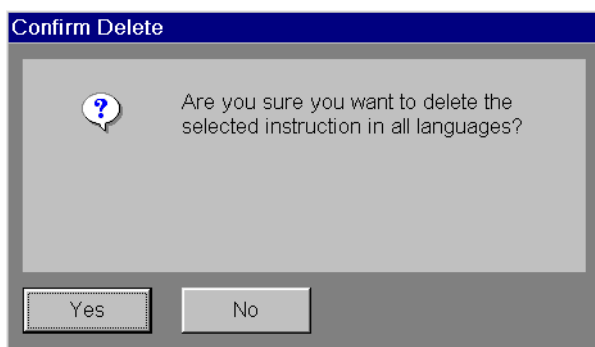


- ✧ Go to the **Instruction** selection list of the **Organize Voice Output** dialog window and select the instructions you want to delete.
→ [Page C.12–9](#)

- ✧ Click **Delete**.

➤ The button is not active, if a default instruction is selected.

The **Confirm Delete** message window opens.



- ✧ Click **Yes**.

The voice output will be deleted from all languages and the label will be removed from the instruction list.

Configuring voice outputs

The default language set for all patients is established in the **Exam UI Configuration - Voice Output** dialog window.

This includes:

- ☐ Standard patient language
- ☐ Standard volume

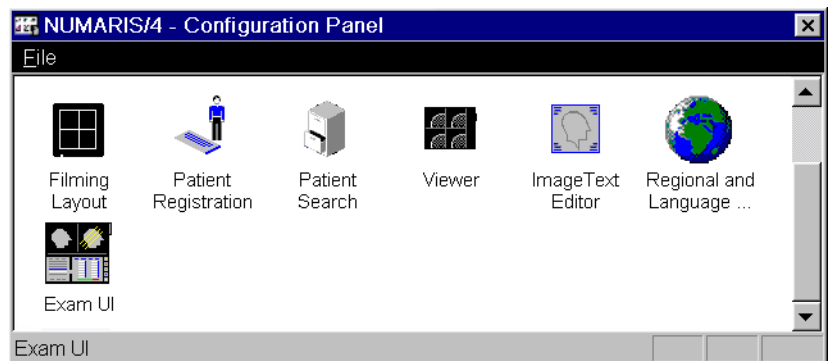
➤ Go to the **Voice Output Properties** to adjust the properties of the voice outputs to the requirements of the patient.

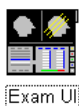
→ [Page C.12–6](#), *Changing the properties of voice outputs*

Open the setting dialog

✧ Select **Options > Configuration**.

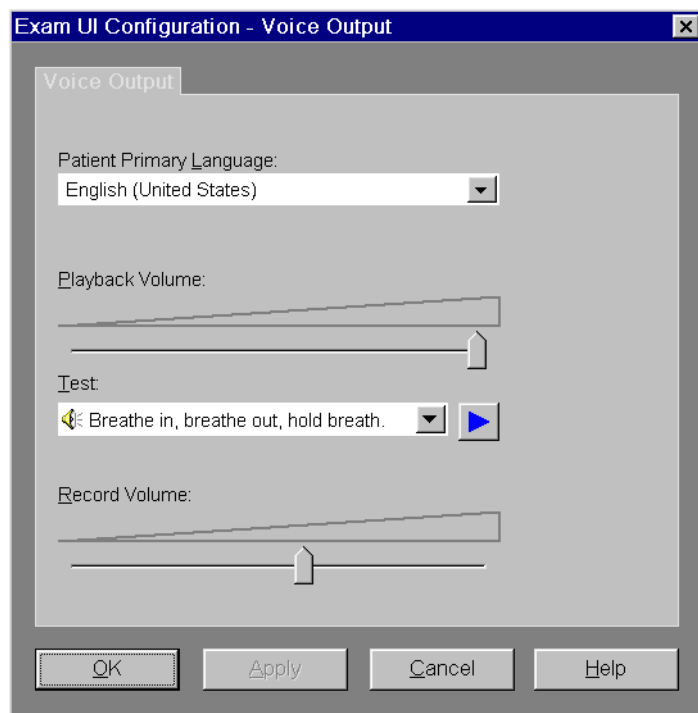
The **NUMARIS/4 - Configuration Panel** dialog window opens.





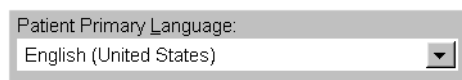
✧ Double-click this icon.

The **Exam UI Configuration - Voice Output** dialog window opens.



Selecting the patient's language

Set the language most commonly used by your patients at your MR system.



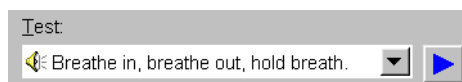
- ✧ Select the required language from the **Patient Primary Language** selection list.
- ✧ You are able to select languages only if your MR system contains at least one patient instruction in this language.

Setting the output volume



- ✧ Slide the **Playback Volume** regulator to the right (left) to increase (decrease) the volume of the voice output in the examination room.

Testing voice output

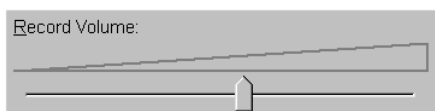


- ✧ Select the required language from the **Test** selection list.
- ✧ Click this button.



Setting the volume

You can set the volume for playback of the voice output you have recorded.



- ✧ Slide the **Record Volume** regulator to the right to increase the volume and to the left to reduce the volume of the voice outputs.

Closing the dialog window

- ✧ Click **OK**.

Your changes are stored and the dialog window closes.

The sequences and parameter settings of frequently performed MR examinations are stored in your system as scan programs. The **Exam Explorer** provides you with an overview of the measurement programs available.

The programs are subdivided into two groups: Siemens programs and customer programs, each with a subdirectory (tree). The SIEMENS tree also contains libraries Siemens created for specific clinical questions. From these, you are able to compose your own measurement programs.

- ⇒ Measurement programs offered as defaults in the SIEMENS tree cannot be changed.
- ⇒ The customer programs are offered for selection on the program card.
→ [Page C.3–14](#), *Selecting a measurement program*

The **Exam Explorer** allows you to manage and maintain your measurement programs. You can utilize it for the following tasks:

- ☐ Copying Siemens measurement programs to the customer tree
- ☐ Changing measurement programs of the customer tree
- ☐ Exchanging measurement programs or protocols with other **syngo MR** systems.



Usually you are not working directly in the **Exam Explorer** during the examination in progress. In some cases, you may use the **Exam Explorer** to search for measurement programs or protocols during an examination and transfer them directly to the queue.

The **Exam Explorer** is primarily a tool for managing and maintaining your measurement programs. If you want to change a measurement program immediately prior to an examination (and for this particular examination only), use the queue in the program control instead.

The **Exam Explorer** provides functions for printing out tables of contents as well as individual protocols including an overview of the parameter settings.

→ [Chapter C.14](#), *Printing Protocols*

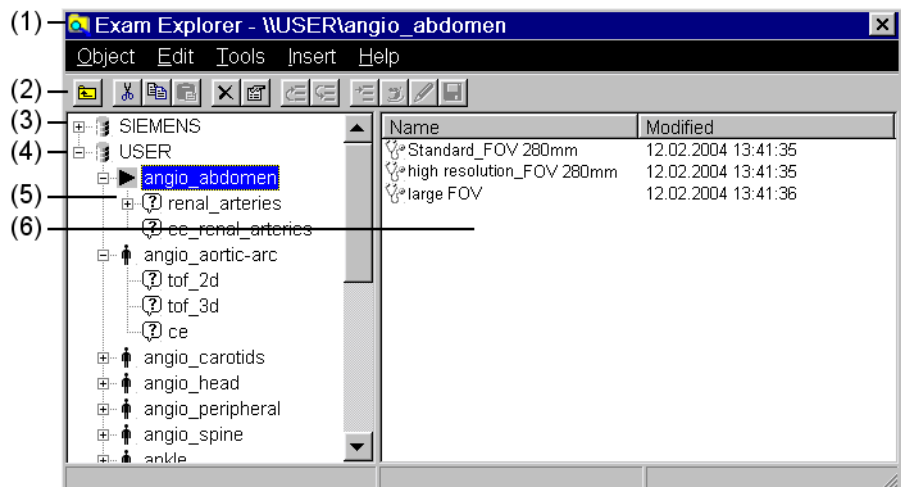
Selecting the Exam Explorer

✧ Select **View > Exam Explorer**.

- Or -

Click this button on the tool bar of the **Exam** task card.

The **Exam Explorer** window is displayed.



- (1) Title bar with path
- (2) Toolbar
- (3) Siemens programs
- (4) Customer programs
- (5) Navigation area
- (6) Content area

Layout of the Exam Explorer

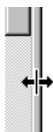
The **Exam Explorer** is similar to the Windows Explorer.

Navigation and content area

The **Exam Explorer** window is divided into two main areas:

- ❑ **Navigation area** (left): Use the mouse to navigate through the various data levels.
→ [Page C.13–5, Data levels](#)
- ❑ **Content area** (right): The navigation area displays the contents of the selected data level.

Moving boundaries



You are able to move the boundaries between the navigation area and the contents area, as required.

- ✧ Drag the boundary line to the left or right, using the mouse.

Data levels

The examinations and measurement programs stored on your system are grouped hierarchically in the **Exam Explorer**.

❑ **Level 1: Customer tree/Siemens tree**

The Siemens customer tree contains a palette of clinically-tested measurement programs installed on your system by Siemens Service. They serve as a basis for preparing your own selection of frequently used measurement programs in the customer tree.

❑ **Level 2: Body part examined**

All examinations for a certain region of the body are grouped together into an examination region (Region).

❑ **Level 3: Examination**

An Examination contains all measurement programs for a certain part of the body or for an anatomical structure of a body region.

❑ **Level 4: Measurement program**

A measurement program (Program) contains a sequence of optimized program steps suitable for clarifying clinical questions.

❑ **Level 5: Program steps**

Program steps may be a measurement protocol (protocol) or a measurement pause (pause). The sequence of protocols and pauses in a measurement program determines the course of the examination.

Showing the contents of a data level

After you selected a data level in the navigation area, its contents are shown in the content area.

(1)	SIEMENS	1	t1_se_sag	03:54
(2)	USER	2	t1 tse sag	03:14
	SequenceRegion	3	t1_tirm_sag_dark-fluid	04:32
(3)	c-spine	4	t1 fl2d opp phase sag	03:52
	general	5	___ ipat ___	
	myelography	6	t1 tse sag pat2	01:44
(4)	library			
(5)	localizer			
	high-res			

View of the contents of a measurement program

- (1) Siemens tree/customer tree
- (2) Region
- (3) Examination
- (4) Measurement program
- (5) Program steps

Symbols of the content types

In the navigation and the contents area of the Exam Explorer as well as in the result list of the search dialog **Search in the database**, the content types are marked with the following icons:

Icon	Content type
	Siemens tree/customer tree
	Region
	Examination
	Measurement program
	Object displayed in the content area
	Pause (displayed in the search dialog only)
	Protocol (displayed in the search dialog only)

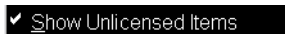
Toolbar

The tool bar provides buttons for selecting frequently used functions:

Icon	Function
	Move up one directory level
	Cut
	Copy
	Paste
	Delete
	Move selection up
	Move selection down
	Edit the protocol
	Display property window
	Insert measurement pause
	Insert steps (not active, if no program is displayed)
	Store the measurement program in the examination database
	Transmit entries to the HIS/RIS
	Receive entries to the HIS/RIS

Showing non-licensed measurement programs

Measurement programs that include non-licensed protocols are normally not displayed in the **Exam Explorer**. However, you may display non-licensed measurement programs and protocols.

A screenshot of a software interface showing a menu item labeled 'Show Unlicensed Items' with a checkmark icon to its left.

✦ Select **Tools > Show Unlicensed Items**.

Non-licensed measurement programs and protocols are now shown as additional gray items in the **Exam Explorer**.

You are able to inform yourself about parameters as well as other properties of the non-licensed protocols.

→ [Page C.13–38](#), *Viewing protocol parameters*

→ [Page C.8–1](#), *Editing the properties for individual protocols*

✦ To hide non-licensed measurement programs and protocols, deselect the menu item.

✦ For an individual, non-licensed measurement program, you are able to establish that it is visible at all times in the directory structure, independent of the setting in the main menu.

→ [Page C.13–25](#), *Editing properties of measurement programs*

Protecting the customer tree

You can write-protect the USER tree to avoid accidental changes to the customer measurement program.

- ✧ Select **Tools > Protect USER tree**.

You are not able to change the measurement programs.

You have to deselect the write protection if you want to work on the customer measurement programs.

- ✧ Deselect **Tools > Protect USER tree**.

Searching for entries

If you know the name of a specific region, examination, measurement program, protocol or pause, you may search for the exact name.

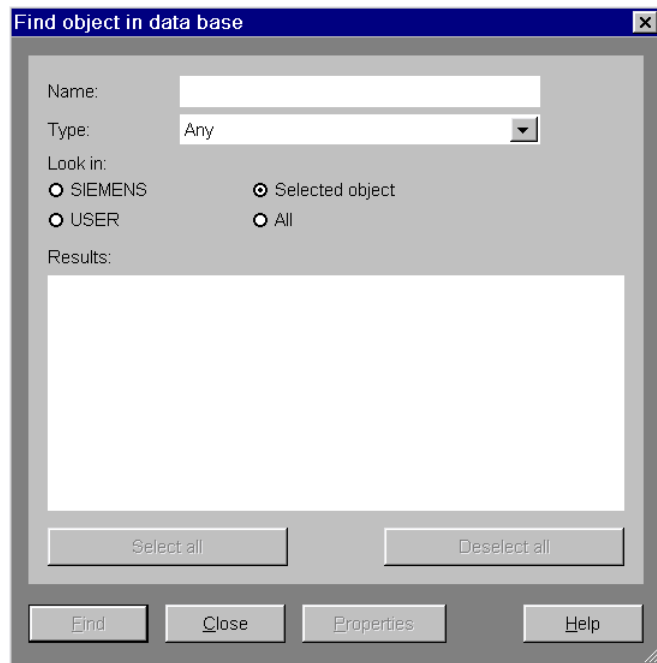
Opening the search dialog

- ✧ Select **Tools > Find**.

- Or -

Press key combination **Ctrl + F**.

The **Find object in data base** dialog window opens.



Making search entries

Entering search terms

✧ Go to the **Name** input field and enter the name of the object you are looking for.

✧ The following characters are not allowed: ^ \ .

If you do not know the entire name or are not sure of its spelling, you may include wildcards in your search.

Wildcard	Effect
*	The asterisk can be inserted at several positions in a search criterion. As a wildcard it stands for a indefinite number of characters.
?	The question mark in a search criterion stands for an unknown character in each case. It can also be inserted at several positions in a search criterion.
Examples:	
Search criterion	Search result
t2*sag	t2_tse_sag Search result always ends with sag
t2*sag*	t2_tse_sag t2_tse_sag_pat2 Search result may also have another ending
t?tse_sag	t1_tse_sag t2_tse_sag The following protocol for instance cannot be found: t1a_tse_sag. "1a" are 2 characters and the wildcard "?" can only stand for one character as wildcard.



Defining the object type

✧ Select the object type (**All** or **Protocol**).

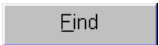
Limiting the search area

The following are available for locating the object of your search:

SIEMENS	Search in the Siemens tree only
USER	In the customer tree only
Selected object	Limit your search only to the marked data levels and their sub-levels
All	Searching the entire examination database

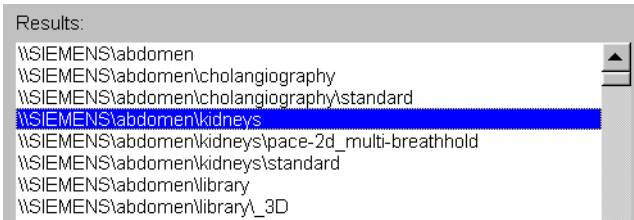
✧ Select the desired option under **Search in**.

Starting a search



✧ Click **Find**.

As a result you receive a list of the objects located including their path:



✧ Double-click the desired entry.



The dialog window closes and the object is displayed in the **Exam Explorer**.

✧ The properties of selected objects in the result list can be changed together.

→ [Page C.8–19](#), *Editing the properties for several protocols*

Editing entries

You are able to copy, delete or rename as well as change the order of the lists in the **Exam Explorer**.



To be able to process customer entries, the USER tree cannot be protected.

→ [Page C.13–10](#), *Protecting the customer tree*

You are not able to make changes in the Siemens tree. You can only look at entries and copy them to the customer tree.

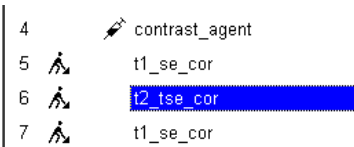
Moving entries

You can define the order of examination regions, examinations and measurement programs displayed in your customer tree, as required.



You may also move individual program steps within a measurement program to modify the program sequence.

- ✧ Select the entry where you would like to rearrange the contents.
- ✧ Go to the contents area and select the entry you want to move.
- ✧ Drag the entry into position with the mouse.



As an alternative, you can move the highlighted entry with the following menu commands or softkeys:

Function	Menu command	Button
Moving it up by one position	Editing > Move Up	
Moving it down by one position	Editing > Move Down	

Copy/cut/or insert entries

You can transfer entries from the Siemens area to your customer tree by copying them.

You can copy, cut, and insert entries in the customer tree.

Copying/cutting

- ✧ From the navigation or contents area, select the entry you want to insert at another location.
- ✧ To transfer a copy of the entry to temporary storage, select **Edit > Copy**.
 - Or -
 - Click this button on the tool bar.



Or

- ✧ To remove the entry and transfer it to temporary storage, select **Edit > Cut**.
 - Or -
 - Click this button on the tool bar.



Paste

- ✧ Select the input in the navigation area where you would like to append the entry.

The entry's contents are displayed in the table of contents area.

- ✧ Select **Edit > Paste**.

- Or -

Click this button on the tool bar.



The entry is appended at the end of the list.

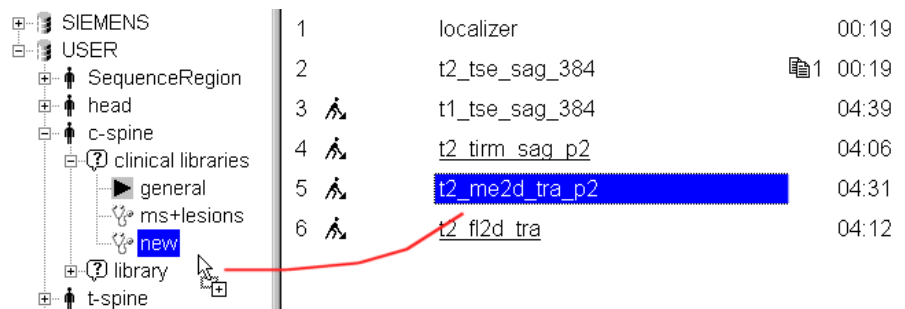
- ✧ If you have marked one of the program steps prior to inserting a program step, the latter will be inserted in front of the marked entry.

Copying/moving with drag & drop

- ✧ To *move* an entry from one object to another, drag the entry to the symbol for the target object in the navigation area.

Or

- ✧ To *copy* an entry from one object to another, drag the entry while pressing the **Ctrl** key to the symbol for the target object in the navigation area.



Transmit entries to the HIS/RIS

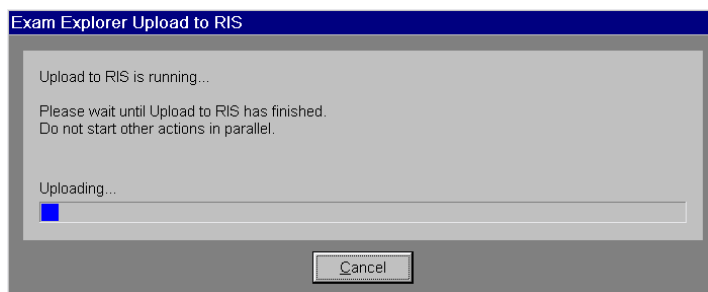
You are able to send inputs from the customer tree to the HIS/RIS. This makes it possible to make measurements which have been designed by you available to all systems connected to the HIS/RIS.

- ✧ Select the entry you would like to send.
- ✧ Select **Edit > Exam Explorer Upload to RIS**.
 - Or -
 - Click this button on the tool bar.



- ✧ If you do not have the **Upload to RIS** menu item or the button, Chorus is configured accordingly. The function is not shown because HIS/RIS does not support transmitting.

The **Exam Explorer Upload to RIS** dialog window opens.



- ✧ Please do not use the following characters for RIS file names:
\\, *, ?, ^.
- ✧ Please refer to the documentation for your HIS/RIS for further details.
- ✧ Smaller entries that you sent to the HIS/RIS can be updated after a modification within the HIS/RIS.
 - [Page C.13–19, Updating entries on the HIS/RIS](#)

Updating entries on the HIS/RIS

Modified single objects or entries (with DICOM code) can be reloaded into the HIS/RIS by a simplified loading process. Thereby the corresponding data is updated within the HIS/RIS.

⇒ Prerequisites:

A Chorus license is available and the protocol update function is activated. The same data must have been previously sent to the HIS/RIS.

✧ Select the entry (or entries) that you want to send.

✧ Call up **Object > Update on RIS....**

The data are updated within the HIS/RIS.

Receive entries from the HIS/RIS

You can load inputs from the HIS/RIS to your customer tree.

✧ Select **Edit > Download from RIS**.

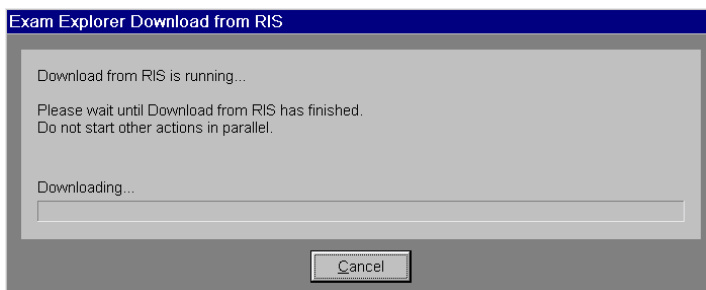
- Or -

Click this button on the tool bar.



✧ Chorus is configured accordingly, if you do not have the **Download from RIS** menu item or the button. The function is not shown because HIS/RIS does not support uploading.

The **Exam Explorer Download from RIS** dialog window opens.



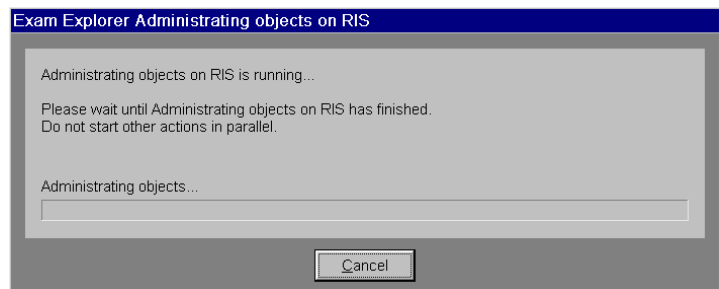
✧ Please refer to the documentation for your HIS/RIS for further details.

Administering inputs in the HIS/RIS system

You are also able to delete inputs in the HIS/RIS or distribute them to other MR systems from the *syngo* Acquisition Workplace

- ✧ Select **Edit > Administrating objects on RIS**.
- ✧ Chorus is configured accordingly, if you do not have the **Administrating objects on RIS** menu item. The function is not shown because HIS/RIS does not support this type of administrative task.

The **Exam Explorer Administrating objects on RIS** dialog window opens.



- ✧ Please refer to the documentation for your HIS/RIS for further details.

Deleting entries

- ✧ Entries with DICOM code cannot be deleted directly from the MR system, as long as they are still stored in the HIS/RIS. These entries can only be deleted from the HIS/RIS and the MR system at the same time.

→ [Page C.13–21](#), *Administering inputs in the HIS/RIS system*

- ✧ Select the entry (without DICOM code) that you want to delete.

- ✧ Select **Edit > Delete**.

- Or -

Click this button on the tool bar.

The selected entry is deleted along with its entire contents.

In case of failed deleting jobs within the HIS/RIS, inconsistencies between the data of the HIS/RIS and the MR system may occur. While entries (with DICOM code) are deleted from the HIS/RIS, they are still available in your MR system.

You can delete these entries manually from the MR system to clear the inconsistencies with the HIS/RIS.

- ✧ Select the entry (the entries) that you want to delete from the MR system.

- ✧ Call up **Edit > Delete RIS objects**.

- Or -

- ✧ Call up **Delete RIS object** (or **Delete RIS objects** if several entries are concerned) using the context menu.

A warning message is displayed.

- ✧ Confirm this message with **OK**.

The highlighted entries are deleted from the MR system.



Deleting inconsistent entries

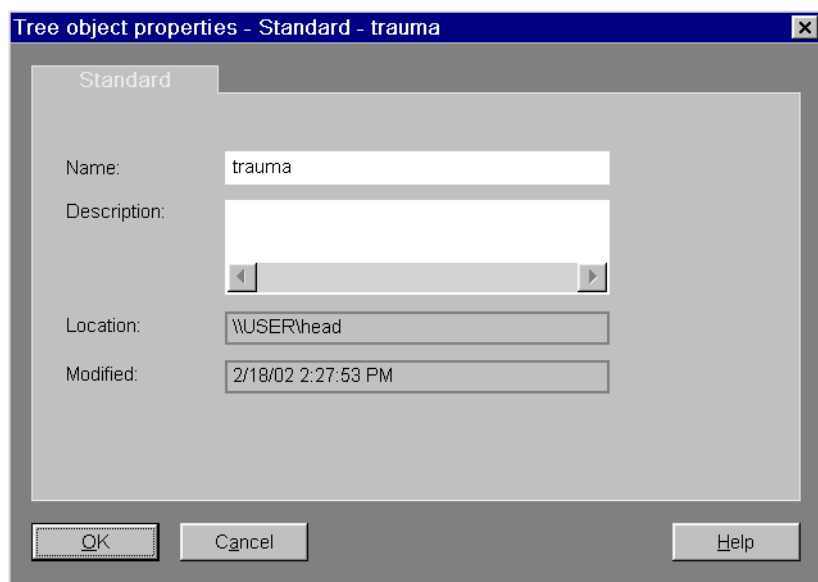
Renaming entries and commenting

You can change the identification of entries in the **Exam Explorer**.

- ✧ Select the entry that you want to rename.
- ✧ Select **Edit > Properties**.
 - Or -
 - Click this button on the tool bar.



The **Tree object Properties** dialog window is displayed with the **Standard** subtask card.



This subtask card shows the name, a brief description, and path of the entry selected. You can also see the date and clock time of the last changes made to this entry.

- ✧ Change the name and/or the short description of the entry.
- ✧ Click **OK**.

The changes are applied and the dialog window closes.

Editing properties of measurement programs

For measurement programs, the **Program properties** dialog windows include two additional checkboxes on the **Standard** subtask card:

- ☐ **Archive**
- ☐ **Visibility - Irrespective of licensing**

The screenshot shows a Windows-style dialog box titled "Program properties - Standard - DefaultProtocols". It has a "Standard" tab selected. The dialog contains the following fields and controls:

- Name:** A text box containing "DefaultProtocols".
- Description:** A large text box that is currently empty.
- Location:** A text box containing the path "\\USER\\SequenceRegion\\UserSequences".
- Modified:** A text box containing the date and time "2/2/2008 4:00:13 PM".
- Archive:** A checkbox that is checked.
- Visibility:** A checkbox labeled "Irrespective of licensing" that is unchecked.
- Buttons:** At the bottom, there are three buttons: "OK", "Cancel", and "Help".

Archive

Archive: ☒

As soon as you change the contents of a measurement program in the **Exam Explorer**, the **Archive** checkbox is automatically activated for the measurement program.

If the **Archive** option is enabled, you may restore measurement programs in case of data recovery or software updates. Measurement programs where the **Archive** option has not been set are usually replaced by original Siemens programs.

You may also manually activate/deactivate the **Archive** function.

- ✧ Select the measurement program.
- ✧ Select the **Program properties** dialog window with the **Standard** subtask card.
 - [Page C.13–25](#), *Editing properties of measurement programs*
- ✧ Activating/deactivating the **Archive** checkbox.

Archive: ☐

Visibility - Irrespective of licensing

When you activate the **Visibility - Irrespective of licensing** checkbox, the measurement program is available in the directory structure of the **Exam Explorer**, even if it is a non-licensed measurement program.

→ [Page C.13–9](#), *Showing non-licensed measurement programs*



However, before you can activate the **Visibility - Irrespective of licensing** option for a non-licensed measurement program, you have to first show *all* non-licensed measurement programs.

✓ Show Unlicensed Items

✧ Select **Tools > Show Unlicensed Items**.

Non-licensed measurement programs are now shown as gray items in the **Exam Explorer**.

✧ Select the required non-licensed measurement program.

✧ Select the **Program properties** dialog window with the **Standard** subtask card.

Visibility: ☒ Irrespective of licensing

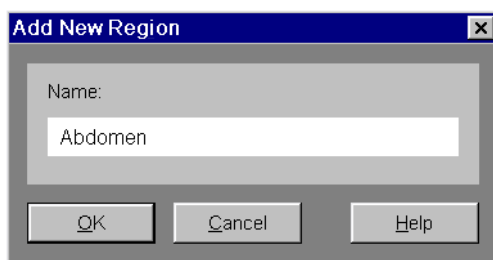
✧ Select the **Visibility - Irrespective of licensing** checkbox.

Generating a new region

You may create new regions in your customer tree.

- ✧ Select the customer tree (**USER**).
- ✧ Select **Object > New**.
 - Or -
 - Select **New** in the context menu.

The **Add New Region** dialog window opens.



- ✧ Enter a name for the new examination region.
- ✧ Click **OK**.

The dialog window is closed and the new region is appended at the end of the list.

Allocating the region to a body part examined

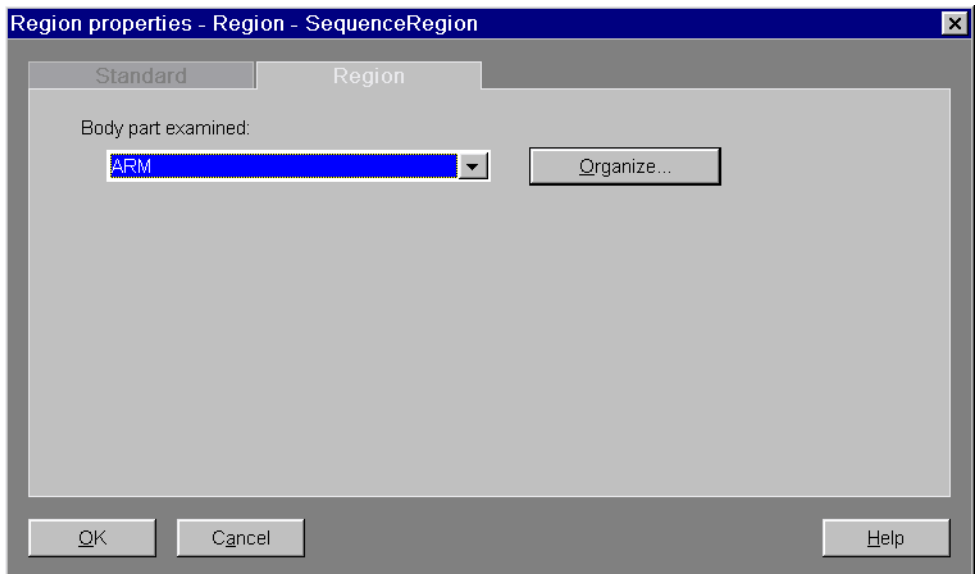
The Region subtask card of the **Region properties** dialog window allows you to assign a newly created examination region or change the assignment of an existing examination region.

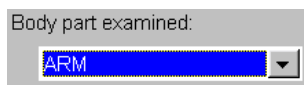
- ✧ Select the examination region.
- ✧ Select **Edit > Properties**.
 - Or -
 - Click this button on the tool bar.



The **Region properties** dialog window opens.

- ✧ Open the **Region** subtask card.





✧ Select the desired examination region from the **Body Part Examined** selection list.

✧ Click **OK**.

Managing the body parts examined list

The **Body Part Examined** selection list shows all the body regions of your system. The three regions are grouped as follows:

☐ **DICOM examination regions**

DICOM regions are regions defined in the DICOM standard.

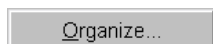
☐ **SIEMENS examination regions**

SIEMENS regions are regions defined by SIEMENS and published in compliance with DICOM specifications.

☐ **User-defined examination regions**

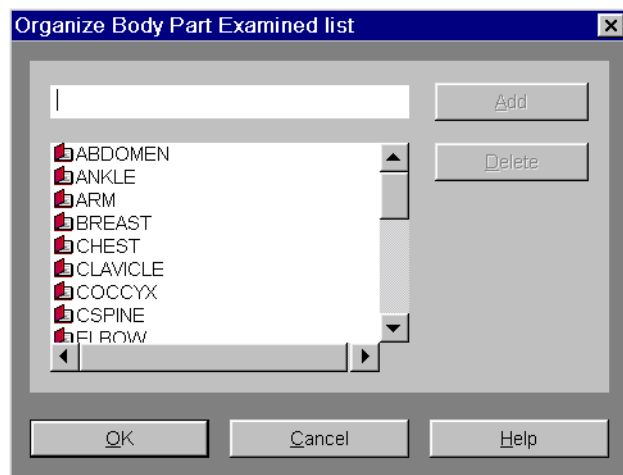
Regions defined by the user.

✧ You may only edit the user-defined exam regions in the list of body parts examined. You may not modify DICOM and SIEMENS regions.



✧ To apply the settings, click **Organize....**

The **Organize Body Part Examined list** dialog window appears.



The icons in the list indicate whether it is a non-modifiable DICOM or SIEMENS region or a user-defined exam region that you can change:



DICOM or SIEMENS body region (cannot be changed)



User-defined exam region (can be changed)

Creating a new examination region

You can add a new user-defined exam region to the list of exam regions.

- ✧ Enter a name for the new exam region in the input field.
- ✧ The name must consist of no more than 16 characters. Acceptable characters are capital letters, figures, blanks, and underscores.

A rectangular button with a light gray background and a thin blue border. The word "Add" is centered in a dark gray font.

- ✧ Click **Add**.

The new body region will appear as a new entry in the list.

Deleting the body part examined

You are able to delete user-defined exam regions.

- ✧ Select the user-defined exam region in the list.
- ✧ Click **Delete**.

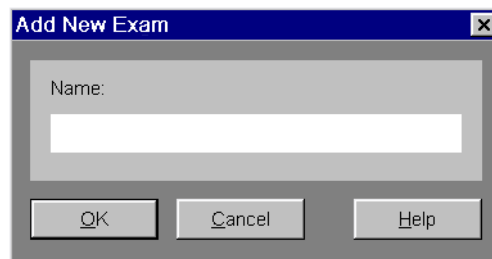
A rectangular button with a light gray background and a thin blue border. The word "Delete" is centered in a dark gray font.

Generating a new examination

You may create new examinations in your customer tree.

- ✧ Select the region for which you want to create a new examination.
- ✧ Select **Object > New**.
 - Or -
 - Select **New** in the context menu.

The **Add New Exam** dialog window opens.



- ✧ Enter a designation in the **Name** input field.
- ✧ Click **OK**.

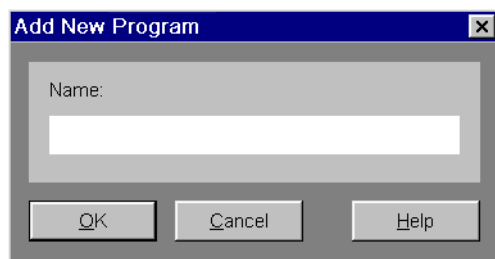
The new examination is appended to the end of the examination list of the respective region.

Creating a new measurement program

You may create new measurement programs in your customer tree.

- ✧ Open the examination for which you want to create a new measurement program.
- ✧ Select **Object > New**.
 - Or -
 - Select **New** in the context menu.

The **Add New Program** dialog window opens.



- ✧ Enter a designation in the **Name** input field.
- ✧ Click **OK**.

The new program is appended to the list of measurement programs of the open examination.

OK

Inserting program steps

You can add individual program steps to a measurement program (protocols or pauses).

If none of the protocols stored on your system is suitable for your new measurement program, you can create a new protocol from a sequence or a reconstructed image and add it to your measurement program.

Selecting program steps

The program steps you insert will usually be from other measurement programs. You can make use of any program steps from customer and Siemens programs.

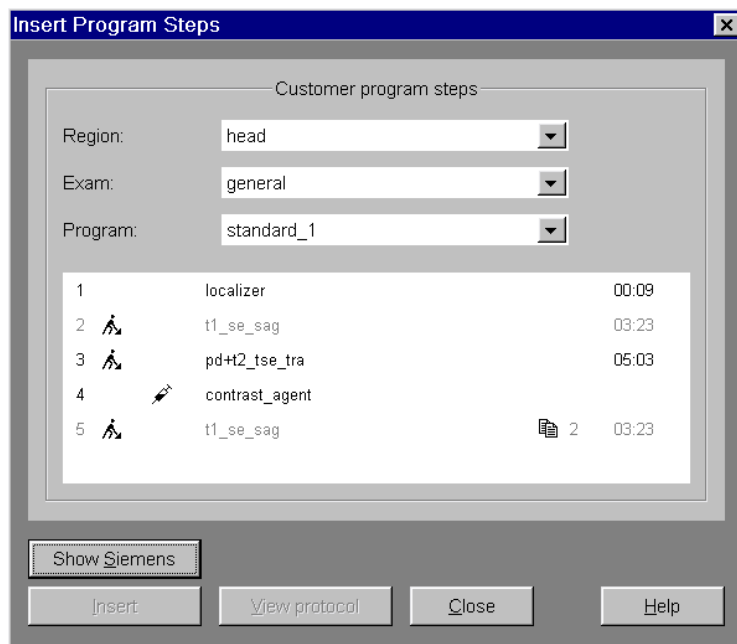
✧ Select **Insert > Program step**.

- Or -

Click this button on the tool bar.

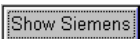


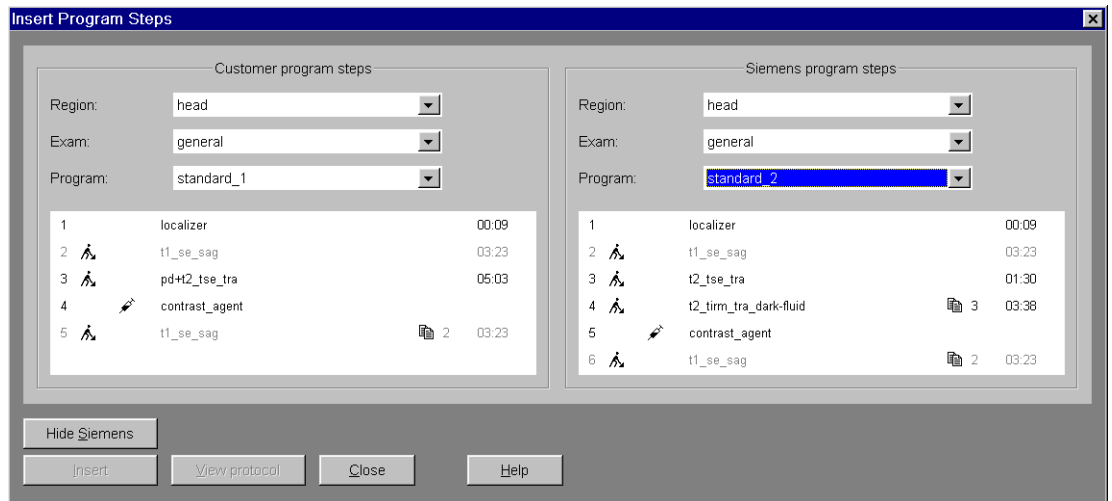
The **Insert Program Steps** dialog window opens.



As far as the open measurement program includes program steps, they will be displayed.

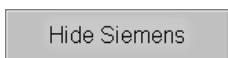
- ✧ Click the **Show Siemens** button to search for program steps in the Siemens tree as well.





✧ Go to another measurement program using the **Region**, **Exam**, and **Program** lists.

✧ **<Select Item>** will be displayed in the selection lists if you have created a new region or examination in your customer tree for which there is no equivalent in the Siemens tree. You now have to search for an appropriate measurement program in the Siemens tree yourself.



✧ Click **Hide Siemens** to hide the Siemens program steps again.

Viewing protocol parameters

✧ During pauses, it is not possible to review parameters.

Before you insert a protocol into your measurement program, you can review the protocol parameters.

- ✧ Select a protocol.
- ✧ Click **View protocol**.

A dialog window opens, in which you can view all measurement parameters in the parameter card stack.

View protocol

The screenshot shows a dialog window titled "\\USER\\SequenceRegion\\SiemensSequences\\DefaultProtocols\\csi_fid (Read-Only)". The window displays various parameters for a protocol, organized into two columns. The parameters include:

- Scan Time: 1:42:30
- Voxel size: 25.0×25.0×50.0 [mm]
- Rel. SNR: 1.00 csi_fid
- Position: Isocenter (dropdown)
- Orientation: Transversal (dropdown)
- Rotation: 0 [deg]
- FoV R >> L: 400 [mm]
- FoV A >> P: 400 [mm]
- FoV F >> H: 400 [mm]
- Slabs: 1
- TR: 1500 [ms]
- TE: 2.3 [ms]
- Averages: 2
- Vol F >> H: 80 [mm]
- Filter: None
- Coil elements: BC

At the bottom, there is a tabbed interface with the following tabs: Routine, Contrast, Resolution, Geometry, System, Physio, BOLD, and Sequence. The 'Sequence' tab is currently selected. Below the tabs are two buttons: 'Close' and 'Help'.

✧ You may not change the parameters in this dialog window.

Inserting a program steps

- ✧ Select the program step you want to transfer to your measurement program.
- ✧ If you want to transfer several program steps from a measurement program, you may extend your selection with the **Ctrl** and **Shift** key.
→ **Basics** in the *syngo* Operator Manual



Insert

- ✧ Click **Insert**.
- or, if you would like to have just one program step marked -
Double-click the program step.

The selected program step will be copied into the measurement program open in the **Exam Explorer**.

- ✧ If a program step is marked in the **Exam Explorer**, the selected program step is inserted ahead of the marking. Otherwise, the selected program step is inserted at the end of the list.

You can now search for additional program steps and copy them into your measurement program.

- ✧ When you have finished working, click **Close**.

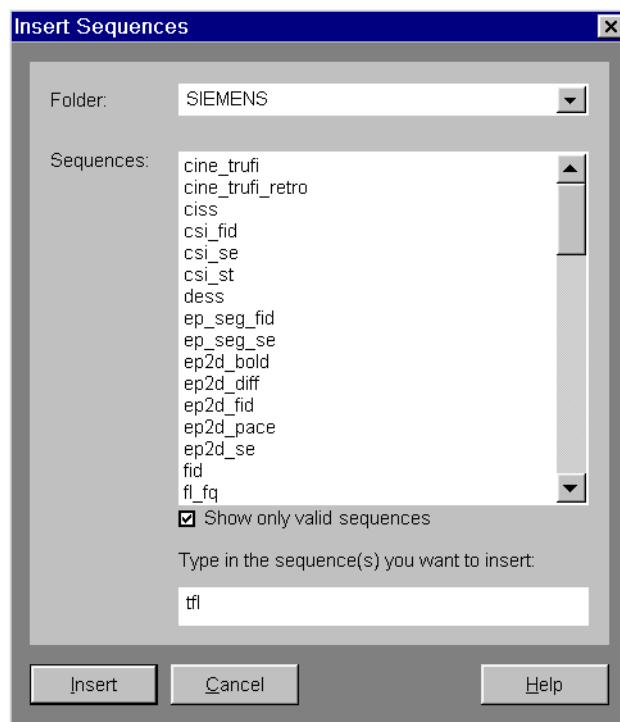
Inserting the sequence as a protocol

A sequence includes the basic command progression for controlling the measurement. Protocols, on the other hand, contain the measurement parameter settings for a sequence.

This is how you take a sequence and create a new protocol for the current measurement program:

- ✧ Select **Insert > Sequence**.

The **Insert Sequences** dialog window opens.





- ✧ Go to the **Folder** selection list. You may either insert a Siemens sequence or a sequence from the customer tree.

The **Sequences** display field now lists all sequences of the selected area in alphabetical order.

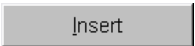
Checking sequences

☒ Show only valid sequences

- ✧ If only licensed as well as executable sequences are to be shown, activate the **Show only valid sequences** checkbox.
- ⇨ This option remains inactive if the folder contains less than 200 sequences.
If there are more than 200 sequences, they are automatically checked.

Selecting sequence(s)

- ✧ Mark the desired sequence in the list.
- Or -
Enter the name of the required sequence in the **Type in the sequence(s) you want to insert** field.
- ⇨ For multiple selections from the list, use the **Ctrl** or **Shift** key.
For multiple selections via the input field, enter the sequences one after the other with a blank between each sequence.


Insert

✧ Click **Insert**.

- or, if you would like to have just one sequence marked -
Double-click the sequence.

The sequence selected is transferred as a protocol to the measurement program.

✧ If you have selected multiple sequences, a protocol is inserted for each sequence.

Generating protocols from a series or an image (Phoenix)

The Phoenix function enables you to reconstruct protocols from previously measured images. You can add protocols to a measurement program in the Exam Explorer or in the queue.

Selecting an image or a series

You can have protocols reconstructed from an image or an entire series.

You have to observe the following differences for Set-n-Go protocols:

- ☐ The corresponding step is reconstructed from an image.
- ☐ The complete protocol is reconstructed from a series.

Phoenix generates all protocols of the Composing group from the resultant image of such a group.



You can get the image and/or series for reconstruction from a variety of locations:

- ☐ From the **Patient Browser**
- ☐ From the image segments in the **Exam** task card (GSP segment or stamp segment of Maestro layout)
- ☐ From the processed part of the queue
- ✧ Select the image or series for reconstruction from the **Exam** task card or the **Patient Browser**.

Reconstructing the protocol

- ✧ To generate the new protocol in the queue, drag the image or series icon with the mouse to the end of the queue.

Or

- ✧ To generate the new protocol in the **Exam Explorer** drag the image and/or series symbol to the requested measurement program in the **Exam Explorer**.

An error message is displayed if the image or series does not meet the requirements. Otherwise, a protocol is generated with precisely the measurement parameters used to measure the image and/or series.

- ✧ A GSP segment provides you with the alternative of selecting **Append to Queue** in the context menu.

Conversion

If you reconstruct protocols from images measured with a different software version or that come from another system, the system performs an additional conversion step. As a prerequisite for successful conversion, the protocol is based on a sequence technique available at your MR system, that is, it is licensed.

Calling up via the **Exam Explorer** automatically triggers conversion, if the protocol cannot be measured. The respective hint is displayed.

When calling up via the **Queue** a dialog is displayed, if the protocol cannot be measured. Either you confirm the conversion job in this dialog or you cancel reconstruction.

If a conversion fails, the protocol will be deleted and an error message will be displayed.

⇒ Changes through conversion will be identified when adapting the version.

→ [Page C.8–17](#), *Displaying upgrade info*

Inserting a measurement pause

You insert a pause, for example, to administer contrast medium to the patient prior to the measurement or to provide him with instructions.

- ✧ From the list, select the program step in front of which you want to insert the pause.
- ✧ If you do not select a program step before you insert the pause, the pause will be appended to the end of the list.
- ✧ Select **Paste > Pause**.
 - Or -
 - Click this button on the tool bar.



The pause is inserted in the list and the **Pause properties** dialog window opens.

→ [Page C.8–26, Editing pause properties](#)

Editing protocol parameters

You edit protocol parameters in the **Protocol Editor**.

- ✧ Select the desired protocol.
- ✧ Select **Edit > Protocol**.
- Or -
- Click this button on the tool bar.



The **Protocol Editor** is displayed with the protocol parameters.

\\USER\SequenceRegion\SiemensSequences\DefaultProtocols\ep2d_se

Scan Time: 0.3 [s] Voxel size: 3.9×3.9×5.0 [mm] Rel. SNR: 1.00 epse

Slice group	1	FoV read	500	[mm]
Slices	1	FoV phase	100.0	[%]
Dist. factor	50	Slice thickness	5.0	[mm]
Position	Isocenter	TR	308	[ms]
Orientation	Transversal	TE	190	[ms]
Phase enc. dir.	A >> P	Averages	1	
Phase oversampling	0	Concatenations	1	
		Filter	None	
		Coil elements	BC	

Routine Contrast Resolution Geometry System Physio Inline Sequence

OK Cancel Virtual Coils... Help

Changing parameter settings

You work in the **Protocol Editor** as you would in the parameter card stack of the **Exam** task card.

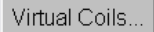
→ [Page C.7–2](#), *Working in the parameter card stack*

Changing a coil configuration

Use the **System - Coils** task card to define the coil elements required for the measurement protocol.

→ [Page P.1–91](#), *Parameter Card Geometry - Inversion*

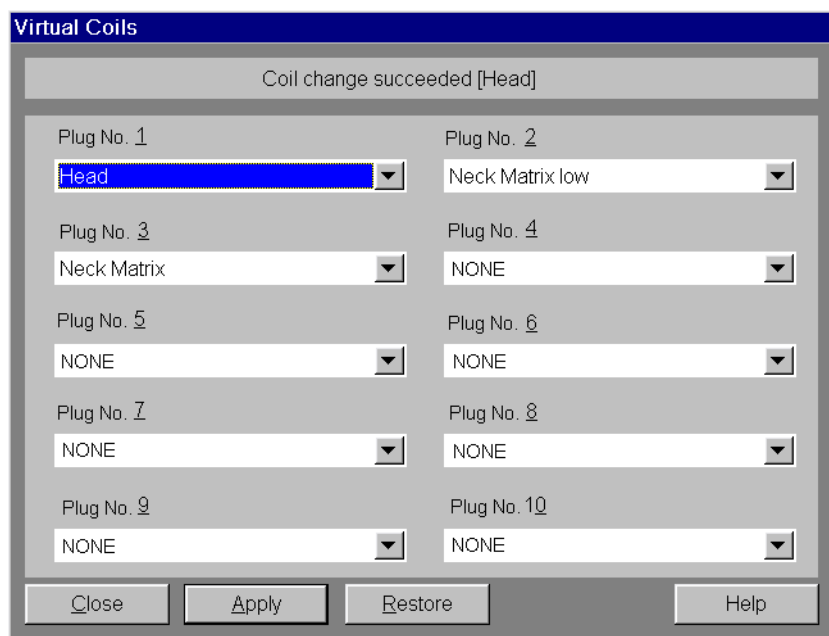
Selecting virtual coils

A rectangular button with a light gray background and a thin black border. The text "Virtual Coils..." is centered on the button in a black, sans-serif font.

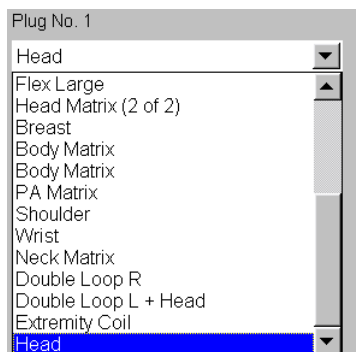
If you are editing a protocol in the **Protocol Editor** and want to select coils different from the ones stored in the protocol, open the **Virtual Coils** dialog window.

✧ Click **Virtual Coils...**

The dialog window is shown.



Use the special selection list to set one coil each for the coil sockets. The list shows only coils that can be connected to the respective coil sockets. The number of coil sockets depends on the system.

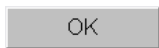


Example: Selection list for coil socket 1

- ✧ Set the required coil configuration using the four selection lists.
- ✧ Click **Apply**.

- ⇒ If you selected array coils as virtual coils, you still have to set the coil elements on the **System - Coils** parameter card.
→ [Page P.1–97](#), *Local coils*
- ✧ Click **Close** to end the dialog.

Applying the changes



- ✧ Click **OK** in the **Protocol Editor** to accept the protocol changes.

All parameter changes are accepted by the protocol and the **Protocol Editor** is closed.

Editing protocol and pause properties

In addition to the measurement parameters that you edit in the **Protocol Editor**, you can set a large number of properties for each protocol and each pause.

→ [Chapter C.8](#), *Protocol and Pause Properties*

Storing measurement programs

Once you have inserted all protocols and measurement pauses into your measurement program, organized them into the desired sequence, and edited your measurement parameters, you will store the results.

- ✧ Open the measurement program that you want to save in the Exam Explorer.
- ✧ Select **Object > Save**.
 - Or -
 - Click this button on the tool bar.



Saving the measurement program under a new name

You are able to store the revised measurement program under a new name and create a new program in this manner. The old measurement program is retained in its original form.

- ✧ Select **Object > Save as**.
 - Or -
 - Select **Save as** in the context menu of the content area.

The **Save Program As [Customer Tree]** dialog window opens.

The dialog window titled "Save Program As [Customer Tree]" contains the following elements:

- Region:** A dropdown menu currently showing "elbow", with a "New Region..." button to its right.
- Exam:** A dropdown menu currently showing "general", with a "New Exam ..." button to its right.
- Programs:** A large, empty list box for selecting a program.
- Name:** A text input field containing the word "standard".
- Buttons:** Four buttons at the bottom: "Save and Edit", "Save", "Cancel", and "Help".

It displays the examination region and examination to which the modified measurement program belongs.

The **Program** list shows all measurement programs stored under this examination in the customer tree. The **Name** text input field shows the previous name of the edited measurement program.

- ✧ The fields **Region**, **Exam**, and **Name** are always filled with values of the original program. **<Select Item>** appears in the **Exam** field, if you have chosen a region containing several programs. In this case you have to select a different examination.

Selecting a region

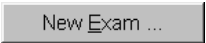
A rectangular button with a light gray background and a thin black border. The text "New Region..." is centered in a black sans-serif font.

- ✧ Go to the selection list **Region** that belongs to your measurement program.
- ✧ If you want to create a new region, click **New Region**.

The dialog window **Add New Region** is shown. You can now create a new examination region

→ [Page C.13–28](#), *Generating a new region*

Selecting an examination

A rectangular button with a light gray background and a thin black border. The text "New Exam ..." is centered in a black sans-serif font.

- ✧ Use the **Exam** selection list to select an examination for your program.
- ✧ If you want to create a new region, click **New Exam**.

The dialog window **Add New Region** is shown. You can now create a new examination region

→ [Page C.13–33](#), *Generating a new examination*

Defining a name

Name	standard
------	----------

- ✧ To create a new measurement program, add a new name into the **Name** input field.
- Or -
- To overwrite an existing measurement program, click the measurement program in the **Program** list.



When you are overwriting an existing measurement program you are replacing it with the new program. You can no longer access the old version of the program.

Save

- ✧ Click **Save** to close the dialog window.

Or

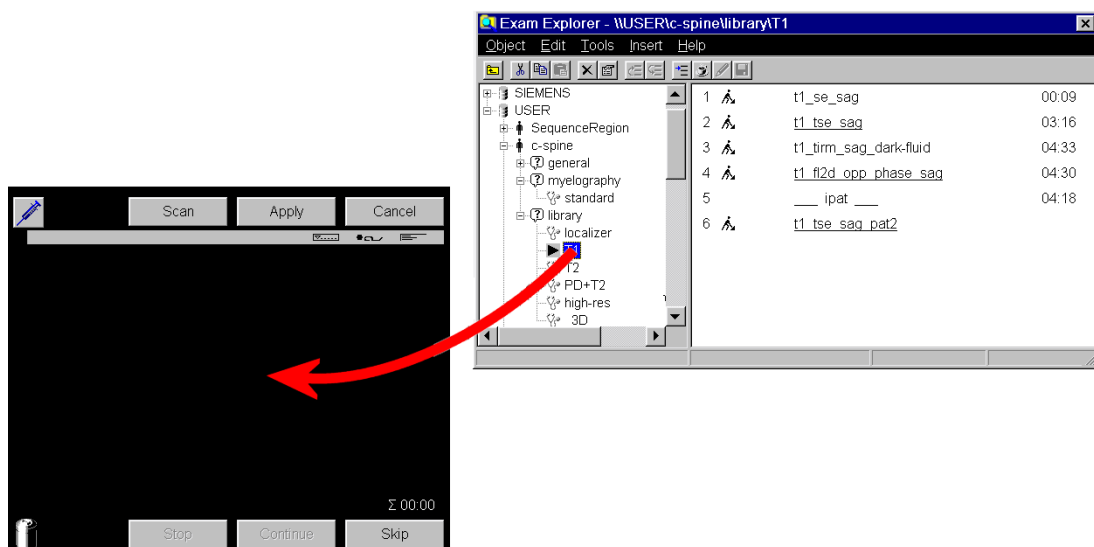
Save and Edit

- ✧ Click **Save and Edit** to save the new measurement program and open it for subsequent editing.

Applying the measurement program to the program control

You can also use the **Exam Explorer** to search for a measurement program and transfer it to the queue of a registered patient.

- ✧ Open the **Exam Explorer**.
→ [Page C.13–3](#), *Selecting the Exam Explorer*.
- ✧ Mark a measurement program or single program steps (one or several) in the navigation or content area.
- ✧ Select **Tools > Append To Queue**.
- Or -
Use the mouse to drag the selection to the queue.



Updating filming layout links

Basics

If you insert images or series in the **Filming** task card, you need to select a suitable film layout for these images. You can automate this task by assigning an existing film layout to all series that have a certain study or series description. When you insert these images or series in the **Filming** task card, the assigned film layout is automatically selected.

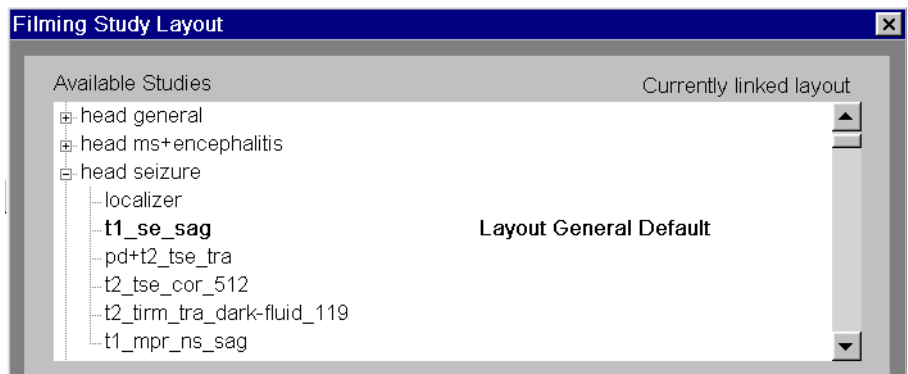
→ **Filming/Layouts with the studies or series** in the *syngo* operator manual.

Updating links

In the **Exam Explorer**, you can automatically update the data contained in the configuration dialog.

✧ Select **Tools > Update Filming Study Layout....**

The data contained in the customer tree of the **Exam Explorer** are transferred to the list of available studies and series descriptions of the **Filming Study Layout** dialog window.





Existing assignments of film layouts for studies and series descriptions are retained, provided that the film layout exists and no other film layout is assigned to these studies and series descriptions.

This means you can make changes in the **Exam Explorer** at any time and update the links without losing your previous allocations.

➤ Updating is only possible if the **Filming** task card is configured.

Exporting / importing entries in the examination database

The Siemens and customer tree are stored in the examination database on your system. You may also access examination regions, examinations, measurement programs and protocols stored on external data media (disk, CD ROM etc.) or on your hospital-wide data network.

You can also export objects from your database to external data media or network addresses, for example, to make your examinations and measurement programs available to other **syngo MR** users.

Importing entries

Via the **Exam Explorer** you are able to import the following into your exam database: examinations, measurement programs or protocols from a network address or an external data carrier.

Open the import dialog

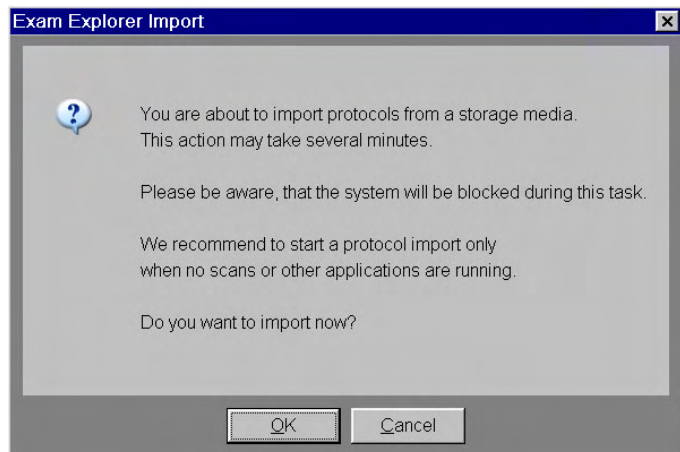
- ✧ Select the input in the navigation area (the directory) where you would like to import data.

Example: Select an examination to import a measurement program.

- ✧ When importing a whole tree, the customer tree has to be selected.

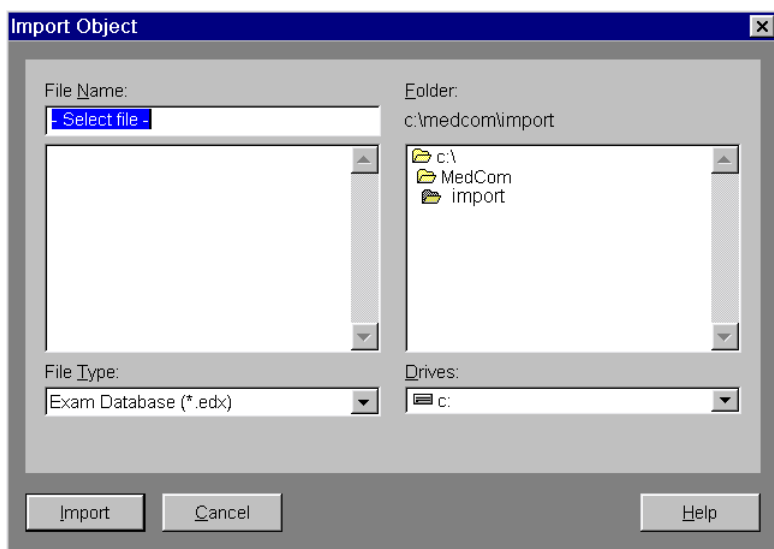
- ✧ Select **Object > Import**.
- Or -
Select **Import** in the context menu.

A message box is displayed.

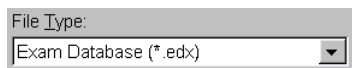


- ⇒ Please note that you cannot perform any other actions during import.
- ⇒ If you want to import now, click **OK**.

The **Import Object** dialog window will be displayed.



Providing import settings

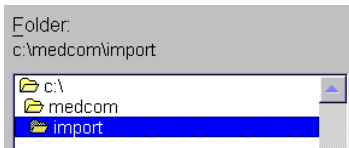


Exam Database (*.edx) is the default setting under **File Type**. No other selection is possible.

- ⇒ The "edx" format is a special file format for the examination database. It is the only format that can be processed by the system.



- ✧ Select the **Drives**/drive or network address from which you want to import data.



- ✧ Under **Folder**, select the directory from which you want to import data.

The contents window now displays all "edx files" of the directory selected.

- ✧ Select the file you want to import.

The name of the selected file is displayed in the **File Name** input field.



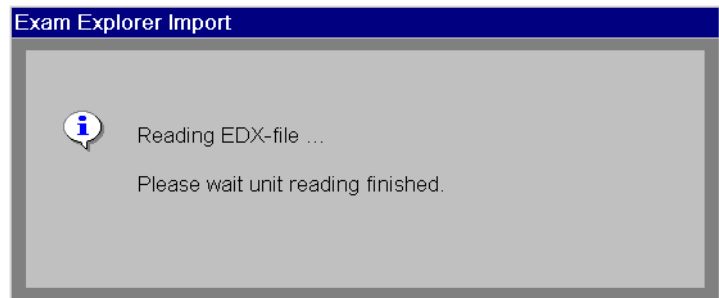
Checking import



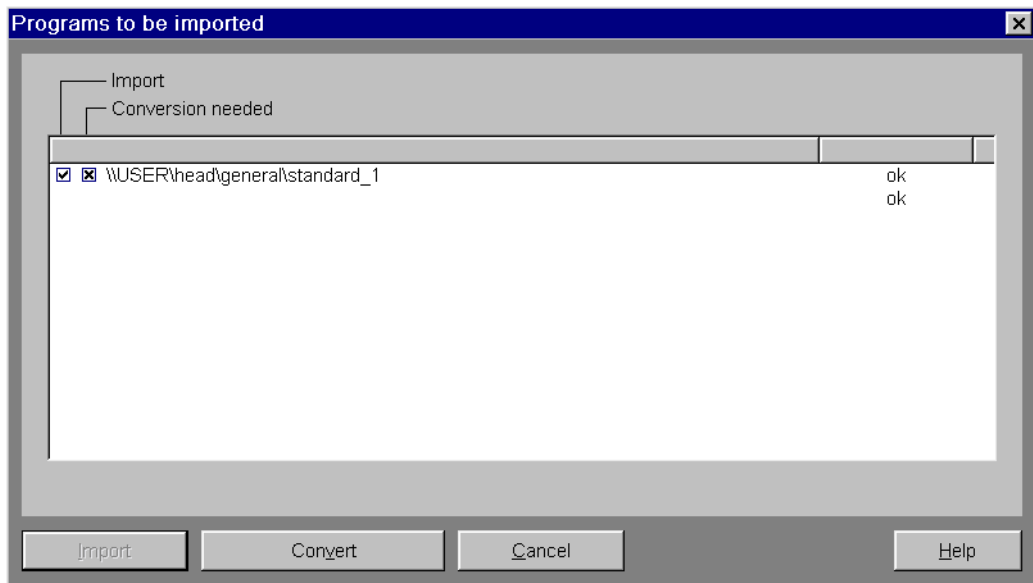
- ✧ Click **Import**.

The system will now check whether importing is possible at the selected position in the navigation area of the **Exam Explorer**.

An error message is displayed when the import fails. Otherwise, the following remark appears:

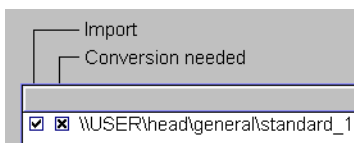


The **Programs to be imported** dialog window opens after importing has finished.



It shows the measurement programs or protocols contained in the read "edx" file.

Performing conversion (special case)



The **Conversion needed** checkbox is active for non-executable measurement programs. These programs have to be converted before they can be imported.

Exclude the measurement programs or protocols that you do *not* want to import. This shortens the time required for the conversion.

✧ To exclude programs from conversion, deactivate the respective **Conversion needed** checkbox.

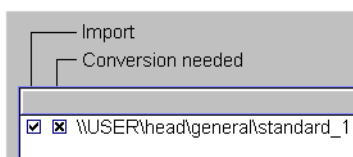


⇒ Subsequently, the program is barred from imports as well.

✧ To start conversion, click **Convert**.

As soon as a measurement program or protocol is fully converted, the cross in the **Conversion Required** checkbox disappears.

⇒ Use the **Stop conversion** button to terminate conversion, e.g., an urgent exam has to be performed.



Go to the **Import** checkbox and select the programs to be imported.

✧ Activate (deactivate) the **Import** checkbox for programs you would like to import (not import).

Starting import



✧ To import the selected measurement program or protocol, click the **Import** function.

⇒ The **Import** button is active only if an importable measurement program is present. It may be necessary that you first successfully convert a measurement program.

The imported measurement programs are appended in the **Exam Explorer** to the end of the directory selected.

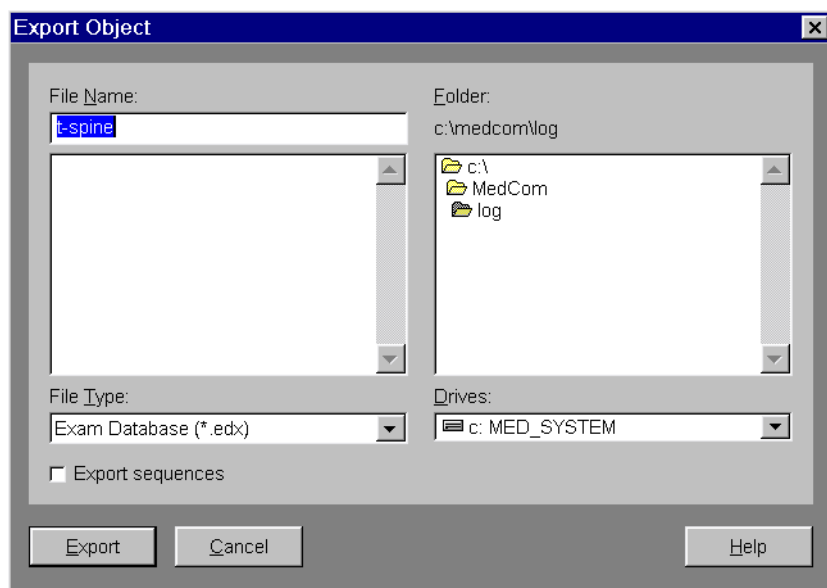
Exporting entries

You can export examination regions, examinations, and measurement programs from your examination database to a network address or to an external data carrier.

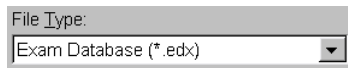
Calling up the export dialog

- ✧ Select the entry you want to export in the navigation area of the **Exam Explorer**.
- ✧ Select **Object > Export**.
- Or -
- Select **Export** in the context menu.

The **Export Object** dialog window opens.



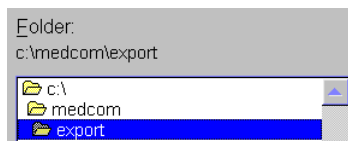
Providing export settings



File Type:
Exam Database (*.edx)



Drives:
c:



Folder:
c:\medcom\export



File Name
TestProtocols

Store objects exported from the examination database must be saved as "edx" files in the target directory. This is the only format that the system can process. This is why **Exam Database (*.edx)** is the default setting under **File Type**.

✧ Choose the selection list from the **Drives** of the target drive and/or the network address.

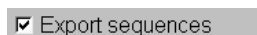
✧ Under **Folder**, select the directory into which you want to export data.

All "edx files" in the directory selected are now displayed in the contents window.

✧ Enter a name for the object to be exported in the **File Name** input field or accept the name suggested by the system.

✧ The object name from the **Exam Explorer** is entered as a suggestion.

Export sequence data (optional)



☒ Export sequences

✧ If you want to export the allocated sequence data as well, activate the **Export sequences** checkbox.

Starting to export



✧ To export the objects selected, click **Export**.



The exported objects are retained on your system. The export function only copies the objects to the target drive or network address.

Printing protocols

From the **Exam Explorer**, you can print out protocols as an overview of all parameter settings and the table of contents of protocols or write them to a pdf file.

The parameter overview enables you to directly compare the parameter settings of the protocols.

→ [Page C.14–1](#), *Printing Protocols*

Closing the Exam Explorer

✧ Select **Object > End**.

- Or -

Click this icon on the title bar of the **Exam Explorer**.



You are able to either print out the parameter overview of protocols, contents of measurement programs, examinations and regions, or write them to a PDF file (export).

The prerequisite for printing is a printer connected to your system either locally or via the network.

The parameter overview enables you to directly compare the parameters of the protocols.

The lists of contents provide you with an overview of the available examinations and scan programs, making selection easier.

⇒ Since printing this data is time-consuming, it is generally not used during clinical examinations.

Protocols are printed out with the
Report-/Druckmodul List & Label® Version 10.0
Copyright *combit*® GmbH 1991-2003.

Printing

This is how you print out protocols:

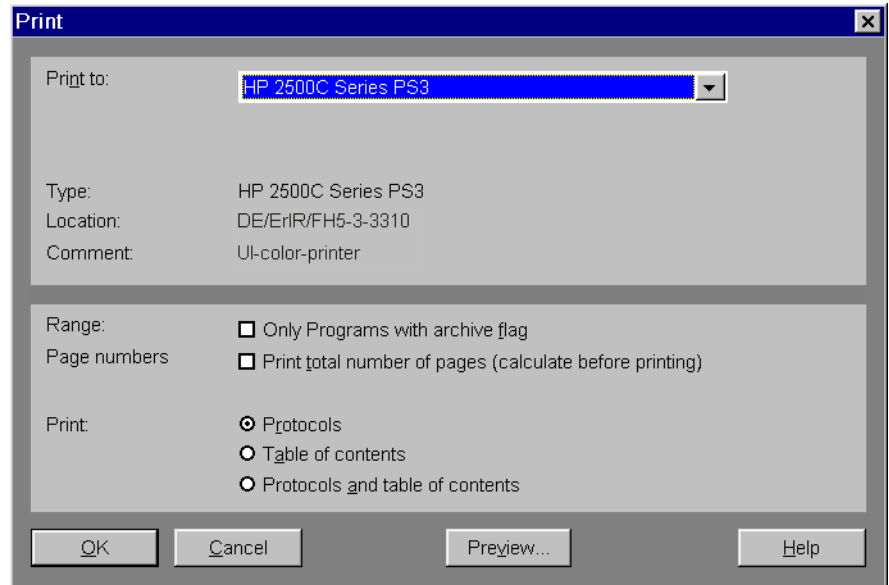
- ☐ Select the print dialog
- ☐ Set the printer
- ☐ Set the print range
- ☐ Select Print Preview (optional)
- ☐ Start printing

Opening the print dialog window

You may select one or more protocols, measurement programs, examinations or regions for printing and then open the print dialog window.

- ⇒ Theoretically, you could print the entire protocol tree in the Exam Explorer. However, preparing such a large volume of data for printing would take a lot of time.
- ✧ Select the requested objects under the **Exam Explorer** or under the **Program** parameter card on the **Exam** task card.
- ✧ Select **Object > Print** in the **Exam Explorer**.
 - Or -
 - Select **Edit > Print Protocol** on the **Program** parameter card.
 - Or -
 - Select **Print** from the context menu.

The **Print** dialog window opens.



Setting the printer

The **Print to** selection list displays all printers available to your system locally or via the network. The standard printer is selected by default.

Printing medium Printer



✧ Select the desired printer from the **Print to** selection list.

Type: HP 2500C Series PS3
 Location: DE/EnR/FH5-3-3310
 Comment: UI-color-printer

The **Type**, **Location** and **Comment** fields provide information about the printer.

- ✧ The default properties of the printers defined in the control panel of the MR system are used for printing.

Setting the print range

You set the print scope and the print options in the lower part of the **Print** dialog window.

Print scope

You can limit the print scope so that only the protocols of *modified* measurement programs are printed:

Range: ☐ Only Programs with archive flag

- ✧ Activate the **Only programs with archive flag** option.
→ [Page C.13–26, Archive](#)
- ✧ If you have did not select a measurement program, examinations, or regions, the checkbox remains inactive.
- ✧ If you do not want to limit the print scope, deactivate the checkbox.

Setting print options

Using the print options you select the data to be printed.

Print: ☒ Protocols
☐ Table of contents
☐ Protocols and table of contents

The print options are as follows:

Print option	Effect
<i>Protocols</i>	An overview of the parameter settings is printed for all selected protocols. The settings are listed in the same way as on the parameter cards.
<i>Table of Contents</i>	A list of contents is printed for the selected measurement programs, examinations, or regions.
<i>Protocols and table of contents</i>	Both the parameter overview and the list of contents are printed.

✧ Select the desired print option.



When setting the print output, remember that printing a large selection of objects can take some time. If necessary, see if you can reduce the number of objects selected.

Setting page numbering

The pages of the printed protocols can be numbered in two ways:

Format	Comment
Page only	Only the page with a plus or minus sign is shown in this format. Example: Page 3+ or Page 5- The plus sign indicates that there are more pages to follow. The minus sign identifies the last page of the print-out. Advantages of this format: Print data using this page numbering format are prepared much faster than page numbering including the running number of total pages.
Page with total pages	This format prints the current page number with the total number of pages. Example: <i>Page 5 / 10</i> .

Page numbers

☐ Print total number of pages (calculate before printing)

✧ When the total number of pages is to be printed out, activate the **Print total number of pages (calculate before printing)**.

Looking at data via print preview

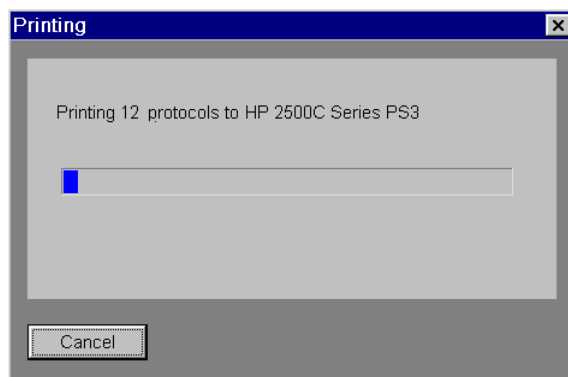
You can check the print files in the print preview.
→ [Page C.14–11](#), *Print preview of protocols*

Starting to print

- ✧ Click the **Print** dialog window and then **OK**.

The selected data are sent to the selected printer. The dialog window closes.

A progress bar is displayed, which you can use, if necessary, to cancel printing.



Exporting data into a file

You are able to print protocols and lists of contents directly into a PDF file. This is useful if the system does not have a printer or if you want to take the data to another location for printing.

- ✧ Open the print dialog window.
→ [Page C.14–2](#), *Opening the print dialog window*

Providing export settings

To export data, the name of the export file and its directory have to be set after the corresponding print medium has been established.

Setting the print medium

Print to: 

- ✧ Select the **Print to** item from the **Adobe PDF File** selection list.

The **File name** input field is active. The target directory of the last data export is the default.

File name: 

As a rule, you can use the following drives:

Drive	Description
T:\	This drive corresponds to the local directory C:\Medcom\User on your MR system.
A:\	Diskette drive
U:\	This drive is generally used to temporarily store data you will later burn on CD-ROM.

⇒ Other network drives may be accessible depending on your MR system.

You can directly enter the drive, directory, and name of the export file as a path in the **File name** input field. As an alternative, you can enter this information in the **Save As** dialog window.

Entering the file name and path directly

⇒ Enter the drive, directory and name of the export file directly into the **File name** input field.

Example: U:\Protokolle\Protokoll1.pdf

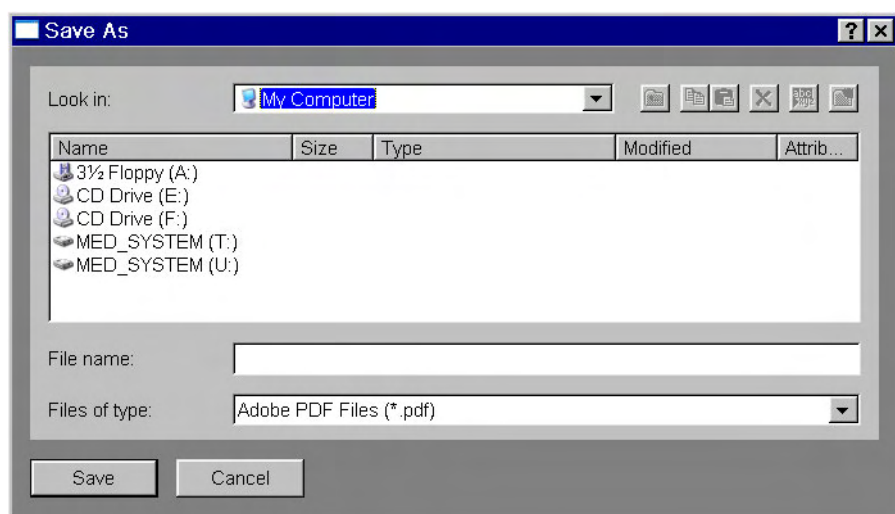
Using the Save As dialog window



Or

⇒ To open the **Save As** dialog window, click this button.

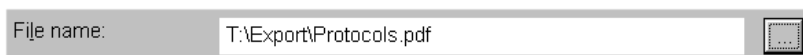
In the **Save As** dialog window you can create subdirectories on drives **A:**, **T:** and **U:**, as well as delete subdirectories and files.



- ✧ Select the target directory for the data export.
- ✧ Enter a name for the export file in the **File name** field.
- ✧ Click **Save** to apply the settings made in the **Print** dialog window.

The **Save As** dialog window closes.

The field **File name** under the **Print** dialog window shows the path and name of the export file. Example:



Performing additional settings

- ✧ Set the print range.
 - [Page C.14–4, Setting the print range](#)
- ✧ You can check the print files in the print preview.
 - [Page C.14–11, Print preview of protocols](#)

Starting to export

- ✧ Click the **Print** dialog window and then **OK**.

The selected data are exported to the desired PDF file. The dialog window closes.

A progress bar is displayed, which you can use, if necessary, to cancel export.

Print preview of protocols

You can check the data under print preview before you start to print or export.

Selecting print preview



- ✧ Click the **Print** dialog window and then **Preview....**

The print data are prepared and displayed in the **Preview** window.

Preview

SIEMENS MAGNETOM Verio syngo MR B15

\\SIEMENS\angiography\Head\TOF-3D-multi-slab\localizer

TA: 9.2 s PAT: Off Voxel size: 2.2x1.1x10.0 mm Rel. SNR: 1.00 SIEMENS: gre

Properties	
Prio Recon	Off
Before measurement	
After measurement	
Load to viewer	On
Inline movie	Off
Auto store images	On
Load to stamp segments	On
Load images to graphic segments	On
Auto open inline display	Off
AutoAlign Spine	Off
Start measurement without further preparation	Off
Wait for user to start	
Start measurements	single

Routine	
Slice group 1	
Slices	1
Dist. factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase enc. dir.	A >> P
Rotation	0.00 deg
Slice group 2	
Slices	1
Dist. factor	20 %
Position	Isocenter
Orientation	Transversal
Phase enc. dir.	A >> P
Rotation	0.00 deg
Slice group 3	
Slices	1
Dist. factor	20 %
Position	Isocenter
Orientation	Coronal
Phase enc. dir.	R >> L
Rotation	0.00 deg
Phase oversampling	0 %
FOV read	280 mm
FOV phase	100.0 %
Slice thickness	10.0 mm
TR	20 ms
TE	8.61 ms
Averages	1
Concatenations	3
Filter	None
Coil elements	HEA/HEP

Contrast	
TD	0 ms
MTC	Off
Magn. preparation	None
Flip angle	40 deg
Fat suppr.	None
Water suppr.	None
SWI	Off
Averaging mode	
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution	
1/4	

Base resolution	256
Phase resolution	50 %
Phase partial Fourier	Off
Interpolation	Off
PAT mode	
Matrix Coil Mode	None
Auto (CP)	
Image Filter	Off
Distortion Corr.	Off
Prescan Normalize	Off
Normalize	Off
Raw filter	Off
Elliptical filter	Off
Geometry	
Multi-slice mode	Sequential
Series	Interleaved
Saturation mode	
Special sat	Standard
None	
Set-n-Go Protocol	
Table position	Off
Table position	H
Table position	0 mm
Inline Composing	Off
System	
Body	Off
HEP	On
HEA	On
SP4	Off
SP2	Off
SP8	Off
SP6	Off
SP3	Off
SP1	Off
SP7	Off
SP5	Off
Positioning mode	
MSMA	REF
Sagittal	S - C - T
Coronal	R >> L
Transversal	A >> P
Transversal	F >> H
Save uncombined	Off
Coil Combine Mode	Adaptive Combine
Auto Coil Select	Default
Shim mode	
Adjust with body coil	Tune up
Confirm freq. adjustment	Off
Assume Silicone	Off
Ref. amplitude 1H	111,000 V
Adjustment Tolerance	Auto
Adjust volume	
Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
R >> L	350 mm
A >> P	263 mm
F >> H	350 mm
Physio	
Tagging	None
Dark blood	Off
Resp. control	Off

Close < > > + - 1:1 Print



Changing the display of the print preview

You can change the display of the print preview via buttons (in the lower part of the dialog window).

Button		Function
		Scrolling forward or backward through the pages
		Scrolling to the first or last page
		Changing the display to twice its size or to half of it
		Returns the display to the default setting

✧ Click the button for the desired function.

Printing from the preview

You can print or export the displayed data directly from the preview.

A rectangular button with a light gray background and a thin black border. The word "Print" is centered in a black, sans-serif font.

- ✧ Click **Print**.

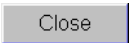
The displayed data is output to the print medium set in the **Print** dialog window.

→ [Page C.14–3, Printing medium Printer](#)

→ [Page C.14–8, Setting the print medium](#)

- ✧ If you did not set an appropriate print medium in the **Print** dialog window, the **Print** button does not work in preview.

Closing the print preview

A rectangular button with a light gray background and a thin black border. The word "Close" is centered in a black, sans-serif font.

- ✧ Click **Close**.

The print preview is closed.

You can use the instruction ranges to define common properties across various program steps in the queue. These are displayed via specific range markers (icons).

There are different types of instruction ranges:

- ☐ Procedure step/Study
- ☐ Body Part Examined
- ☐ Image comment

Procedure step/Study



If you have the necessary license and your system is connected to a RIS, you can obtain work requests via the HIS in the form of procedure steps for your MR system. During patient registration, a note regarding the procedure step is created in the queue. The information for the procedure step is now available for further processing within the hospital.

→ **MPPS** in the *syngo* Operator Manual

Procedure steps and studies can be differentiated via the tool tip.

Studies can be created automatically.

→ [Page C.3–2, Study Split](#)

Body Part Examined



Via the body part instruction range, the images to be measured are provided with an invisible input regarding the body part. This input can be evaluated, for example, via a hospital information system.

The body part to be examined is defined by selecting the measurement program during patient registration.

Image comment



Instruction ranges also include the display of image comments to be saved with images in the image database.

Editing the instruction range Body Part Examined

When you select your measurement program, you automatically set the first range marker for the Body Part Examined. As a result, the images to be measured obtain an invisible entry regarding the Body Part Examined.

You can change the presettings for the examination region in the **Exam Explorer**.

→ [Page C.13–29](#), *Allocating the region to a body part examined*

Changing the instruction range of the Body Part Examined

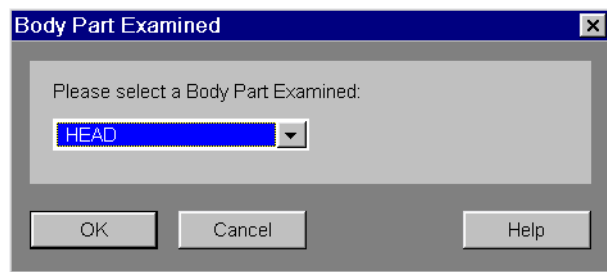
If you determine during the examination that you would like to examine another body part, you can change the region set.

✧ Select **Queue > Change Body Part Examined**.

- Or -

Select **Change Body Part Examined** from the context menu of the program control.

The following dialog window will be displayed:



✧ Select a new examination region from the selection list.

✧ Click **OK**.



A range marker for the new Body Part Examined is inserted.

1	localizer	
2	t1_se_sag	
3	pd+t2_tse_tra	
4	t1_se_cor_3mm	05:40
5	t2_tse_cor_3mm_512	4 07:19



- When you place the mouse pointer on the range marker, the designation for the Body Part Examined is superposed temporarily.

Deleting the range marker for the Body Part Examined

- Select the range marker of the Body Part Examined you want to delete.
- Select **Delete** from the context menu of the program control.
The invisible input for the Body Part Examined is no longer entered into the images.
- The images determined by selecting a measurement program are entered again into the Body Part Examined as invisible input.
If the first range marker is deleted, the invisible input in the Body Part Examined is empty.

Editing the instruction range Procedure step

During patient registration, the range marker for a procedure step is automatically generated. The procedure step ensures that all series and consumables used for a study are available in the hospital information system for further editing.



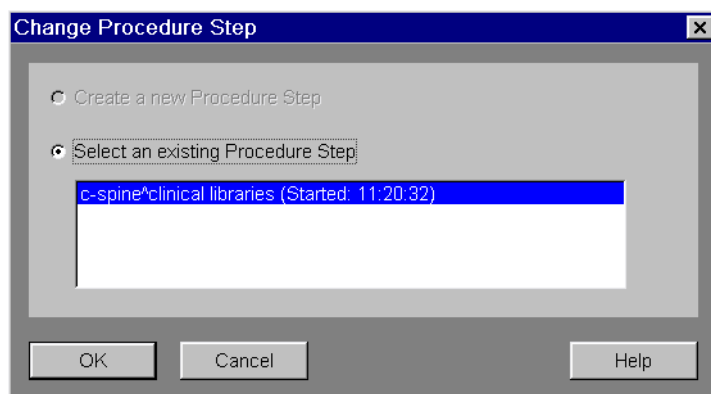
- ✦ When you place the mouse pointer on the range marker of the procedure step, the starting time for the measurement as well as information regarding the procedure step are displayed temporarily.
- ✦ If you have to repeat measurements or insert new protocols, they will be assigned to the existing procedure step.

Adding the range marker for the procedure step

Use the program control to insert an additional procedure step.

- ✦ Select **Queue > Change procedure step**.
 - Or -
 - Select **Change Procedure Step** from the context menu of the program control.

The **Change procedure step** dialog window opens.



The dialog window contains a list of predefined procedure steps.

The option **Select an existing Procedure Step** is active.

✧ Click **OK**.

A new range marker is inserted in the queue.

When you marked a program step, the range marker that you have selected is inserted into the queue above the program step. Otherwise, the range marker is appended at the end of the queue.

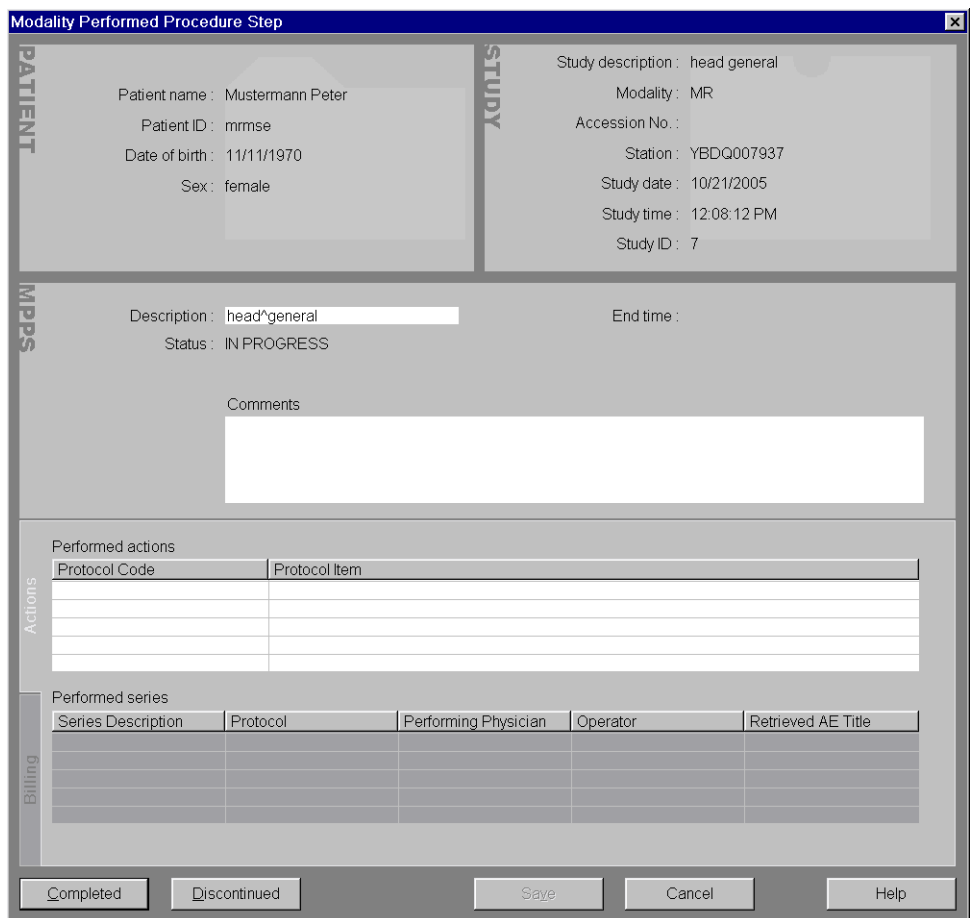
This procedure step records subsequent examination steps for billing purposes.



Editing a procedure step

- ✧ Mark the range marker.
- ✧ Select **Properties** from the context menu.

The **Modality Performed Procedure Step** dialog window is displayed.



The dialog window is titled "Modality Performed Procedure Step" and contains the following sections:

- PATIENT** (left sidebar):
 - Patient name : Mustermann Peter
 - Patient ID : mmmse
 - Date of birth : 11/11/1970
 - Sex : female
- STUDY** (right sidebar):
 - Study description : head general
 - Modality : MR
 - Accession No. :
 - Station : YBDQ007937
 - Study date : 10/21/2005
 - Study time : 12:08:12 PM
 - Study ID : 7
- MPPS** (main area):
 - Description : head*general
 - Status : IN PROGRESS
 - End time :
 - Comments
- Actions** (bottom left sidebar):

Performed actions
Protocol Code Protocol Item
- Billing** (bottom right sidebar):

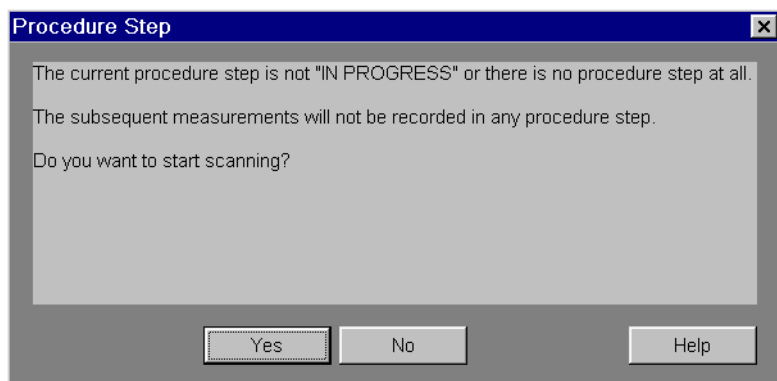
Performed series				
Series Description	Protocol	Performing Physician	Operator	Retrieved AE Title

Buttons at the bottom: Completed, Discontinued, Save, Cancel, Help.

- ✧ Enter the designation in the **Description** field that will be subsequently used for billing purposes.
- ✧ Record all services to be included for billing.
 - **MPPS** in the *syngo* Operator Manual

Response to a missing procedure step

A message box will appear when you start a protocol, if no procedure step exists or if the procedure step has already been completed.



Yes

No

- ✧ To start the measurement, click **Yes**.
- Or
- ✧ In order not to start the measurement, click **No**.

Terminating or closing a procedure step

You can close the procedure step after all measurements of the examination have been completed and all planned post-processing has been finalized (e.g., copying series to the film sheet or printing them). However, cancel the procedure step if the examination cannot be executed as planned.

- ✧ Select the range marker.
- ✧ Select **Properties** from the context menu.

The **Modality Performed Procedure Steps** dialog window is displayed.

- ✧ To cancel the procedure step, click **Discontinued**.



The procedure step icon is shown with a red cross. Subsequent measurements are no longer entered in the procedure step.

Or

- ✧ To end the procedure step, click **Completed**.



The procedure step icon is marked with a green checkmark. Image calculations still in progress are no longer recorded for this procedure step.

- ✧ If you are deregistering a patient who has been examined or you are registering a new patient, you are *not* informed that the procedure step is not yet complete.
 - **MPPS** in the *syngo* Operator Manual

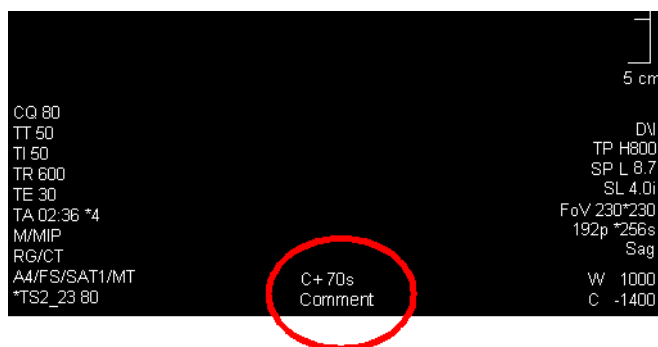
Deleting the procedure step range marker

- ✧ Select the range marker you want to delete.
- ✧ Select **Delete** from the context menu of the program control.

The procedure step is deleted but is retained in the patient database.

Editing image comments

When registering a patient, an empty range marker is automatically generated for image comments. You can enter a comment for all protocols pending measurement. The comment will appear as image text in the series.

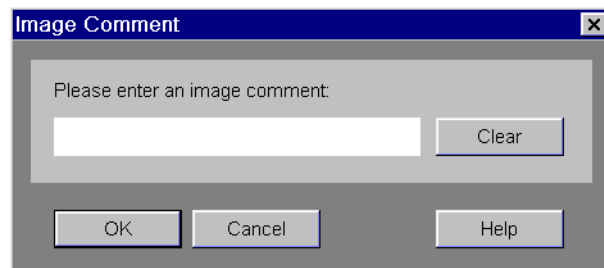


- ✧ If you do not enter an image comment in the program control, the comments entered in the patient registration will be applied.

Inserting an image comment

- ✧ Select the program step where you want the image comment to begin.
- ✧ Select **Queue > Set Image Comment**.
 - Or -
 - Select **Set image comment** in the context menu in the outer margin of the program control.

The **Image Comment** dialog window appears.



- ✧ Enter the desired comment in the field.
- ✧ Click **OK**.

The dialog window closes. An icon for image comments is inserted in front of the program step in the queue. All subsequent images to be measured are provided with this image comment.

- ✧ You can change the image comment by inserting a new image comment at certain locations in the queue.
- ✧ If you place the mouse pointer on an icon for image comment, the comment is displayed temporarily.



Changing the image comment

You can change image comments for program steps that are not yet processed.

- ✧ Select the icon for image comments.
- ✧ Select **Properties** from the context menu.

The **Image Comment** dialog window appears.

- ✧ Click **Delete**.

The input image is emptied.

- ✧ Enter the new text.

- ✧ Click **OK**.

The new image comment is used for images still pending measurement.

Deleting an image comment

- ✧ Mark the icon for the image comment you would like to delete.
- ✧ Select **Delete** from the context menu.

The icon for the image comment is removed from the queue of the program control.

After you have deleted the icon for image comments, no comment is entered for images taken in the subsequent measurements. Images of a measurement in progress are not affected.



Contrast agents may be required for a variety of MR examinations. Paramagnetic contrast agent enhances the signal of many pathological structures.

If you want to run a protocol with contrast agent, use a measurement program that already contains pauses for administering contrast agent.

→ [Page C.3–14](#), *Selecting a measurement program*



All protocols *after* this pause are marked with a syringe icon in the job list.

If you have not yet planned contrast agent pauses in the measurement program, you can insert them into the queue while the examination is in progress. Or, you may utilize other unrelated pauses in the examination sequence to administer contrast agent to the patient and document it accordingly.



The physician is responsible for contrast administration. The recommendations of the pharmaceutical company and expert literature have to be considered. Due to medical advances, these recommendations can change quickly.

The examples provided in this publication with respect to contrast administrations are used only for the purpose of illustrating the operation of the system and the software. They are not a recommendation with respect to use or dosage of contrast agents.

Procedure for a contrast agent measurement

This is the usual procedure for examinations using contrast agent:

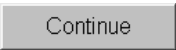
- ✧ Perform a native measurement.
- ✧ Stop the examination prior to the first contrast agent protocol (e.g., by inserting a pause just before it)
- ✧ Ensure that the contrast agent protocol is assigned the "complete" status (no construction worker icon).
- ✧ Document contrast agent administration.
 - [Page C.16–4](#), *Documenting that you administered contrast agent*
- ✧ Move the patient table out of the magnet.



If you have not registered a new patient while you are loading the next protocol after the table reached the HOME position, a dialog window appears querying you whether you forgot the registration or repositioned the patient.

→ [Page C.10–6](#)

Do not lower the table. Otherwise, the examination will be considered complete. You cannot continue the measurement. Instead you have to start the entire examination again.


Continue

- ✧ Administer contrast agent (you can use the injector as well).
- ✧ Return the patient table to the magnet isocenter.
- ✧ Click **Continue** in the pause dialog to start the first contrast agent protocol.

After the pause, all program steps are identified as contrast agent measurements by showing a syringe icon to the left of the protocol name.

This syringe icon is shown next to the patient name.

- ✧ When you notice during the measurement that the contrast agent no longer affects image contrast, reset the syringe icon.

→ [Page C.16–9](#), *Resetting the contrast agent icon*

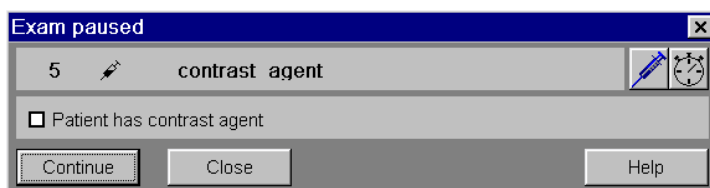
All additional protocols no longer show the contrast agent icon.

Documenting that you administered contrast agent

You identify the measurement as one performed with contrast agent. In addition, you may want to include the type and amount of contrast agent administered.

Documenting the contrast agent in the exam pause

If the measurement program already includes a contrast agent pause, the examination is paused at the required time, and the **Exam paused** dialog window is shown.



- ✧ Activate the **Patient has contrast agent** checkbox.

If the system includes a stimulation monitor, a notice appears on it.

→ [Page C.10–26, Automatic Stimulation Monitoring](#)



- Use this button for documenting the type and quantity of contrast agent administered.

→ [Page C.16–7](#), *Providing contrast agent details*

Starting a contrast agent measurement

Continue

- ✧ Administer contrast agent to the patient.

- ✧ To start the measurement, close the dialog window by clicking **Continue**.

Or

Close

- ✧ Click **Close** and start the contrast agent measurement in the program control or use **F12**.

- The contrast agent dialog can be selected only, when no measurement is active. Entries are dimmed for measurements in progress.

Documenting contrast agent administration without pause

If the program does not include a contrast agent pause and/or it is not possible to insert a pause (e.g., for multiload start measurements), use a pause in the measurement sequence for the contrast agent examination.

- ✧ Select **Queue > Contrast Agent**.

- Or -

Click this button next to the queue.

The **Apply Contrast Agent** dialog window opens.



Apply Contrast Agent

☒ Patient has contrast agent

Volume: 10 ml

Agent name: GADOLINIUM [Catalog...]

Total dose: 50 ml

Active ingredient: Gadolinum

Dilution: 1:1

Concentration: 370 mg / ml

OK Cancel Help

- ✧ Administer the contrast agent (also possible via injection in the magnet).
- ✧ Activate the **Patient has contrast agent** checkbox.
The input fields for volumes and contrast agent names are activated.
- ✧ Enter the name and quantity of the contrast agent administered.
- Or -
Click **Catalog** to select the required contrast agent in the contrast agent catalog.
→ [Page C.16–7, Providing contrast agent details.](#)

Catalog...

- ✧ Click **OK**.

The measurement continues.

- ✧ To document a contrast agent measurement, it is not mandatory to specify the name and quantity of the contrast agent. Contrast agent administration is identified by the respective image text.

Providing contrast agent details



☒ Patient has contrast agent

- ✧ Click this button in the **Exam paused** dialog window.
- Or -
Click this button next to the queue.

The **Apply Contrast Agent** dialog window opens.

- ✧ Activate the **Patient has contrast agent** checkbox.

Catalog...

✧ Click **Catalog**.The **Contrast Agent Catalog** dialog window opens.

Contrast Agent Catalog [X]

	Contrast Agent	Code Value	Active Ingredient	Concentration [mg/ml]	Dilution	Comment
1	Dotarem		GADOLINIUM			
2	Gadolinium		GADOLINIUM			
3	Multihance		GADOLINIUM			
4	Omniscan		GADOLINIUM			
5	Prohance		GADOLINIUM			

Insert agent Delete agent

OK Cancel Apply Help

✧ Select a contrast agent.

✧ Click **OK**.

The dialog window is closed and you return to the dialog window used by you to start the contrast agent catalog (**Apply contrast agent** dialog window or **Examination pause** dialog window).

✧ Click **OK**.

Your selections are applied and the dialog window closes.

The syringe icon is active again and you can start the measurement.

Resetting the contrast agent icon

If you notice during the measurement that the contrast agent no longer affects image contrast, hide the syringe symbol for all subsequent measurements.

- ✧ Select **Queue > Contrast Agent**.

- Or -

Click this button next to the queue.

The **Apply Contrast Agent** dialog window opens.

- ✧ Deactivate the **Patient has contrast agent** checkbox.

All additional protocols no longer show the contrast agent icon.

- ✧ The contrast agent icon is reset as well, when you deactivate the **Patient has contrast agent** checkbox under the **Exam Paused** dialog window.

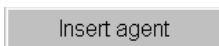


☐ Patient has contrast agent

Editing the contrast agent catalog

You are able to amend or reduce the contrast agent catalog during the examination.

Adding contrast agent



- ✧ Click **Insert agent**.

A new line is inserted into the catalog.

- ✧ Enter a name for the contrast agent, the code value, the active component, concentration, dilution and, possibly, a comment.

- ⇒ You need the code if you want to document the contrast agent used via the HIS (Hospital Information System). The data are entered as consumables.

→ [Page C.15–1, Procedure step/Study](#)

Apply

- ✧ Click **Apply**.

The newly entered contrast agent is now included in the catalog.

Deleting contrast agent

Delete agent

- ✧ Select the contrast agent you want to delete.
- ✧ Click **Delete agent**.
- ✧ Click **OK**.

The contrast agent is removed from the catalog.

Inserting and planning contrast agent pauses

You document the administration of contrast agent in the **Pause properties** dialog window. This applies to new as well as already existing contrast agent pauses. You are able to document while the examination is in progress.

Inserting a new contrast agent pause

When you want to insert a new contrast agent pause, open the dialog window as follows.

✧ Select **Queue > New pause**.

- Or -

Select **New pause** in the context menu in the left margin of the program control.

The **Pause properties** dialog window opens.

Editing the existing contrast agent pause

To edit contrast agent pauses that are already positioned in the queue, open the dialog window as follows:

✧ Select the pause in the part of the queue that has not been measured so far.

✧ Select **Edit > Properties**.

- Or -

Select **Properties** from the context menu of the program control.

The **Pause properties** dialog window opens.

- ✧ Open the **Contrast agent** subtask card of the dialog window.



Planning contrast agent administration



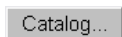
- ✧ Activate the **Contrast agent administration** checkbox.

The **Contrast agent** selection field and the **Catalog** button become active.

- ✧ Choose the required contrast agent from the selection list.

- Or -

Click **Catalog** and select the required contrast agent in the **Contrast Agent Catalog** dialog window.



- ⇒ How to add or delete contrast agent to/from the catalog, is described under
→ [Page C.16–9, Editing the contrast agent catalog](#)

- ✧ Click **OK**.

The dialog window closes. If you inserted a new contrast agent pause, it will be appended to the end of the queue.

- ⇒ It is possible to shift the contrast agent pause to the required position within the part of the queue that still has to be processed.

→ [Page C.6–32, Changing the sequence of program steps](#)

As soon as the pause in the queue is ready to be processed, the measurement stops. The **Exam paused** dialog window opens.

- ✧ Administer the contrast agent.

- ✧ Start the contrast agent measurement with **Continue**.

The dialog window closes.

Continue



Introduction

Physiological movements in the body, e.g., heart beat or respiration, may cause motion artifacts. Such artifacts are recognizable in the image as e.g., noise or blurring.

You can avoid these motion artifacts by synchronizing physiological signals (ECG signal, pulse, respiration) with the measurements. Physiologically-triggered measurements also create cine studies of cyclic motion, such as the function of a heart valve.

Measurement procedure

Two measurement methods are currently used, namely triggering or retrospective gating.

Triggering involves using a characteristic position in the physiological signal curve as the trigger for acquisition. Triggering is possible using ECG signals, pulse signals, respiratory signals, and external trigger signals.

Retrospective gating involves retrospectively correlating the acquired image data with the curve of a physiological signal. Retrospective gating can be applied using ECG signals, pulse signals, and external trigger signals.

Operating elements

The **Physiological Display** window enables you to observe the physiological signal curve and to select the desired triggering procedure. You can also check long-term statistical values.

The signal curves of the sensors are shown on the **PMU display** (optional) which has been integrated into the cover of the magnet.

The special parameters for triggering or retrogating measurement methods are set on the **Physio** parameter card.

Physiological Display window

Opening the Physiological Display

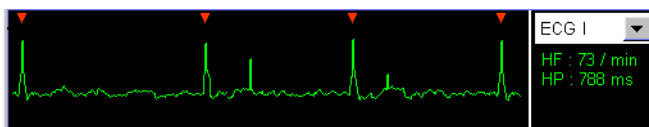


✧ Select **View > Physiolog. Display**.

- Or -

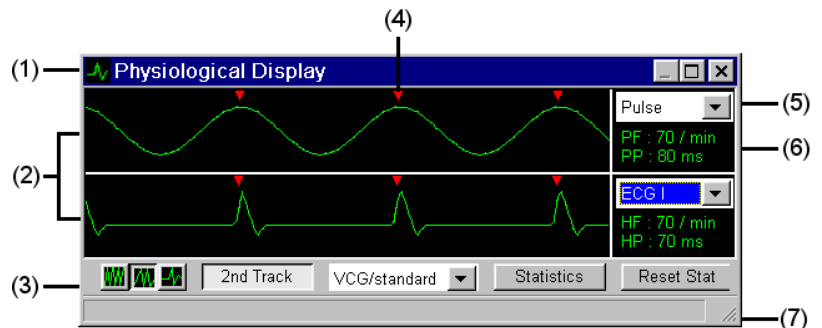
Click this button on the tool bar of the **Exam** task card.

The **Physiological Display** dialog window opens.



Physiological display without a frame

Configuration of the window



- (1) Title bar
- (2) Image area for display of signal curves
- (3) Buttons for setting the speed and operating the physiological display
- (4) Icon for a standard trigger
- (5) List for selecting the physiological signal
- (6) Online statistical values
- (7) Status bar

Activating/deactivating the frame

The window can be displayed with or without a frame. When the frame is hidden, the title line, keys and status line are not visible.

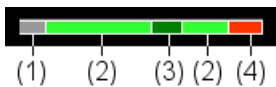
- ✧ To activate or deactivate the frame, select or deselect the **Frame** option in the context menu.

Display of physiological signals

The physiological signals are displayed in the image area of the dialog window. They move from right to left. If you enlarge or reduce the window, the signals in the image area will be displayed enlarged or reduced accordingly.

Displaying the time domain

The various time domains of a measurement are indicated by a bar below the first signal curve in the **Physiological Display** window. Each time domain is assigned a different color.



- (1) Delay time (gray)
- (2) Measurement interval and/or measurement acquisition window (green)
- (3) Measurement pause, e.g., sending saturation pulses (dark green)
- (4) Timeout (red)

The following restrictions apply to the display of time domains:

- ☐ Time domains are only shown for the first signal curve.
- ☐ With display rate **Fast** no time domains are shown.
- ☐ With display rate **Slow** time domains are shown if the selected signal is a respiratory signal or an external signal.

→ [Page C.17–7, Displaying of time domain](#)

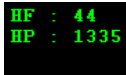
Displaying trigger information

Trigger information is displayed above the physiological curve.



- ☐ Standard trigger - is shown during triggering and retrogating

Online statistical values



HF : 44
HR : 1335

The online statistical values are displayed to the right of the curve. They show the values currently being measured by the system. For an ECG, for example, the current cardiac rate and cycle are displayed.

Message when sensors show a contact error

If sensors show contact errors, the respective error messages are shown in the status bar:

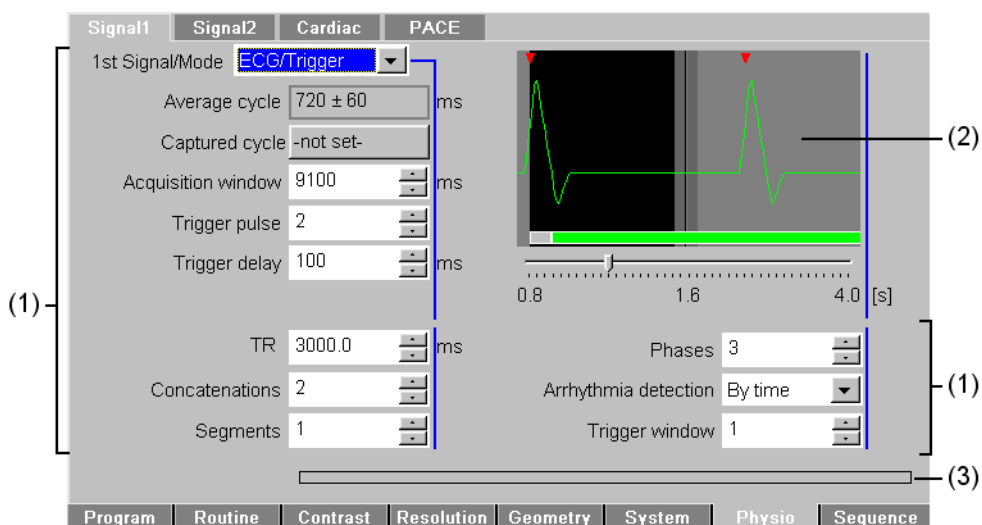
- ☐ Faulty electrodes (for ECG)
- ☐ Sensor not connected (with pulse sensor)

Parameter card Physio - Signal 1

To perform physiologically-controlled measurements, you have to set special measurement parameters in the **Physio - Signal 1** parameter card.

- ✧ Open the **Physio** parameter card.
- ✧ Go to the parameter card **Physio** and open the **Physio - Signal 1** subcard.
- To obtain the most current value possible for the **Average Cycle** parameter, we recommend that you reset the long-term statistics value in the **Physiological Display** dialog window prior to setting the measurement parameters.
→ [Page C.17–12, Resetting the long-term statistical values](#)

Configuration of the parameter card



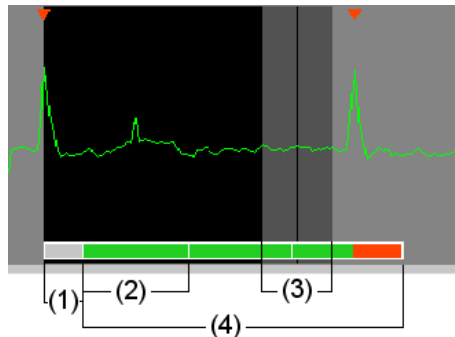
Parameter card Physio - Signal 1 in the Ext/Trigger mode

- (1) Parameters
- (2) Graphic display of the time domains
- (3) Range display

→ [Page P.1–108](#), Parameter Card *Physio - Signal 1*

Displaying of time domain

The top right areas of the parameter card shows the physiological signal and the time domains resulting from the parameters set.



- (1) Delay time
- (2) Repetition time
- (3) Twice the standard deviation
- (4) Measurement acquisition window

⇒ If e.g., the measurement acquisition time set is longer than the time between two trigger events, the acquisition window will overlap the following trigger section. The time overlap is shown in red.

→ [Page C.17–4, Displaying the time domain](#)

Measurements using physiological triggering

For physiologically-controlled measurements proceed as follows:

- ☐ You prepare the measurement by attaching the sensors, setting the volume of the trigger indication, and loading the measurement program.
- ☐ The **Physiological Display** window enables you to observe the long-term statistical values and to select the desired triggering procedure.
- ☐ You have to set additional measurement parameters in the **Physio - Signal 1** parameter card.
- ☐ Start the measurement.

Preparing a physiologically-controlled measurement

For physiologically controlled measurements, you go through a number of special preparations and have a choice of additional setting and control possibilities.

Applying sensors

Position sensors on the patient prior to a physiologically triggered measurement. These sensors pick up the signals used for the physiologically-triggered measurement control.

Refer to the System Manual

- ☐ ECG electrodes for acquiring the ECG
- ☐ Pulse sensor for acquiring the pulse curve
- ☐ Respiratory cushion for acquiring the respiratory curve

Setting the volume of the trigger indicator

You can listen to the trigger signal (a short beep) via the intercom system. Use the slider at the syngo Acquisition Workplace to adjust the volume of the beep.

Refer to the System Manual

Loading the measurement program

✧ Select the required measurement program from the **Program** card.

✧ Transfer the measurement program into the program control.

On completion of the localizer, the first protocol is opened. Adjust the slices graphically and set the measurement parameters.

→ [Page C.5–43](#), *Graphic Slice Positioning (GSP)*

→ [Chapter C.7](#), *Adapting the Measurement Parameters*

Monitoring signals on the physiological display

Opening the Physiological Display



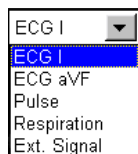
- ✧ Select **View > Physiolog. Display**.
 - Or -
 - Click this button on the tool bar of the **Exam** task card.
- The **Physiological Display** dialog window opens.
- [Page C.17–3, Configuration of the window](#)

Setting the Plotting Speed



- You can choose one of the following three options:
- ☐ Fast
 - ☐ Medium
 - ☐ Slow
- ✧ Click the respective button to select the plotting speed.
 - Or -
 - Select **Fast**, **Medium** or **Slow** in the context menu.

Selecting a signal



- ✧ Go to the upper right in the selection list and choose the physiological signal.
- The signal curve is displayed in the image area.

Displaying a second signal

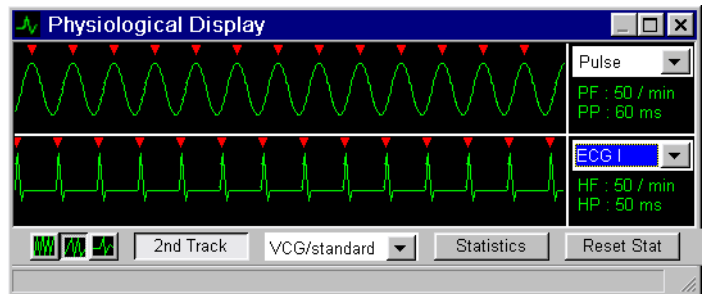
You are able to display two signals simultaneously. The two signals are synchronized in time.

A rectangular button with a light gray background and a thin black border, containing the text "2nd Track" in a black sans-serif font.

✧ Click the **2nd Track**.

- Or -

Select **Second track** in the context menu.



Physiological display with signal 1 (top) and signal 2 (bottom)

Showing the long-term statistical values

In addition to current data, the system also collects signal data over a longer period to calculate the average value, standard deviation, minimum, maximum, as well as the system acquisition window.

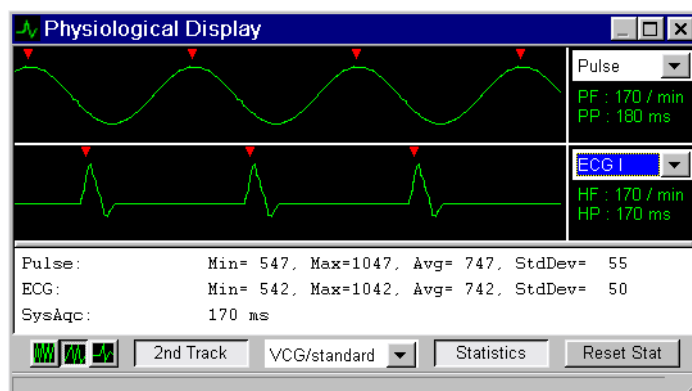
A rectangular button with a light gray background and a thin black border, containing the text "Statistics" in a black sans-serif font.

✧ Click **Statistics**.

- Or -

Select **Statistics** from the context menu.

The long-term statistical data are shown underneath the signal display.



Resetting the long-term statistical values

After the patient is familiar with the measurement situation, the long-term statistical values should be reset so that the system can calculate the current values again. The system acquisition window will also be updated.

Reset Stat

✧ Click **Reset Stat**.

- Or -

Select **Reset statistics** from the context menu.

Selecting the ECG trigger procedure

Various trigger procedures are available for ECG triggered measurements:

Procedures	Comment
VCG/standard	Both ECG channels are used to determine the trigger pulse (default setting).
Channel 1 only	Only ECG channel 1 is used to determine the trigger pulse.
Channel aVF only	Only channel aVF of the ECG is used to determine the trigger pulse.



✧ Select the trigger procedure required from the selection list below the signal curve.

- Or -

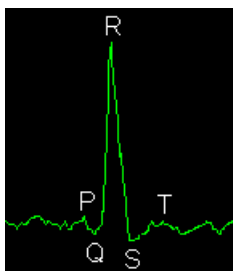
Select the trigger procedure required from the context menu.

☞ You cannot change trigger procedures during the measurement.

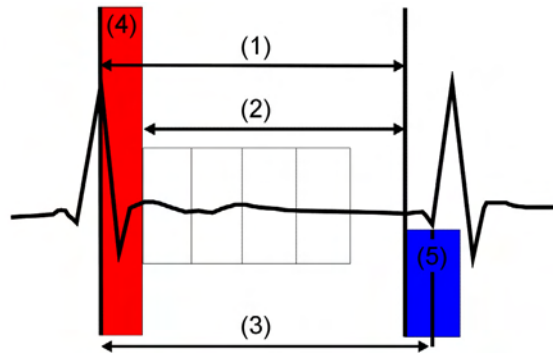
Setting ECG triggering

ECG triggering is especially useful for thoracic and cardiac measurements to avoid image blurring caused by the heartbeat during standard measurements.

The **ECG Signal** shows the action potential of the heart in the form of a curve. The individual curve phases correspond to the respective contraction or relaxation phases of the heart. The R-wave in the QRS complex is used as the trigger point for the measurement.



The signal curve for ECG triggering may be illustrated as follows:



- (1) System acquisition window
- (2) Measurement acquisition window
- (3) Average interval (average cycle)
- (4) Delay time (delay trigger)
- (5) Standard deviation of the average cycle

Set the parameters

1st Signal/Mode ECG/Trigger

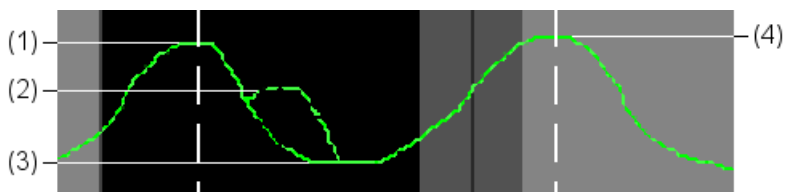
- ✧ Open the parameter card **Physio - Signal 1**.
→ [Page C.17–5](#), *Parameter card Physio - Signal 1*
- ✧ Go to the selection list and choose **ECG/Trigger** from the selection list **1st Signal/Mode**.

The parameter card lists all parameters relevant for measurements with ECG triggering.

- ✧ Set the parameters.

Setting pulse triggering

Pulse triggering is especially useful for suppressing motion and flow artifacts caused by pulsating blood or CSF.



- (1) First pulse wave
- (2) Second pulse wave
- (3) Diastolic blood pressure
- (4) Systolic blood pressure

The first pulse wave (premature pulse wave) is used for triggering. This wave corresponds to the systolic blood pressure.

Set the parameters

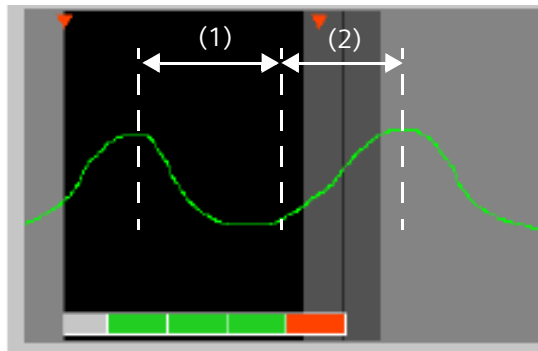


- ✦ Open the parameter card **Physio - Signal 1**.
→ [Page C.17–5](#), *Parameter card Physio - Signal 1*
- ✦ Select **Pulse/Trigger** under **1st Signal/Mode** in the selection list.

The parameter card lists all parameters relevant for measurements with pulse triggering.
- ✦ Set the parameters.

Setting respiratory triggering

Respiratory triggering is used to avoid motion artifacts caused by breathing.



(1) Expiration

(2) Inspiration

Set the parameters

1st Signal/Mode Resp./Trigger

- ✧ Open the parameter card **Physio - Signal 1**.
→ [Page C.17–5](#), *Parameter card Physio - Signal 1*

- ✧ Select **Resp./Trigger** under **1st Signal/Mode** in the selection list.

The parameter card lists all parameters relevant for measurements with respiratory triggering.

- ✧ Set the parameters.

Setting an external trigger signal

An external signal may be used for functional measurements to trigger measurements of the stimulated series.

The external digital trigger signal is supplied via the connection at the column cover of the table.



The rising edge of the signal is used to start the measurement.

Set the parameters



- ✧ Open the parameter card **Physio - Signal 1**.
→ [Page C.17–5](#), *Parameter card Physio - Signal 1*
- ✧ Select **Ext./Trigger** under **1st Signal/Mode** in the selection list.

The parameter card lists all parameters relevant for measurements with an external trigger signal.

- ✧ Set the parameters.

Setting retrospective gating

Some special sequences enable retrospective gating.

The measurement runs without triggering. The acquired image data are retroactively sorted according to cardiac phases.

Contrary to triggering, retrospective gating allows you to measure all cardiac phases including the late diastole.

Retrospective gating may be applied to ECG signal curves, pulse signal curves, and external trigger signal curves. The **1st Signal/Mode** selection list on the **Signal 1** subtask card contains the following additional entries:

- ☐ ECG/Retro
- ☐ Pulse/Retro
- ☐ Ext/Retro

Set the parameters



- ✧ Open the parameter card **Physio - Signal 1**.
→ [Page C.17–5](#), *Parameter card Physio - Signal 1*
- ✧ Select Retrospective Gating under **1st Signal/Mode** in the selection list.

The parameter card shows all relevant parameters.

- ✧ Set the parameters.

Starting the measurement

After you have set all measurement parameters on the parameter cards, transfer them to the MR system.

Apply

✧ Click **Apply** in the program control.

The settings are transferred to the MR system. The protocol is now complete. It starts automatically.

If there are still incomplete protocols in the job list (construction-worker icon), open and edit them like the first protocol of the program so that they can be started automatically as well.

→ [Page C.6–10](#), *Measurement process for routine examinations*

You can use **Inline Display** for the following applications:

- ❑ You are able to view the reconstructed images during the measurement in progress. This allows you to control the images generated and to know when to intervene in a measurement.
- ❑ You interactively localize the requested slice positioning and slice orientation of a real time protocol.
- ❑ You can trace the contrast agent signal in a CARE Bolus measurements before starting the high-resolution protocol.
- ❑ You are able to trace the navigator signal (respiratory curve).
- ❑ You can start a multiple breath-hold measurement, as well as pause and resume it interactively for some sequence types, e.g. *TimCT Oncology*.
- ❑ You can trace the t-test for BOLD measurements on the EPI mosaic images.
- ❑ You are able to filter the display of BOLD images (does not apply to Mosaic BOLD images) to display just the images of a certain slice.
- ❑ For *TimCT Oncology* measurements, you can filter either the reconstructed images or a projection progress image.

Starting Inline Display

You may start **Inline Display** only if you have full access rights to the data of the patient being examined.

⇒ Detailed information regarding all security settings in *syngo* are included in the *syngo* Security Package brochure.

You can change the display and size of **Inline Display**.

⇒ The display includes only images that are loaded automatically from image reconstruction. It is not possible to load manual images into **Inline Display** and then to scroll back to previously displayed images.

Starting Inline Display automatically

The **Inline Display** opens automatically when a protocol with the **Auto open inline display** property is about to run in the job list.

→ [Page C.8–9](#), *Auto open Inline Display*

Starting Inline Display manually



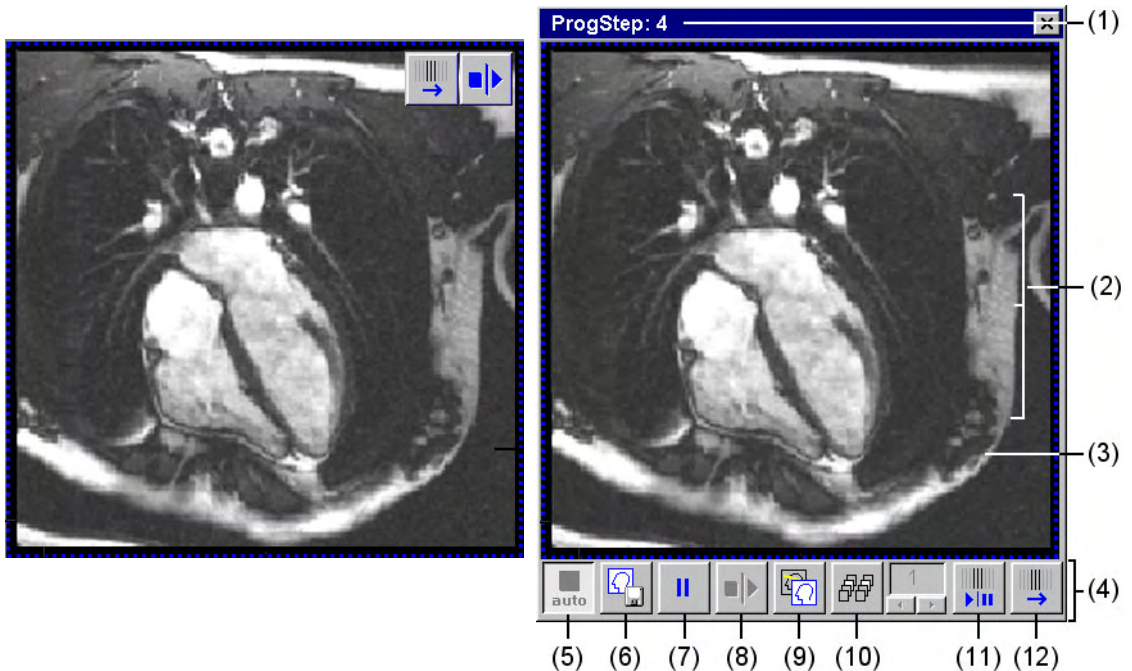
✧ Select **View > Inline Display**.

- Or -

Click this icon in the tool bar of the **Exam** card.

Without a border, **Inline Display** appears at the position of the right GSP segment. If the Zoom/Pan mode is deactivated, you can move **Inline Display** freely.

You are able to show the **Inline Display** with or without a frame
(→ [Page C.18–4](#)).



Inline Display with or without a frame

- (1) Number of the program step
- (2) Scale bar
- (3) Image area
- (4) Control area with icons
- (5) **Auto Store Images** icon
- (6) **Save** icon
- (7) **Display Pause** icon
- (8) **Stop & Continue** icon
- (9) **Copy Image Position** icon
- (10) **Display Filter On/Off** icon
- (11) **Scan/pause** icon
- (12) **Scan breathhold** icon



- ✧ To superimpose the number of the program step without a frame in the **Inline Display**, keep the mouse pointer briefly at the lower left corner (*drag & drop* area).

Changing the Inline Display

- ✧ To activate or hide the frame of **Inline Display**, select the option **Frame On** or off in the context menu.
- ✧ To change the position of **Inline Display**, use the mouse and shift it to the location required.

Setting the size of the Inline Display

The size of **Inline Display** is freely selectable or you can select one of three predefined sizes.

Setting the size freely

- ✧ Place the mouse pointer along the edge of the Inline Display.
- ✧ Drag the Inline Display to the required size holding the left mouse button down.

Image Display is temporarily stopped.



Selecting a predefined size

You can select one of the following predefined sizes:

Inline	The Inline Display has the size of a GSP segment in the three-segment layout.
Large	The Inline Display fills the height of the screen This setting is useful for interventional examinations.
Full screen	The Inline Display fills the entire monitor.

- ✦ Make sure **Zoom/Pan On** mode is switched off in the context menu.
- ✦ To switch between **Inline** and **Large**, double-click the Inline Display.
- Or -
Set the required size with **Set Size** in the context menu.
- ✦ Exit **Full screen** size again by pressing the **Esc** key.

Switching to another task card

- Inline Display closes if you switch to another task card (e.g., **Viewing**).
- ✦ If you return to the **Exam** task card, Inline Display will be open again with the same settings as applicable for manual start-ups. Only the last **Frame On** setting is retained.

Image display

Inline Display shows the images in sequence, similar to a fast slide show.

How reconstructed images are displayed depends on the sequence used and the protocol settings:

- ❑ If image reconstruction provides images very quickly, it may not be possible to display all images. In this case, you will only see the most up-to-date image.
- ❑ If the slices are measured with interleaving, **Inline Display** is initially empty. The images are reconstructed only when the data have been acquired.

Pause the display



- ✧ Click this icon to begin or end a pause.
 - Or -
 - Select/deselect **Display Pause On or Off** in the context menu.

The image visible at the beginning of the pause remains until you end the pause.

All images reconstructed during the pause are not shown. After the pause has ended, the display with the recently reconstructed images appears again.

✧ It is not possible to pause the display if automatic image storage is deactivated.

→ [Page C.8–8](#), *Automatically storing and loading*

Changing the image display

The images are transferred from image reconstruction with standardized window values and aspect ratios. To optimize the display, you can window, zoom, and pan the images.



If displayed very quickly in **Inline Display**, changes made through windowing, zooming, or panning may not be visible.

✧ For this reason, stop **Inline Display** when changing your settings.



Display changes in **Inline Display** are not saved.

Windowing an image

Windowing with the center mouse button allows you to modify both brightness and contrast of the displayed image.

Windowing images up to the end of the series

Using the mouse, you are able to automatically apply the new window values of an image to all subsequent images in the respective series.

✧ Select **Window On Succeeding On** from the context menu.

Windowing images automatically



✧ Double-click the image with the center mouse button.

- Or -

Press this key.

→ **Basics/Windowing images** in the syngo Operator Manual

Zooming and panning an image

The size of the displayed images is automatically adjusted to that of the **Inline Display**. You can change the zoom factor and the position of the image contents, as required.

→ **Basics/Zooming and panning images** in the syngo Operator Manual

Zooming/panning images up to the end of the series

Using the mouse, you are able to automatically apply changes of the visible image section to all subsequent images in the respective series.

✧ Select **Zoom/Pan On Succeeding** from the context menu.

Resetting changes

This is how you cancel changes via zoom/pan in the **Inline Display**:

- ✧ Double-click the image.

The images are adjusted again to the segment size.

Activating/deactivating image text

- ✧ Select/deselect **Image Text On** in the context menu.

The image texts in the transferred images are either activated or deactivated.

Removing an image

The image shown last remains displayed if you, e.g., interrupted a measurement. To delete the image:

- ✧ Select **Clear Segment** in the context menu.

Saving images

Usually the images shown in **Inline Display** are automatically stored in the **Patient Browser**. In special cases, however, you can switch off automatic saving.

Automatically storing images

For each protocol, the system defines whether automatic image storage may be switched off or not.

For protocols involving offline image calculation (e.g., spectroscopy), it is not possible to switch off automatic image storage. For real-time measurements, on the other hand, it makes sense to switch off automatic image storage and just save single images manually.

If automatic image storage is switched off, the images will be deleted after their one-time display in **Inline Display**.

Activating/deactivating automatic image storage



Depending on whether you need all images for diagnostic purposes, you can switch on/off image storage.

→ [Page C.8–8](#), *Automatically storing and loading*

✧ Click to activate/deactivate the image icon.

- Or -

Select/deselect **Store Images On** in the context menu.

After the function is switched off, the images of your current measurement are no longer automatically stored in the **Patient Browser**. They are only displayed in the **Inline Display**.



When you switch off automatic image storage, your image data will be lost!



- ⇒ Automatic image storage can be disabled for a current measurement only.
- ⇒ You are not able to pause the display when automatic image storage is switched off. The respective icon is disabled.
- ⇒ The **Store Images On/Off** key is disabled in case image reconstruction is completed or the **Auto store images** option cannot be deselected.

Manually storing images



- ✧ Click this icon to store the image shown in the **Inline Display**.
- ⇒ If the **Inline Display** does not contain an image, this icon is not active.

Copying the image position

You are able to copy the slice position and orientation of the current image in the **Inline Display** to the protocol opened in the program control.

If you select a graphic object in the open protocol, the position and orientation will be transferred to this object. If no graphic object is selected, the position and orientation will be transferred to the first slice group of the protocol.

Slice position and orientation are simultaneously saved in the buffer even when no protocol is opened. Subsequently, you can apply these settings for a protocol via **Tools > Paste Image Position**.



✧ Click this icon.

- Or -

Select **Copy Image Position** from the context menu.

When a protocol is open, you are able to transfer the image position with *drag & drop* as well:



✧ Click the lower left corner of the image (*drag & drop* area) with the mouse and drag it to a GSP Segment.

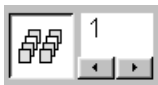
Selective image display

BOLD

For **BOLD** measurements that do not use the mosaic format, you can show a particular slice in the **Inline Display**.

The slice is selected by using its image number.

→ [Page C.6–36](#), *Numbering reconstructed images*



✧ To select image display, click this icon.

✧ To display specific images, enter the number of the slice into the field right next to the **Display Filter On/Off** icon.

Use the arrow keys to scroll through the slices during the measurement in progress.

TimCT

For **TimCT** measurements you can display either the reconstructed images or a projection progress image. The **slice number** field will not be available in this case.



✧ To toggle between the two display options, click this icon.

Interactive scanning in real-time mode (optional)

Your system contains various protocols that support real-time mode similar to fluoroscopy.

During this type of measurement, you are able to change the slice position and orientation of the slices measured.

This allows you to interactively locate a slice and transfer its position and orientation to another (high-resolution) measurement protocol.

Interactive measurements in real-time mode run as continuous loops. After locating the desired slice position and orientation, you may close the measurement in the program control.

Selecting the interactive measurement procedure

- ✧ Select a real-time protocol from the program card.
→ [Page C.3–14, Selecting a measurement program](#)
- ✧ Transfer the protocol to the program control.
→ [Page C.3–16, Applying the measurement program to the program control](#)
- ✧ Go to the properties of the protocol and set automatic opening in the **Inline Display**.
→ [Page C.8–8, Automatically storing and loading](#)
- ✧ Deactivate automatic image storage in the properties of the protocol.
→ [Page C.8–8, Automatically storing and loading](#)
- ✧ Prepare the protocol for the measurement.
→ [Page C.5–1, Positioning Slices](#),
→ [Page C.7–1, Adapting the Measurement Parameters](#).
- ✧ Click **Apply** in the program control to start the protocol.
- ✧ View the first reconstructed images in the **Inline Display**.
- ✧ When needed, change the slice position or orientation during the measurement in progress.

The next images are already reconstructed with the new slice position and orientation.

Apply



- ✧ To transfer the image position, click this icon.
→ [Page C.18–12](#), *Copying the image position*

- ✧ To end the measurement, click **Stop** in the program control.

As a next step, prepare the high-resolution protocol.

- ✧ To get the slice position and slice orientation of the real-time protocol from intermediate storage, select **Tools > Paste image position**.
→ [Page C.5–77](#), *Transferring the position of reference images*

Inline display for CARE Bolus measurement

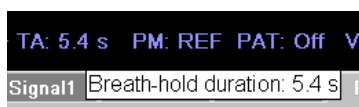
Inline Display allows you to check acquired images parallel to the measurement in progress. When you monitor the inflow of the contrast agent, terminate the current measurement as soon as the contrast agent is visible in the image. You can immediately start the subsequent, high-resolution imaging sequence.



- ✧ To end the current measurement and start the next one, click this icon in the **Inline Display**.
- Or -
Press **F12**.
- Using the keyboard provides you with the advantage of focusing on the **Inline Display** instead of the mouse pointer.

Starting multiple breathhold measurements

A multiple breathhold protocol is started in the **Inline Display** and measured in several steps.



After you have positioned the slices in the GSP, you can define the number of breathhold intervals by entering the **Concatenations**. The acquisition time required per breathhold interval is superposed when you leave the mouse pointer on the measurement time in the protocol info line.

After each breathhold command, the image rows of slices that belong to *one* concatenation are measured.

→ [Page P.1–114](#), *Parameter Concatenations*

You will start the measurement per concatenation (= breathhold interval) from the **Inline Display**.

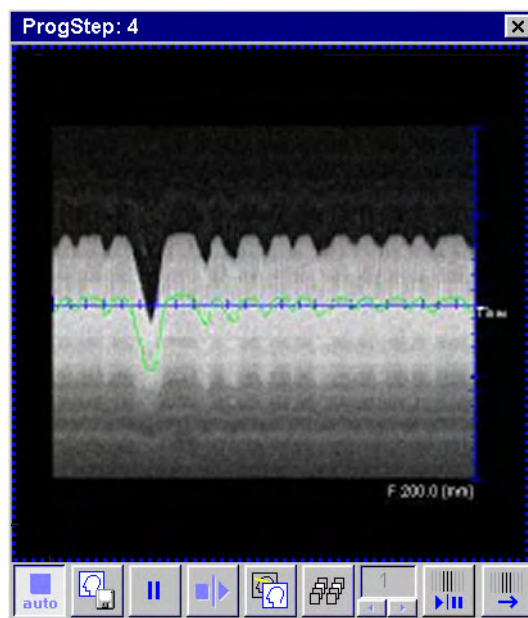
With navigator protocols, you can also observe the respiratory movement of the diaphragm in the **Inline Display**.

- ✧ Prepare a navigator protocol for the measurement (select the respiratory control mode and plan the preparation phase).
→ [Page P.1–79](#), *Parameter Navigator*
- ✧ Open **Inline Display**.

Apply

✧ Click **Apply** in the program control to start the protocol.

The position of the diaphragm (navigator signal) is displayed as a function of time in the **Inline Display**.



✧ Monitor the respiratory curve in the **Inline Display**.



- ✧ Apply the breathhold command.
- ✧ To start the measurement, click this icon.
- Or -
Press **F12**.

Navigator signals are *not* recorded during the measurement. Only afterward, the navigator signal is measured and displayed again. The position of the diaphragm during the first measurement is displayed as a yellow line.

- ✧ After the measurement, inform the patient that he can breathe again.

The patient should draw several normal breaths.

- ✧ Give the breathhold command for measuring the image rows of the next concatenation.
- ✧ Start the next measurement.



To stop multiple breathhold measurements:

- ✧ Click **Stop** in the program control.

The results of fully measured slices are stored. All other slices are lost.

! If you cancel a protocol with multiple measurements, all images of the current measurement may be lost.

Pausing and resuming breath-hold measurements



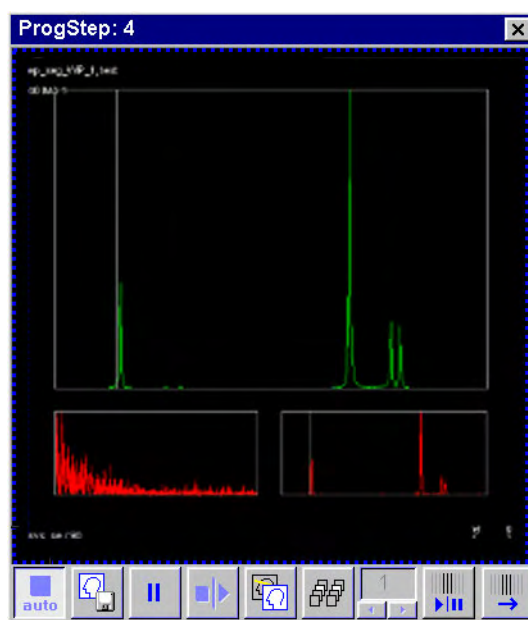
For some sequence types, e.g. *TimCT Oncology*, the **Scan breathhold** icon will start the measurement for a duration which can be set in the **breath-hold duration** parameter of the **Physio - Pace** parameter card.

→ [Page P.1–143](#), Parameter *Breath-hold duration*

✧ Click this icon to pause and resume the breath-hold measurement

Display of the spectroscopy time signal

You may use **Inline Display** to track the MR spectrum while a measurement is in progress. The current and accumulated spectroscopy data are displayed as a vector graphic in **Inline Display**.



- ⇒ During spectroscopy signals you are not able to deselect **Auto store images**.

Ending Inline Display

Inline Display may be ended either manually or automatically.

Closing it manually



- ✧ Click this icon in the upper right corner of the **Inline Display**.
- Or -
- Click this icon on the tool bar of the **Exam** task card.

Inline Display is closed.

Closing automatically

Use the context menu to automatically close the **Inline Display** as soon as image reconstruction is completed.

- ✧ Select **Close Inline Display** from the context menu.
- ⇒ **Inline Display** is automatically cleared when protocol reconstruction is complete and a new measurement protocol has been enabled.



Tips on optimum performance with real-time scans

- ✧ Check whether one or two receiver coils are connected.
- ✧ Deactivate the image text display.
- ✧ Switch off auto image storage.
- ✧ Finish all other processes, e.g., archiving and sending.
- ✧ Optimize the protocol parameters, e.g., minimize TE and TR, reduce phase-encoding steps, and deactivate fat saturation.
- ✧ The smaller the **Inline Display** window, the less performance is required for image display.

AutoAlign is the method for automatic positioning and alignment of slices for an MR examination.

AutoAlign provides alignment algorithms for different anatomical regions:

☐ **Head**

→ [Page C.19–2](#), *AutoAlign Head Examination*

☐ **Spine**

→ [Page C.19–11](#), *AutoAlign Spine Examination*

☐ **Knee**

→ [Page C.19–9](#), *AutoAlign Knee Examination*

Licensing

For each anatomical region, a separate AutoAlign license is available.

Explanation of terms

The *AutoAlign scout* is the first step of an AutoAlign program. It scans a 3D volume and computes the AutoAlign matrices.

The *AutoAlign region* and *reference* specify the focus of the AutoAlign computation.

The *AutoAlign matrix* defines the shift, translation, and rotation of the detected anatomy relative to the patient coordinate system.

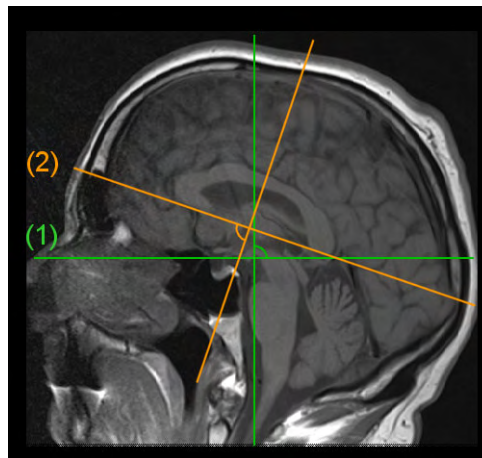
AutoAlign Head Examination

AutoAlign Head is based on the automatic recognition of specific anatomical bone landmarks on the skull. These landmarks are used to position the slices for a standardized and reproducible scan. The algorithm allows examinations independent of the patients' age or diseases.

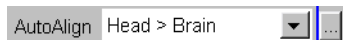
AutoAlign head matrices

The AutoAlign scout for a head examination (**AAHScout**) computes two different types of matrices.

- (1) **Head > Basis** is typically used for sag/cor and brainstem examinations
- (2) **Head > Brain** is typically used for tra/cor, pituitary, and IAC examinations.



Examination workflow



An AutoAlign head program starts with an AutoAlign scout protocol. This is followed by multiple clinical AutoAlign protocols using the computed AutoAlign results.

With the **AutoAlign** parameter of the **Routine** parameter card the user selects the AutoAlign matrix to be used for the clinical protocol.

A predefined AutoAlign program can be found in the **Exam Explorer** under `\\SIEMENS\head\programs`.

1	AAHScout		00:19
2	t1_se_sag		02:25
3	t2_tse_tra_512		03:47
4	t2_tirm_tra_dark-fluid		04:32
5	contrast agent		
6	t1_se_sag		02:25
7	t1_se_tra		02:25

The AutoAlign scout acquires a 3D volume, which is used for automatically reconstructing MPR localizers. The MPR localizers are saved in a second series.

The geometry (e.g. slice positions) of the clinical AutoAlign protocols are adapted accordingly by the referenced AutoAlign matrix. With a working man in the clinical AutoAlign protocols, the user can predefine whether to review and perhaps adapt the aligned slice positions or to allow the system to scan the aligned protocol automatically.

AutoAlign supports *scan@center*. For this, the AutoAlign scout has to be measured with distortion correction, and the clinical AutoAlign protocols have to be preset with the **ISO** positioning mode.

→ [Page C.9–8, Isocentric measurements](#)

AutoAlign feedback



This symbol in the program step indicates the successful computation of the AutoAlign matrices.



A crossed out symbol indicates that the AutoAlign matrix does not map to the anatomy of the patient or that the computation of the AutoAlign matrix failed.



If the AutoAlign computation did not succeed, a user notification is displayed, and the construction worker icon is automatically activated in the pending clinical AutoAlign protocols. To continue the examination you have to:

- ✧ Open the protocol.
- ✧ Adapt the slice positions manually.
- ✧ Click **Apply**.

Apply

The protocol is closed. The scanning is started using the adapted slice positions.

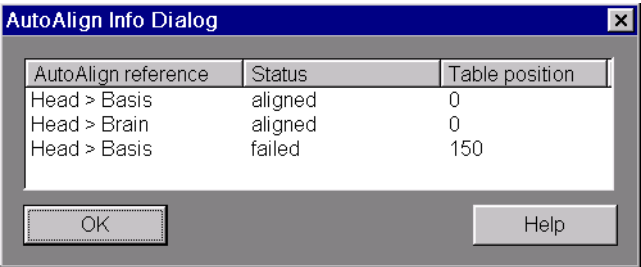
Status of the calculated matrices



You can view the status of the calculated matrices at any time within an AutoAlign examination.

✧ Click the **AutoAlign** icon above the queue.

The **AutoAlign Info Dialog** window is displayed.



Status	Description
aligned	The AutoAlign matrix could be detected with high reliability. No further action is necessary.
to be checked	The AutoAlign matrix could be detected, but only approximately. The slice positioning (proposed by AutoAlign Head) should be checked.
failed	The AutoAlign matrix could not be detected, not even approximately. Succeeding protocols are not aligned and have to be positioned manually.

✧ Click **OK** to close the dialog window.

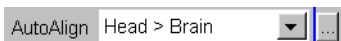
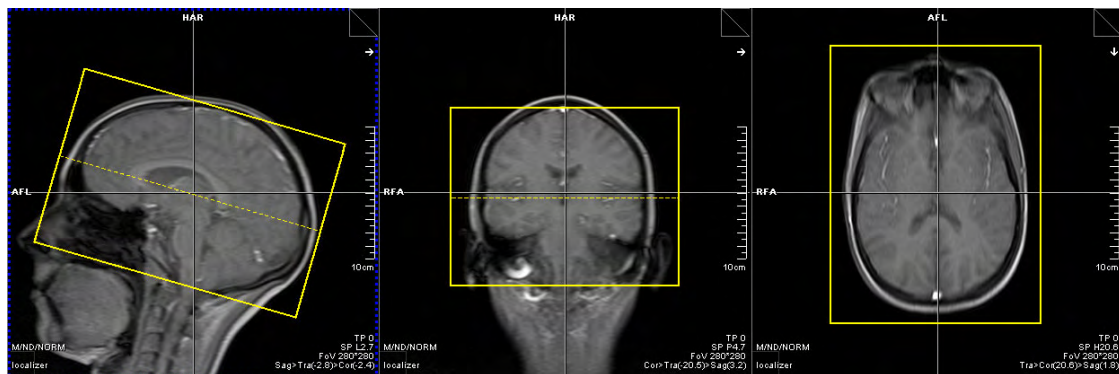
Inline localizer MPR

An additional series of MPR localizer images is automatically created from the given 3D scout volume. These images are automatically displayed in the reference image segments.

Adapting the following protocols

Once an AutoAlign matrix is calculated successfully, it is automatically applied to the clinical protocols in the queue that refer to this matrix.

The initial position and orientation of the predefined protocol graphics (e.g., slices, spectroscopy voxels) are shifted and tilted relative to the referenced AutoAlign matrix.



- In a protocol, you can select an AutoAlign matrix or deselect AutoAlign, using the **AutoAlign** parameter.
- If you do not agree with the auto-aligned slices, you can position the slices manually.
→ [Page C.5–1, Positioning Slices](#)

AutoAlign Head (Atlas-based)

AutoAlign Head (Atlas-based) is a former AutoAlign algorithm for AutoAlign head examinations. It uses a brain atlas for the alignment instead of bone landmarks. It is based on an automatic segmentation algorithm for grayscale values and CSF (cerebral spine fluid) in 3D volume data sets. The AutoAlign scout protocol is named **AAScout**.

The resulting AutoAlign matrix can be referenced by any clinical AutoAlign protocol and is called **Head > Brain Atlas**.

Please note the following restrictions for **AutoAlign Head (Atlas-based)**:

In certain situations, large anatomical differences between the current patient and the internal **AutoAlign Atlas** may lead to unreliable **AutoAlign** results or failure. Therefore, **AutoAlign Head (Atlas-based)** should not be used

- ☐ on patients younger than 17 years (developmental changes)
- ☐ on geriatric patients (degenerative changes)
- ☐ on patients with severe brain deformations
- ☐ on patients after major brain surgical interventions

These restrictions largely do not exist with the newer **AutoAlign Head** principle.

→ [Page C.19–2](#), *AutoAlign Head Examination*

Coil selection

AAScout: body coil

AAScout (Trio a Tim System): body coil or TxRx head coil

Workflow

The differences between **AutoAlign Head (Atlas based)** and the **AutoAlign Head** standard principle described above, are:

- ❑ An AAScout computes only one single AutoAlign matrix, (**Head > BrainAtlas**) focusing on the brain anatomy and not other head structures.
- ❑ An AAScout does not create Inline MPR localizer images. Therefore, an extra localizer protocol should be scanned immediately after the AAScout.
- ❑ You can create user-defined AutoAlign programs using a special planning patient.
→ [Page C.19–22, Planning AutoAlign head programs with atlas images](#)

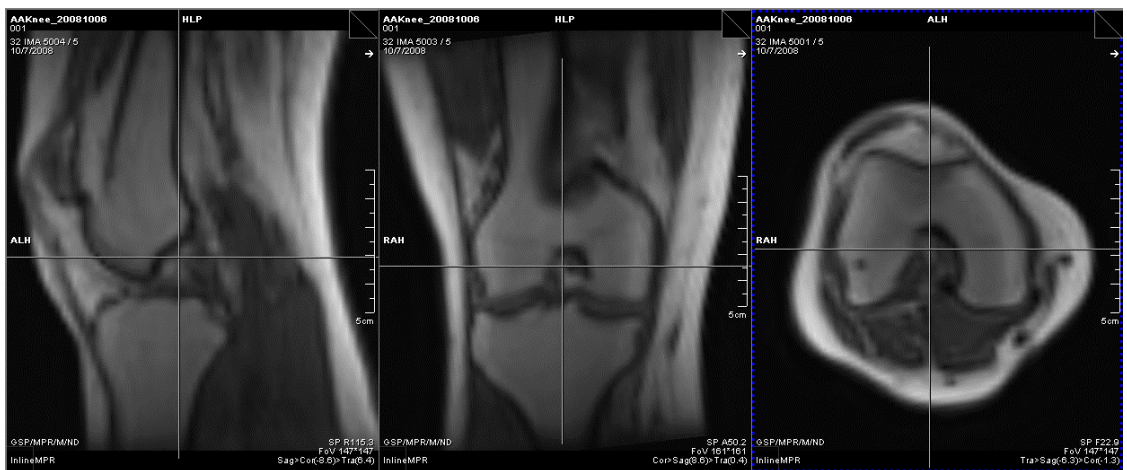
Program setup

An older AutoAlign program based on a previous software version has to be adjusted once, before being used:

- ✧ If you want to use the **Atlas based AutoAlign Head principle (AAScout)**, set the **AutoAlign** parameter for all following clinical protocols to **Head > Brain Atlas**.
- ✧ If you switch from the **Atlas based AutoAlign Head principle (AAScout)** to the newer **AutoAlign Head** principle, replace the **AAScout** plus the **localizer** with the **AAHScout** from the Siemens tree (\\SIEMENS\\head\\localizer).
- ✧ If you use the newer **AutoAlign Head** principle, set the **AutoAlign** parameter for all following clinical protocols to the right AutoAlign matrix (**Head > Basis** or **Head > Brain**).
→ [Page C.19–2, AutoAlign head matrices](#)
This can be done in one run via the **Find object in data base** dialog window.
→ [Page C.13–10, Searching for entries](#)
- ✧ Test the new workflow with a proband and adjust the workflow properties, if necessary (Coils, AutoCoilSelect, iPAT, PrescanNormalize, AutoLoad, Construction worker, ...).

AutoAlign Knee Examination

AutoAlign Knee is based on the automatic recognition of specific anatomical landmarks on the knee. These landmarks are used to determine an AutoAlign matrix **Knee > Basis** for standardized and reproducible scans.



The patient can be examined in the **Head first - supine** position and in the **Feet first - supine** position.

Examination workflow

The workflow of an **AutoAlign Knee** examination is similar to **AutoAlign Head**.

→ [Page C.19–2, AutoAlign Head Examination](#)

Example of an AutoAlign Knee program:

1		AAKScout_8ch	
2		pd_tse_fs_tra	02:51
3		t1_vibe_we_sag	04:17
4		contrast agent	
5		t1_vibe_we_sag	 3 04:17

- ☐ The examination starts with the respective **AAKScout** protocol.
- ☐ For all clinical protocols which are adjusted automatically, the **AutoAlign** parameter has to be set to **Knee > Standard**. You can define an offset (shift or tilt) relative to the AutoAlign matrix.
- ☐ AutoAlign Scout protocols suitable for the respective knee coils can be found in the **Exam Explorer** under \\SIEMENS\\knee\\localizers.
- ☐ A predefined **AutoAlign Knee** program can be found in the **Exam Explorer** under \\SIEMENS\\knee\\programs.

AutoAlign Spine Examination

Basics

AutoAlign Spine automatically positions and orients transverse slice groups on intervertebral disk layers. The correction for the scoliosis angle is taken into account for the entire spine.

A license is required for the use of **AutoAlign Spine**.

Algorithm

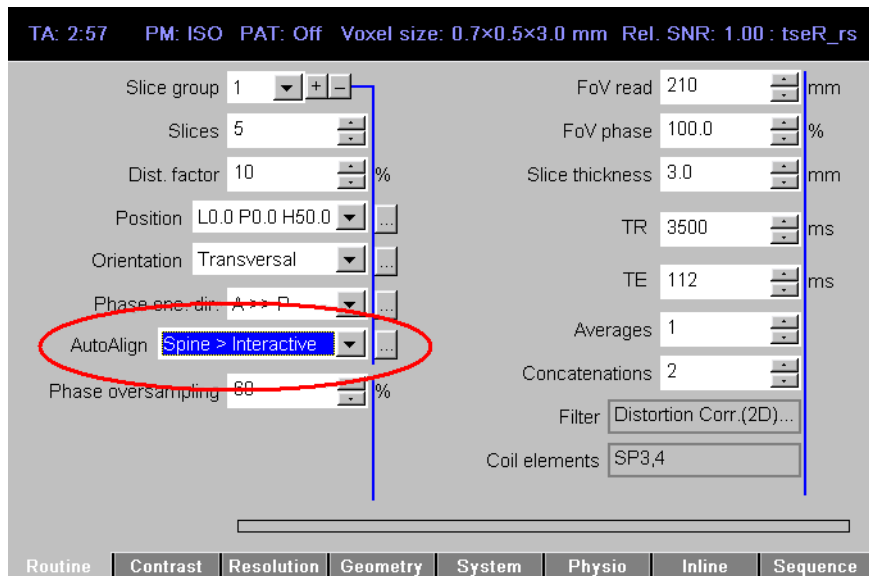
The search algorithm calculates the position and double-orthogonal orientation of transverse slice groups based on the anatomical characteristics of the intervertebral disk.

The position of a coronal saturation band is automatically aligned with transverse slice groups.

AutoAlign Spine uses a search algorithm based on T2-weighted sagittal images. Good quality T2-weighted sagittal images are the prerequisite for accurate and reliable calculations.

Switching AutoAlign Spine on/off

In AutoAlign spine protocols, the parameter **AutoAlign** on the **Routine** parameter card is set to **Spine > Interactive**.



→ [Page P.1–6, AutoAlign](#)

You can also enable or disable the AutoAlign spine functionality at any time via the main menu:

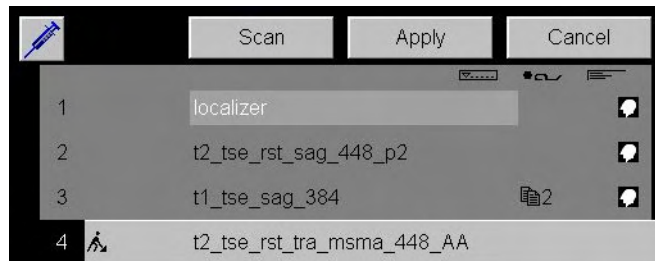
✧ Select **Tools > AutoAlign Spine**.

Preparing the automatic slice positioning

- ✧ Transfer the AutoAlign Spine program into the queue.

The localizer and a sagittal T2-weighted protocol will be scanned. The resultant images of the T2 weighted sagittal spine protocol have to be loaded into the graphic segments as localizers for the interactive AutoAlign Spine positioning of transversal slices.

- ✧ Open the Auto Align spine protocol in the queue.



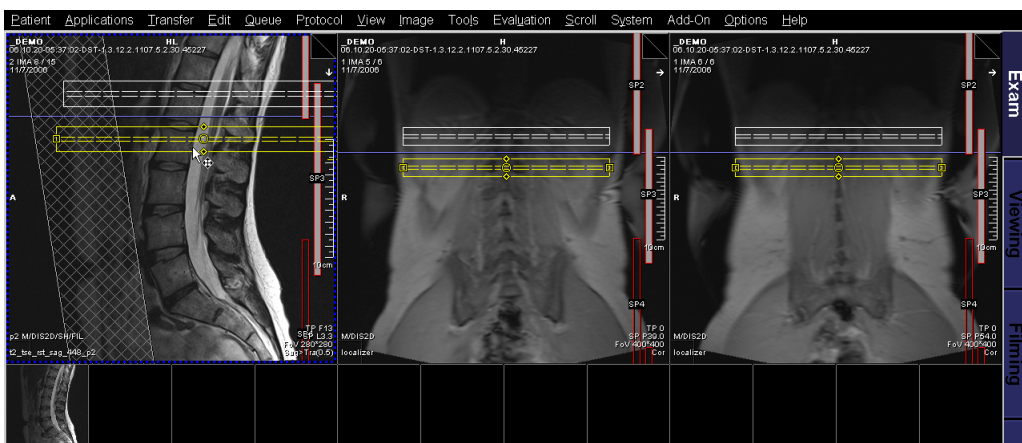
The user can immediately start positioning the slice groups to the intervertebral disks as soon as the protocol with the **AutoAlign** setting **Spine > Interactive** is opened.

Automatic positioning of slice groups and sat regions

During automatic positioning of the slice groups, a coronal view is reconstructed with sagittal images. The transverse slice groups are automatically adjusted to it. For each intervertebral disk, the center of the slice group is defined exactly between the vertebral bodies above and below the intervertebral disk.

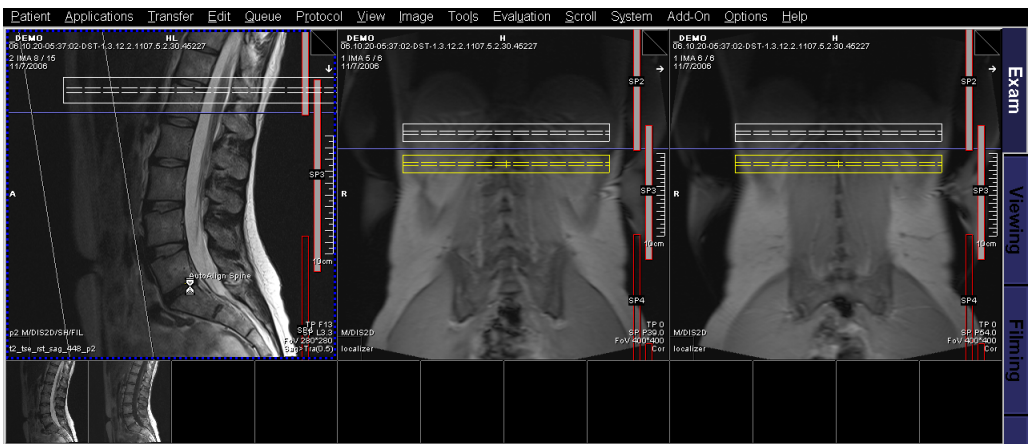
AutoAlign Spine algorithm

- ✧ Press and hold the mouse button down and drag the transverse slice group to any position within an intervertebral disk. Subsequently, release the mouse button.



- ✧ The position has to be located within an intervertebral disk or in one of the end plates lying above or below it so that the algorithm can calculate the correct position and angulation of the intervertebral disk.

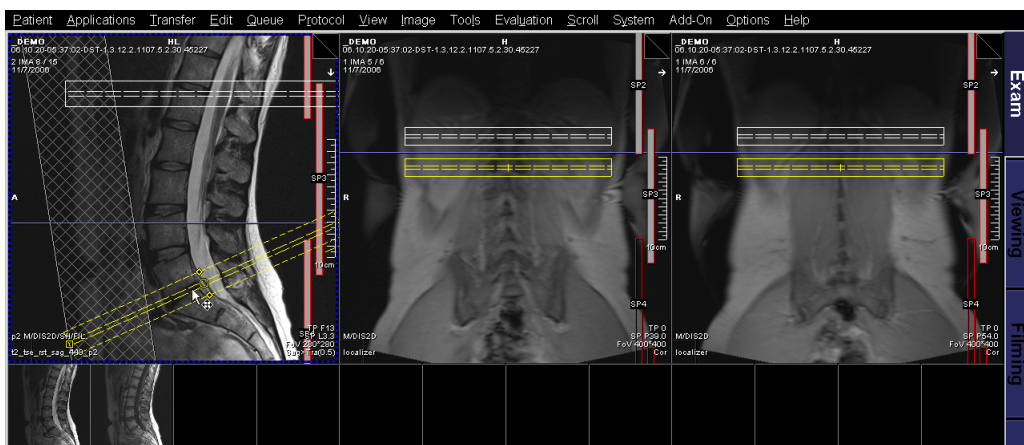
- If you drag the transverse slice group into an image area that is outside the intervertebral disk, the system is not able to calculate this position. In this case, the message "Spinal disk not found" is displayed in the status bar.
- ✧ Repeat this procedure for all transverse slice groups of your protocol.



The AutoAlign Spine algorithm starts automatically and tries to find the contours of an intervertebral disk in the immediate vicinity where you released the mouse button.

Adapting the position of the slice groups

After an intervertebral disk has been located, AutoAlign Spine aligns the slice group with the intervertebral disk.



Not applying the position of the slice groups

If you do not agree with the automatic slice positioning, you can position the slices manually.

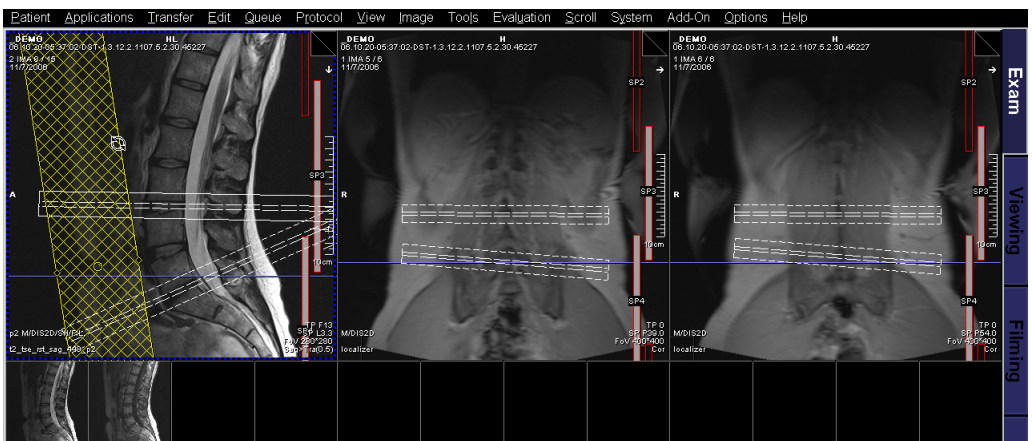
→ [Page C.5–1, Positioning Slices](#)

- ✧ Keep the **Shift** key pressed to temporarily deactivate AutoAlign Spine during manual slice positioning.
- ✧ As soon as you release the **Shift** key, AutoAlign Spine is activated again.

Adapting the saturation region

If the protocol includes a sat region, AutoAlign Spine automatically calculates its position and angulation. During this process, all slice groups in the protocol are taken into account. The algorithm adjusts the saturation region to the anatomy to be shown (vertebral body) without covering it.

→ [Chapter C 5, Positioning Slices](#)



Editing the AutoAlign program

→ [Page C.6–25](#), *Editing the queue during measurements in progress.*

Inserting program steps

You can insert program steps or add completed programs to the queue, even if they are from other body regions.

→ [Page C.6–28](#), *Inserting new program steps*

Adapting added protocols

Check the measurement parameters of the added protocols and adapt them to your needs.

→ [Page C.7–1](#), *Adapting the Measurement Parameters*

Check the **AutoAlign reference** in the clinical protocols.

→ [Page C.19–19](#), *Editing the AutoAlign parameters of a clinical protocol*

⇒ We recommend preparing all head and knee protocols in the **Exam Explorer** with an AutoAlign reference. If you measure such a protocol without AutoAlign scout, the reference will be ignored automatically.

Setting copy references

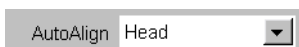
You use a copy reference to apply parameters from one protocol of the AutoAlign program to another protocol or vice versa.

If the copy reference contains the slice positions, the AutoAlign matrix is applied to the target protocol.

→ [Page C.8–12](#), *Setting a copy reference*

→ [Page C.7–18](#), *Parameter groups*

Editing an AutoAlign scout



The AALScout sequence can be used for AutoAlign head (**AAHScout**) and AutoAlign knee (**AAKScout**) scout protocols. This is selectable by the AutoAlign parameter on the Routine parameter card.

→ [Page P.1–6](#), *Parameter AutoAlign*

Editing the AutoAlign parameters of a clinical protocol

For clinical AutoAlign protocols, you set the AutoAlign matrix to be used and several AutoAlign geometry parameters which are matrix-related. The AutoAlign parameters are found on the **Routine** parameter card.

→ [Page P.1–6](#), *Parameter AutoAlign*

☞ In spectroscopy protocols, the **AutoAlign** parameters are found on the **System - Miscellaneous** parameter card.

Reconstructing AutoAlign programs (PhoenixZIP)

If you want to perform a follow-up **AutoAlign** examination for a patient, you can recall an already performed AutoAlign examination by using PhoenixZIPs.

→ [Page K.5–1](#), *PhoenixZIP*

Creating new AutoAlign programs

You have a number of choices to create a new AutoAlign program and to store it as a new AutoAlign program in the customer area of the **Exam Explorer**.

Generating an AutoAlign program

You are able to generate your own AutoAlign program in the **Exam Explorer**.

- ✧ Insert an AutoAlign scout in any head measurement program.
- ✧ Set the AutoAlign parameter of the AutoAlign scout.
- ✧ Set the **AutoAlign** parameters and the other measurement parameters (e.g., protocol status or copy references) of the clinical protocols according to your needs.



✧ Save the measurement program under a new name.

→ [Page C.13–34](#), *Creating a new measurement program*

→ [Page P.1–6](#), *Parameter AutoAlign*

Exporting measured programs into the Exam Explorer

You can save the queue of an AutoAlign Examination as a new program. It can be used for other patient examinations and follow ups.

✧ Select **Queue > Save As Program**.

- Or -

Select **Save As Program** from the context menu of the queue.

→ [Page C.13–50](#), *Saving the measurement program under a new name*

Planning AutoAlign head programs with atlas images

AutoAlign Head (Atlas-based) allows you to plan slices on the basis of statistical images for certain diagnostic problems. As a result, you can create a user-defined measurement program.

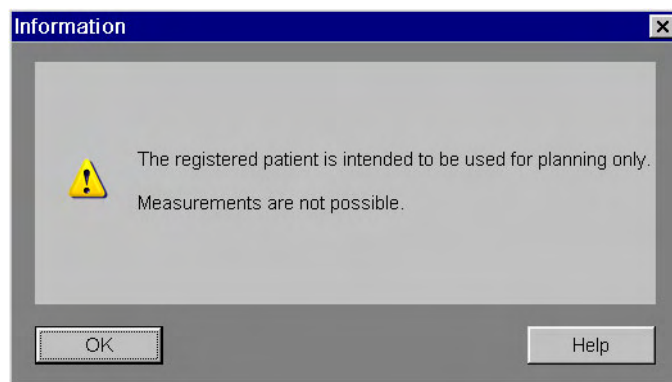
On the basis of atlas images you predefine the slice positioning once. Then you save the positioned measurement program in the **Exam Explorer**. After that, the measurement program is available for measurements which use the **Auto Align Head (Atlas-based)** principle.

Registering a planning patient

To plan an AutoAlign program, a special planning patient is available in the **Patient Browser**.

- ✧ Select the **AutoAlign** patient installed by the system and open the **Patient Registration** dialog window.
- ✧ Select the study that you want to position and register the **AutoAlign** patient in the position **Head first - supine**.

A message window is displayed.





- ✧ Click **OK** to close the message window.

The planning mode is active.

Planning mode

The images for planning are automatically loaded into the image segments of the **Exam** task card.

Scanning is not possible in planning mode. However, protocols can be edited without restriction.

Adapting protocols

- ✧ Load the program into the queue.
- ✧ Open the protocols and position the slices on the atlas images.



- ✧ Click **Apply**.

The changes are applied to the opened protocols.

Saving new protocols

You can save the edited protocols in the **Exam Explorer**.

→ [Page C.6–36](#), *Storing the queue as a new program*

Exiting planning mode

- ✧ Select **Patient > Close Patient**.
 - Or -
- Register a real patient.



Dynamic Analysis

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D.2 *Starting the Evaluation Function*

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D.4 *Functions for Dynamic Analysis*

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The **Dynamic Analysis** function provides **syngo MR** with a number of arithmetic and statistical functions for evaluating dynamic processes and images.

Depending on the function, you can evaluate single images, more than one image, one series, or more than one series.

A separate dialog window is shown for every evaluation function selected. The procedure for each operation in **Dynamic Analysis** is the same. We will begin by describing a few general steps before going into the specific aspects of each analysis.

The result images are stored in the database. You can view these images via the **Viewer** or use them as initial images for additional evaluations.

Requiring access rights for dynamic analysis

Dynamic analysis requires that the user have full access rights to the patient data to be evaluated.

Detailed information regarding user administration are included in the *syngo* Security Package brochure.

Overview for evaluation

The following functions are available for analyzing dynamic processes and post-processing images:

- ❑ Addition → [Page D.4–1](#)
- ❑ Subtract → [Page D.4–3](#)
- ❑ Arithmetic Mean → [Page D.4–7](#)
- ❑ Differentiation → [Page D.4–8](#)
- ❑ ADC → [Page D.4–10](#)
- ❑ Division → [Page D.4–13](#)
- ❑ Integration → [Page D.4–16](#)
- ❑ Logarithm → [Page D.4–18](#)
- ❑ Multiplication → [Page D.4–20](#)
- ❑ Standard deviation → [Page D.4–22](#)
- ❑ Slope → [Page D.4–25](#)
- ❑ T2/T2* → [Page D.4–27](#)
- ❑ T1 → [Page D.4–29](#)
- ❑ Time To Peak → [Page D.4–31](#)
- ❑ TTest → [Page D.4–34](#)
- ❑ Editing Images with Image Filters → [Page E.1–1](#)

The functions for **Dynamic Analysis** are started from the **Patient Browser** and the **Viewing** task card.

Selecting images

⇒ Only MR images may be loaded into **Dynamic Analysis**. If you load images from other modalities, e.g., CT or AX images, a message is displayed indicating that these image types are not supported.

In the Patient Browser

- ✧ Open the **Patient Browser**.
- ✧ Select images or series in the navigation or content area.

Or

In the Viewer

- ✧ Select the required images or series in the image area of the **Viewing** task card.

Starting image analysis

Depending on where you selected the images (**Patient Browser** or **Viewing**), you start the evaluation function.

- ✧ Select the function required under **Evaluation > Dynamic Analysis**.
- ✧ The **Viewing** task card lets you select 7 frequently used functions (of a total of 16) via the respective symbol on the **Eval.** subtask card:



Addition



Subtraction

Arithm.
MeanDiffusion
coefficient

T1



T2



Image filter

As soon as you have started the evaluation function, the images selected are loaded into the respective function dialog. It may take several seconds to load the images. A percentage indicator in the footer shows the current progress made.

The evaluations are performed identical to mathematical functions. The series selected first is the first operand of the function, all series selected after that are the other operands.

Checking whether images are suitable

For dynamic evaluation, the images have to meet the following criteria:

- ☐ Originated from the same examination
- ☐ Same field of view (FoV)
- ☐ Corresponding matrix size
- ☐ Same slice orientation
- ☐ Same slice position
- ☐ Same slice thickness or times (for T1, T2, T2*, and TTest only)
- ☐ Same intensity correction (all corrected or not corrected)
- ☐ Same type of distortion correction (none, 2D, 3D)

A corresponding system message is displayed during loading if the selected images do not meet a particular criterion.

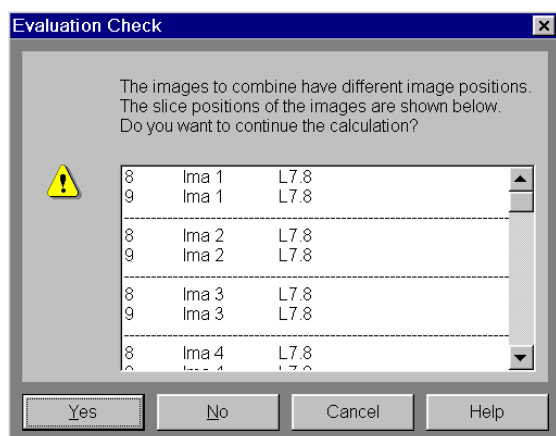
- ⇒ The distortion correction can be encoded within the frame of reference. In such cases, a message is displayed indicating that the frame of reference is not different, but rather that distortion-corrected images have been combined with images that have not been corrected.

Checking image positions and slice thicknesses

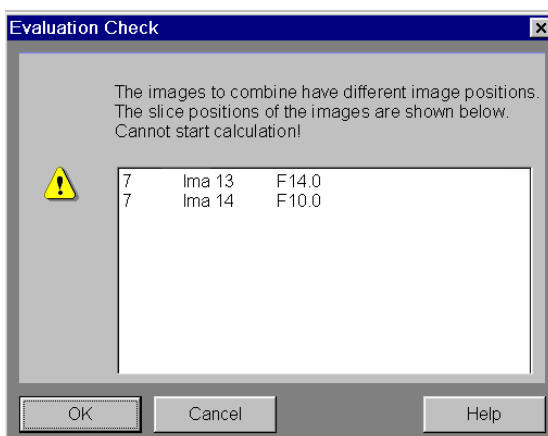
If different image positions or slice thicknesses are present, the **Evaluation Check** dialog window is displayed. It lists the images affected.

➤ If you selected several series, the images are listed together according to their image numbers.

You are able to continue evaluation only when the calculation planned does *not necessarily* require the same image position or slice thickness.

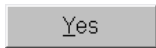


Example 1 for a calculation that can be continued despite different slice positions



Example 2 for a calculation that cannot be continued due to different slice positions

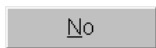
Example 1: Calculation can be continued



- ✧ Click **Yes** if you want to continue the calculation.

Calculation is continued if it does *not necessarily* require images with the same image position.

Or



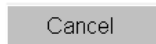
- ✧ Click **No** if you do not want to continue the calculation because you want to return to the function dialog window.

Example 2: Calculation can be continued

- ✧ Click on **OK**.

The program returns to the function dialog without performing the calculation when only images with the same image position, slice thickness, or similar are acceptable.

Cancelling calculation in general



- ✧ Click on **Cancel**.

The calculation is cancelled and the function dialog is closed.



After you selected an evaluation function, the function dialog opens and the images selected are loaded.

Displaying initial images

All images or series loaded are included in the operand list of the function dialog.

The sequence of the series in the operand list is usually based on the succession used during loading. The series that is loaded first is the first operand, the series after that are the subsequent operands.

Exception: During **Subtraction** and **Division**, the images or series loaded are listed in the temporal sequence of their generation.

Operands:	Series Description	Image Ran...	Increment
	97 2_ep2d_se_tra	12	
	102 TTest_S15_2	12	

Operand list with two operands

Examples for image areas

If you selected only part of the series, the **Image Area** column shows you the images transferred for evaluation.

Image area	Images
3	Fig. 3
2-5	Images 2 to 5
4 - < Images. Last image >	Image 4 to the end
2, 5, 8	Images 2, 5, 8

Changing the image selection in the operand list

You are provided with the following possibilities for correcting the operand list:

- ☐ Adding images
- ☐ Assign increment (evaluate only every x image)
- ☐ Deleting series
- ⇒ When you want to evaluate just part of a series, select only the required images of the series via the **Patient Browser** or in the **Viewing** task card and start the evaluation function.

Adding images

- ✧ Select the images in the **Patient Browser** and drag them with the mouse to the operand list in the function dialog window.
- ⇒ When you add images from a series where you have already loaded part of it, the respective operand is added. The information in the **Image Area** is updated and the operand moves to the top of the list.

Entering an increment

For example, if you want to perform **Dynamic Analysis** only for every third image of the series, set the increment to the value **3** for the series.

- ✧ Select the series to be processed from the operand list.
- ✧ Double-click the selected series in the **Increment** column.

An input field is displayed.

Series Description	Image Range	Increment
6_2_t2_tse_tra	1-21	

- ✧ Enter the increment.
- ✧ To complete the input, click the mouse outside the input field.

Removing series

- ✧ Select the series to be processed from the operand list.
- ✧ Select **Delete** from the context menu.
 - Or -
 - Press **Del**.

Setting the operating mode

Go to the function dialog underneath the operand list and select the operating mode.

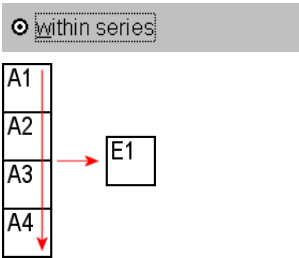
Evaluating images within a series

If you evaluate images within a series, only the images of the initial series will be used as operands for the evaluation function.

In this mode, you generated a result series from the initial series. The result series contains one or several images.

The **within series** mode is not possible when

- ☐ The series loaded contains a single image only
- ☐ Several series have been loaded

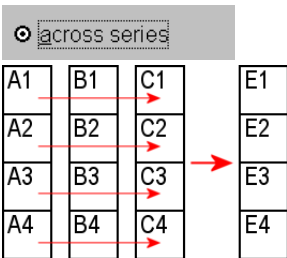


Evaluating images across series

You can use evaluation functions for images from several different series.

In this case, you are generating result series from the individual images of the original series by combining the n-th images of the series. When you assign an increment, the result series contains fewer images.

With subtraction, the result series will usually contain the same number of images as the smallest initial series.



The **across series** mode is not possible when:

- ☐ Only one image or series has been loaded
- ☐ When the number of images to be evaluated differs in the series loaded

Applying the constant to images

Many evaluation functions can be performed with a constant, e.g., adding a constant to the images of a series.



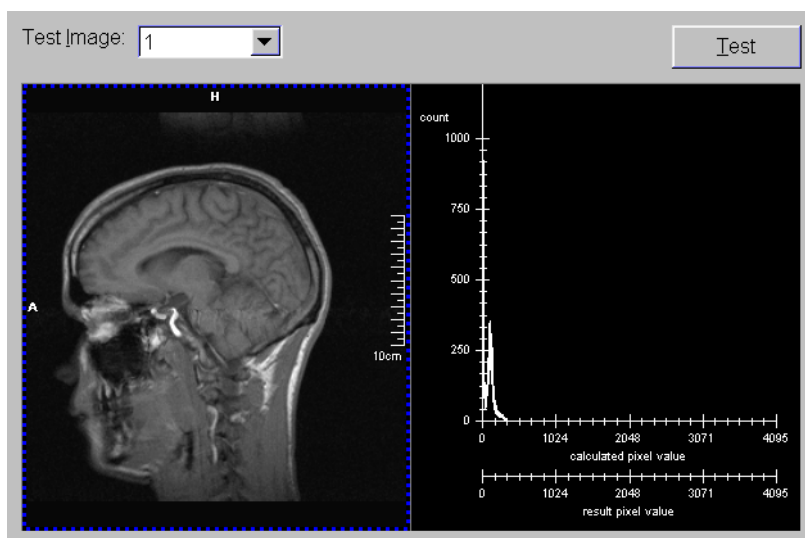
The image shows a user interface element with a radio button icon to the left of the text "_constant". To the right of this is a rectangular text input field containing the number "100".

- ✧ Select the **constant** option.
- ✧ Enter the constant desired in the field.

The operation with the constant is applied to each image in the series.

Generating preview images

You can generate test (preview) images for most evaluation functions. These test images are used to determine whether you want to evaluate the entire series with these settings.



Test image (left) with associated histogram (right)

The **Test Image** selection list displays all image combinations possible for a test evaluation. The image numbers of the combinable images are displayed for each image combination. Images numbers are separated by slashes. If only one image number is displayed, the test image is created from a single image.

Test Image: 1|2|3|4|5|6|7| ▼

Test

✧ Select an image combination from the selection list.

✧ Click **Test** to display the test image and the histogram in the image area.

If the test image and histogram displayed do not meet your expectations, select another test combination and perform a new test.

Limiting the display area

You can limit the range displayed in the histogram (lower X axis) to the relevant part by defining a lower and an upper threshold. Only the grayscale values within the thresholds will be used for the evaluation (refer to upper X axis).

✧ Select **Properties** from the context menu of the histogram.
- or, for **Subtraction** or **Division** -
Click **Scaling** in the function dialog.

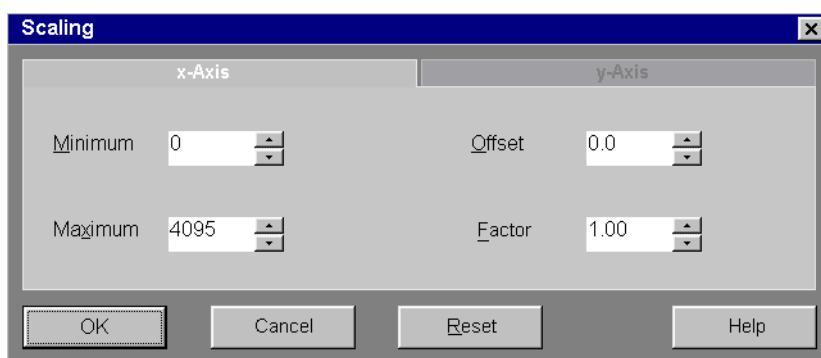
Scaling

The **Scaling** dialog window is displayed.

Limiting grayscale ranges

For scaling, select the range of the calculation result data that is mapped onto the grayscale range (0-4095) or onto the range given on the lower X-axis.

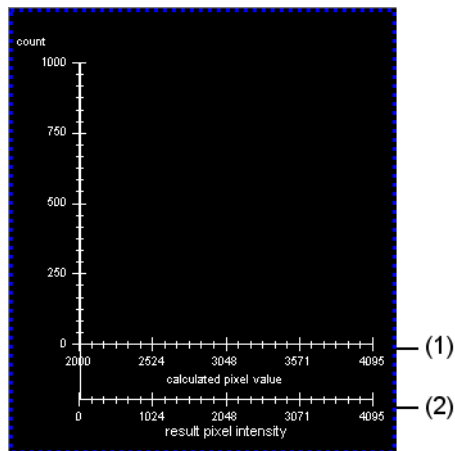
- ✧ Activate the **x-Axis** subtask card.



The screenshot shows a 'Scaling' dialog box with a blue title bar and a close button (X). It has two tabs: 'x-Axis' (selected) and 'y-Axis'. The 'x-Axis' tab contains four input fields with up/down arrows: 'Minimum' (0), 'Maximum' (4095), 'Offset' (0.0), and 'Factor' (1.00). The 'y-Axis' tab is currently inactive. At the bottom of the dialog are four buttons: 'OK', 'Cancel', 'Reset', and 'Help'.

- ✧ Enter the lower threshold in the **Minimum** field.
- ✧ Enter the upper threshold in the **Maximum** field.
- ✧ You can also enter the **Offset** and **Factor** function instead of the two threshold values. These two values have a linear relationship to the **Minimum** and **Maximum**.

The set interval (upper X-axis) is mapped to the grayscale area (lower X-axis, generally 0 to 4095) when displaying the test image.



- (1) Grayscale range defined by scaling
- (2) Grayscale range displayed in the result image

Limiting the number of pixels

- ✧ Activate the **Y-axis** subtask card.
- ✧ Enter the lower threshold for the number of pixels in the **Minimum** field.
- ✧ Enter the upper threshold for the number of pixels in the **Maximum** field.

Reset

Noise Level: 0



Changes to the Y-axis scaling only affect the histogram. It does not affect the result images

✧ To apply the settings, click **OK**.

✧ Use **Reset** to return to the original settings.

Setting a noise level

You can set a noise level for some functions. This threshold value determines the pixels used for evaluation.

✧ Enter a value in the **Noise Level** input field.

Naming the result series

A name that is automatically generated for the **Result Series** is entered in the result series field. You can accept this name or change it as desired.

Result Series Description: ADD_S1_C0_1

The suggested name comprises the following sections, separated by underscores:

- ❑ Abbreviation for the function used (e.g., ADD)
- ❑ Abbreviation for the operand ("S" for series, "I" for image, and "C" for constant), generally together with the series number, image number, or value of the constant

- ⇒ If you are using more than two series or images, the word "diverse" will be entered instead of a series or image number.

Example: You add the grayscale values of image 34 to those of image 45. The system produces the following name: "ADD_I34_I45".

Changing the series name

If the name of the result series does not meet your requirements, change the name:

- ✧ Overwrite the name suggested in the **Result Series Description** field with the name desired.
- ⇒ You will not be able to store the series if its name contains unauthorized characters. The program displays a message indicating that the series name is invalid and requests that you review and correct the text entered.

Selecting results series

At times when you select more than one series (e.g., **Logarithm**, **Multiplication**, **Subtraction**, or **Division**), you can use the selection list next to the **Result Series** input field to select series, check names, or change them, if necessary.

Display in the Viewer

When you have completed the evaluation, the results are displayed in the background on the **Viewing** task card. Here, you can process images further.

Starting calculation

Once you have completed all preliminary steps in the function dialog window, start the calculation:

- ✧ Click on **OK**.
- Or -
- Press **Enter**.

Steps when correcting the initial data during the evaluation

Corrections may be made to the initial data during **Dynamic Analysis**.

→ **Patient Browser/Correcting data** in the *syngo* Operator Manual

Data correction affects evaluation in the following cases:

Situation	Type of corrected data	Consequences
Function dialog is open, images are selected, however, calculation has not yet been started.	Image or series data	Function dialog closes
	Patient or examination data	View of function dialog is updated
Evaluation interrupted (Pause)	Image or series data	Images have to be reloaded and the evaluation has to be restarted

Special features of individual functions that are different in operation will be described in the following.

Addition

Image addition is used to add the grayscale values of images or to add a constant.

Adding the grayscale values of images is equivalent to image overlay. For example, you can overlay an image with the result image of a T-image with exceptional anatomical details.

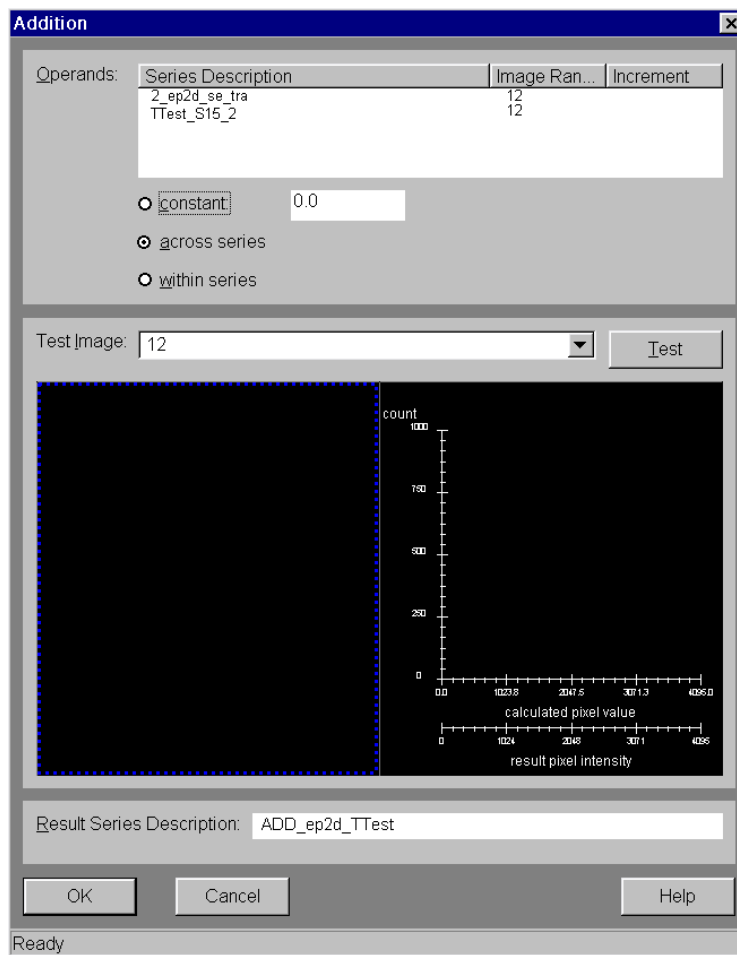
Adding a constant shifts the entire grayscale range.

You can add:

- ☐ Images from a series with a constant
- ☐ Images within a series
- ☐ Multiple series with the same number of images

- ✧ Start the evaluation function **Add....**
→ [Page D.2–1](#), *Starting the Evaluation Function*

Function dialog Addition



- ✧ Change the image selection in the operand list (optional).
→ [Page D.3–2](#)
- ✧ Correct the operating mode (optional).
→ [Page D.3–4](#)

- ✧ Generate preview images.
 - [Page D.3–6](#)
- ✧ Correct the display area (optional).
 - [Page D.3–7](#)
- ✧ Correct the name of the result series (optional).
 - [Page D.3–10](#)
- ✧ Start the calculation.
 - [Page D.3–12](#)

Subtract

During subtraction, you subtract the grayscale values of images from other images or subtract a constant.

For example, you could use subtraction to clearly visualize changes in the image after administering contrast agent. For this purpose, you subtract the grayscale values of images generated without a contrast agent from those generated with a contrast agent. The results are images showing only the changes caused by the contrast agent.

You can subtract:

- ☐ Image from image
 - ☐ Image from series
 - ☐ Images/series from constants
 - ☐ Constants from images/series
 - ☐ Series from series
-
- ✧ Start the **Subtract** evaluation function.
 - [Page D.2–1](#), *Starting the Evaluation Function*

Function dialog Subtract

Subtract Series 8 from Series 9

Operands:	Series Description	Image Ran...	Increment
	11 11_t1_f3d_cor_dyn_BIOPSY_1	16, 21	

- 8 t2_tse_sag | 1-15 |

☐ constant: 0.0 Exchange

☒ across series

☐ within series

Test Image: 9ima1 - 8ima1 Test

☐ Auto Scale Scaling

Result Series Description: SUB: S11-S10_1

OK Cancel Help

Ready

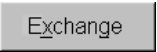
The subtrahend is shown in a separate field below the list of operands.

- ✧ Change the image selection in the operand list (optional).
→ [Page D.3–2](#)

During subtraction, you can change the sequence of operands if you do not want to, e.g., subtract image 4 from image 5, but rather image 5 from image 4.

You can swap the following types of subtraction:

- ☐ Constant of image
- ☐ Image x from image y
- ☐ Series x from series y
- ☐ A single series from all other series (the new single series has to be selected prior to swapping)

A rectangular button with a light gray background and a thin black border. The word "Exchange" is written in a black, sans-serif font, centered within the button.

- ✧ Click **Exchange**.

The new operation is shown in the title bar.

- ✧ Correct the operating mode (optional).
→ [Page D.3–4](#)

- ✧ During the **across series** mode, the **Subtrahend** is shown in its own field below the list of operands. As a default, the older series is used as the **Subtrahend**.

✧ Generate preview images.

→ [Page D.3–6](#)

✧ Correct the display area (optional)

→ [Page D.3–7](#)

Or

✧ While you are in the function dialog, activate the **Auto Scale** checkbox.

✧ Some evaluations require current pixel values as input even if they are negative. In these cases, the result images should be scaled automatically. The grayscale range will now start with the value **-500**.

✧ Correct the name of the result series (optional).

→ [Page D.3–10](#)

✧ Start the calculation.

→ [Page D.3–12](#)

Examples for subtraction:

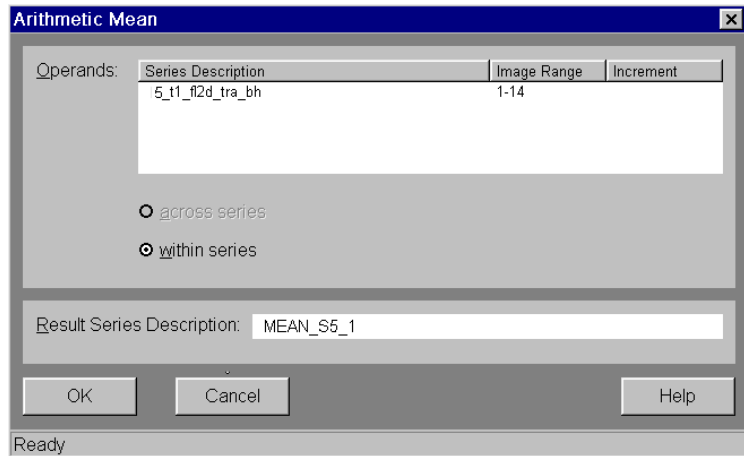
Selection	Subtraction possibilities
2 images (one series each with 1 image or one series with 2 images)	Subtract image from image Subtract the constant of both images
2 series with the same number of images	Image of a series is subtracted from an image of another series. Subtract the constant from the images of both series
Several series, of these at least one series with one image only (series with different number of images)	Subtracting a single image of a series from the images of all other series

Arithmetic Mean

Calculating the mean value of the grayscales you can average large intensity differences in the images. For parallel images with different slice positions, this method is equivalent to an MPR Thick display. With a temporal sequence of images, averaging the images improves the signal-to-noise ratio or reduces image artifacts.

- All images of the initial series must have the same slice position.
- ✧ Start the evaluation function **Arithmetic Mean**.
 - [Page D.2–1](#), *Starting the Evaluation Function*

Function dialog Arithmetic Mean



- ✧ Change the image selection in the operand list (optional).
→ [Page D.3–2](#)
- ✧ The operation mode cannot be changed directly. It depends on the number of series or images loaded.
→ [Page D.3–4](#)
- ✧ Correct the name of the result series (optional).
→ [Page D.3–10](#)
- ✧ Start the calculation.
→ [Page D.3–12](#)

Differentiation

The derivative determines the intensity increase (rise) of grayscale values within a series.

Restrictions

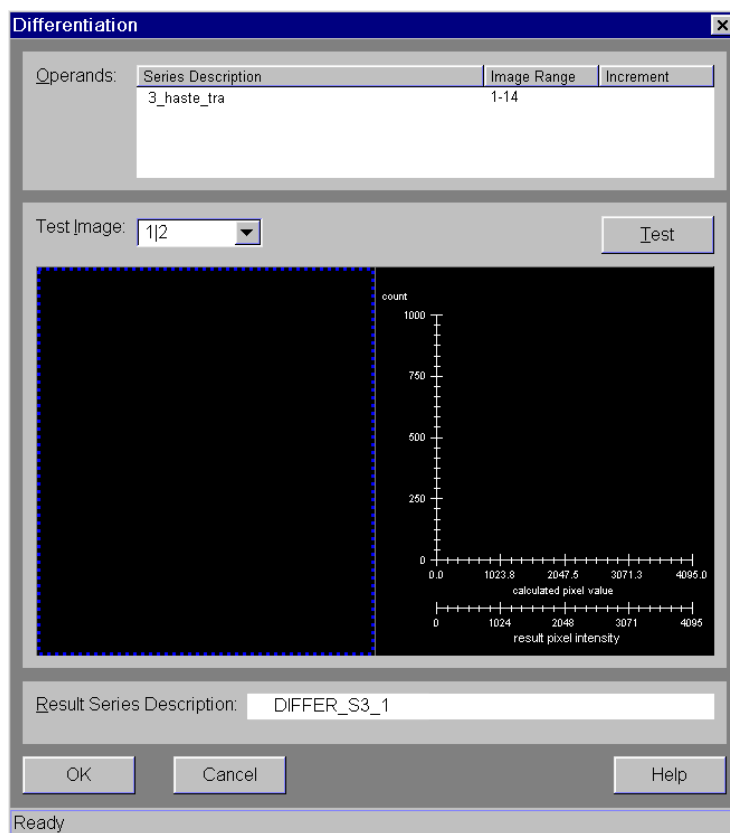
- ☐ You can take the derivative of one series only.
- ☐ All images of this series must have the same slice position.

As a result of differentiation, the new series has one less image than the original series.

The grayscale value in the result images is a direct measure of the tendency of grayscale values to increase or decrease in the initial series.

- ✧ To start the evaluation function, select **Evaluation > Dynamic Analysis > Differentiate...**

Function dialog Differentiation



- ✧ Change the image selection in the operand list (optional).
→ [Page D.3–2](#)
- ✧ Generate preview images.
→ [Page D.3–6](#)
- ✧ Correct the display area (optional)
→ [Page D.3–7](#)

- ✧ Correct the name of the result series (optional).
→ [Page D.3–10](#)
- ✧ Start the calculation.
→ [Page D.3–12](#)

ADC

The Apparent Diffusion Coefficient (ADC) is a measure of the mobility of water molecules between different tissue types. Before you can determine the apparent diffusion coefficients, you must have measurements with various diffusion weightings (b-values). The following section of this manual provides you with additional information regarding diffusion imaging:
→ [Part H, syngo Perfusion](#)

The grayscale values in the result images corresponds to the apparent diffusion coefficients at the respective locations.

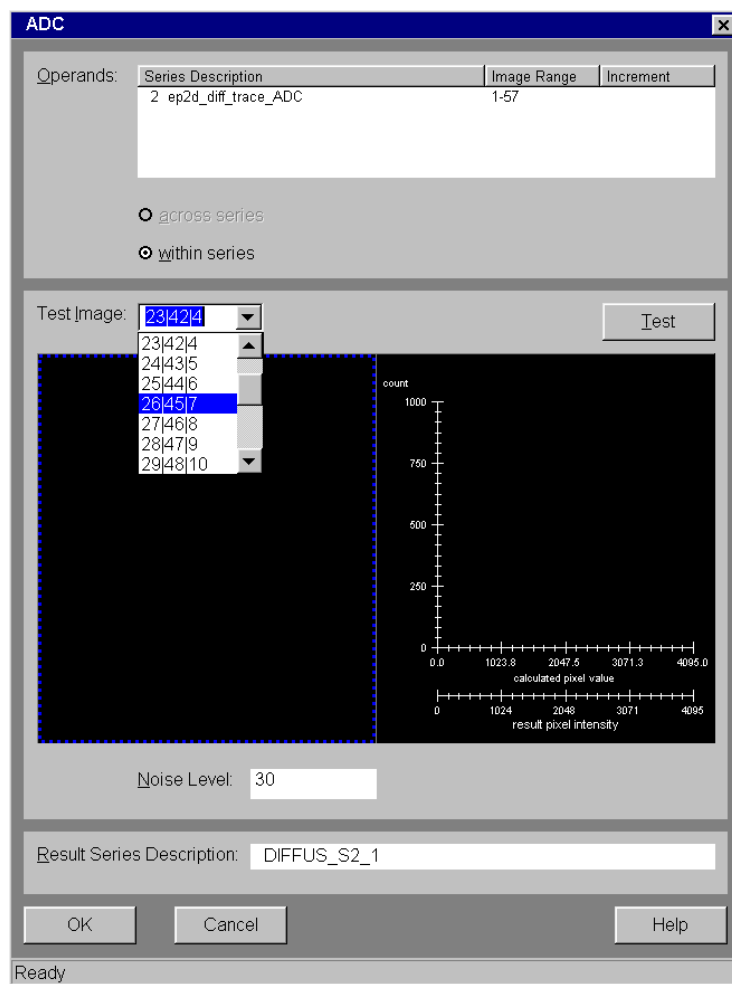
Special characteristics of the operating mode:

Operating mode	Comment
Within series	The series has to include at least 2 images for evaluation within a series. The images of a series are combined into groups that have the same slice position and direction of diffusion. An image with b-value = 0 is assigned to all groups. All images of a group must have a different b-value (diffusion weighting). A result image is calculated for each group.
Across series	As a prerequisite for evaluating between series, each series includes images of all slice positions that have the same b-value. A result image is calculated for each slice position. It is not possible to take different diffusion directions into account.

✧ Start the evaluation function **ADC**.

→ [Page D.2–1](#), *Starting the Evaluation Function*

Function dialog ADC



✧ Change the image selection in the operand list (optional).

→ [Page D.3–2](#)

- ⇒ The operation mode cannot be changed directly. It depends on the number of series or images loaded.
→ [Page D.3–4](#)
- ✧ Generate preview images.
→ [Page D.3–6](#)
- ✧ Correct the display area (optional)
→ [Page D.3–7](#)
- ✧ Correct the name of the result series (optional).
→ [Page D.3–10](#)
- ✧ Start the calculation.
→ [Page D.3–12](#)

Division

You can divide:

- ☐ An image by a second image
- ☐ A series by an image
- ☐ A series and/or images by a constant
- ☐ A series by a second series
- ✧ To start the evaluation function, select **Evaluation > Dynamic Analysis > Division...**

Function dialog Division

Division

Operands:	Series Description	Image Range	Increment
	45 PERFUSION 10 SLICES	1-10	

divide by: 44 PERFUSION 10 SLICES | 1-10 |

☐ constant: 1,0 Exchange

☒ across series

☐ within series

Test Image: 45Bild1 / 44Bild1 Test

count: 1000, 750, 500, 250, 0

0.0 1023.8 2047.5 3071.3 4095.0

calculated pixel value

0 1024 2048 3071 4095

result pixel intensity

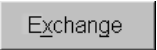
Scaling

Result Series Description: DIV S45/S44 1

OK Cancel Help

Ready

The divisor is shown in a separate field below the list of operands.

A rectangular button with a light gray background and a thin black border. The word "Exchange" is written in a black, sans-serif font, centered within the button.

- ✧ Change the image selection in the operand list (optional).
→ [Page D.3–2](#)

You can change the order of the operands, that is, swap the dividend and divisor of the division.

- ✧ Click **Exchange**.

The new operation is shown in the title bar.

- ✧ Correct the operating mode, as needed.
→ [Page D.3–4](#)
- ⇨ During the **across series** mode, the divisor is shown in a field below the list of operands. By default, the older series is used as the **Divisor**.
- ✧ Generate preview images.
→ [Page D.3–6](#)
- ✧ Correct the display area (optional)
→ [Page D.3–7](#)
- ✧ Correct the name of the result series (optional).
→ [Page D.3–10](#)
- ✧ Start the calculation.
→ [Page D.3–12](#)

Examples for the division mode:

Selection	Possible divisions
2 images	Dividing image by image Dividing both images by the constant
2 series with the same number of images	The image of a series is divided by an image of another series Dividing the images of both series by a constant
Several series, of these at least one series with a single image only	Dividing the images of all other series by a single image of a series

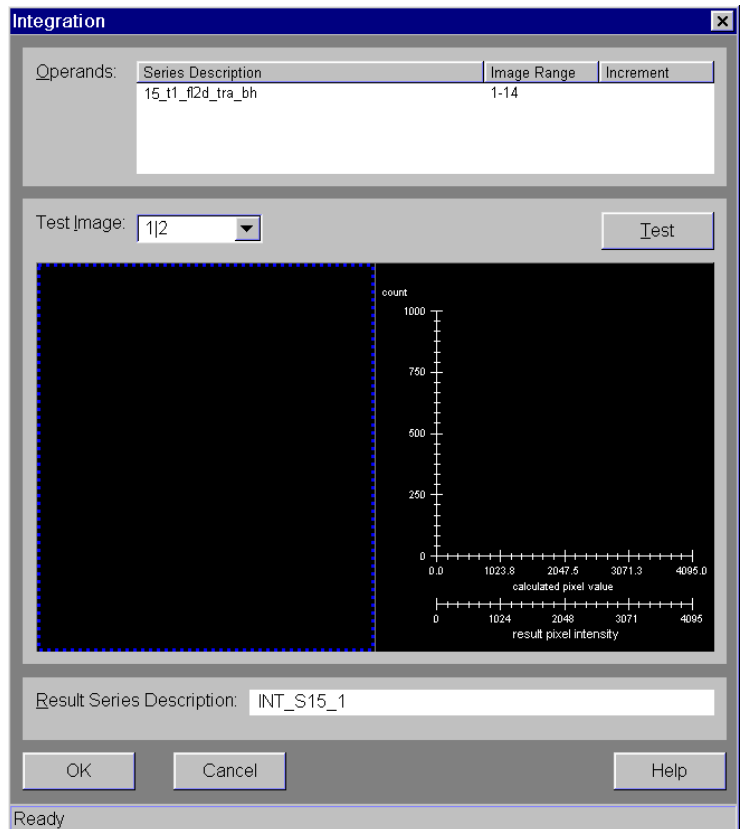
Integration

Integrating the grayscale values of a series provides information on enhancement when using contrast agent studies.

Restrictions

- ☐ You can only integrate via the images of a series.
- ☐ All images of this series must have the same slice position.
- ✧ To start the evaluation function, select **Evaluation > Dynamic Analysis > Integrate....**

Function dialog Integration



- ✧ Change the image selection in the operand list (optional).
→ [Page D.3–2](#)
- ✧ Generate preview images.
→ [Page D.3–6](#)
- ✧ Correct the display area (optional)
→ [Page D.3–7](#)

✧ Correct the name of the result series (optional).

→ [Page D.3–10](#)

✧ Start the calculation.

→ [Page D.3–12](#)

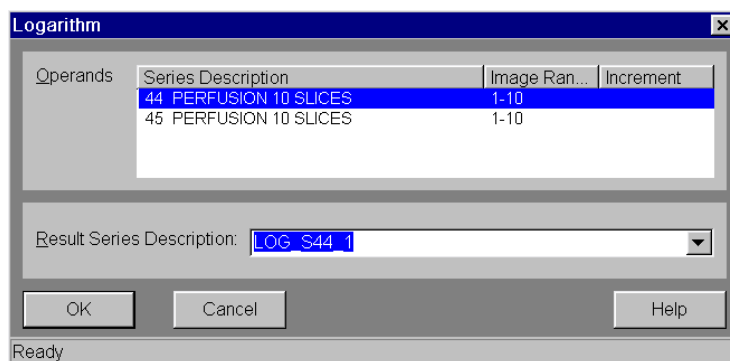
As a result of integration, the new series has one less image than the original series.

Logarithm

Calculating the logarithm of the grayscale values eliminates large differences in MR image intensity. Large intensity gradients are common in MR images of the shoulder and the lungs, for example.

✧ To start the evaluation function, select **Evaluation > Dynamic Analysis > Logarithm....**

Function dialog Logarithm



- ✧ Change the image selection in the operand list (optional).
→ [Page D.3–2](#)
- ✧ Correct the name of the result series (optional).
→ [Page D.3–10](#)
- ✧ Start the calculation.
→ [Page D.3–12](#)

Multiplication

Multiplying the grayscale values by a constant rescales the grayscale values.

You can multiply:

- ☐ Images by a constant
- ☐ A series by a constant
- ☐ Two or more series by a constant

✧ To start the evaluation function, select **Evaluation > Dynamic Analysis > Multiply...**

Function dialog Multiplication

Operands:	Series Description	Image Range	Increment
	44 PERFUSION 10 SLICES	1-10	
	45 PERFUSION 10 SLICES	1-10	

Constant: 1,0

Scaling: ☐ automatic ☒ none

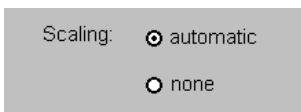
Result Series Description: MULT S44 C1 1

OK Cancel Help

Ready



Constant 50



Scaling: ☒ automatic
☐ none

- ✧ Change the image selection in the operand list (optional).
→ [Page D.3–2](#)

- ✧ Enter a value for the constant during the operating mode.

No test image is generated for multiplication. For this reason, result images cannot be scaled manually. However, you can select automatic scaling.

- ✧ Click the **automatic** option under **Scaling** if you want automatic scaling of the grayscale values.
- ✧ If you deselected automatic scaling (option **none**), all grayscale values above 4095 will be set to this value. The scaling of result images corresponds to the scaling of initial images.

- ✧ Correct the name of the result series (optional).
→ [Page D.3–10](#)

- ✧ Start the calculation.
→ [Page D.3–12](#)

Standard deviation

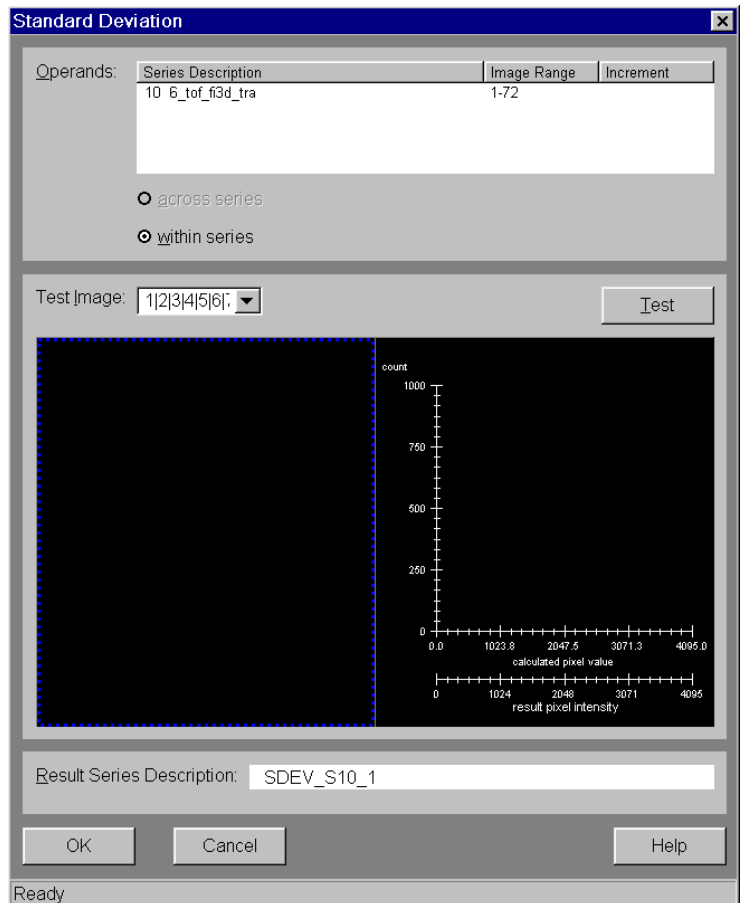
The standard deviation describes the average deviation of the values from the mean value. This function visualizes image modifications.

The grayscale values scatter proportionally to standard deviations which you can determine within a series or between series.

The grayscales in the result images correspond to standard deviations.

✧ To start the evaluation function, select **Evaluation > Dynamic Analysis > Standard Deviation...**

Function dialog Standard Deviation



- ✧ Change the image selection in the operand list (optional).
→ [Page D.3–2](#)

- ⇒ The operation mode cannot be changed directly. It depends on the number of series or images loaded.
→ [Page D.3–4](#)
- ✧ Generate preview images.
→ [Page D.3–6](#)
- ✧ Correct the display area (optional)
→ [Page D.3–7](#)
- ✧ Correct the name of the result series (optional).
→ [Page D.3–10](#)
- ✧ Start the calculation.
→ [Page D.3–12](#)

Slope

You can determine the slope of the linear regression within a series (a result image) or between series (a result series).

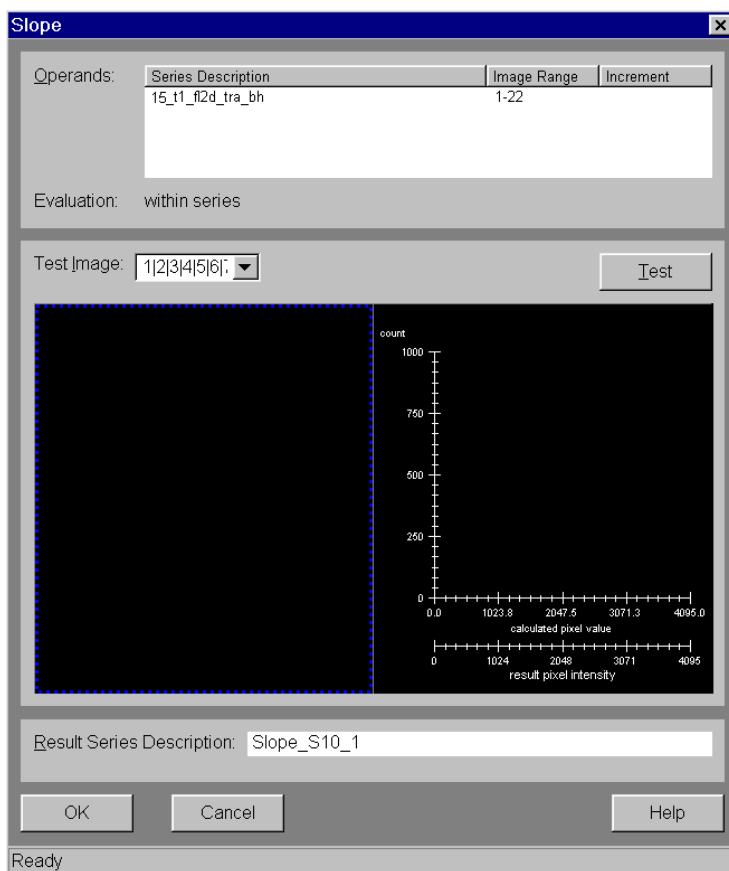
The slope of linear regression describes the rate at which the intensity of the grayscale values increases or decreases within a time series. As in the case of differentiation, the different absorption rates of contrast agent in tissue can be visualized.

⇒ Time series have to contain at least two images. Otherwise, evaluation will not be possible.

The intensity of grayscale values in the result images corresponds to the rising slope of the linear regression. The brighter the grayscale value, the greater the slope.

✧ To start the evaluation function, select **Evaluation > Dynamic Analysis > Slope....**

Function dialog Slope



- ✧ Change the image selection in the operand list (optional).
→ [Page D.3–2](#)

- ⇒ The operation mode cannot be changed directly. It depends on the number of series or images loaded.
→ [Page D.3–4](#)
- ✧ Generate preview images.
→ [Page D.3–6](#)
- ✧ Correct the display area (optional).
→ [Page D.3–7](#)
- ✧ Correct the name of the result series (optional).
→ [Page D.3–10](#)
- ✧ Start the calculation.
→ [Page D.3–12](#)

$T2/T2^*$

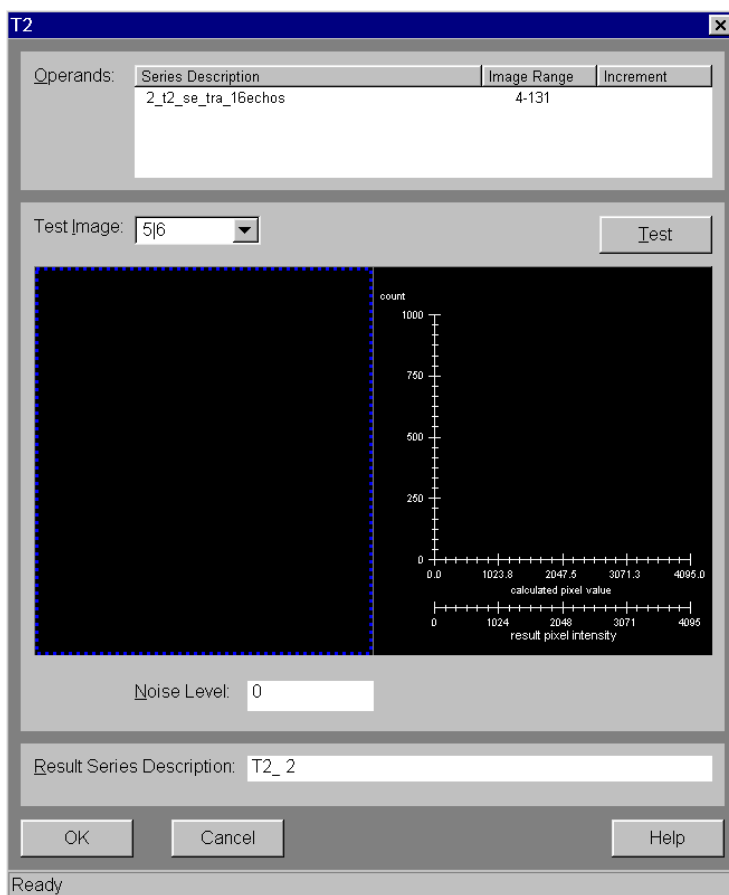
T2* images are especially suitable for showing pathological lesions in tissue.

To calculate T2/T2* weighted images, you must have images with the same repetition time TR, however, with different echo times TE which you generated with multi-echo sequences. These images must have the same slice position. You will find a multi-echo protocol in the **Exam Explorer** under the **Multi-Echo** protocol file (select in the **Exam Explorer: Head > Library > Multi-Echo**).

The result will be a T2/T2* and a T1 weighted image.

- ✧ Start the evaluation function **T2**.
→ [Page D.2–1, Starting the Evaluation Function](#)

Function dialog T2



- ✧ Change the image selection in the operand list (optional).
→ [Page D.3–2](#)
- ✧ Generate preview images.
→ [Page D.3–6](#)
- ✧ Correct the display area (optional).
→ [Page D.3–7](#)
- ✧ Correct the name of the result series (optional).
→ [Page D.3–10](#)
- ✧ Start the calculation.
→ [Page D.3–12](#)

T1

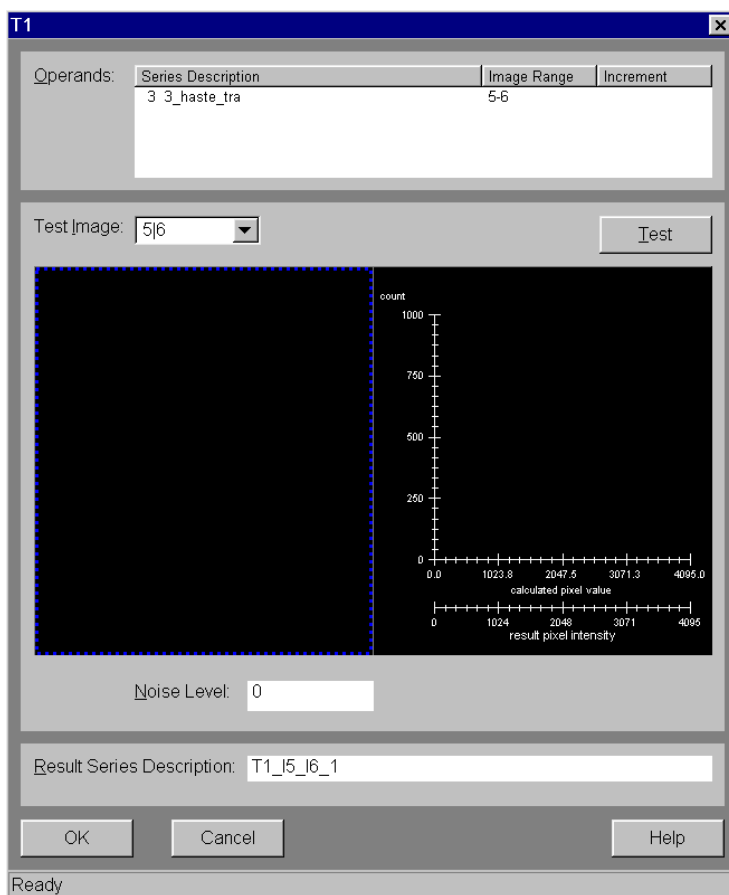
T1 weighted images are suitable for displaying anatomical structures. The system provides you with two ways of calculating T1 weighted images:

- ☐ For the inversion recovery method, you require two images with different inversion times TI. However, the repetition time TR and the echo time TE are the same.
- ☐ For the saturation recovery method, you require two images with different repetition time TR yet with the same echo time TE.

The result will be a T1 and a T2-proton density weighted image.

- ✧ Start the evaluation function **T1....**
→ [Page D.2–1, Starting the Evaluation Function](#)

Function dialog T1



- ✧ Change the image selection in the operand list (optional).
→ [Page D.3–2](#)
 - ✧ Generate preview images.
→ [Page D.3–6](#)
 - ✧ Correct the display area (optional)
→ [Page D.3–7](#)
 - ✧ Correct the name of the result series (optional).
→ [Page D.3–10](#)
 - ✧ Start the calculation.
→ [Page D.3–12](#)
- The result is two images.

Time To Peak

The time up to maximum contrast agent uptake can be evaluated with **Time To Peak**

You can either calculate a minimum grayscale value based on T2/T2* weighted EPI images or a maximum grayscale value based on T1 weighted images.

The result of Time to Peak evaluation is either an image (when using one slice thickness only) or a series of images (when using different slice positions). The grayscale values on the images correspond to the uptake times starting with the first image.

- ✧ The *later* tissue absorbs contrast agent, the *brighter* it appears in the image.

- ✧ To start the evaluation function, select **Evaluation > Dynamic Analysis > Time To Peak...**

Function dialog Time to Peak

Time To Peak

Operands:

Series Description	Image Range	Increment
4 ep2d_perf_tra	1-600	

☐ across series
☒ within series

☐ T1 based
☒ T2* based

Test Image: 1|2|3|4|5|6| Test

Noise Level: 0

Result Series Description: TimeToPeak_S4_1

OK Cancel Help

Ready

☐ T₁ based

☒ T₂* based

✧ Change the image selection in the operand list (optional).

→ [Page D.3–2](#)

✧ The operation mode cannot be changed directly. It depends on the number of series or images loaded.

→ [Page D.3–4](#)

For **Time to Peak** you decide whether evaluation will be made on the basis of T₁ or on the basis of T₂/T₂* weighted images.

✧ Select the respective option below the operand list.

✧ Generate preview images.

→ [Page D.3–6](#)

✧ Correct the display area (optional).

→ [Page D.3–7](#)

✧ Correct the name of the result series (optional).

→ [Page D.3–10](#)

✧ Start the calculation.

→ [Page D.3–12](#)

TTest

This special method is used to evaluate examinations performed with functional neuro imaging. These examinations generally comprise the following series:

- ❑ **Stimulated series** (functional brain activity, acquired during stimulation)
- ❑ **Native series** (no functional brain activity, acquired without stimulation)

You can perform a **TTest** calculation once the series are assigned to stimulated or native series. In the resulting image data, high pixel values indicate that the signal waveform in the measured data takes a course similar to the stimulation paradigm. Anatomical information is no longer contained in the data.

- You can superimpose these images with Anatomy images (e.g., ep2d)
→ [Page D.4–1, Addition](#)

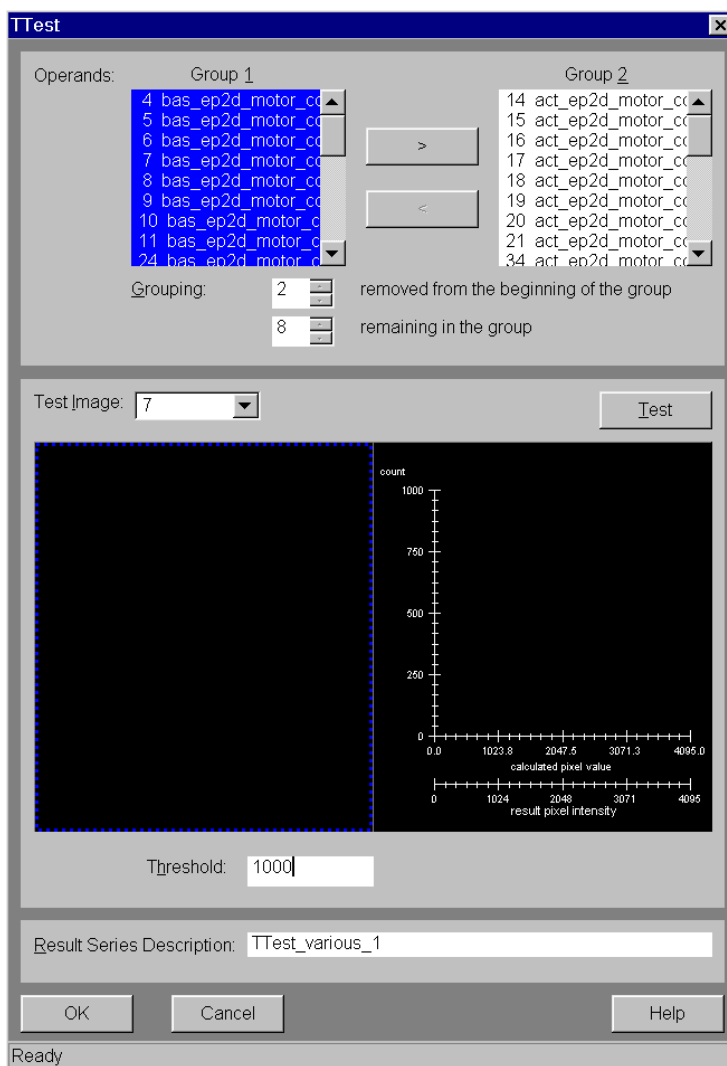
-
- ! • If you superimpose **TTest** images on non-EPI images, compare these images with the superimposed EPI images for safety reasons. Image distortion may occur with EPI measurements.
-

Selecting images

TTest evaluation is started via the **Patient Browser** only. For these evaluations, you need the same number of stimulated and non-stimulated series.

- ✧ Open the **Patient Browser** and select the appropriate series.
- ✧ To start the evaluation function, select **Evaluation > Dynamic Analysis > TTest...**
- ✧ Wait in the function dialog **TTest** until the series in the **Group 1** list is fully displayed.

Function dialog TTest



Grouping series

Grouping series is explained using the following example of a typical BOLD examination with block paradigms:

The paradigm consists of 10 stimulated and 10 non-stimulated series. The procedure is repeated five times.

Initially, all series appear in the **Group 1** list.

You can exclude any series from the evaluation that was acquired at the beginning of the activity or during relaxation, because they only show the time required to reach a steady state of the deoxygenation/oxygenation of blood.

removed from the beginning of the group

✧ Enter **2** in the upper field to exclude the first two series from being evaluated.

- Or -

remaining in the group

Enter **8** in the lower field.

The first two series in **Group 1** and **Group 2** are removed.

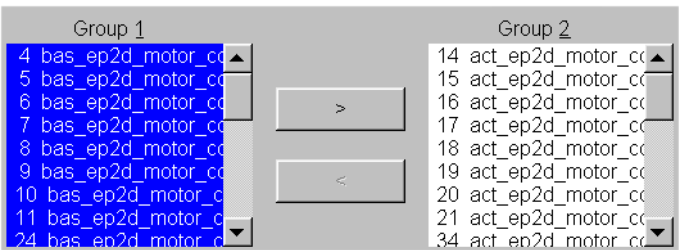
✧ The numeric sum in the two fields corresponds to the total number of series in the groups. If you increase the number of series excluded, the number of remaining series is automatically reduced by this value.

✧ Mark *All* series in the **Group 1** list.



✧ Click this button to group the series in **Group 1** and **Group 2**.

After a few seconds the "act" series (stimulated) are moved to the **Group 2** list, the "bas" series (not stimulated) remain in **Group 1**.



✧ Check whether the lists contain the same number of stimulated and non-stimulated series.



If the numbers in the two lists are not the same, you are able to shift selective series of a group list to the other list by using these buttons.

Additional settings

- ✧ Generate preview images.
→ [Page D.3–6](#)

The grayscale values in the test images are compared with the threshold value (default = 0). Grayscale values below this threshold are not displayed.



- ✧ Enter the required pixel value in the **Threshold** input field.
- ✧ Correct the additional display area (optional)
→ [Page D.3–7](#)
- ✧ Correct the name of the result series (optional).
→ [Page D.3–10](#)
- ✧ Start the calculation.
→ [Page D.3–12](#)

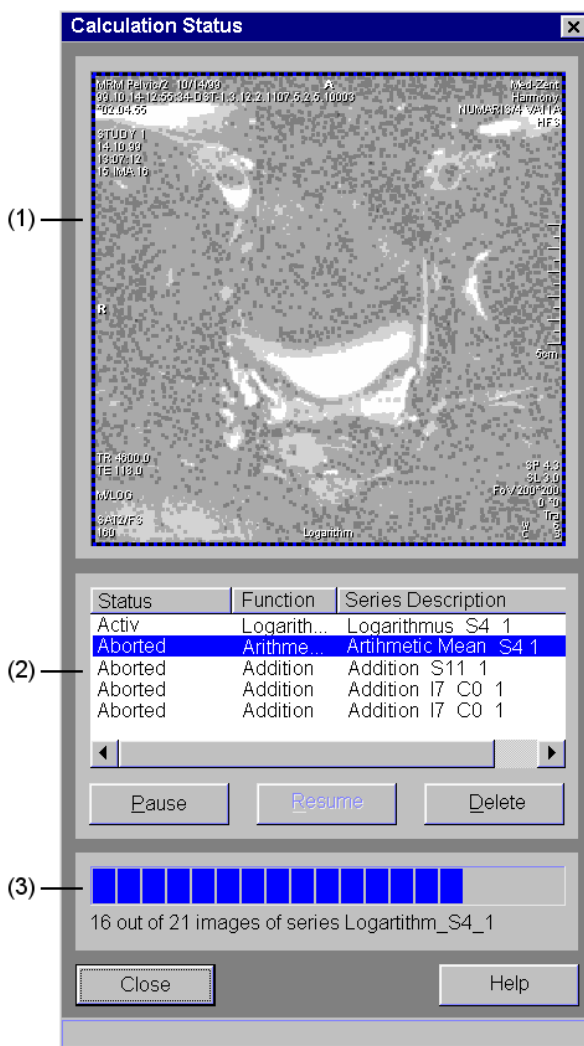
Displaying the evaluation progress

If you are evaluating a large number of series or very large series, you can monitor the progress of evaluation, as well as pause, resume, or cancel it.



- ✧ Click the **Dynamic Analysis** symbol on the status bar of the system.

The **Calculation Status** dialog window opens.



- (1) Image area
- (2) Status display of the evaluation jobs
- (3) Progress indicator

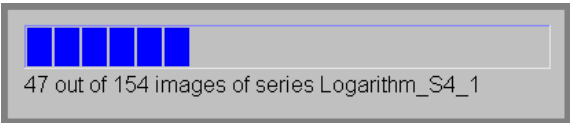
Status display

The evaluation jobs can have the following processing statuses:

Status	Comment
Active	The job is currently being processed
Paused	The job has been interrupted
Aborted	The job was ended by the MR system for, e.g., calculation errors.
Waiting	The job is still waiting for evaluation

Progress indicator

The progress indicator shows the progress of the current evaluation. It does not show the status of the selected job in the list.



- ! If there is a change of operators during the **Calculation Status** dialog and the new operator has no access authorization to the data of the reconstructed image, the data will be hidden.
The series name is replaced by the name of the user who generated the image.
The new user can stop and then resume evaluation. But he cannot delete the job.

Starting the next job

Once a job has been processed, the images will be displayed on the **Viewing** task card.

The next job with the "Waiting" status (top of the list) is started automatically.

Closing the dialog window



✧ Click **Close** to close the dialog window.

Pausing evaluation

You can pause the evaluation at any time, for example, to view an evaluated image and to check whether the evaluation meets your requirements.



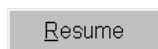
✧ Click **Pause**.

Current image reconstruction will be completed and the image is displayed in the image area.

The evaluation status is set to "Paused".

If the list contains a job with the status "Waiting", it will be started automatically.

Resuming evaluation



✧ Click **Resume**.

If you have activated another job, this job will go into the queue.

Cancelling and deleting the evaluation

You can stop evaluation from the queue, for example, if you find that the evaluation does not meet your requirements. You have to stop an evaluation prior to deleting it.

✧ Select the requested active evaluation from the list.

✧ Click **Pause** to stop the evaluation.

✧ Click **Delete** to clear the selected job from the list.

The canceled job is then deleted from the queue.

⇒ Images already reconstructed are stored and may be displayed on the **Viewing** task card.

A rectangular button with a light gray background and a thin black border. The word "Pause" is centered in a dark gray, sans-serif font.A rectangular button with a light gray background and a thin black border. The word "Delete" is centered in a dark gray, sans-serif font.



Filtrate, Correct and Convert Images

E.1 *Editing Images with Image Filters*

Selecting images	E.1–1
Setting the filter	E.1–3
Selecting a filter	E.1–3
Defining customer filters	E.1–4
Naming a result series	E.1–4
Starting filtering	E.1–5

E.2 *Correcting Distortion Artifacts*

Basics	E.2–1
Using 2D distortion correction	E.2–3
Selecting images	E.2–3
Starting calculation	E.2–3
Using 3D Distortion Correction	E.2–4
Selecting images	E.2–4
Starting calculation	E.2–4
Monitoring the progress of distortion correction	E.2–5
Results	E.2–5
Result series	E.2–5
Visualization of image information that was not measured	E.2–6
Resetting 2D distortion correction	E.2–7
Selecting images	E.2–8
Starting calculation	E.2–8

E.3 *Converting Images to RGB Images*

Basics	E.3–1
Converting images	E.3–2



Use the **Offline Image Filter** to post-process blurred images or those with strong noise. These are adaptive filters that search for conspicuous structures in the image. Subsequently, the filter checks if these are either random or part of other structures. Filters take into account the local distribution of signal intensity and the course of local tissue structures.

The images selected in the **Patient Browser** or the **Viewing** task card are used for reconstruction.

Siemens provides three default filters and three freely definable filters.

Prerequisites

It is not possible to postprocess images or series with an **Offline Image Filter** that have been processed with another filter already.

Selecting images

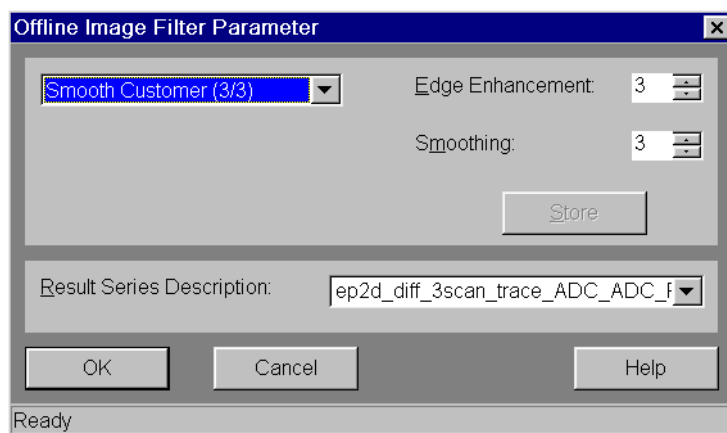
Select the images or the series in the **Patient Browser** or in the **Viewing** task card.

- ☐ Advantages of **Patient Browser**:
 - Fast image selection
- ☐ Advantages of **Viewing** task card:
 - You can look at the images prior to filtering and edit them, if necessary.

- ✧ Select all series and images that you want to filter in the **Patient Browser** or **Viewing**.
- ✧ Select **Evaluation > Dynamic Analysis > Image Filter** to start the filter.
 - Or -
 - Click this button on the tool bar of the **Patient Browser** or in the **Evaluation** subtask card in the **Viewing**.



The **Offline Image Filter Parameter** dialog window opens.



Setting the filter

Select the image filter you want to use in the **Offline Image Filter Parameter** dialog window.

In the case of a customer filter, define the following properties.

- ☐ **Edge enhancement:**
Degree of edge and line enhancement
- ☐ **Smoothing:**
Degree of smoothing

Selecting a filter

Three default Siemens filters are available: **Smooth**, **Medium** and **Sharp**. In addition, there are three freely-selectable customer filters: **Smooth Customer**, **Medium Customer** and **Sharp Customer**.

- ✧ The actual filter properties are displayed in parentheses after the name:

Filter name (edge enhancement/smoothing)



- ✧ Select a filter in the upper selection list.

Defining customer filters

Edge Enhancement:

- ✧ Enter a whole number between 1 and 5 in the **Edge Enhancement** input field.

Smoothing:

- ✧ Enter a whole number between 1 and 5 in the **Smoothing** input field.

- ✧ Click **Store**.

✧ The values cannot be changed when using a Siemens filter.

Naming a result series

A result image or result series is generated for each original image or series.

Name of the result series

An automatically generated name for the **Result Series Description** is entered in the result series field. You can accept this name or change it as desired.

Result Series Description:



The suggested name comprises the following sections, separated by underscores:

- ❑ Name of the initial series
- ❑ "FIL" abbreviation for Offline image filter
→ [Page P.2–17, List of image types](#)
- ❑ A running number
- ✧ The result series name may only contain valid characters and may not exceed 64 characters in length. If necessary, the name of the initial series is truncated automatically.

Renaming

- ✧ If the name suggested does not meet your expectations, enter the description preferred by you.

Selecting result series

If you loaded several series, the **Result Series Description** input field functions as a selection list as well. This allows you to check and/or change the names for the result series that have been loaded.

Starting filtering

After you have selected your filter, you can start image processing.

- ✧ Click **OK**.

The dialog window closes.



This icon in the status bar indicates that a filtering process is running in background.

Monitoring filtering



You can monitor the progress of filtering in the **Calculation Status** dialog window. You can also interrupt or cancel filtering and delete filter jobs from the list.

✧ Click the icon on the status bar.

The **Calculation Status** dialog window opens.

→ [Page D.5–1, Tracking the Progress of Evaluation](#)

Displaying the results

Once the jobs have been processed, the images will be displayed on the **Viewing** task card.

The following image types indicate filtered images; the values displayed in the user-defined filters correspond to the values set for edge enhancement and smoothing:

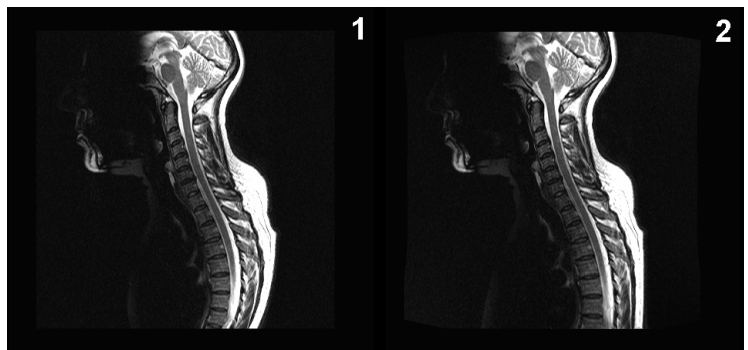
Filter (Edge enhancement/ Smoothing)	Image type
Smooth (3/3)	FSM (Filter Smooth)
Medium (3/3)	FM (Filter Medium)
Sharp (3/3)	FSH (Filter Sharp)
Smooth Customer (1/4)	FSM1_4
Medium Customer (2/5)	FM2_5
Sharp Customer (3/2)	FSH3_2

→ [Page P.2–17, List of image types](#)

Basics

Geometric distortions (barrel-shapes/pin cushion) may occur at the edge of images with a large FoVs or with off-center slices. These image errors can be eliminated using **Distortion Correction**.

Image artifacts



- (1) Image with distortion artifacts
- (2) Image after distortion correction

Cause

Gradient coils are used to encode spatial information. Under ideal circumstances, the gradient field rises linearly. In reality, however, deviations from linearity may occur at the image margin.

Remedies

If you are planning quantitative evaluations of images, it is essential to correct distortion artifacts.

You have two possibilities:

- (1) Before measuring, switch on the **Distortion Corr.** filter. The images resulting from this measurement are distortion-corrected.
→ [Page P.1–52](#), *Parameter Card Resolution - Filter*
- (2) You are correcting the images retroactively by selecting distortion correction in the **Patient Browser**. This method is described on the next pages.

A subsequent distortion correction generates a new series. The uncorrected original images are retained and can be used to position slices for additional measurements.

2D and 3D distortion correction

During 2D distortion correction, image distortions are individually corrected in each slice.

During 3D distortion correction, not only the voxels from the current slice, but also those in neighboring slices are taken into account. The correction results are more precise, but require a longer reconstruction time. 3D distortion correction may only be used for specific data sets.

→ [Page E.2–4](#), *Prerequisites*

Using 2D distortion correction

Requirements

The images have to meet the following prerequisites for 2D distortion correction:

- ☐ The images were acquired with **syngo MR**
- ☐ The images still need to be distortion-corrected

Selecting images

You can select individual images, one or more series or single images of different series.

- ✧ Select **Patient > Browser**.
- ✧ Select the images or series in the Patient Browser.

Starting calculation

- ✧ Select **Evaluation > 2D Distortion Correction** from the Patient Browser.

Distortion correction is started.

- ✧ You can monitor distortion correction.
→ [Page E.2–5](#)

Using 3D Distortion Correction

Prerequisites

3D distortion correction is applicable only for entire series, it is not suitable for individual images. These series have to meet the following prerequisites:

- ☐ Only entire series are selected
- ☐ The images in the series were acquired with **syngo MR**
- ☐ The series do not contain any images that have already undergone distortion correction
- ☐ The series has to have at least 2 slices
- ☐ The slices have to be the same size
- ☐ The slices have to be the same slice thickness
- ☐ The distance between slices may not exceed the slice thickness, if the data were measured with a 2D sequence

Selecting images

You can select one or more series.

- ✧ Select **Patient > Browser**.
- ✧ Select the series from the Patient Browser.

Starting calculation

- ✧ Select **Evaluation > 3D Distortion Correction** from the Patient Browser.

Distortion correction is started.

- ✧ You can monitor distortion correction.
→ [Page E.2–5](#)

Monitoring the progress of distortion correction



As soon as you've started distortion correction, the icon for image post-processing jobs appears. If calculation fails, the icon appears crossed out.



✧ Click the icon on the status bar.

The **Post-processing Queue** dialog window is displayed. This is where you monitor the progress of correction distortion.

→ [Page G.4–5](#), *Monitoring post-processing*

Results

Result series

You receive new series as a result of distortion correction. Distortion-corrected images can be recognized by image type: DIS2D for 2D and DIS3D for 3D corrected images.

→ [Page P.2–17](#), *List of image types*

Example

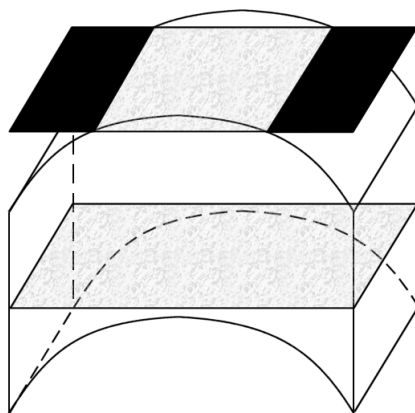
You have selected three images from series S1, four images from series S2, and two images from series S3 for 2D distortion correction.

Three new result series are created: Series E1 with three images, series E2 with four images, and series E3 with two images.

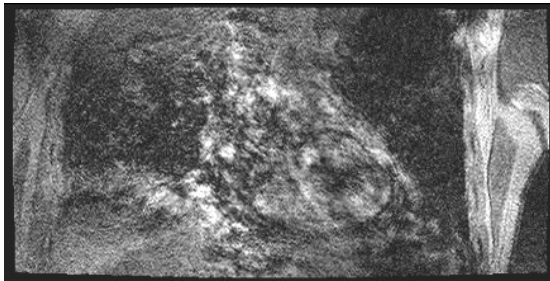
Visualization of image information that was not measured

The expansions and compression of original images during distortion correction may create areas in the result images without data acquisitions. These areas are visible as black surfaces.

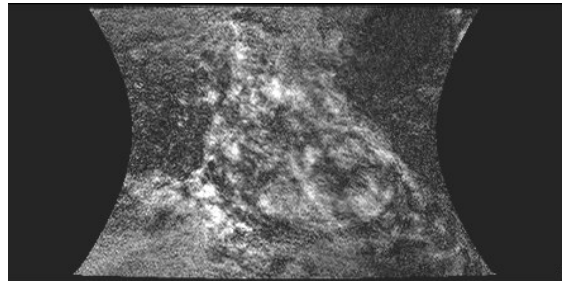
2D distortion corrections may show black areas along the edges of the result images. The larger the field of view and the larger the distance of the acquired image from the magnet isocenter, the stronger the effect will be. With 3D distortion corrections, the slices along the edge may occasionally show larger black areas with circular or parabola-shaped boundaries. This happens when additional (non-existing) slices in the original image would have been needed for reconstructing the image areas.



Location of 3D slabs in reference to the volume measured (with exaggerated distortion correction)



Phenomena with DIS2D



Magnification of phenomenon with DIS3D

Resetting 2D distortion correction

This function is required in special cases when you must have non-distortion corrected images (e.g., for positioning during a spectroscopy examination) but you only have 2D distortion corrected images available.

Prerequisites

The images have to meet the following prerequisites to **Undo 2D Distortion Correction**:

- ☐ The images were acquired with **syngo MR**
- ☐ The images have undergone 2D distortion correction

Selecting images

You can select individual images, one or more series or single images of different series.

- ✧ Select **Patient > Browser**.
- ✧ Select the images or series in the Patient Browser.

Starting calculation

- ✧ Select **Evaluation > Undo 2D Distortion Correction** from the Patient Browser.

Non-distortion corrected images identified with ND are reconstructed retroactively from distortion-corrected images.

- ⇒ This is where you monitor the progress of reconstruction. Proceed as you do when monitoring the progress of distortion correction.
→ [Page E.2–5](#), *Monitoring the progress of distortion correction*

Converting Images to RGB Images

Basics

syngo-based systems allow to link images with overlay graphics. These overlay graphics are not based on the DICOM format but rather on a Siemens-own vector format.

To display these overlay graphics with a non-*syngo*-based DICOM Viewer (e.g., PACS), they have to be converted into another, pixel-oriented image format.

Traditional conversion

During traditional conversions, the overlay graphics are converted into DICOM bitmaps. However, diagnostic information may get partially lost during this conversion. The following disadvantages are present when displaying these images in standard DICOM viewers:

- ☐ No color support
- ☐ Poor scalability
- ☐ Blurring with small image matrix

RGB converter

However, when converting data into DICOM-conform RGB images, the diagnostic image is retained. Data exchange of secondary capture images in the RGB pixel format is always ensured, because most medical systems support RGB images.

You can go to the **Patient Browser** and convert all images or image series into RGB images or RGB series via the appropriate function.

Converting images

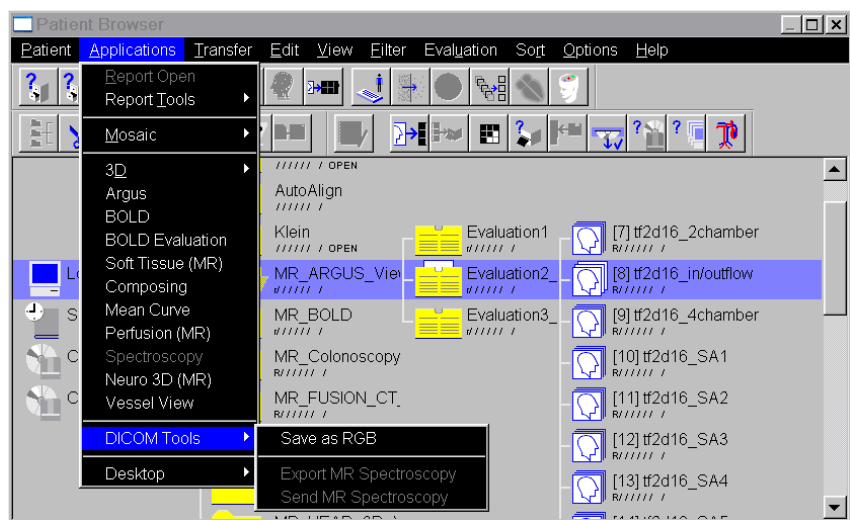
You are able to select a series or, as an alternative, either individual or several images within a series.

Selecting images

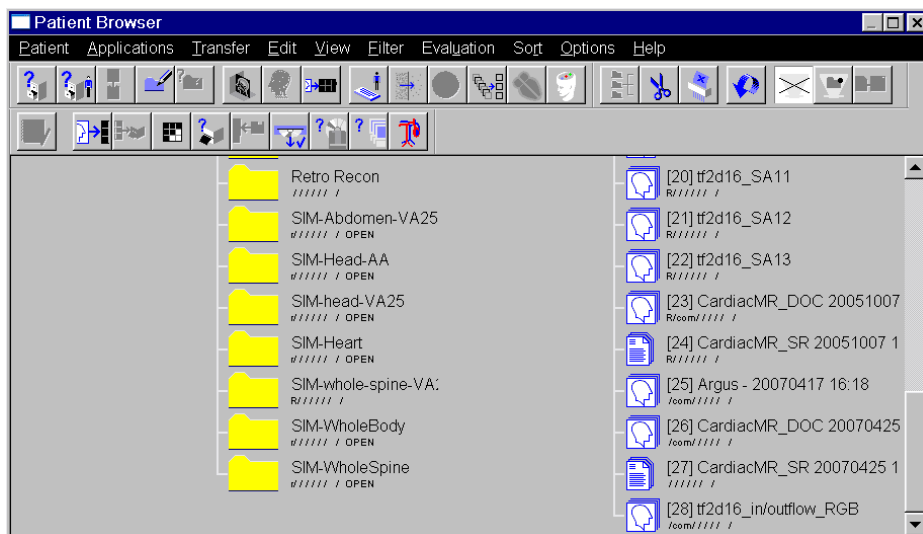
- ✧ Select **Patient > Patient Browser**.
- ✧ Select the desired images or series in the **Patient Browser**.

Starting the conversion

- ✧ Select **Applications > DICOM Tools> Save as RGB**.



The converted images or series are shown as a new image series in the **Patient Browser**. Their names are extended by "_RGB".



You can use the RGB pixel images of this series for exchanging data with other systems or DICOM viewers.



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F.3 *Performing Evaluations*

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The **Mean Curve** task card allows for statistical evaluation of signal changes as a function of time and place. For example, you use **Mean Curve** to examine how the average grayscale value changes in a region of interest (**ROI**) depending on slice position, trigger time, or image number within measured series or results series after post-processing (e.g., MPR/MIP or Dynamic Analysis).

The result of a **Mean Curve** evaluation appears as a curve that plots the mean grayscale value against a selectable second variable (X-axis).

A **Mean Curve** evaluation is frequently performed for perfusion measurements, flow measurements, or angiography measurements.

Mean Curve task card

Starting Mean Curve

You can open the **Mean Curve** task card from any other task card or from the **Patient Browser**.

✧ Select **Applications > Mean Curve**.

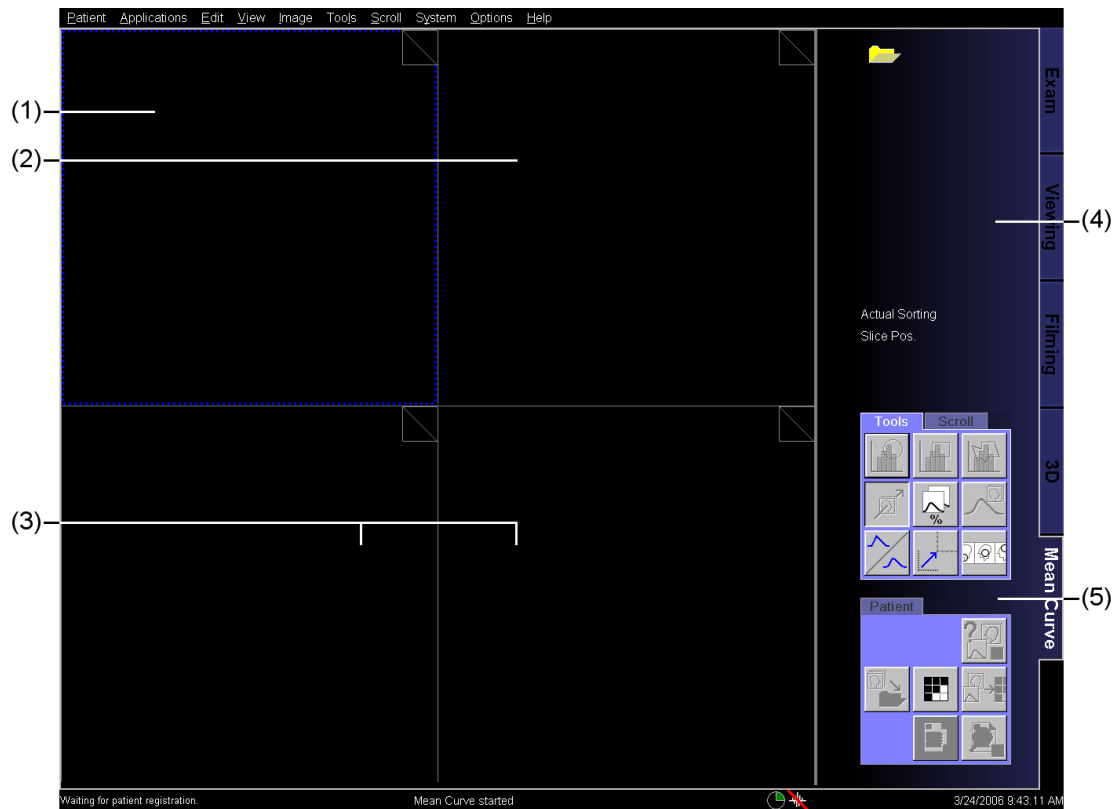
Mean Curve task card opens.

Closing Mean Curve

✧ Select **Applications > Close Mean Curve**.

The **Mean Curve** task card closes.

Configuration of task card



- (1) First segment (evaluation segment)
- (2) Second segment (result segment)
- (3) Third and fourth segment
- (4) Display of sort criteria
- (5) Control area with subtask cards

Loading and Viewing Images

Loading images

For evaluation with **Mean Curve**, load the images and series of an examination into the first, third, and fourth segment of the task card. The second segment is reserved for displaying results.



Only MR images can be loaded into **Mean Curve**. It is not possible to load images of other modalities, such as CT or AX images.

⇒ It is not possible to load a complete examination or patient on the **Mean Curve** task card.



For **Mean Curve** evaluations, the user has to have full access rights to the patient data to be evaluated. You also require the privilege to store data.

Detailed information with respect to security settings in *syngo* are included in the *syngo* Security Package brochure.

Loading images to the first segment

Select the images and series in the **Patient Browser** or in the **Viewing** task card and load them into the **Mean Curve** task card.

Prerequisites for Loading

To *load* images into the first or third segment, they have to meet the following criteria:

- ☐ Same matrix size
- ☐ Same field of view (FoV)
- ☐ Same orientation
- ☐ Same series block and same table position
→ [Page C.1–6, Series block](#)
- ☐ Same Frame of Reference
→ [Page C.1–7, Frame of Reference \(FoR\)](#)
- ☐ Same type of distortion correction (2D, 3D, none)
- ☐ Uniform motion correction (yes or no)

An appropriate system message is displayed during loading if the selected images do not meet a criterion.

↗ Radial series cannot be evaluated with **Mean Curve**.

Meeting prerequisites for evaluation

The images in the first segment have to meet the following additional criteria for *evaluation*:

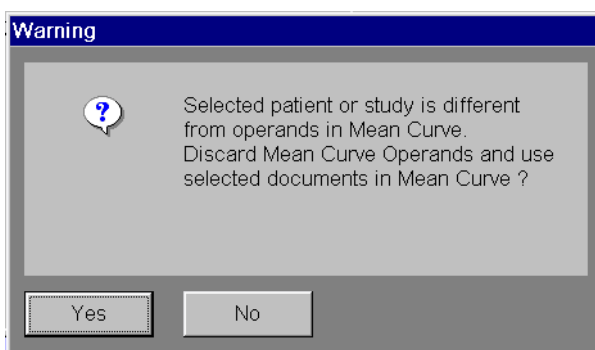
- ☐ Same slice position
- ☐ Same slice thickness

Loading images via a menu	✧ Select the desired images or series in the Patient Browser or the Viewing task card. ✧ Select Applications > Mean Curve. The Mean Curve task card is opened and the images are loaded.
Loading images via drag & drop	How to load images from the Patient Browser into the open Mean Curve task card. ✧ Open the Patient Browser. ✧ Drag the series required to the first segment with the mouse. - Or - - Double-click the series required.
Image display	The images and series loaded are displayed as an image stack. ❑ You will see the center image of the series if you have loaded one series only. ❑ You will see the center image of the middle series if you have loaded several series. The images are ordered according to the default sorting rules. → Page F.2–7, Determining the sort ✧ Graphic objects contained in the images (e.g., from a previous Mean Curve evaluation) are not loaded.

Adding images

You may load additional images of the same examination into the first segment at any time. All images in the first segment are sorted according to the default sort.

You cannot load images from another examination. The following warning is displayed if you load images from another examination:



✧ To replace the loaded images, click **Yes**.

Or

✧ Click **No** to cancel loading.

Loading images into the third and fourth segments

The images in the third and fourth segment will *not* be taken into account in the actual evaluation. However, the images have to belong to the same patient and to the same examination as the images in the first segment.

Prerequisites

- ❑ You may load, for example, the subtraction series or the MIP series of the examination into the third segment. To copy ROIs from the third segment to the first segment, the images in the third segment must meet the same conditions as the images in the first segment.
→ [Page F.3–5](#), *Drawing ROIs in the third segment*
- ❑ Images in the fourth segment may have a different orientation from the images in the first segment. For example, you can load reference images of the exam or images showing the lesion in a different orientation into the fourth segment.
→ [Page F.2–14](#), *Searching for an original image by scrolling in the fourth segment*

Loading images

- ✧ Open the **Patient Browser** and select the series.
- ✧ Drag the selected series into the third or fourth segment with the mouse.

Image display

The same display rules apply to the loaded images as in the first segment.

Exception:

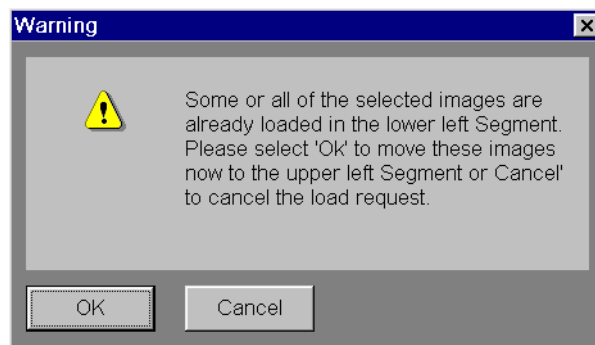
You have already loaded images or series in the first segment and you are now in the process of loading another series into the third segment. The third segment shows *the* image with the same default sort criteria as the current image in the first segment.

Moving images to another segment

A series or image cannot be simultaneously displayed in multiple segments. However, you can move images from one segment to another.

- ✧ Select the same images again that are already displayed in the **Patient Browser**.
- ✧ Drag the selected images to the required **Mean Curve** segment.

The following warning may appear.



- ✧ Click **OK**.



Determining the sort

By sorting the loaded images, you establish the scrolling sequence in the first and third segments and define the X-axis of the **Mean Curve** evaluation.

In the fourth segment, the images are always sorted by the image number and the series number.

Evaluation mode

A distinction is made between evaluation *within* a series and evaluation *between* the corresponding images of several series.

Images in 1st segment	Evaluation mode	Sorting the loaded images
One series	 Within:	Via the X-axis sort criteria → Page F.2–9
Multiple series	 Across:	Across the series via the sort criterion of the X-axis within the series via a second sort criterion → Page F.2–10

↗ The current sort order is displayed in the control area.

Opening the Scaling dialog window



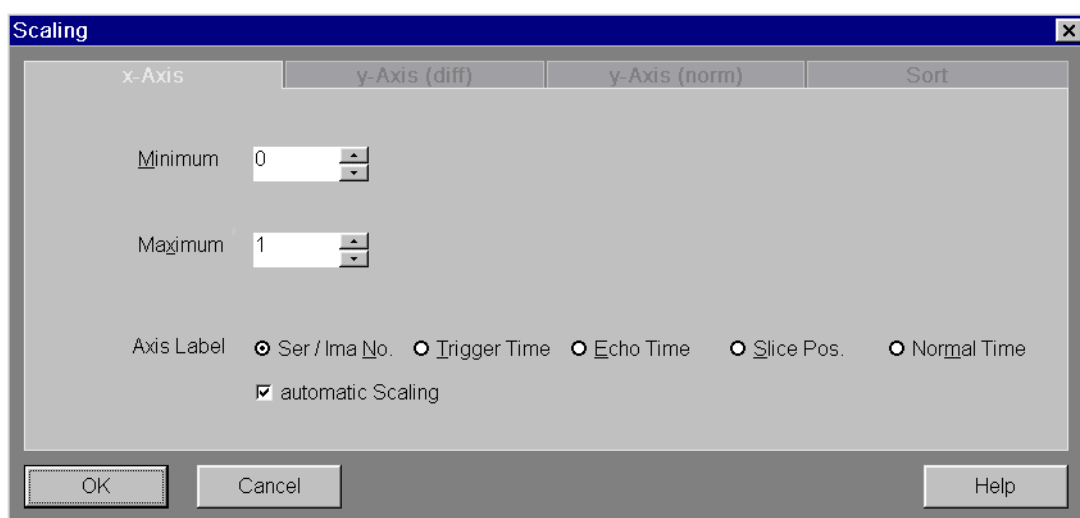
Establish the sort criteria in the **Scaling** dialog window.

✧ Select **Tools > Scaling/Sorting**.

- Or -

Click the **Tools** subtask card using this button.

The **Scaling** dialog window opens.



Sorting in the evaluation mode *Within*

Establish the sort criteria in the *Within* mode via the X-axis. The following criteria are available:

Sort criteria (X-value)	Comment
Series / Image No.	Sorting by image or series number
Trigger Time	Sorting according to trigger time. Suitable for displaying physiologically triggered measurements, e.g., cardiac series
Echo time	Sorting by echo time suitable for displaying multi-echo sequences, e.g., multi-echo spin-echo sequences
Slice Pos.	Sorting by slice position
Normal time	Sorting by excitation time suitable for dynamic and/or motion studies. For the evaluation mode <i>Within</i> , the normal time of an image is the difference between the acquisition time of the image and the shortest acquisition time for an evaluation of relevant images.

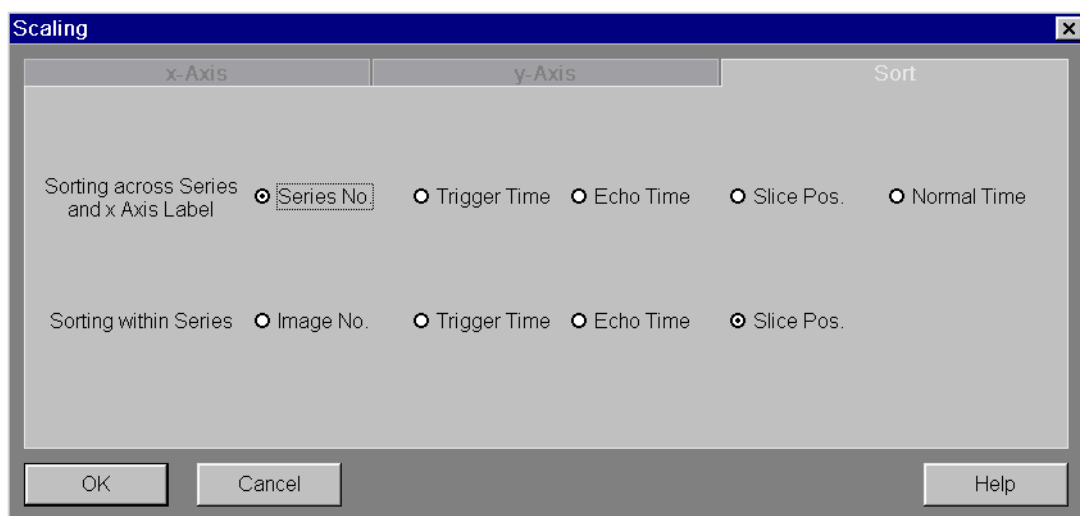
- ✧ Click the **X-axis** subtask card in the **Scaling** dialog window.
- ✧ Select the desired option in the **Axis** group.

Sorting in the evaluation mode Across

You set the sort criteria in the *Across* mode using the **Sort** subtask card.

✧ This subtask card is not available in the *Within* evaluation mode.

✧ Click the **Sort** subtask card in the **Scaling** dialog window.



Sorting	Comment
Across series	X-axis parameter; establishes the evaluation sequence for the corresponding images
Within series	Establishes the sequence of images within the series

- ✧ Select the desired option for sorting across the series.
- ✧ Select the desired option for sorting within the series.
- ✧ The two sort criteria have to be different.
- ✧ Click **OK**.

The sort criteria will be applied and the **Scaling** dialog window closes.

Selecting images when using the same sort size

The sort order of images in the first segment is checked and updated every time images are loaded, deleted, or when the sort criterion is switched.

If multiple images have the same value for a sort criterion, the image selection with the **Same sort size dialog** window appears.



Function	Comment
All oldest (newest) images	Selects the oldest/newest image from all images with the same sort size for evaluation
Oldest (newest) image	Selects from images with the same sort size displayed, takes the oldest (newest) image for evaluation (a new selection has to be made for each match)
Cancel	Cancels the procedure, resulting in the display of images with the same sort size (e.g., loading of additional images)

✧ Click the button with the desired function.

Searching for an original image for evaluation

You can scroll through the loaded images and series to look for a suitable image.

Sorting through images

Sorting via the menu using dog ears is described in detail in the *syngo* Operator Manual.

→ **Viewing/Changing image settings**

The scrolling functions described below are available in the **Scrolling** subtask card in the control area of the **Mean Curve** task card.



Button	Function
 	Scrolls within the series to the next / previous image
 	Scrolls to the next / previous series

Special issues

- ☐ Images that are not evaluated because of the same sort criterion will not be displayed as you scroll.
- ☐ Once you have loaded a series into the first and third segments and are scrolling through one of them, the other segment will look for and display the corresponding image with the same *within* sort criterion. If an image with the same *within* sort criterion is not loaded in the segment, the next image meeting this sort criterion is displayed.
- ☐ Images that have the same *within* criterion will be displayed as you scroll across series. A series is placed last in the scrolling order and an empty segment is displayed along with a message, if an image with the corresponding *within* sort criterion is missing in a series. This series will then be ignored during evaluation.

Searching for an original image by scrolling in the fourth segment

In some cases it is simpler to look for the most suitable image in the first segment by using the reference images.

For this reason, **Mean Curve** provides you with the possibility of loading the reference images of the images in the first segment into the fourth segment.

→ [Page F.2–5, Prerequisites](#)

In the reference images loaded, move the cut line that represents the position of the current image in the first segment.

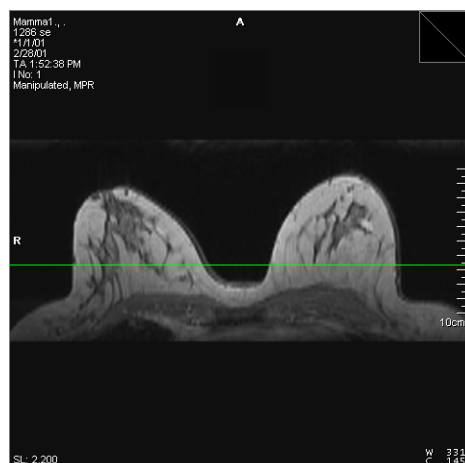
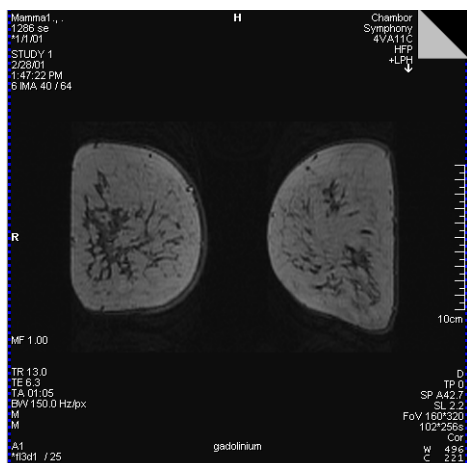
- ✧ Load the reference images of the examination into the fourth segment.
- ✧ Look for the reference image that best represents the anatomy to be evaluated.
- ✧ Use the mouse and move the cut line in the reference image to the anatomy to be evaluated.

The first segment displays the image the position of which best matches the position of the cut line. Segment three automatically displays the image with the identical slice position (as shown in the first segment).

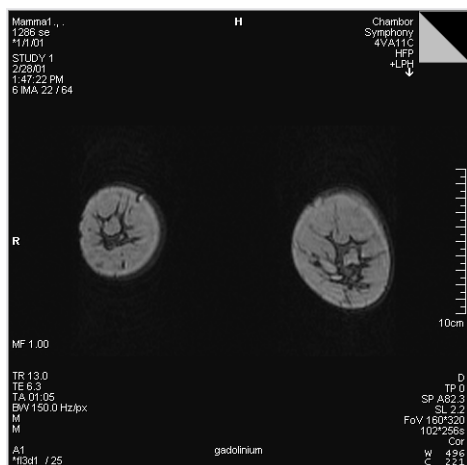
- ✧ The cut line is not displayed, if the image in the first segment and the reference image have the same orientation. In this case, look for another reference image.

Example

Starting situation in the first and fourth segment:



After having moved the cut line:



Changing image display

You may:

- ☐ Change the brightness and contrast (windowing)
- ☐ Zoom/reduce/pan
- ☐ Zoom/reduce to segment size
- ☐ Invert grayscale values

These actions are described in detail in the *syngo* Operator Manual.

→ **Viewing**/Changing image settings



During scrolling, the **Mean Curve** task card automatically transfers each change in display to all images of the respective segment.

- ✧ Select the desired function from the **Image** menu.
 - Or -
 - Select the function desired from the context menu for the corresponding segment.

A **Mean Curve** evaluation involves the following work steps:

- ❑ Drawing ROIs
- ❑ Establishing the type of evaluation and starting it
- ❑ Optimizing the display of results

Drawing ROIs

After you have optimized the image display and selected the starting image for evaluation in the first segment, draw ROIs (Regions of Interest). With ROIs you define the areas of the image that will be evaluated.

➤ You can draw up to four ROIs in the first or third segment.

Drawing ROIs in the first segment

The following tools are available in the **Mean Curve** task card for drawing ROIs:

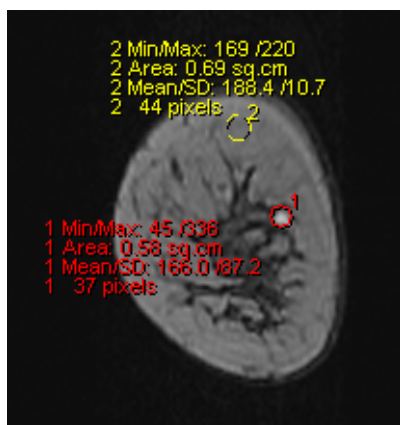
Button	Tool
	Rectangular ROI
	Circular ROI
	Freehand ROI

Drawing and editing ROIs is described in detail in the syngo Operator Manual.

→ **Basics**/2D- and 3D evaluation

- ✧ Draw one or more ROIs in the original image in the first segment.
- ⇒ Ensure that the ROIs do not extend beyond the edge of the image. In case it is not possible to perform a calculation, "???" will be displayed in place of ROI measurement values.

The ROIs differ in color and line style. Each ROI is also identified by a number. The text belonging to the respective ROI can be moved.



Applying ROIs

Within the first segment, you apply ROIs by scrolling. The result depends on the ROI mode set.

ROI modes

There are two different ROI modes:

- ☐ **Static ROI:** The ROIs of the initial image also apply to all other images in the first segment (default after starting **Mean Curve**).
- ☐ **Dynamic ROI:** ROIs in the initial image can be adapted to the other images in the first segment.

Changing the ROI mode



- ✧ Select **Tools > Dynamic ROI**.
- Or -
- Click the **Tools** subtask card and select this button.

Or



- ✧ Select **Tools > Static ROI**.
- Or -
- Click the **Tools** subtask card and select this button.

Scrolling with static ROIs

As you scroll, the ROIs drawn in the initial image will be applied without changes to the other images in the first segment. ROIs drawn in the other images are overwritten.

Scrolling with dynamic ROIs

While scrolling, ROIs from the previous image are copied to the current image. However, existing ROIs are not overwritten. You have the opportunity to review the ROIs and adapt them to the anatomy to be evaluated, if necessary.

Example:

You have drawn four ROIs in the initial image. The next image already contains ROI 1 and 2. As you scroll, the next image contains only ROI 3 and 4 from the initial image.

Adjust ROI 2 to the anatomy to be evaluated in the next image. The next image does not include any ROIs. As you continue scrolling, this image will contain all four ROIs from the previous image but not from the initial image.

Deleting ROIs

✧ Select the ROI that you want to delete.

✧ Select **Edit > Delete Graphics**.

- Or -

Press **DEL**.

The ROI selected is deleted in all images from the first and third segment.

Filling in missing ROIs

All images required for the current evaluation must have the same number of ROIs.

Mean Curve adds missing ROIs automatically by copying them from the nearest images. For example, this is the case when you load images into the first segment after drawing ROIs.

Drawing ROIs in the third segment

Mean Curve provides you with the possibility of loading additional series to the third segment, e.g., subtraction series, parameter maps or MIP series of the examination. These images frequently allow for easier anatomic evaluations and a more accurate drawing of ROIs.

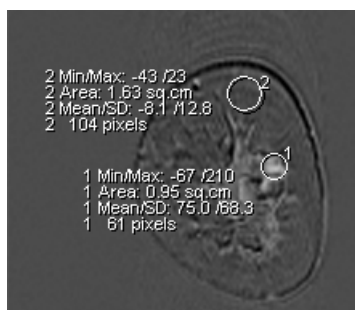
The ROIs of the first segment are displayed automatically in the third segment. All changes to ROIs in the third segment (drawing, changing, or deleting) are applied automatically to the first segment.

Loading images

- ✧ Load suitable images into the third segment.

The third segment shows the image that has the same sort criterion (*within*) as the current image in the first segment.

- ☐ If the slice position of the images in both segments is identical, the ROIs in the third segment are displayed with the line type and color from the first segment.
- ☐ If the slice positions in the first and third segment differ, the ROIs in the third segment are displayed in white with a solid line going through them (refer to figure).



Drawing or changing ROIs

As in the first segment, you may draw a maximum of four ROIs in the third segment. If there are already four ROIs, you have to delete one before drawing a new one.

→ [Page F.3–4, Deleting ROIs](#)

✧ Search in the third segment for the image where you want to draw or change ROIs.

✧ Change the desired ROIs.

✧ Draw the desired ROIs.

→ **Basics**/Evaluating regions in the *syngo* Operator Manual

If the images in the first and third segment have the same slice position, modified and new ROIs are applied automatically to the first segment.

If the slice positions of the images in the two segments differ, an appropriate message is displayed. You then have to decide whether to apply the ROIs to the first segment.

✧ To apply ROIs, click **Yes**.

Or

✧ Click **No** and the ROIs will not be applied.

If you do not apply the ROIs, the new ROIs and any changes in the third segment are deleted.

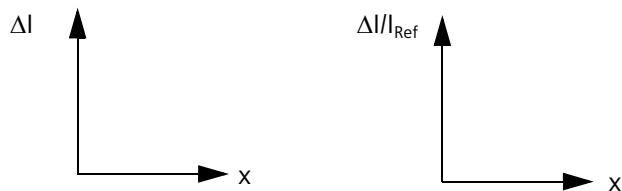
Setting and starting the evaluation

Before you start the evaluation, define whether you want to perform an absolute or relative evaluation.

➤ Absolute evaluation is the default setting after you have started **Mean Curve**.

The signal intensity is assigned to the X-axis during absolute evaluation.

Relative evaluation generates two curves for each ROI. The difference between the signal intensity and a reference value as well as the ratio of the difference to the reference value are applied to the X-axis.



- ☐ Select a single image as the reference image if you are evaluating within a series.
- ☐ Select a whole series as the reference if you are evaluating across series.

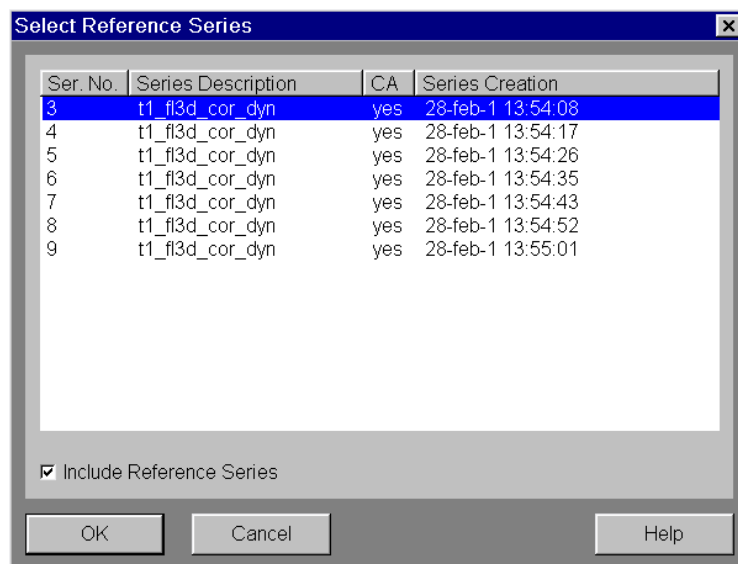
Setting the relative evaluation

- ✧ Select the **Tools > Evaluation mode: relative**.

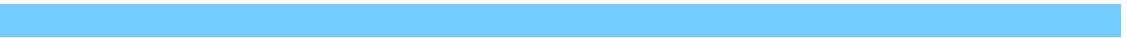
- Or -

Click the **Tools** subtask card and select this button.

The dialog window **Select Reference Image** (*within mode*) or **Select Reference Series** (*across mode*) opens.



The list contains the images or series loaded in the first segment.



✧ Select the desired reference image or reference series.

☒ Include Reference Series

✧ When the reference image or the reference series are not to be included in the evaluation, please disable the checkbox.

✧ No references are required when evaluating cardiac images as a function of the trigger time. The references are required, however, when you evaluate contrast agent enhancement in dynamic series.

✧ Click **OK**.

The dialog window closes.

✧ The selected reference is used for all subsequent evaluations. As soon as you switch evaluation modes (*within* or *across*), the selected series is no longer valid.

Starting evaluation

- ! If you want to change patient or examination data (via **Correct** in the Patient Browser) that are loaded in **Mean Curve**, change them prior to starting the evaluation. Otherwise, inconsistent result images are generated with **Mean Curve**, because the changes could not be incorporated into result images that are stored or transferred after **Filming**.

→ **Patient Browser**/Correcting data in the *syngo* Operator Manual

- ✧ Select **Tools > Start Evaluation**.

- Or -

Click the **Tools** subtask card using this button.



The results of **Mean Curve** evaluation will be displayed in diagram and table format in the second segment (result segment).

- ✧ The evaluation cannot be started, if you have not drawn any ROIs or if (due to the current sort criteria) only one image is available for evaluation. In this case, the button and menu item are dimmed.

Automatic recalculation

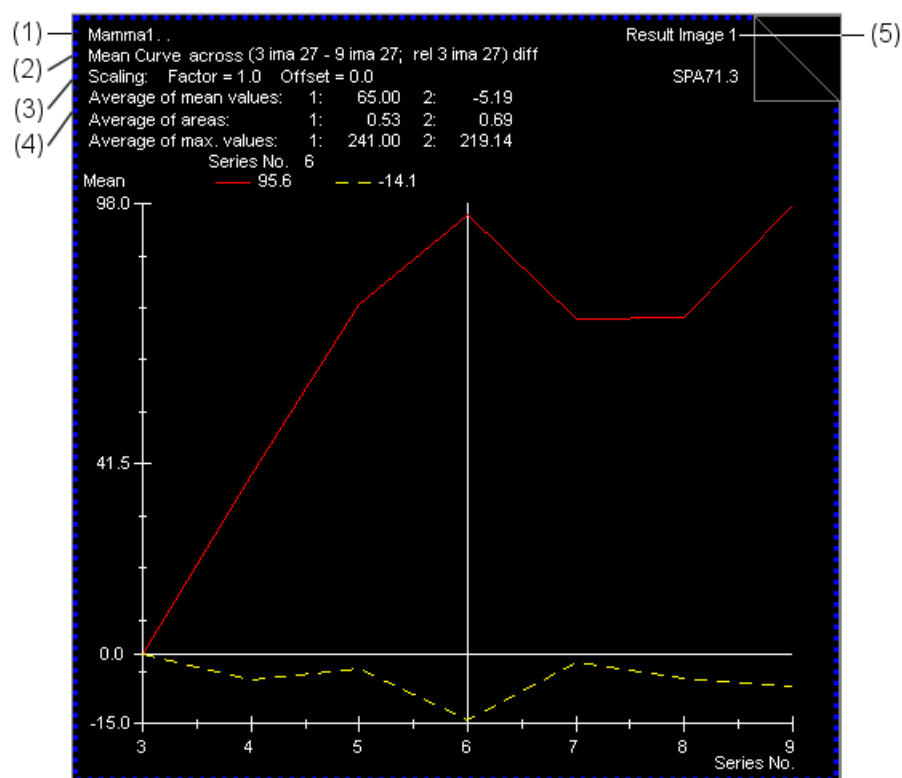
The results will be rejected and removed from the second segment as soon as you add new images, delete images, manipulate ROIs, or change the sort criterion. You are able to go to the menu and establish that in cases such as these, recalculation will be performed automatically.

✧ Select **Tools > Auto Recalculation**.

- ☞ If you have loaded a large number of series into **Mean Curve**, you should deselect this option. This will save time.
- ☞ If you switch from **Dynamic ROI** evaluation to **Static ROI**, the evaluation is automatically recalculated.

Result display in the diagram

The diagram shows a curve for each ROI. Using the color and line style, you can assign the ROIs in the first segment to the corresponding curves in the diagram.



- (1) Patient name
- (2) Changing the image comment (can be changed)
- (3) Information about scaling
- (4) Mean values for measurement results (for absolute evaluation only)
- (5) Number of the table or diagram



Result display in the table

A table is generated for each diagram.

Mammal1 . . .											
Mean Curve across (3 ima 27 - 9 ima 27; rel 3 ima 27) diff											
Scaling: Factor = 1.0 Offset = 0.0											
Series No.	ROI 1			ROI 2			ROI 3			ROI 4	
	Mean	SDev	Area	Mean	SDev	Area	Mean	SDev	Area	Mean	SDev
3	0.0	31.1	0.5	0.0	7.5	0.7					
4	39.3	44.6	0.5	-5.6	7.7	0.7					
5	76.0	79.5	0.5	-3.0	8.1	0.7					
6	95.6	90.9	0.5	-14.1	9.5	0.7					
7	72.8	46.8	0.5	-1.7	7.7	0.7					
8	73.5	46.5	0.5	-5.0	8.9	0.7					
9	97.8	54.6	0.5	-7.0	13.0	0.7					

The left column lists the values on the X-axis, e.g. the serial numbers. Next to that you will find the mean value, standard deviation, and area content of each ROI.

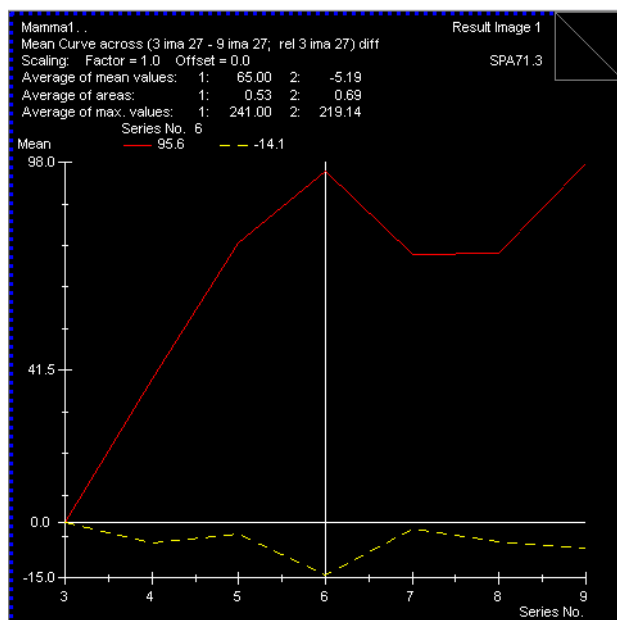
Changing the display of the results

You can use the following options for changing the display of the results in the diagram:

- ☐ Using a vertical measurement line
- ☐ Scaling axes
- ☐ Smoothing curves
- ☐ Changing the image comment
- ☐ Using an image from the first segment as background

Using a vertical measurement line

For a more precise evaluation of the curve, you can have the intensity of the signal displayed for a defined X-value. For this purpose, you can shift the vertical line in the image along the X-axis.



✧ Move the line using the mouse.

The X-value (e.g., series number) and the mean values of the intensities for each ROI are displayed above the curves.

✧ If the vertical line is not visible, it is superimposed directly onto the Y-axis.

Scaling axes

After starting **Mean Curve**, automatic scaling is activated, which scales both axes according to the input data.

Manual scaling helps you adjust the graduation of the axes more accurately to the section of interest.

You are able to scale the grayscale values.

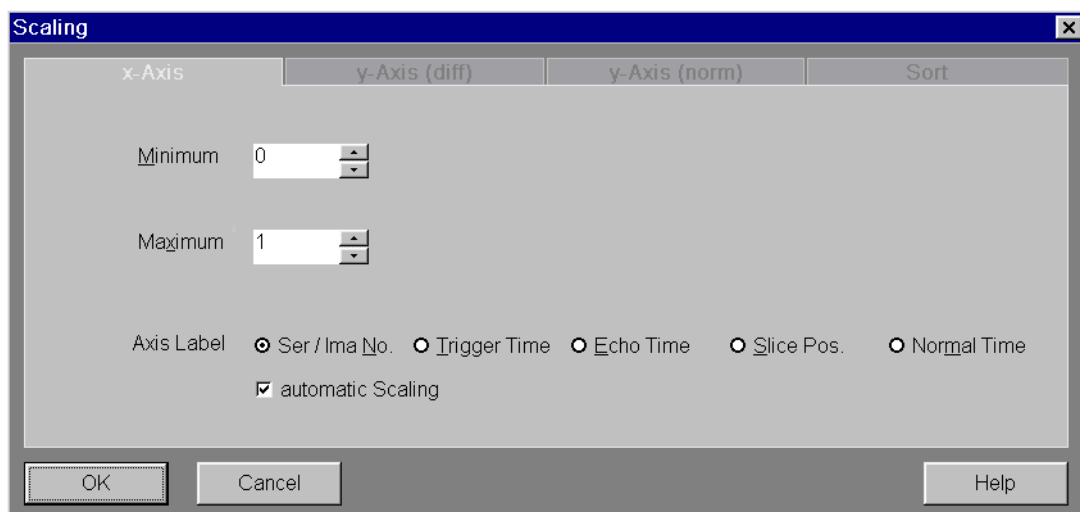
✧ Select **Tools > Scaling/Sorting**.

- Or -

Click the **Tools** subtask card using this button.

The **Scaling** dialog window opens.





Scaling the X-axis

- ✧ Click on the **X-axis** subtask card.
- ✧ Enter the left boundary of the axis into the **Minimum** input field.
- ✧ Enter the right boundary of the axis into the **Maximum** input field.

The **automatic Scaling** checkbox is automatically disabled. The scaling set applies to all subsequent evaluations.

- ✧ **Automatic Scaling** can be enabled together only for the X- and Y-axis.

Scaling the Y-axis

Depending on the type of evaluation, different subtask cards are available for scaling the Y-axis:

Evaluation	Subtask cards
Absolute	Y-axis
Relative	Y-axis (diff) for signal differentiation curves Y-axis (norm) for normalized curves

Minimum

Maximum

☒ Linear

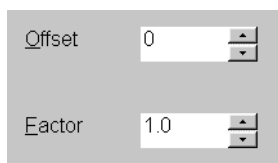
☐ Logarithmic

- ✦ Click the **Y-axis** (or **Y-axis (diff)** or **Y-axis (norm)**) subtask card.
- ✦ Enter the left boundary of the axis into the **Minimum** input field.
- ✦ Enter the right boundary of the axis into the **Maximum** input field.

In the **Scale** group, you determine whether the Y-axis should be subdivided linearly or logarithmically.

- ✦ Select the required option.

Scaling grayscale values



A screenshot of a software dialog box for scaling grayscale values. It contains two input fields: 'Offset' with the value '0' and 'Factor' with the value '1.0'. Each field has a small up/down arrow button to its right.

Scaling the measured grayscale values is useful if the grayscale values are proportional to a physiologically relevant value and the proportionality factor is known (e.g., flow velocity of the blood). Instead of the previous grayscale values, the scaled grayscale values are displayed in the result images.

- ✧ Enter the number by which you want to multiply all Y-values in the **Factor** input field.
- ✧ Enter the number you want to add to all Y-values in the **Offset** input field.

Example: Factor = 2, Offset = 20

All Y-values will be multiplied by 2 and shifted 20 units toward the positive Y-axis.

- ✧ To apply the values entered, click **OK**.

The dialog window closes. The graphs and tables are updated.

- ✧ The scaling set applies to all subsequent evaluations.

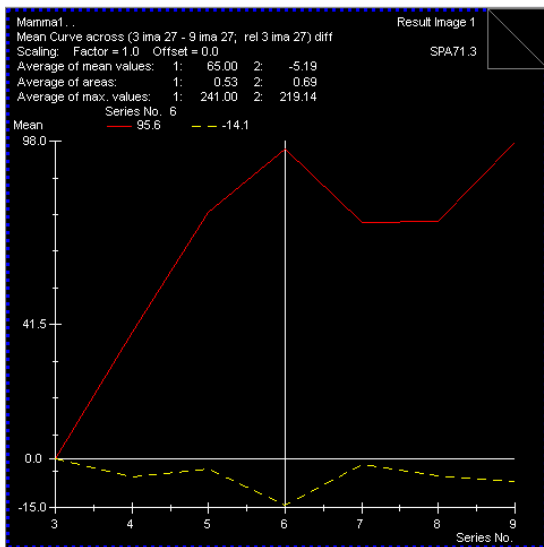
Smoothing curves

During smoothing, the individual measurement points can either be connected via straight lines or via a smooth curve.

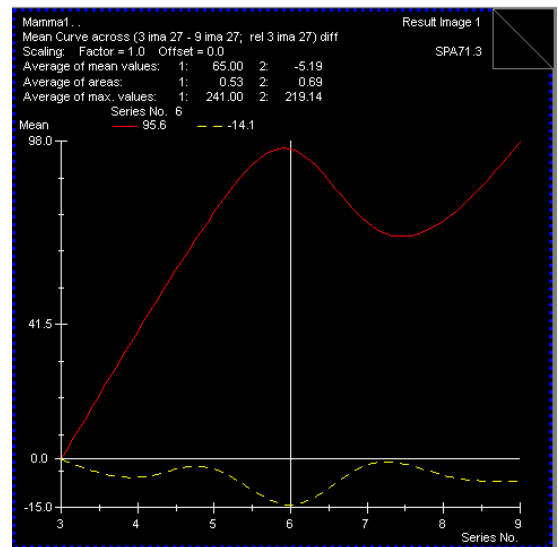
✧ Select **View > Smoothing On**.

- Or -

Click the **Tools** subtask card using this button.



Curve is not smoothed



Smoothed curve

Changing the image comment

Example of an automatically generated commentary line:

Mean Curve across (4IMA7-11IMA7; rel4IMA7) norm

Component	Comment
4 IMA7 - 11 IMA7	Evaluation of image 7 across series 4 to 11
rel 4ima 7	Relative evaluation with image 7 from series 4 as the reference
norm	Normalized curve
diff	In the case of signal difference curves, ratio of the signal difference to the reference image

✧ Double-click the commentary line.

The comment line is opened for editing.

✧ Enter the new comment.

✧ Press **ENTER**.

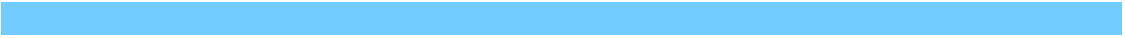
The comment is updated in all result images.

☞ If a portion of the standard text changes, e.g., the image number or reference, the standard text is displayed again in the comment line.

Using the background image for diagrams

You can use the current image in the first segment as the background for curves.

- ✧ If necessary, adjust the contrast and brightness in the first segment for an optimally visible curve in the second segment. Restart the evaluation.
- ✧ In the **View** menu, select the **Image with Graphics** option.
- ✧ You can reset a black background using the **Graphics only** option in the **View** menu.



There are various ways to document the results of the **Mean Curve** evaluation:

- ☐ Saving images
- ☐ Saving results in ASCII format
- ☐ Filming images
- ☐ Creating a Report



If evaluated data are still loaded in **Mean Curve** where the patient or examination data were changed after evaluation, **Mean Curve** displays a message that data may be inconsistent.
→ **Patient Browser**/Correcting data in the syngo Operator Manual

Saving images

In **Mean Curve**, you can save all images for an evaluation (result images and relevant initial images) in a series without having previously selected them.

- ⇨ If you only want to store part of the images or append them to an existing series, proceed as described in the **Viewing** task card.
→ **Viewing**/Saving images in the syngo Operator Manual

Saving the evaluation as a new series



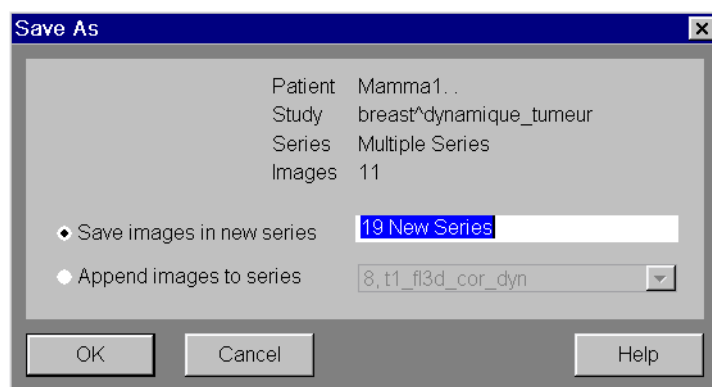
- ✧ Select **Patient > Save Evaluation As.**

- Or -

Click the **Patient** subtask card using this key.

- ✧ The menu item and the key remain dimmed as long as evaluation results are not displayed in the second segment.

The first and second segments are selected. The **Save As** dialog window is displayed.



- ✧ Enter a name for the new series in the input field.

- ✧ Base the name on the one for the original study.

- ✧ Click **OK**.

You will not be able to store the series if its name contains unauthorized characters. The program displays a message indicating that the series name is invalid and requests that you review and correct the text entered.

The new series includes all initial images used for the evaluation with ROIs, windows and Zoom/Pan values from the first segment and the result images from the second segment. The ROIs for the 3. segment are not automatically stored.

You are able to automatically save and/or transmit the new series.

⇒ The ROIs of images stored in **Mean Curve** can no longer be processed on the **Viewing** task card.

Saving results in ASCII format

✧ Select **Patient > Save As ASCII**.

- Or -

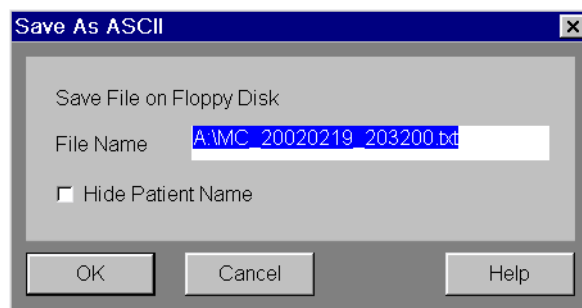
Select **Save as ASCII** in the context menu for the second segment.



When user administration is active, this function is available only to users with export privileges.

Additional information is included in the *syngo* Security Package.

The **Save As ASCII** dialog window opens.



The ASCII file can be saved to a disk (drive **A:**) or in directories especially set up for import/export by Siemens Service.

- ✧ If necessary, change the directory and name of the file using the **File Name** input field.
- ✧ If you do not want to save the patient's name for data security reasons, click the checkbox.
- ✧ Click **OK**.

The data are stored. A message window is superposed which includes the most important storage data (e.g., drive or file name).

- ✧ For a relative evaluation, the suggested file name for a signal differential table ends in **_diff**, and for a normalized table it ends in **_norm**.
- ✧ Click **OK** to close the message window.

☒ Hide Patient Name

Opening the ASCII file

You are also able to open the ASCII files of the **Mean Curve** evaluation and process them further, e.g., in **Microsoft Excel**.

- ✧ While searching for the file to be opened, ensure that you are looking for a **Text file (*.txt)**. In addition, ensure that you set a decimal point under **Regional Settings/Numbers** on your PC.

Filming images

The manual film function described in the **Examination** part can be used with **Mean Curve** as well.

→ [Page C.4–35](#), *Copying images to the film sheet*

In addition, you are able to transfer the two current images from the first and second segment to the virtual film sheet without making a selection.

- ✧ Select **Patient > Film Evaluation**.

- Or -

Click the **Patient** subtask card using this key.



The images will be transferred to the virtual film sheet and sent to the camera either immediately or after a delay depending on the default setting.

For additional information, refer to the syngo Operator Manual.

→ **Filming**

Creating and editing reports

You can use the results of the current **Mean Curve** evaluation to create a structured medical report in DICOM format. The report can be edited, printed, and transmitted.

Creating a report

You can create a report as soon as the evaluation results are displayed in the second segment.

✧ Select **Patient > Create Report**.

- Or -

Click the **Patient** subtask card using this key.

The report is created and saved as a separate series in the current patient folder. The report name is derived from the name of the current series.

The menu item and button are deactivated. Multiple reports cannot be created for an evaluation.



Displaying report

✧ Select **Patient > View Report**.

- Or -

Click the **Patient** subtask card using this key.

The **Report - Viewer** is shown:



Report - Viewer

Current Procedure Description Findings Interpretation of findings Conclusions

Institution: Symphony Quantum FH4/13
Institution Address: 6Karl-Schallstr. , Erlangen, Sued, 91052, Germany

Patient Name: MR_BOLD
Patient ID: 123
Birth Date/Sex: 1/1/1980 / M

Study: HEAD / 9/18/2002 4:05:53 PM / 1
Accession Number:
Series: MeanCurveMR_SR 20070425 16:09:04 / 4/25/2007 4:09:04 PM / 27
Report: 4/25/2007 4:09:04 PM / 1

Completion Flag: PARTIAL

MeanCurve Report

Observer Name: n.a.
Language of Content Item and Descendents: English

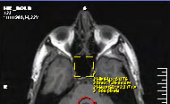
Current Procedure Description

Procedure Description: n.a.

Findings

Analysis: Dynamic Signal Analysis

Finding



Adding data

You can add report data for additional evaluations to a report that has already been created. Use this function, for example, when you have drawn and evaluated new ROIs.

Restrictions

- ☐ It is not possible to append individual images to the existing report.
- ☐ Data cannot be added to reports that were edited, saved, or transmitted in the **Report - Editor (Patient Browser)**. The report has to be set to **PARTIAL** status.

Adding data

- ✧ In the first segment, select the images that you want to add to the report, along with their corresponding evaluation data.
 - ✧ Select **Patient > Append to Report**.
 - Or -
 - Click the **Patient** subtask card using this key.
- The selected data are added to the report.



Editing a report

You can start the **Report - Editor** in the **Patient Browser** and edit the report retroactively.

→ [Page K.2–2, Editing a report](#)

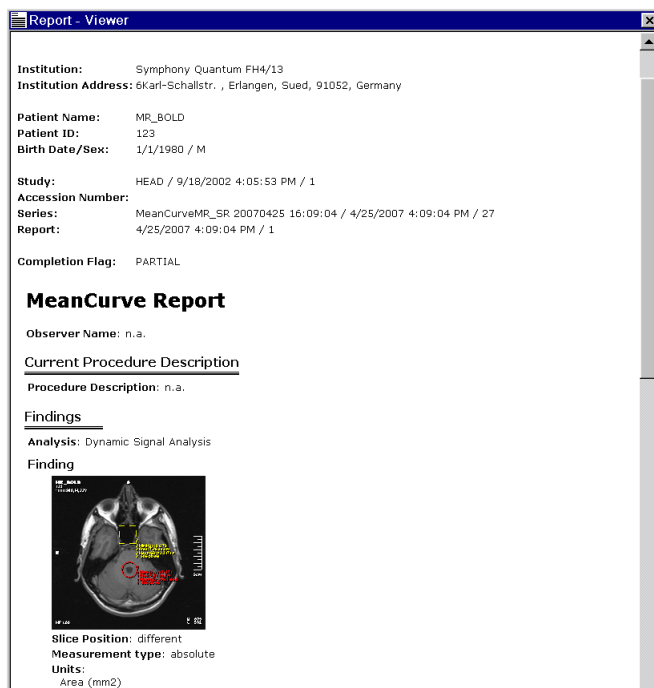
Printing a report

- ✧ Open the report in the **Report - Viewer**.
→ [Page F.4–7, Displaying report](#)

Print Mode

- ✧ Click **Print Mode**.

The Print Preview is displayed in the **Report - Viewer** window.



A rectangular button with a dark gray background and the word "Print" in white text.

- ✧ Click **Print**.

The **Print** dialog window opens.

- ✧ Make the settings desired and confirm with **OK**.

The report is printed.

Deleting images and results

You can either delete images or results of *one* segment or delete *all* images loaded in **Mean Curve** in a single step.

Deleting images of a segment

- ✧ Explicitly select the images in the segment to be deleted.
→ **Basics/Selecting images** in the *syngo* Operator Manual

- ✧ Select **Edit > Delete Document(s)**.

- Or -

In the context menu for the segment selected, use **Delete Documents**.

The selected images are deleted from the segment.

Deleting all images

- ✧ Select **Patient > Close Patient**.

- Or -

Click the **Patient** subtask card using this key.

All segments will be cleared.



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The **BOLD** task card enables you to superimpose anatomical images and parameter images to obtain alpha images. You can also retroactively calculate parameter maps.

Definition of terms

- ❑ **Parameter images** are post-processed images whose pixel carry functional information.
For **BOLD** use parameter maps from an online or offline **BOLD** calculation or generated on the basis of a statistical evaluation with the image **Dynamic Analysis > TTest...** function.
- ❑ The **Alpha image** is an overlay image (magnitude image) created by combining a currently displayed anatomical image with a currently displayed parameter map. The same slice position is the prerequisite for creating an overlay.

Access rights

Detailed information regarding all security settings in *syngo* are included in the *syngo* Security Package brochure.

You are required to have access authorization for the **BOLD** task card. If you log on without access authorization, you will not be able to perform protected functions.

If you do not have access to a certain patient, his data will not be shown in the **Patient Browser** when you are logged into the system.

The BOLD task card

Starting BOLD

- ✧ Go to the **Patient Browser** or one of the task cards to select **Applications > BOLD**.

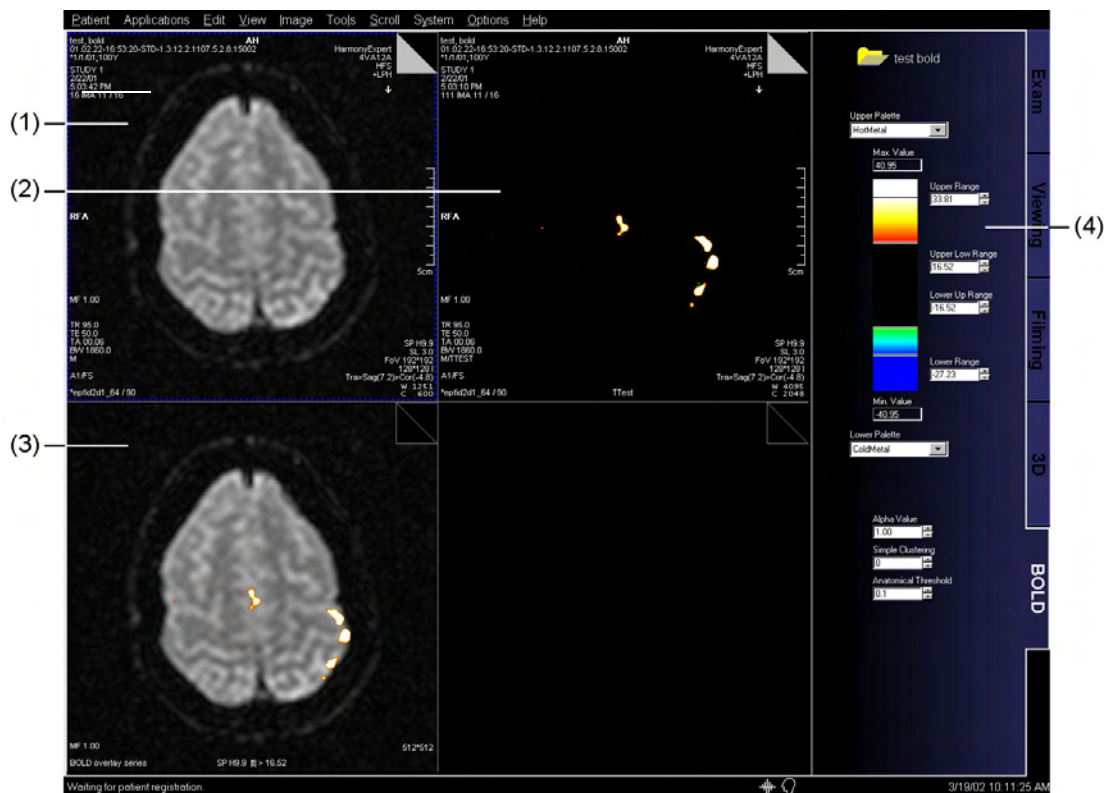
The **BOLD** task card is opened.

Closing BOLD

- ✧ Select **Applications > Close BOLD**.

The **BOLD** task card closes.

Configuration of task card



- (1) Anatomical segment (segment 1)
- (2) Parameter image segment (segment 2)
- (3) Alpha image segment (segment 3)
- (4) Control area



Loading images

You are transferring anatomical images and parameter images to the **BOLD** task card. The sequence is not important. You can choose single images as well as series or complete studies.

This automatically creates overlay images, known as alpha images.

→ [Page G.2–5](#), *Automatic calculation of the alpha images*

Restrictions

- ☐ A complete set of patient images cannot be transferred to the **BOLD** task card.
- ☐ Distortion-corrected images may not be loaded together with images without distortion correction.
- ☐ Images may only be loaded from the local database, not from CD-ROM.
- ☐ Images with data loss due to compression may *not* be transferred to the **BOLD** task card. An appropriate system message is displayed if you attempt to load lossy images.
- ☐ During retroactive calculation of parameter images, lossy images will be removed from the list of series to be processed. An appropriate note is displayed.
- ☐ Images from modalities other than **MR** will not be transferred to the **BOLD** task card.



A message appears when a user takes over who has no access authorization to the data currently loaded.

You can then decide whether to cancel logon of the new user or to continue. If you decide on the latter, the current data will be unloaded without saving them and the task card is closed.

Loading series into the empty task card

Load the anatomical series and parameter image series from the **Patient Browser** to the **BOLD** task card.

- ✧ Select the desired study, series, or individual images in the **Patient Browser**.
→ **Patient Browser**/Searching for and selecting patient data in the syngo Operator Manual
- ✧ Select **Applications > BOLD**.
- Or -
Use the mouse to drag and drop the images from the **Patient Browser** to the **BOLD** image area.
- ✧ You may also load an individual series by double-clicking.

Display rules

- ❑ Anatomical series are loaded to the first segment. Parameter image series are loaded to the second segment.
- ❑ Images without a clear slice plane are loaded into the first segment and placed at the end of the series.
- ❑ The segments show the first image of the first series. If the conditions for an alpha image are met, the alpha image segment will show a single alpha image.
- ❑ If the anatomical images are not from EPI measurements, special conditions apply.
→ [Page G.2–12, Activating/deactivating blocked areas.](#)
- ❑ If different methods for distortion correction are used, anatomical images and parameter images cannot be transferred or superimposed.

Loading additional series

You are able to load additional series to those already loaded in the **BOLD** task card.

- ✧ How to load additional series into the **BOLD** task card.
→ [Page G.2–2, Loading series into the empty task card](#)

Results obtained by loading additional series:

- ❑ If the selected series consist of anatomical images, they will be transferred to the first segment. Additionally, the first image of the first new series you have loaded will be displayed.
- ❑ If the new series originate from the same study, they will be sorted according to their series number. In this case, you are able to scroll between series.
→ [Page G.2–8, Scrolling series by series](#)
- ❑ If the new series you have selected are from another study, a dialog window will be displayed. Here you decide whether the currently loaded series should be replaced by the new series.

Loading mosaic images

You are able to transfer mosaic images to the **BOLD** task card as well.



BOLD imaging generates thousands of maps. To simplify and accelerate the process, special sequences generate mosaic images.

A mosaic image combines the slices measured within a TR interval. The matrix of the mosaic is defined by the number of slices.

The images are sorted in anatomical order starting to the left at the top of the matrix.

→ [Page A.3–13, Splitting mosaic images](#)

Automatic calculation of the alpha images

Ensure that the source images loaded meet the following requirements prior to automatically calculating alpha images:

- ❑ At least one slice position of the anatomical series and parameter image series has to match
- ❑ The parameter image must cover at least half the anatomical image at the matching slice position

If the parameter image covers less than half the anatomical image, it is not possible to calculate the alpha image. The third segment remains black.

- ⇒ Only the alpha image for the currently displayed image pair (anatomical image and parameter image) is calculated and displayed.
- ⇒ Image numbers are not displayed in the alpha image.

The image text of the alpha image provides the slice position and the magnitude of the threshold value (**Upper Low Range, Lower Up Range**) from the color scale, e.g., SP H 29.7 |t| > 4.0. This also applies to mosaic images. In this case, the slice position of the first mosaic image is used.

Rules for mosaic

- ❑ If both source images (anatomical image and parameter images) are in the mosaic format, the alpha image is displayed in mosaic format as well. The slice positions have to match.
- ❑ If only one source image is in mosaic format, (for example, the anatomical image is a single slice image while the parameter image is in mosaic format) BOLD will search for a slice position in the mosaic image that matches the anatomical image. If the search is successful, the alpha image will be computed and displayed.
- ❑ Anatomical mosaic images that are not available in EPI format cannot be superimposed.



CAUTION

Use of mosaic images for displaying the slice position!

Incorrect diagnosis due to erroneous display of slice position and/or angle

- ✧ Do not use mosaic images for displaying the slice position.
-

Scrolling through images and series

The anatomical segment and parameter image segment contain image stacks that you can scroll through image by image or series by series.

Scrolling from image to image

- ✧ Scroll from image to image using the menu or with the help of dog ears or symbol keys.
 - **Viewing** *Changing image settings* in the syngo Operator Manual

If several anatomical series have been loaded, scroll beyond the boundaries of the individual series.

When you select a slice position while scrolling through the parameter image series that matches an image in the anatomical series, the image will be displayed, and the program calculates the associated alpha image. Otherwise, a message appears informing you that it is not possible to calculate an alpha image from either of the two initial images and that the third segment remains blank.

Scrolling series by series

✧ Scroll from series to series via the menu or the symbol keys.

If at least one slice position of the anatomical series and the parameter images series matches, the first suitable image pair is displayed in the first and second segment. The corresponding alpha image is calculated and displayed automatically. Otherwise, a message appears informing you that it is not possible to calculate an alpha image from either of the two initial images and that the third segment remains blank.

Optimizing image display

Depending on the image type, there are a number of ways to change the display in the **BOLD** task card.

Image type	Changing the display
Anatomical image	Changing window values
Parameter image	Activate/deactivate interpolation Setting colors
Alpha image	Changing window values (indirectly) Activate/deactivate interpolation Setting colors Changing the anatomical threshold Activating/deactivating blocking areas Changing the alpha value Changing the cluster size

Windowing images

You can window the anatomical images. The windowing values are transferred automatically to the alpha image. Parameter images cannot be windowed.

- ✧ In the first segment, select the image you want to window.
- ✧ Window the image via the mouse or keyboard.
→ **Basics/Windowing images** in the syngo Operator Manual

Deactivating interpolation

The pixels of parameter images from EPI measurements are clearly visible due to the small image matrix used. To improve image display, the images are interpolated by default, that is, flowing transitions are simulated. You can deactivate this function.

- ✧ Deselect **View > Interpolation On**.

Alpha images and parameter images are displayed without interpolation.

Setting colors

To visualize the functional information in parameter images and alpha images, different colors are assigned to different signal intensities. You determine the color palette to be used as well as the range of values for the signal intensities.

Selecting a color palette



If the parameter image contains only positive values, you just need the upper color palette. Similarly, only the lower color palette is required if the parameter image contains only negative values. Deactivate the color palette that does not apply.

- ✧ In the **Upper Palette** selection list, select the color palette for positive values.

Or

- ✧ Deactivate the upper palette by selecting **none**.

- ✧ In the **Lower Palette** selection list, select or deactivate the color palette for negative values in the same manner.

The color display for the parameter images and alpha image is adapted to the new settings.

- ✧ If you deactivate a color palette although corresponding values are available, the respective image areas will not be displayed.

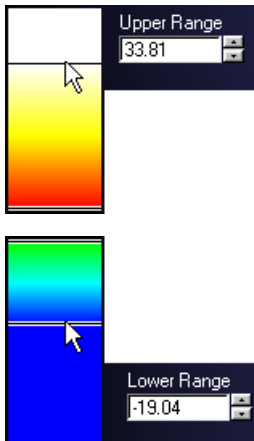


If you want to expose alpha images and have a black-and-white printer, select **GrayScale** as the color scale (black-and-white scale).

To rule out possible ambiguities in grayscale values, window the grayscales values in the first segment.

Changing the range of values

Using the input field or lines in the selected color palette, you establish the functional values assigned to a color. Values without an assigned color are not displayed.

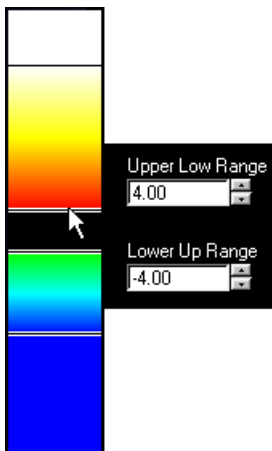


- ✧ Enter the new value for the **Upper Range** of the positive range of values.
- Or -
- Use the mouse to drag the upper line in the color palette to the location required.

- ✧ Enter the new value for the **Lower Range** of the negative range of values.
- Or -
- Use the mouse to drag the lower line in the color palette to the location required.

The color display for the parameter and alpha image is adapted to the new settings.

Changing the functional threshold



By establishing a functional threshold, you suppress the display of low intensity signals, and as a result, the noise in the parameter images.

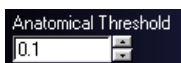
The threshold value creates a symmetrical gap around the value 0 between the negative and positive range of values.

- ✧ Enter the required threshold into the **Lower Up Range** or **Upper Low Range** input field.
 - Or -
 - Use the mouse to drag the corresponding line to the location required.
- If you display both positive and negative values, both ranges of values are changed symmetrically.

The color display for the parameter and alpha image is adapted to the new settings.

Changing the anatomical threshold

In the alpha image, you can also hide anatomical pixels when their intensity falls below a specific threshold. The default value is a threshold of 0.10. This corresponds to 10% of the mean value of all pixels in the anatomical image.



- ✧ Enter the required threshold in the **Anatomical Threshold** input field.

The alpha image is updated.

- ✧ The setting is retained when scrolling from image to image.

Activating/deactivating blocked areas

If you have transferred parameter images and anatomical images of different resolution to the **BOLD** task card (e.g., anatomical spin-echo images), the alpha image is adjusted in size to the anatomical image.

Due to the measurement method, echoplanar measurements are the basis for functional information and may result in distortions and local signal loss in some image areas. In these areas, unambiguous superimposition of functional information on anatomical images cannot be ensured.



By using additional information from the field map measurement (generates a map of the B0 field), it is possible to block these critical areas. Masking or blocking indicates that the positional assignment of functional information to the anatomy has to be checked carefully in these areas as well as their immediate vicinity.

Activating/deactivating blocked areas

✧ Select or deselect **View > Show Fieldmap**.

The magenta colored blocked areas are shown or hidden.

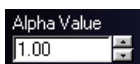
If you deactivate the blocked areas (field maps), a system message warns you regarding possible misinterpretation of the alpha image. The message is only displayed once per series.

✧ Acknowledge the warning with **OK**.

This message is displayed in the alpha image as a remark. It can only be hidden by reactivating the blocked areas.

Changing the alpha value

The alpha value defines how clearly the anatomical image is overlaid by the parameter image. The higher the alpha value (maximum 1), the greater the degree to which the pixels of the activation map cover the anatomical image. If the alpha value is 0, only the anatomy is visible.



- ✧ Enter the desired value in the **Alpha value** input field.

The alpha image is updated.

- ✧ The setting is retained when scrolling from image to image.

Changing the cluster size

The cluster size is specified in order to suppress randomly occurring pixels in the alpha image. For example, if you enter cluster size 3, all pixel clusters that are smaller than 3 pixels are hidden.

The default cluster size is 1 pixel.



- ✧ Enter the required cluster size in the **Simple Clustering** input field.

The alpha image is updated.

- ✧ The setting is retained when scrolling from image to image.

The following images can be stored in **BOLD**:

- ☐ Whenever their slice positions match, all alpha images of the anatomical images and parameter images are loaded
- ☐ Current parameter image with associated color palette
- ☐ Current alpha image as a color image

Selecting the save function

Storing all alpha images

If you want to compute and store all the alpha images of the loaded anatomical images and parameter images:

- ✧ Select **Patient > Save All Alpha As...**
 - Or -
 - Select **Save All Alpha As...** from the context menu.
- ✧ A message appears if there are more than 30 alpha images to be calculated. You have the opportunity to cancel the procedure.

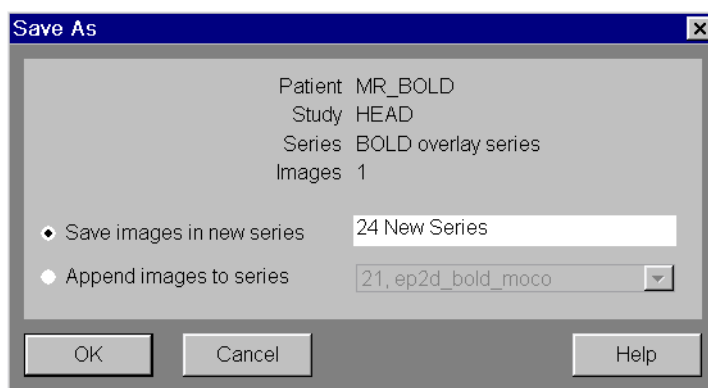
Storing an individual alpha image or parameter image

If you just want to store an individual alpha image or parameter image including the associated color palette, proceed as follows:

- ✧ In the appropriate image segment, select the parameter image or alpha image desired.
- ✧ Select **Patient > Save As...**

Determining the storage setting

After selecting a storage function, the dialog window **Save As...** is opened.



Here you define whether to append the images to an existing series or save them as a new series.

- ✧ Select the desired save option.
- ✧ Enter the series name (new series) or select the name of an existing series.
 - **Viewing/Saving images** in the syngo Operator Manual
- ✧ Click **OK**.

If you store all alpha images, the program searches one-by-one for a matching parameter image for each anatomical image. If successful, the respective alpha image is computed, displayed, and stored.



You can monitor the progress during calculation and storage. You can terminate the process at any time.

➤ All alpha images are stored including their current display characteristics (colors, limits, alpha values, cluster size, anatomical threshold, field map on/off, interpolation on/off).



If a user without access authorization to the program or the data loaded takes over during the save mode, the data will be hidden on the task card.

Saving is completed in background. All data are then unloaded from the task card and the task card is closed.



Calculating Parameter Images Retroactively

You can retroactively calculate parameter images on the **BOLD** task card. The images will be saved in mosaic format and automatically displayed in the **BOLD** task card.

Suitable series

The series have to meet the following criteria before they can be post-processed:

- ☐ Same series block and same table position
→ [Page C.1–6, Series block](#)
- ☐ Same Frame of Reference
→ [Page C.1–7, Frame of Reference \(FoR\)](#)
- ☐ Same slice position
- ☐ Same number of images



CAUTION

BOLD post-processing with image data generated by Numaris 3 or 3.5!

Incorrect diagnosis through erroneous superposition of anatomical and functional image

- ✧ Do not use *syngo* MR for BOLD processing with image data generated with Numaris 3 or 3.5.
-



Preparing for post-processing

BOLD calculation can be performed only via the **Patient Browser**.

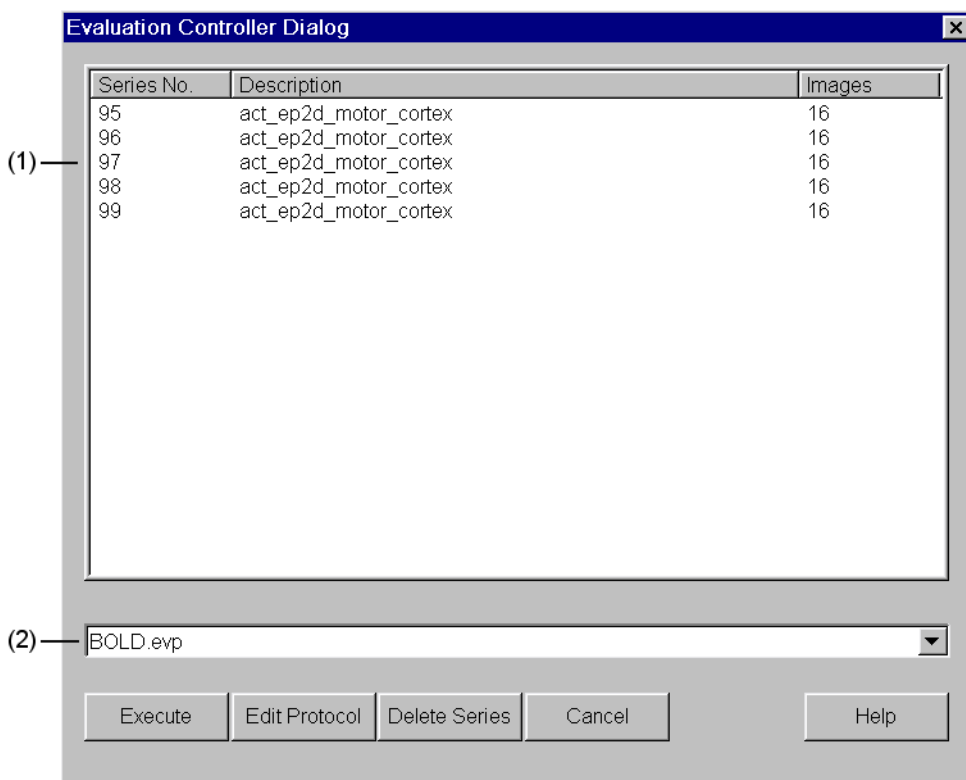
Selecting a series

- ✧ Select the functional series (EPI measurements) in the **Patient Browser**.

Opening the post-processing dialog

- ✧ Select **Applications > BOLD Evaluation**.
The **BOLD** task card is opened.

At the same time, the **Evaluation Controller Dialog** dialog window opens.



(1) Selected series

(2) Post-processing protocol last used

The MR system checks whether the series selected meets all pre-requisites.

→ [Page G.4–1](#)

Deleting series

If unsuitable series have been selected, they are identified in the list and a message is displayed. You have to delete these series from the list:

A rectangular button with a light gray background and a thin black border. The text "Delete Series" is centered in a dark gray font.

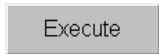
- ✧ Click **Delete Series**.
- ✧ If the series were selected in error, identify and delete them from the list.

Selecting the protocol

A horizontal dropdown menu with a light gray background and a thin black border. The text "BOLD.evp" is displayed on the left, and a small downward-pointing arrow is on the right.

- ✧ Select the required post-processing protocol from the selection list.
- ✧ Experienced users have the opportunity to edit the post-processing protocol.
 - [Page G.4–7, Editing the post-processing protocol](#)

Starting post-processing



- ✧ Click **Execute** in the **Evaluation Controller Dialog** dialog window.

The dialog window closes and post-processing starts.



The calculation runs in background. The image post-processing job icon appears in the status bar.

After calculation has been successfully completed, the calculated parameter images are displayed as mosaic images in the second segment of the **BOLD** task card. Additionally, the alpha image is calculated and displayed.

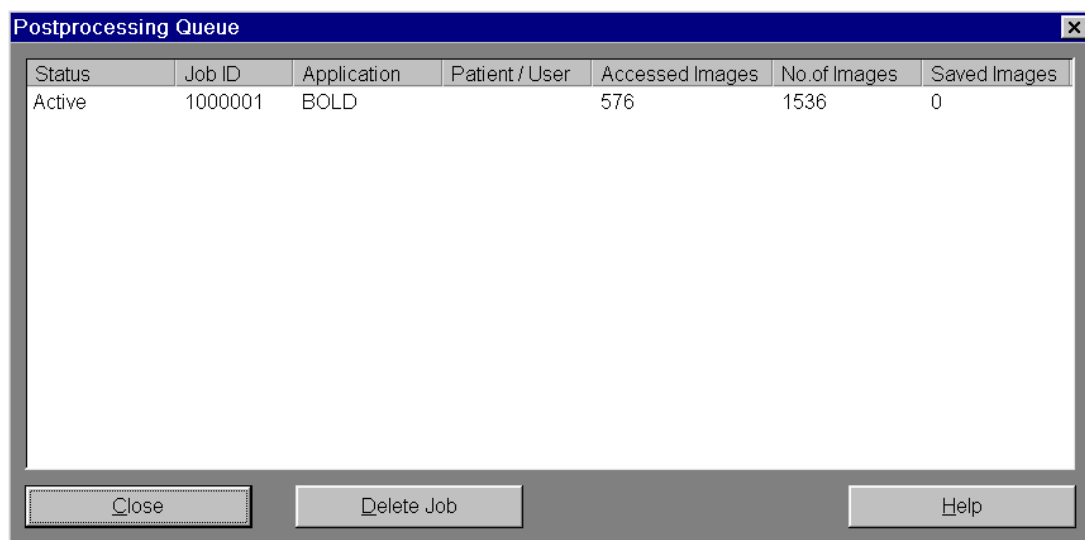
Monitoring post-processing

You can monitor the progress and status of image post-processing in the **Postprocessing Queue** dialog window.



- ✧ Click this icon in the status bar.

The **Postprocessing Queue** dialog window is displayed.



Stop the calculation



✧ Select the job with the "active" status.

✧ Click **Delete Job**.

Calculation is stopped and the job is deleted from the list.

Status after ending the calculation

After calculation is successfully completed, the "complete" status is assigned. If not, the "failed" status is shown.

Editing the post-processing protocol

The sequence and properties of the individual post-processing steps are defined by the post-processing protocol. Experienced users can adapt protocols to their requirements and generate new protocols.

Open the BOLD post-processing protocol

After you started the post-BOLD calculation with selected series, the **Evaluation Controller Dialog** dialog window is opened.

→ [Page G.4–3](#)

✧ Select the required post-processing protocol from the selection list above the buttons.

✧ Click **Edit**.

The **Edit Protocol** dialog window with the parameters relevant for BOLD post-processing opens.

✧ As soon as you select a different protocol in the **Evaluation Controller Dialog** dialog window, this protocol will be loaded in place of the previously opened one.

Editing protocol parameters For details regarding the individual parameters, refer to
→ [Page P.1–156](#), Parameter Card *BOLD*.

✧ Depending on the protocol you are processing, the **Edit Protocol** dialog window contains different parameters.

Edit Protocol: t-test_10B10A_1.evp

GLM Statistics	Motion correction
Starting ignore meas: 0	Motion correction: ☒
Ignore after transition: 0	Interpolation: 3D-K-space
Model transition states: ☐	Spatial filter: ☒
Temp. highpass filter: ☐	Filter setting: 1.0
Threshold: 4.00	
Paradigm size: 20	
Meas [1]: Baseline	

Buttons: Save, Save As ..., Delete, Cancel

✧ Adapt the parameters to your requirements.

Creating a new post-processing protocol

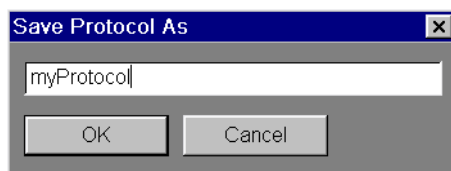


Only experienced users should edit post-processing protocols!

You are able to store an open post-processing protocol under a different name.

- ✧ Click **Save As**.

The **Save Protocol As** dialog window opens.

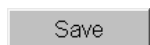


- ✧ Enter a file name.
- ✧ Click **OK**.

A new post-processing protocol including the selected name is generated.

The protocol selection list in the **Evaluation Controller Dialog** dialog window is updated. The new protocol is automatically selected.

Saving the changes



You can save a changed customer protocol and overwrite the old version.

✦ Click **Save**.

⇒ You have to store the Siemens post-processing protocols under a new name before you can edit them as required.

Deleting a customer protocol



You can delete an open customer protocol.

✦ Click **Delete**.

⇒ This function is inactive when you open a Siemens post-processing protocol.

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H.2 *Loading and Displaying Images*

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 Loading images for evaluation H.2–1

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H.3 *Storing and Filming Results*

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H.4 *Calculating Parameter Images*

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The **Perfusion** evaluation function allows the display and evaluation of perfusion-weighted images, for example, for tumor diagnosis.

You can calculate the following parameters in the **Perf MR** task card:

- ☐ Relative cerebral blood volume (relCBV)
- ☐ Relative cerebral blood flow (relCBF)
- ☐ Time to bolus maximum (Time To Peak, TTP)
- ☐ Relative mean transit time (relMTT)

Parameter calculations are performed using well established methods of analysis, primarily based on the technique described by Østergaard et al. (Magn. Reson. Med. 36, 715–725 (1996)).



The use of such parameter images has been shown to be highly beneficial in a number of perfusion studies. Nevertheless, you should only attempt it if you have sufficient experience interpreting parameter images and consider possible limitations.

The Perf MR task card

Starting perfusion

You can open the **Perf MR** task card from any other task card or from the **Patient Browser**.

- ✧ Go to the **Patient Browser** or one of the task cards to select **Applications > Perfusion (MR)**.

The **Perf MR** task card opens.

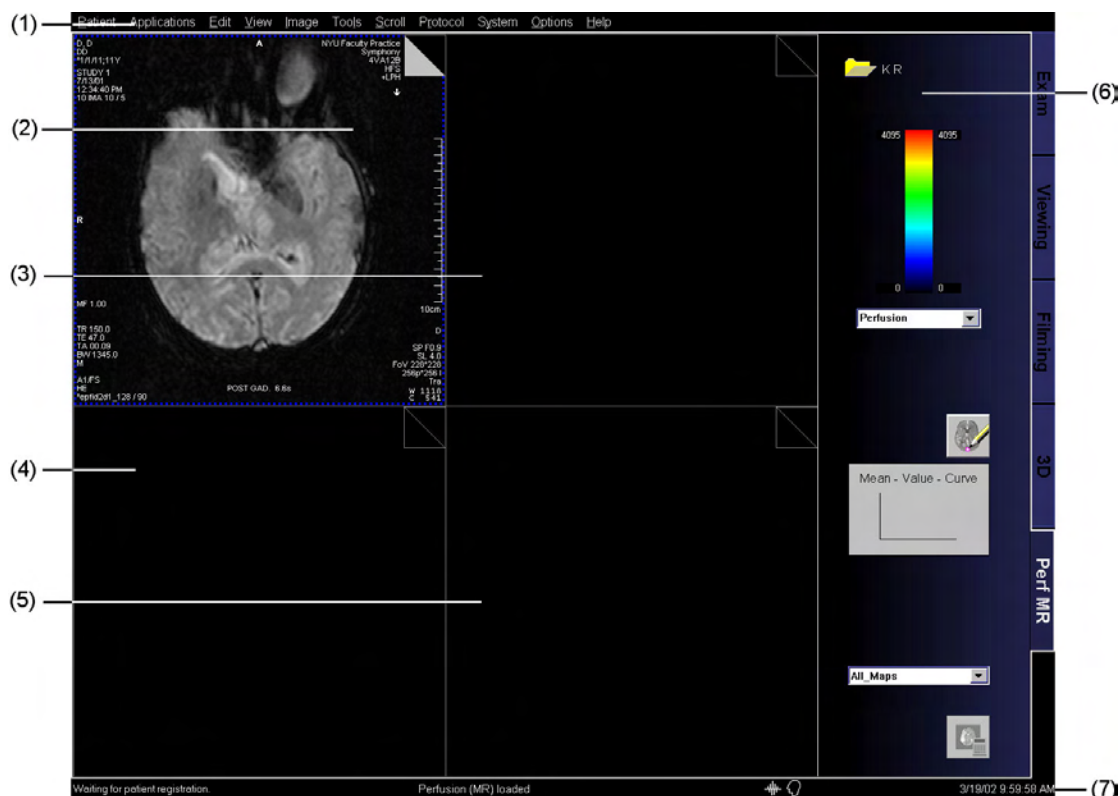
If images, series, or a study are selected in the **Patient Browser**, they are automatically loaded into **Perf MR**.

Ending perfusion

- ✧ Select **Applications > Close Perfusion (MR)**.

The **Perf MR** task card is closed.

Configuration of task card



- (1) Menu bar
 - (2) Base image segment
 - (3) Viewing area
 - (4) Parameter image segment (TTP and relMTT)
 - (5) Parameter image segment (relCBV and relCBF)
 - (6) Control area
 - (7) Status bar
- (2), (4), and (5) represent the processing area.



Loading images

You can load individual images, series, or a complete study in the **Perf MR** task card. You cannot load images from different studies.

⇒ A perfusion evaluation requires that you have full access rights to the patient data to be evaluated. Detailed information regarding all security settings in *syngo* are included in the *syngo* Security Package brochure.

Loading images for evaluation

For the evaluation, you are typically using time series $T2^*$ - or $T2$ -weighted single-shot EPI images. These base image series can contain several hundred images which were acquired in fast succession for dynamic display (measurement every 1 – 2 seconds over 1 – 2 minutes).

Requirements

For calculation of parameter images, suitable base images are grouped into a "volume block" and evaluated. These base images have to meet the following criteria:

- ☐ Same series block and same table position
→ [Page C.1–6, Series block](#)
- ☐ Same Frame of Reference
→ [Page C.1–7, Frame of Reference \(FoR\)](#)
- ☐ Same FOV
- ☐ Same slice orientation
- ☐ Same matrix size
- ☐ Same number of images for a slice position
- ☐ The slice position can only be moved vertically to the slice plane

- ✧ Several "volume blocks" can be loaded in the base image segment. However, the only volume block that is evaluated is the one that contains the current image in the base image segment.
- ✧ It is also possible to use distortion-corrected images of the same type.
- ✧ Select one or more series in the **Patient Browser**.
- ✧ Select **Applications > Perfusion (MR)**.
 - Or -
 - Use the mouse to drag and drop the images from the Patient Browser to the **Perf MR** task card processing area.
- ✧ You may also load an individual series by double-clicking the open **Perf MR** task card.

The images are transferred to the **Perf MR** task card and are automatically sorted into the segments of the processing area.

Image type	Segment
Anatomical images	Base image segment (upper left)
TTP images and relMTT images	Parameter image segment (lower left)
relCBV images and relCBF images	Parameter image segment (lower right)

Loading images for viewing

For example, you are able to load the diffusion images of the study in the top right segment (viewing area) to view and compare the anatomy being evaluated.

- ✧ Open the **Patient Browser**.
- ✧ Select the series and images you want to view in the Patient Browser.
- ✧ Drag the images to the top right-hand segment with the mouse.

Viewing images

The loaded images are displayed in the segments as an image stack. You may scroll separately image-by-image or series-by-series in each of the segments.

Scrolling gives you a general view of the scan, allowing you to find a suitable basic image for positioning the "Region of Interest" (ROI).

You optimize image display using various procedures.

Automatic sorting of images

The image sort that applies in the **Patient Browser** does *not* apply to the **Perf MR** task card.

Sorting

The sort in the basic image segment is different from the sort in the other segments:

Segment	Sorting
Base image segment (upper left)	in anatomical sequence image copies are sorted according to the originals.
All other segments – when scrolling from image to image	first according to the serial number followed by according to the image number and then according to the time of image generation (image copies)
All other segments – when scrolling from series to series	first according to the image number then followed by according to the series number and then according to the time of image generation (image copies)

Automatic sorting when loading more images

You can load additional images to previously loaded images. These images are correctly sorted in the segments by image type, series, and image number.

Scrolling through images and series

Scrolling image by image allows you to look for the slice position that provides the best view of the anatomy to be evaluated.

Subsequently, the different measurement times of the images for this slice position are shown by scrolling from series to series.

In this manner, you are able to locate the image with the most optimal anatomical display and the best vessel contrast in the bolus.

Sorting via the menu using dog ears is described in detail in chapter **Basics** of the *syngo* Operator Manual.

Optimizing image display

You can use the following methods:

- ☐ Windowing (changing brightness and contrast)
- ☐ Zoom/reduce/pan
- ☐ Zoom/reduce to segment size
- ☐ Invert grayscale values
- ☐ Showing/hiding image text
- ☐ Inserting an annotation

These actions are described in the sections **Basics** and **Viewing** of the *syngo* Operator Manual.



The calculated parameter images are automatically stored in the database without a color palette. The series have the following designation:

[series number] ep2d_perf_Parameter image type.

The type of parameter images are shown to the left in the image text.

→ [Page P.2–1](#), *Text Information in Medical Images*

Storing selected images

You can store the reference image and select additional parameter images.

- ✧ Select the images explicitly.
 - **Viewing**/Changing image settings in the syngo Operator Manual
- ✧ Select **Patient > Save As...**

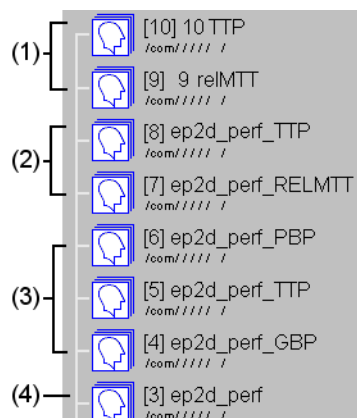
The **Save As** dialog window opens. Proceed with the storage setting and start the storage process.

→ **Viewing**/Saving images in the syngo Operator Manual

- ☞ The selected images are saved with the color palette.

Parameter image series in the Patient Browser

Examples:



- (1) Parameter images generated "offline"; explicitly stored with the color palette
- (2) Parameter images generated "offline"; automatically stored without the color palette
- (3) Parameter images generated "inline"
- (4) Original series; input for post-processing

Filming results

You may copy images to the film sheet, define film layouts, start filming, and organize film jobs on the **Perf MR** task card. Detailed information is included in the **Filming** section of the syngo Operator Manual.

- ✧ Select the images you would like to transfer to the film sheet.
- ✧ Select **Patient > Copy to Film Sheet**.

Calculating Parameter Images

The following parameter images may be calculated offline via post-processing protocols:

- ☐ TTP image
- ☐ relMTT image
- ☐ relCBV image
- ☐ relCBF image

Steps involved

- (1) Selecting a post-processing protocol
- (2) Calculate the arterial input function (AIF) and set the time range for the evaluation
- (3) Start the post-processing protocol
- (4) Optional: Assign a suitable color display for reconstructed parameter images and optimize windowing values
- (5) Optional for experienced users: Edit the post-processing protocol

Selecting a post-processing protocol

The lower part of the control area includes a selection list with the post-processing protocols available.



- ✧ Choose the required post-processing protocol from the selection list.
- ✧ You are able to edit post-processing protocols and select the requested parameter images.
 - [Page H.4–13, Editing the post-processing protocol](#)

Calculating mean AIF

Arterial Input Function (AIF)

The AIF is calculated from the time curve of the contrast agent (CA) concentration in the artery. The measurement method used is the same as for CA concentration in tissue. The AIF is measured together with the tissue concentration and determined in a specific slice of the slab. This slice should contain sufficiently large vessels to prevent partial volume effects.

Calculation of the relCBF and relMTT involves deconvolution with the signal time curve selected for the AIF. This means that the quality of the parameter images is degraded by the reduced SNR of the basic images used to calculate the parameter images and to determine the AIF.



CAUTION

Wrong selection of pixels when determining the Arterial Input Function (AIF)!

Incorrect computation of parameter images

- ✧ Only experienced users should select image pixels for determining the Arterial Input Function.
-



Recommendation: Do not perform diagnostic evaluation of the result images (parameter images TTP, relMTT, relCBV and relCBF), if the SNR is too low in the base images or noticeable head movements of the patient can be detected.

Selecting AIF curves

You determine the mean AIF curve from the individual AIF curves within an ROI.

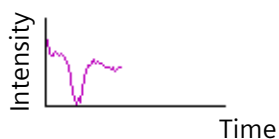
- ✧ Look for a suitable basic image for positioning the AIF-ROI in the top left segment.

→ [Page H.2–5](#), *Scrolling through images and series*

- ✧ Click this button in the control area.
- Or -
Select **Tools > AIF**.



An AIF-ROI is drawn in the basic image. The AIF curves are shown for all pixels within the AIF ROI.



The AIF curves represent the change over time in the signal intensity for each pixel in the AIF ROI. All AIF curves have the same scaling and are therefore comparable.

The position and number of AIF curves shown match the position and number of pixels in the AIF ROI.

Time Points : 50

In the footer of the **Step 1: Select AIF** subtask card shows the number of images used for determining the AIF.

Positioning an AIF ROI

To calculate the mean AIF, select pixels exhibiting a signal over time curve that is typical of healthy arteries.



Only personnel trained in perfusion diagnostics should position the AIF-ROI and select suitable AIF curves to calculate the mean AIF.

- ☐ The AIF ROI should be located close to the perfusion anomaly.
- ☐ Position the AIF ROI with great care!
- ☐ over healthy arteries, such as the A. cerebri media, but also over smaller arteries that are not explicitly recognizable as vessels in the basic image.

✧ Left-click the AIF ROI and drag it to its new position in the basic image.

The associated AIF curves are displayed in the viewing area when you release the mouse button.

Selecting AIF curves

Select suitable AIF curves for determining the mean AIF.

⇒ AIF curves with a very low signal minimum may have a distorted shape if the SNR of the associated pixel in the basic image is too low. Do *not* use such AIF curves for evaluation.

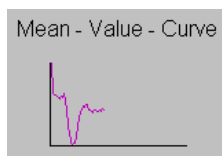
⇒ Select the AIF curves in the **Step 1: Select AIF** subtask card.

The selected AIF curves are displayed with a white background.

⇒ For multiple selections use the **Ctrl** and/or **Shift** key.

The mean AIF curve in the control area is refreshed. It indicates the arithmetic mean of the selected AIF curves.

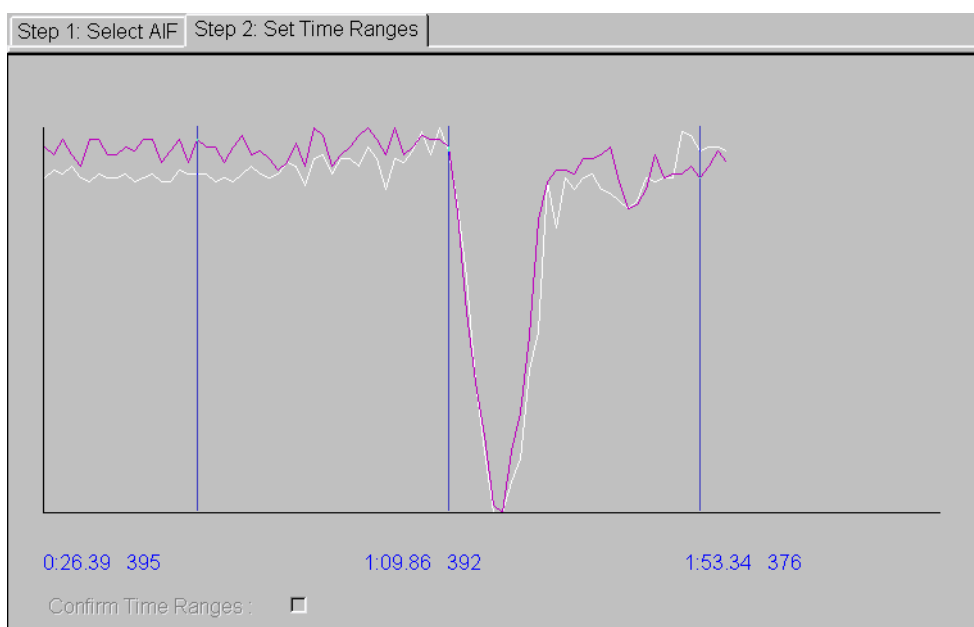
⇒ The selection of AIF curves is retained until you move the AIF ROI to another point in the basic image or until you select an image outside the time series displayed.



Setting the time range

It is mandatory to set the time range for perfusion analysis.

- ✧ Move the subtask card **Step 2: Set time ranges** in foreground.



The mean AIF is magnified (purple curve). In addition, the Global Bolus Plot (GBP) is shown (white curve).

You define the time range to be used for perfusion analysis. The image range corresponds to the basic images to be used.

- ✧ Drag the right line with the mouse to the end of the First Pass.
- ✧ Drag the center line with the mouse to the end of the base line.
- ✧ Drag the left line with the mouse to the start of the base line.
- ✧ Observe the GBP curve. The bolus arrives in the tissue later than in the vessel. Do not define the end of the first pass to the far left. The bolus in the tissue could be cut off.
- ✧ Select the **Confirm Time Ranges** checkbox.
The post-processing protocol may now be started.

Confirm Time Ranges



Performing reconstructions

You can start the reconstruction after you have selected a post-processing protocol, calculated the mean AIF and set the time range.

✧ Select **Protocol > Execute**.

- Or -

Click this button in the lower control area.



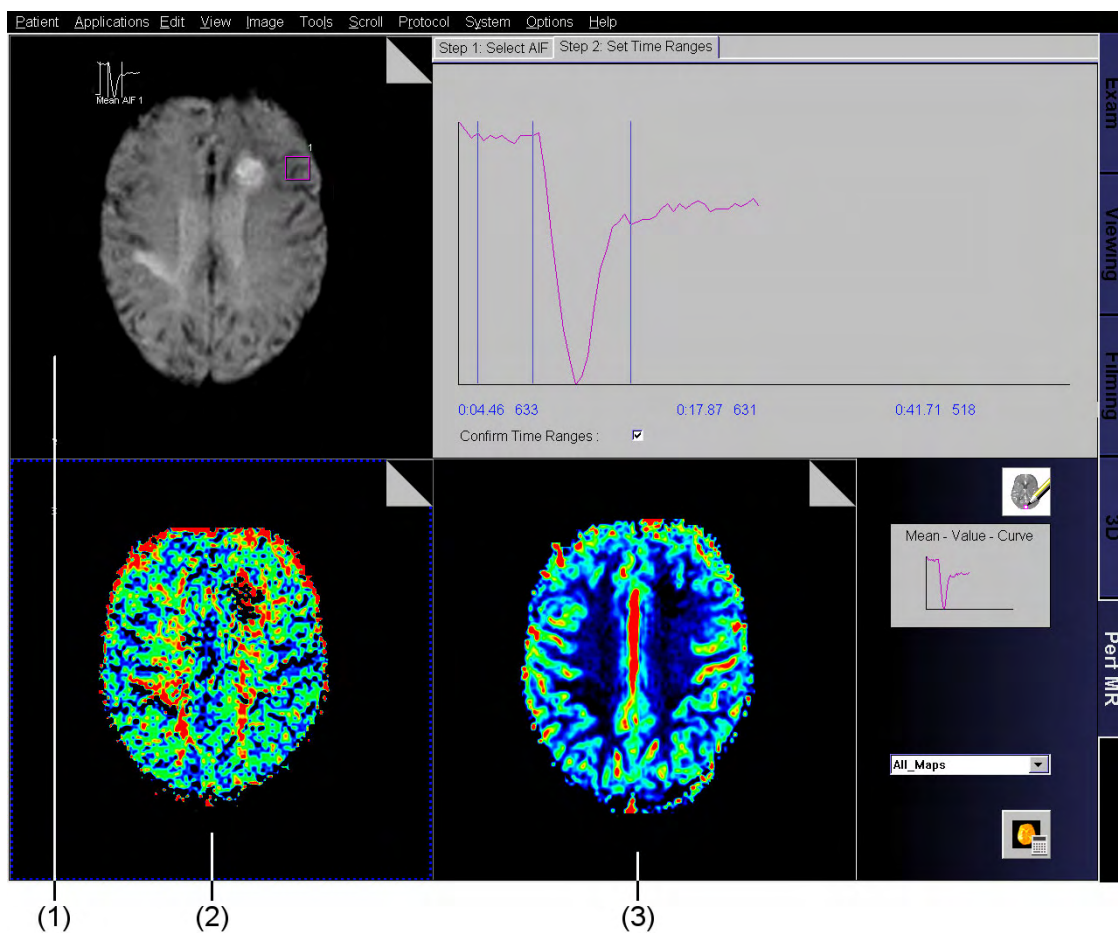
A message appears informing you that reconstruction is running. The image post-processing job icon appears on the status bar.

→ [Page H.4–11](#), *Monitoring calculation*

As soon as the reconstruction is completed, a message is displayed on-screen and the parameter images are shown in the two lower segments.

Images already loaded in the lower segments are retained and can be viewed via scrolling.

The top left segment shows the mean AIF with the time range set in the basic image in addition to the AIF ROI (reference image).



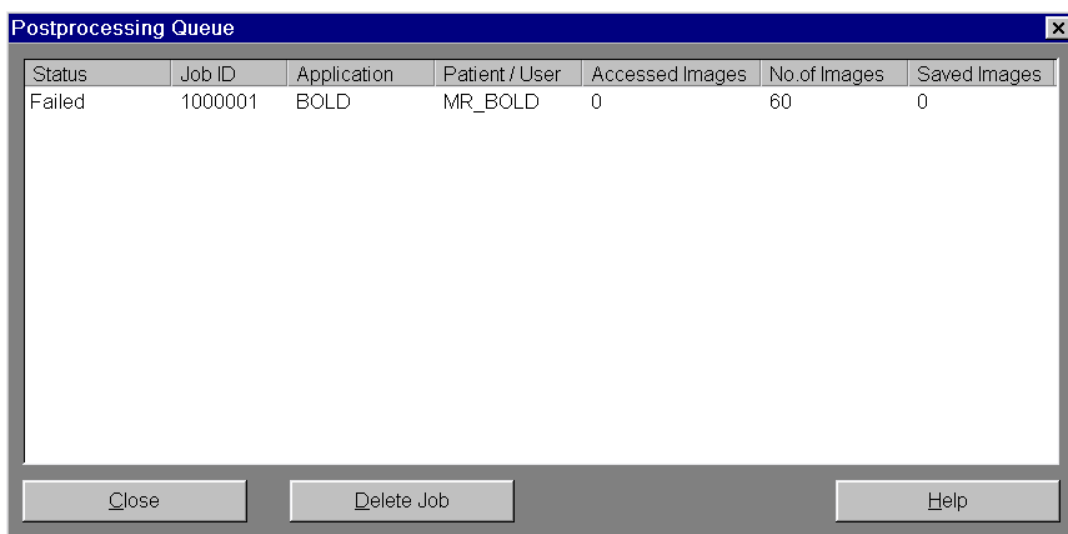
Monitoring calculation

You can monitor the progress and status of image post-processing in the **Postprocessing Queue** dialog window.



✧ Click this icon in the status bar.

The **Postprocessing Queue** dialog window is displayed.



Stop reconstruction

A rectangular button with a light gray background and a thin black border. The text "Delete Job" is centered in a dark gray font.

✧ Select the job with the "active" status.

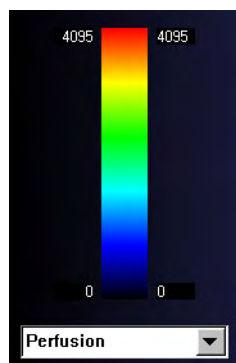
✧ Click **Delete Job**.

Reconstruction is stopped and the job is deleted from the list.

Status after ending reconstruction

After reconstruction is successfully completed, the "complete" status is assigned. If not, the "failed" status is assigned.

Changing the color display



The color display of parameter images may be defined via the color palette. The color settings apply to all images loaded in the parameter image segments.

✧ Select the desired color in the selection list located in the control area.

The color palette shows the selected color scheme. The color display in the two lower parameter image segments is updated to the color palette selected.

Windowing color parameter images

- ✧ Window the image via the mouse or keyboard.
→ **Basics/Windowing images** in the *syngo* Operator Manual
- The minimum and maximum values in the color palette are adjusted to the current window values.
- ✧ If optimal windowing of the color parameter images proves to be difficult, set the parameter images first to black and white. Window the black and white parameter images and select the desired color palette again.

Editing the post-processing protocol

The sequence and properties of the individual post-processing steps are defined by the post-processing protocol. You can adjust protocols to your requirements or create new protocols.

Opening a protocol



This is how you open a post-processing protocol for processing:

- ✧ Go to the selection list in the lower control area of the **Perf MR** task card and select the required post-processing protocol.

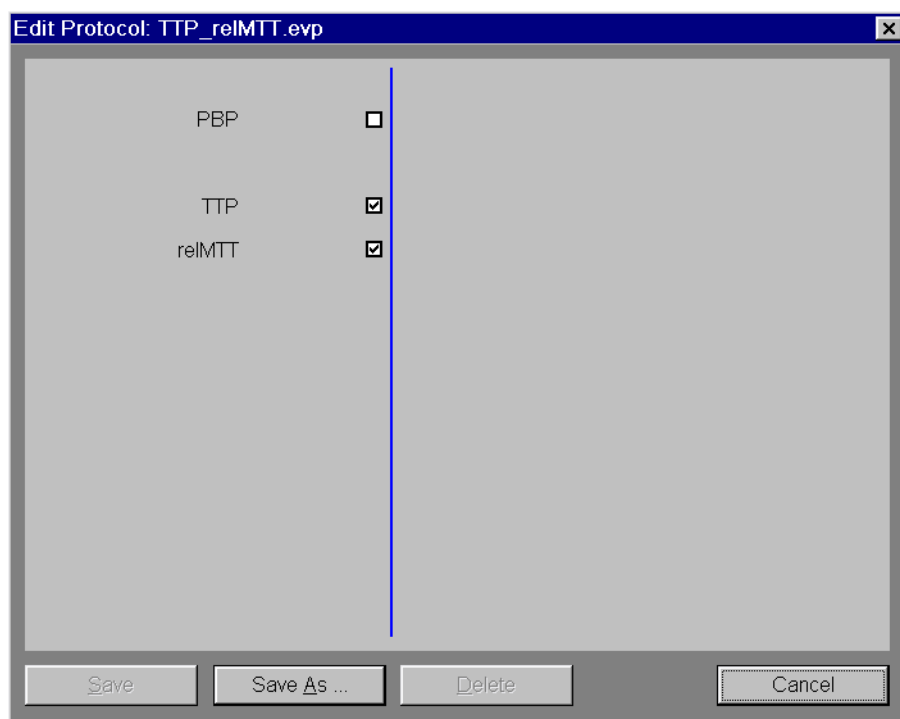
- ✧ Select **Protocol > Edit**.

The **Edit Protocol** dialog window with the parameters relevant for perfusion post-processing opens.

- ✧ As soon as you select a different protocol in the protocol selection list, this protocol will be loaded in place of the previously opened one.

Editing protocol parameters

For details regarding the individual parameters, refer to
→ [Page P.1–171](#), Parameter Card *Perf*.



- ✧ Adapt the parameters to your requirements.

Creating a new post-processing protocol

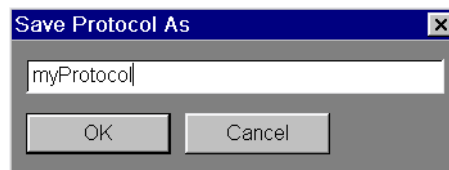


Only experienced users should edit post-processing protocols!

You are able to store an open post-processing protocol under a different name.

✧ Click **Save As**.

The **Save Protocol As** dialog window opens.



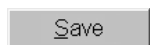
✧ Enter a file name.

✧ Click **OK**.

A new post-processing protocol including the selected name is generated.

The protocol selection list in the **Perf MR** task card is updated. The new protocol is automatically selected.

Saving the changes



You can save a changed customer protocol and overwrite the old version.

✦ Click **Save**.

☞ You are able to store Siemens post-processing protocols under a different name.

Deleting a customer protocol



You can delete an open customer protocol.

✦ Click **Delete**.

This function is inactive when you open a Siemens post-processing protocol.

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J.2 *Calculating Parameter Maps*

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J.3 *Calculating Combined Images*

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The **Soft Tissue Evaluation (MR)** application dialog provides you with the following evaluations:

- ☐ Edit and save the protocol parameters used in the protocols for calculation
- ☐ Calculate motion-corrected images from anatomical images
- ☐ Calculate result images (parameter maps or combined images) from loaded images

Prerequisites

The images to be processed must meet the following prerequisites:

- ☐ Same series block and same table position
→ [Page C.1–6, Series block](#)
- ☐ Same Frame of Reference
→ [Page C.1–7, Frame of Reference \(FoR\)](#)
- ☐ Same type of distortion correction (none, 2D, 3D)
- ☐ Same motion correction (yes, no)



The image numbering set in the measurement protocol of the **System – Miscellaneous** parameter map does not affect the type of parameter maps calculated with **Soft Tissue Evaluation (MR)**.

When changing the standard measurement sort on the parameter map mentioned, a different image sort for the anatomical images and the parameter maps calculated from them may be used during **Soft Tissue Evaluation (MR)**.

Image types for soft tissue evaluations

Image type	Definition
Anatomical image	Anatomical images are acquired during a patient exam. They are used as the basis for calculating parameter maps.
Parameter map	Parameter maps contain functional rather than anatomical information. Using algorithms, they are generally calculated based on anatomical images from several measurements.
Combined image	Combined images are calculated by combining several parameter maps.

Access authorization

Detailed information regarding all security settings in *syngo* are included in the *syngo* Security Package brochure.

If you do not have access to certain patients, their data will not be shown in the **Patient Browser** when you are logged into the system.



If you do not save the data before a change in users, a message indicating that data have not been saved will appear, if the new user does not have the required read privileges. You may then decide whether to continue or cancel the action.

Note: If you go through with the action, the evaluation dialog window will close and all settings made so far will be lost.

Selecting the evaluation dialog window

Go to the **Patient Browser**, select the images to be evaluated and open the evaluation dialog.

Selecting images

Select anatomical or parameter maps, depending on the type of image you want to calculate.

- ☐ From anatomical images you calculate motion-corrected images and/or parameter maps.
- ☐ From parameter maps you calculate combined images.

You are not able to simultaneously load images from both types.

- ✧ Select **Patient > Browser**.

The **Patient Browser** is displayed.

- ✧ Go to the **Patient Browser** and select the requested examination, series or individual images.
 - **Patient Browser**/Searching for and selecting patient data in the syngo Operator Manual



Open the evaluation dialog

If you have selected at least one series with anatomical images or parameter maps, you are able to select an evaluation dialog:

- ✧ Select **Applications > Soft Tissue (MR)** in the **Patient Browser**.
- The menu command is inactive, if you did not select images.

Only one image type is available

If the examinations or series contain images of one type only (anatomical images or parameter maps), the dialog window opens automatically. This also activates the subtask card relevant for evaluation of the loaded image type.

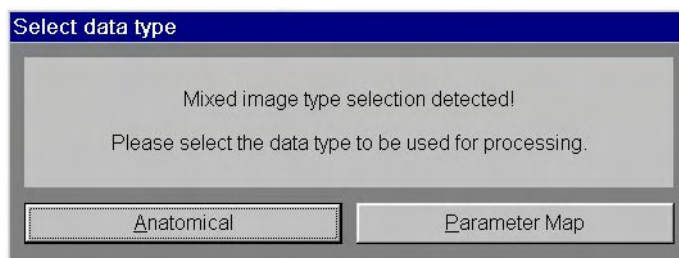
Image type	Associated evaluation/subtask card
Anatomical images	Calculate parameter map
Parameter maps	Calculate combined images



Only series containing magnitude images are taken into account in anatomical series. Other series (e.g., other modalities or compressed images) are automatically filtered out when the dialog window starts. A suitable message is displayed.

Both image types are available

When you select a series that contains both anatomical images as well as parameter maps, the dialog window **Select data type** appears prior to loading the images.

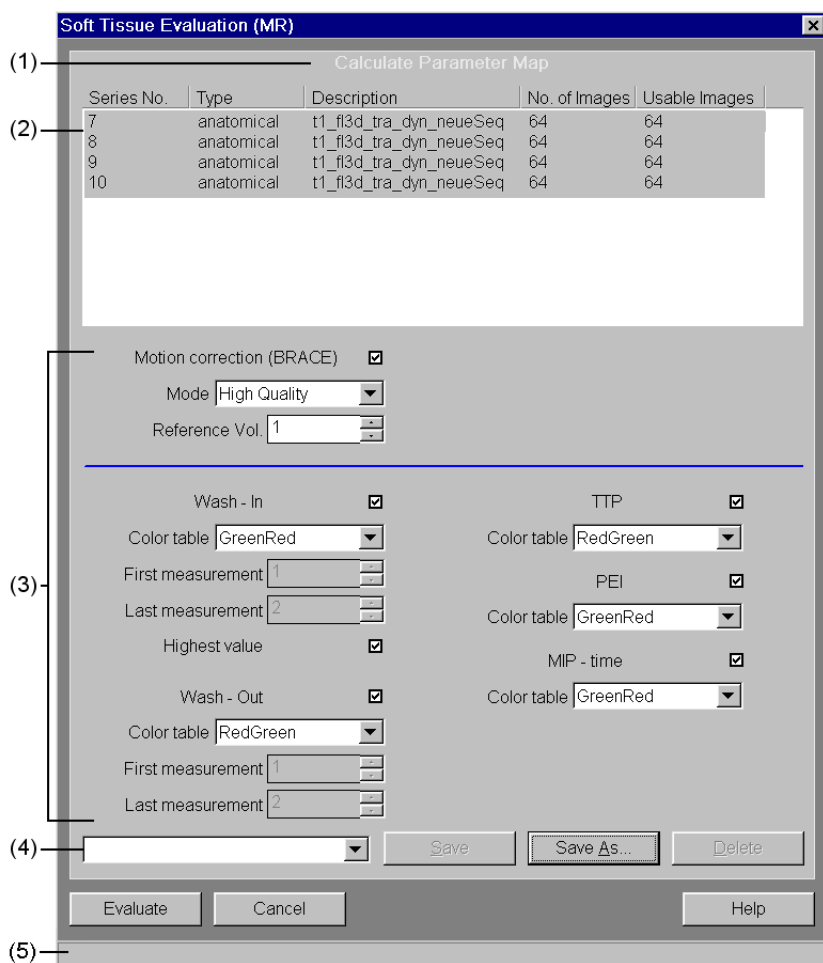


You have to decide the image type.

✧ Click the button to start the required evaluation.

The evaluation dialog window opens; the subtask card for the selected evaluation is automatically active.

Configuration of dialog window



- (1) Type of calculation (active subtask card)
- (2) List of loaded series
- (3) Parameters for the selected post-processing protocol (the **Calculate Parameter Map** task card includes motion correction)
- (4) Managing post-processing protocols
- (5) Status bar

Calculating Parameter Maps

For calculation, you have to load at least two series with the same number of usable anatomical images in the evaluation dialog window. The images have to meet the following conditions:

- ☐ same series block
→ [Page C.1–6, Series block](#)
- ☐ Same table position
- ☐ Same type of distortion correction (none, 2D, 3D)
- ☐ Same motion correction (yes, no)

A series of parameter maps is subsequently calculated from the loaded series using a post-processing protocol.

Selecting anatomical series

When loading the series, the number of usable images is determined automatically. The first image of the first series serves as the benchmark for image usability. The series with the same number of usable images are marked in the list. They are preselected for calculation.

- Series with a different number of usable images are excluded from the calculation.



Proceed as follows if you do *not* want use the automatically pre-selected series for calculation:

- ❑ To eliminate individual series from the calculation, drop them from the preselection.
- ❑ You can choose new series with a different number of usable images.

Deselecting series from the preselection

- ✧ Hold the **Ctrl** key down and consecutively select the series you want to eliminate.

The series are no longer marked.

Making a new selection

- ✧ Click the series in the list that will serve as the benchmark for selection.

The previous selection is cancelled.

All series in the list having the same number of usable images as the selected series are automatically marked.

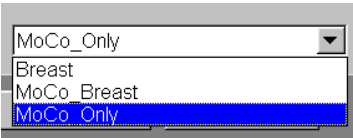
- ✧ You can also deselect unnecessary series from this new selection.

Selecting a post-processing protocol

Do you need post-processing protocol for image calculation. The **Calculate Parameter Map** subtask card lets you select several different protocol types:

Protocol type	Effect
Breast Image calculation only	Calculates parameter maps from either motion-corrected or non-motion corrected anatomical images. A mixed series calculation containing both motion-corrected and non-motion corrected images is not possible.
Moco_Breast Motion correction and image calculation	Calculates a defined motion correction when non-motion corrected anatomical images have been loaded. These motion-corrected images are then used as the basis for determining the parameter images of the result series.
Moco_Only Motion correction only	Calculates the corresponding motion-corrected images from the loaded anatomical images. If the anatomical images have already been corrected, they are rejected.

The selection manages both Siemens and user-defined customer protocols.



✧ Choose the required post-processing protocol from the selection list.

The task card shows you the protocol parameters for motion correction and/or for image calculation. If necessary, you may change them prior to calculation.

Determining motion correction (optional)

The image quality of anatomical images is improved with the BRACE motion correction (BREast Acquisition CorrEction). These images serve as the basis for calculating parameter maps. Both calculations can be combined using suitable post-processing protocols.

✧ Motion correction may only be applied to images that have not previously been corrected.

Motion correction estimates the relative motion of two volumes and removes it from images via multiple iterations. For this reason, a reference volume is determined and serves as the basis for all other volumes.

Enabling motion correction

Motion correction (BRACE) ☒

✧ Select **Motion Correction (BRACE)**.

Establish the mode

Mode

For motion correction, you can select between the modes **Fast** and **High quality**.

✧ Select the mode desired from the **Mode** selection list.

Determining the reference volume

The position of the series required for the reference volume has to be established in the series list. Example: If you want to set the second series from the top as the reference, enter **2**.



Reference Vol.

- ⇒ The serial number shown in the list does not indicate the position.
- ✧ Enter the position of the required reference series within the series list in the **Reference Vol.** input field.

Editing protocol parameters

Determining the types of evaluation

You are able to select the following evaluations:

Evaluation	Definition
Wash - In	Signal change at the start of the dynamic measurement sequence
Wash - Out	Signal change at the end of the dynamic measurement sequence
TTP	(Time To Peak) Time required to reach the signal peak
PEI	(Positive Enrichment Integral) – Area under a signal intensity time curve
MIP time	(Maximum Intensity Projection)– highest pixel value in the measurement interval



Wash - Out

☒

The evaluations listed can be activated via checkboxes. Preselection may have already been met depending on the protocol selected.

✧ Select or deselect the evaluation.

If an evaluation is deactivated in the protocol, the corresponding parameter image is not reconstructed in the evaluation.

Determining the parameters

You may set one or more of the following parameters for each evaluation type:

Parameters	For evaluations	Comment
Color table	all	Color palette used to display the parameter map.
First measurement	Wash - In Wash - Out	First measurement needed to calculate the parameter map. For example, the value 2 indicates that the second series from the top is to be used for the first measurement.
Last measurement	Wash - In Wash - Out	Last measurement needed to calculate the parameter map. For example, the value 2 indicates that the second series from the top is to be used for the last measurement.
Highest value	Wash - In	When the checkbox is activated, the parameter map is calculated by using the highest value of the input images in the "first measurement - last measurement". The parameter map is not calculated by using the value from the last measurement.

- ✧ Check the default values for the activated evaluations.
- ✧ If necessary, select a different **Color table**.
- ✧ If necessary, correct the **First measurement** and **Last measurement** parameters.
- ✧ Select or deselect the **Highest value** option.

Saving protocol data

When storing protocols, you have to differentiate between Siemens protocols and protocols created by the customer:

☐ **Customer protocols:**

You can save changes to the same protocol, overwriting previous settings. Alternatively, you can save the changes as a new customer protocol.

☐ **Siemens protocols:**

Save changed Siemens protocols under a new name as a customer protocols.



When a user logs off, unsaved data will be lost irretrievably.

Check whether you need to save data. Save all data you want to keep.

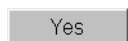
→ [Page J.1–2](#)

Overwriting a current customer protocol



- ✧ Click **Save**.

A security question is displayed.



- ✧ Click **Yes**.

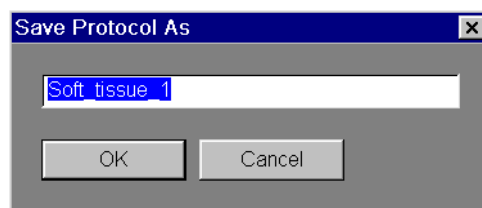
The changes are saved in the current customer protocol.

Saving a protocol as a new customer protocol



- ✧ Click **Save As**.

The **Save Protocol As** dialog window opens.



- ✧ For a changed Siemens protocol, the system automatically suggests a new name for the protocol in the text input field.
- ✧ Enter the desired name in the input field.
- ✧ Click **OK**.

The protocol data are saved as a new customer protocol.

Deleting the customer protocol

While you are able to delete customer protocols, you cannot delete Siemens protocols.

✧ Mark the protocol to be deleted in the selection list.

✧ Click **Delete**.

A security question is displayed.

✧ Click **Yes**.

The protocol is deleted.

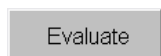
A rectangular button with a light gray background and a thin black border. The word "Delete" is written in a dark gray, sans-serif font.A rectangular button with a light gray background and a thin black border. The word "Yes" is written in a dark gray, sans-serif font.

Performing calculation



For calculations with motion correction of anatomical images, it is not possible to start calculation if the images loaded do not meet the prerequisites for motion correction.

The **Evaluate** button remains dimmed and a corresponding message is displayed.



Start the calculation after all the required settings have been made.

✧ Click **Evaluate**.

Calculation starts and the evaluation dialog window closes. A safety query is displayed if the protocol data have not been saved yet.

Both, motion correction and parameter map calculation run in background. During calculation, the image post-processing icon is displayed in the status bar.

→ [Page G.4–5, Monitoring post-processing](#)

Calculated series of parameter maps are stored to the database and loaded into the **Viewing** task card.

- ✧ During image calculation, you can define and subsequently start additional calculations. To this end, you load new images from the Patient Browser, individual calculations are processed sequentially.
- ✧ Motion-corrected images that do not meet the established minimum quality standard are discarded. Otherwise, they are stored as a new series in the database and loaded into the **Viewing** task card.

Calculating Combined Images

For calculations, you have to load at least two parameter maps series with the same number of usable images.

Only parameter images can be combined if the following characteristics match:

- ☐ same series block
→ [Page C.1–6, Series block](#)
- ☐ Same table position
- ☐ Same type of distortion correction (none, 2D, 3D)
- ☐ Same motion correction (yes, no)



Combinations are based on **Wash - In**, **Wash - Out**, **TTP**, **MIP**, and **PEI** parameter images only.

Selecting parameter map series

When you select parameter images for calculating combined images, proceed the same way as for selecting anatomical images for computing parameter maps.

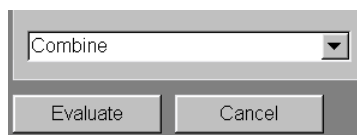
→ [Page J.2–1, Selecting anatomical series](#)

Selecting a post-processing protocol

To calculate combined images, you require a *Combine* protocol.

- ✧ Choose the required post-processing protocol from the selection list.

The protocol parameters are loaded into the fields of the dialog window.



Add Wash-In ☒	Add PEI ☒
Wash-In weight 1.0	PEI weight 1.0
Wash-In window 800	PEI window 500
Wash-In center 850	PEI center 3000
Add Wash-Out ☒	Add MIP ☒
Wash-Out weight -1.0	MIP weight 1.0
Wash-Out window 400	MIP window 500
Wash-Out center 0	MIP center 1500
Add TTP ☒	
TTP weight -1.0	
TTP window 50	
TTP center 150	
	Combine color table GreenRed



Editing protocol parameters

The image types for the series selected for the calculation are automatically activated in the dialog window. Image types that are not included in the selected series are deactivated automatically.

Correcting weighting parameters

For image types **Wash-In**, **Wash-Out**, **TTP**, **PEI** and **MIP**, you are able to determine the weighting parameters for calculating combined images.

→ [Page J.2–5](#)

For all activated image types, you may set the following parameters:

Parameters	Effect on combined images
Weighting	Weighting with which the parameter maps of the corresponding type will be included in the combined image.
Center	Contrast value with which the parameter maps of the corresponding type will be windowed for calculation.
Windows	Brightness value with which the parameter maps of the corresponding type will be windowed for calculation.

- ⇒ Use the optimal settings for **Center** and **Window** from the windowing values shown in the **Viewing** task card or set fixed values. When using fixed values, the following rules apply:

Pixel value $\geq$ Center - window/2 = Smallest gray or color value

Pixel value $\leq$ Center + window/2 = Highest gray or color value

- ⇒ Change the settings as required.

Assigning a color table



Via the **Combine color table** selection list, allocate a color palette to the combined window.

- ⇒ Select the color table.

Saving protocol data

Refer to → [Page J.2–7](#), *Saving protocol data*

Deleting the customer protocol

Refer to → [Page J.2–9](#), *Deleting the customer protocol*

Performing the calculation

Start the calculation after you have made the required settings.

- ✧ Click **Evaluate**.

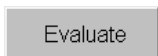
Calculation starts and the evaluation dialog window closes. A safety query is displayed if the protocol data have not been saved yet.

The calculation of combined images runs in background. During the calculation, the image post-processing icon is displayed in the status bar.

→ [Page G.4–5](#), *Monitoring post-processing*

- ✧ During image calculation, you can define and subsequently start additional calculations. To this end, load new images from the Patient Browser. The individual calculations are processed sequentially.

The result images are stored as a new series in the database and loaded into the **Viewing** task card.



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CHAPTER

K.1

Introduction

Results of evaluations (images, diagrams, etc.) can be stored as structured reports (SR) in DICOM format.

Objective

You are able to edit and transmit the reports.

Contents of a report

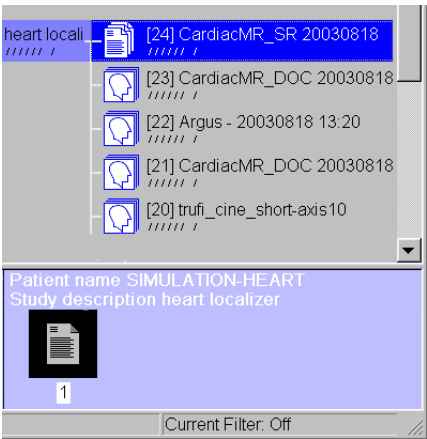
The following information can be recorded in a report:

- ☐ Data relating to the institute (name, address, ...)
- ☐ Demographic patient data
- ☐ Quantitative results of the examination
- ☐ Qualitative results of the examination
- ☐ Summary and conclusions comprising individual diagnoses, values measured, and medical codes
- ☐ Recommendations for further diagnosis or therapy
- ☐ Links to medical images or graphics
- ☐ Links to curve plots

You can select, edit, and print reports in the **Patient Browser** or in the applications where they were created. Additionally, the **Patient Browser** sends reports via the network.

Reports in the Patient Browser

Reports in the **Patient Browser** are stored on the same level as the measurement or evaluation results.



Reports are identified in the **Patient Browser** as follows:

Symbol	Definition
	Report icon in the directory structure.
	Report icon in the contents area of the most current version
	Report icon in the contents area of the oldest version
	Report icon in the contents area of secondary captures
	Report icon in the contents area (study selected).

Basics

The reports are coded according to the DICOM standard. Unstructured handling of the report is greatly limited, providing the following advantages:

- ❑ Each item of information is classified and incorporated in a hierarchical structure.
- ❑ Subsequent evaluation of the data from the report is easier due to the DICOM structure.
- ❑ Use of the standard enables communication with the DICOM world. Data exchange of reports from different systems or applications is possible without negative effects. The same applies to displaying and evaluating external reports.

Structure of a report

Each item of information in a report is a so-called content item.

A content item comprises a name and a value.

The name is always a code.

There are various types of values:

- ❑ Free text
- ❑ Code
- ❑ Numeric value with unit
- ❑ Person's name, date, time
- ❑ Reference to a DICOM object (image, diagram, or unformatted text)

Content items

Relationship between content items

A DICOM report provides individual information in a hierarchical structure.

The individual information "document title" is of highest priority. It is followed by additional individual information in a complex tree structure with different relationships to one another.

For example, the following relationships are possible between content items of a report:

- ☐ Configuration within the hierarchy
- ☐ Property of a higher-level individual item
- ☐ Conclusions from higher-ranking individual information
- ☐ Reference for a higher-ranking information item

Codes in general

Most of the report codes are automatically generated by the system. Only a diagnostic code can be set or defined for the respective report.

→ [Page K.2–7](#), *Enter the diagnostic code (not for Vessel View reports)*

Diagnostic codes (not for Vessel View reports)

Coded texts are usually integrated into reports to facilitate searches and evaluation. These are obtained from special dictionaries of code scheme designators.

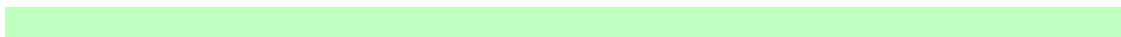
A DICOM code entry consists of three parts:

- ❑ **code value** (value),
e.g., *N11.1*
- ❑ **code scheme designator** (code catalogs)
e.g., *I9* or *UCUM*
- ❑ **code meaning** (comprehensive description of code),
e.g., *Chronic obstructive pyelonephritis*

syngo MR allows you to select a code catalog for reporting/diagnostics or for defining your own code catalog.

Examples of diagnostic code catalogs:

- ❑ SNOMED ("SNM3" and "SRT")
- ❑ ACR Index for Radiological Diagnosis ("ACR")
- ❑ ICD9 and ICD10 ("I9" and "I10")
- ❑ ICD9-CM and ICD10-PCS ("I9C" and "I10P")



Creating and Editing Reports

Reports can be generated from the **Argus**, **Vessel View** and **Mean Curve** task cards. They are stored in the local database.

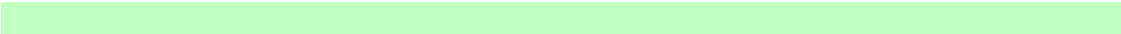
Supplements to a report are also handled exclusively via the appropriate task card. However, all remaining work for a report has to be handled strictly via the **Report - Editor**.

Creating a report

A report is always generated via the respective task card.

Detailed information:

to Argus	→ Page 93 in the Argus Operator Manual
to Mean Curve	→ Page F.4–6
to Vessel View	→ Page 79 in the Vessel View Operator Manual



Supplementing a report

You can add to a loaded report via the task card used for generating the report. As a prerequisite this report has never been post-processed, stored, transmitted or printed with the **Report - Editor**.

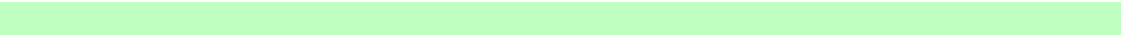
Detailed information

to Argus	→ Page 93 in the Argus Operator Manual
to Mean Curve	→ Page F.4–8
to Vessel View	→ Page 79 in the Vessel View Operator Manual

Editing a report

You can edit a report in the **Report - Editor** as follows:

- ☐ Set the Completion flag
- ☐ Record the report author
- ☐ Create or edit diagnosis and comments
- ☐ Enter the diagnostic code (not for Vessel View)
- ☐ Delete parts of the report



The edited report can be stored. The changes in the current report may either be stored in the same report or in a new report depending on the editing status set.

- ✧ Make sure that you have saved your changes before you log off. The dialog window will be closed and unsaved changes will be lost.

Opening a report

- ✧ Mark the requested report in the table of contents in the **Patient Browser**.
- ✧ Select **Applications > Report Open**.
 - Or -
 - Double-click the report.

The **Report - Editor** dialog window opens: Example of an Argus report:

The screenshot displays the 'Report - Editor' window with a dark blue header and a light blue sidebar. The main content area is divided into several sections. On the left, a vertical list of numbered callouts (1) through (6) points to specific elements in the interface.

(1) Points to the patient information section, which includes the following fields:

- Institution: ZRI
- Institution Address:
- Patient Name: SIM_Heart
- Patient ID: 002
- Birth Date/Sex: 3/3/1984 / M
- Study: Heart_PP / 10/13/2005 6:31:50 PM / 5
- Accession Number:
- Series: CardiacMR_SR 20051017 16:50 / 10/19/2005 5:30:16 PM / 16
- Report: 10/20/2005 5:30:16 PM / 1

(2) Points to the 'Completion Flag' dropdown menu, which is currently set to 'PARTIAL'.

(3) Points to the 'Cardiac Report' section, which includes the 'Observer Name' (Proband 1) and the 'Language of Content Item and Descendants' (English (eng, ISO639_2)).

(4) Points to the 'Current Procedure Description' section, which contains a '+ Procedure Description' button.

(5) Points to the 'Conclusions' section, which includes a '+ Conclusion' button, a 'Select Conclusion:' dropdown menu, and a text input field for 'or Add Code:' followed by a '(Code, Meaning)' label.

(6) Points to the bottom navigation bar, which contains buttons for 'Save', 'Reset', 'Close', 'Print Mode', and 'Help'.

Legend

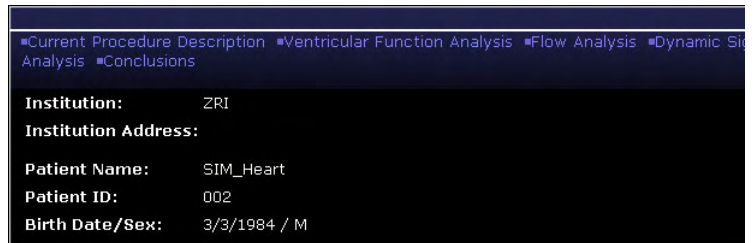
- (1) Data from the patient ID
- (2) Report header: Report author
- (3) Information units: Individual diagnoses and diagnostic code (for Argus only)
- (4) Toolbar of the window.

Navigating in the report

Navigation allows you to easily branch to all parts of the report.

Jumping to information items

The information units included are listed below the header of the report. The names of the items listed are directly linked to the information unit in the report.



- ✧ Click the name of an information unit, e.g., *Ventricular Function Analysis*.

The report display jumps to that item.

⇒ Alternative possibilities:

1. Shift the scroll bar of the report
2. Page through the report with the respective keyboard keys.

Returning to the beginning of the report



Arrows are positioned above large information units.

⇒ Click this arrow.

The report display jumps back to the beginning.

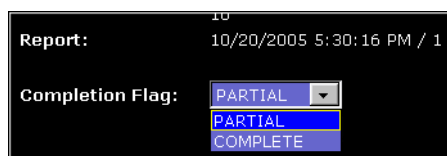
Changing a report

The opened report is automatically displayed in edit mode.

Setting the Completion flag

By selecting a **Completion Flag** you can indicate the progress of the report. Two functions are available for this:

- ☐ COMPLETE
- ☐ PARTIAL



⇒ Set the status in the **Verification Flag** selection list.

⇒ Once a report has been assigned the **COMPLETE** status, it has to be saved as a new report. It is no longer possible to save changes to the existing report.

Recording the author of the report

The author has to be entered in the **Observer Name** input field.

A screenshot of a software interface showing a text input field labeled "Observer Name:". The field contains the text "Dr. XY".

✧ Enter your name in the **Observer Name** field.

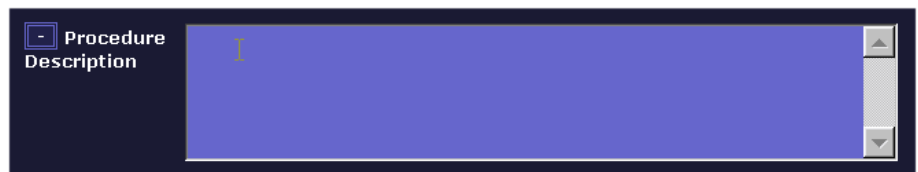
Acquiring diagnoses and comments

Some parts of the report contain input fields for recording additional comments.

A screenshot of a software interface showing a dark blue button labeled "Procedure Description" with a small plus sign icon to its left.

✧ Instead of empty text input fields, a plus sign is shown in front of the name of the input field.

✧ To open the input field, click the plus sign.

A screenshot of a software interface showing a dark blue button labeled "Procedure Description" with a small minus sign icon to its left. To the right of the button is a large, empty text input field with a vertical scrollbar on the right side.

✧ Enter a new comment or overwrite the existing comment.

Enter the diagnostic code (not for Vessel View reports)

The diagnostic codes are organized in catalogs in DICOM format. **Argus** and **Mean Curve** reports do not provide any default catalogs. The diagnostic code catalog has to be entered and managed under **Configuration** by the reporting physician or the hospital.

→ [Page K.4–4](#), *Create and edit diagnostic code catalogs*

In addition to entering catalogs with various diagnostic codes, you are able to set a general catalog for reporting in the **Configuration**. For this reason you only need to select the code from a selection in the catalog.

- ✧ A diagnostic code catalog has to be preset in the report configuration before a report is edited for the first time. This enables you to define the appropriate code by either selecting it or entering it manually.

→ [Page K.4–4](#)

- ✧ Select the desired entry from the **Consequences** selection list.

The diagnostic code is now displayed in the selection list. The abbreviation for the catalog is shown at the right end of the list.



Conclusion:	or Add Code:
Nephritis	N11.1

If a diagnostic code is not included in the catalog set, enter the code manually.

- ✧ Select the empty entry from the **Consequences** selection list.

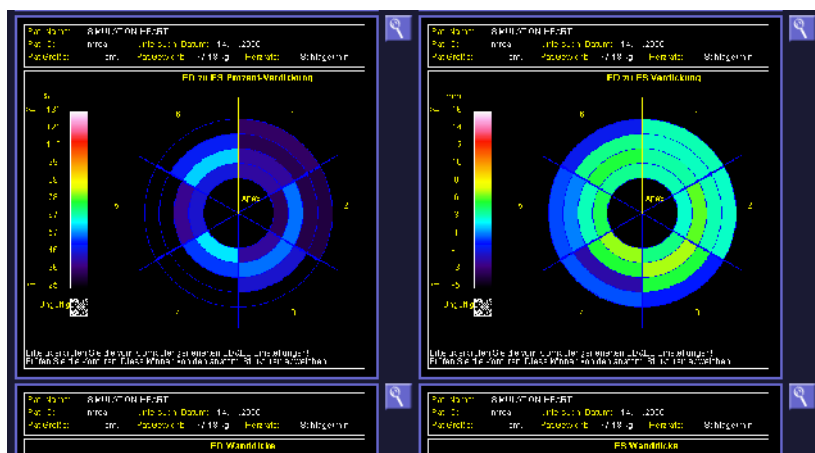
- ✧ Enter the diagnostic code and the description of the code into the two fields of the **Add Codes** line.

Select		N11.1
Conclusion:		
or Add Code:	N11.2	Hydronephrosis with Obstruction

- ✧ When the report is saved, the diagnostic code that has been entered manually is assigned to the code catalog that you pre-set under **Configuration**. The code can now be edited again in the configuration.

Enlarged display of images and tables

The default display shows images and tables in reduced size.



- ✧ Click this button to display the image in its original size.

Delete parts of the report

You can only select one information unit in the report for deleting. When you delete an information unit in the report, this unit as well as all subordinate units are deleted.

✧ Click the tab of the information unit to be deleted.

- Or -

Click directly on images, diagrams, or tables.

The unit is shown with a blue border. An **X** appears in the upper left edge of the marker.



✧ You can undo the selection by clicking it again or by clicking another item.

✧ Click the **X** in the checkbox.

- Or -

Press the **Del** key.

The content of the information unit and all lower-level information units are deleted from the report.

✧ You cannot delete several information units simultaneously.



Undoing changes



You can undo all changes made in the report as long as they have not been saved.

✧ Click **Reset**.

The entries and changes made in the report are undone.

➤ You cannot individually undo changes in the report, you have to do them all.

Saving a report

However, the way the report is saved depends on its completion status.

Status	Storage
<i>Partial</i>	The changes are saved in the current report.
<i>Completed or Verified</i>	Changes are saved in a new version of the report with a new version number.

Additionally, a new version of the report is created as well, if the report in the database is marked with one of the following statuses:

- ☐ Printed
- ☐ Archived
- ☐ Archived + Verified
- ☐ Archived + Committed
- ☐ Sent
- ☐ Sent + Committed
- ☐ Exported
- ☐ Received

- ⇒ The existing report is not changed if one of these statuses is set.

How to save a changed customer protocol:

A blue rectangular button with the word "Save" in white text.

- ✧ Click **Save**.
- ⇒ Instead of report images, references to the DICOM images are saved.



Save the changed report before printing it out. If you subsequently close the print preview without saving it in edit mode, all changes to the report are lost.

Closing the edit dialog

A blue rectangular button with the word "Close" in white text.

To end displaying the report, close the **Report - Editor**. Unsaved changes to the report are lost.

- ✧ Click **Close**.
- Or -
- Click the **X** icon in the top right corner of the window.

The following possibilities are available for documenting reports:

- ☐ Printing a report
- ☐ Sending reports via the network
- ☐ Exporting reports

Printing a report

Prior to printing you can preview the report.

If you want to make changes to the report, you can go back to the edit window from the print preview at any time.

→ [Page K.3–3](#), *Closing the print preview*

Selecting print preview

The print preview is opened from the edit window.

- ✧ The print mode is available only for users with printing privileges. Otherwise the **Print Mode** button is inactive.
- ✧ Click **Print Mode** in the edit window.

The current report is shown in the print preview. You cannot change the report in this dialog window.

Print Mode

Report - Editor

Institution: ZRI
Institution Address:

Patient Name: SIM_Heart
Patient ID: 002
Birth Date/Sex: 3/3/1984 / M

Study: Heart_PP / 10/13/2005 6:31:50 PM / 5
Accession Number:
Series: CardiacMR_SR 20051017 16:50 / 10/19/2005 5:30:16 PM / 16
Report: 10/20/2005 5:30:16 PM / 1

Completion Flag: PARTIAL

Cardiac Report

Observer Name:

Current Procedure Description

Ventricular Function Analysis

Finding

Left Ventricle Absolute

Patient Height: 70.87 in Patient Weight: 176.21 lb Heart Rate: 65.00 {H.B.}/min

		Normal Range: M (MRI)		
LV Absolute Ejection Fraction	EF	56.94	56.00..78.00	%
LV Absolute End Diastolic Volume	EDV	53.20	77.00..195.00	ml

Closing the print preview

There are two ways of closing the print preview:

- ☐ Switching to edit mode
- ☐ Closing print preview which closes the entire report as well

Switching to edit mode



If you do not want to print out the report or want to make changes to the report before printing, return to the edit mode from the print mode.

- ✧ Click the **Edit Mode** in the print preview.

Closing the print preview



This is how you close the print preview as well as the report:

- ✧ Click **Close**.

- Or -

Click the **X** icon in the top right corner of the window.

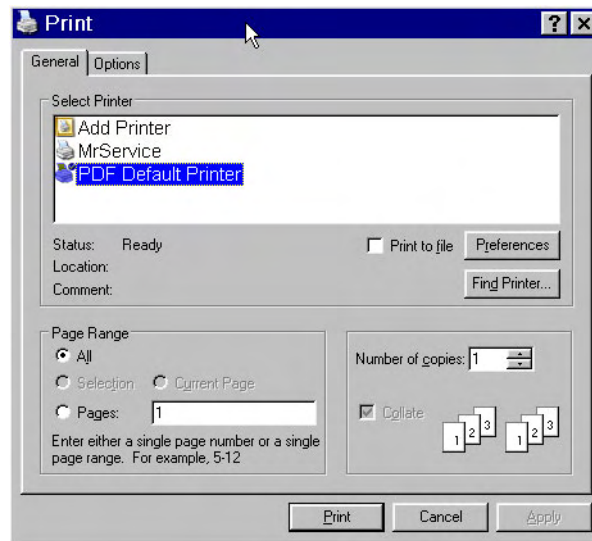
Unsaved changes to the report are lost.

Starting to print

Print

- ✧ Click **Print** in the print preview.

The **Print** dialog window opens. You can make the necessary print settings here.



- ✧ Select a printer.
- ✧ Define the print range.
- ✧ Set the number of copies.
- ✧ Click **Print**.

Printing starts at the printer set.

Print

Sending a report

If your system is connected to a network, you can send patient reports to other workstations via the network with the function **Send To....**

→ **Patient Browser**! *Sending data via the network* in the syngo Operator Manual

- ⇒ Reports are sent in the "Structured Report" DICOM format. You will find more detailed information in the DICOM Structured Report.
- ⇒ Referenced images are not automatically sent with the report. They must be sent separately.
- ⇒ You are not able to save changes in a transmitted report. A report that was sent and subsequently changed has to be saved under a new name.

Exporting reports

For data exchange, reports of a patient may be exported to an external data carrier (CD or DVD).

Basics

An exported report consists of two parts:

- ❑ **DICOM component:**
DICOM data directory with linked report data in DICOM-Format and displayed on the DICOM-Viewer.

☐ **HTML part:**

Reports on Xhtml basis and jpg images derived from DICOM data of the respective data directories.

Exported report data of a patient can be displayed via the supplied DICOM Viewer or via an Internet browser.

Making the report available for export

- ✧ Select the report in the **Patient Browser**.
- ✧ Select **Applications > Report Tools > Export HTML to Off-line**.

The report data are collected in a CDR_Offline_Directory until you burn the data to a data carrier (CD or DVD).

Burning the report to CD or DVD

All data collected in the CDR_Offline_Directory can be stored together to an external data carrier. If you only want to save report data to a data carrier, ensure that the CDR_Offline_Directory is empty before you export report data from the local database to the CDR_Offline_Directory.

- ✧ Select **Transfer > Export to Off-line**.

Burning is started in the single session mode.

All data burned to the external medium are automatically deleted in the CDR_Offline_Directory.

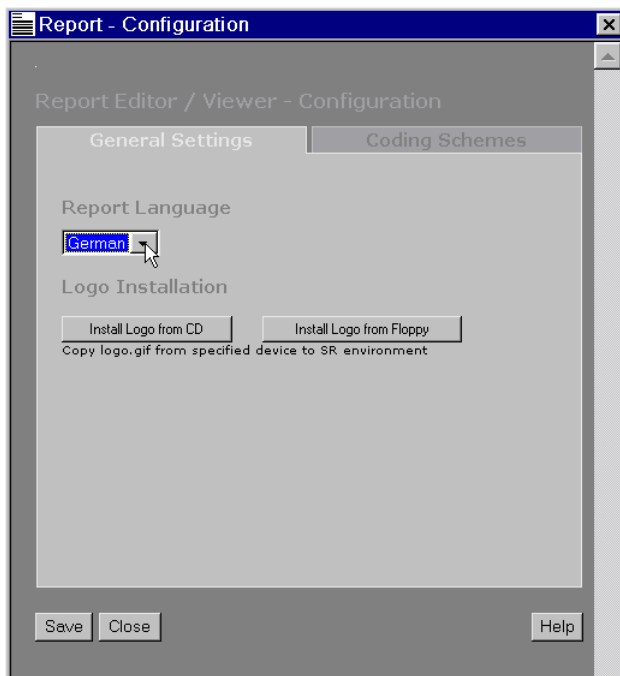
For a report, you use defaults on two subtask cards for generating reports via the configuration dialog:

- ☐ **General settings** (language/user logo)
- ☐ **Code catalogs** (Diagnostic code catalog for Argus or Mean Curve reports)

Open the configuration dialog window

- ✧ Select **Applications > Report Tools > Report Configuration** in the **Patient Browser**.

The **Report Configuration** dialog window opens.



Enter general settings

The general settings include the report language and the optional transfer of your own logo into the report.

Setting the report language

You are able to define the language used in the report on the **General Settings** subtask card. This configuration setting applies to all subsequent new reports.

- ✧ Click the **General Settings** subtask card into foreground, if necessary.
- ✧ Select the language from the **Report Language** selection list.
- ⇒ The language set only applies to the reports and may differ from the language set on the system. You can manually change the configuration setting in each individual report.



Transferring the user logo

- ✧ The logo has to be stored in gif format and named **Logo.gif** to be transferred to the report.

This is how you install the logo:

- ✧ Save the logo on a suitable data medium (CD or floppy disk).
- ✧ Insert this data medium into the appropriate drive on your station.



- ✧ Click the **Install Logo from ...** button on the data carrier.
The logo is read in and you receive a message about successful installation.
When the logo was installed successfully, it will be automatically used for all saved reports.
- ✧ Save the settings prior to performing changes.
→ [Page K.4–10, Saving and ending the configuration](#)

Create and edit diagnostic code catalogs

⇒ Diagnostic codes are not used in Vessel View reports.

The code catalogs used for the reports can be set in the **Coding Schemes** subtask card of the **Report - Configuration** dialog window. You can also set up complete code catalogs with the relevant diagnostic codes.

✦ Open the **Coding Schemes** subtask card.



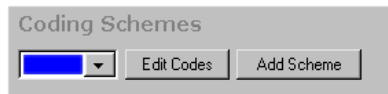
Configuration options:

- ☐ Set the requested code catalog by default
- ☐ Changing or creating a new diagnostic code
- ☐ Creating a new code catalog

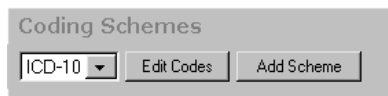
Predefining the encoding scheme

As a rule, the code catalog has to be set in the configuration before a report is edited in the Patient Browser for the first time. Otherwise, it is not possible to assign a diagnostic code in the report.

The first time the configuration is called, no code catalog has been set:



- ✧ Select the required catalog from the **Coding Schemes** selection list.
- ✧ By default, the code catalogs ICD-9 and ICD-10 are empty (without codes).



You can create additional code catalogs as required.

→ [Page K.4–9, Generating code catalogs](#)

Changing or creating a new diagnostic code

A code catalog may contain different diagnostic codes. This enables you to quickly select diagnostic codes in the report.

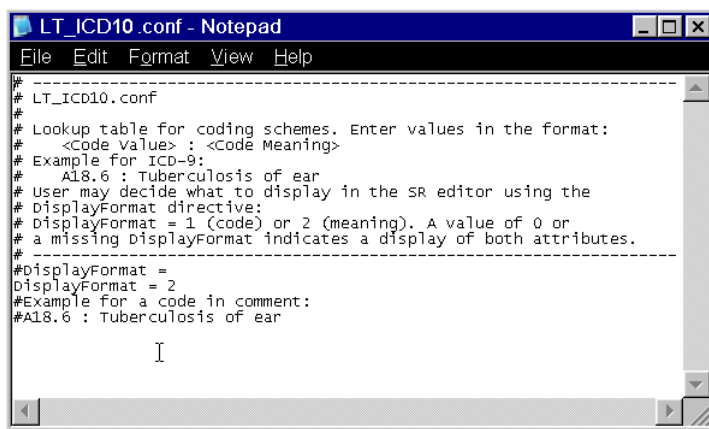
→ [Page K.2–7](#), *Enter the diagnostic code (not for Vessel View reports)*

✧ Select the code catalog you want to edit.

✧ Click **Edit Codes**.

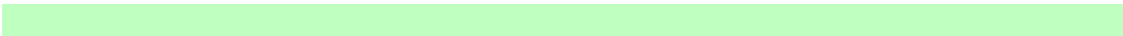
The text editor for the codes of the active catalog opens.

Edit Codes



```
# LT_ICD10.conf
#
# Lookup table for coding schemes. Enter values in the format:
# <Code value> : <Code Meaning>
# Example for ICD-9:
# A18.6 : Tuberculosis of ear
# user may decide what to display in the SR editor using the
# DisplayFormat directive:
# DisplayFormat = 1 (code) or 2 (meaning). A value of 0 or
# a missing DisplayFormat indicates a display of both attributes.
#
#DisplayFormat =
DisplayFormat = 2
#Example for a code in comment:
#A18.6 : Tuberculosis of ear
```

You are provided with explanations and an example of editing diagnostic codes. Lines preceded by a "#" are comment lines.



In all other lines, you can define the codes you require, one code per line.

Input example:

N11.1:chronic obstructive pyelonephritis

The complete code consists of the code value (**N11.1**) followed by a colon and a short description of the code (**chronic obstructive pyelonephritis**).

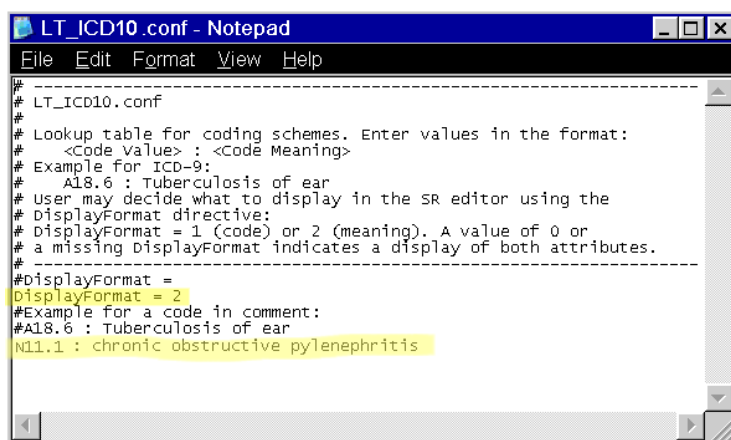
✧ Enter the required code line by line.

You also define how the diagnostic code will appear in the report. The following display formats are possible:

Display format	Identifier
Code value and code description, e.g., N11.1:Chronic obstructive pyelonephritis	0
Code value only, e.g., N11.1	1
Code description only, e.g., Chronic obstructive pyelonephritis	2

✧ Enter the identifier for the requested display format at the end of the line **DisplayFormat =**.

Example: Entering a diagnostic code and a display identifier:



```
# LT_ICD10.conf
#
# Lookup table for coding schemes. Enter values in the format:
# <Code value> : <Code Meaning>
# Example for ICD-9:
# A18.6 : Tuberculosis of ear
# User may decide what to display in the SR editor using the
# DisplayFormat directive:
# DisplayFormat = 1 (code) or 2 (meaning). A value of 0 or
# a missing DisplayFormat indicates a display of both attributes.
#-----
#DisplayFormat =
#DisplayFormat = 2
#Example for a code in comment:
#A18.6 : Tuberculosis of ear
#N11.1 : chronic obstructive pylenephritis
```

To activate the settings made in the text editor, you have to save them in the text editor (not to be confused with the **Report -Configuration** dialog window).

✧ Open **File > Save** in the text editor.

The inputs are saved and you can close the text editor.

✧ Click the "close" icon in the upper right corner.

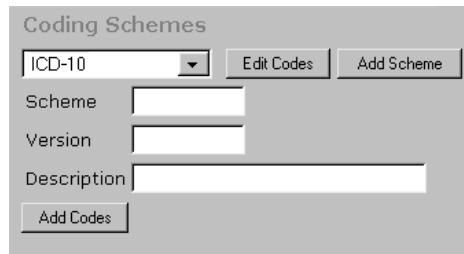


Generating code catalogs

Add Scheme

- ✧ Click **Add Scheme**.

These additional edit boxes are displayed:



The screenshot shows a dialog box titled "Coding Schemes". At the top, there is a dropdown menu currently set to "ICD-10", followed by "Edit Codes" and "Add Scheme" buttons. Below this, there are three input fields: "Scheme", "Version", and "Description". The "Scheme" field is empty, while "Version" and "Description" contain some text. At the bottom of the dialog, there is an "Add Codes" button.

- ✧ Enter the name for the new catalog in the **Scheme** input field.
- ✧ Go to the **Version** and **Description** field and enter the respective data of the catalog.

If necessary, you are now able to define further codes for a new code catalog.

Add Codes

- ✧ Click **Add Codes**.
- ✧ Enter the code in the text editor of the new scheme.
 - [Page K.4–6](#), *Changing or creating a new diagnostic code*

Saving and ending the configuration

If you want to apply the configuration settings made in the reports, save them.

A rectangular button with a thin black border and the word "Save" in a sans-serif font.

✧ Click **Save**.

All settings will be saved. The **Report - Configuration** dialog window opens.

You may then close the **Report - Configuration** dialog window:

A rectangular button with a thin black border and the word "Close" in a sans-serif font.

✧ Click **Close**.

⚠ If you close the dialog window without saving first, you will lose all changes made in the dialog window.

PhoenixZIP in the HIS/RIS workflow

The *PhoenixZIP* or the *Phoenix Evidence Document (PED)* is a structured report in the DICOM format as described for the reports in this section of the Operator Manual. *PhoenixZIP* is generated in the **Exam** task card.

→ [Chapter K.1, Introduction](#)

Background

PhoenixZIP is used for the efficient storage and transport of protocol information.

Follow-up examinations

To ensure optimal follow-up examinations within the workflows of hospitals, beginning with software version **B15A**, follow-up examinations can be performed without looking for them and importing them manually in the archive or creating them again in the program map.

→ [Page C.3–1, Automatic loading \(Chorus\)](#)

Information carrier PhoenixZIP

PhoenixZIP plays a central role in the reconstruction of examinations. When initiating a new examination, a *PhoenixZIP* is automatically generated per protocol and stored together with the measurement results as soon as the last image reconstruction of the measurement has been completed.

PhoenixZIP contains important information used to perform follow-up examinations with the help of Phoenix.

→ [Page K.5–2, Protocol-specific information](#)

Special characteristics of PhoenixZIP

Special characteristics of PhoenixZIP

The following prerequisites are required for efficient follow-up via the HIS/RIS:

- ☐ Efficient/compact storage of protocol data without the need to transfer the entire study
- ☐ Exact reconstruction of examinations performed

To this end, *PhoenixZIP* includes the following main characteristics:

- ☐ Efficient handling of storage capacities
- ☐ Protocol-specific information

Protocol-specific information

As compared to other structured reports, *PhoenixZIP* contains all protocol data of the measurement protocols and protocol properties required to accurately reconstruct an examination. As an overview, the **Report Viewer** shows the following protocol parameters:

- ☐ Modality
- ☐ Performed Procedure Step
- ☐ System type
- ☐ Software version
- ☐ Field strength
- ☐ Protocol name
- ☐ Name of sequence
- ☐ FoV read
- ☐ FoV phase
- ☐ Acquisition time
- ☐ Echo time
- ☐ Repetition time
- ☐ Flip angle
- ☐ Number of slices
- ☐ Image type of the images measured

PhoenixZIP during the examination

First examination

Creating a new examination

- ✧ Register the patient.
→ [Page A.3–7, Patient registration](#)
- ✧ Load the measurement programs for the examination requested.
- Or -
Select the individual protocols for the examination requested.
→ [Page C.3–12, Checking information according to the region and examination](#)
→ [Page C.3–14, Selecting a measurement program](#)
→ [Page C.3–18, Transferring individual program steps](#)

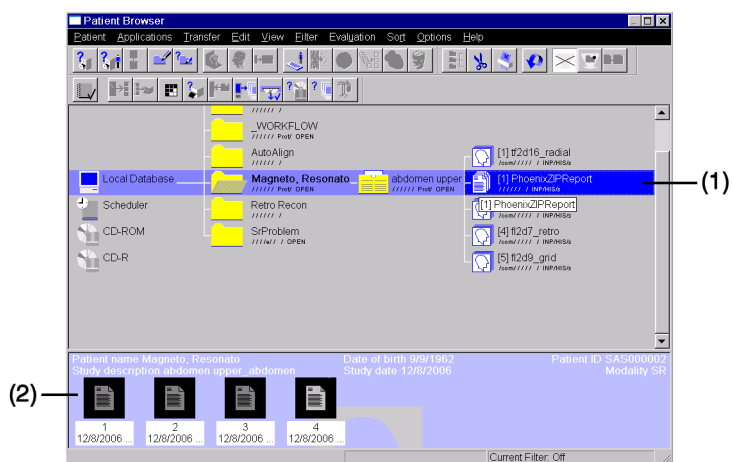
Automatically generating PhoenixZIPs

After completing the examination, a *PhoenixZIP* is created for each protocol measured after finishing the last image reconstruction. PhoenixZip is saved in the **Patient Browser** parallel to the measurement results.

- ⇨ For Set-n-Go protocols *one PhoenixZIP* per Set-n-Go-protocol is generated.

Automatically storing PhoenixZIPs

In the **Patient Browser**, a document series per examination will be generated at the series level. In this document series, the *PhoenixZIPs* of all protocols measured are stored.



- (1) Document series on the series level
- (2) *PhoenixZIPs* of the individual protocols at the instance (image) level

The results of the initial examination are automatically sent via the *PhoenixZIP* series to the PACS (Picture Archiving and Communication System). They are available to a HIS/RIS as well as long as a Chorus license is used.

- Automatic transmission to PACS functions only, when PACS has been configured as the default archive by service technicians.

Follow-up examinations

If you want to perform a follow-up examination for a patient, you reconstruct it from the examination by using *PhoenixZIPs*.

There are two possibilities:

- ❑ The *Chorus* function is available:
Automatic transport and loading of *PhoenixZIPs* in order to perform the follow-up.
- ❑ The *Chorus* function is not available:
Manual loading of *PhoenixZIPs* from the PACS and manual entry of *PhoenixZIPs* into the Program Control in order to perform the follow-up.
- ⇒ In both cases, there is no need to manually prepare the follow-up examination in the program card.

Performing follow-up examinations with Chorus

If your RIS system supports *Chorus*, you can plan and automatically run follow-up examinations in the RIS.

→ [Page C.3–1, Automatic loading \(Chorus\)](#)

Performing follow-up examinations without Chorus

- ✧ Select the **Patient Browser** and open the **Scheduler**.
- ✧ Select the requested follow-up exam.
- ✧ If you do not have the required *PhoenixZIP* series on your system, you import it from the archive.
→ **Patient Browser**//*Importing data from storage disks* in the *syngo* Operator Manual
- ✧ Mark the *PhoenixZIP* series on the series level and pull it to the program control with the mouse.

The follow-up examination is reconstructed.

- ✧ If protocol conversion is necessary, a dialog appears *prior* to reconstruction, which informs you that the protocols have to be converted.
- ✧ Click **OK** to perform conversion and to continue with reconstruction.
- Or -
- ✧ Click **Cancel** to terminate conversion.

Viewing and printing PhoenixZIP

- ✧ At the instance (image) level of the **Patient Browser**, double-click the PhoenixZIP you want to view.

The **Report - Viewer** opens.

You can also print the PhoenixZIP from the **Report - Viewer**.

→ [Page K.3–1, Printing a report](#)

Report - Viewer

Accession Number: U49SA1P000000004
Series: PhoenixZIPReport / 12/8/2006 10:38:45 PM / 1
Report: 12/8/2006 10:38:45 PM / 4
Completion Flag: PARTIAL

Phoenix ZIP Acquisition Report

Document Title Modifier: Phoenix ZIP Document
Language of Content Item and Descendents: English
Observer Type: Person
Observer Name: n.a.

Current Procedure Description

Modality: Magnetic Resonanz
Performed Procedure Step: n.a.

Acquisition Protocol

Protocol Header

System Type: Avanto
Software Version: syngo MR B15
Field Strength: 1.5
Protocol Name: t2_tse_tra_mbh_pat2
Sequence Name: %SiemensSeq%\tse

FoV Read	350 mm
FoV Phase	75.0 %

Acquisition Time: 0:32

Echo Time	103.0 msec
Repetition Time	4000 msec
Flip Angle	150 deg
Number of Slices	20

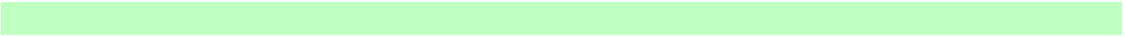
Image Acquired

Image Type: n.a.

Comment: n.a.

Close **Print Mode** **Help**

✧ Click **Close** to exit the **Report Viewer**.



syngo MR Composing

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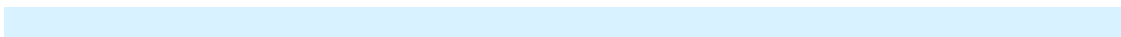
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CHAPTER

L.1

Introduction

You have obtained several *overlapping* volume data sets from a tomographic examination and you want to combine them into overview images. For this purpose, you are using the **Composing** task card.

Use

You are generating overviews, e.g., of the spinal column.

Reconstruction

Once you have started **Composing**, load the series with the **Patient Browser**. The overview images are reconstructed (composed).

Visualization

The **Composing** task card allows you to display the original, detail and reconstruction images in various layouts, compare two reconstruction images for evaluation of dynamic processes and film the results in various layouts.

Measurements

Subsequently, you can perform measurements:

- ☐ Basic functions: Circle, distance, angle, freehand ROI, arrow
- ☐ Orthopedic functions: Scoliosis angle according to Cobb, kyphosis angle, vertical distance measurement, difference in height

Hardcopy

The overview images as well as individual images can be shown via hardcopy cameras (film sheet) or paper printers using the **Filming** task card.



CAUTION

Dislocation during manual MR-image adjustment!
Viewing of combined MR images for diagnostic purposes!

Incorrect diagnosis

- ✧ Always additionally refer to the original images for diagnostic purposes.

-
- ! As the user you are responsible for accurately selecting original images for reconstruction as well as manual adjustment of the image margins.

Reconstructed images differ from original images with respect to image information, especially at the image transitions. For this reason, reconstructed images of bones and vascular structures are not always suitable for diagnosis.

When transmitting one or several reconstructed images through the network or on a data medium, please inform the recipients of the above facts.

Starting and ending Composing

Selecting Composing

You are able to start the **Composing** task card from any of the other task cards or from the **Patient Browser**.

⇒ **Composing** can be started only if you have obtained and installed the license.

✧ Select **Applications > Composing**.

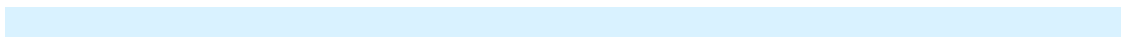
The **Composing** task card is opened.

⇒ If too many task cards are active, please close one of them.

Closing Composing

✧ Select **Applications > Close Composing**.

The **Composing** task card will be closed. Another task card is moved into foreground.



Reconstruction

Start reconstruction by selecting a series in the **Patient Browser**.

When loading the **Composing** task card, images are checked according to certain criteria and then composed into overview images. Depending on the type of images, the reconstruction algorithm is automatically set or selected by you.

Manual correction

After reconstruction, you can manually shift images in the image plane or in the slice volume.

Selecting the series and starting reconstruction

Select the required series of images for reconstruction in your local database using the **Patient Browser**.

Selecting studies/series

- ✧ Open a study and click a range of suitable series.
- ✧ Select *more than one* series for composing. Otherwise the slice position information of the series is corrupted by composing.
- ✧ Press and hold **Shift** while you select several consecutive studies or press and hold **Ctrl** to select several individual series.
- ✧ Select only matching individual *Series*, because in practical application, a study always includes additional series/images that could cause a complete study to cancel while you are loading it.

Loading the selection

- ✧ Select **Applications > Composing** in the menu of the **Patient Browser**.
- Or -
- Use the mouse to drag the selection to the image area of the **Composing** task card.
- ✧ To load multiple selections via double-click, keep the **Shift** or **Ctrl** key pressed as well.

Reconstruction

MIP and subtraction images can be composed, in addition to original images measured.

- ✧ You can only compose MIP images that were measured inline. You cannot use MIP images generated with *syngo* 3D.

Prerequisites

The images selected for reconstruction via **Composing** have to meet the following prerequisites:

Criterion	Description
Image type	Only MR images acquired with syngo MR are acceptable.
Image matrix	All images of a volume must have the same matrix.
Pixel spacing	All images of a volume have to have the same pixel spacing. Pixel spacing = FoV / image matrix
Image layout	All images of a volume have to have the same size and slice center.
Database storage	All images (or series of images) must belong to one study. All images of a volume must come from one series.
Series block	All images must be part of the same series block.
Slice thickness	All images of a volume have to have the same slice thickness.
Distortion correction	The same distortion correction procedure will be/was used for all images. It is not possible to mix already corrected images with uncorrected ones or 2D corrected with 3D corrected images.

Criterion	Description
Slice distance	The images within a volume must have the same slice distance.
Slice orientation	The images within a volume must have the same slice orientation.
Slice inclination (based on the first volume)	If the slices were acquired in coronal direction (primary direction), only an inclination from coronal to transversal is permitted (max. $\pm 45^\circ$). Inclinations in other directions are not permitted (tolerance max. $\pm 1^\circ$). If the slices were acquired in sagittal direction (primary direction), only an inclination from sagittal to axial is permitted (max. $\pm 45^\circ$). Inclinations in other directions are not permitted (tolerance max. $\pm 1^\circ$). See graphic representation → Page L.2–4, Resulting slice plane Images acquired in axial direction cannot be composed.

If the input data do not meet one of these criteria, the data are rejected and a corresponding message is displayed.

- The prerequisites that have to be identical within a volume according to the list above may differ from volume to volume.

Distortion correction

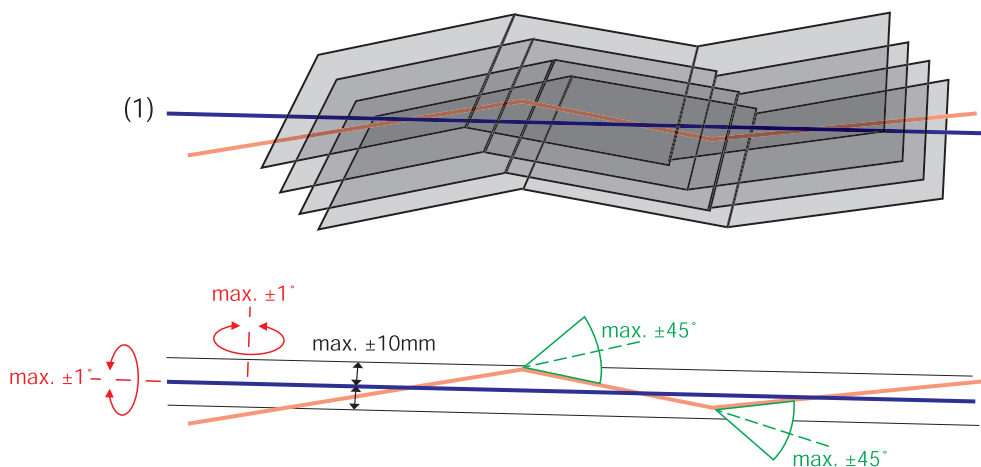
Before images are composed, they may have to be geometrically corrected. The images are first checked to see if they are already corrected.

If they are not, they are corrected now. In this case, a relevant message is shown in the status bar.

The corrected images are stored each time in a new series.

Resulting data

The spatial deviation of a pixel in the composed image may deviate by max. 10 mm from the resulting slice plane. The resulting slice plane is computed for opposing slabs in a way that the criteria is maintained as far as possible.



(1) Resulting slice plane

- ❑ MR images are generated if this criterion is met. The composed images may be used to plan additional measurements.
- ❑ If this criterion is not met, SC (Secondary Capture) images are created. SC images may not be used to plan additional measurements. In the image text they are characterized by the letters SC.
- Due to the changed criterion for generating MR-images it is possible that a data set, used to compose MR images with **syngo MR 2004A, 2004V, 2005E, 2006T or LEONARDO 2004**, only allows you to create SC images with **syngo MR B15A**.

Process of reconstruction

After selecting images for reconstruction, the software of the **Composing** task card searches for similar structures in the overlapping areas of individual images.

Automatic alignment

Individual images are shifted until the best alignment is obtained.

Manual alignment

In cases where transition areas have not been optimally reconstructed by the computer, you can adjust them manually. Proceed by shifting the image part below the transition edge relative to the image part above the transition edge until the best alignment is achieved in the overlapping area.

- The first time you perform manual alignment, a dialog window appears with a warning.

Original images

In certain cases, for example, when diagnosing bone structures, it will be necessary to use the original image information. To this end, you can use individual images, or as an alternative, a special display mode, the Cut mode (**Hard Image Transitions**).

In Cut mode, the images are joined at the cut lines without overlapping. This displays an overall image with the unchanged information of the individual images.

→ [Page L.3–15](#), *Switching between blend mode and cut mode*

Important Note

The reconstruction algorithms detect similar prominent features in the images to be joined and match them to one another.

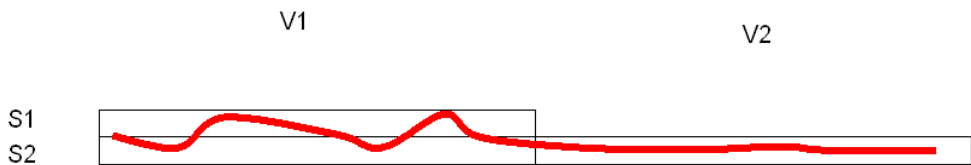


Before using a composed image to support a diagnosis, visually verify the image transitions using the original images with an appropriate resolution.

Independent of how reliable the transitions between individual images may appear, only a physician can ensure diagnostic accuracy using the image margins of the original image for visual inspection.

✧ Use the **View > Single with Original** function to compare the composed image with the original images.

Same volume position	If volumes are selected that were acquired at the same position, reconstruction will not be performed.
Volumes do not overlap	If selecting volumes that do not overlap, "holes" will be present during reconstruction. Reconstruction is performed. Missing areas are shown in black in the composed images. ⇒ Image margins cannot be manually adjusted.
Incomplete slices	During distortion correction, areas may be added to the image where no data were measured. Reconstructions with a Spine algorithm <i>do not</i> include incomplete slices. However, reconstructions with an Angio algorithm do include incomplete slices. This is shown in the following display:



A blood vessel runs through volumes V1 and V2. V1 includes both slices S1 and S2. V2 only includes S2. Slice S1 is incomplete, because it is not included in V2. If the **Angio** algorithm uses the complete slice S2 only, the information from S1 would be missing.

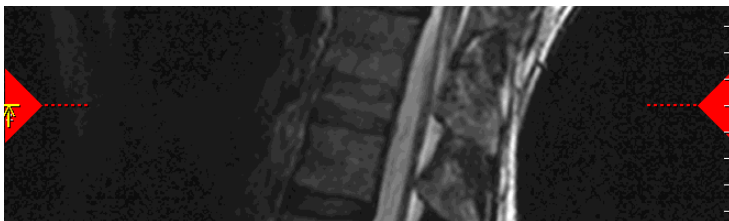
Too many images

If you select volumes with too many images, reconstruction will not be performed.

- ✧ Depending on the quantity of images, some of the images that were loaded may have to be unloaded before a new data set can be composed.

Transition does not match

If one or more transition areas between images are not detected satisfactorily by the reconstruction algorithm, a message appears after reconstruction.



- ✧ Transition areas that do not match exactly are marked with a red triangle each to the left and right. In addition, the text "image sections do not match" is shown at the bottom of the image.
- ✧ Check the transition areas marked.

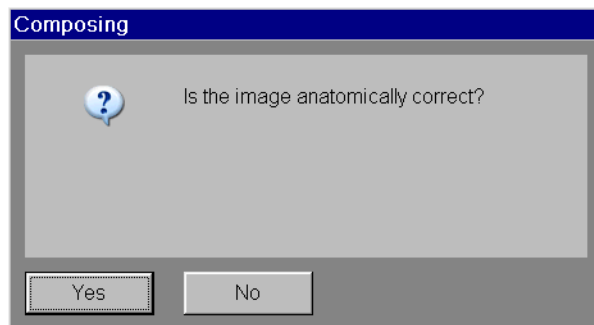


The note "image sections do not match!" is an image text that can be blanked out.

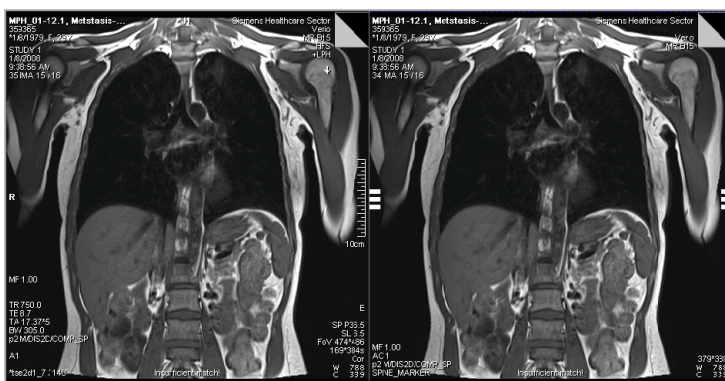
- ✧ Ensure that the image text is activated during diagnoses.
→ [Page L.3–22, Switching image text on/off](#)

Anatomically correct?

If it was not possible to match the image sections correctly (red triangles), or the image margin was manually corrected (**M**), a query appears prior to storage asking if the composition is anatomically correct.



- ✧ Click **Yes** or **No**.



(1)

(2)

- (1) When you click **Yes**, the red triangles and the note "Image sections do not match!" or the markers for manual adjustments (**M**) are removed.
- (2) In case of **No** the transition areas are marked with black and white stripes.



As the user, you are responsible for correct evaluation of the reconstruction and the diagnostic conclusions drawn from it.

Change in users

If the new user has access rights to the images loaded with **Composing**, the function remains available for use.

If a reconstruction is in progress at the time the users change places, **Composing** is blocked for the new user until reconstruction is ended.

If the new user does not have rights for the current patient, the images are automatically unloaded.

If **Composing** is blocked for the new user while reconstruction is in progress, the contents of the **Composing** task card are deleted.

The new user has to close **Composing** and start it again so that he can work with it.

Refer to the brochure *syngo* Security Package.

Reconstruction

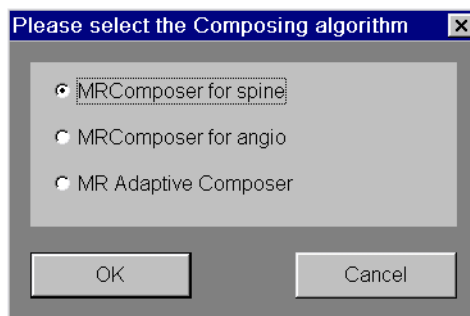
By accepting the series, the **Composing** task card moves into foreground.

If the input images are type Inline MIP, reconstruction starts automatically with the MIP algorithm.

Selecting the algorithm

Otherwise, a dialog window for selecting the algorithm is opened. You select one of the two following composing algorithms:

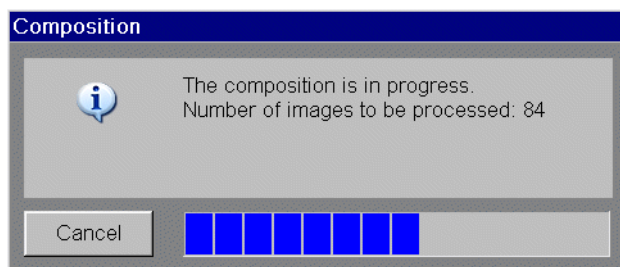
- ☐ The **Spine** algorithm is based on the bone structure of the images. Composed images of this kind may be measured and evaluated, e.g., scoliosis, kyphosis, pelvic obliquity.
- ☐ The **Angio** algorithm is based on the vessel structure in the images. This allows for overall display of the vessels.
- ☐ The **Adaptive** algorithm is based on elastic matching. It especially corrects B0 induced inhomogeneities in image areas beyond the magnetic homogeneity volume.



✧ Click the required algorithm and confirm with **OK**.

The reconstruction is started.

The **Composing** window shows the reconstruction process.



Canceling reconstruction

Reconstruction can be canceled if, for example, you have selected a series with a large number of images and the reconstruction process is taking too long:

✧ Click **Cancel**.

Reconstruction stops within the current process.

The images composed so far are displayed.

Adjusting reconstruction manually

Normally, all parameters required for image reconstruction are determined automatically.

Occasionally, the automatic algorithm does not provide optimum results. In this case, you may manually determine the position of the images relative to one another (pixel-by-pixel shifting).

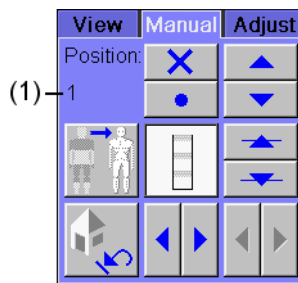
-
- ! By changing the reference parameters, the graphics shown in the composed image are no longer valid following manual adjustment. They have to be redrawn accordingly.
-

⇒ For a stored composed image that was loaded again, the position of individual images can be changed only, when the original images are still in the local database.

-
- ! As the user, you are responsible for correct evaluation of the reconstruction and the diagnostic conclusions drawn from it.
-

Tools

The tools for manual positioning are located under the **Manual** subtask card of the **Composing** task card.



(1) Position number (position of the orientation label)

Selecting the image position to be shifted

First, select the image in the series you want to pan:

- ✧ Double-click the required position in the *overview image* (on the right).
→ [Page L.3-2, The Composing task card](#)



The position of the image margin to be corrected is shown via an orientation label in the left margin of the *detail image* (on the left).

The position number is displayed.

→ [Page L.3-4, Displays in images](#)

Setting the number of pixels to be shifted

For subsequent corrections you can set the number of pixels per mouse click in the **Image** menu.

You can use one of the following values: 1, 3, 5, 10, 15, 20, 30, 40, 50, 100, 200 pixels

When you start the system, 3 pixels per mouse click is the default setting.

✧ Select the required number of pixels in **Image > Manual Adjustment**.

The setting remains the same until you change it again or restart the application.

Shifting the images relative to each other in a slice

You can now move the image identified with the orientation mark. This image as well as all subsequent images are moved relative to the images above the marked image.

✧ The top image cannot be moved.

-
- ! The shift should be performed in Cut mode (**hard image transitions**) because pixel-by-pixel shifting is easier to see in this mode.
→ [Page L.3–15](#), *Switching between blend mode and cut mode*
-

Moving up/down



- ✧ Click the up or down arrow.

The images below are moved up or down by the set number of pixels.

Moving left/right



- ✧ Click the right or left arrow.

The images below are moved to the right or left by the set number of pixels.

Moving a cut line



- ✧ Click the up or down arrow with the line symbol.

The cut-line (image margin) is moved up or down by the set number of pixels.

Shifting slices relative to each other

You can move from slice to slice in the composed volume, maintaining the composition.

Selecting a slice

- ✧ Use the dog ears and scroll to the required slice.
 - Or -
 - Press the **Image - (Num 1)** or **Image + (Num 2)** on the keyboard.

Shifting the image



- ✧ Click the end of the arrow (cross) or arrowhead (circle) once or repeatedly.

The slice is shifted to the front or the back.

The composed slice images affected are recalculated. Composed slice images that are not affected remain unchanged.

Normalizing image brightness

You can normalize the brightness impression of image sections for volumes included in the composed volume (caused by measurements performed with different coils).



✧ Click this button.

Or

Select **Image > Normalize**.

Brightness of the individual image sections is normalized for the current composed volume.

✧ During manual correction **Normalize** is automatically switched off again.

Display



When a single image has been moved manually, it is identified with an "M" as a composed image. At the bottom of the image "Performed manual adjustment" is superposed.



Please note that after a manual correction in **Composing** while you determine the film layout, **All** is always selected for the **image text** selection, so that the note "manual adjustment performed" is exposed to film.

→ **Basics/Image text** in the *syngo* Operator Manual

Returning to automatic positioning

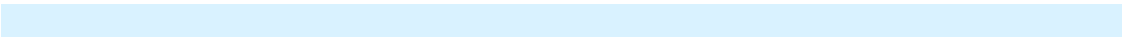
✧ Select **Auto Adjustment** from the context menu.

- Or -

Click this button.



All manual positioning is canceled and the status after automatic reconstruction is restored.



CHAPTER L.3

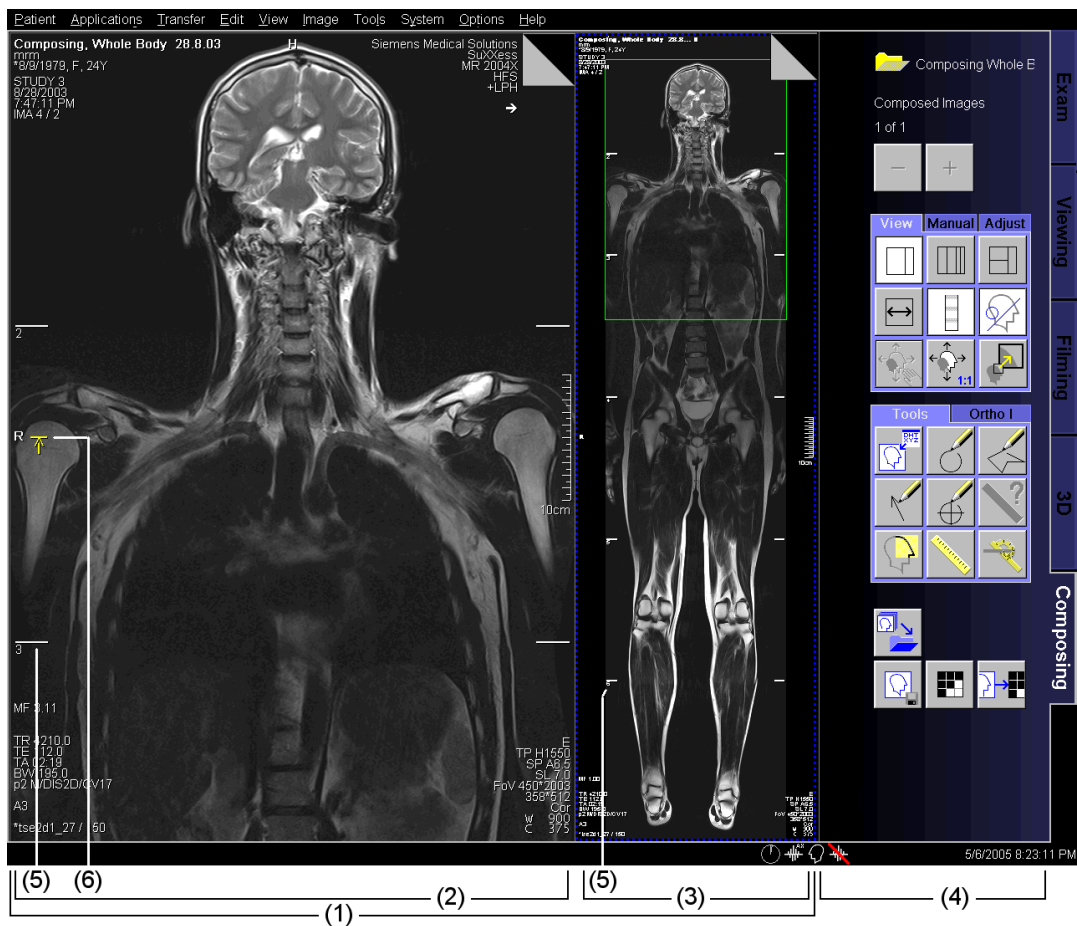
Viewing Images

The **Composing** task card allows you to view and post-process *composite images* after reconstruction.

You can switch between different layouts and scroll among several composed images and slices.

You can load and view images that have already been composed and stored. (Manual alignment is possible only if all original images are available.)

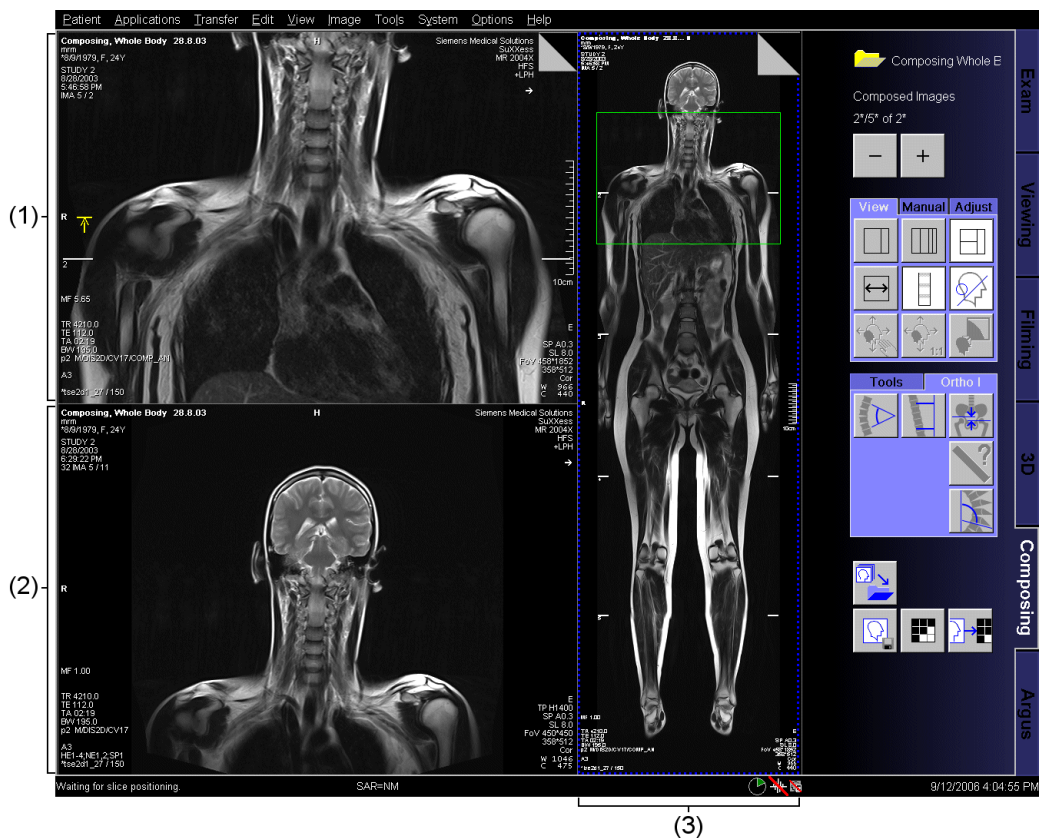
The Composing task card



- (1) Image area
- (2) Detailed image
- (3) Overview image
- (4) Control area
- (5) Section markers and image position numbers
- (6) Orientation mark

Image segments

The image area of the **Composing** task card can be divided into different segments which can be switched on or off.



- (1) Detail image segment: Display of composed image
- (2) Original image segment: Display of the corresponding original image
- (3) Overview image segment: Overview of the entire composed image

Displays in images

The following special displays appear in the image area for the **Composing** task card:



Dog ears for scrolling through the image stack



Image position numbers



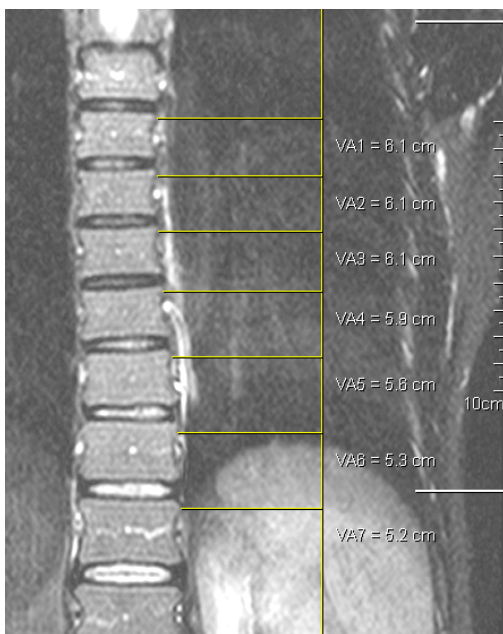
Image position number with orientation mark



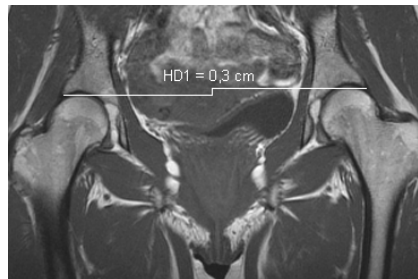
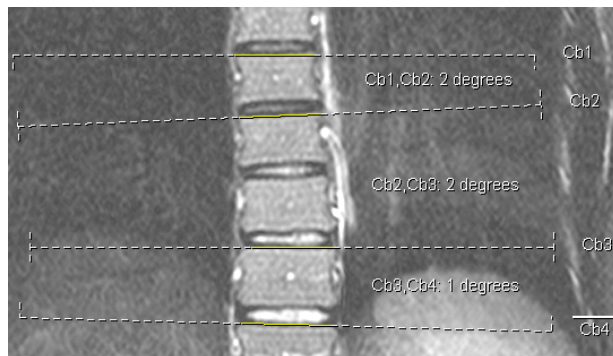
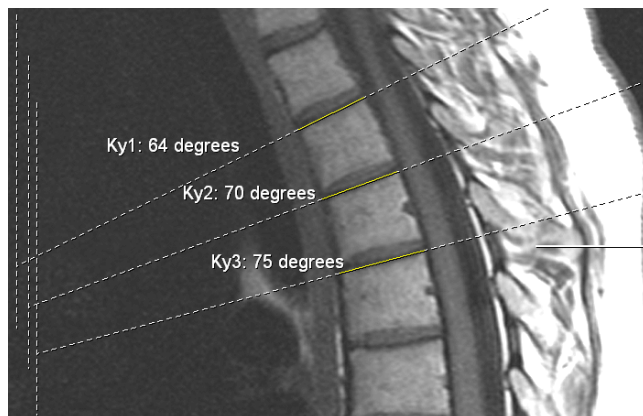
Image position number identified with "Manual"



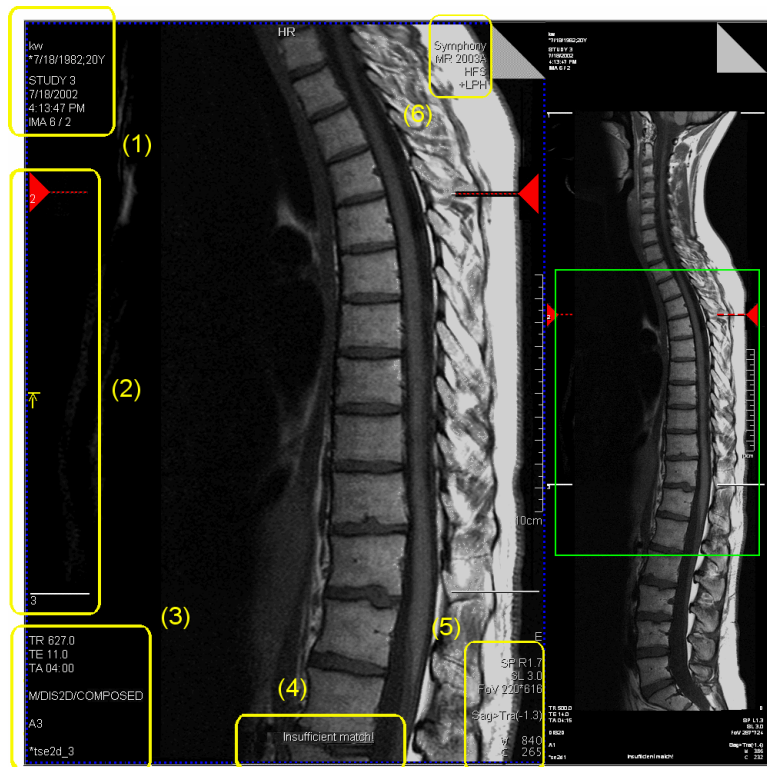
Image position number with identification "image segments do not match!"



Vertical Distance measurement

**Height Difference****Cobb angle****Kyphosis angle**

Text information in images



- (1) Patient and examination data
- (2) Image position numbers with orientation mark and identification
- (3) Examination and image parameters
- (4) Notes
- (5) Slice information and window values
- (6) System-specific information

Loading and displaying images

For viewing and post-processing, load the series and images into the **Composing** task card.

- ✧ Select **Patient > Browser**.
- ✧ Select the requested patient/requested study.
→ **Patient Browser**/Searching for and selecting patient data in the syngo Operator Manual
- ✧ Click the tab of the **Composing** task card.

The **Composing** task card is placed in foreground.



You can only load images of *one* patient into the **Composing** task card.

Images of *one* series block only may be loaded per load operation (same frame of reference).

If you load images of another patient, the current patient is automatically closed if you confirmed the message accordingly.

Loading images for reconstruction

- ✧ Select the series in the navigation or content area of the **Patient Browser**.
→ [Page L.2–1](#), *Selecting the series and starting reconstruction*

Loading a composite image/volume or single images/volumes

You are able to load a composed as well as saved image/volume or just individual images/volumes in the **Composing** task card.



In the case of a composed image/volume that has been stored and reloaded, you can change the position of single images relative to each other *only if*:

- ☐ All original images are available
- ☐ The images are not the result of a canceled reconstruction
- ☐ The composed image or the original images were not changed outside **Composing** or provided with an electronic shutter.

Loading a volume

- ✧ Select the required series in the content area of the **Patient Browser** which already contains composed images.
- ✧ Select **Applications > Composing** in the menu of the **Patient Browser**.
 - Or, in case **Composing** is open -Double-click the series containing composed images.

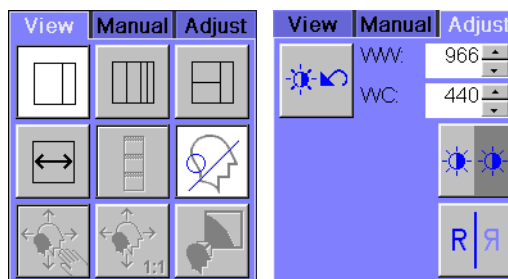
The composed volume is loaded into the **Composing** task card.
- ✧ If you double-click a series that does not contain composed images, only this particular series is loaded.

Display after loading

For a volume that has just been composed, the center image of the result volume is shown. The focus is in the center volume.

An already composite image is displayed as it was when it was last saved.

Changing image display



The tools are located in the **View** and **Window** subtask cards as well as in the **View** menu.

→ [Page L.2–5, Process of reconstruction](#)

→ [Page L.2–6, Important Note](#)

Effect of functions

Some functions affect the selected image segment (focus) only, in particular in the **Single with Original** image view; other functions affect all image segments.

Function	Only affects the selected image segment	Affects all image segments
Windowing	x	
Reset window values		x
Inverting grayscale values		x
Flip vertically		x
Zoom/Pan	x	
Zoom factor 2.0 or Zoom factor 0.5	x	
Acquired image size	x	
Determining the area	x	
Home Zoom/Pan		x

Views

Detailed image or comparison



The **Composing** task card provides you with three different kinds of image displays (no film layouts):

☐ **Single Layout** (default setting)

Only the composed image is displayed. In this display you will see a composed image, if necessary, with an overview reduced in size.

→ [Page L.3–12, Selecting single layout](#)



☐ **Single with Original**

The composed image and the original image with the original image information (if available) are displayed one below the other. This display mode can be used for checking reconstruction and performing diagnoses.

→ [Page L.3–13](#), *Select single with original image view*



☐ **Double Layout**

Two composed images are displayed next to each other. This display can be used to compare composed images/volume, for example, to assess the course of an illness.

→ [Page L.3–14](#), *Selecting double layout*

Full image or overview



In each case you can switch between:

☐ **Full image display**

Only the detail image or the detail image with the original image in maximum width is displayed. The overview is hidden.



☐ **Overview** (default setting)

In addition to the detailed/original image(s), one or two overviews of the composed images is/are displayed. In this display you can navigate easily with the support of the overviews.

Switching between full image and overview

- ✧ To display (hide) the overview image, deselect/select **View > Hide Overview**.

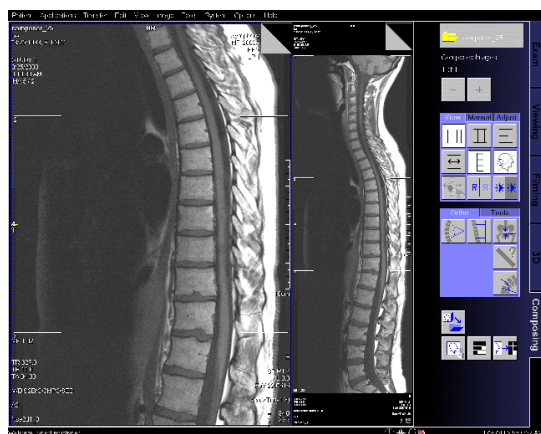
- Or -

Click the **View** subtask card and select/deselect this button.

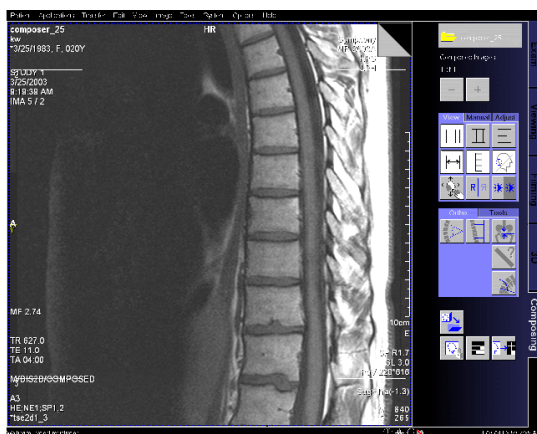


Selecting single layout

If you just want to view only one composed image of a volume, select **Single Layout**.



(1)



(1)

Single Layout with/without overview display

- (1) Detailed image
- (2) Overview image

- ✧ Select **View > Single Layout....**

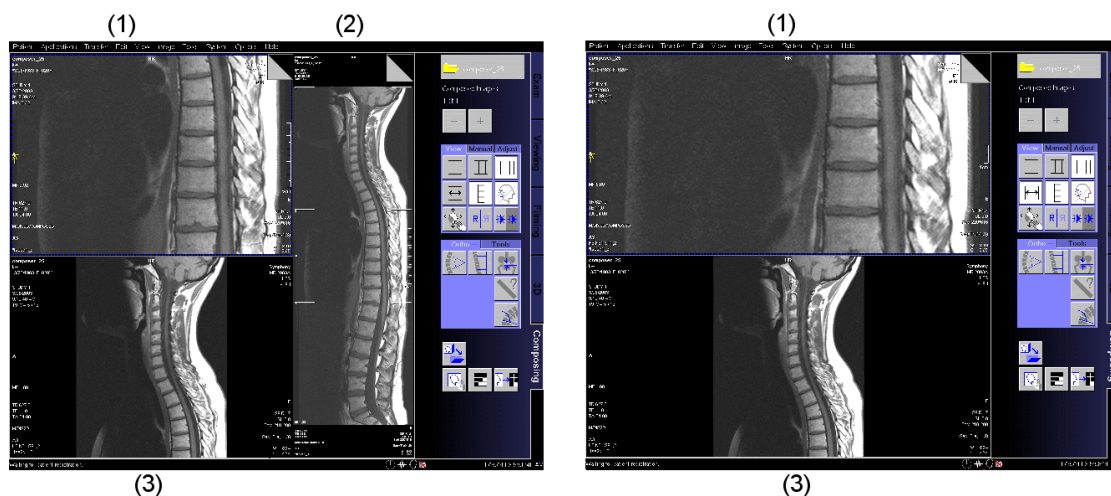
- Or -

Click this button on the **View** subtask card.



Select single with original image view

The original image should be used for diagnosis. For this purpose, the composed image and an original image can be displayed simultaneously and compared. Select the image view **Single with Original**.



Single with Original with/without overview display

- (1) Detailed image
- (2) Overview image
- (3) Original image

✧ Select **View > Single with Original**.

- Or -

Click this button on the **View** subtask card.

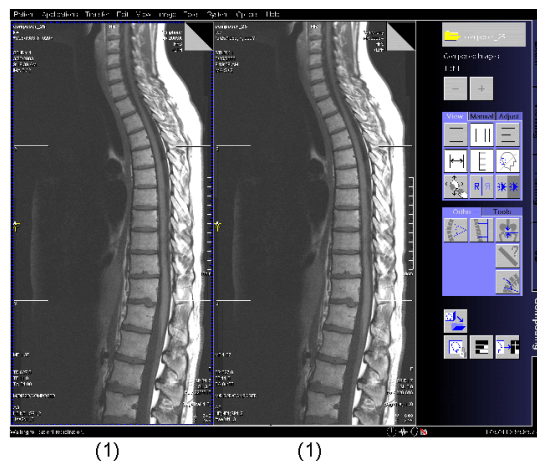
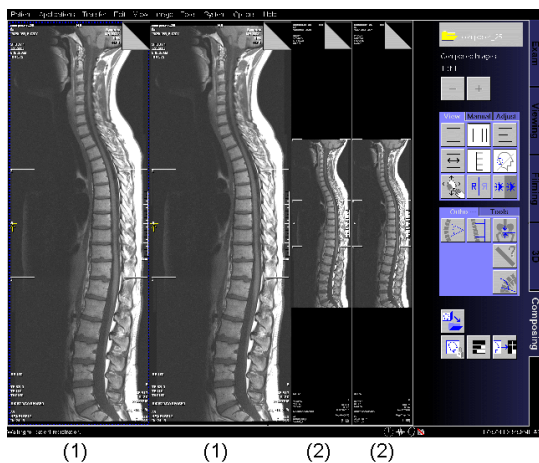


Selecting double layout

When you want to compare two composed images (of two volumes), select **Double Layout**.

Prerequisite:

Several volumes have been composed and loaded.



Double layout with/without overview display

- (1) Detailed image
- (2) Overview image

✦ Select **View > Double Layout**.

- Or -

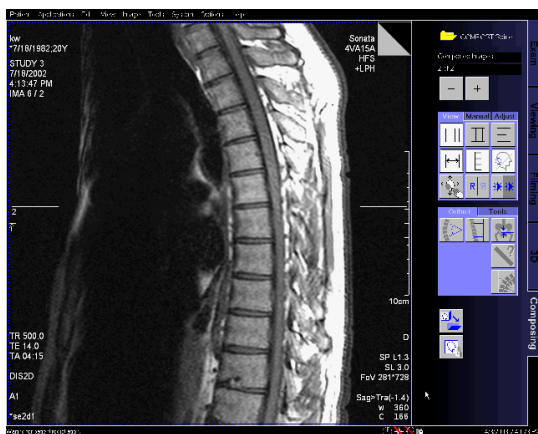
Click this button on the **View** subtask card.



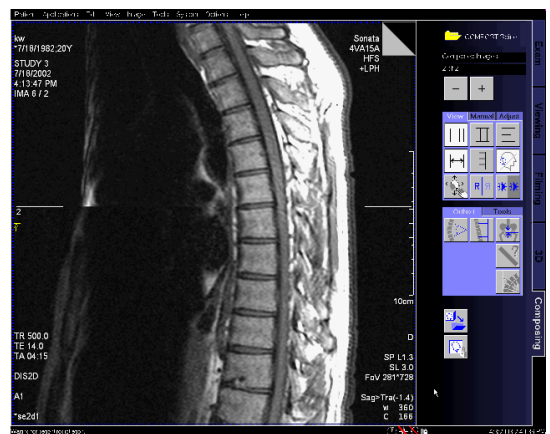
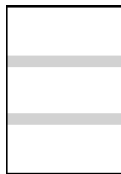
✦ If only one composed volume is loaded, a composed image is displayed twice. In this case, you cannot use all functions in the right image.

Switching between blend mode and cut mode

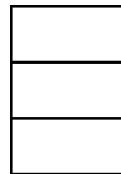
Usually, composed images are shown with "soft" transitions. For diagnostic purposes you can use images with "hard" cuts and display the original image information as well:



Blend mode (soft image transitions; default): The single images are blended at the transitions between each pair of images.



Cut (hard image transitions): The single images are joined without blending.



Switching blend mode on/off



⇒ Image transitions for a saved and reloaded composed image can be changed only if the original images are still available.

⇒ To switch on (off) soft image transitions, select/deselect **View > Soft transitions**.

- Or -

Click the **View** subtask card and select/deselect this button.

The display changes.

⇒ If the reconstruction algorithm does not identify sufficiently similar features at an image transition, a warning will be displayed, and the image transition will be marked.

⇒ The images can be repositioned manually.



When **Soft transition** mode is used, there are transition areas between the single images that were interpolated from two images. Although an image in the **Soft transition** display appears more pleasing to the eye, *only* original images (**single images with original image view** or **hard transitions**) may be used for *diagnosing* bone and vessel structures.

Scrolling through composite volumes

If several volumes have been composed and loaded, you can scroll through them one by one.

✧ Select the required image segment in the **double layout**.

✧ Click – or +.

- Or -

Press the **Study - (Num 7)** or **Study + (Num 8)** on the keyboard.

The images of the previous or next composed volume are displayed.

⇨ In overview mode, the overview also changes.



Scrolling through individual images

You can also "scroll" among single volumes within a reconstructed volume.

✧ Click the required image segment.

This segment is identified with a blue (light) border.

✧ Press the **Series - (Num 4)** or **Series + (Num 5)** on the keyboard.

The displayed image is moved to the next/previous partial image/volume.

⇨ In overview mode, the overview also changes.



Scrolling in the image stack

You can scroll through the image stack of composed images using the dog ears or icons.

→ **Viewing**/Changing image settings in the syngo Operator Manual

Selecting an image section / navigating in the detailed image

There are several ways of navigating in the composed detailed image:

- ☐ Shifting the image position
- ☐ Selecting / changing an image section
- ☐ Continuously zooming/panning an image
→ **Basics**/Zooming and panning images in the syngo Operator Manual
- ☐ Enlarging or reducing the image by a factor of 2
- ☐ Setting 1:1 pixel display

Shifting the image position



You determine the image position by setting an orientation marker in the detailed image.

- ✧ Double-click the required position *in the overview image*.

The position is marked on the left with the orientation marker in the *detailed image*.

The detailed image is shifted so that the marked position is in the center in the vertical direction.

If applicable, the corresponding original image is displayed in the lower part.

Or

- ✧ Press one of the following keys on the keyboard.

Page up: The orientation marker is moved to the next lower image number.

Page down: The orientation marker is moved to the next higher image number.

Home: The orientation marker is moved to the first image number.

End: The orientation marker is moved to the last image number.

The detailed image is shifted accordingly.

Selecting an image section



You can also select an image section (ROI).

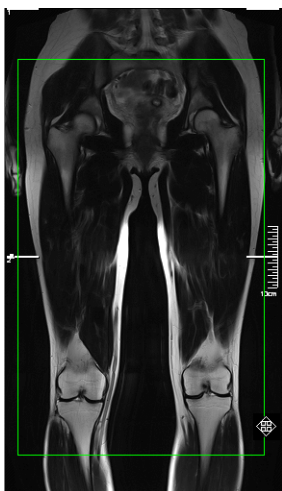
- ✧ Click this button.
Or
Select **Tools > Capture Area** in the main menu.

- ✧ Click and draw a frame in the detailed image.

The *detailed image* is enlarged according to the image section and centered.

The image section is shown in the *overview image*.

Modifying the image section



In the *overview image* (right), the portion of the image shown in the detailed image (left) is marked with a green border.

You can change this border and therefore the image section.

Moving an image section:

- ✧ Move the mouse pointer to one of the four sides of the green border.
- ✧ Drag the green border to the desired position.

Resizing the image segment:

- ✧ Move the mouse pointer to one of the four corners of the green border.
- ✧ Enlarge or reduce the size of the green border.

The image section shown in the detailed image is modified accordingly.

Enlarging or reducing the image by a factor of 2

You can also enlarge or reduce the image step-by-step (each time by a factor of two).

- ✧ Go to the main menu and select **Image > Zoom in by 2.0** or **Zoom in by 0.5** in the main menu.

The *detailed image* is enlarged/reduced by a factor of 2.

The *overview image* does not change.

Setting 1:1 pixel display



To achieve optimal image display without interpolation, you can define that each image pixel is assigned a monitor pixel.

- ✧ Click this button.

The *detailed image* is magnified or reduced so that every image pixel corresponds to a monitor pixel.

Resetting the image display

You can reset the image to the default setting for zoom and position.

- ✧ Select **Image > Home Zoom/Pan** in the main menu.

The original image display is restored.

Switching image text on/off

The *image text* contains the following information:

- ❑ Text input at the four corners of the image (patient data, etc.)
- ❑ System messages in the lower image center (e.g., "Image sections do not match")

This text can be hidden if, for example, you want to film an image anonymously.

! If you film a composed image with the image text and/or graphics switched off, non-matching or manually adjusted transitions between the individual images cannot be identified as such!

- ✧ Switch on image text and graphics for diagnostic applications.

-
- ✧ Select/deselect **View > Image Text On**.

The image text is switched on or off accordingly.

Activating/deactivating graphic elements

You are able to temporarily hide graphic elements (lines, circles...), if, e.g., you would like to view details underneath them.

This includes:

- ☐ All graphic elements drawn (lines, circles...)
- ☐ The image segment markings and image position numbers
- ☐ Special identifications for transitions (e.g., red triangles for non-matching transitions or "M" for transitions set manually)

Activating/deactivating graphic elements



- ✧ To switch on (off) graphic elements, select **View > Image Graphics On**.

- Or -

Click the **View** subtask card and select/deselect this button.

Image display/windowing

Set the windowing values in the **Composing** task card.

- ⇒ During scrolling, windowing values are automatically applied to other composed images.

Windowing with the mouse or keyboard

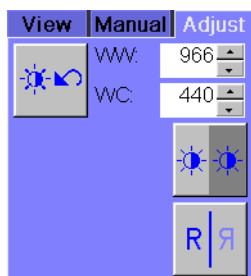
Details for windowing with the mouse or the keyboard are included in the *syngo* Operator Manual.

→ **Basics/Windowing images**

Setting the windowing values numerically

Set the window values numerically by using the **Window** sub-task card:

- ✧ Enter the desired window values with the keyboard into the **WW** (= window width) and **WC** (= window center) fields.
- Or -
Set the required values using the arrow keys.



Resetting the window values



- ✧ Click this button to reset the window values to the values last saved.

Additional display modes

Function	Button	Description
Mirroring an image		→ Viewing /Changing image settings in the syngo Operator Manual
Inverting grayscale values		→ Viewing /Changing image settings in the syngo Operator Manual
Editing graphics	→ Viewing /2D evaluation in the syngo Operator Manual	

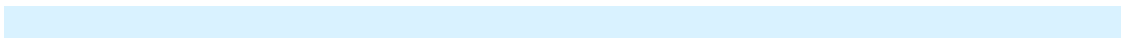


Image is mirrored

Confusing left/right leads to incorrect diagnosis!

The examiner is responsible for correct application of functions and/or interpretation of the images as well as the consequences resulting from his actions.

✧ Minimize the risk of incorrect interpretations by e.g., providing notes/comments.



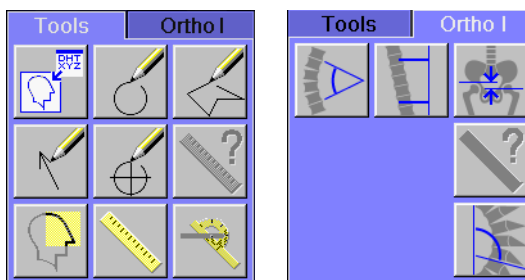
CHAPTER L.4

Evaluations

The **Composing** task card allows you to perform evaluations and apply post-processing functions.

Tools

The tools are located on the **Tools** and **Ortho I** subtask card as well as in the **Tools** menu.



Important notes

Distance measurements

When performing distance measurements, please keep the following in mind:

- ☐ Distance measurements made across image boundaries in composed images (in the vertical direction on the monitor) are inaccurate. This error is additive across the sum of the original images.
- ☐ Digits appearing after the decimal point in the measurement results are *not* representative of the accuracy of the measurement taken. When using a measurement to support diagnosis, the physician has to consider all sources of errors.



CAUTION

Distance measurements across image boundaries of combined images are error-prone. This is an accumulated error depending on the number of original images!

Incorrect distance measurement

- ✧ Perform measurements only in the original images, if possible.
- ✧ Use only the original images for diagnostic purposes.
- ✧ Do not use geometrical measurement values as a sole basis for diagnosis. Do not perform measurements across image borders marked as Insufficient match!

Angle measurements

When performing angle measurements, please keep the following in mind:

- ❑ The main sources for error when determining Cobb and Kyphosis angles are the *subjective positioning of the tangent* to the vertebrae and the *position* of the patient's body.
- ❑ Digits appearing after the decimal point in the measurement results are *not* representative of the accuracy of the measurement taken.
- ❑ The accuracy of Cobb angle or kyphosis angle measurements is comparable with measurements made by hand of filmed images.
- ❑ In the case of angle measurements on projection images, the projected plane must be parallel to the angle plane.

Graphic functions

Availability of graphic functions

Graphic functions are still available when red triangles are shown in the images.

→ [Page L.2–8](#), *Transition does not match*

However, measurements across image transitions (e.g., distance measurements) should be performed only when all image sections fit together (no red triangles).



Graphic elements may be drawn in the detailed image only, not in the overview.

The functions described in what follows can be selected via the buttons on the **Tools** subtask card as well as via the **Measurement** menu.

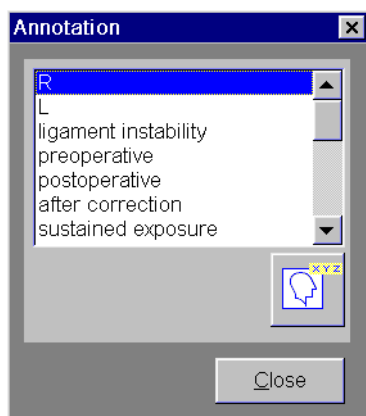
Annotating images (comments)

You can enter predefined text or freely formulated texts on the **Composing** task card.

✧ Click this button on the **Tools** subtask card.

The **Annotation** dialog window opens.





Using a predefined text

- ✧ Mark the desired sequence in the list.
- ✧ Click the required position in the image.
The text is inserted immediately.

Entering free text



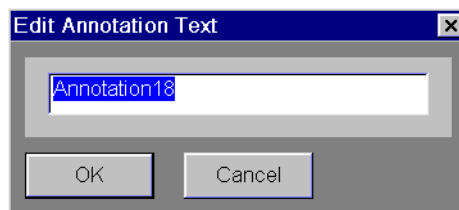
- ✧ Click this key in the **Annotation** dialog window.
- ✧ Click the required position in the image.
A text field is displayed.
- ✧ Enter the text.
- ✧ Press the **Enter** key.

Defining text

20 predefined texts are available. You can change the texts at any time:

- ✧ Double-click the entry you want to change in the **Annotation** dialog window.

The **Edit Annotation Text** dialog window opens.



- ✧ Overwrite the text as required.
- ✧ To apply new text, click **OK**.

Positioning text



- ✧ Click the text.

The text is identified by small squares.

- ✧ Move the text to the desired position.

Editing text



- ✧ Double-click the text.

The text is applied to an input field.

- ✧ Change or supplement the text.
- ✧ Press the **Enter** key.

Deleting text



- ✧ Click the text.

The text is identified by small squares.

- ✧ To remove the text, press the **Del** key.

→ **Viewing/2D evaluation** in the *syngo* Operator Manual

Ending text input, switching to another image

- ✧ When you have completed adding comments, close the **Annotation** dialog window.

Or

- ✧ Select the next image where you would like to add annotations and continue with your work.

Switching text on/off

→ [Page L.3–22](#), *Switching image text on/off*

Drawing circles and lines/polygons

The **Composing** task card lets you draw lines/polygons the same way circular and freehand ROIs are drawn on the **Viewing** task card. However, this is not associated with statistical evaluations.

Drawing function	Button	Description / cross-reference
Drawing circles		→ <i>Basics/Evaluating regions</i> in the syngo Operator Manual
Drawing circles with a center		You draw circles with a center the same way you draw other circles. Using this method you can easily perform distance measurements starting from the center of the circle, e.g., the center of a ball-and-socket joint. As compared to the circle line, you are not able to select the "crosshair" in the circle.
Drawing lines/polygons		→ <i>Basics/Evaluating regions</i> in the syngo Operator Manual

Drawing arrows

You can emphasize image details with arrows:



- ✧ Click the button.
- ✧ Click the point in the image where you want to position the arrowhead, and drag the mouse to the end point of the arrow to create the line.

The arrow is drawn.

Changing the size

- ✧ To change the size of the arrow, drag one of the two arrow end points with the mouse.

Additional graphic functions

Graphic function	Button	Description / cross-reference
Shutter		—
Distance line		→ Basics /2D- and 3D evaluation in the syngo Operator Manual
Angle		→ Viewing /Measuring angles in the syngo Operator Manual
Switching graphics on/off		→ Page L.3–23 , Activating/deactivating graphic elements
Editing graphics	→ Viewing /2D evaluation in the syngo Operator Manual	

Evaluations of the spine and pelvis

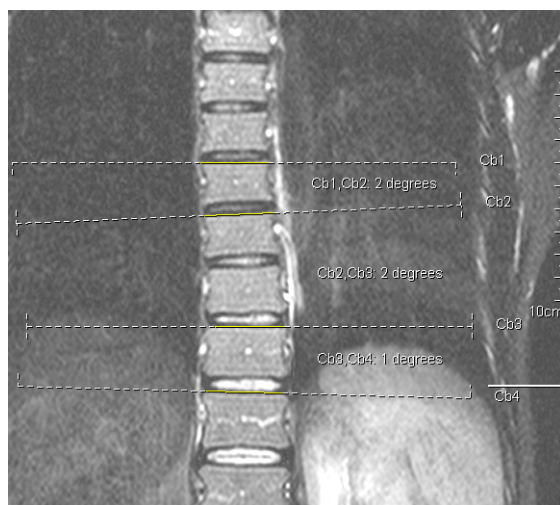
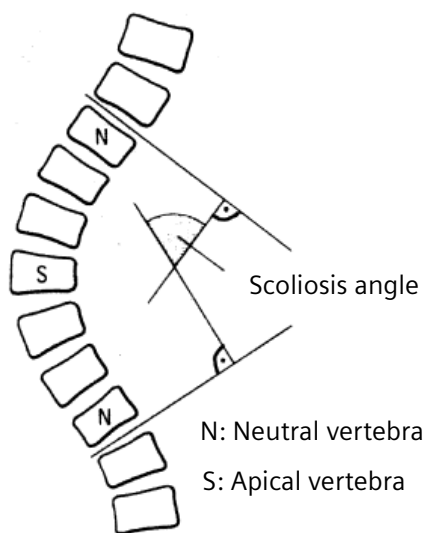
Depending on the reconstruction used, you are able to perform the following evaluation with composed images:

Tool	Evaluation
	Scoliosis angle according to Cobb: Measuring one or several curvatures of the spine. → Page L.4–10, Determining the curvature of the spine (scoliosis angle according to Cobb) .
	Kyphosis angle: Measuring the angle of the basic spine tangent relative to the vertical. → Page L.4–14, Determining kyphosis angles of vertebrae
	Vertical distance measurement: Measuring the spine's deviation from the vertical. → Page L.4–16, Measuring the deviation of the spine from the vertical
	Difference in height: Measuring the difference in height, e.g., between the left and right side of the pelvis. → Page L.4–20, Measuring the difference in height

Determining the curvature of the spine (scoliosis angle according to Cobb)

There are different methods for determining the scoliosis angle of the spine. The most widely accepted is the method according to Cobb.

This method involves drawing the tangent to the dorsal plate of the upper neutral vertebra and to the basal plate of the lower neutral vertebra. From these, the system calculates the scoliosis angle:



Scoliosis angle: Schematic representation¹ (left), example (right)

1. Zilch, H.; Weber, U.: Orthopädie mit Repetitorium. Berlin: de Gruyter, 1989

Drawing the scoliosis angle

One or several scoliosis angles may be drawn:

- ✧ Select the required image.
→ [Page L.3–7, Loading and displaying images](#)
- ✧ Select detailed display.
→ [Page L.3–9, Changing image display](#)
- ✧ You can only draw the angle tangents in a detailed image; however, the results are also displayed in the overview image.
- ✧ Select **Tools > COBB angle**.
- Or -
Click the button.



First tangent

- ✧ Move the mouse pointer into the detailed image.
The cursor changes shape.
- ✧ Drag a line from the starting point to the end point of the first tangent with the mouse.
The tangent is drawn.

Second tangent

- ✧ If necessary, move the image area by double-clicking in the overview (new position of the orientation mark).
- ✧ Draw the second tangent in the same way.
The tangents are marked according to their position, for example, **Cb1**, **Cb2**, etc.
The scoliosis angle is drawn in the vicinity of the apical vertebra.

Additional tangents

If there are additional curvatures, more angles can be defined:

- ✧ Draw additional tangents in the same way.

In each case, scoliosis angles are drawn in the vicinity of the apical vertebrae.

Changing the scoliosis angle drawn

Previously drawn angles can be changed at any time:

Changing angles or tangents

- ✧ Click on the required tangent.

The tangent is marked with small squares at the corner points.

- ✧ Drag an end point to the new position with the mouse.

The relevant scoliosis angles are updated.

Moving the annotation

- ✧ Click the annotation to be moved.

The characters are marked with a small square.

- ✧ Drag the annotation to the desired new position.

Removing angles

To remove an angle, delete a tangent:

- ✧ Click on the required tangent.

The tangent is marked with small squares at the corner points.

- ✧ Select **Edit > Delete Selected Graphics**.

- Or -

Press the **Del** key.

If it was the lowest or topmost tangent, the lowest or topmost scoliosis angle is deleted.

If it was the middle tangent, the scoliosis angles above and below it are deleted and a new angle is drawn instead.

If it was one of two tangents, both tangents including the angle will be deleted.

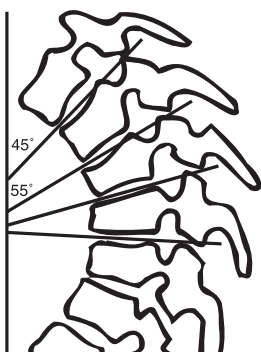
Ending editing

- ✧ Select another function.

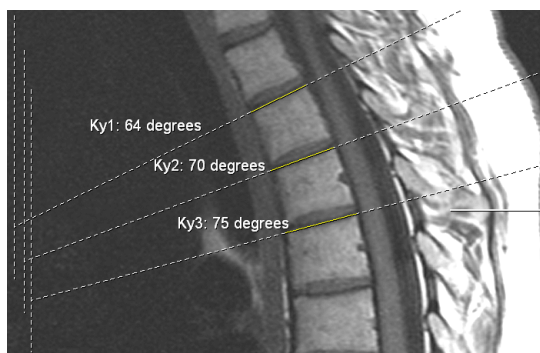
If only one tangent (but no angle) is drawn, it is deleted.

Determining kyphosis angles of vertebrae

A kyphosis angle is the angle between a vertebra and the vertical in a lateral acquisition:



Kyphosis angle: Schematic representation



Example

Drawing kyphosis angles

You can draw one or several kyphosis angles:

- ✧ Select the required image.
→ [Page L.3–7, Loading and displaying images](#)
- ✧ Select detailed display.
→ [Page L.3–9, Changing image display](#)
- ☞ You can only draw the angle tangents in a detailed image; however, the results are also displayed in the overview image.



✧ Select **Tools > Kyphosis**.

- Or -

Click this button.

Drawing an angle

✧ Drag a line from the first point to the second point on the vertebra with the mouse.

The line is extended. A vertical line and a Kyphosis angle are drawn.

Changing the scoliosis angle drawn

Changing angles or lines

✧ Click on the required line.

The line is marked with small squares at the end points.

✧ Drag one of the end points to the position desired.

The relevant kyphosis angle is updated.

Removing angles

To remove an angle, delete the associated line:

✧ Click on the required line.

The line is marked with small squares at the end points.

✧ Select **Edit > Delete Selected Graphics**.

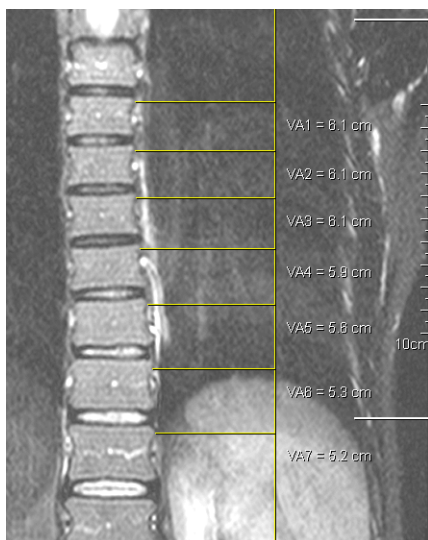
- Or -

Press the **Del** key.

The line and the associated kyphosis angle are deleted.

Measuring the deviation of the spine from the vertical

An additional tool for evaluating scoliosis is to determine the deviation of the spine from an imaginary vertical line.



Example

Drawing lines

- ✧ Select the required image.
→ [Page L.3–7, Loading and displaying images](#)
- ✧ Select detailed display.
→ [Page L.3–9, Changing image display](#)
- ✧ Select **Tools > Vertical Alignment**.
- Or -
Click the button.



Vertical line



As a first step, determine the line to be used for measuring the deviation from the vertical line:

- ✧ Move the mouse pointer into the detailed image.
The cursor changes shape.
- ✧ Click the image and drag a short line up or down with the mouse.
As soon as you move the mouse, a line is displayed.
- ✧ Release the mouse button.
A vertical line is drawn through the entire image.

Reference point

As soon as you have drawn the vertical line, draw distances to this line:

- ✧ Use the mouse to draw a short line from the desired point toward the vertical line.

A plumb line is dropped to the vertical line and the length of the plumb line is shown in cm.

Changing drawn lines

Changing a line

- ✧ Click on the required line.

The line is marked with a small square at the end point.

- ✧ Drag the end point to the new position.

The distances are updated accordingly.

Removing a line

- ✧ Click on the required line.

The line is marked with small squares at the end points.

- ✧ Select **Edit > Delete Selected Graphics**.

- Or -

Press the **Del** key.

The line and the given distance are deleted.

Moving the annotation

The annotation displayed (distance) can be moved if, for example, it hides important details:

- ✧ Click the annotation to be moved.

The characters are marked with a small square.

- ✧ Drag the annotation to the desired new position.

Moving a vertical line

- ✧ Click the line.

The line is marked with small squares at the end points.

- ✧ Drag the line to the desired position.

All distances are updated.

Removing entire graphics

- ✧ Click the vertical line.

The vertical line is marked with small squares at the end points.

- ✧ Select **Edit > Delete Selected Graphics**.

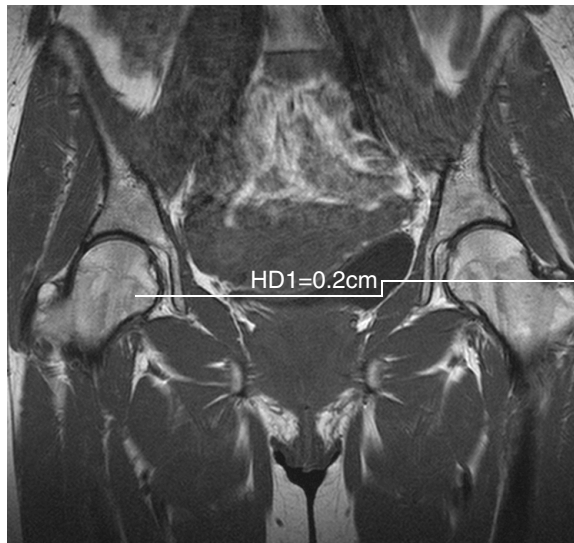
- Or -

Press the **Del** key.

The lines as well as all distances are deleted.

Measuring the difference in height

The system offers a difference in height measurement which can be used for measuring the difference in height between two points.



Example of difference in height measurement

Drawing the difference in height

You can draw one or several difference(s) in height:

- ✧ Select the required image.
→ [Page L.3–7, Loading and displaying images](#)
- ✧ Select detailed display.
→ [Page L.3–9, Changing image display](#)
- ✧ Select **Tools > Height Difference**.
- Or -
Click this button.



- ✧ Move the mouse pointer into the detailed image.
The cursor changes shape.
- ✧ Position the mouse pointer on the first reference point.
- ✧ Drag a line from the first reference point to the second reference point with the mouse.
The line indicating the difference in height as well as the actual difference in cm are displayed.

Changing the drawn differences in height

Modifying the line indicating the difference in height

- ✧ Click on the required line.
The line is marked with small squares at the end points.
- ✧ Drag one of the end points to the position desired.
The relevant line is updated.

Moving the annotation

- You can move the displayed annotation (height difference):
- ✧ Click the annotation to be moved.
The characters are marked with a small square.
 - ✧ Drag the annotation to the desired new position.

Removing difference in height

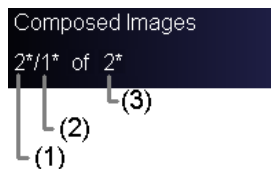
- ✧ Click the required line indicating the difference in height.
The line is marked with small squares at the end points.
- ✧ Select **Edit > Delete Selected Graphics**.
- Or -
Press the **Del** key.
The line and the given difference in height are deleted.

The following functions are available under **Composing** for documenting composed images:

- ☐ Save to database
- ☐ Transfer to the virtual film sheet
- ☐ Sending to other workstations
- The **Copy Image to Clipboard** function cannot be used at a *syngo* Acquisition Workplace or a *syngo* MR Workplace. It is reserved for the **Composing** version at the *syngo* MultiModality Workplace.

Display of changes

If an image was newly composed or composed images were changed, this is indicated by an asterisk * above the – and + keys.



- (1) Current volume
* means: At least one image of the current volume was changed.
- (2) Current image
* means: The image currently displayed was changed.
- (3) Total number of the composed volume loaded
* means: At least one image of all volumes loaded was changed.

The following changes are taken into account:

- ❑ After a new composition
- ❑ Manual adjustment, including resetting the manual adjustment
- ❑ Mirroring
- ❑ Inverting
- ❑ Changes in the image text
- ❑ Changing graphic elements
- When performing **Save**, **Save As ...** or **Save All As ...**, the display is reset.

Saving images

To keep composed images with the graphics, they have to be *saved before* you select a new patient.

- When **Composing** is to be closed and data are loaded that have not been saved yet, a query appears asking the user whether the respective images should be stored or scrapped.



CAUTION

Missing identification of critical areas (for example, seams) in composed images (composer)!

Incorrect diagnosis

- ✧ When storing composed images, ensure that critical areas (seams) in images corrected manually or in images that have not been optimally reconstructed by the computer are marked.
- ✧ To mark critical areas, confirm in the pop-up window that the composed image does not correspond with the anatomy of the patient. The critical areas are not marked if you select in the pop-up window that the composed image corresponds with the anatomy of the patient.

Saving all

When you want to store all composed images of the current volume displayed:

- ✧ Select **Patient > Save All As**

Saving individually

When you want to save the currently displayed, composed image:

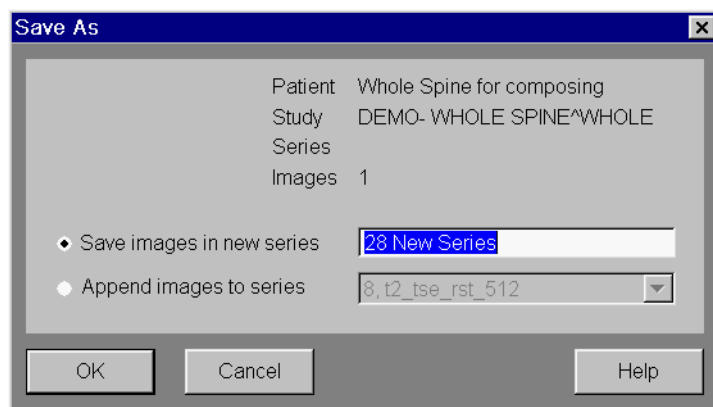
- ✧ Select the requested image of the volume.
 - [Page L.3–17, Scrolling through composite volumes](#)
 - [Page L.3–17, Scrolling through individual images](#)

- ✧ Select **Patient > Save**.

Only the currently displayed composed image is saved as a new series under the name **SPINE, ANGIO, or ADAPTIVE**.

Or

- ✧ Select **Patient > Save As**



- ✧ Select the requested save option and the name of the series.
→ **Viewing**/Saving images in the syngo Operator Manual

- ✧ Click **OK**.

The composed image currently displayed is saved with all its settings.

Filming images

The complete overview image (unzoomed) is always sent or filmed from the **Composing** task card.

- ✧ The 2x1 layout is very suitable for filming composed images.

For additional information, refer to the syngo Operator Manual.
→ **Filming**



Filming volumes

- ✧ Select the required volume.
→ [Page L.3–17](#), *Scrolling through composite volumes*
- ✧ Select **Patient > Copy to Film Sheet**.

Filming a single image

- ✧ Select the required image.
→ [Page L.3–18](#), *Scrolling in the image stack*

The image segment is given a colored (bright) border.



✧ Select **Patient > Copy to Film Sheet**.

- Or -

Click this button.

Only the currently displayed composed image is copied to the virtual filmsheet (copied with the **Fit to Segment** setting).

Filming all composed images

✧ Select **Patient > Film all....**

All composed images are copied to the virtual filmsheet (copied with the **Fit to Segment** setting).

Sending images

You are able to send composed images as you do with other applications.

Note

When the receiving system is an older **syngo MR** version or equipped with a different software platform than the transmitting system, you have to consider the following:

- ☐ Annotations cannot be changed after transmission.
- ☐ However, new annotations can be added.



When you loaded composed images into the **Viewing** task card, only the visible section is transmitted for non-square, zoomed images.

Complete images are sent from the **Composing** task card.



- ✧ Scroll to the required image.
 - [Page L.3–17, Scrolling through composite volumes](#)
 - [Page L.3–18, Scrolling in the image stack](#)

- ✧ To start the measurement, click this button.

For additional information, refer to the *syngo* Operator Manual.

→ **Patient Browser**/Storing and sending data

Closing the patient

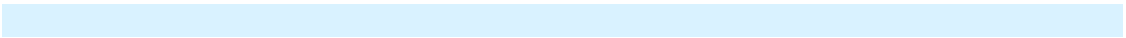


- ✧ Select **Patient > Close Patient**.

- Or -

Click this button.

- ✧ If you want to load images of another patient, the current patient is automatically closed if you confirm the message accordingly.



M.1 *Introduction*

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M.2 *Tissue 4D*

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Purpose

The aim of the **Tissue 4D** application is to facilitate the detection of tumors in organs such as the prostate or the liver. **Tissue 4D** is used during post-processing after the MR scan. It is composed of several workflow steps, such as the application of pharmacokinetic (PK) modelling for the calculation of parameter images.

Background information

The application of non-linear fitting with pharmacokinetic models allows the calculation of parameter maps. The exchange of contrast agents between blood and tissue is described by the following parameters:

- ❑ **Ktrans** (transfer constant)
- ❑ **Ve** (extra-vascular extra-cellular volume fraction)
- ❑ **Kep** (reflux constant)

The software provides functionalities for the application of different model input functions and for visualization of parametric maps as color-coded images. User-defined ROIs allow the calculation and comparison of values for specific tissues.



The **Tissue 4D** task card supports the following functionalities:

- ☐ Loading of and navigating in 4D dynamic series, pre-contrast data, and morphological data
- ☐ Operation of the UI layout and the navigation mode by using the control area
- ☐ Application of an automatic deformable registration algorithm
- ☐ Manual segmentation of 3D ROIs (for curve calculation)
- ☐ Definition of an elliptical VOI (for parameter calculations)
- ☐ Application of different pharmacokinetic models
- ☐ Visualization of parameter images
- ☐ Saving of parameter images
- ☐ Exporting results

Data types

The following three types of raw data can be loaded into the **Tissue 4D** task card:

- ☐ Pre-contrast data
- ☐ Dynamic series
- ☐ Morphological data

You can load the data separately or simultaneously by selecting a complete study from the **Patient Browser**. When loading the data simultaneously, they are sorted automatically.

Pre-contrast and dynamic data must be acquired with identical distortion filter settings.



The application requires the loading of dynamic series which are acquired with fixed flip angle.

Dynamic series must be acquired with a Flash3D protocol or a variant such as VIBE or TWIST.

The pre-contrast series must be acquired with a Flash 3D protocol or a variant such as VIBE. At least two volumes with variable flip angle (i.e. 2° , 15°) are required for T1 map calculation. Pre-contrast geometrical and spatial parameters must match the dynamic series (resolution, dimensions, orientation). If pre-contrast series are not available, an estimation of T1 can be applied. The loading of morphological data is optional. If loaded, the morphological data can serve as a background for the parametric images.



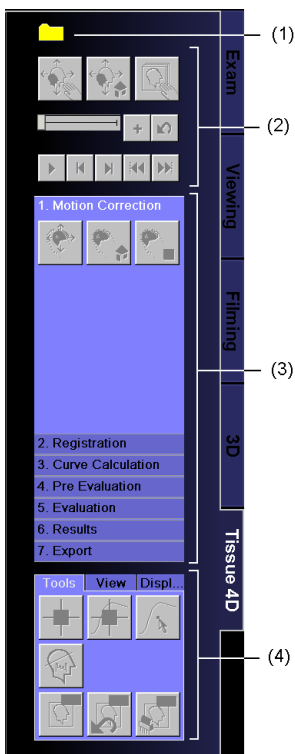
The Tissue 4D task card

Control Area

The control area contains control elements for the common functionalities.

The control area is subdivided as follows:

- (1) Patient folder
- (2) View Mode functions frequently used in the clinical workflow
- (3) Blinds for every specific workflow step
- (4) **Tools**, **View**, and **Display** subtask cards



Display Area

This is the main area for the display of:

- ☐ MPR segments
- ☐ Slice stack segments
- ☐ Plot segments

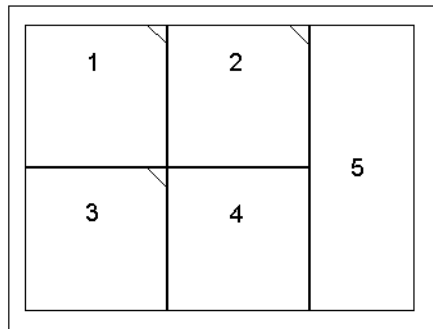


Figure 1: Basic layout with display area

Data in the segments:

- ☐ Segment 1:
Morphologic or pre-contrast data, original data
- ☐ Segment 2:
Dynamic series, original data
- ☐ Segment 3:
Parameter maps, derived data
- ☐ Segment 4:
Curves or histograms
- ☐ 5 Control Area

Slice Stack Segments

Segments 1/2/3 display the slice stacks of pre-contrast data, dynamic data, and parametric images respectively. Segment 3 can additionally contain text/tables with the median and mean standard deviation values for the selected ROI.

Segment 2 offers the possibility to navigate in time and slice direction. Segment 2 is particularly helpful in the identification of suspicious enhancing regions. It can also help to evaluate the motion correction result on the dynamic series by allowing navigation to different time points of the same slice position and using subtraction mode.

MPR Segments

Morphological images are displayed as MPR images at the positions of dynamic images.

Segment 1/2/3

Only during task 2 (morphological registration) in registration display, can the user interact with MPRs to visualize morphological data. The reference is provided by the dynamic series and is used for the visual check (the dynamic series is already motion-corrected).

Interaction inside the MPR views is similar to the **3D** task card.

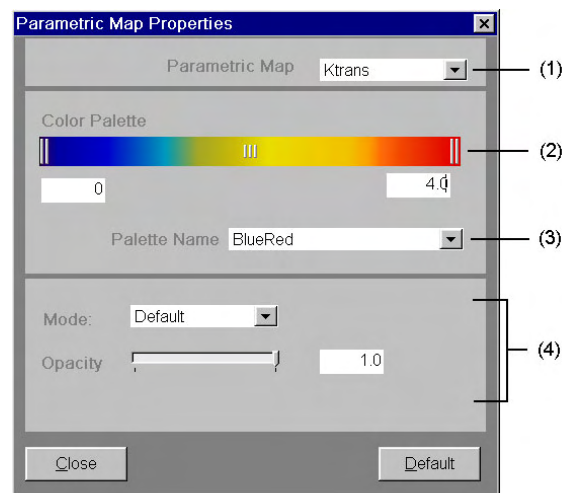
Plotting

Segment 4 displays the following curves and histograms:

- ☐ A plot of relative enhancement curves for all ROIs of dynamic series.
A maximum of four curves can be shown. A ruler overlaps the graph which allows the definition of the arrival time of the contrast agent (available in step 5). Legends of ROIs can be shifted within the graph area to avoid overlaps with the graph.
The color of the signal enhancement curves match the color of the ROIs for curve computation.
Similarly, the color of the histogram matches the color of the ROI selected in the parametric images.
- ☐ A fit curve for one ROI label resulting from the application of pharmacokinetic models.
Only one fit curve can be displayed. All enhancement curves are calculated at once, but the fit curve is calculated for the selected ROI only.
- ☐ Histograms for the selected ROI label of the computed parametric image.

Parametric Map Properties dialog box

The **Parametric Map Properties** dialog box provides two areas containing functions to adjust the visual properties of the parametric map images in the lower left segments. Additionally, windowing mode properties can be changed. The **Parametric Map Properties** dialog box is window specific. It is accessible by selecting **Display Properties** on the **Displ...** subtask card.



- (1) Selection list to choose between different parametric maps such as **Ktrans**, **Ve**, and **Kep**.
- (2) Edit functions for fine-tuning the minimum and maximum windowing of the parametric map and a min./max. slider bar to change the color mapping and the threshold values for the parametric map values.
- (3) Color selection list to select predefined color palettes.
- (4) **Mode** selection list and opacity slider to control the opacity of the parametric maps. A value of 0 corresponds to a fully transparent map while 1 defines a full opaque map.



Display and Navigation

You can load a high-resolution anatomical series, i.e. T1-weighted or T2-weighted images into the upper left segment. Anatomical MPR images matching the slice positions and the geometrical planes of the dynamic images are calculated and displayed.

You can toggle between pre-contrast and anatomical images.

The following View Mode Functions are available:

- ☐ Zoom / Pan
- ☐ Home Zoom / Pan
- ☐ Scroll In Time / Slice
- ☐ Discard Volume
- ☐ Undo All
- ☐ Slider
- ☐ Play / Stop (Auto Cine)
- ☐ Previous Volume
- ☐ Next Volume
- ☐ First Volume
- ☐ Last Volume



Most of these listed functions are well-known or self-explanatory, except

Scroll in Slice / Time

This function enables you to scroll both in time and slice direction. If you move the cursor to the left or to the right, you can scroll in time. By moving the cursor up and down you scroll in slice direction.



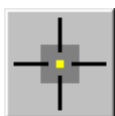
Discard Volume

Activate this function if you want to exclude a certain point in time from pharmacokinetic modeling.

⇒ Please note that **Close Patient** can only be called up via the menu.

The subtask cards

Tools subtask card



The following icons are located on the **Tools** subtask card:

Pixel Lens

This function enables you to view the value of the raw pixel data at the location of the mouse cursor.

- The behavior of the Pixel Lens here is different from *syngo*: The Pixel Lens is active for the current segment. If you want to change to another segment, you must click into this segment.



Move Cursor and Display Curve

Click this icon to move the mouse in the parametric image (segment 3). The corresponding concentration time course for the voxel under the mouse cursor and the fitted curve are displayed in segment 4. If you move the mouse cursor in the dynamic segment (2), the signal intensity time course is displayed in segment 4.



Click and Display Curve

Use this function to click on a point in the parametric image (segment 3). The corresponding concentration time course for the voxel under the mouse cursor and the fitted curve are displayed in segment 4. If you click into the dynamic segment (2), the signal intensity time course is displayed in segment 4.



Profile

This function enables you to draw a line in the parametric segment (segment 3) and display the intensity profile.



Annotation

This function supports overlay of the annotation text strings at defined positions in any segment. You can select and move the annotation text along the x and y axes of the active quadrant. In contrast to *syngo*, you can write several lines (start a new line by using the return key).



Undo Annotation

This function enables you to remove the last annotation text.



Erase All Annotations

This function enables you to remove all annotation texts.

View subtask card



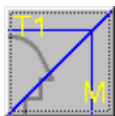
Subtraction

This function is used for subtracting the reference volume and displaying the result as a magnitude image in the same segment. The reference volume is the first volume.



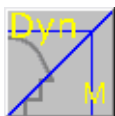
Blow Up

This function enables you to toggle between one large view of the active segment and 2x2 view.



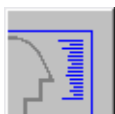
Toggle T1 / Morphological

This function enables you to toggle in the upper left segment between the T1 map (resp. pre-contrast) and the morphological view.



Toggle Dynamic / Morphological

This function enables you to toggle the background image (in gray) in the lower left segment (the parametric segment). There are two modes: a) the last volume from the dynamic series as background, b) the morphological volume as background.



No Text / All Text

This function enables you to switch on and off the display of annotation text for all slice segments and the labels for the pre-contrast, dynamic and plot segments. Only the orientation marks and the orientation cube (where it is displayed) are unaffected by this function. They are continuously displayed.

Display subtask card



Auto Windowing

This function enables you to automatically adjust the image contrast in the active segment.



Display Properties

This icon is used to call up the **Parametric Map Properties** dialog window.

In cine mode, you can view dynamic data as a movie. Combining cine mode with subtraction mode allows fast navigation through the 4D dynamic data set. The speed for cine mode is preset and cannot be changed by the user.

Workflow

The functionalities of the **Tissue 4D** task card are not restricted to one segment, but are context-sensitive. Manual adjustment is not possible. You can select workflow steps by using the control elements explained in this section.

Motion Correction (step 1)

The principle of motion correction consists of a deformable 3D registration.

- ⇨ The currently displayed volume of the dynamic series is selected as reference volume.

You can correct motion in all other volumes with this reference volume.



Figure 2: **Motion Correction** blind



The functions for motion correction are represented by the three following icons:

Registration of dynamic series

Starts the registration of the dynamic data.

The new motion-corrected series automatically replaces the original dynamic series. It is displayed in the same segment.



Clear registration

Clears the new series created during the registration. You can undo the registration for unsatisfactory results. You can evaluate the quality of the motion correction algorithms by using the **Subtraction** functionality, presented in the **View** subtask card.



Save registration

Saves motion-corrected series in the database. The name of the new series is automatically generated and appended to the series list in the database.

Registration (step 2)

The **Registration** blind enables you to register morphological and pre-contrast data to the dynamic series. If a morphologic volume is loaded, segments 1, 2, and 3, display three fused MPRs for the registration process.

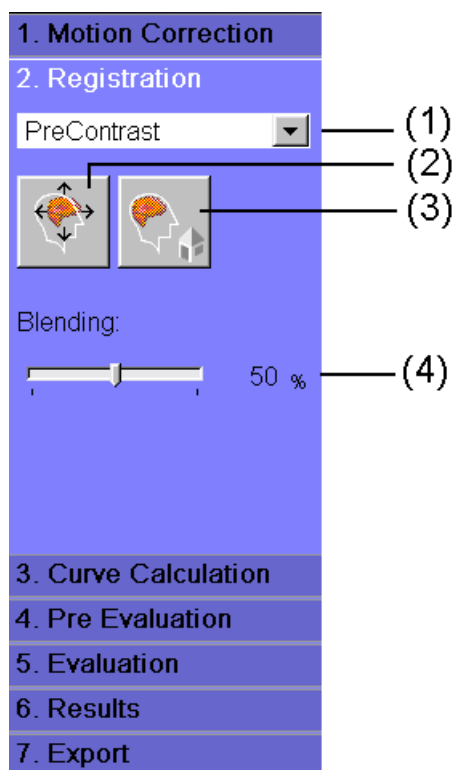


Figure 3: **Registration** blind

- (1) Selection of the registration mode
- (2) **Registration** icon
- (3) **Clear Registration** icon
- (4) Slider to adjust the blending of the morphological or pre-contrast data with the selected dynamic volume



To evaluate the registration quality of the morphologic data, fused MPR (three orthogonal MPRs) renderings of the data are provided. The reference is a volume of the dynamic series which you can select by navigating through the time. The currently selected time point is chosen. The reference volume is shown as colored images. Each fused MPR contains two types of data sets with the same orientation. You can use the **Blending Slider** to control the blending process of the data sets.

The **Registration** blind can be accessed even without performing motion correction in advance. If motion-corrected images were discarded, you can also use original data for the registration. If a data set was registered in a previous session or if the data set does not show motion artifacts, you can skip the first registration step.

Curve Calculation (step 3)

The **Curve Calculation** blind allows you to calculate enhancement curves and segment structures manually by drawing ROIs in the dynamic series. Once entered, motion correction (step 1) and registration (step 2) controls are disabled.

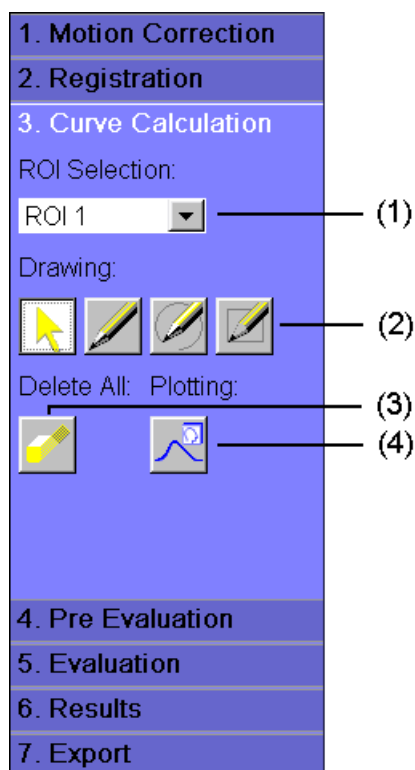
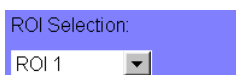


Figure 4: **Curve Calculation** blind

- (1) **ROI Selection** list
- (2) Drawing contours tools
- (3) **Delete All** icon
- (4) **Plotting** icon

In this workflow step, segment 2 is the only segment where overlays/ROIs can be drawn to segment structures manually. Once ROIs are drawn, the enhancement curves for the ROIs are calculated and displayed in segment 4. The curves are shown as relative enhancement curves, the first volume serves as reference.

You can use the control elements on the blind to define ROIs and calculate the signal enhancement curves for selected ROIs:



✧ Click this icon to enable/disable the selection of ROIs in segment 2.



✧ Click this icon to draw closed contours.



✧ Click this icon to draw an ellipse.



✧ Click this icon to draw a rectangle.



✧ Click this icon to delete all ROIs in segment 2.



✧ Click this icon to plot the signal enhancement curves after the ROIs have been drawn.

✧ In contrast to *syngo*: Multiple 2D ROIs can be drawn in the same slice or in different slices to form a 3D ROI, as long as the selection is not changed. ROIs cannot be shifted to individual points in time.

Pre Evaluation (step 4)

The T1 map calculation of pre-contrast data is a prerequisite for pharmacokinetic modeling. It runs automatically when you click the **Pre Evaluation** blind.

The image shows a software interface for the 'Pre Evaluation' step. It consists of a vertical list of steps on the left, with '4. Pre Evaluation' highlighted. To the right of this step are several input fields and icons. Numbered callouts point to specific elements: (1) points to the 'Noise Level' input field containing the value '20'; (2) points to the 'MR Protocol' dropdown menu showing 'T1 and C'; (3) points to a group of three input fields: 'Contrast Ag.' (dropdown showing 'Contrast'), 'Molarity' (text field '0.50' with unit 'mmol/ml'), and 'Relaxivity' (text field '4.30' with unit 'l/mmol/s'); (4) points to the 'Volume' input field containing '18.00' with unit 'ml'. Below these fields are three icons for 'VOI' (Volume of Interest): a yellow mouse cursor icon, a yellow sun icon, and a yellow pencil icon. The bottom of the interface shows steps 5 through 7: '5. Evaluation', '6. Results', and '7. Export'.

Step	Parameter/Icon	Value/Unit
1. Motion Correction		
2. Registration		
3. Curve Calculation		
4. Pre Evaluation	Noise Level	20
	MR Protocol	T1 and C
	Contrast Ag.	Contrast
	Molarity	0.50 mmol/ml
	Relaxivity	4.30 l/mmol/s
	Volume	18.00 ml
	VOI Icons	Cursor, Sun, Pencil
5. Evaluation		
6. Results		
7. Export		

Figure 5: **Pre Evaluation** blind

- (1) T1 fitting parameter
- (2) Measurement protocol
- (3) Contrast agent parameters
- (4) **VOI** icons

T1 fitting is restricted to pixels with values above a noise level value. Once T1 map is calculated, the pre-contrast is replaced.

The T1 maps masks define the areas to be adjusted for the pharmacokinetic fit. For a better performance of the non-linear modeling fit, all noise pixels will be excluded from the mask.

You can use a variety of functionalities to define a set of pre-required parameters for the application of pharmacokinetic models:

Noise Level 20

- ✧ In the **Noise Level** selection list, you can define a threshold value for masking out noise. The default is 20.

MR Protocol T1 and C

- ✧ In the **MR Protocol** selection list, you can select the MR acquisition technique. The default is T1 + Dynamic. If T1 map is not available, dynamic 3D is chosen and an estimated value for T1 can be set.

Contrast Ag.

- ✧ In the **Contrast Ag.** selection list, you can select the contrast agent used. An application specialist can configure a list of preset contrast agents.

Molarity mmol/ml
Relaxivity l/mmol/s
Volume ml

- ✧ **Molarity, Relaxivity and Volume** are used to perform PK modeling. The respective values are automatically selected by the system as a function of the contrast agent entered.



✧ Click this icon to select a VOI.



✧ Click this icon to draw a VOI.



✧ Click this icon to delete a VOI.

Click one of the above icons for the manual segmentation of an organ (like the prostate) or of a subvolume. You can define an ellipse in a single slice which is extended to form an ellipsoid in a 3D structure.

Evaluation (step 5)

The **Evaluation** blind allows pharmacokinetic modeling for the mean curves or voxelwise for the selected ROI label. You can select different model arterial input functions (AIF) on this card. Currently, the two-compartment Tofts model is provided.

The results of fitting are displayed in segments 3 and 4. Segment 4 displays fitting curves, and segment 3 displays parameter map results blended on source images.

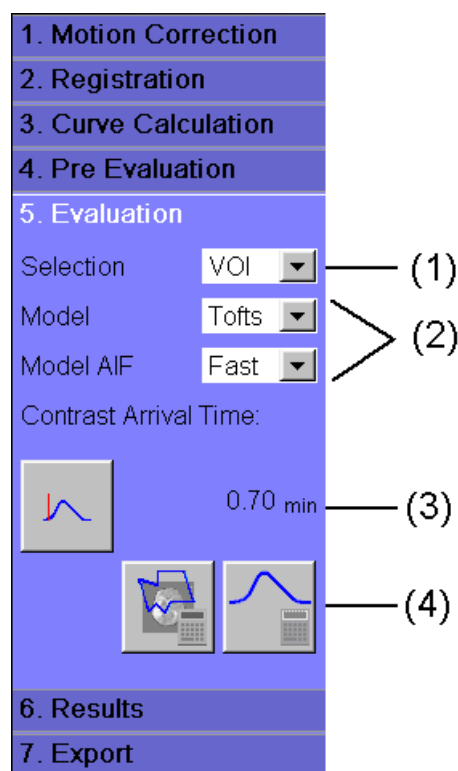


Figure 6: **Evaluation** blind

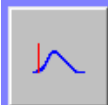
- (1) ROI / VOI selection list
- (2) Pharmacokinetic model input function
- (3) Onset time of enhancement
- (4) Fitting ROI tools

Selection VOI

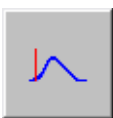
Model Tofts

Model AIF Fast

Contrast Arrival Time:



0.70 min



You can use a variety of functionalities to perform pharmacokinetic modeling and to select between different fitting parameters:

- ✧ In the **Selection** selection list, you can select the ROI or VOI whose parameters should be fitted.
- ✧ In the **Model** selection list, you can select the PK model Tofts. More models will be available in the future.
- ✧ In the **Model AIF** selection list, you can select the AIF function which gives information about the concentration of the contrast enhancing media in the blood plasma. You can choose between fast, intermediate, and slow.
fast - high temporal resolution, high kinetics
intermediate - moderate temporal resolution and/or kinetics
slow - slow temporal resolution, slow kinetics
- ✧ In the **Contrast Arrival Time** field, the read-only arrival time value is displayed.
- ✧ Click this icon to adjust the onset time of enhancement. After a second click, the arrival time is updated. The arrival time is automatically calculated when curves have been calculated before.
- ✧ Click this icon to perform pixelwise pharmacokinetic modeling of the selected ROI label. The results are displayed in segment 3.
- ✧ Click this icon to perform pharmacokinetic modeling for mean ROI curves of the selected ROI label. The results are displayed in segment 4.

Results (step 6)

The ROI functionalities on the **Results** blind comply with the functionalities on the **Curve Calculation** blind. The setting affects only the ROIs displayed in segment 3 (parameter images).

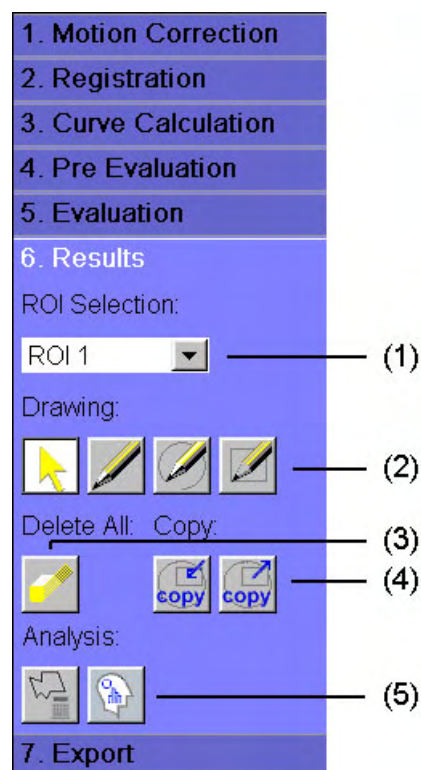
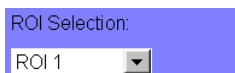


Figure 7: **Results** blind

- (1) **ROI Selection** list
- (2) **Drawing** tools
- (3) **Delete** icon
- (4) **Copy** icons
- (5) **Analysis** icons

This blind is used for the analysis of results from PK modeling, either by adding new ROIs or by copying ROIs from segment 2. Segment 3, in which overlays can be drawn, is the only segment containing parametric data.

You can use the following functions to draw and copy ROIs:



- ✧ In the **ROI Selection** list you can define different ROI labels in different colors.



- ✧ Click this icon to enable/disable selection of ROIs in segment 2.



- ✧ Click this icon to draw closed contours.



- ✧ Click this icon to draw ellipses.



- ✧ Click this icon to draw rectangles.



- ✧ Click this icon to delete all ROIs in segment 2.



- ✧ Click this icon to copy ROIs from segment 2 to segment 3.



- ✧ Click this icon to copy ROIs from segment 3 to segment 2.



- ✧ Click this icon to calculate the median, mean and standard deviation for the selected ROI. The results are displayed in segment 3.
- ✧ Click this icon to display the histogram for the selected ROI label in segment 3.

Export (step 7)

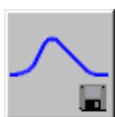
The **Export** blind is the last card of the workflow. It is used to save or export results.



Figure 8: **Export** blind



- ✧ Click this icon to save all parameter values from segment 4 for the selected ROI to text file. The relevant values come from the parameters **Ktrans**, **Kep**, **Ve**, **Chi²**, **FitCode**, and **iAUC**.



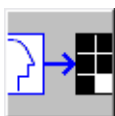
- ✧ Click this icon to save the currently displayed curves for the selected ROI to a text file.



- ✧ Click this icon to append the current volume of the parameter image as a new series to the existing series.



- ✧ Click this icon to save the segments to the database or as bit-map to the filesystem.



- ✧ Click this icon to send the currently selected image to the **Filming** task card.
Please note that annotations and graphics cannot be hidden, since they are firmly attached to the images.



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Safety

General safety information



During examinations, you are required to adhere to the safety measures as described in the “Safety” chapter of the system manual.



CAUTION

Insufficient image quality of the remote computer!

Incorrect interpretation of the MR images

✧ Do not use the remote computer for diagnoses or reports.



CAUTION

Remote user does not adhere to the currently valid safety guidelines!

Hazard. Patient injury due to uncontrolled actions of the remote user.

Possible injury to patient due to missing control or absence of local user.

- ✧ As the local user, you are responsible for the patient during the entire examination.
 - ✧ Remain continuously at the MR console and monitor the actions of the remote user.
 - ✧ Remain continuously in touch with the remote user via telephone.
 - ✧ Stop the patient table if hazardous conditions are present for the patient or terminate the entire measurement sequence (e.g. via the **table STOP /sequence termination** at the intercom).
 - ✧ Interrupt the remote connection in case of uncontrolled actions by the remote user.
-

Description

The *syngo* Expert-i software supports local users at the MR console via experts (e.g. physicians) outside the examination room (for example, at a remote networked PC).

Functionality

Remote access is handled via a suitable computer (remote computer) connected to the local network.

For remote access, the remote user has to register at the MR console. For this purpose, he needs a password.

This password is generated by the *syngo* Expert-i software and is only applicable for the registration planned.

In this case, the system can be operated at the MR console as well as the remote computer.

HIPAA audit trail

Each time you use the *syngo* Expert-i software, the protocol inputs are entered into the HIPAA audit trail.

The inputs are generated when you perform the following actions:

- ☐ The local user *syngo* Expert-i activates the software.
- ☐ The remote user successfully established the remote connection.
- ☐ The remote connection was not established within the time reserved for this purpose.
- ☐ The local user or the remote user disconnected the remote connection.



Software installation

The *syngo* Expert–i software can only be used in conjunction with the released versions of the *syngo* software.

The released versions are listed on the front page of this document.

The *syngo* Expert–i software needs to be installed and configured by Siemens Service on the MR system as well as the designated remote computer. The *syngo* Expert–i software is integrated in the MR system by Numaris/4 installation.

Minimum requirements

The following minimum requirements must be met for the remote computer and the local network:

Computer hardware	Current standard hardware
Processor	850 MHz
Operating system	Windows NT4.0, XP
Primary storage	256 MB
Network card	100 Mbit/s
Graphic card, monitor	1280x1024
Color	32 bit
Local network	100 Mbit/s or higher
Min. transfer rate	> 600 kB/s
Response time	<30 ms

- !

- When falling beyond the minimum requirements, the performance and display of the remote computer may be adversely affected.

Operation

Perform the following steps to access the MR console:
syngo Expert-i

- ☐ Activate the *syngo* Expert-i software at the MR console
- ☐ Transfer the password to the remote user
- ☐ Establish a connection at the remote computer.

! As the local user, you are responsible for the patient during the entire examination.

Do not leave the MR console at any time during remote connection and monitor the actions of the remote user.

Local user: Activate software

To ensure that the remote user can access the MR console, the local user must first activate the *syngo* Expert-i software at the MR console.

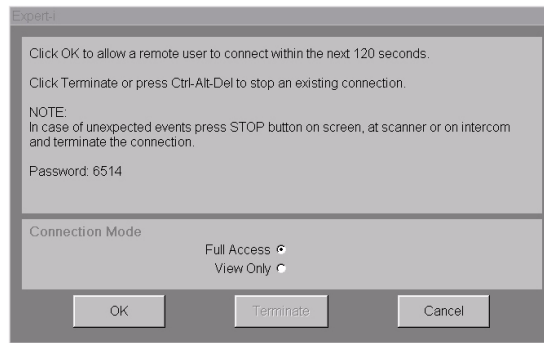
Prerequisite: The *syngo* Expert-i software is installed on the MR system.

- ✧ Call the remote user to request the initiation of a remote connection.



- ✧ Activate the *syngo* Expert–i software by clicking the tray icon.

The “*syngo* Expert–i” dialog window is opened; a four-digit password is displayed.



Dialog window of the MR console

-
- ! The password displayed is only valid once - for the next registration. The *syngo* Expert–i software generates a new password for every registration.
-

Local user: Transferring the password



- ✧ Confirm the message with **OK** and activate the *syngo* Expert-i software with this message.
- ✧ Communicate the generated password to the remote user.

While the local user is waiting for registration of the remote user, the tray icon is changing.

Local user: Establishing the connection

After the local user has activated the *syngo* Expert-i software at the MR console, the remote user has 2 minutes to establish the remote connection.

The local user must select whether the remote user should have **Full Access** or **View Only** access to the MR console.



If he exceeds the allocated time, the local user has to request a new password. In this case, you have to start the *syngo* Expert-i software again.

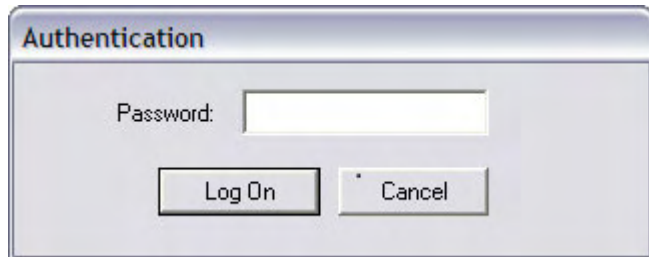
Prerequisites:

- ☐ The *syngo* Expert-i software is installed at the remote computer.
- ☐ You received the password from the local user.
- ✧ Start the *syngo* Expert-i software via the program icon or a link at the remote computer.



The location of the link depends on the installation of the software on the remote computer.

Remote user: Entering a password and registering at the MR system



Entering the password at the remote computer

- ✧ Enter the password and click **Log On**.

The active connection is shown by the tray icon of the *syngo* Expert-i software at the MR console.

The screen of the remote computer shows the user interface of the MR console. The remote user is now able to access the MR console.



Local user: Disconnecting the remote connection

The local user can disconnect the remote connection at any time.

- ✧ Press the key combination **Ctrl+Alt+Del** to disconnect the remote connection immediately.
–Or–
Click the tray icon of the *syngo* Expert-i software to display the dialog window.
- ✧ Click “*syngo* Expert-i” in the dialog window and press **Terminate**.

The remote connection is disconnected. The tray icon shows that the connection is down.

Remote user: Disconnecting the active connection

The remote user can disconnect the active connection at any time.

- ✧ Click the symbol to close the window which is shown when you move the mouse in the upper right corner of the screen.
–Or–
Click the tray icon of the *syngo* Expert-i software to display the dialog window.
- ✧ Click “*syngo* Expert-i” in the dialog window and press **Terminate**.

The remote connection is disconnected.



Local user: Installing Expert-i Viewer

The local user can also install the Expert-i Viewer.

- ✧ Click the Expert-i client link in the Expert-i entry in the **programs** menu.
The **installation** dialog windows opens.
- ✧ Click **Next**.

The dialog window for the user opens. You can browse to the right directory or type into the entry field.

- ✧ Click **Next** after the configuration.

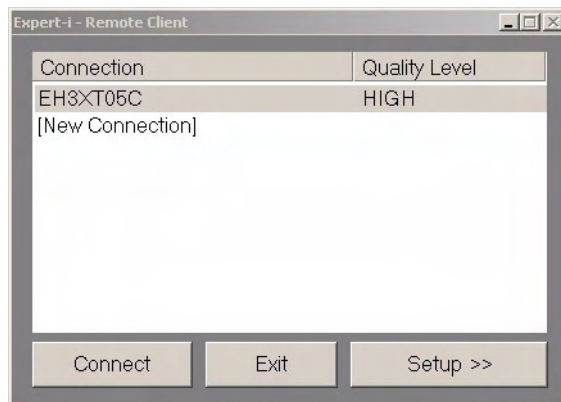
The installation is done.

Remote user: Using the Expert-i Remote Client

The remote user can use the **Expert-i Remote Client** to connect to a specific workplace.

✧ Click the Expert-i client link in the **startup** menu.

The **syngo Expert-i - Remote Client** dialog window opens.



✧ Select **New Connection** and click **Setup** to create a new connection.

–Or–

Select an existing connection and click **Setup** to edit an existing connection.

If you have problems connecting

If the connection on the remote computer fails, the reasons for it may be as follows:

- ❑ The local user did not activate the syngo Expert-i software at the MR console.

The following message appears at the remote computer:

"Your connection has been rejected. Please contact local user first."

- ❑ The remote user entered the wrong password or the time allowed for entering the password exceeded:

The following message appears at the remote computer:

"Authentication failed!"

- ❑ The remote computer is unable to establish a connection to the MR console.

The following message appears at the remote computer:

"Failed to connect to server."

- ✧ Check the error sources described and repeat the registration.



Establishing a connection at the remote computer failed.

The remote user entered the wrong password or he exceeded the time for registration:

- ✧ Get in touch with the local user and request a new password.
 - ✧ Repeat your registration.
-



Establishing a connection at the remote computer failed.

The local user did not activate the *syngo* Expert-i software at the MR console.

- ✧ Ensure that the local user activated access to the *syngo* Expert-i software.
 - ✧ Request that the local user provide you with a new password.
 - ✧ Repeat your registration.
-

-
- ! Establishing a connection at the remote computer failed.
- The remote computer is unable to establish a connection to the MR console.
 - ✧ Check the network connection from the remote computer to the MR console.
 - ✧ Get in touch with your system administrator.
-

- ! If it is still not possible to establish a connection, get in touch with your system administrator or Siemens Service.
-

Description

The *syngo* RemoteAssist software supports local users at the MR console via Siemens Customer Service.



Do not use the *syngo* RemoteAssist software function **Take over remote control** during patient examination!

During patient clinical routine it is not allowed to let 3rd persons take over control via the *syngo* RemoteAssist software.

Functionality

Remote access is handled via a suitable computer (remote computer) connected to the local network.

For remote access, the remote user has to register at the MR console. For this purpose, he needs a password.

This password is generated by the *syngo* RemoteAssist software and is only applicable for the registration planned.

In this case, the system can be operated at the MR console as well as the remote computer.

HIPAA audit trail

Each time you use the *syngo* RemoteAssist software, the protocol inputs are entered into the HIPAA audit trail.

The inputs are generated when you perform the following actions:

- ☐ The local user *syngo* RemoteAssist activates the software.
- ☐ The remote user successfully established the remote connection.
- ☐ The remote connection was not established within the time reserved for this purpose.
- ☐ The local user or the remote user disconnected the remote connection.

Software installation

The *syngo* RemoteAssist software can only be used in conjunction with the released versions of the *syngo* software.

The released versions are listed on the front page of this document.

The *syngo* RemoteAssist software is installed and configured by Siemens Service on the MR system as well as the respective remote computer.

Minimum requirements

The following minimum requirements must be met for the remote computer and the local network:

Computer hardware	Current standard hardware
Processor	850 MHz
Operating system	Windows NT4.0, XP
Primary storage	256 MB
Network card	100 Mbit/s
Graphic card, monitor	1280x1024
Color	32 bit
Local network	100 Mbit/s or higher
Min. transfer rate	> 600 kB/s
Response time	<30 ms

- !

- On failing to meet the minimum requirements, the performance and display of the remote computer may be adversely affected.

-
- ! The Remote Assistance Software must be installed on your MR system.
-

Siemens service technicians can support the local user at the MR system e.g. if problems arise or new software has to be installed.

A service technician can access your system only if you enable him to do so. Unauthorized access is blocked.

Operation

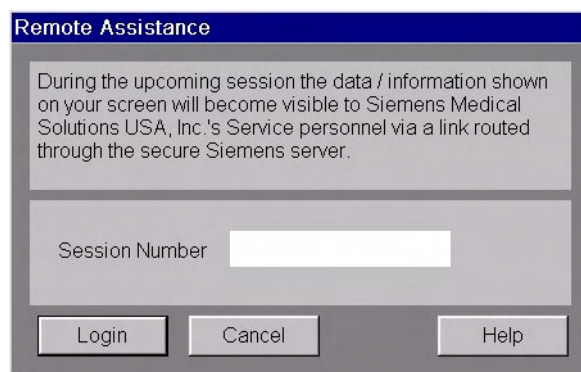
Usually your service technician informs you if he wants to access your MR system.

You can activate a single connection with a service technician via a menu.

Activating a single connection

✧ Call up **Options > Service > Remote Assistance**.

The following dialog window is displayed:



In this dialog window, you enter a non-assigned session number.

✧ Enter the session number in the input field.

If there are sensitive data on your system, a corresponding message is displayed, referring to privacy rules for these data.



✧ Click the **Logon** button to activate the connection.

If remote assistance has been successfully started, the icon (**single connection**) appears in the status bar. Remote user access to your MR system is in **View Only** mode with another cursor. The remote user cannot operate your MR system.



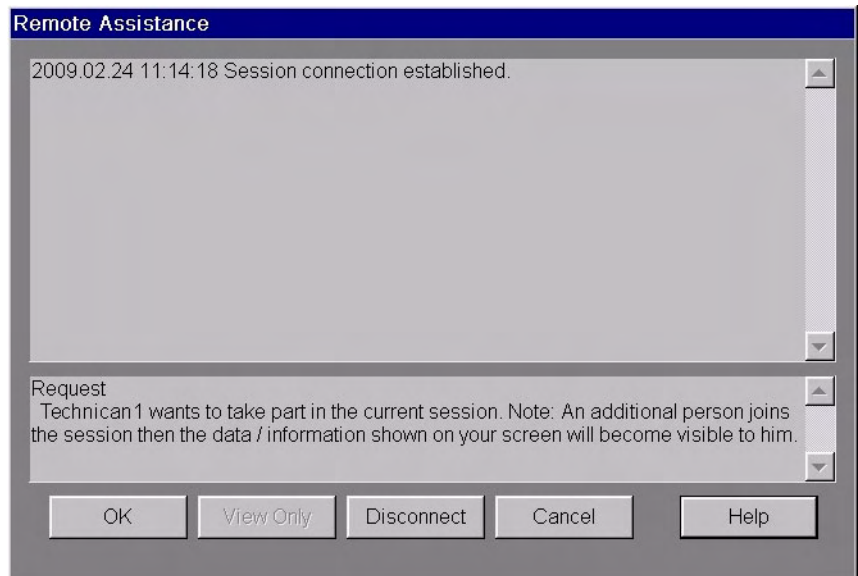
If a remote connection could not be established, a corresponding message is displayed on your screen. In this case, contact your service technician.

Activate third-party access

If another service technician wants to access your MR system, the remote assistance icon in the status bar blinks.

✧ Click the **Remote Assistance** icon.

A corresponding request appears in this dialog window:



✧ If you agree to further access, click **OK**.

–Or–

Click **Cancel** to cancel the dialog and refuse the request.

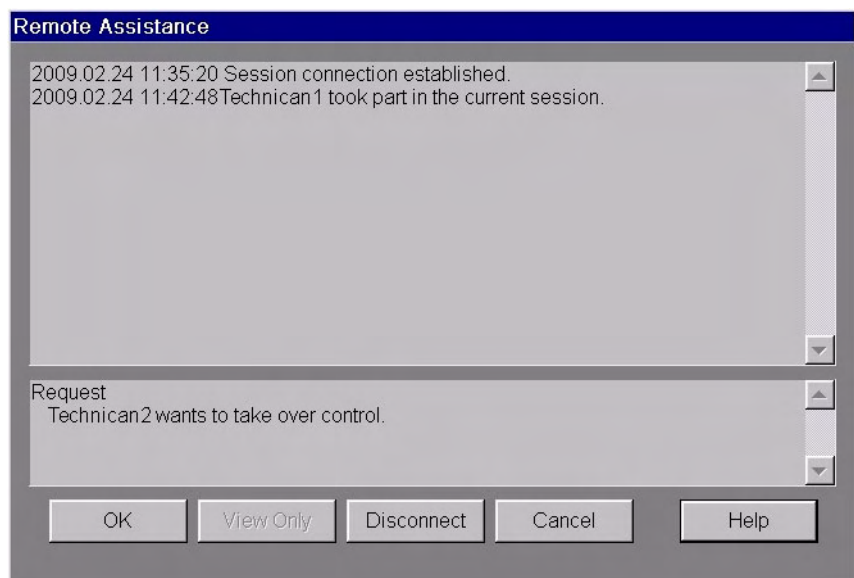
If remote assistance is started successfully, an icon (**multiple connections**) is displayed in the status bar. A third party can access your MR system in **View Only** mode with another cursor and cannot operate your MR system.

Transferring Remote Assistance

Even if several service technicians can access your workstation, there are - at most - two remote users who can access your MR system and you can control the system all the time. If any technician wants to take control of your system, the **Remote Assistance** icon in the status bar blinks.

✧ Click the **Remote Assistance** icon in the status bar.

The following request is displayed in this dialog window:



✧ If you agree to transfer control, click **OK**.

–Or–

Click **Cancel** to cancel the dialog and refuse the request.

If transferring control access is started successfully, the remote user can operate on the MR system. At the same time, there is only one remote user who can operate the MR system. If one remote user is allowed to operate the MR system, another remote user who is operating the MR system will lose control automatically.

Restricting access

If necessary, you can restrict the access of the service technician to your workstation. For instance, if you switch to **View Only** mode, you can call functions by yourself. In the meantime the service technician is restricted to **View Only** mode.



View Only

- ✧ Click the **Remote Assistance** icon to open the **Remote Assistance** dialog window.
- ✧ Click the **View Only** button.

The service technician has **View Only** access to your workstation.

Disconnecting Remote Assistance

You can disconnect the Remote Assistance at any time.



Disconnect

- ✧ Click the **Remote Assistance** icon to open the **Remote Assistance** dialog window.
- ✧ Click the **Disconnect** button.

The connection to the service technician is disconnected.



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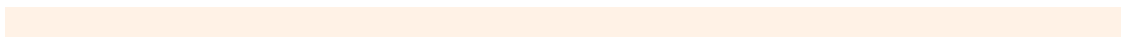
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Parameters of the Parameter Cards

On the following pages you will find a detailed description of all parameter cards and their measurement parameters. You can change the measurement parameters within the framework of your different measurement protocols in **syngo MR**.

An introduction on how to work and enter data on the parameter cards is included in the **Exam** section of this Operator Manual.

→ [Page C.7–1, Adapting the Measurement Parameters](#)

A detailed description of the parameters for system adjustment is included in the **Managing and adjusting the system** section of this Operator Manual.

→ [Page B.2–1, Adjusting the System](#)

The Spectroscopy Manual provides information regarding measurement parameters relevant only for spectroscopy measurements.



Information (tool tips) are superposed for most of the parameters when you keep the mouse pointer on it for a short time.

Routine

Slab group	1					FoV read	180	mm
Slabs	3					FoV phase	100.0	%
Dist. factor	-20.00				%	Slice thickness	0.70	mm
Position	L2.0 A25.0 H0.0				...	TR	25	ms
Orientation	Transversal				...	TE	7.00	ms
Phase enc. dir.	R >> L				...	Averages	1	
AutoAlign	---				...	Concatenations	3	
Phase oversampling	0				%	Filter	None	
Slice oversampling	20.0				%	Coil elements	BC	
Slices per slab	40							

Program Routine Contrast Resolution Geometry System Physio Angio Sequence

e.g.: Routine parameter card (3D)

The **Routine** parameter contains all parameters of your protocol that you need to control and adjust frequently while preparing measurements.

The **Routine** parameter card shows different parameters depending on whether your protocol uses 2D or 3D measurements.

(1) *Slab group*

The slabs measured within the framework of a protocol are combined into groups.

Here you can see the number of the slab group currently displayed. All slice parameters, currently seen on the **Routine** or **Geometry** card refer to this slab group.

(2) *Slabs*

This parameter determines the number of slabs in this group.

(3) *Dist. factor*

This parameter determines the gap between slices and/or slabs of a group in percentages.

At a 100% value the gap corresponds exactly to one slice and/or slab thickness. Negative values lead to overlapping.

⇒ Negative distance values may not be entered for 3D measurements.

(4) *Position*

This parameter defines the position of the object center.

The following information is displayed when you position the mouse pointer on the field:

☐ **Table position**

Table position of the protocol, referenced to the whole body patient coordinate system.

☐ **Phase offcenter**

Shift in phase-encoding direction

☐ **Read offcenter**

Shift in readout direction

☐ **Slice shift**

Shift in slice-selection direction

⇒ The selection list is dimmed if the position matches the iso-center.



Use this key to open the **Position** dialog window for the slice selected.

→ [Page P.1–220](#), *Position dialog window*

(5) Orientation

This parameter indicates the position of the object in space based on the whole body patient coordinate system.



Use this key to open the **Orientation** dialog window for the slice selected.

→ [Page P.1–219](#), *Orientation dialog window*

(6) Phase enc. dir.

The current phase-encoding direction (direction of the phase-encoding gradient) is indicated in the main orientations of the whole body patient coordinate system.

You can change the phase-encoding direction. This will swap the phase-encoding and readout direction. Using this method allows you to prevent aliasing artifacts in the phase-encoding direction or change the direction of flow and motion artifacts.

⇒ In the **Phase enc. dir.** selection list you are only offered configurations suitable to the current orientation.

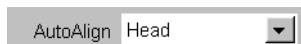


Use this button to open the **Inplane Rotation** dialog window for the slice selected.

→ [Page P.1–222](#), *Inplane Rotation dialog window*

(7) *AutoAlign*

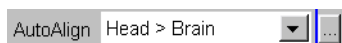
This parameter is used in two different variants:



AutoAlign scout protocols:

In AutoAlign scout protocols the **AutoAlign** parameter is used to specify the anatomical region where the AutoAlign matrices shall be computed.

- ⇒ Predefined AutoAlign scouts are found in the localizer libraries of the Siemens protocol tree of the **Exam Explorer** (e.g. \\SIEMENS\\head\\library\\localizers).



AutoAlign clinical protocols:

AutoAlign clinical protocols are protocols which use only one of the AutoAlign matrices. Almost all protocols can be used as an AutoAlign clinical protocol.

The AutoAlign parameter specifies which of the AutoAlign matrices will be used to align this clinical protocol. The displayed value is a composite of the licensed AutoAlign region and an AutoAlign reference.

- ⇒ The content of the **AutoAlign** parameter is ignored, if the referenced AutoAlign matrix is not yet computed.

You can select the required entry from the **AutoAlign** selection list.



For detailed settings, use this button to open the **AutoAlign** dialog window. There you can set the region, the reference, and different geometry parameters.

→ [Page P.1–223, AutoAlign dialog window](#)

(8) *Phase oversampling*

This parameter increases the phase-encoded area symmetrically at both sides of the field of view (FoV). The expanded FoV area is not shown. It is used to avoid overfolding artifacts.

Phase oversampling is shown as a percentage of the FoV in the phase-encoding direction.

- ⇒ Phase oversampling increases the measurement time. The signal-to-noise ratio is improved
- Oversampling is always automatically performed in the read-out direction so it does not lead to an increase in measurement time.

(9) *Slice oversampling*

This parameter enlarges the phase-encoding area symmetrically on both sides of the slab in the slice selection direction. This extended FoV area is not shown. It is used to avoid overfolding artifacts.

Slice oversampling is indicated as a percentage of the slab thickness.

- ⇒ The measurement time is prolonged through slice oversampling. The signal-to-noise ratio is improved.

(10) *Slices per slab*

This parameter determines the number of slices per slab:

Changes of the **Slices per slab** affect the following:

- ☐ Thickness of slab
- ☐ The absolute slice oversampling as percentage of the slab.

(11) *FoV read*

This parameter determines the size of the anatomical region to be displayed (extension of the measurement in the readout and phase-encoding direction) and its resolution (pixel size).

- When the **Tim CT mode** is active, this parameter determines the anatomical region required during RF excitation.

(12) *FoV phase*

The FoV in the phase-encoding direction (**FoV phase**) is shown as a percentage of the FoV in the readout direction (**FoV read**).

(13) *Slice thickness*

Together with the number of slices, this parameter determines the number of extension of the measurement area in the slice selection direction.

2D measurements

In the case of 2D measurements, the slice thickness corresponds to the thickness of a slice within a slice group.

When you change the slice thickness for 2D measurements, you also change the distance between the slices.

3D measurements

For 3D measurements, the slice thickness is the thickness of a slice within a slab (partitions).

When you change the slice thickness with 3D measurements, you change the thickness of the slab as well.

⇒ Increasing the slice thickness improves the signal-to-noise ratio, but degrades the spatial resolution in the slice-selection direction.

(14) *TR (repetition time)*

This parameter determines the repetition time TR that elapses between two successive excitations. (At times an alternative designation *Repetition time* is used).

Changing the repetition time affects image contrast and measurement time.

If you would like to enter several repetition times, additional keys are superimposed for scrolling between the individual times.

(15) *TE (echo time)*

This parameter determines the echo time (TE) that elapses between the RF excitation pulse and the pulse for the echo to be measured.

- ⇒ This parameter is not active for all sequences.
- ⇒ You may enter several echo times for multi-echo sequences. You scroll through the echo times below the field names by using the arrow keys. When you change a given echo time, the following echo times will be adjusted accordingly.

(16) *Averages*

This parameter determines the number of repeat measurements to improve the signal-to-noise ratios. The results of repeat measurements are averaged by the system.

- ⇒ Please note that the greater the number of measurements, the longer the measurement time.

(17) *Concatenations*

This parameter defines into how many repetition times TR the measurement of the planned slices should be divided. The system then determines across how many single sequential measurements the slices will be distributed.

Using this method, you can acquire many slices with a short repetition time (T1 weighted imaging) and prevent slice crosstalk.

- ⇒ When you selected a measurement in multiple breathhold mode and decided to choose one of the inputs **Breath-hold**, **Breath-hold & Monitor** or **Breath-hold & Follow** under **Respiratory Control**, the number of breathhold intervals is determined via the **Concatenation** parameter.
- ❑ In the **Interleaved** multi-slice mode, the number of breathhold intervals results from the product of the parameter values **Measurements** and **Concatenations**.
- ❑ In the **Single Shot** multi-slice mode, the number of breathhold intervals is the product of the parameter values **Measurements**, **Concatenations** and **Averages**.
- ⇒ With triggered multislice measurements ("interleaved" excitation sequence), it is sometimes not possible to acquire all slices in one measurement. The slices missing from one measurement will then be acquired in the next measurement.
- ⇒ This field is active only, when the **Interleaved** or **Single Shot** option was selected in the **Multi-slice mode** on the **Geometry - Common** parameter card.

(18) *Filter*

This field shows the filter selected.

- ⇒ You are setting filters on the **Resolution - Filter** parameter card.

(19) *Coil elements*

This field shows the coils for this protocol.

➡ You set the coils on the **System - Coils** parameter card.

(20) *Slice group*

Slices measured within the framework of a protocol are combined into a group.

This parameter shows the number of the currently displayed slice group. All slice parameters that are currently seen on the **Routine** or **Geometry** task card refer to this slice group.

(21) *Slices*

This parameter determines the number of slices in this group.

Contrast - Common

The screenshot displays the 'Contrast - Common' parameter card. The 'Common' tab is selected, showing various MRI parameters. The 'Dynamic' tab is also visible. The parameters are organized into two columns. The left column includes TR (3000.0 ms), TE 1 (20.1 ms), MTC (checked), Magn. preparation (None), TI 1 (136 ms), and Flip angle 1 (90 deg). The right column includes Fat suppr. (Water excit. norr), Fat sat. mode (Weak), Water suppr. (None), Restore magn. (unchecked), and SWI (unchecked). A green slider at the bottom indicates the TR value, ranging from 200.0 to 10000.0, with a marker at 3000.0. The bottom navigation bar shows tabs for Program, Routine, Contrast (selected), Resolution, Geometry, System, Physio, and Sequence.

The settings for image contrast are included on the **Contrast** parameter card.

Depending on medical indications, you can generate differently weighted images via these settings. Additional effects can be obtained by spin preparation.

T1, T2, and proton density contrast

With spin-echo sequences, you obtain T1, T2 or proton density contrast weighting by setting the parameters TR (repetition time) and TE (echo time). The following applies:

- ❑ Short TR and short TE produces T1 contrast.
- ❑ Long TR and long TE produces T2 contrast.
- ❑ Long TR and short TE produces proton density contrast.

Spin preparation

You are planning an RF pulse (so-called spin preparation) before the actual measurement when you want changes in contrast or want to suppress certain signals (e.g., inversion recovery sequence).

Signal suppression

The MR signal comprises the sum of signals from water and fat protons. This may result in chemical shift artifacts. Motion artifacts may be enhanced, and contrast may degrade.

Signal suppression may be used to decrease these effects.

Magnitude, phase, and real images

The **Contrast** parameter card provides a list of the image types for reconstruction:

- ⇒ Reconstruction of image types is not possible in every protocol and every sequence.

☐ **Magnitude images**

Show the intensity of the MR signal.

☐ **Phase images**

Show the phase positions of the spins

☐ **Real images**

Show the amplitude plus algebraic sign of the longitudinal magnetization after an inversion pulse.

Suppression of fat and water signals

Using the Dixon method, you can split the MR signal into a fat and a water signal and display the two signals separately.

(1) *TR (repetition time)*

This parameter determines the repetition time TR that elapses between two successive excitations. (At times an alternative designation *Repetition time* is used).

Changing the repetition time affects image contrast and measurement time.

If you would like to enter several repetition times, additional keys are superimposed for scrolling between the individual times.

(2) *TE (echo time)*

This parameter determines the echo time (TE) that elapses between the RF excitation pulse and the pulse for the echo to be measured.

- ⇒ This parameter is not active for all sequences.
- ⇒ You may enter several echo times for multi-echo sequences. You scroll through the echo times below the field names by using the arrow keys. When you change a given echo time, the following echo times will be adjusted accordingly.

(3) *MTC (magnetization transfer)*

This parameter determines whether the signal of tissue with a high portion of macro molecules is weakened by a special RF excitation pulse.

- ⇒ By activating the **MTC** checkbox, you obtain better contrast in images for e.g., vessel examinations.

(4) *Magn. preparation*

During inversion recovery (IR) and saturation recovery (SR), this parameter determines whether an RF pulse is transmitted prior to each measurement to affect contrast.

You can send the inversion pulse as slice-selective or non-selective. Consider whether your current sequence is an inversion recovery sequence (IR) or a saturation recovery sequence (SR).

The following settings are available:

Adjustment	Description
Slice sel./slab sel. (IR or SR)	The measurements are performed slice by slice.
Non-sel. (IR or SR)	The RF pulses stimulate the entire volume independent of the current slice position or measurement sequence.
Non-sel. DIR	DIR stands for "Double Inversion Recovery" and allows the suppression of signals from two tissue types with different T1 relaxation times. Two inversion pulses are applied at defined times TI1 and TI2. While you can set the first inversion time TI1, the second inversion time TI2 is fixed.
T2 preparation	A special preparation pulse is transmitted for the entire volume to enhance T2 contrast. The preparation pulse suppresses signals from tissue with short T2 times. Go to the Contrast parameter card and determine the duration of the preparation pulse in the T2 Prep. duration field.
TI Scout	This scout generates a series of images with different inversion time. Based on this series, you can determine the most suitable inversion time and enter the value in the TI parameter.
T2 sel IR	A special inversion preparation pulse is transmitted for the entire volume. This pulse effects that only tissue with a long T2-time (fluids) is fully inverted. Tissue with short T2 is only inverted in part. In a subsequent excitation, it is possible to display it with a higher signal (e.g. for protocols with dark fluid contrast).
None	No inversion pulse is sent.

⇒ When you select the **None** option in the **Magn. preparation** list, you are usually not offered the **TI** parameter on the **Contrast - Common** card.

(5) *TI (Inversion time)*

This parameter determines whether an additional T1 contrast is added with spin preparation. For this purpose, a 180° RF pulse is used (inversion pulse) that inverts the spins.

☐ **For inversion recover sequences:**

In this case, TI is the time between sending the inversion pulse and the excitation pulse of the subsequent measurement (e.g., a spin-echo pulse sequence). Depending on the TI, certain signals are suppressed (e.g., fat) and additional T1 contrast is applied to the signal.

☐ **For Turbo FLASH sequences:**

TI describes the time between sending the inversion pulse and reading out the echo signal, sorted in the center of the raw data matrix (this echo determines the contrast).

(6) *Flip angle*

This parameter determines the extent the RF pulse deflects the longitudinal magnetization from the Z-direction of the magnetic field. The flip angle directly affects image contrast.

With a 90° excitation pulse, for example, the longitudinal magnetization is rotated by 90° out of the Z-direction of the magnet. If a smaller flip angle is used, the magnetization returns to equilibrium more quickly allowing you to reduce the repetition time TR (for gradient echo sequences).

For spin echo sequences, you can enhance the T1 contrast by reducing the flip angle.

(7) *Fat suppr.*

This parameter suppresses the signal from fat protons in a targeted way.

The following settings are available:

Adjustment	Description
Fat sat.	Suppresses the fat signal and has no effect on TE, TR is greatly prolonged. This extends the measurement time as well.
Water excit. normal	Suppresses the fat signal, effects moderate extension of TE and TR.
Water excit. fast	Suppresses the fat signal and causes slight extension of TE and TR. The effect on fat saturation is slightly less than in Water excit. normal mode.
Q-fat sat.	Suppresses the fat signal. This mode is only available when selecting Quick mode in the Saturation mode .
SPAIR	Prior to image saturation, the fat signal is suppressed via a frequency-selective adiabatic inversion pulse. However, this option is only available with certain sequences for 1.5 T and 3 T systems.
None	Does not suppress the fat portion in the MR signal.

⇒ To shorten the measurement time in connection with fat sat, select **Quick** Mode in the **Saturation Mode** parameter.

(8) *Fat sat. mode*

This parameter determines the extent of fat suppression.

(9) *Water suppr.*

This parameter suppresses the signal of water protons in a targeted manner.

The following settings are available:

Adjustment	Description
Water sat.	Suppresses the water signal, has no effect on TE, TR is greatly prolonged. This extends the measurement time as well.
Fat excit.	Suppresses the water signal and has a moderate effect on TE and TR. This leads to a slightly longer measurement time.
None	Does not suppress the water portion in the MR signal.

When you select **Quick** setting in the **Saturation mode** parameter in connection with fat saturation, you are able to shorten the measurement time.

(10) *Blood suppr.*

This parameter determines whether the blood signal is suppressed by increased gradient spoiling. As a result, the minimum possible echo time TE is increased.

The following settings are available:

Adjustment	Description
On	Blood suppression by increased gradient spoiling.
Off	No blood suppression.

⇒ This parameter is only available for the se sequence.

(11) *Restore magn.*

This parameter accelerates restoring the longitudinal magnetization. This increases the signal for T2 weighted acquisitions (even if short repetition times are used).

(12) *SWI*

The parameter activates reconstruction of susceptibility-weighted images.

(13) *T2 prep. Duration*

This parameter defines the duration of the T2 preparation pulse train to increase T2 contrast.

⇨ This parameter is active only when the **T2 Preparation** option has been selected in the **Magn. Preparation** selection list.

(14) *Dixon*

This parameter determines whether the fat and water signal are displayed separately. For this purpose, the Dixon method separates the MR signal into a fat and water signal.

The following settings are available:

Adjustment	Description
No Dixon	The Dixon method for separation of the MR signal is <i>not</i> applied.
Water image	The Dixon method is performed and an image is reconstructed that contains the water signal only.
Fat image	The Dixon method is performed and an image is reconstructed that contains the fat signal only.
Water + fat image	The Dixon method is performed and two images are reconstructed: one image contains only the water signal, the other only the fat signal.

(15) *Lines per shot*

This parameter defines the number of slices to be measured with one shot. For example, the number of lines that are acquired after a fat sat pulse.

⇒ This parameter is available only, when the **Q-fat sat.** or **SPAIR** mode has been selected under the **Fat suppr.** parameter.

(16) *Save original images*

This parameter determines whether the original in-phase and opposed-phase images are saved.

⇒ This parameter has to be activated when the **Dixon** parameter is enabled.

(17) *Freeze suppressed tissue*

When selecting this parameter, the system computes the T1-value of the tissue that is currently suppressed by the TR/TI combination selected and saves it. Effective immediately, a change in the repetition time (TR) automatically adjusts the inversion time (TI), so that the tissue with the saved T1-value is suppressed by the new TR/TI combination as well.

(18) *Minimize Inflow*

Liquor-suppressed head measurements may be affected by liquor from the neighboring slices. Since this inflow of liquor was not inverted at the inversion time, it results in undesirable signal contributions. This signal contribution is minimized as much as possible when selecting this parameter.

- When activating this parameter, the max.possible number of slices per TR may be reduced.

Contrast - Dynamic

The screenshot displays the 'Contrast - Dynamic' parameter card. The 'Dynamic' tab is active. The parameters are as follows:

Parameter	Value
Averages	4
Averaging mode	Short term
Infinite measurement	☐
Measurements	13
Delay in TR	0.001 ms
Breath-hold meas.	10
Reconstruction	Magnitude
Multiple series	Off

The bottom navigation bar includes the following tabs: Program, Routine, Contrast, Resolution, Geometry, System, Physio, and Sequence.

(1) *Averages*

This parameter determines the number of repeat measurements to improve the signal-to-noise ratios. The results of repeat measurements are averaged by the system.

- Please note that the greater the number of measurements, the longer the measurement time.

(2) *Averaging mode*

This parameter determines the method for averaging measurements when several measurements are performed.

- ☐ **Short term**
effects a better signal-to-noise ratio while maintaining optimal resolution
- ☐ **Long term**
effects a better signal-to-noise ratio while maintaining optimal resolution

(3) *Reconstruction*

This parameter determines the image type(s) that have to be reconstructed.

Image type	Description
Magnitude images	MR images are usually so-called magnitude images. The MR signal intensity is displayed directly in these images.
Phase images	The grayscales in phase images correspond to the spin phases (between -180° and $+180^\circ$). These depend on the velocity with which spins move in the body (e.g., blood flow velocity). Spins with the same phase position (i.e., velocity) have the same grayscale value in this type of image.
Real images	The grayscale distribution in real images shows the real alignment of the longitudinal magnetization after an inversion pulse.

- Real images can be selected only when an inversion pulse has been chosen.

(4) *Infinite measurement*

For real-time sequences, this parameter sets the number of measurements to the max. value.

⇒ When the **Infinite measurement** checkbox is activated, the **Measurement** parameter is hidden.

(5) *Measurements*

This parameter determines how often a measurement is to be performed.

(6) *Delay in TR*

This parameter determines the time between two subsequent measurements. The parameter is relevant for all ep2d sequences.

⇒ The delay time set applies to all measurements.

(7) *Breath-hold measurements*

In the case of triggered respiratory control (respiratory control = Breath-hold & Trigger or Breath-hold & Trigger & Follow) this parameter defines how many measurements to perform during the first breathhold. All subsequent measurements are triggered. The number of measurements taken in total are determined by the **Measurement** parameters.

- This parameter is active only when selecting **Breath-holds & Trigger** or **Breath-hold & Trigger & Follow** under **Resp. control** of the **Physio - PACE** parameter card.

(8) *Multiple series*

This parameter determines whether the images for each measurement cycle are stored in their own series.

The following settings are available:

Adjustment	Description
Each measurement	A separate series is generated for each series.
Each slice (not for BOLD sequences)	A separate series is generated for each slice.
Each slice and measurement (not for BOLD sequences)	A separate series is generated for each measurement and each slice.
Off	All images of a measurement are stored together in one series.

(9) *Pause after Meas.*

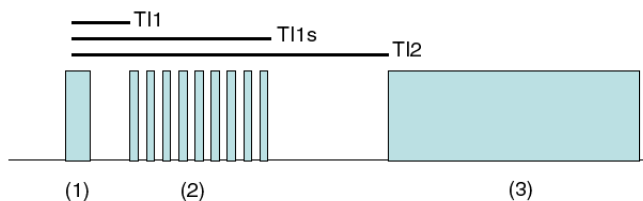
During dynamic measurements, this parameter determines a delay time (in seconds) between the individual measurements. You separately determine the pause for each measurement:

- ✧ Click the lower arrow keys to get to the pause the duration of which you want to change.
- ✧ Enter the desired value in the input field.
- ⇒ Up to 64 individual pauses are possible. In most cases, you are setting the same pause time for all measurements or set the pause time to zero. Beginning with 65 pauses, you can only enter a general pause time.

Contrast - ASL

Common	Dynamic	ASL
Perfusion mode PICORE Q2T		
Inversion time 1 700 ms		
Saturation stop time 1800 ms		
Inversion time 2 1800.0 ms		
Flow limit 100.0 cm/s		
Inversion time 1 25 700 1600		
Program	Routine	Contrast
Resolution	Geometry	System
BOLD	Sequence	

Diagram of TI parameters



- (1) Inversion
- (2) Saturation
- (3) Image acquisition

(1) *Perfusion mode*

This parameter determines the ASL perfusion preparation.

The following settings are presently available:

Adjustment	Description
PICORE Q2TIPS	The PICORE label scheme is used in combination with periodic saturation pulses (Q2TIPS). FOCI pulses are used together with inversion.

(2) *Inversion time 1*

This parameter (TI1) determines the time between the inversion pulse and the first saturation pulse.

→ [Page P.1–30](#), *Diagram of TI parameters*

(3) *Saturation stop time*

This parameter (TI1s) determines the stop time of the periodic saturation.

⇨ When the saturation stop time is set to a minimum, saturation is switched off

→ [Page P.1–30](#), *Diagram of TI parameters*

(4) *Inversion time 2*

This parameter (TI2) determines the time between the inversion pulse and excitation of the first slice.

→ [Page P.1–30](#), *Diagram of TI parameters*

(5) *Flow limit*

This parameter is used in ASL to limit flow in order to attenuate the signal from large arteries.

⇨ The flow limit is switched off when set to maximum.

Resolution - Common

Common	iPAT	Filter
FoV read	350	mm
FoV phase	87.5	%
Slice thickness	6.0	mm
Base resolution	256	
Phase resolution	70	%
Phase partial Fourier	Off	
Interpolation		☒

Program Routine Contrast Resolution Geometry System Physio Inline Sequence

With image resolution you determine how large and how detailed the images will be displayed that have been reconstructed from the raw data. The higher the image resolution, the longer the acquisition time.

All parameters of importance for image resolution (in part different for 2D and 3D measurements) are listed on the **Resolution - Common** parameter card.

(1) *FoV read*

This parameter determines the size of the anatomical region to be displayed (extension of the measurement in the readout and phase-encoding direction) and its resolution (pixel size).

➤ When the **Tim CT mode** is active, this parameter determines the anatomical region required during RF excitation.

(2) *FoV phase*

The FoV in the phase-encoding direction (**FoV phase**) is shown as a percentage of the FoV in the readout direction (**FoV read**).

(3) *Slice thickness*

Together with the number of slices, this parameter determines the number of extension of the measurement area in the slice selection direction.

2D measurements

In the case of 2D measurements, the slice thickness corresponds to the thickness of a slice within a slice group.

When you change the slice thickness for 2D measurements, you also change the distance between the slices.

3D measurements

For 3D measurements, the slice thickness is the thickness of a slice within a slab (partitions).

When you change the slice thickness with 3D measurements, you change the thickness of the slab as well.

- ⇒ Increasing the slice thickness improves the signal-to-noise ratio, but degrades the spatial resolution in the slice-selection direction.

(4) *Base resolution*

This parameter determines the number of pixels in the readout direction. This determines the spatial resolution in the readout direction.

The base resolution is also the reference value for specifying the percentage of resolution in the phase-encoding direction.

The basic resolution also determines the size of the image matrix that can be doubled by selecting interpolation.

(5) *Phase resolution*

This parameter determines the resolution of the slices in the phase-encoding direction and is provided in percentages referenced to the basic resolution.

If phase resolution is 100%, the resolution in the readout and phase-encoding directions will have the same value and the pixels are square. At e.g., 75%, the pixels are rectangular, and resolution decreases.

(6) *Slice resolution*

In 3D measurements, phase-encoding is performed in addition to the slice selection and phase-encoding directions.

This parameter determines the resolution in the slice-selection direction. The resolution ratio is expressed as a percentage of the readout steps.

With slice resolution at 100%, the phase-encoding table in the slice-selection direction has as many steps as the number of slices (partitions) to be reconstructed.

For < 100% slice resolution, fewer phase-encoded steps are measured in the slice-selection direction. This reduces the measurement time. Additional slices are subsequently calculated using interpolation. The slice thickness displayed in the image text does not change. An "i" is simply appended to its numeric value to identify it (e.g., **SL 2.0i**).

(7) *Phase partial Fourier*

If partial Fourier matrices are used for 2D measurements, only part (at least half) of the phase-encoding steps are acquired.

This parameter determines that the k-space is asymmetrically sampled in the slice selection direction. As a result, the measurement is reduced at an unchanged spatial resolution. However, the signal-to-noise ratio is reduced.

The following settings are available:

Adjustment	Description
4/8 (half Fourier), 5/8 , 6/8 , 7/8	Number of acquired phase-encoding steps
Allowed	The sequence automatically computes and uses the most optimal setting of the phase partial Fourier.
Off	The entire image matrix is used.

➤ Different modes are available depending on the sequence.

(8) *Slice partial Fourier*

During 3D measurements, you are able to change the image matrix and measurement time not only via the phase-encoding and readout direction, but also via the slice-selection direction.

This parameter determines that the k-space is asymmetrically sampled in the slice selection direction. As a result, the measurement is reduced at an unchanged spatial resolution. However, the signal-to-noise ratio is reduced.

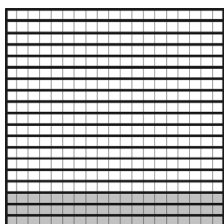
The following settings are available:

Adjustment	Description
4/8 (half Fourier), 5/8 , 6/8 , 7/8	Number of acquired phase-encoding steps
Allowed	The sequence automatically computes and uses the most optimal setting of the phase partial Fourier.
Off	The entire image matrix is used.

(9) Trajectory

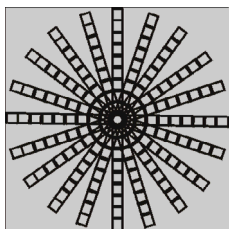
This parameter defines the geometric shape to be sampled in the k-space.

The following shapes are possible:



Cartesian

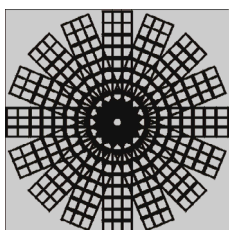
The k-space is sampled as a matrix of rows and columns. The k-space is built up line by line, for example, from the bottom left to the top right.



Radial

The k-space is read out in individual lines. The lines form the shape of a star.

☞ This mode is only available, if Advanced Cardiac is installed.



BLADE

Data are acquired in so-called blades. Every blade comprises parallel phase-encoding lines. The individual blades are rotated in order to cover a circle in the raw data space.

☞ The number of lines per blade is determined in the **Turbo Factor** parameter on the **Sequence - Part 2** parameter card.

(10) *View sharing*

This parameter defines whether k-space data already acquired in a measurement should be reused, for example to obtain higher temporal resolution.

The following settings are available:

Adjustment	Description
Shared phases	During a movie measurement, data from the outer k- spaces are copied into the following (cardiac) phase.
Off	No reuse of previously acquired data.
TWIST	The data from the outer k-space are dynamically determined from the data of several measurements.

(11) *Interpolation*

This parameter magnifies the image matrix to twice its size without increasing the measurement time.

After interpolation, the transitions in the image will be softer because the pixels are now smaller. You need four times the memory to store the image in the database.



➤ In an interpolated image, the interpolated matrix, *not* the scan matrix, is shown bottom right in the image text. Interpolation is indicated by the I after the matrix.

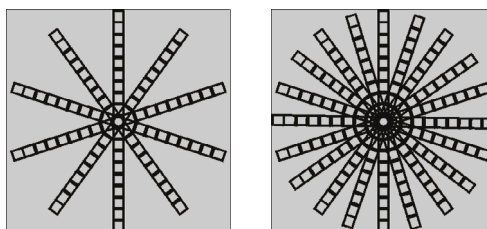


Interpolation doubles the size of the image matrix in the read-out and phase-encoding directions. No interpolation is performed in the slice-selection direction.

If you increase the base resolution, interpolation is deactivated automatically. Otherwise, the reconstruction time will increase considerably. You can activate interpolation at any time.

(12) *Radial views*

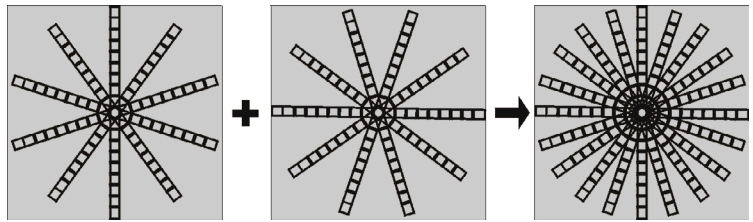
For radial sequences, this parameter determines the number of lines for radial k-space sampling.



Example: k-space sampling with 5 and 10 radial lines

(13) *Radial Interleaves*

This parameter determines the number of TR intervals required to radially sample all lines of the k-space.



Example: For **Radial Interleaves** = 2 all requested lines of the k-space are sampled in two TR intervals.

(14) *BLADE coverage*

This parameter calculates the number of blades of varying rotation angles for k-space sampling.

The value is indicated as a percentage.

☐ **100%**

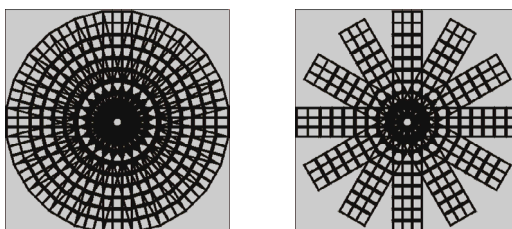
The number of blades is selected so that a circle in the k-space is completely covered.

☐ **Higher than 100%**

More blades are measured than necessary for complete coverage. The measurement time increases and the signal-to-noise ratio improves.

☐ **Less than 100%**

Less blades are measured than necessary for complete coverage. Measurement time is reduced and image quality degraded.



Example: *BLADE coverage* = 100% (in this case 12 blades) and
BLADE coverage = 50% (in this case 6 blades)

(15) *Central region A*

This parameter determines the size of the k-space center that is acquired in full resolution.

This value is provided as a percentage of the sample points of the total k-space that will be measured.

☞ This parameter is only active when you selected the **TWIST** mode under the **View sharing** parameter.

(16) *Sampling density B*

This parameter determines the sampling density used to sample the individual temporal phases in the outer k-space.

☞ This parameter is only active when you selected the **TWIST** mode under the **View sharing** parameter.

Resolution - iPAT

The screenshot displays the 'iPAT' parameter card within a software interface. The card is divided into two main sections by a vertical blue line. The left section contains parameters for PAT mode, acceleration factors, and reference lines for both 2D and 3D. The right section contains parameters for Matrix Coil Mode and Reference scan mode. At the bottom, a horizontal bar contains tabs for different parameter categories.

Common	iPAT
PAT mode	GRAPPA
Accel. factor PE	2
Max. recomm. factor PE	1
Ref. lines PE	24
Accel. factor 3D	2
Max. recomm. factor 3D	1
Ref. lines 3D	24
Matrix Coil Mode	Auto (Triple)
Reference scan mode	Undefined

Program Routine Contrast Resolution Geometry System Physio Sequence

These are the parameters for PAT reconstruction that shorten the measurement time (PAT = parallel acquisition technology).

(1) *PAT mode*

This parameter determines the PAT reconstruction mode. The following modes are available:

- ☐ **None:** The PAT reconstruction technology is not being used.
- ☐ **GRAPPA:** PAT reconstruction mode based on the GRAPPA algorithm.
- ☐ **mSENSE:** PAT reconstruction mode based on the mSENSE algorithm.
- A PAT mode can be selected only when at least two coil elements or RF receive channels are available.

The number of RF receive channels depends on the selected coils. With Matrix Coils, the matrix coil mode has to be considered as well.

- ☐ Matrix Coil mode **CP:** one RF receive channel per selected Matrix Coil element cluster
- ☐ Matrix Coil mode **Dual:** two RF receive channel per selected Matrix Coil element cluster
- ☐ Matrix Coil mode **Triple:** three RF receive channel per selected Matrix Coil element cluster

(2) *Accel. Factor PE*

During PAT reconstruction, this parameter determines the acceleration factor in the phase-encoding direction.

The max. acceleration factor in phase-encoding direction corresponds to the number of receive channels used.

(3) *Max. recomm. factor PE*

This field shows the max. **Accel. factor PE** recommended by the system.

The system determines the value of the table position, the position of the slices, and the position of the selected coil elements.

➤ The parameter is visible in the Online Editor only.



Values higher than the max. recommended **Accel. Factor PE** may have a negative effect on image quality.

(4) *Ref. lines PE*

During PAT reconstruction, this parameter determines the number of reference lines in the phase-encoding direction.

The maximum number of reference lines in the phase-encoding direction corresponds to the number of lines in the phase-encoding direction. It is provided dynamically depending on the sequence and protocol parameters.

(5) *Accel. factor 3D*

During 3D PAT reconstruction, this parameter determines the acceleration factor in the slice-selection direction.

The max. acceleration factor in the slice-selection direction corresponds to the number of receive channels used.

- ⇒ When changing the parameter **Accel. factor 3D**, slice oversampling may have to be slightly adjusted depending on the number of selected slices (adjustment is automatic via the **Scan Assistant** dialog window).

(6) *Max. recomm. factor 3D*

This field shows the max. **Accel. factor 3D** recommended by the system.

The system determines the value of the table position, the position of the slices, and the position of the selected coil elements.

- ⇒ The parameter is visible in the Online Editor only.



Values higher than the max. recommended **Accel. Factor 3D** may have a negative effect on image quality.

(7) *Ref. lines 3D*

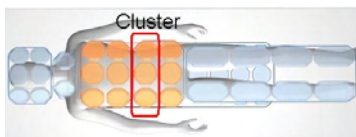
This parameter determines the number of reference lines in the slice-selection direction.

- ⇒ This parameter may be changed only when the protocol is based on a 3D sequence that supports PAT in 3D.

The maximum number of reference lines in the slice selection direction corresponds to the number of slices of the slabs.

- ⇒ The maximum number of reference lines is provided dynamically according to sequence and protocol parameters.

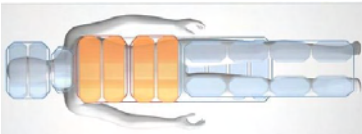
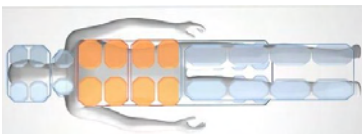
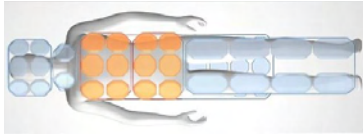
(8) *Matrix Coil Mode*



Matrix coils comprise one or several coil clusters in the cranial-caudal direction. Every coil cluster generally contains three individual coil elements arranged in the left-right direction.

Matrix Coils can be operated in different Matrix Coil modes. The Matrix Coil mode affects each coil cluster.



CP Mode (Circularly Polarized)	Dual Mode	Triple Mode
		
The three coil elements of the cluster behave like a CP coil element. The coil cluster is read via an RF receive channel.	The three coil elements of the cluster behave like two CP coil elements. The coil cluster is read via two RF receive channels.	The three coil elements of the cluster behave like three individual coil elements. The coil cluster is read via three RF receive channels.
Lowest data volume	PAT factor 2 is possible in the left-right direction.	PAT factor 3 is possible in the left-right direction.
One RF receive channel per coil cluster	Two RF receive channels per coil cluster	Three RF coil channels per coil cluster

As a function of the PAT mode selected, the system changes the Matrix Coil mode, as required. The currently used Matrix Coil mode appears in brackets after the **Auto** entry.

Matrix Coil mode recommended by the system **Auto (...)**:

PAT Mode	Matrix Coil Mode
None	The Matrix Coil mode is set to Auto (CP) for optimizing the reconstruction times.
GRAPPA mSENSE	The Matrix Coil mode is set to Auto (Triple) . If you set the Matrix Coil mode Dual prior to enabling Auto (...) , this setting remains.



The Matrix Coil mode applies to matrix coils only - other coils are not affected.

(9) *Reference scan mode*

The PAT reconstruction mode requires that every measurement contains reference lines for calibration.

This parameter determines the way in which the reference lines are measured.

The following settings are available:

Adjustment	Description
Integrated	The reference lines are part of the measurement procedure of the sequence. This method has the advantage of being extremely robust, for example in regions with fast motion.

Adjustment	Description
Separate	The reference lines are measured together with the sequence, however, separately just prior to actually measuring the image data. This method is faster, e.g., for measurements with high acceleration factors and low resolution. With 3D sequences this method also supports fat saturation in the Quick saturation mode (Geometry - Saturation parameter card). The method also supports physiologically-triggered measurements for 3D sequences.
Self-calibrated	<i>For special measurements only when averaging > 1</i> Additional reference lines are not measured because the lines from the repetitions are reconstructed. The PAT factor has to be smaller or equal to the value of the Averages parameter.
tPat	<i>For special measurements only when averaging > 1 or phases > 1</i> Additional reference lines are not measured because the lines from the repetitions are reconstructed. The PAT factor has to be smaller or equal to the value of the Measurements or Phase parameters.

☞ This parameter is not available for all PAT sequences.

(10) *tPAT average all frames*

This parameter determines if all temporary frames are used to calculate the sensitivity cards of tPAT reconstruction.

- ⇒ This parameter is available only, when the **tPAT** mode has been selected under the **Reference measurement mode** parameter.

(11) *tPAT averaged frames*

This parameter determines if all temporary frames are used to calculate the sensitivity cards of tPAT reconstruction.

- ⇒ This parameter is available only, when the **tPAT** mode has been selected under the **Reference measurement mode** parameter and the **tPAT average all frames** parameter is not activated.

Resolution - Filter

Common	iPAT	Filter
Image Filter ☒ Intensity Medium Edge Enhancement 3 Smoothing 3 Unfiltered images ☒ Distortion Corr. ☒ Mode 2D Unfiltered images ☐		
Prescan Normalize ☐ Normalize ☒ ... Unfiltered images ☐ B1 filter ☐ Raw filter ☐ Elliptical filter ☐ POCS Read & Phase		
Program	Routine	Contrast Resolution Geometry System Physio Angio Sequence

(1) Raw filter

The outer lines of the raw data matrix contain the sharpness information of an image. You can weight specific lines with the raw data filter, for example, to suppress edge oscillation.

(2) *Intensity*

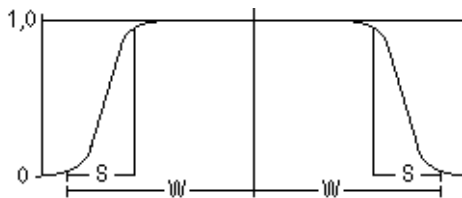
This parameter determines the intensity of the filter.



The image will be more homogeneous by increasing the filter intensity. However, image contrast may be compromised at the same time. This means that weak filters should be used whenever possible.

(3) *Slope*

This parameter shows the slope along the flanks of the filter.



S: Slope at the flanks of the filter

W: filter width

(4) *Normalize*

If you are using surface coils, the area in the vicinity of the coil will appear lighter in the images and darker in the areas farther from the coil. The signal intensity is greater in the vicinity of the coil.

Use the normalization filter to reduce the brightness of areas in the vicinity of the coil and increase the brightness in areas farther away from the coil.

-
- ! Use of the normalization filter may lead to a loss of contrast and an increase in background noise.
-

(5) *Cut off*

This parameter determines the level of brightness below which the pixels are no longer corrected with the normalization method, but rather with a fixed value.

- ⇒ The threshold is given as a percentage of the maximum intensity profile. For a value of **20**, e.g., all pixels with an intensity that lies below 20% of the maximum value, a factor of 5 (=1/20%) only is used for correction.

(6) *Width*

This parameter determines the width of the filter

- ⇒ When you selected **Free** in the **Intensity** selection list, you can determine the increase yourself.

(7) *Unfiltered images*

This parameter determines whether unfiltered images should be saved as well.

- ⇒ If you reconstructed and stored unfiltered as well as filtered images in Inline, all other inline functions can no longer be accessed.
Filtered images are always shown in the Inline display.

(8) Prescan Normalize

This filter corrects for signal intensity decays caused by the coil profiles. It works the similar way as the **Normalize** filter but the data used for homogenization are acquired through an adjustment measurement.

The following settings are available:

Adjustment	Description
Normal	This is the default mode, which will correct for signal intensity decays caused by the coil profiles.
Moderate	In Normal mode, noise in the center of the image might be enhanced too much. The Moderate mode does a progressively weakened normalization, so that the optical impression can be less noisy. However, the resulting images will show remainders of the original coils sensitivity profiles.
Broad Range	The Normal mode utilizes a signal threshold, which is used to avoid noise enhancement in regions far from the coils. In Broad Range mode, the threshold is decreased, so that a wider range of the object is normalized. Possible applications are spine images, if the spine is distant from the coil.

➤ Use the **Normalize** filter instead of the **Prescan Normalize** filter if :

- ☐ you use the Body Coil for the measurement,
- ☐ you use other nuclei as H-nuclei for image generation,
- ☐ the **Prescan Normalize** checkbox is disabled.

(9) *B1 filter*

B1 filter is a homomorphous filter that can be used to reduce signal differences in MR images caused by dielectric resonances at field strengths of 3T and higher.

The following filter intensities can be set if the parameter is activated:

- ☐ Weak
- ☐ Medium
- ☐ Strong

(10) *Distortion Corr.*

This parameter activates 2D or 3D distortion correction. This compensates for the pillow-shaped distortions at the edge of the image. These distortions occur in images with a large FoV or eccentric slices (Offcenter).

The following settings are available:

Adjustment	Description
2D	With 2D distortion correction , image distortion is corrected individually in each slice.
3D	With 3D distortion correction , the voxel in the current slice as well as those in the surrounding slices are taken into account. The correction results are more precise, but require a longer reconstruction time.

(11) *Image filter*

The image filter is used to set the intensity, edge enhancement, and smoothing.

⇒ Beginning with software version VB 15A this filter is activated by default for every protocol. For changes on the default setting please contact our customer service.

(12) *Edge enhancement*

This parameter determines the edge enhancement value of the filter.

(13) *Smoothing*

This parameter determines the limit for filter smoothing.

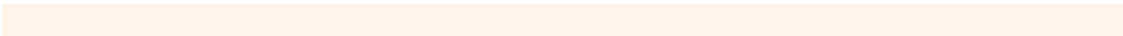
(14) *Elliptical filter*

With the elliptical filter, you will only use the center of the raw data space (the corners are set to 0).

In this way you improve the signal-to-noise ratio up to 10% (without loss in resolution).

The following settings are available:

Adjustment	Description
In the plane	This filter is only used within the image planes.
Volume	The filter is applied to the entire volume.



(15) *POCS (Projection Onto Convex Sets)*

This parameter improves the edge sharpness for Partial Fourier sampling. Missing k-space pixels are not set to zero but are extrapolated.

The following settings are available:

Adjustment	Description
Off	No POCS.
Read slice	POCS in readout and slice direction
Read phase	POCS in readout and phase-encoding direction

Geometry - Common

Common	Saturation	Navigator	Tim Planning Suite
Slab group	1		FoV read 300 mm
Slabs	18		FoV phase 100.0 %
Dist. factor	50 %		Slice thickness 2.00 mm
Position	Isocenter		TR 3000.0 ms
Orientation	Transversal		Multi-slice mode Interleaved
Phase enc. dir.	A >> P		Series Interleaved
Phase oversampling	25 %		Concatenations 2
Slice oversampling	25 %		
Slices per slab	64		

Program Routine Contrast Resolution **Geometry** System Physio Sequence

e.g.: Parameter card Geometry - Common (3D)

During routine examinations you graphically plan the orientation of your slice and slab groups. You are able to numerically enter the graphic objects in the **Geometry** parameter card.

The parameter card is subdivided into three sub-cards:

❑ **Common**

Here you will find all parameters pertaining to the position and size of the slices or slabs to be measured (may be different for 2D and 3D measurements).

❑ **Saturation**

These are the parameters relevant when planning saturation regions and/or fat /water saturation.

❑ **Navigator**

These are the parameters for navigator objects.

(1) *Slab group*

The slabs measured within the framework of a protocol are combined into groups.

Here you can see the number of the slab group currently displayed. All slice parameters, currently seen on the **Routine** or **Geometry** card refer to this slab group.

(2) *Slabs*

This parameter determines the number of slabs in this group.

(3) *Dist. factor*

This parameter determines the gap between slices and/or slabs of a group in percentages.

At a 100% value the gap corresponds exactly to one slice and/or slab thickness. Negative values lead to overlapping.

✧ Negative distance values may not be entered for 3D measurements.

(4) *Position*

This parameter defines the position of the object center.

The following information is displayed when you position the mouse pointer on the field:

☐ **Table position**

Table position of the protocol, referenced to the whole body patient coordinate system.

☐ **Phase offcenter**

Shift in phase-encoding direction

☐ **Read offcenter**

Shift in readout direction

☐ **Slice shift**

Shift in slice-selection direction

✧ The selection list is dimmed if the position matches the iso-center.



Use this key to open the **Position** dialog window for the slice selected.

→ [Page P.1–220](#), *Position dialog window*

(5) *Orientation*

This parameter indicates the position of the object in space based on the whole body patient coordinate system.



Use this key to open the **Orientation** dialog window for the slice selected.

→ [Page P.1–219](#), *Orientation dialog window*

(6) *Phase enc. dir.*

The current phase-encoding direction (direction of the phase-encoding gradient) is indicated in the main orientations of the whole body patient coordinate system.

You can change the phase-encoding direction. This will swap the phase-encoding and readout direction. Using this method allows you to prevent aliasing artifacts in the phase-encoding direction or change the direction of flow and motion artifacts.

⇒ In the **Phase enc. dir.** selection list you are only offered configurations suitable to the current orientation.



Use this button to open the **Inplane Rotation** dialog window for the slice selected.

→ [Page P.1–222](#), *Inplane Rotation dialog window*

(7) *Phase oversampling*

This parameter increases the phase-encoded area symmetrically at both sides of the field of view (FoV). The expanded FoV area is not shown. It is used to avoid overfolding artifacts.

Phase oversampling is shown as a percentage of the FoV in the phase-encoding direction.

- ⇒ Phase oversampling increases the measurement time. The signal-to-noise ratio is improved
- Oversampling is always automatically performed in the read-out direction so it does not lead to an increase in measurement time.

(8) *Slice oversampling*

This parameter enlarges the phase-encoding area symmetrically on both sides of the slab in the slice selection direction. This extended FoV area is not shown. It is used to avoid overfolding artifacts.

Slice oversampling is indicated as a percentage of the slab thickness.

- ⇒ The measurement time is prolonged through slice oversampling. The signal-to-noise ratio is improved.

(9) *Slices per slab*

This parameter determines the number of slices per slab:

Changes of the **Slices per slab** affect the following:

- ☐ Thickness of slab
- ☐ The absolute slice oversampling as percentage of the slab.

(10) *FoV read*

This parameter determines the size of the anatomical region to be displayed (extension of the measurement in the readout and phase-encoding direction) and its resolution (pixel size).

- When the **Tim CT mode** is active, this parameter determines the anatomical region required during RF excitation.

(11) *FoV phase*

The FoV in the phase-encoding direction (**FoV phase**) is shown as a percentage of the FoV in the readout direction (**FoV read**).

(12) *Slice thickness*

Together with the number of slices, this parameter determines the extension of the measurement area in the slice selection direction.

2D measurements

In the case of 2D measurements, the slice thickness corresponds to the thickness of a slice within a slice group.

When you change the slice thickness for 2D measurements, you also change the distance between the slices.

3D measurements

For 3D measurements, the slice thickness is the thickness of a slice within a slab (partitions).

When you change the slice thickness with 3D measurements, you change the thickness of the slab as well.

⇒ Increasing the slice thickness improves the signal-to-noise ratio, but degrades the spatial resolution in the slice-selection direction.

(13) *TR (repetition time)*

This parameter determines the repetition time TR that elapses between two successive excitations. (At times an alternative designation *Repetition time* is used).

Changing the repetition time affects image contrast and measurement time.

If you would like to enter several repetition times, additional keys are superimposed for scrolling between the individual times.

(14) *Multi-slice mode*

This parameter determines the measurement modes for multi-slice mode:

- ☐ **Sequential** measurement, slice by slice
All lines (phase-encoding steps) of the first slice are measured step-by-step. This is followed by all lines of the second slice, etc.
- ☐ **Interleaved** measurement, line-by-line
All first lines (phase-encoding steps) of all interleaved slices are measured one after the other in repetition time TR. This is followed by all second lines of all interleaved slices, etc.
- ☐ **Single shot** - Special mode for fast sequences.
All lines (phase-encoding steps) of a slice after excitation are measured simultaneously. Then all lines of the second slice are measured together, etc.

(15) Series

This parameter determines the order in which the slices are processed.

- The images are displayed in ascending order according to the image numbers after reconstruction, regardless of the sequence for slice processing.

The following excitation sequences are available:

☐ Ascending (1)

The slices are excited at the beginning of the slice or slab group (beginning → end).

☐ Descending (2)

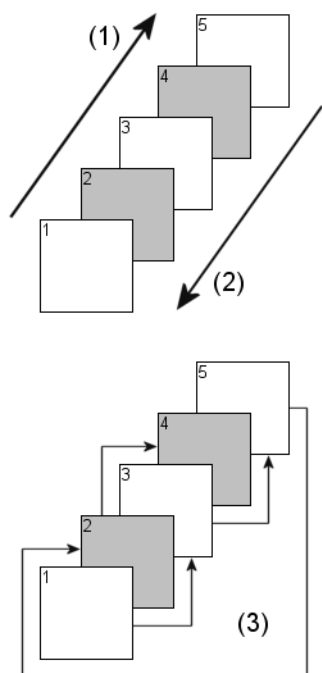
The slices are excited at the end of the slice or slab group (end → beginning).

☐ Interleaved (3)

☐ Interleaved in A. A. (without display)

(Interleaved in breathhold interval). The slices are measured separately in the **Interleaved** mode for each breathhold interval of a multiple breathhold measurement.

- Input **Interleaved in A. A.** is superposed only when you are planning a measurement in a multiple breathhold mode and selected one of the inputs **Breath-hold**, **Breath-hold & Monitor** or **Breath-hold & Follow** under **Breath-hold** on the **Physio - PACE** parameter card.



- ⇒ You can combine the **Multi-slice mode Interleaved** and **Series Interleaved** settings for a very short TR and small slice distance to avoid cross-talk.
- ⇒ Determine the main orientation of the slices (e.g., sagittal-coronal-transverse) using the **MSMA** parameter.

(16) *Concatenations*

This parameter defines into how many repetition times TR the measurement of the planned slices should be divided. The system then determines across how many single sequential measurements the slices will be distributed.

Using this method, you can acquire many slices with a short repetition time (T1 weighted imaging) and prevent slice crosstalk.

- ⇒ When you selected a measurement in multiple breathhold mode and decided to choose one of the inputs **Breath-hold**, **Breath-hold & Monitor** or **Breath-hold & Follow** under **Respiratory Control**, the number of breathhold intervals is determined via the **Concatenation** parameter.
- ❑ In the **Interleaved** multi-slice mode, the number of breathhold intervals results from the product of the parameter values **Measurements** and **Concatenations**.
- ❑ In the **Single Shot** multi-slice mode, the number of breathhold intervals is the product of the parameter values **Measurements**, **Concatenations** and **Averages**.

- ⇒ With triggered multislice measurements ("interleaved" excitation sequence), it is sometimes not possible to acquire all slices in one measurement. The slices missing from one measurement will then be acquired in the next measurement.
- ⇒ This field is active only, when the **Interleaved** or **Single Shot** option was selected in the **Multi-slice mode** on the **Geometry - Common** parameter card.

(17) *Slice group*

Slices measured within the framework of a protocol are combined into a group.

This parameter shows the number of the currently displayed slice group. All slice parameters that are currently seen on the **Routine** or **Geometry** task card refer to this slice group.

(18) *Slices*

This parameter determines the number of slices in this group.

Geometry - Saturation

Common	Saturation	Navigator	Tim Planning Suite
Saturation mode: Standard		Fat suppr.: Water excit. norm	
		Fat sat. mode: Weak	
		Water suppr.: None	
		Restore magn.: ☐	
Sat. region: 1		Special sat.: Parallel F	
Thickness: 8 mm		Gap: 10.0 mm	
Position: Isocenter		Thickness: 50 mm	
Orientation: Transversal			

Program | Routine | Contrast | Resolution | **Geometry** | System | Physio | Sequence

Saturation regions

Saturation regions are areas whose signals are saturated by special RF pulses so as to avoid motion artifacts.

Standard and tracking saturation regions

Standard saturation regions are numbered and may be positioned as required.

Parallel or tracking saturation regions are always linked to a slice or slab group. As a result, it is not possible to change the position and orientation of these saturation regions. Instead, the position of these regions is automatically adjusted by the system whenever you move or rotate the slice or slab group.

The position and orientation of parallel or tracking saturation regions are always linked to a slice or slab group. The system automatically adjusts these regions as soon as you shift or rotate the respective slice or slab group.

Prior to each measurement, **tracking saturation regions** are excited at a given distance either in front or behind the slice and at the thickness specified, that is, they track the current slice to be measured within the slice group.

Parallel saturation regions are positioned either on one side or on both sides parallel to the slice group. These sat regions do not travel with the slice, but are rather located at the distance specified either in front or in back of both sides of the slice group.

(1) Saturation mode

This parameter determines how often the saturation pulses are being sent:

Mode	Description
Standard	A saturation pulse is sent prior to each measurement in the Standard saturation mode.
Quick	Only the RF pulses actually required for saturation are sent in the Quick saturation mode. The frequency of sending sat. pulses is added to the run time.

Selecting the **Quick** saturation mode lets you reduce the measurement time with fat/water saturation and regional saturation (regular and parallel saturation regions).

➤ To obtain shorter measurement times, select the **Quick** mode, e.g., for breathhold studies.

(2) Saturation region

This parameter shows you the number of saturation region displayed. Via the **Thickness**, **Orientation** and **Position** parameters, you determine the position of the saturation regions.

You add a new saturation region to the measurement protocol by using the **Plus** button next to the **Sat. region** selection list. You delete the current saturation region by using the **Minus** button.

(3) *Thickness*

This parameter determines the expansion of the slice in millimeters.

(4) *Position*

This parameter defines the position of the object center.

The following information is displayed when you position the mouse pointer on the field:

☐ **Table position**

Table position of the protocol, referenced to the whole body patient coordinate system.

☐ **Phase offcenter**

Shift in phase-encoding direction

☐ **Read offcenter**

Shift in readout direction

☐ **Slice shift**

Shift in slice-selection direction

➤ The selection list is dimmed if the position matches the iso-center.



Use this key to open the **Position** dialog window for the slice selected.

→ [Page P.1–220](#), *Position dialog window*

(5) Orientation

This parameter indicates the position of the object in space based on the whole body patient coordinate system.



Use this key to open the **Orientation** dialog window for the slice selected.

→ [Page P.1–219](#), *Orientation dialog window*

(6) Lipid Suppr.

This parameter suppresses the signal from fat protons in a targeted way.

The following settings are available:

Adjustment	Description
Fat sat.	Suppresses the fat signal and has no effect on TE, TR is greatly prolonged. This extends the measurement time as well.
Water excit. normal	Suppresses the fat signal, effects moderate extension of TE and TR.
Water excit. fast	Suppresses the fat signal and causes slight extension of TE and TR. The effect on fat saturation is slightly less than in Water excit. normal mode.
Q-fat sat.	Suppresses the fat signal. This mode is only available when selecting Quick in the Saturation mode selection list.

Adjustment	Description
SPAIR	Prior to image saturation, the fat signal is suppressed via a frequency-selective adiabatic inversion pulse. However, this option is only available with certain sequences for 1.5 T and 3 T systems.
None	Does not suppress the fat portion in the MR signal.

⇒ To shorten the measurement time in connection with fat sat, select **Quick Mode** in the **Saturation Mode** parameter.

(7) *Fat Sat. mode*

This parameter determines the extent of fat suppression.

(8) *Water suppr.*

This parameter suppresses the signal of water protons in a targeted manner.

The following settings are available:

Adjustment	Description
Water sat.	Suppresses the water signal, has no effect on TE, TR is greatly prolonged. This extends the measurement time as well.
Fat excit.	Suppresses the water signal and has a moderate effect on TE and TR. This leads to a slightly longer measurement time.
None	Does not suppress the water portion in the MR signal.

When you select **Quick** setting in the **Saturation mode** parameter in connection with fat saturation, you are able to shorten the measurement time.

(9) *Restore magn.*

This parameter accelerates restoring the longitudinal magnetization. This increases the signal for T2 weighted acquisitions (even if short repetition times are used).

(10) *Special sat.*

The parameter establishes *exactly one* parallel or tracking saturation region.

- ⇒ Special sats. may be used if there is one slice/slab group only.
- ⇒ Tracking saturation regions cannot be planned in Quick mode.

The information whether the tracking saturation region is located in front or behind the slice or slice group, is shown in the main orientation of the slice group:

- ☐ For **transverse** main orientation: **H** (in front)/**F** (in back)
- ☐ For **sagittal** main orientation: **L** (in front)/**R** (in back)
- ☐ For **coronal** main orientation: **P** (in front)/**A** (in back)

In this case *in front* means that the saturation region is tracking in front, while *back* means that the saturation region is tracking in back.

(11) *Gap*

This parameter determines the distance to the associated slice and/or slab group.

(12) *Thickness*

This parameter determines the expansion of the slice in millimeters.

Geometry - Navigator

Navigator Objects

You can define navigator objects for special navigator sequences, for example, to measure respiratory movement.

(1) *Navigator*

This parameter determines the type of currently displayed navigator objects. All parameters that you can currently see on the **Geometry - Navigator** card refer to this navigator object.

You position the navigator exactly on the subphrenic space. For optimal positioning use coronal as well as transverse images.

(2) *Position*

This parameter defines the position of the object center.

The following information is displayed when you position the mouse pointer on the field:

☐ **Table position**

Table position of the protocol, referenced to the whole body patient coordinate system.

☐ **Phase offcenter**

Shift in phase-encoding direction

☐ **Read offcenter**

Shift in readout direction

☐ **Slice shift**

Shift in slice-selection direction

⇒ The selection list is dimmed if the position matches the iso-center.



Use this key to open the **Position** dialog window for the slice selected.

→ [Page P.1–220](#), *Position dialog window*

(3) *Orientation*

This parameter indicates the position of the object in space based on the whole body patient coordinate system.



Use this key to open the **Orientation** dialog window for the slice selected.

→ [Page P.1–219](#), *Orientation dialog window*

(4) *Rotation*

This parameter determines the angle used to rotate the navigator object. The angle refers to the slice plane determined in the **Orientation** parameter.

Only 90° or 0° are possible as angle of rotation for navigator objects. The 90° angle corresponds to swapping the readout direction with the phase-encoding direction.

(5) *Base size phase*

This parameter determines the expansion of the navigator object in the phase-encoding direction.

(6) *Base size read*

This parameter determines the expansion of the navigator object in the readout direction.

(7) *Thickness*

This parameter determines the expansion of the navigator object in millimeters.

Geometry - Tim Planning Suite

e.g.: Geometry - Tim Planning Suite (Set-n-Go)

Large examination regions cannot be covered by a normal field of view (FoV) without distortions. However, you can divide large measurement regions into several smaller regions at different table positions that fit fully into the isocenter of the system. The individual images can then be composed into a single overall image.

In order to facilitate planning and evaluating multi-slice measurements, group the individual measurements or combine them in a Set-n-Go protocol. Make the required settings on the **Geometry - Tim Planning Suite** parameter card.

(1) *Set-n-Go protocol*

This parameter indicates whether the measurement protocol is part of a Set-n-Go protocol.

Notes:

- ☐ Distortion correction is a prerequisite for Set-n-Go protocols. The **Distortion Corr.** parameter is automatically set to **2D**.
- ☐ The **ISO** or **FIX** mode has to be selected on the **Positioning mode** parameter under the **System - Miscellaneous** parameter card.

(2) *Measurement step*

This parameter defines the step of the Set-n-Go protocol whose parameters can currently be edited.

(3) *Table position*

This parameter determines the table position where the protocol will be measured. The zero point is defined by the initial table position of the first measurement of a series block.

→ [Page C.4–26](#), *Dependencies between the table position, reference image, and protocol*

The selection list defines the direction of movement.

- ☐ **H**: In the head direction
- ☐ **F**: In the foot direction

The input field defines the distance in mm.

(4) *Table position memory*

The list displays all table positions already used for the current series block. You can choose a table position from the list and apply it to your protocol.

(5) *Inline Composing*

This parameter indicates that the images of the steps or the protocol of a composing group can be composed into a complete image.

- ⇒ **Inline Composing** can be enabled only, when **2D** or **3D** is activated in the **Distortion Corr.** field.
- ⇒ The **GLM Statistics** parameter on the **BOLD** parameter card is automatically deactivated.

(6) *Composing algorithm*

This parameter establishes the methods used to compose the images.

The following settings are available:

Adjustment	Description
Spine	This algorithm uses the bone structures in the images as a basis. For example, it generates images for the measurement and evaluation of scoliosis, kyphosis, and pelvic obliquity.
Angio	This algorithm uses the vessel structures of the images as a basis. This allows for overall display of the vessels.
Adaptive	This algorithm is based on elastic matching. It especially corrects B0 induced inhomogeneities in image areas beyond the magnetic homogeneity volume.

Geometry - Tim CT

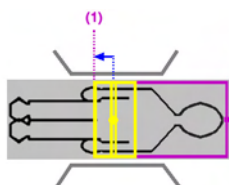
Common	Saturation	Tim Planning Suite	Tim CT
Tim CT mode	On		Recon mode
Range start	H	1255	mm
Total FoV	H >> F	1230	mm
Slabs	1		
Slices per slab	88		
Slice thickness	1.30		mm
Dist. factor	20		%
FoV read	450		mm
FoV phase	68.8		%
			Table Speed
			17.1
			mm/s

Program Routine Contrast Resolution **Geometry** System Angio Sequence

(1) Tim CT mode

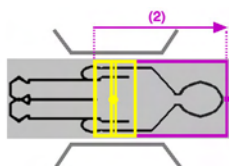
This parameter activates the TimCT mode.

An extended measurement region is determined in the TimCT mode, which is measured in continuous table move.



(2) *Range start*

This parameter defines the start position of the measurement region (1) (measured in the Whole Body Patient Coordinate System).



(3) *Total FoV*

This parameter determines the total length of the measurement region (2).

(4) *Slices*

This parameter determines the number of slices in this group.

(5) *Slice thickness*

Together with the number of slices, this parameter determines the number of extension of the measurement area in the slice selection direction.

2D measurements

In the case of 2D measurements, the slice thickness corresponds to the thickness of a slice within a slice group.

When you change the slice thickness for 2D measurements, you also change the distance between the slices.

3D measurements

For 3D measurements, the slice thickness is the thickness of a slice within a slab (partitions).

When you change the slice thickness with 3D measurements, you change the thickness of the slab as well.

⇒ Increasing the slice thickness improves the signal-to-noise ratio, but degrades the spatial resolution in the slice-selection direction.

(6) *Dist. factor*

This parameter determines the gap between slices and/or slabs of a group in percentages.

At a 100% value the gap corresponds exactly to one slice and/or slab thickness. Negative values lead to overlapping.

⇒ Negative distance values may not be entered for 3D measurements.

(7) *FoV read*

This parameter determines the size of the anatomical region to be displayed (extension of the measurement in the readout and phase-encoding direction) and its resolution (pixel size).

⇒ When the **Tim CT mode** is active, this parameter determines the anatomical region required during RF excitation.

(8) *FoV phase*

The FoV in the phase-encoding direction (**FoV phase**) is shown as a percentage of the FoV in the readout direction (**FoV read**).

(9) *Table speed*

This field shows the table movement speed.

(10) Recon. Mode

This parameter determines how the images are saved.

The following settings are available:

Adjustment	Description
Total FoV	The measurement results are saved as an overall image.
Frames	The measurement results are saved as several individual images. These images fit into one image segment each.
Total FoV and individual images	The measurement results are saved as an overall image. In addition, it is also divided and stored as several individual images.

(11) Variable resolution

This parameter enables you to increase the resolution in the phase-encoding and slice-selection direction to the maximum permissible value in part of the examination region, independent of the values for the **Phase resolution** and **Slice resolution** parameters.

→ [Page P.1–33, Resolution - Common](#)

The following settings are available:

Adjustment	Description
Off	Variable resolution
End of exam region	Increased resolution at the end of the examination region

☞ If the setting is not **Off**, the reconstructed images display variable levels of resolution. In addition, the image series from the respective areas of the examination region are generated with the higher resolution only.

Geometry - Inversion

Common Saturation **Inversion** Navigator Tim Planning Suite

Graph. Sel. Inversion 1

TI 700 ms

Thickness 50 mm

Position Isocenter

Orientation Transversal

Routine Contrast Resolution **Geometry** System Physio Inline Sequence

This parameter card is used for the graphical planning of slice-selective inversion.

(1) *Graph. Sel. Inversion*

This parameter activates the graphical planning of the inversion, if the value is > 0 .

(2) *TI*

This parameter determines the inversion time for the graphical inversion.

The value range is 0 to 10000 ms. The default value is 700 ms.

⇒ This parameter is only active if **Graph. Sel. Inversion** is set to a value > 0.

(3) *Thickness*

This parameter determines the slice thickness for the graphical inversion in millimeters.

The value range is 0 to 150 mm. The default value is 50 mm.

⇒ This parameter is only active if **Graph. Sel. Inversion** is set to a value > 0.

(4) *Position*

This parameter defines the position of the object center.

The following information is displayed when you position the mouse pointer on the field:

☐ **Table position**

Table position of the protocol, referenced to the whole body patient coordinate system.

☐ **Phase offcenter**

Shift in phase-encoding direction

☐ **Read offcenter**

Shift in readout direction

☐ **Slice shift**

Shift in slice-selection direction

➤ The selection list is dimmed if the position matches the iso-center.



Use this key to open the **Position** dialog window for the slice selected.

→ [Page P.1–220](#), *Position dialog window*

(5) *Orientation*

This parameter determines the orientation of the object for graphical inversion.

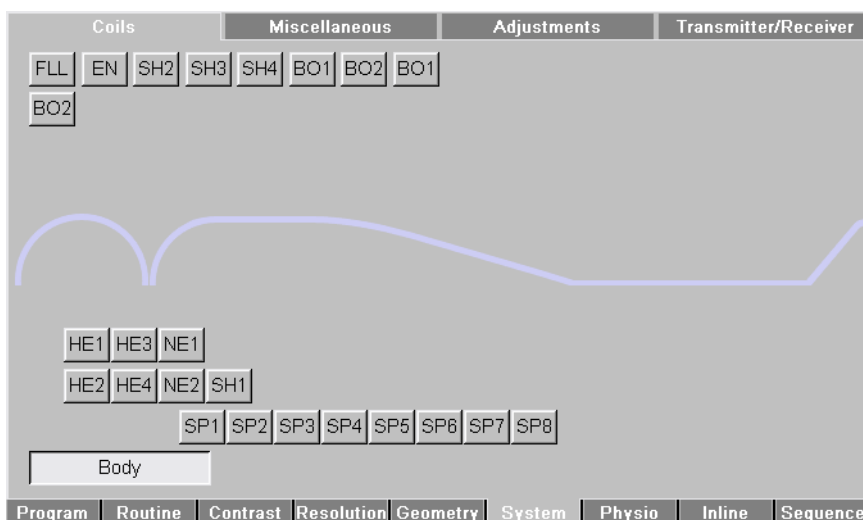
The following settings are available:

✧ sagittal

✧ coronal

✧ transverse

System - Coils

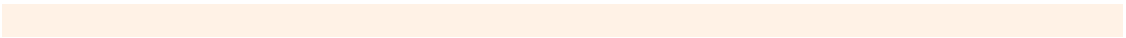


Here you find the positions of the coils connected and their position to the patient.

- ☐ **Coil elements with unknown positions** are displayed in the upper area of the parameter card.
- ☐ **Coil elements with known positions** are displayed in the lower area of the parameter card. These coil elements are also visible in the GSP.

The figure illustrates the position of the patient:

- ☐ Head left: Head first
- ☐ Head right: Feet first



All connected coil elements are shown on the parameter card as buttons without overlaps (maximum 64 coil elements). The user can select or deselect them for the measurement.

Several fully identical coils (e.g., Body Array coils) may be connected to the system. The coils are uniquely identified and differentiated by the software. To superpose the coil name, the name of the coil element and the coil socket number, keep the mouse pointer briefly on the appropriate button.

The *Coil Operator Manual* provides you with a detailed description of the coils and their implementation.

Coil memory

Coil selection is linked to table position. As a result, consecutive protocols with similar table positions automatically accept previously edited and completed predecessor protocols when opened.

You can also switch off coil memory:

✧ For that purpose deselect **Queue > Coil memory**.

Automatic coil selection

When this setting is activated, the coils in the vicinity are automatically selected when opening a protocol automatically.

The algorithm is based on a cuboid-shaped envelope across all slices of the protocol. All coils in the area of this envelope are selected. Other parameters are kept constant whenever possible.

⇒ The maximum number of coils used is limited by the number of system channels.

For large measurement regions on systems with a low number of channels, it may happen that automatic coil selection does not supply the most optimal results.

You can switch off automatic coil selection as well:

✧ Deselect **Queue > Auto Coil select**.

⇒ To ensure that automatic coil selection functions smoothly when using several Body Matrix Coils, the following has to be observed with respect to the numbers assigned to the coil sockets: Coils closer to the magnet have to be connected to a socket with a lower number.

Copying with copy reference

When you copy using copy reference, the target protocol inherits the coil configuration of the source protocol. This new coil configuration is stored at the respective table position.

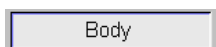
Local coils

Matrix and array coils consist of several elements which you may select independently of one another.

Depending on the examination region relevant to your diagnostic problem, you may activate different numbers of coil elements.

- ⇒ To superpose the coil name, the name of the coil element and the coil socket number, keep the mouse pointer briefly on the appropriate button.

Body coil Spine Matrix



All local coils will be deactivated if you want to perform a measurement with the Body Coil.

- ✧ Click **Body**.
- ⇒ When you deselect the Body Coil again, you activate all previously active local coils and coil elements.



Select the coil elements that are inside the region under examination. Otherwise, image quality may be adversely affected.

Coil elements and iPAT acceleration factor

The number of coil elements or modes corresponds to the max. iPAT acceleration factor. The Matrix Coil mode is taken into consideration for the Matrix Coils.

After the coil elements have been deselected and the acceleration factor set is technically no longer feasible, the acceleration factor is automatically set to the lowest value possible (adjustment using the **Scan Assistant** dialog window).

System - Miscellaneous

The screenshot displays the 'System - Miscellaneous' parameter card. The interface is divided into four main sections: Coils, Miscellaneous, Adjustments, and Transmitter/Receiver. The 'Miscellaneous' section is currently selected and contains the following parameters:

- Positioning mode:** ISO
- Table position:** H, 0 mm
- Table position memory:** (empty dropdown)
- Image Numbering:**
 - MSMA: C - T - S
 - Sagittal: R >> L
 - Coronal: A >> P
 - Transversal: F >> H
- Matrix Coil Mode:** Auto (CP)
- Coil Combine Mode:** Sum of Squares
- Save uncombined:** (checkbox)
- AutoAlign:** ---
- Auto Coil Select:** Default

At the bottom of the interface, there is a horizontal bar with tabs for different parameter categories: Program, Routine, Contrast, Resolution, Geometry, System (highlighted), Physio, Angio, and Sequence.

This is where you set the table position. You can perform the measurement at the current table position or you can define a different position.

You are able to define the numbering for the reconstructed images in the selection list for image numbering.

(1) *Positioning mode*

This parameter determines how the table position of the protocol will be determined or which reference images should be loaded for the protocol.

The following settings are available:

Adjustment	Definition
FIX	The table position is determined by the protocol, suitable reference images are being searched for.
REF	The table position of the protocol is determined by the focused reference image.
ISO	Both the table positions of the reference image and protocol are independent. The common center of all positioned slices indicates the table position. Distortion-corrected reference images (image type DIS2D / DIS3D) have to be used in this mode.

A detailed description of the dependencies between the positioning mode, table position, and reference images is included in
→ [Page C.4–26](#), *Dependencies between the table position, reference image, and protocol*

(2) *Table position*

This parameter determines the table position where the protocol will be measured. The zero point is defined by the initial table position of the first measurement of a series slab.

→ [Page C.4–26](#), *Dependencies between the table position, reference image, and protocol*

The selection list defines the direction of movement.

☐ **H:** In the head direction

☐ **F:** In the foot direction

The input field defines the distance in mm.

(3) *Table position memory*

The list displays all table positions already used for the current series block. You can choose a table position from the list and apply it to your protocol.

(4) *Image numbering*

This parameter determines the image numbering. MSMA (Multi Slice Multi-Angle) is used to establish the primary sequence according to orientation.

Example:

Using setting **S - C - T**, the reconstructed images are numbered in the following sequence:

- ☐ All sagittal images
- ☐ All coronal images
- ☐ All transverse images

In the **sagittal**, **coronal**, and **transverse** selection lists, you can set the secondary sequence. This sequence determines whether the images are numbered in ascending or descending order. The patient coordinates inform about the order (L, R, P, A, H, F). The conventions of the DICOM patient coordinate system apply.

→ [Page A.2–1](#), *Coordinate System*

☐ **Sagittal R >> L or L >> R, Med >> Lat or Lat >> Med**

The sagittal images are numbered in ascending (**R >> L** from right to left or **L >> R** from left to right).

For sagittal slice the images can also be numbered according to their distance from the isocenter (**Med >> Lat** or **Lat >> Med**). Sorting is performed according to the sagittal position of the slices in the LPH *whole body patient coordinate system*.

❑ **Coronal A >> P or P >> A**

With respect to their position, coronal images are numbered in either ascending order (**A >> P** from anterior to posterior) or in descending order **P >> A** from posterior to anterior).

❑ **Transverse F >> H or H >> F**

With respect to their position, transverse images are numbered in either ascending order (**F >> H** from foot to head) or in descending order **H >> F** from head to foot).

➤ In every case, the images are numbered by slice groups.



When you activated the **Inherit image numbering** setting of the **queue** menu, the settings for image numbering via protocols are inherited as soon as they are opened for editing.

This applies only to sequential protocols with the same or similar table position.

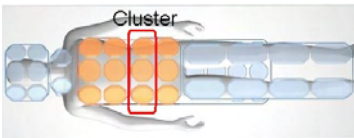
The inherit mode cannot be switched off.

(5) *Save uncombined*

The parameter determines whether, in addition to the combined images of an array coil, the output images of the individual coil elements are to be saved.

➤ It is not possible to combine this parameter with the **Adaptive combine** mode under the **Coil Combine Mode** parameter.

(6) *Matrix Coil Mode*



Matrix Coils comprise one or several coil clusters in the cranial-caudal direction. Every coil cluster generally contains three individual coil elements arranged in the left-right direction.

Matrix Coils can be operated in different Matrix Coil modes. The Matrix Coil mode affects each coil cluster.

CP Mode (Circularly Polarized)	Dual Mode	Triple Mode
The three coil elements of the cluster behave like a CP coil element. The coil cluster is read via an RF receive channel.	The three coil elements of the cluster behave like two CP coil elements. The coil cluster is read via two RF receive channels.	The three coil elements of the cluster behave like three individual coil elements. The coil cluster is read via three RF receive channels.
Lowest data volume	PAT factor 2 is possible in the left-right direction.	PAT factor 3 is possible in the left-right direction.
One RF receive channel per coil cluster	Two RF receive channels per coil cluster	Three RF coil channels per coil cluster

As a function of the PAT mode selected, the system changes the Matrix Coil mode, as required. The currently used Matrix Coil mode appears in brackets after the **Auto** entry.

Matrix Coil mode recommended by the system **Auto (...)**:

PAT Mode	Matrix Coil Mode
None	The Matrix Coil mode is set to Auto (CP) for optimizing the reconstruction times.
GRAPPA mSENSE	The Matrix Coil mode is set to Auto (Triple) . If you set the Matrix Coil mode Dual prior to enabling Auto (...) , this setting remains.

- !

- The Matrix Coil mode applies to Matrix Coils only - other coils are not affected.

(7) *Coil combine mode*

This parameter determines the algorithm used to combine the signals of several receive coils into one measurement.

The following algorithms are available:

☐ **Sum of squares**

☐ **Adaptive Combine**

⇒ **Adaptive Combine** improves the results for most protocols.

⇒ It is not possible to combine this parameter with the **Adaptive combine** mode under the **Save uncombined** parameter.

(8) *AutoAlign*

The **AutoAlign** parameters are found on the **System - Miscellaneous** parameter card in spectroscopy protocols only.

For a detailed description, please refer to the **Routine** parameter card.

→ [Page P.1–6](#), Parameter *AutoAlign*

(9) *Auto Coil Select*

This parameter allows a protocol-specific setting of the **Automatic coil selection**.

→ [Page P.1–96](#), *Automatic coil selection*

Adjustment	Definition
Default	Accepts the global Automatic coil selection settings made by the system. → Page P.1–96 , <i>Automatic coil selection</i>
On	Activates Auto Coil Select for the <i>respective</i> protocol.
Off	Deactivates Auto Coil Select for the <i>respective</i> protocol.

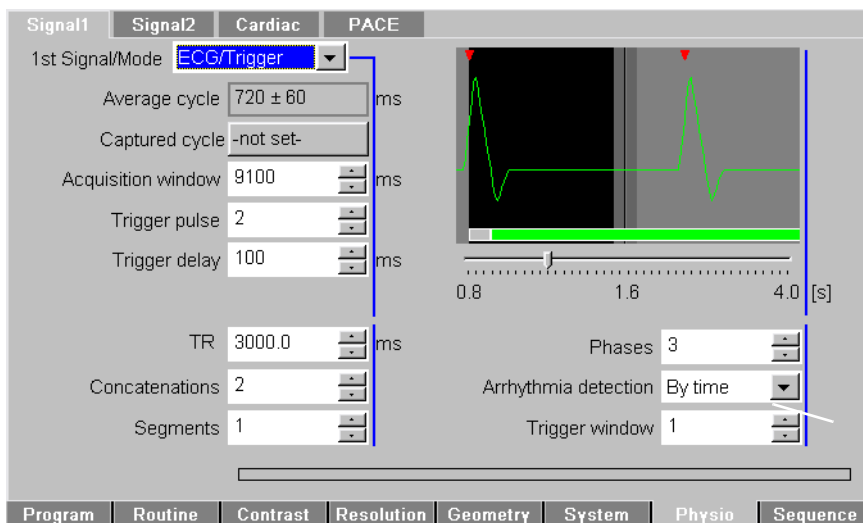
Adjustments and transmitter/ receiver

These two parameter cards include the settings for system adjustment.



Routine operations do not require changes in adjustment parameters. They should be adjusted in exceptional cases and by highly experienced users only
→ [Chapter B.2, Adjusting the System](#).

Physio - Signal 1



Physio - Signal 1 parameter card in the **ECG/Trigger** mode

Here you find the special measurement parameters for physiologically triggered measurements. Physiologically triggered measurement synchronize physiological signals sensed on the patient's body. This avoids motion artifacts caused by the heart-beat or respiration.

You may also retrospectively assign acquired image data to physiological signal curves.

- ⇒ Prior to setting the measurement parameters, we recommend to reset the long-time statistics in the **Physiological Display** window to obtain the most current value for the **Average cycle** parameter.

→ [Page C.17–12](#), *Resetting the long-term statistical values*

(1) 1st Signal/Mode

This parameter determines the physiological signal applied and the measurement mode for physiologically triggered measurements.

Mode	Description
ECG signal	ECG triggering is especially suitable for measurements of the chest and heart, because such images would be blurred by the heartbeat when using a standard measurement mode. The ECG signal is detected on the skin surface with electrodes. The signal shows the action potential of the heart as a curve. The R-wave in the QRS complex is used as the trigger point for the measurement.
Pulse signal	You use pulse triggering especially if you want to suppress motion or flow artifacts generated by pulsing blood or CSF. You can take the pulse signal, for example, from the middle finger of the patient with the pulse sensor. The first pulse wave ("premature pulse wave") is used for triggering. This wave corresponds to the systolic blood pressure.

Mode	Description
External signal	You use an external triggering signal, for example, for functional measurements to trigger measurement of a series. You can input an external, digital triggering signal via the PMU strip at the foot end of the patient table. The rising edge of the signal is used to start the measurement.
Respiratory signal	Respiratory triggering enables you to avoid motion artifacts caused by the patient's breathing. The respiratory signal is obtained with the respiratory belt. The cyclic expansion and contraction of the thorax generates the respiratory curve.
ECG/Retro, Pulse/Retro, Ext/Retro	Some special sequences allow retrospective gating. In this case, the measurement is performed with triggering. Instead, the acquired image data are retrospectively sorted and subsequently correlated with the characteristics of a physiological signal. Retrospective gating may be applied to ECG signal curves, pulse signal curves, and external trigger signal curves.

(2) *Average cycle*

This acquisition time shows you the average time between two trigger events.

The value of the average cycle is used to compute the system acquisition window:

Signal	Calculation
ECG, Pulse, External	System acquisition window = Average cycle - 2 x standard deviation
Respiratory signal	System acquisition window = Average cycle / 2 - standard deviation

(3) *Captured cycle*

You press this button to compute the acquisition window again from the current average cycle.

The button shows you the value of the average cycle used for the calculation. Protocol parameters may have to be adjusted.

(4) *Acquisition window*

This parameter determines the data acquisition time, that is, the time used after the trigger pulse for a measurement following a physiologically-triggered pulse. In this way you define the scan acquisition window.

From the size of the acquisition window, the delay time, number of phases and repetition time are calculated automatically for a number of cardiac sequences.

If you keep the mouse pointer for some time over the **Acquisition window** input field, the value recommended by the system appears.

- ⇒ The acquisition window should be approx. 10% shorter than the average signal period (average cycle).

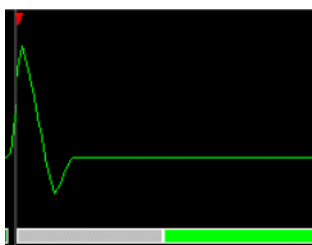
(5) *Trigger pulse*

This parameter determines which trigger events release a measurement (1 = every trigger, 2 = every second trigger, etc.).

- ⇒ This parameter is active only for the ECG, pulse, and external trigger signals.

(6) *Trigger delay*

You are able to acquire images anywhere within the signal cycle by entering a delay time into the **Trigger delay** input field for the **ECG/Trigger** signal.



The value entered corresponds to a delay time (gray bar) between the trigger signal and the start of the measurement.

For an adult patient with a pulse of 70 beats per minute, you need a delay time of 0 ms to obtain systolic images with ECG triggering, and a delay time of 250 - 350 ms to obtain diastolic images.

➤ This parameter is active only for the ECG, pulse, and external trigger signals.

(7) *TR (repetition time)*

This parameter determines the repetition time **TR** that elapses between two successive excitations. (At times an alternative designation *Repetition time* is used).

Changing the repetition time affects image contrast and measurement time.

If you would like to enter several repetition times, additional keys are superimposed for scrolling between the individual times.

(8) Concatenations

This parameter defines into how many repetition times TR the measurement of the planned slices should be divided. The system then determines across how many single sequential measurements the slices will be distributed.

Using this method, you can acquire many slices with a short repetition time (T1 weighted imaging) and prevent slice crosstalk.

- ⇒ When you selected a measurement in multiple breathhold mode and decided to choose one of the inputs **Breath-hold**, **Breath-hold & Monitor** or **Breath-hold & Follow** under **Respiratory Control**, the number of breathhold intervals is determined via the **Concatenation** parameter.
- ❑ In the **Interleaved** multi-slice mode, the number of breathhold intervals results from the product of the parameter values **Measurements** and **Concatenations**.
- ❑ In the **Single Shot** multi-slice mode, the number of breathhold intervals is the product of the parameter values **Measurements**, **Concatenations** and **Averages**.
- ⇒ With triggered multislice measurements ("interleaved" excitation sequence), it is sometimes not possible to acquire all slices in one measurement. The slices missing from one measurement will then be acquired in the next measurement.
- ⇒ This field is active only, when the **Interleaved** or **Single Shot** option was selected in the **Multi-slice mode** on the **Geometry - Common** parameter card.

(9) *Flow sensitivity*

This parameter determines the flow sensitivity of the sequence by modifying the flow-spoiling gradient.

Can be used to adjust a syngo NATIVE SPACE acquisition to the expected blood flow conditions.

The following settings are available:

Adjustment	Definition
Default	The default spoiler settings of the SPACE sequence are used.
Weak	Weak flow spoiler gradients are used.
Medium	Medium flow spoiler gradients are used.
Strong	Strong flow spoiler gradients are used.

⇨ Available only for SPACE sequences.

⇨ Available only if the parameter **NATIVE** is set to **3D Mode** or **TD scout**.

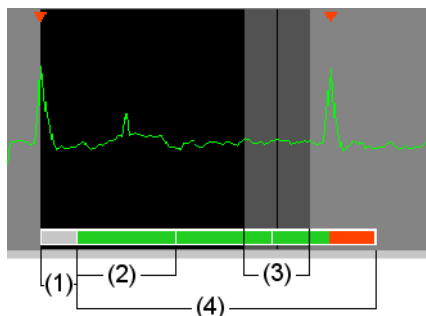
(10) *Segments*

This parameter defines the number of rows in the k-space that are measured for an image during a TR interval.

⇨ The parameter is especially relevant to physiologically-triggered measurements.

(11) *Displaying the time domain*

The physiological signal as well as the time domains resulting from the parameters set are shown.



- (1) Delay time
- (2) Repetition time
- (3) Twice the standard deviation
- (4) Measurement acquisition window

⇨ If e.g., the measurement acquisition time set is longer than the time between two trigger events, the acquisition window will overlap the following trigger section. The time overlap is shown in red.

→ [Page C.17–4, Displaying the time domain](#)

(12) *Phases*

This parameter determines how many phases of the heart beat or how many respiratory phases you want to require. You use this for, e.g., multi-slice/multi-phase measurements of the heart.

- ⇒ The number of heart phases or respiratory phases depends on the selected repetition time TR. Please observe the limits.

(13) *Arrhythmia detection*

This parameter determines the mode for automatically detecting of arrhythmia.

Some sequences include an automatic detection of arrhythmia. This detection is based on a time-controlled recognition of additional systoles (**By time**).

(14) *Trigger window*

This parameter establishes the size of the acceptance window when extra systolic detection is activated.

- ⇒ This parameter is only active for arrhythmia detection according to time.

(15) *Threshold*

This parameter determines at what time the measurement will be triggered within the respiratory cycle. When the respiratory curve reaches this threshold value, the signal is triggered.

The threshold value is expressed as a percentage of the respiratory curve. 100% corresponds to the maximum expansion of the rib cage.

⇒ This parameter is only shown for **Respiratory triggering**.

(16) *Calculated phases*

This parameter determines the number of reconstructed images per heart interval.

⇒ The parameter is shown when you selected a **Retro** mode in the **1st Signal/Mode** selection list.

(17) *Resp. phase*

This parameter determines whether **inspiration** or **expiration** is being used for triggering.

(18) *Target RR*

This parameter establishes the patient's average heart rate for the system. Based on this parameter, arrhythmia is detected with some sequences and the data acquired with arrhythmia are discarded. The value should correspond to the average heart rate of the patient.

⇒ This parameter is not available for CV-NAV sequences.

⇒ The value of this parameter effects the **Segments** parameter.

(19) *NATIVE*

This parameter is used to generate arterial and venous images without contrast agent.

The following settings are available:

Adjustment	Definition
Off	The NATIVE functionality is not used.
3D Mode	Used for 3D non-contrast enhanced angiography. Typically, two ECG-triggered data sets are acquired (in systole and diastole). The parameters TD peak flow and TD min flow must be set properly. Resulting angiographic images are computed via Inline subtraction.
TD scout	Used to determine suitable TD values. In this mode, a series of thick-slab projection images with different trigger delays is acquired from the slab of interest. The timing can be adjusted with the two parameters TD first and TD increment .

- Available only for SPACE sequences.
- Available only if the parameter **1st Signal/Mode** is set to **ECG/Trigger, Pulse/Trigger** or **Ext./Trigger**.

(20) *TD peak flow*

This parameter determines the trigger delay for the first acquisition.

After the trigger (R-wave), the system waits for the defined delay time before the measurement starts.

A typical TD value that provides good results is obtained by subtracting approximately 30 ms from the peak flow time within the cardiac cycle.

⇒ Available if **NATIVE** is set to **3D Mode**.

(21) *TD min flow*

This parameter determines the trigger delay for the second acquisition.

After the trigger (R-wave), the system waits for the defined delay time before the measurement starts.

A typical TD value that provides good results is 0 ms.

⇒ Available if **NATIVE** is set to **3D Mode**.

(22) *TD first*

This parameter determines the trigger delay for the first acquisition of a series of TD scout acquisitions.

After the trigger (R-wave), the system waits for the defined delay time before the measurement starts.

➤ Available if **NATIVE** is set to **TD scout**.

(23) *TD increment*

This parameter determines the trigger delay increment for a series of TD scout acquisitions.

The trigger delay increases from one measurement to the next.

➤ Available if **NATIVE** is set to **TD scout**.

(23) *Measurements*

This parameter determines how often a measurement is performed.

➤ In the context of **NATIVE**: These measurements are carried out at different trigger delays. Typically, two measurements are done in **3D Mode**, and a series of multiple measurements are done for a **TD scout**.

Physio - Cardiac

Signal1	Signal2	Cardiac	PACE
Tagging		Line tag	
Distance		20	mm
Angle		0	
Magn. preparation		None	
TI 1		136	ms
Fat suppr.		Water excit. norr	
Dark blood		☒	
Dark blood thickness		10	%
Dark blood flip angle		90	deg
FoV read		300	mm
FoV phase		100.0	%
Phase resolution		40	%
Cine		Off	
Trajectory		Cartesian	
View sharing		Off	

Program | Routine | Contrast | Resolution | Geometry | System | Physio | Sequence

Here you will find the parameters for a cardiac examination. This parameter card is available only when the current measurement protocol is based on a sequence which supports the cardiological measurement. The resulting image data can be evaluated using the **Argus** task card.

(1) *Tagging*

It is possible to excite tagging lines in the image plane. This provides for clearly visible differences in heart wall thickness.

The following settings are available:

Adjustment	Description
Grid tag	A grid of lines is shown as an orientation aid. It is used to clearly visualize regional and global cardiac wall movements.
Line tag	Parallel stripes are shown as orientation aid. They are used to clearly visualize cardiac wall motion in the long-axis view or in the four-chamber view.
none	No orientation aid is shown

(2) *Distance*

This parameter determines the distance between taggings.

(3) *Angle*

When you selected **Line tag**, enter the angle of the orientation aid into the phase-encoding direction.

(4) *Magn. preparation*

During inversion recovery (IR) and saturation recovery (SR), this parameter determines whether an RF pulse is transmitted prior to each measurement to affect contrast.

You can send the inversion pulse as slice-selective or non-selective. Consider whether your current sequence is an inversion recovery sequence (IR) or a saturation recovery sequence (SR).

The following settings are available:

Adjustment	Description
Slice sel./slab sel. (IR or SR)	The measurements are performed slice by slice.
Non-sel. (IR or SR)	The RF pulses stimulate the entire volume independent of the current slice position or measurement sequence.
Non-sel. DIR	DIR stands for “Double Inversion Recovery” and allows the suppression of signals from two tissue types with different T1 relaxation times. Two inversion pulses are applied at defined times T11 and T12. While you can set the first inversion time T11, the second inversion time T12 is fixed.
T2 preparation	A special preparation pulse is transmitted for the entire volume to enhance T2 contrast. The preparation pulse suppresses signals from tissue with short T2 times. Go to the Contrast parameter card and determine the duration of the preparation pulse in the T2 Prep. duration field.

Adjustment	Description
TI Scout	This scout generates a series of images with different inversion time. Based on this series, you can determine the most suitable inversion time and enter the value in the TI parameter.
T2 sel IR	A special inversion preparation pulse is transmitted for the entire volume. This pulse effects that only tissue with a long T2-time (fluids) is fully inverted. Tissue with short T2 is only inverted in part. In a subsequent excitation, it is possible to display it with a higher signal (e.g. for protocols with dark fluid contrast).
None	No inversion pulse is sent.

- ⇒ When you select the **None** option in the **Magn. preparation** list, you are usually not offered the **TI** parameter on the **Contrast - Common** card.

(5) *TI (Inversion time)*

This parameter determines whether an additional T1 contrast is added with spin preparation. For this purpose, a 180° RF pulse is used (inversion pulse) that inverts the spins.

☐ **For inversion recover sequences:**

In this case, TI is the time between sending the inversion pulse and the excitation pulse of the subsequent measurement (e.g., a spin-echo pulse sequence). Depending on the TI, certain signals are suppressed (e.g., fat) and additional T1 contrast is applied to the signal.

☐ **For Turbo FLASH sequences:**

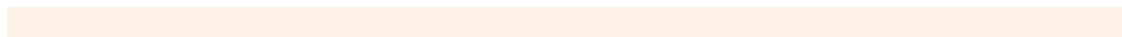
TI describes the time between sending the inversion pulse and reading out the echo signal, sorted in the center of the raw data matrix (this echo determines the contrast).

(6) *Fat suppr.*

This parameter suppresses the signal from fat protons in a targeted way.

The following settings are available:

Adjustment	Description
Fat sat.	Suppresses the fat signal and has no effect on TE, TR is greatly prolonged. This extends the measurement time as well.
Water excit. normal	Suppresses the fat signal, effects moderate extension of TE and TR.



Adjustment	Description
Water excit. fast	Suppresses the fat signal and causes slight extension of TE and TR. The effect on fat saturation is slightly less than in Water excit. normal mode.
Q-fat sat.	Suppresses the fat signal. This mode is only available when selecting Quick mode in the Saturation mode .
SPAIR	Prior to image saturation, the fat signal is suppressed via a frequency-selective adiabatic inversion pulse. However, this option is only available with certain sequences for 1.5 T and 3 T systems.
None	Does not suppress the fat portion in the MR signal.

⇒ To shorten the measurement time in connection with fat sat, select **Quick** Mode in the **Saturation Mode** parameter.

(7) *Dark Blood*

This parameter determines if a non-selective preparation pulse which saturates the blood will be selected for cardiac sequences. This displays the heart muscle exceptionally well and the blood appears dark in the image.

(8) *Dark blood thickness*

This parameter determines the thickness of the saturation slab used to reduce the intensity of the blood. The thickness is expressed as a percentage of the slice thickness.

⇒ A typical value is 200%.

(9) *Dark blood flip angle*

This parameter determines the flip angle of the preparation pulse for saturating the blood.

⇒ The default value is 200.

(10) *FoV read*

This parameter determines the size of the anatomical region to be displayed (extension of the measurement in the readout and phase-encoding direction) and its resolution (pixel size).

⇒ When the **Tim CT mode** is active, this parameter determines the anatomical region required during RF excitation.

(11) *FoV phase*

The FoV in the phase-encoding direction (**FoV phase**) is shown as a percentage of the FoV in the readout direction (**FoV read**).

(12) *Phase resolution*

This parameter determines the resolution of the slices in the phase-encoding direction and is provided in percentages referenced to the basic resolution.

If phase resolution is 100%, the resolution in the readout and phase-encoding directions will have the same value and the pixels are square. At e.g., 75%, the pixels are rectangular and resolution decreases.

(13) *Cine*

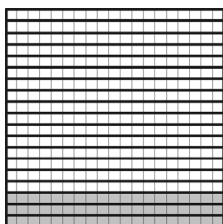
The parameter indicates whether this is a Cine sequence for the display of dynamic processes.

(14) *Trajectory*

This parameter defines the geometric shape to be sampled in the k-space.

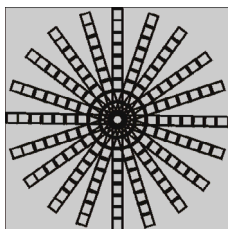
The following shapes are possible:

Cartesian



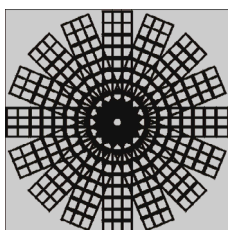
The k-space is sampled as a matrix of rows and columns. The k-space is built up line by line, for example, from the bottom left to the top right.

Radial



The k-space is read out in individual lines. The lines form the shape of a star.

BLADE



Data are acquired in so-called blades. Every blade comprises parallel phase-encoding lines. The individual blades are rotated in order to cover a circle in the raw data space.

➤ The number of lines per blade is determined in the **Turbo Factor** parameter on the **Sequence - Part 2** parameter card.

(15) *View sharing*

This parameter defines whether k-space data already acquired in a measurement should be reused, for example to obtain higher temporal resolution.

The following settings are available:

Adjustment	Description
Shared phases	During a movie measurement, data from the outer k- spaces are copied into the following (cardiac) phase.
Off	No reuse of previously acquired data.
TWIST	The data from the outer k-space are dynamically determined from the data of several measurements.

(16) *Inline ventricular analysis*

This parameter activates Inline ventricular analysis. Segmentation of the left ventricle in short-axis images of the heart is calculated. The resulting contour lines are shown as graphic overlays in the image. The contours can be edited later on in Argus.

Physio - PACE

Signal1

Cardiac

PACE

Resp. control

Gate & Follow

Scout mode

☒

Scout duration

19

s

Scout TR

100

ms

Accept window ±

4.0

mm

Accept position (green)

128.0

mm

Search window ±

32.0

mm

Search position (red)

128.0

mm

Tracking factor

0.60

Chronologic position

Before echo

RF pulse type

Crossed pai

Concatenations

1

Program

Routine

Contrast

Resolution

Geometry

System

Physio

Angio

Sequence

Here you'll find parameters for suppressing respiratory artifacts.

(1) *Resp. control*

This parameter determines the method used for suppressing respiratory artifacts.

Respiratory control mode	Description
Off	Navigator control is switched off.
Breath-hold	Slices of one concatenation are measured as soon as you press the Scan Breathhold button in the Inline display. The number of manual starts required for the complete measurement equals the number of concatenations set. For some sequence types, e.g. <i>TimCT Oncology</i>, the duration of a breath-hold scan can be set in the Breath-hold duration parameter. → Page P.1–143, Parameter <i>Breath-hold duration</i> The Inline display also provides an icon for continuing the scan until the end of the measurement. → Page C.18–1, <i>Inline Display</i>
Breath-hold & Monitor	Same as Hold Breath . However, in addition, the respiratory curve of the patient before the first breathhold as well as during pauses between breathhold intervals is monitored using the Navigator.
Breath-hold & Follow	Same as Breath-hold & Monitor . In addition, however, the calculated offset of the diaphragm position between breathhold intervals is used to adjust the positions of the slices to be measured in real time.

Respiratory control mode	Description
Breath-hold & Trigger	After a learning phase of approximately 5 respiratory phases, the first measurements can be started as soon as you press the Scan Breath-hold button in the Inline display. All subsequent measurements are started automatically (as in Trigger mode) once the measured diaphragm position indicates the end of the expiration phase. Set the number of measurements to be measured during the first breathhold phase: → Page P.1–28, Breath-hold measurements
Breath-hold & Trigger & Follow	Same as Breath-hold & Trigger . However, in addition, the positions of the slices to be measured are offset in accordance with the navigator result.
Gate	The image data are only accepted if the diaphragm position is within the acceptance window.
Gate & Follow	If the navigator result is within the acceptance window, the positions of the slices to be measured are offset in accordance with the navigator result and the measurement is resumed with the next iteration of the loop structure. Otherwise, the current loop is repeated.
Trigger	After a learning period of approximately 5 respiratory phases, a block of image data is acquired once the measured diaphragm position indicated the end of the expiration phase. Respiratory triggering reduces motion artifacts by synchronizing measurement of the image data with the respiratory cycle of the patient.
Trigger & Follow	The same as Trigger . However, in addition, the positions of the slices to be measured are offset in accordance with the navigator result.
Monitor only	The navigator signals are calculated and displayed in the usual way, however, they are not used to control the measurement.

(2) *Scout mode*

This parameter determines a preparation phase which only measures the navigator signal. In this way you are able to check whether or not the navigator records the respiratory signal as required.

(3) *Scout duration*

This parameter determines the duration of the preparation phase.

☞ This parameter is active only, when the **Scout mode** parameter is activated.

(4) *Scout TR*

This parameter determines the repetition time of the navigator pulse.

(5) *Accept window $\pm$*

This parameter is set when you selected one of the modes shown in the list under **Resp. control**.

Respiratory control mode	Effect of the Accept window $\pm$ parameter
Breath-hold & Follow	The deviation of the diaphragm position from the position immediately before the first breath-hold is determined. If the deviation is greater than the value specified in the Accept window $\pm$, the slice positions are not changed for safety reasons. If the diaphragm deviation is greater than the limit value, the result is implausible.
Gate Gate & Follow	The image data are acquired as soon as the deviation of the diaphragm position (relative to the reference position) is less than the value specified in the Accept window $\pm$.
Trigger Trigger & Follow	Determines the vertical width of the yellow acceptance window shown in the Inline display. The trigger algorithm determines the end of expiration as soon as the sequence of measured diaphragm positions (green curve) falls within the acceptance window. The acceptance window is not displayed while the patient is breathing in.

(6) *Accept. position (green)*

In the **Accept.** input field **Position (green)**, you enter the requested center position of the accept window.

⇒ This parameter is shown only for respiratory control **Gate** or **Gate & Follow**.

(7) *Search window ±*

For **Gate** or **Gate & Follow** you enter the size in millimeters for the **Search window ±**.

The search window is displayed as a red box surrounding the tolerance center in the **Inline Display**.

(8) *Search position (red)*

For **Gate** or **Gate & Follow** you enter the center position of the search window in millimeters in the **Search position (red)** input field.

(9) *Tracking factor*

This parameter established the correlation between diaphragm movement and the resulting shift in anatomy.

⇒ This parameter is superposed when you set the **Breath-hold & Follow**, **Trigger & Follow** or **Gate & Follow** under **Resp. Control**.

(10) *Chronologic position*

Select the time when to trigger the navigator signal in the **Gate** or **Gate & Follow** respiratory control mode.

- When selecting the **Gate & Follow** respiratory control mode and the chronological position **before & after** of the echo train of the image, the slice follow algorithm of the information is based on the first navigator signal and the gating algorithm on a combination of the first and second navigator.

(11) *RF pulse type*

This parameter determines the length and envelope of the RF excitation pulse.

The following settings are available:

Adjustment	Description
Fast	Short RF excitation pulse which could cause cross-talk between the slices /slabs. For this purpose, we recommend this setting only for rapid measurements with distance factors that are not too small, e.g., for respiratory breathhold techniques.
Normal	An RF excitation pulse with a good slice profile allows measurements with a small distance factors and little cross-talk.
Low SAR	Extended RF excitation pulse with good slice profile and reduced specific absorption rate. You can select this setting for SAR reduction. This reduces measurement performance.
Optimized	Optimized RF excitation pulse to reduce slice cross-talk.

(12) *Position accept window*

If the respiratory control is set to **Trigger** or **Trigger & Follow**, the **Position accept window** appears.

This parameter determines whether the system automatically sets the center of the acceptance window during a learning phase, or if you enter it manually in the **Accept position** input field.

(13) *Trigger pulse*

This parameter determines which trigger events release a measurement (1 = every trigger, 2 = every second trigger, etc.).

⇒ This parameter is active only for the ECG, pulse, and external trigger signals.

(14) *Card. trig. per resp. cycle*

This parameter determines the number of cardiac trigger pulses per respiratory cycle.

⇒ During double triggering, this parameter is superimposed in place of **Slices per resp. cycle**.

(15) Concatenations

This parameter defines into how many repetition times TR the measurement of the planned slices should be divided. The system then determines across how many single sequential measurements the slices will be distributed.

Using this method, you can acquire many slices with a short repetition time (T1 weighted imaging) and prevent slice crosstalk.

- ⇒ When you selected a measurement in multiple breathhold mode and decided to choose one of the inputs **Breath-hold**, **Breath-hold & Monitor** or **Breath-hold & Follow** under **Respiratory Control**, the number of breathhold intervals is determined via the **Concatenation** parameter.
- ❑ In the **Interleaved** multi-slice mode, the number of breathhold intervals results from the product of the parameter values **Measurements** and **Concatenations**.
- ❑ In the **Single Shot** multi-slice mode, the number of breathhold intervals is the product of the parameter values **Measurements**, **Concatenations** and **Averages**.
- ⇒ With triggered multislice measurements ("interleaved" excitation sequence), it is sometimes not possible to acquire all slices in one measurement. The slices missing from one measurement will then be acquired in the next measurement.
- ⇒ This field is active only, when the **Interleaved** or **Single Shot** option was selected in the **Multi-slice mode** on the **Geometry - Common** parameter card.

(16) *Store profile images*

The temporal change of the navigator signal is displayed graphically in the **Inline Display**. These images can be saved in a separate series.

(17) *Accept. Position*

This parameter determines the center of the acceptance window for **Trigger** and **Trigger & Follow**. 0% of the center position corresponds to the center position at the end of expiration during the learning phase and 100% correspond to the center position at the end of inspiration during the learning phase.

(18) *Slices per resp. cycle*

This parameter determines the number of slices acquired per respiratory cycle.

☞ This parameter is shown only during **haste**, **trufi** and **tfl** sequences.

(19) *Resp. Motion Adaptation*

This parameter determines whether the position of the accept window should adjust to the changes in amplitude of the respiratory curve. This prevents, for example, a shallower respiratory curve leading to infinite measurements.

⇒ This parameter is available only when the **Gate & Follow** or **Gate** mode has been selected under the **Resp. control** parameter.

(20) *Breath-hold duration*

This parameter determines a patient specific duration of one breath-hold for certain sequence types (e.g. *TimCT Oncology*) as an alternative to the duration of one concatenation.

Angio - Common

Common Inline Composing

TONE ramp 70 %

Flow direction F >> H

Flip angle 18 deg

MTC ☒

Measurements 1

3D centric reordering ☒

TONE ramp 20 70 100

Program Routine Contrast Resolution Geometry System Physio Angio Sequence

e.g.: Angio - Common (Time-of-Flight angiography)

Common		Inline	
Flow mode	Single vel.		
Encodings	3		
Velocity enc.			
Velocity enc.	20		
Direction			
Direction 1	R >> L		
Direction 2	Through plane		
Direction 3	A >> P		
		Rephased images	☒
		Magnitude images	☒
		Magnitude sum	☒
		Phase images	☐
Program	Routine	Contrast	Resolution
Geometry	System	Physio	Angio
Sequence			

e.g.: Angio - Common (Time-of-Flight angiography and flow quantification)

The **Angio** parameter card is available only when the sequence used as the basis for the current measurement protocol supports the following examinations:

- ☐ Time-of-flight angiography
- ☐ Contrast-enhanced angiography
- ☐ Phase contrast angiography (2D and 3D)
- ☐ Flow quantification

Time-of-flight angiography	In time-of-flight angiography sequences, unsaturated spins flow into the slice or volume to generate especially high signal intensity.
Contrast-enhanced angiography	Contrast-enhanced angiography (Contrast Enhanced Angiography) takes advantage of the fact that the contrast agent (gadolinium compound) shortens T1 in blood.
Phase contrast angiography and flow quantification	2D phase contrast angiography is used to display vessels within large measurement volumes. With 3D phase contrast angiography, you are able to further process the entire measured data volume with the MIP technique. → 3D/display mode in the <i>syngo</i> Operator Manual.

(1) *Flip angle*

This parameter determines the extent the RF pulse deflects the longitudinal magnetization from the Z-direction of the magnetic field. The flip angle directly affects image contrast.

With a 90° excitation pulse, for example, the longitudinal magnetization is rotated by 90° out of the Z-direction of the magnet. If a smaller flip angle is used, the magnetization returns to equilibrium more quickly allowing you to reduce the repetition time TR (for gradient echo sequences).

For spin echo sequences, you can enhance the T1 contrast by reducing the flip angle.

(2) *TONE ramp*

This parameter adjusts the form of the RF excitation pulse to the velocity and direction of the blood flow (slow, medium, fast) to avoid saturation effects of blood when passing through a slab.

This parameter specifies the ratio of the respective flip angle at the two edges of the slab in percent. The nominal flip angle is used at the center of the slab. For example, a flip angle of 20° together with a TONE pulse value of 60% result in an excitation profile with a flip angle increasing from 15° to 25° across the slab.

In this way, even signal distribution of the blood vessels of a slab is achieved.

⇒ TONE-Technique = **T**ilted **O**ptimized **N**onsaturating **E**xcitation

(3) *Flow direction*

The parameter determines the direction of blood flow to be shown. The direction always refers to the patient coordinate system.

For example, in a mainly transverse slice you can select the Head-Feet or Feet-Head flow direction.

⇒ When you change directions, the tracking sat regions may be automatically positioned on the side of the slab from where the arterial blood exits.

(4) *MTC (magnetization transfer)*

This parameter determines whether the signal of tissue with a high portion of macro molecules is weakened by a special RF excitation pulse.

⇒ By activating the **MTC** checkbox, you obtain better contrast in images for e.g., vessel examinations.

(5) *3D centric reordering*

This parameter determines whether the center of the raw data space will be measured as quickly as possible after contrast agent inflow with contrast-enhanced 3D angiography. This ensures optimal contrast of the arterial vessels.

(6) *Time to center*

This field shows the measurement time required to reach the center of the k-space.

This information is required for timing the contrast agent bolus in contrast agent enhanced angiography.

(7) *Flow mode*

This parameter determines the flow encoding mode:

Adjustment	Description
Single vel.	One flow sensitivity (flow sensitivity encoding): Measures the velocity of the blood in 3 spatial directions with one flow sensitivity. This mode provides vessel display independent of the direction of flow.
Single dir.	One direction for several flow sensitivities: Measures the blood flow as a function of several flow sensitivities, however, in one spatial direction only. This mode is used to acquire large variations in flow velocities (e.g., in the area of peripheral arteries).
Free	Allows for freely-selectable flow sensitivities and spatial directions.

(8) *Encodings*

This parameter determines how many flow sensitivities will be distinguished.

(9) *Velocity enc.*

This parameter determines the different flow sensitivities.

(10) *Direction*

This parameter determines the location of the flow-sensitive axis for the different flow sensitivities.

With the **Through plane** option, you choose a setting where e.g., the flow sensitive axis is located vertically to the image plane. Only the flow vertically to the image plane will be acquired.

(11) *Rephased images*

This parameter determines if flow-phased images (magnitude images) will be reconstructed.

(12) *Magnitude images*

This parameter determines whether a set of magnitude images are to be reconstructed (1 image per flow direction or flow sensitivity).

(13) *Magnitude sum*

This parameter determines whether a magnitude sum is to be reconstructed from the magnitude image of a slice.

(14) *Phase images*

This parameter determines whether a set of phase images are to be reconstructed (1 image per flow direction or flow sensitivity).

(15) *MIP images*

This parameter defines whether an MIP images (maximum intensity projection) should be reconstructed from all magnitude sum images of a measurement.

This procedure is used primarily for MR angiography. As a result, blood vessels appear lighter than the rest of the image.

(16) *Std.Dev. images*

This parameter defines whether the standard deviations from the magnitude sums should be calculated in order to reconstruct an image.

(17) *Central region A*

This parameter determines the size of the k-space center that is acquired in full resolution.

This value is provided as a percentage of the sample points of the total k-space that will be measured.

⇒ This parameter is only active when you selected the **TWIST** mode under the **View sharing** parameter.

(18) *Sampling density B*

This parameter determines the sampling density used to sample the individual temporal phases in the outer k-space.

⇒ This parameter is only active when you selected the **TWIST** mode under the **View sharing** parameter.

(19) *Dynamic reconstruction mode*

This parameter determines in which way the k-space is dynamically reconstructed for several measurements.

The following settings are available:

Adjustment	Description
Forward share	The k-space is fully sampled for all measurements, in-between only the center is sampled. The data of the full sampling are propagated to all subsequent images.
Backward share	The k-space is fully sampled for all measurements, in-between only the center is sampled. The data of the full sampling are propagated to all previous images.
Linear Interpolation	The measurement center is fully measured for each measurement. The partially measured outer k-space is completed by the linear interpolation of measurements adjacent in time.
Spline Interpolation	The measurement center is fully measured for each measurement. The partially measured outer k-space is completed by the cubic spline interpolation of measurements adjacent in time.

➤ This parameter is only active when you selected the **TWIST** mode under the **View sharing** parameter.

(20) *Temporal interpolation of MIP series*

This parameter determines how the images are interpolated linearly. The number of MIP images is increased by an interpolation factor.

⇒ This parameter is only active when you selected the **TWIST** mode under the **View sharing** parameter.

(21) *Temporal resolution / Virt. temporal resolution*

This field shows the time between two consecutive MIP images of the reconstructed 3D image series.

This value is the "virtual" temporal resolution resulting from the time between two measurements and the interpolation factor set under **Temporal interpolation of the MIP series**.

With an interpolation factor greater than 1, the parameter is described as **Virt. temporal resolution**.

⇒ This parameter is only active when you selected the **TWIST** mode under the **View sharing** parameter.

(22) *Burn time-to-center*

This parameter determines if the value from the **time to k-space center** is burned into the pixel data of the image so that this information can be seen in the **Exam** task card.

⇒ This parameter is only active when you selected the **TWIST** mode under the **View sharing** parameter.

Angio - Inline

Here you find parameters for dynamic image evaluation of angio examinations.

The parameter card **Angio - Inline** corresponds to the **Inline - Common** parameter card.

→ [Page P.1–174](#), *Inline - Common*

BOLD

GLM Statistics		Motion correction	
GLM Statistics	☒	Motion correction	☒
Dynamic t-maps	☐	Interpolation	3D-K-space
Starting ignore meas	0	Spatial filter	☒
Ignore after transition	0	Filter setting	1.0
Model transition states	☐	Measurements	20
Temp. highpass filter	☐	Delay in TR	0 ms
Threshold	4.00	Multiple series	☐
Paradigm size	20		
Meas[1]	Baseline		

Program | Routine | Contrast | Resolution | Geometry | System | Physio | BOLD | Sequence

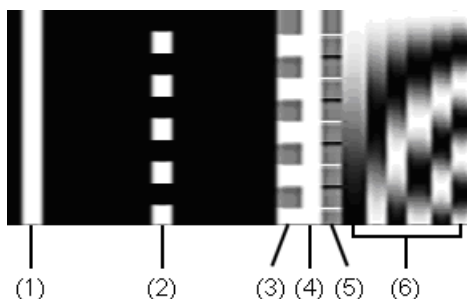
The BOLD abbreviation stands for **B**lood **O**xygenation **L**evel **D**ependent Contrast. BOLD imaging displays the change in the oxygenation state of blood. Generally, T_2^* -weighted EPI sequences are used for this purpose.

GLM Statistics

Using the General Linear Model (**GLM**), the signal changes are modeled using an expanded hemodynamic response function of the brain. The enables better description of the measurement data. The method provides a reliable model for the activity over time curve.

The hemodynamic response function is a curve that represents the brain's response, and therefore the resulting MR signal changes to a short, needle-shaped stimulus.

The model can also be utilized for data acquired shortly after a change in stimulation. Additionally, temporal changes between the model function and measurement data can be modeled. This is necessary because the measurement of individual slices does not take place at the same time.



Design matrix of a GLM evaluation

- (1) Display of all measurements used for evaluation.
white = measurements used
black = measurements not used
- (2) Paradigm - Active/Baseline
- (3) Paradigm convoluted with HRF
(Model transition state = on)
- (4) Offset component
- (5) Evolution of paradigm over time
- (6) Model function for modelling oscillations over time

The following parameter settings effect an analysis to be performed with the hemodynamic response function:

- ☐ Ignore after transition = 0
- ☐ Model transition states = on
- ☐ Temp. highpass filter = on

(1) *GLM Statistics*

This parameter determines whether the data are acquired with the GLM statistics mode (GLM = General Linear Model).

The General Linear Model enables a comprehensive statistical analysis of fMRI data. The measurement data are modeled using a variety of functions, for example the hemodynamic response function.

☞ If the parameter is deactivated, all images are saved without analysis.

(2) *Dynamic t-maps*

This parameter determines whether t-maps are being stored during the measurement. This is required, for example, for an Inline evaluation on the **Neuro 3D** task card.

A "StartFMRI" series is also created at the start of the measurement.

(3) *Starting ignore meas.*

This parameter determines the number of initial measurements that were not used for evaluation to avoid start-up artifacts.

(4) *Ignore after transition*

This parameter determines the number of measurements that are to be ignored after changing the stimulation.

Example: With **Ignore after transition** = 2 and the first change in paradigm for measurement 11, measurements 1,2,11 and 12 are ignored.

(5) *Model transition states*

This parameter determines that the hemodynamic response function of the brain will be included in the computation.

The paradigm is linked to the hemodynamic response function of the brain to obtain a sensible model of the activity over time curve.

⇒ If the **Ignore after transition parameter** is = 0, the first temporal derivation of this model is added to the design matrix to show the time offset between the measured data and the model, since slices of a series are acquired during different points in time.

(6) *Temp. highpass filter*

This parameter determines whether low frequency oscillations over time can be eliminated via a high-pass filter. The number of components of the high-pass filter is determined automatically.

The filter components are displayed in the last column of the design matrix (no. 6).

(7) *Threshold*

With this parameter you determine at which intensity the pixels from t-test images will be used for reconstructing overlay images.

☞ This threshold does not apply for reconstructing pure t-test images.

(8) *Paradigm size*

This parameter determines the number of measurements per paradigm. An input is generated in the paradigm table for each measurement.

(9) *Paradigm table*

This is the table of all measurements within a paradigm. You make the following settings for every measurement:

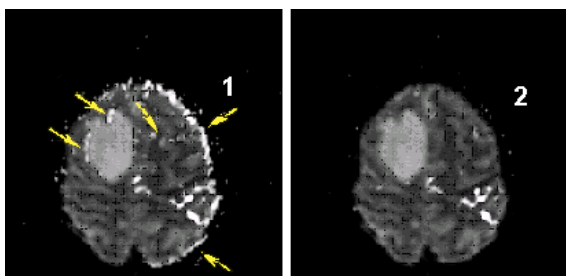
Adjustment	Description
Active	The measurement is performed <i>with</i> stimulation.
Baseline	The measurement is performed <i>without</i> stimulation.

To change the entries in the right column of the table, click the respective field. A selection list including all possible settings is displayed.

(10) *Motion correction*

This parameter determines whether motion correction will be performed when reconstructing images.

Even small movements of the patient (0.3 mm) may lead to significant artifacts in the functional information. Image 1 uses yellow arrows to mark these pseudo activations. Using 3D motion correction, you can reduce the relative motion (translation and rotation) between measured volume data sets (Fig. 2).



The motion correction is displayed in the image text.

→ [Page P.2–1](#), *Text Information in Medical Images*

Motion: -3.98,-0.27,4.04,-2.48,3.12,-0.35

The comment line shows the correction parameters. The first three values show the translation in the x, y, z direction (millimeter) and the following three values show the rotation about the x, y, z axis (degree).

⇒ The images of the first repetition exclusively contain the comment "Reference volume for motion correction".

(11) *Interpolation*

This parameter determines the interpolation method for motion correction.

- ☐ **Linear:** Linear Interpolation. Fast method for real-time post-processing.
- ☐ **3D K-space:** Interpolation in the Fourier space. Good quality method for real-time post-processing.
- ⇒ This parameter is active only, when **Motion Correction** is activated.

(12) *Spatial filter*

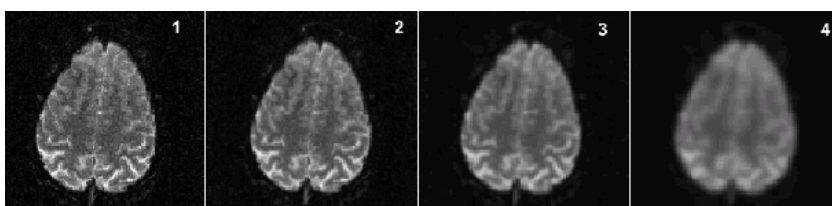
This parameter activates a low pass filter for smoothing images.

The spatial filter leads to an increase in the signal-to-noise ratio at the expense of spatial resolution.

(13) *Filter setting*

This parameter determines the width of the Gaussian filter used. It defines the filter strength.

⇒ This parameter is active only, when the **Spatial filter** checkbox is activated.



- (1) No filtering
- (2) Weak (2.0)
- (3) Medium (1.0)
- (4) Strong (0.5)

(14) *Measurements*

This parameter determines how often a measurement is to be performed.

(15) *Delay in TR*

This parameter determines the time between two subsequent measurements. The parameter is relevant for all ep2d sequences.

⇒ The delay time set applies to all measurements.

(16) *Multiple series*

This parameter determines whether the images for each measurement cycle are stored in their own series.

When the parameter is activated, a separate series is generated for each measurement.

Diff

Common

Diffusion mode1-Scan Trac

Diff. weightings5

b-value s/mm²

b-value0

Please change val0

Please change val0

Please change val0

Diff. weighted images

Trace weighted images

Average ADC maps

Individual ADC maps

FA maps

Tensor

Noise level40

Diff. directions6

Program

Routine

Contrast

Resolution

Geometry

System

Physio

Diff

Sequence

Here you will find the parameters for a diffusion-weighted examination.

(1) *Diffusion mode*

This parameter describes the measurement method and the diffusion-sensitive orientation.

1-Scan Trace	Only one measurement is required per image. Advantage: shorter acquisition time than in the 3-scan Trace .
3-Scan Trace	Three measurements are required per image. Advantage: shorter acquisition time and improved SNR than in the 1-scan Trace . One diffusion-weighted image per slice position and b-value is calculated. The diffusion weighting depends on the trace of the diffusion sensor (mean value of the diagonal elements Dxx, Dyy, and Dzz).
Orthogonal	Three images per slice position and b value (when $b \neq 0$) are acquired, one image each in diffusion weighting in the slice, read-out and phase-encoding direction. For b-value $b = 0$, only one image per slice position is acquired.
Slice	One image is acquired per slice position and b-value, diffusion weighting is in the slice-selection direction.
Phases	One image is acquired per slice position and b-value, diffusion weighting is in the phase-encoding direction.
Read	One image is acquired per slice position and b-value, diffusion weighting is in the readout direction.
Diagonal	In the HASTE sequence with a diffusion weighting, the diffusion weighting points in the direction of the spatial diagonal.
MDDW	MDDW (Multi-directional diffusion weighting). One diffusion-weighted image is acquired per slice position, per b-value, and (for $b > 0$) per diffusion encoding direction. The number of directions is defined with the Diff. Directions parameter.

(2) *Diff. weightings*

This parameter determines the number of diffusion-weightings that are acquired during a measurement.

The individual weightings have to be defined in the **b-value** input field prior to the measurement. You can enter up to 16 b values.

(3) *b-value*

The nominal b value is a measure of diffusion-weighting. It is provided in s/mm^2 .

The greater the value, the stronger the diffusion weighting. The b-value increases with the intensity, duration, and time interval of the diffusion-sensitive gradient pulses.

b value = 0 corresponds to one T2*-weighted image.

The number of possible b values is defined by the **Diff. weightings** parameter.

(4) *Diff. weighted Images*

This parameter determines if original images with diffusion weighting are reconstructed. These images contain T1, T2, and diffusion-weighted portions.

Diffusion weighting is performed in the direction set with the **Diffusion mode** parameter (e.g., if **Slice** in the slice-selection direction applies).

(5) *Trace weighted Images*

This parameter determines if an isotropic diffusion-weighted image is reconstructed. In this type of image, diffusion-weighting is applied in all three spatial directions.

☞ This checkbox is available for **1-scan Trace** and **3-scan Trace** diffusion modes.

(6) *Average ADC maps*

This parameter determines whether ADC maps are reconstructed where averaging is performed in different spatial directions and the b-values used.

ADC Maps (**A**pparent **D**iffusion **C**oefficient) show the diffusion coefficient as grayscale values. The grayscales are derived from measurements with different diffusion weighting (b values). ADC Maps are free of T1 and T2 portions.

☞ The checkbox can be activated only, when at least two b-values have been set.

(7) *Individual ADC maps*

This parameter determines whether ADC maps are reconstructed that show the ADC along the gradient axis.

☞ The checkbox can be activated only, when at least two b-values have been set.

(8) *FA maps*

This parameter determines whether FA maps (**F**ractional **A**nisotropy Maps) are reconstructed.

FA maps show isotropic diffusion characteristics as dark, while anisotropic diffusion characteristics are shown as bright.

(9) *Tensor*

This parameter determines whether diffusion tensor data are saved in the database. This enables diffusion weighting in the **Neuro 3D** task card.

(10) *Noise level*

This parameter determines the intensity at which pixels for reconstructing ADC maps are being used.

(11) *Diff. directions*

This parameter determines the number of diffusion-encoding directions. Possible directions are **6, 10, 12, 20, 30, 64** or **256**.

The image text of diffusion-weighted images shows the b-value and the diffusion direction.

Example: **b1000#3**

Definition: b = 1000 s/mm², third direction of 6, 10, 12, 20, 30, 64 or 256

⇒ This parameter is active only if you deselected the **MDDW** diffusion mode.

(12) *Diff. moment*

This parameter is a measure of the strength of diffusion weighting. It is the amplitude multiplied by the duration of the diffusion gradient.

⇒ This parameter is used only for the **psif** sequence.

(13) *Mosaic*

This parameter determines whether the acquired MDDW data are saved as a mosaic image or individually.

Perf

GBP ☒

PBP ☒

TTP ☒

Original images ☒

Starting ignore meas

Measurements

Program Routine Contrast Resolution Geometry System Physio **Perf** Sequence

Here you will find the parameters for perfusion measurements.

One method of MR perfusion imaging is to determine the signal change in images as a function of time while injecting intravenous contrast agent. Generally, T_2^* -weighted EPI sequences with a gadolinium contrast agent are used for this purpose.

(1) GBP (*Global Bolus Plot*)

This parameter determines whether a global time-density curve has been determined for evaluating the bolus passage.

- ✦ The parameter cannot be activated at less than 15 measurements.

(2) *PBP (Percentage of Baseline at Peak Map)*

This parameter determines whether a percental signal image should be reconstructed for every slice. This image shows the signal change of the bolus peak relative to the base line.

The brighter an image area, the less contrast agent arrived on-site.

- ✧ The parameter cannot be activated at less than 15 measurements.

(3) *TTP (Time-to-Peak-Map)*

This parameter determines whether a time-to-peak image is reconstructed for every slice. The pixel intensity value in the image shows the time that expired until the signal peak was reached.

The brighter an area in the grayscale image, the more time expired until the signal peak was reached.

- ✧ For the **Perf** card the parameter can be activated beginning with the 15th measurement.
For **Inline - Breast** card, the parameter can be activated after the second measurement.
- ✧ In colored perfusion images, the TTP display depends on the color palette selected.

(4) *Original images*

With perfusion measurements, original images are *always* reconstructed and saved. You can use them for additional post-processing.

☞ This checkbox cannot be deactivated.

(5) *Starting ignore meas*

This parameter determines the number of initial measurements that were not used for evaluation to avoid start-up artifacts.

(6) *Measurements*

This parameter determines how often a measurement is to be performed.

Inline - Common

Common	
Subtract	☒
Save images	☒
Autoscaling	☐
Scaling factor	1
Offset	0
Subtrahend	1
Measurements	3
Std-Dev-Sag	☐
Std-Dev-Cor	☐
Std-Dev-Tra	☐
Std-Dev-Time	☐
MIP-Sag	☐
MIP-Cor	☐
MIP-Tra	☐
MIP-Time	☐
Save original images	☒

Program Routine Contrast Resolution Geometry System Physio **Inline** Sequence

Here you find parameters for dynamic image evaluation.

- With measurement protocols for angio examinations, the **Inline** parameter card as well as the **Angio - Inline** subtask card are shown.

(1) *Subtract*

This parameter defines whether a subtraction evaluation should be performed with a series of the current measurement. In this way, images are generated that e.g., show the changes after contrast agent administration.

If the measurement is performed only once, the measured series is used for subtraction with a series from other protocols (cross-protocol subtraction):

- ☐ If the protocol is marked as a contrast agent measurement by a syringe icon, the series last buffered on the image reconstruction system will be subtracted from the current series.
- ☐ If the protocol is not marked with the syringe icon, the current series will be loaded into the buffer of the image reconstruction system and subtracted from the series of a subsequent protocol.
- In the case of Set-n-Go protocols, the relationship is between the steps of the same indices.

(2) *Save images*

This parameter determines whether the result images of subtraction are stored automatically.

- This parameter is superposed only when the **Subtract** checkbox is activated.

(3) *Autoscaling*

This parameter determines whether the display range of the computed subtraction values will be adjusted automatically.

- ⇒ This parameter is superposed only when the **Subtract** checkbox is activated.

(4) *Scaling factor*

This parameter determines the factor for scaling the display area of the computed subtraction factor.

- ⇒ This parameter is shown only, when the **Autoscaling** checkbox is activated.

(5) *Offset*

This parameter determines the offset for the lower and upper thresholds of the display range for the computed subtraction values.

- ⇒ This parameter is shown only, when the **Autoscaling** checkbox is activated.

(6) *Subtrahend*

If the measurement is performed multiple times, indicate which of the measured series is to be subtracted from the other series (subtraction within the protocol).

- ☞ The **Subtrahend** parameter is shown only when the **Subtract** checkbox is activated.
- ☞ The **Subtrahend** parameter can be selected only, when the parameter is **Measurements** > 1.

(7) *Measurements*

This parameter determines how often a measurement is to be performed.

(8-10) *Std-Dev-Sag / Std-Dev-Cor / Std-Dev-Tra*

These parameters determine whether standard deviation images with sagittal, coronal, and transverse orientation are reconstructed from slabs measured with the current protocol. In these you can see the scatter of the pixel values in the respective orientation.

- ☞ The checkboxes can be activated only for a 3D measurement with at least 4 slices in the slabs.

(11) *Std-Dev-Time*

When the **Std-Dev-Time** checkbox is activated, standard deviation result images measured in the orientation of the slice groups and/or slabs are reconstructed from the slices groups and/or slabs of the actual protocol. You can see the scatter of the pixel values within the measurement period. Both multiple phases and multiple measurements can be evaluated.

(12-14) *MIP-Sag / MIP-Cor / MIP-Tra*

These parameters determine whether MIP images with sagittal, coronal or transverse orientation should be reconstructed from the measured slabs of the actual protocol.

- ⇒ The checkboxes can be activated only for a 3D measurement with at least 4 slices in the slabs.

(15) *MIP-Time*

This parameter determines whether MIP images can be reconstructed in the orientation of the measured slice groups and/or slab from the slice groups and/or slabs measured of the actual protocol. In the calculation, the highest pixel value along the time axis is taken into account.

- ⇒ The checkbox can be activated only if at least one repetition measurement is made (parameter **Averages** > 1) or at least two phases were measured for triggered measurements.

(16) *Save original images*

This parameter determines whether the unprocessed images are stored in the database as well.

- ⇒ The parameter can be deactivated only if one of the following parameters is active: **Subtraction**, **MIP**, **StdDev**, **DynaVIBE**, **T1-map** or **T2-map**.

(17) DynaVIBE

This parameter activates 3D liver recognition.

- ⇨ This parameter is available only when a value larger than 2 is stated in the **Measurement** parameter.



CAUTION

Use of substracted images for diagnostic purposes!

Fine anatomical structures are lost; incorrect diagnosis

- ⇨ Always use both for diagnostic purposes, the substracted images as well as the original images.
-

Inline - Breast

Common **Breast**

Wash - In ☒

Color table GreenRed

First measurement 1

Last measurement 2

Highest value ☐

Wash - Out ☒

Color table RedGreen

First measurement 1

Last measurement 2

TTP ☒

Color table RedGreen

PEI ☒

Color table GreenRed

MIP - time ☒

Color table GreenRed

Measurements 3

Pause after meas. 1 0.0 s

Program Routine Contrast Resolution Geometry System Physio **Inline** Sequence

(1) *Wash - In*

This parameter determines whether signal changes are to be activated at the beginning of the dynamic measurement sequence.

(2) *Color table*

This parameter determines the color table for displaying different image values.

(3) *First measurement*

This parameter determines which measurement should be used as the first measurement of a sequence.

(4) *Last measurement*

This parameter determines which measurement should be used as the last measurement of a sequence.

(5) *Highest value*

This parameter determines if, in place of the last measurement, the highest value between the first and last measurement should be used for reconstructing the wash-in parameter image.

(7) *TTP (Time-to-Peak-Map)*

This parameter determines whether a time-to-peak image is reconstructed for every slice. The pixel intensity value in the image shows the time that expired until the signal peak was reached.

The brighter an area in the grayscale image, the more time expired until the signal peak was reached. For this reason, arrival of the contrast agent is delayed.

- ⇒ For the **Perf** card the parameter can be activated beginning with the 15 measurement.
For **Inline-Breast** card, the parameter can be activated after the second measurement.
- ⇒ In colored perfusion images, the TTP display depends on the color palette selected.

(8) *PEI (Positive Enhancement Integral)*

This parameter determines whether an image of the positive enhancement integral should be reconstructed.

- ⇒ Positive enhancement interval (PEI): Area under a signal intensity time curve

(9) *MIP time*

This parameter determines whether MIP images can be reconstructed in the orientation of the measured slice groups and/or slabs from the slice groups and/or slabs measured of the actual protocol. In the calculation, the highest pixel value along the time axis is taken into account.

- ⇒ The checkbox can be activated only if at least one repetition measurement is made (parameter **Averages** > 1) or at least two phases were measured for triggered measurements.

(10) *Measurements*

This parameter determines how often a measurement is to be performed.

(11) *Pause after meas.*

During dynamic measurements, this parameter determines a delay time (in seconds) between the individual measurements. You separately determine the pause for each measurement:

- ✧ Click the lower arrow keys to get to the pause the duration of which you want to change.
- ✧ Enter the desired value in the input field.
- ⇒ Up to 64 individual pauses are possible. In most cases, you are setting the same pause time for all measurements or set the pause time to zero. Beginning with 65 pauses, you can only enter a general pause time.

Inline - Composing

The screenshot shows a software interface with a tabbed menu at the top containing 'Common' and 'Composing'. The 'Composing' tab is active. On the left side of the panel, there are four settings: 'Inline Composing' with a checked checkbox, 'Composing Function' with a dropdown menu showing 'Angio', 'Composing Group' with a numeric input field showing '1', and 'Last Step' with a checked checkbox. On the right side, there are two settings: 'Distortion Corr.' with a checked checkbox and 'Mode' with a dropdown menu showing '2D', followed by 'Unfiltered images' with a checked checkbox. At the bottom of the interface is a horizontal bar with ten tabs: 'Program', 'Routine', 'Contrast', 'Resolution', 'Geometry', 'System', 'Physio', 'Inline', and 'Sequence'. The 'Inline' tab is currently selected.

Here you will find the parameters for Inline composing.

You can combine several protocols to composing groups. As a result, the images of a protocol are automatically combined into an overall image.

(1) *Inline Composing*

This parameter indicates that the images of the steps or the protocol of a composing group can be composed into a complete image.

- **Inline Composing** can be enabled only, when **2D** or **3D** is activated in the **Distortion Corr.** field.
- The **GLM Statistics** parameter on the **BOLD** parameter card is automatically deactivated.

(2) *Composing algorithm*

This parameter establishes the methods used to compose the images.

The following settings are available:

Adjustment	Description
Spine	This algorithm uses the bone structures in the images as a basis. For example, it generates images for the measurement and evaluation of scoliosis, kyphosis, and pelvic obliquity.
Angio	This algorithm uses the vessel structures of the images as a basis. This allows for overall display of the vessels.
Adaptive	This algorithm is based on elastic matching. It especially corrects B0 induced inhomogeneities in image areas beyond the magnetic homogeneity volume.

(3) *Composing Group*

This parameter determines to which composing group the protocol belongs.

You set this parameter when you use **Inline Composing** to compose the images of all protocols of a composing group into an overall image.

(4) *Last step*

The chronologically last measurement of a Composing Group has to be identified as the Last Step. For this purpose you activate this checkbox.

The subsequent protocols belong to a new Composing Group, even if they have the same group number.



Protocols of a Composing Group not completed with a *Last Step* flag are identified by a post-processing icon that has been crossed out.

(5) *Distortion Corr.*

This parameter activates 2D or 3D distortion correction. This compensates for the pillow-shaped distortions at the edge of the image. These distortions occur in images with a large FoV or eccentric slices (Offcenter).

The following settings are available:

Adjustment	Description
2D	With 2D distortion correction , image distortion is corrected individually in each slice.
3D	With 3D distortion correction , the voxel in the current slice as well as those in the surrounding slices are taken into account. The correction results are more precise, but require a longer reconstruction time.

(6) *Unfiltered images*

This parameter determines whether unfiltered images should be saved as well.

⇒ If you reconstructed and stored unfiltered as well as filtered images in Inline, all other inline functions can no longer be accessed.

Filtered images are always shown in the Inline display.

Inline - MapIt

Common	Inline	Breast	Composing	MapIt
MapIt T1 map				Noise threshold 15
☐ Auto angle calculation				☒ Save original images
T1 estimate 1000 ms				
Flip angle 1 2 deg				
Flip angle 2 12 deg				
Measurements 2				
Contrasts 1				
TE 1.22 ms				
TR 4.19 ms				

Program Routine Contrast Resolution Geometry System Physio **Inline** Sequence

The parameters of this parameter card are only valid for the currently selected step.

(1) MapIt

This parameter determines whether inline parametric maps are being calculated.

The following settings are available:

Adjustment	Description
None	No parametric maps are calculated.
T1 map	Activates the Inline computation of T1 maps. The Measurement parameter is set to the value 2 and cannot be changed. The Contrasts parameter is set to the value 1 and cannot be changed. This mode is only available for VIBE sequences.
T2 map	Activates the Inline computation of T1 maps. The Contrasts parameter is set to value 2 .It can assume a value from 2 to 12 . The Measurement parameter is set to the value 1 and cannot be changed. This mode is only available for spin-echo sequences.
T2* map	Activates the Inline computation of T2* maps. The Contrasts parameter is set to value 2 .It can assume a value from 2 to 12 . The Measurement parameter is set to the value 1 and cannot be changed. This mode is only available for GRE sequences.

(2) *Auto angle calculation*

This parameter activates automatic calculation for the two flip angles for T1 cards.

- ⇒ This parameter is available only, when the **T1 map** mode has been selected under the **Parametric map** parameter.

(3) *T1 estimate*

You enter an estimated value for T1 in this field. This value is required for automatically calculating the flip angle.

- ⇒ This parameter is only available when the **Auto angle calculation** parameter has been activated.

(4) *Flip angle 1*

*When you selected the **T1 map** mode in the **MapIt** parameter:*

This parameter determines the first flip angle for T1 maps.

- ⇒ You can edit this parameter only when the **Auto angle calculation** parameter has been deactivated.

*When another mode was selected in the **MapIt** parameter:*

The parameter indicates the flip angle used.

(5) *Flip angle 2*

This parameter determines the second flip angle for T1 maps.

- ⇒ This parameter is available only, when the **T1 map.** mode has been selected under the **MapIt** parameter and the **Auto angle calculation** parameter is deactivated.

(6) *Measurements*

This parameter determines how often a measurement is to be performed.

(7) *Contrasts*

This parameter determines the number of images to be generated with different T2 (e.g., se-tse sequences) or T2*-weighting (**gre** sequences) within a measurement (number of image contrasts).

- ⇒ The **Contrasts** parameter is available only for a few sequences.
- ⇒ You have to define an echo time for each contrast.

(8) *TE (echo time)*

This parameter determines the echo time (TE) that elapses between the RF excitation pulse and the pulse for the echo to be measured.

- ☞ This parameter is not active for all sequences.
- ☞ You may enter several echo times for multi-echo sequences. You scroll through the echo times below the field names by using the arrow keys. When you change a given echo time, the following echo times will be adjusted accordingly.

(9) *TR (repetition time)*

This parameter determines the repetition time TR that elapses between two successive excitations. (At times an alternative designation *Repetition time* is used).

Changing the repetition time affects image contrast and measurement time.

If you would like to enter several repetition times, additional keys are superimposed for scrolling between the individual times.

(10) *Noise threshold*

This parameter determines the threshold below which measurement values are ignored. This suppresses noise.

(11) *Save original images*

This parameter determines whether the unprocessed images are stored in the database as well.

⇒ The parameter can be deactivated only if one of the following parameters is active: **Subtraction**, **MIP**, **StdDev**, **DynaVIBE**, **T1-map** or **T2-map**.

Sequence - Part 1

Part 1	Part 2
Introduction ☒	Contrasts 32
Dimension 3D	Bandwidth 1 510 Hz/Px
Elliptical scanning ☐	Flow comp. No
Phase stabilisation ☐	Readout mode Monopolar
Compensate T2 decay ☐	Optimization None
Averaging mode Short term	Allowed delay 0.000 s
Multi-slice mode Interleaved	Free echo spacing ☐
Reordering Linear	Echo spacing 0 ms
Asymmetric echo Weak	Sequence type

Program | Routine | Contrast | Resolution | Geometry | System | Physio | Sequence

Here you will find the parameters specific to a sequence type. The card is divided into **Part 1** and **Part 2** subtask cards.

A number of different sequence types are available for examinations:

- ☐ Gradient echo sequences (GRE, FLASH, Turboflash, TrueFISP, MEDIC, PSIF, CISS, DESS)
- ☐ Spin echo sequences (se)
- ☐ Turbo spin echo sequences (tse)
- ☐ HASTE sequences
- ☐ Single-shot EPI sequences
- ☐ Segmented epi sequences (ep_seg_fid, ep_seg_se)

The protocol info line above the parameter card includes an abbreviation which helps you recognize the sequence of the current protocol.

→ [Page C.7–2](#), *Protocol info line*

(1) *Introduction*

This parameter determines whether the patient should be made aware that the measurement has been started by the short knocking noise made by the gradient system.

(2) *Dimension*

The parameter indicates whether a 2D or 3D measurement has been planned.

- ⇨ If you switch from 2D to 3D, the slice group parameters on the different parameter cards change into slab group parameters.
- ⇨ For sequences where this dimension cannot be changed, this parameter is either inactive or hidden.

(3) *Elliptical scanning*

This parameter determines whether the corners of the k-space that contribute only minimally toward resolution should be ignored during data acquisition.

This allows you to reduce the measurement time by up to 25% for 3D sequences without compromising resolution.

(4) *Phase stabilization*

This parameter avoids phase errors generated by e.g., respiration.

Phase stabilization improves image quality especially for gradient-echo sequences.

(5) *Compensate T2 decay*

This parameter avoids negative effects of T2 decay while acquiring long echo trains.

(6) *Averaging mode*

This parameter determines the method for averaging measurements when several measurements are performed.

- ☐ **Short term**
effects a better signal-to-noise ratio while maintaining optimal resolution
- ☐ **Long term**
effects a better signal-to-noise ratio while maintaining optimal resolution

(7) *Multi-slice mode*

This parameter determines the measurement modes for multi-slice mode:

- ☐ **Sequential** - measurement, slice by slice
All lines (phase-encoding steps) of the first slice are measured step-by-step. This is followed by all lines of the second slice, etc.
- ☐ **Interleaved** - measurement, line-by-line
All first lines (phase-encoding steps) of all interleaved slices are measured one after the other in repetition time TR. This is followed by all second lines of all interleaved slices, etc.
- ☐ **Single shot** - Special mode for fast sequences.
All lines (phase-encoding steps) of a slice after excitation are measured simultaneously. Then all lines of the second slice are measured together, etc.

(8) *Reordering*

This parameter determines the acquisition sequence of raw data lines.

☐ **Linear**

The k-space is stepped through linearly.

☐ **Centric**

The first measured raw data lines are sorted into the center of the k-space.

⇒ In the **Centric** mode, you achieve good fat saturation with single shot sequences, because the center of the k-space is measured early on after the fat sat pulse.

(9) *Asymmetric echo*

This parameter allows for echo asymmetry in the readout direction. It is used to reduce the echo time. The maximum asymmetry not to be exceeded is defined in the sequence.

The following settings are available:

Adjustment	Description
Allowed	An asymmetric echo is used automatically, if necessary. If the echo time is long enough, the echo remains centered.
Weak	A weak asymmetric echo is used.
Strong	A strong asymmetric echo is used. ⇒ Strong asymmetries may lead to truncation artifacts.
Half echo	A completely asymmetric echo is used. As a result, extremely short echo times below 1 ms can be obtained. ⇒ This mode is only available for UTE sequences.

⇒ You can display the asymmetry of the echo in a tooltip.

(10) *Contrasts*

This parameter determines the number of images to be generated with different T2 (e.g., se-tse sequences) or T2* weighting (**gre** sequences) within a measurement (number of image contrasts).

- The **Contrasts** parameter is available only for a few sequences.
- You have to define an echo time for each contrast.

(11) *Bandwidth*

This parameter determines the fat-water-pixel offset and the signal-to-noise ratio.

In some sequences it is possible to assign a readout bandwidth to each contrast.

Value for bandwidth in image text compared to protocol parameter:

For two reasons, the MR scan is performed with a bandwidth value that can slightly differ from the value entered in the parameter card:

First, the bandwidth is internally converted to the dwell time which must be a multiple of the ADC sampling time type. The basis for the scan is 100 nsec and not the bandwidth itself. The real bandwidth used internally is always some fractional value within the rastered values as used in the parameter card. The data in the image text is provided with an accuracy of 1 Hz while the increments in the parameter card are sequence-specific and can be much larger.

Second, the dwell time can be different for each frame. The image text should not be a copy of the protocol parameter but reflect the actual value of each frame.

(12) *Flow comp.*

This parameter prevents signal loss and smearing caused by moving spins. For this purpose, additional gradient pulses are switched with a suitable time duration and amplitude.

Some sequences enable you to specify flow compensation separately for each contrast. Use the down arrow key to scroll from contrast to contrast and perform the following setting:

Adjustment	Description
Yes	Compensation in readout and slice encoding direction, as well as in phase-encoding direction for some sequences (e.g., tfl, CV).
Read	Compensation in readout direction only
Slice	Compensation in slice-selection direction only
No	No flow compensation

(13) *Readout mode*

This parameter determines the read-out mode for gradient echo sequences (GRE).

The following modes are available:

Mode	Description
Bipolar	In this mode, the MR signals of the different contrasts are alternately read out with a positive or negative gradient. This enables narrower echo spacing than in the monopolar mode. However, the fat-water shift alternates from contrast to contrast.
Monopolar	In this mode, the MR signals of all contrasts are read out with a positive gradient. Echo spacing is greater than in the bipolar mode. The fat-water shift is identical in all contrasts.

(14) Optimization

This parameter is used for time optimization.

The following modes are available:

Mode	Description
Min. TE	The time for TE is optimized to the minimum.
Min. TR	The time for TR is optimized to the minimum.
Min. TR TE	The times for TE and TR are optimized to the minimum.
Min. echo spacing	Echo spacing is optimized to the minimum.
In phase	You determine the shortest time interval in which fat and water are in phase.
Opposed phase	You determine the shortest time interval in which fat and water are out-of phase.
None	Time optimization is not performed.

- ➡ It is no longer possible to manually change the optimized parameters. For example, you are not able to edit the **TE** field for **Minimize TE**.

(15) *Allowed delay*

This parameter establishes the maximum permissible delay time after completing the measurement.

The delay time is used to reduce the specific absorption rate (SAR). The required delay time is automatically calculated by the system and ranges between 0 seconds and the maximum delay time.

(16) *Free echo spacing*

This parameter determines whether you can determine the echo spacing in the **Echo spacing** input field.

(17) *Echo spacing*

This parameter determines the spacing between the echoes in the pulse train, e.g., for Turbo spin echo or EPI sequences.

Smaller echo spacing effects a more compact sequence timing and reduces image artifacts. This means better resolution in the phase-encoding direction for Turbo spin echo sequences or lower susceptibility distortion for EPI sequences.



High echo spacing may lead to susceptibility distortion for EPI sequences.

(18) *Sequence type*

The parameter establishes the type of sequence.

The following types are possible:

Type	Description
Gre	The sequence measures with the gradient echo method.
Trufi	The sequence measures with the TrueFisp method.

(19) *Reduced motion sens.*

This parameter determines if modified phase-encoding gradients should be used. This makes the sequence more robust against slight motions which would normally lead to segmentation artifacts.

➤ This mode is only available for tse sequences.

(20) *Adiabatic-mode*

This parameter determines whether a non-sensitive B₁ excitation pulse (adiabatic pulse) should be used to reduce the signal variations via the field of view.

Sequence - Part 2

Part 1	Part 2	Special	Nuclei
Define Turbo factor		RF pulse type Low SAR	
Turbo factor 64		Gradient mode Fast	
Slice turbo factor 10		Excitation Slab-sel.	
Echo trains per slice 2		Flip angle mode Constant	
Echo train duration 10		RF spoiling ☒	
EPI factor 64		Cine Off	
Segments 1		Motion correction Off	
Combined echoes 3			
Trufi delta freq. 90 Hz			

Program	Routine	Contrast	Resolution	Geometry	System	Physio	Sequence
---------	---------	----------	------------	----------	--------	--------	----------

e.g.: **Sequence Part 2** for sequences with **Define = echo trains**
or **Turbo factor**

(1) *Define*

The **Echo trains/Turbo factor** parameters or **Shots/Segment** parameters are reconstructed as a function of one another.

This parameter determines which of the two parameters is freely selectable.

Adjustment	Description
Echo trains	Determines that the Turbo factor is computed according to the desired number of echo trains per slice.
Turbo factor	Determines that the number of echo trains per slice are computed on the basis of the Turbo factor.
Shots	Determines that the segment size is computed according to the desired number of shots per slice.
Segment	Determines that the number of shots per slice is computed on the basis of the segment size.

(2) *Turbo factor*

This parameter determines the number of refocused spin echoes per RF excitation pulse that contribute to an image. It therefore determines the gain in measurement time as compared to a conventional spin echo sequence with comparable parameters.

☞ When the **BLADE** mode is selected in the parameter card **Resolution - Common** or **Physio - Cardiac** in the **Trajectory** parameter, the turbo factor determines the lines per blade.

(3) *Slice turbo factor*

This parameter defines the number of slices to be measured with an echo train.

- ⇒ This parameter is active only when the echo train is long enough to measure several slices and when the **Echo trains** mode was selected in the **Define** selection list.
→ [Page P.1–208](#), *Define*

(4) *Echo trains per slice*

This parameter defines the number of echo trains with which a slice is to be measured.

(5) *Echo train duration*

In this display field you check whether the echo train is long enough for a sensible measurement.

(6) *EPI factor*

This parameter determines the number of refocused gradient echoes per RF excitation pulse that contribute toward an image.

- ⇒ For single-shot EPI sequences, the number of lines to be measured is used as the EPI factor.

(7) *Segments*

This parameter defines the number of rows in the k-space that are measured for an image during a TR interval.

- ✦ The parameter is especially relevant to physiologically-triggered measurements.

(8) *Combined echoes*

This parameter determines the number of echoes of a measurement that can be combined into an image (e.g., for a MEDIC sequence).

You obtain flow-compensated T2*-images with a high signal-to-noise ratio.

(9) *Trufi delta freq.*

Use this parameter to move banding artifacts to areas of the image where they are not a bother.

- ✦ This field is only active for **TrueFisp** sequences.

(10) *RF Pulse Type*

This parameter determines the length and envelope of the RF excitation pulse.

The following settings are available:

Adjustment	Description
Fast	short RF excitation pulse which could cause cross-talk between the slices /slabs. For this purpose, we recommend this setting only for rapid measurements with distance factors that are not too small,e.g., for respiratory breathhold techniques.
Normal	an RF excitation pulse with a good slice profile allows measurements with a small distance factors and little cross-talk.
Low SAR	Extended RF excitation pulse with good slice profile and reduced specific absorption rate. You can select this setting for SAR reduction. This reduces measurement performance.
Optimized	optimized RF excitation pulse to reduce slice cross-talk.

(11) *Gradient mode*

This parameter determines the rise time and maximum gradient strength that can be used to switch gradients during the course of the sequence.

The possible settings depend on the gradient system:

Adjustment	Description
Fast	Gradient rise time and strength are fully utilized. This mode may cause peripheral nerve stimulation in the patient.
Normal	For many sequences this setting represents a good compromise between performance and noise development.
Whisper	Ensures rather quiet gradients at an acceptable performance level.

When adjusting the gradient mode with the stimulation monitor, the setting is identified by an asterisk *, e.g., **Fast***. This means the gradient rise times have been adjusted automatically to prevent them from exceeding the stimulation limit.

→ [Page C.10–28](#), *Stimulation limit has been reached*

You can modify the mode by changing **Fast*** to **Fast**.

(12) *Excitation*

This parameter determines how the RF excitation pulse is being sent: **slice-sel.** or **slab sel.** or **non-sel.**



If you have selected the **non-sel.** option, aliasing artifacts may occur in the slice-selection direction.

(13) *Flip angle mode*

This parameter provides for a more precisely defined flip angle of the refocusing pulse in the echo train.

The following settings are available:

Adjustment	Description
Constant	The flip angle of the refocusing pulse remains constant across the entire echo train. You set it here: → Page P.1–18, Flip angle
Hyperecho	The flip angles of the refocusing pulse vary across the echo train when generating a so-called hyper echo. This mode enables T2 weighted contrast with a high signal-to-noise ratio, and is optimized for SAR.
Variable	The flip angles of the refocusing pulse increase across the echo train. This mode enables optimum FLASH contrast. It is used for cardiovascular sequences, for example.

Adjustment	Description
T1 var	The flip angles of the refocusing pulse vary across the echo train. This mode enables T1 weighted imaging with short echo times and is optimized for SAR.
T2 var	The flip angles of the refocusing pulse vary across the echo train. This mode enables very long echo trains for fast T2 weighted measurements, and is optimized for SAR.
PD var	The flip angles of the refocusing pulse vary across the echo train. This mode is particularly suited to proton density measurements, and is optimized for SAR.

(14) *RF spoiling*

This parameter destroys the remaining phase coherence of the spins. It is used for gradient echo sequences to produce FLASH contrast.

(15) *Cine*

The parameter indicates whether this is a Cine sequence for the display of dynamic processes.

(16) *Motion correction*

The parameter indicates whether a BLADE motion correction is performed. The motion correction corrects for patient movement in the image plane that occurs between the acquisition of two echo trains.

The following settings are available:

Adjustment	Description
On	Patient motions in the image plane are corrected.
Off	Motion correction is not performed.
On+Off	Two image series are reconstructed, one with motion correction and one without motion correction.

⇒ This parameter is active only when you go to the parameter card **Resolution - Common** and select the **BLADE** mode in the **Trajectory** field.

(17) *Shots per slice*

Number of shots necessary for a slice/slab to completely fill the k-space.

⇒ This parameter is active only when the **Define** parameter of this parameter card is set to **Shots**.

(18) *Stereotactic*

This parameter reduces image artifacts caused by the use of stereotactic frames.

⇒ This parameter should be activated only when a stereotactic frame is used.

(19) *WARP*

This parameter reduces susceptibility artifacts if the patient has MR conditional implants.

The sequence uses high bandwidth RF pulses to reduce artifacts caused by off-resonance effects occurring in the vicinity of MR conditional implants.



Please adhere to all safety instructions regarding implants.
(→ Operator Manual – System)

MR imaging of patients with metallic implants brings specific risks. However, certain implants are approved by the governing regulatory bodies to be MR conditionally safe. For such implants, the previously mentioned warning may not be applicable. Please contact the implant manufacturer for the specific conditional information. The conditions for MR safety are the responsibility of the implant manufacturer, not of Siemens.

(20) VAT

Off-resonance effects (for example from MR conditional implants) may lead to severe image distortions: image signal may be displaced by many pixels in the frequency-encoding direction.

The VAT technique (View Angle Tilting) reduces such distortions by applying additional frequency-encoding gradients.

☐ 0% = off

☐ 100% = maximal effect

⇒ The appropriate VAT value depends on the level of artifacts and their effect on the anatomic region of interest. In general, a higher level of artifacts requires a higher VAT value.



This parameter may lead to image blurring, especially of object edges.

Blurring can be reduced by

☐ high readout bandwidth

☐ thin slices

☐ reduced VAT

⇒ This parameter is only available if **WARP** is activated.

(21) *Phase correction*

Defines whether the sequence performs a phase correction.



In most cases the mode **Auto** will provide good results!

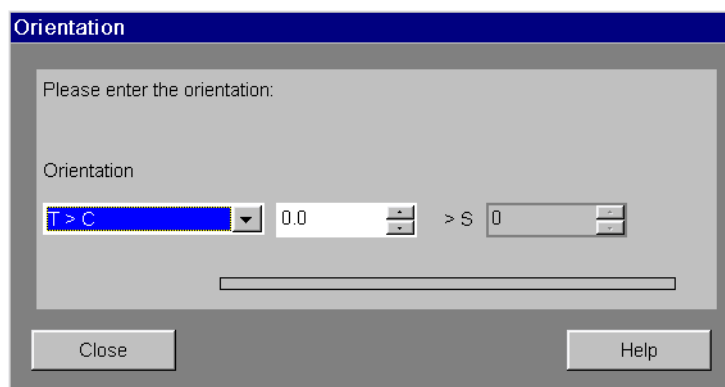
The following settings are available:

Adjustment	Description
On	Phase correction is performed.
Off	No phase correction
Auto	Whether or not a phase correction is performed, depends on the type of the system and the type of the protocol.



This parameter should only be adapted by experienced users!

Orientation dialog window



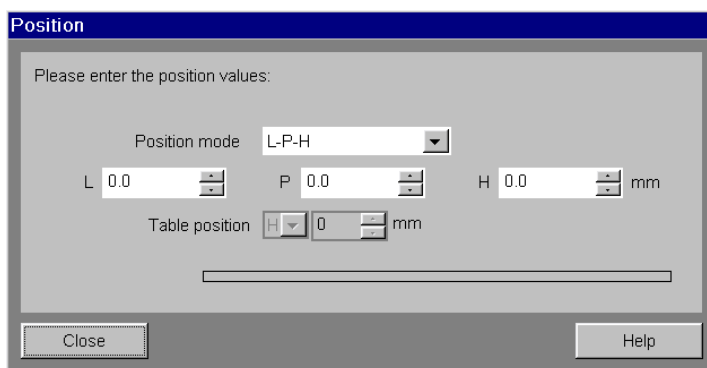
(1) *Orientation*

This parameter determines the orientation of the slice and/or slab referenced to the *Whole Body Patient Coordinate System*.

→ [Page A.2–5](#), *Whole Body Patient Coordinate System*

- ☐ Selection list, left:
Determines the direction of the tilt
- ☐ Input field, center
Determines the flip angle (for single oblique slices)
- ☐ Input field, right:
Determines the flip angle for the third orientation plane (for double-oblique slices)

Position dialog window



(1) *Position mode*

This parameter determines the direction for shifting the center of the object out of the magnet isocenter.

- ☐ In the **L-P-H** mode, enter the offset based on the *Whole Body Patient Coordinate System*:

→ [Page A.2–5, Whole Body Patient Coordinate System](#)

L – to the left

P – to posterior

H – towards the head

Enter a negative value to move in the opposite direction (right, anterior, and feet).

- ☐ Enter the shift of the slice from the planned isocenter in the gradient direction in the **Offcenter-Shift** mode:
 - Phase** – in the phase-encoding direction
 - Read** – in the readout direction
 - Shift** – in the slice-selection direction
- During volume positioning (e.g., VOI, adjustment volume), you *always* work in **L-P-H** mode. The **Position mode** selection list is *not* available.
- The selected position mode is maintained after the current positioning. When you open the **Position** dialog window the next time (even from another protocol), the position mode previously selected is still applied.

(2)

The screenshot shows a horizontal bar with three input fields. The first field is labeled 'L' and contains the value '3.9'. The second field is labeled 'P' and contains the value '10.9'. The third field is labeled 'H' and contains the value '0.2'. To the right of the 'H' field is the unit 'mm'. Each input field has a small square button with a plus/minus sign to its right.

These parameters determine the direction for shifting the center of the object out of the magnet isocenter.

(3) *Table position*

This parameter determines the table position where the protocol will be measured. The zero point is defined by the initial table position of the first measurement of a series block.

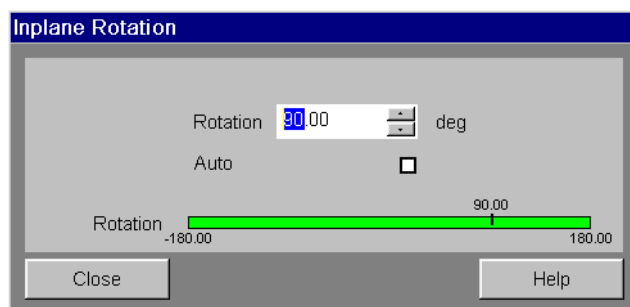
→ [Page C.4–26](#), *Dependencies between the table position, reference image, and protocol*

The selection list defines the direction of movement.

- ☐ **H**: In the head direction
- ☐ **F**: In the foot direction

The input field defines the distance in mm.

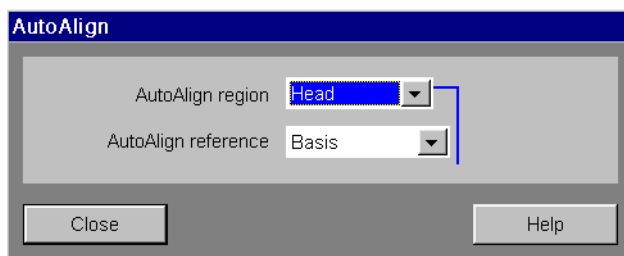
Inplane Rotation dialog window



When activating the **Auto** checkbox, rotation in the slice plane is automatically calculated when the **Rotation** input window is dimmed.

- The **Auto** checkbox is available only when another mode than **Coronal** is selected in the **Orientation** parameter.

AutoAlign dialog window



(1) *AutoAlign region*

This parameter defines the anatomical region for **AutoAlign**, e.g., Head, Spine, Knee.

➤ The adaptable regions depend on your AutoAlign license.

(2) *AutoAlign reference*

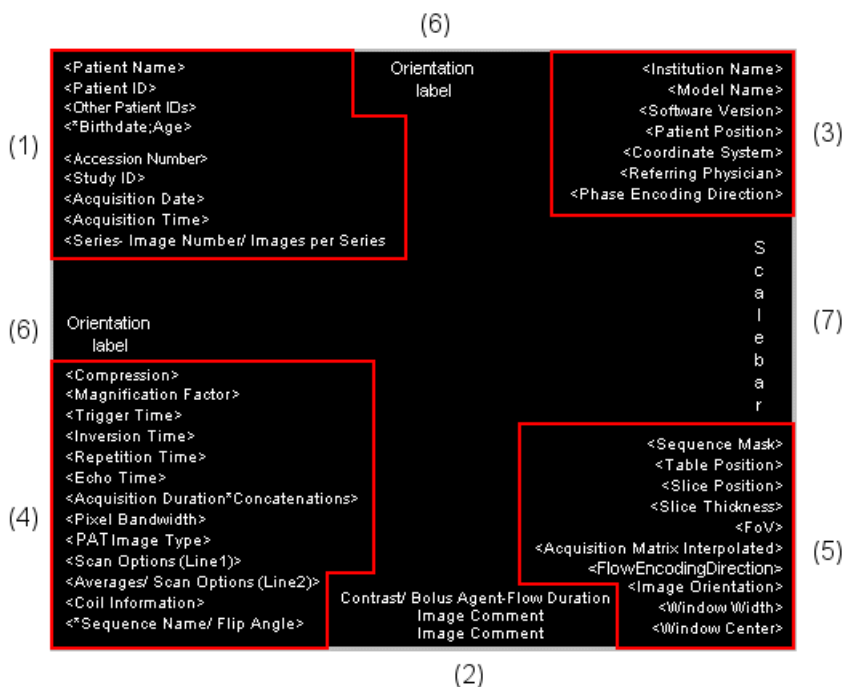
At least one AutoAlign reference is predefined for every anatomical region. Usually, there are several predefined references for the head region.

For example see → [Page C.19–2](#), *AutoAlign head matrices*.

Text Information in Medical Images

Various information is shown as image text on medical images. This information is used to identify the patient and to document measurement and image parameters.

The image text is sorted by topic and positioned in the four corners of the image.



- (1) Patient and examination data
- (2) Comment lines
- (3) System-specific information
- (4) Examination and image parameters
- (5) Data regarding the MR image
- (6) Orientation mark
- (7) Scale bar

Content of the image text components

Here is a list of the individual items of information in the image text components.

- (1) Patient and examination data

The top left corner shows the patient and examination data.
➤ Text lines *Other Patient IDs* and *Accession Number* are suppressed when empty. All subsequent lines are moved upward.

Image text	Example	Description
Patient name	Peter Patient	
Patient ID	02081948	
Additional patient number	0123456789	
Date of birth, age	*11/11/1970; 30Y	
Accession number	# 12345	
Examination Number	STUDY 1	
Date acquired	17.10.01	
Time acquired	13:47:46	
Series and image number and total number of images	2 IMA 5/7	Series 2, a total of 7 images in series 2, image 5 is displayed

(2) Comment lines

At the lower edge next to the measurement and image parameters, data regarding contrast agent administration as well as an image comment are provided.

➤ The second line of the image comment is optional.

Image text	Example	Description
Contrast medium/time after injection in sec	GADOLINIUM 1.1	1.1 s after administering GADOLINIUM
Image comment	Motion: 0.27, -0.07, 1.29, -0.47, 0.11, -0.01	Motion correction parameter

(3) System-specific information

The top right corner shows the name of your hospital or practice, the system name, the software version as well as the patient position and direction of viewing.

Image text	Abbreviation	Example
Institution	-	St. Mary's, New York
System name	-	MAGNETOM Verio
Software version	-	MR B15V
Patient position (orientation and position) and viewing orientation	HFS - Head First Supine HFP - Head First Prone HFDR - Head First Decubitus Right HFDL - Head First Decubitus Left FFS - Feet First Supine FFP - Feet First Prone FFDR - Feet First Decubitus Right FFDL - Feet First Decubitus Left	

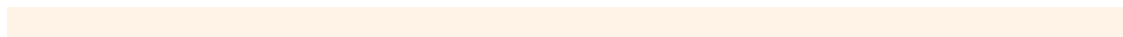


Image text	Abbreviation	Example
Coordinate system	LPH (patient coordinate system left-posterior-head)	+LPH
Referring physician		Dr. Mustermann
Phase-encoding direction		→

(4) Examination and image parameters

In the bottom left corner you can see the parameters used to generate the image.

Image text	Abbreviation	Example
Rate of compression	CR (<i>Compression Ratio</i>)	CR 100
Loss of quality through compression	How much storage space is saved (bits) 100 means that: an image that is normally 512 KB, is now only 5.12 KB This applies to lossless compression as well. CQ (<i>Compression Quality</i>) Measure for loss on resolution 100 = no loss, 0 =greatest loss on resolution CQ! Image is compressed, however, the numbers for CR or CQ are missing ⇒ Either CR or CQ is shown.	CQ 75 CQ!
Current zoom factor	MF	MF 1.51
Trigger time	TT in msec	TT 50
Inversion time	TI in msec	TI 50
Repetition time	TR in msec	TR 600
Echo time	TE in msec	TE 30
Measurement duration * concatenations	TA HH:MM:SS (hours range); MM:SS (minutes range); SS:HS (seconds range)	TA 02:36*4
Pixel bandwidth	BW in Hertz/pixels	BW 1000.0
Image type	→ Page P.2–17 , <i>List of image types</i> p(x) - Grappa with acceleration factor x P(x) - Sense with acceleration factor x	p2 M/MIP
Measurement options (1st line)	RG - Respiratory Gating CG - Cardiac Gating RT - Respiratory Trigger CT - Cardiac Trigger EXT - External Trigger	RG/CT

Image text	Abbreviation	Example
Averages/measurement options (2nd line)	A(y) - Number of averages FS - fat saturation WS - Water saturation WE - Water excitation SAT (y) - Number of saturation regions DB - Dark blood IR - Inversion recovery SR - Saturation recovery G(x) - Grid tag with x mm grid distance or L (x) - Line tag with x mm line distance MT - MTC pulse PPF - Phase Partial Fourier	A4/FS/SAT1/MT
Coil information	→ Page P.2–8 , <i>List of coil abbreviations in the image text</i>	NE1;SP1
Sequence name/flip angle	A * in front of the sequence name means that an original Siemens sequence was used.	*TS2_23 80

(5) Data about the MR image

The bottom right corner shows data regarding the position, orientation, and thickness of the acquired slice, about the extent (FoV), as well as the window values.

⇒ The text line *Flow Encoding Direction* is suppressed when empty. All subsequent lines are moved upward.

Image text	Abbreviation	Example
Sequence mask	D - Door open I - Interpolation R - Raw data filter E - Elliptical filter	DII
Table position (H,F)	TP H/F in mm	TP H300
Slice position (+LPH)	SP in mm	SP L8.7
Slice thickness	SL in mm	SL 4.0
Interpolated slice thickness	SL i in mm	SL 4.0 i
FoV (field of view)	FoV in mm	FoV 230×230
Acquisition matrix interpolated	I - Interpolation	
Acquisition matrix not interpolated	p - partial Fourier s - swap	192p×256s 192p×256s
Image orientation	Sag, Tra, Cor	Tra>Cor(-20)>Sag(-30)
Flow-encoding direction	v(XXX)_inplane_xy - in-plane flow encoding v(XXX)_through - through-plane flow encoding XXX - flow velocity in cm/sec xy - direction, e.g., fh (feet to head)	v150_inplane_fh
Window width (contrast)	W	W -100
Window center (brightness)	C	C -200

List of coil abbreviations in the image text

The coils and coil elements used are shown in the lower left image text.

<Patient Name>	Orientation	<Institution Name>
<Patient ID>	label	<Model Name>
<Other Patient IDs>		<Software Version>
<*Birthdate;Age>		<Patient Position>
<Accession Number>		<Coordinate System>
<Study ID>		<Referring Physician>
<Acquisition Date>		<Phase Encoding Direction>
<Acquisition Time>		
<Series> Image Number/ Images per Series		S
		c
		a
		l
		e
		b
		a
		r
Orientation		
label		
<Compression>		
<Magnification Factor>		
<Trigger Time>		
<Inversion Time>		<Sequence Mask>
<Repetition Time>		<Table Position>
<Echo Time>		<Slice Position>
<Acquisition Duration*Concatenations>		<Slice Thickness>
<Pixel Bandwidth>		<FoV>
<PAT Image Type>		<Acquisition Matrix Interpolated>
<Scan Options (Line1)>		<FlowEncodingDirection>
<Averages/ Scan Options (Line2)>		<Image Orientation>
<Coil Information>	Contrast/ Bolus Agent Flow Duration	<Window Width>
<*Sequence Name/ Flip Angle>	Image Comment	<Window Center>
	Image Comment	

The following notation is used:

- The matrix coil mode is located in front of the coils and is separated by a colon.
- The coils are sorted alphabetically in ascending order and separated by a semi-colon.
- The elements of the coils are sorted numerically in ascending order and separated by commas.

Example

C:BO1,3;SP1,3,5-7
The system automatically selected the *CP* matrix coil mode. Coils *B* and *SP* were used for the measurement. For coil *B*, elements 1 and 3 were selected, for coil *SP* elements 1,3,5,6,7 were selected.

! Please note: The coil abbreviations for Biograph mMR are listed in the syngo MR B20P Operator Manual.

Matrix coil mode (all MAGNETOMs)

Image text	Matrix coil mode
C c	CP Mode selected by the system selected manually
D d	Dual mode selected by the system selected manually
T t	Triple Mode selected by the system selected manually
(MAGNETOM ESSENZA) V v	Various matrix coil modes selected by the system selected manually

MAGNETOM Verio

Image text	Coil name
BA1 to BA4	16CH_BodyArray_Anterior
BL and BR	4Ch_Breast
BO1 and BO2	BodyMatrix
BP1 to BP4	16CH_BodyArray_Posterior
EN	Endorectal coil
EX	TxRx_Extremity
FA	8Ch_FootAnkle
FL	FlexLarge
FS	FlexSmall
HE	TxRx_Head
HEA and HEP	HeadMatrix 32Ch_Head
KN	8Ch_Knee TxRx_15Ch_Knee
L4	Loop_4
L7	Loop_7
L11	Loop_11
LBR and MBR	7Ch_Breast
NE1 and NE2	NeckMatrix
PL1 to PL6 and PR1 to PR6	PAMatrix
SC	4CH Special Purpose
SHL	ShoulderLarge
SHS	ShoulderSmall
SP1 to SP8	SpineMatrix

Image text	Coil name
WR	8Ch_Wrist

MAGNETOM Trio

Image text	Coil name
BA1 to BA4	16CH_BodyArray_Anterior
BL and BR	4Ch_Breast
BO1 and BO2	BodyMatrix PAMatrix
BP1 to BP4	16CH_BodyArray_Posterior
CL and CR	Carotid
EN	Endorectal
EX	CP_Extremity
FA	8Ch_FootAnkle
FL	FlexLarge
FS	FlexSmall
HE	TxRx_Head
HEA and HEP	HeadMatrix 32Ch_Head
KN	8Ch_Knee
L4	Loop_4
L7	Loop_7
L11	Loop_11
LBR and MBR	7Ch_Breast
MBR	7Ch_Breast
NE1 and NE2	NeckMatrix
PH	8Ch_Head

Image text	Coil name
PL1 to PL6 and PR1 to PR6	PAMatrix
SC	4CH Special Purpose
SHL	ShoulderLarge
SHS	ShoulderSmall
SP1 to SP8	SpineMatrix
TEM	TEM_3000
WR	CP_Wrist 8Ch_Wrist

MAGNETOM Avanto and Symphony

Image text	Coil name
1H and 31P	HeartLiver
BA1 to BA4	16CH_BodyArray_Anterior
BL	4Ch_Breast (head first) 4Ch_Breast_Machnet BreastMatrix
BO1 and BO2	BodyMatrix
BP1 to BP4	16CH_BodyArray_Posterior
BR	4Ch_Breast (head first) 4Ch_Breast_Machnet BreastMatrix
BRL and BRR	BreastArray
CL and CR	Carotid
DLL	DoubleLoopLeft
DLR	DoubleLoopRight
EN	Endorectal

Image text	Coil name
EX	CP_Extremity Extremity_MAI
FA	8Ch_FootAnkle
FL	CP_FlexLarge FlexLarge
FLL	FlexLoopLarge
FLS	FlexLoopSmall
FS	CP_FlexSmall FlexSmall
HE	CP_HeadArray TxRx_Head
HE1 and HE3	HeadMatrix (upper part)
HE2 and HE4	HeadMatrix (lower part)
HEA and HEP	32Ch_Head
HL1_P2	HeartLiver 31P
HRH	HR_Head
KN	HR_Knee
L4	Loop_4
L7	Loop_7
L11	Loop_11
LBR	7Ch_Breast
LEF	LowerExFlex
MBR	7Ch_Breast
NE1 and NE2	NeckMatrix
NH	NeonateHead
PH	8Ch_Head
PH1	LargePediatric SmallPediatric

Image text	Coil name
PL1 to PL3	LargePediatric PAMatrix
PL4	PAMatrix
PR1 to PR4	PAMatrix
PS1 to PS3	SmallPediatric
RBR	7Ch_Breast
SC	4CH Special Purpose
SEA	SmallExtremity
SH	ShoulderArray
SP1 to SP8	SpineMatrix
UEF	UpperExFlex
WR	8Ch_Wrist WristArray
WR2	WristArray
WT	WristArray_MAI

MAGNETOM Espree

Image Text	Coil name
BA1 to BA4	16CH_BodyArray_Anterior
BL	4CH_Breast_Confirma 4CH_Breast_Machnet BreastMatrix
BO1 and BO2	BodyMatrix
BP1 to BP4	16CH_BodyArray_Posterior
BR	4CH_Breast_Confirma 4CH_Breast_Machnet BreastMatrix
BRR and BRL	CP_BreastArray
CL and CR	CarotidArray
DLL	DoubleLoopArrayLeft
DLR	DoubleLoopArrayRight
EN	Endorectal
EX	CP_Extremity Extremity_MAI
FA	8Ch_FootAnkle
FL	CP_FlexLarge FlexLarge
FLL	FlexLoopLarge
FLS	FlexLoopSmall
FS	CP_FlexSmall FlexSmall
HE	CP_HeadArray TxRx_Head
HE1 and HE3	HeadMatrix (upper part)
HE2 and HE4	HeadMatrix (lower part)

Image Text	Coil name
HEA and HEP	32Ch_Head
HEB	OR_HeadBottom
HET	OR_HeadTop
KN	HR_Knee
L4	Loop_4
L7	Loop_7
L11	Loop_11
LBR MBR	7Ch_Breast
NE1 and NE2	NeckMatrix
NH	NeonateHead
PH1	LargePediatric
PL1 to PL3	LargePediatric PAMatrix
PL4	PAMatrix
PR1 to PR4	PAMatrix
RBR	7Ch_Breast
SC	4CH Special Purpose
SEA	SmallExtremity
SH	ShoulderArray
SP1 to SP8	SpineMatrix
WR	8Ch_Wrist WristArray

List of image types

The image type is shown in the lower left image text.

<Patient Name>	Orientation	<Institution Name>
<Patient ID>	label	<Model Name>
<Other Patient IDs>		<Software Version>
<*Birthdate;Age>		<Patient Position>
<Accession Number>		<Coordinate System>
<Study ID>		<Referring Physician>
<Acquisition Date>		<Phase Encoding Direction>
<Acquisition Time>		
<Series- Image Number/ Images per Series>		S
		c
		a
Orientation		l
label		e
		b
<Compression>		a
<Magnification Factor>		r
<Trigger Time>		
<Inversion Time>		<Sequence Mask>
<Repetition Time>		<Table Position>
<Echo Time>		<Slice Position>
<Acquisition Duration*Concatenations>		<Slice Thickness>
<Pixel Bandwidth>		<FoV>
<PAT Image Type>		<Acquisition Matrix Interpolated>
<Scan Options (Line1)>		<FlowEncodingDirection>
<Averages/ Scan Options (Line2)>		<Image Orientation>
<Coil Information>	Contrast/ Bolus Agent-Flow Duration	<Window Width>
<*Sequence Name/ Flip Angle>	Image Comment	<Window Center>

An image may include more than one image type, e.g., after several post-processing steps. These are appended to one another and separated by a forward slash. If the maximum line length is exceeded, the last image types are omitted.

Example

Assuming the defined maximum length is 18 characters:

From **TYP1/TYP2/TYP3/TYP4** (19 characters)

will be truncated to **TYP1/TYP2/TYP3** (14 characters).

Abbreviation	Image type	Topic
3D EDITOR	Image resulting from 3D editor processing	3D reconstruction
3D FUSION	Fusion image	3D reconstruction
ADC	Apparent Diffusion Coefficient Map	Diffusion, Perfusion
ADD	Image resulting from an addition	Dynamic Analysis
AVERAGE	VRT image	3D reconstruction
BLACK IMAGE	Graphic image	3D reconstruction
BLACK IMAGE	Graphics and table image	Mean Curve
COMB	Combination image	Diffusion, Perfusion
COMPOSED	Composed Images	Image Composing
COR	Correlation image	Functional imaging Dynamic Analysis
CV(xx)	Filtered Images (xx stands for filter strength, integer value)	Dynamic Analysis
DIFFER	Differentiation image	Dynamic Analysis
DIFFUS	Diffusion image	Dynamic Analysis
DIS2D	2D distortion corrected image	Inline post-processing
DIS2D*	2D distortion-corrected image. Cannot be used as a reference image for positioning, because measurement is too far off-center.	Inline post-processing
DIS3D	3D_distortion-corrected image	Inline post-processing
DIV	Division image	Dynamic Analysis
E1	Diffusion image that shows the weighting of the eigenvector E1	Diffusion
E1-E2	Diffusion image that shows the weighting of the eigenvector ratio E1-E2	Diffusion
E1-E3	Diffusion image that shows the weighting of the eigenvector ratio E1-E3	Diffusion
E2	Diffusion image that shows the weighting of the eigenvector E2	Diffusion

Abbreviation	Image type	Topic
E2-E3	Diffusion image that shows the weighting of the eigenvector ratio E2-E3	Diffusion
E3	Diffusion image that shows the weighting of the eigenvector E3	Diffusion
EXP	Exponential Map	Diffusion
FA	Fractional Anisotropy Map	Diffusion
FIL	Inline Image Filter	Inline post-processing
FILTERED	BOLD or perfusion image with spatial filtering	Inline post-processing
FS(x_Y), FSH(x_y), FM(x_y), FS, FSH or FM	Offline Image Filter	Dynamic Analysis
GBP	Global time-density curve (Global Average Bolus Curve Plotted; passage of bolus)	Diffusion, Perfusion
GSP	Reference image with graphic objects	Graphic slice position
INT	Integration image	Dynamic Analysis
LIN	Barycentric Map, category "Linear"	Diffusion
LOG	Logarithmic function image	Dynamic Analysis
LOW_B	Image with low diffusion weighting	Diffusion
M	Rephased image (magnitude image)	Flow quantification
MAG	Magnitude image	Phase contrast Angio
MEAN	Mean value image	Functional imaging Dynamic Analysis
MIP	Maximum Intensity Projection image	3D reconstruction Inline post-processing
MIP THIN	MIP Thin image	3D reconstruction
MIP_COR	Coronal image from MIP reconstruction	Inline post-processing
MIP_SAG	Sagittal image from MIP reconstruction	Inline post-processing
MIP_TRA	Transverse image from MIP reconstruction	Inline post-processing
MITP	Maximum Intensity Time Projection image	Diffusion, Perfusion

Abbreviation	Image type	Topic
MOCO	BOLD or perfusion image with motion correction	Inline post-processing
MOSAIC	Mosaic image	Functional imaging
MPR	Multiplanar reconstruction image (MPR)	3D reconstruction
MPR CURVED	Curved MPR image	3D reconstruction
MPR THICK	MPR Thick image	3D reconstruction
MPR THIN	MPR Thin image	3D reconstruction
MSUM	Sum image from magnitude images of a slice	Phase contrast Angio
MULT	Multiplication image	Dynamic Analysis
ND	non-distortion corrected image	Inline post-processing
NORM	Image after normalization algorithm	Inline post-processing
OTHER	Converted N3/3.5 image or SC (secondary captured) image	Import/Export
P	Phase contrast image	Flow quantification
PBP	Percentage signal image (Percent-of-Baseline at Peak; bolus peak relative to baseline).	Diffusion, Perfusion
PDT1	T1-weighted proton density image	Dynamic Analysis
PDT2	T2-weighted proton density image	Dynamic Analysis
PEI	Positive Enhancement Integral image	Diffusion, Perfusion
PLAN	Barycentric Map, category "Planar"	Diffusion
POSDISP	Position display image	Position display
PROJECTION IMAGE	Projection image	Inline post-processing
R	Real image	Flow quantification
RA	Relative Anisotropy Map (relative FA)	Diffusion
RELCBF	Relative CBF image (relat. cerebr. blood flow)	Diffusion

Abbreviation	Image type	Topic
RELCBV	Relative CBV image (relat. cerebr. blood volume)	Diffusion
RELMTT	Relative MTT image (relat. average transit time)	Diffusion, Perfusion
RESAMPLED	Zooming/panning image	3D reconstruction
RETRO	Retro-Gating image	Inline post-processing
SCM	Signal Change image	Diffusion, Perfusion
SDEV	Standard deviation image	Dynamic Analysis
SLOPE	SLOPE evaluation result	Dynamic Analysis
sMRI	Structural MRI (Anatomy)	Diffusion
SPEC	Spectrum	Spectroscopy
SPHER	Barycentric Map, category "Spherical"	Diffusion
SSD	SSD reconstruction (Shaded Surface Display)	3D reconstruction
STDDEV_COR	Coronal standard deviation image	Inline post-processing
STDDEV_SAG	Sagittal standard deviation image	Inline post-processing
STDDEV_TRA	Transverse standard deviation image	Inline post-processing
SUB	Subtraction image	Dynamic Analysis
SUBTRACT	Subtraction image	3D reconstruction
T1	T1 image	Dynamic Analysis
T2	T2 image	Dynamic Analysis
TCS	Time Course of Signal	Diffusion, Perfusion
TG	Tensor graphic	Diffusion
TMIP	Temporal MIP, MIP reconstruction of time sequence of images	Inline post-processing
TMIP_COR	Coronal TMIP image	Inline post-processing
TMIP_SAG	Sagittal TMIP image	Inline post-processing
TMIP_TRA	Transverse TMIP image	Inline post-processing

Abbreviation	Image type	Topic
TRACE	Trace weighted image	Diffusion
TSTDDEV	Temporary standard deviation image	Inline post-processing
TTEST	Student's t-test image (for BOLD images)	Functional imaging Dynamic Analysis
TTP	Time To Peak image	Diffusion, Perfusion
TU	The protocol parameters were automatically changed by a tool (e. g. AutoAlign Spine)	General
TU M	Protocol parameters were automatically changed by a tool (e. g. AutoAlign Spine) and then one more time manually.	General
VR	Volume Ratio Map	Diffusion
VRT	VRT (Volume Rendering Technique) image	3D reconstruction
WI	Wash In image	Diffusion, Perfusion
WO	Wash Out image	Diffusion, Perfusion

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© Siemens AG 2009-2012
Order No.
MR-05000.621.19.01.24
12/2012

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